Synopsis 2026
§-LANG s1 Translation

Peer-Reviewed Volumes 0-IV · Canonical IDs 1-5000

Trasgo symbolic transport · version 1.1.0-s1

Round trip: 5000/5000 exact source records · hard edges: 6924 preserved · evidence: A=2268 M=2364 S=365 H=3 P=0
Authority: symbolic transport only; no evidence promotion and no asserted runtime semantics for custom SYNOPSIS_ENTRY blocks.

Transport record

T[source Synopsis -> §-LANG s1](record) = (SYNOPSIS_ENTRY, preserved IDs/statements/evidence/dependencies/provenance, transformed field layout and dialect chart, lost PDF typography/page geometry, active remainder, exact source_b64 witness).

§-LANG strata used here: L_syn for addressing/compression; L_form for exact statement/assumption/dependency surfaces; L_gov for evidence and correction history; L_op for bounded M/A operator fixtures; L_sem/L_ctl where architecture or orchestration is explicitly part of the source record.

§|LANG|SYNOPSIS_PEER_REVIEW

decision = Accept_for_unified_reference_use_with_corrections
graph = 5000_objects / 6924_hard_edges / 1_root / 1_component / 0_cycles / 0_dangling
metadata_remainder = 1196_A_records_with_empty_assumptions_in_Volumes_I_III
corrections = 5; evidence_promotion = forbidden

§CORRIGENDUM[S26-T-00002158] mathematical correction

new := For sigma-finite positive measures mu,nu with nu << mu, there exists a nonnegative measurable density f, unique mu-a.e., with nu(E)=integral_E f dmu for every measurable E; if nu is finite, then f is in L^1(mu).

remainder := Sigma-finiteness alone does not imply a globally L1 density; Lebesgue measure with respect to itself on R has derivative 1, which is not in L1(R).

§CORRIGENDUM[S26-T-00002262] scope correction

new := If U is a connected open subset of C and f:U->C is holomorphic and nonconstant, then f is an open map. On a disconnected domain, apply the conclusion componentwise on components where f is nonconstant.

remainder := Global nonconstancy on a disconnected domain is insufficient when f is constant on one component.

§CORRIGENDUM[S26-T-00003053] foundation-scope clarification

new := In the classical ZFC reading used for this theorem, an arbitrary product of compact spaces is compact in the product topology.

remainder := The arbitrary-product theorem requires an explicit classical choice convention in a project that distinguishes foundation contexts.

§CORRIGENDUM[S26-T-00004062] mathematical correction

new := Starting from the base field K, k Cayley-Dickson doublings produce a K-vector space of dimension 2^k. More generally, starting from a finite-dimensional involutive K-algebra A_0 of dimension d, k doublings produce K-dimension d*2^k. Algebraic laws retained or lost depend on the starting algebra and doubling parameters, not on geometric dimension n.

remainder := The released statement mixed a general starting algebra with the special 2^k dimension formula, which is exact only when the starting K-dimension is one.

§CORRIGENDUM[S26-D-00004341] terminology/scope clarification

new := A building mesh uses vertices, chambers and incidence inherited from the chosen Bruhat-Tits object. In this registry, a combinatorial geodesic means a shortest path in the declared vertex graph or a shortest gallery in the declared chamber metric; this terminology is kept distinct from geodesics in a geometric realization of a higher-rank building.

remainder := The released wording could conflate graph/gallery geodesics with metric geodesics in a geometric realization for higher-rank buildings.

§PACK Synopsis_2026_Volume_0_sLANG_S1

volume := 0 · Foundations, Language, Registry & Epistemic Protocol

canonical_range := 1-1000

§CLAIM[S26-N-00000001|N|S|Proof+Compile+Meta] Object language and metalanguage

§FORM := A mathematical assertion is interpreted only relative to a declared ambient language or formal context. Mentions of expressions are distinguished from the objects denoted by those expressions.

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§EVIDENCE := S · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000002|D|A|Proof+Compile+Meta] Proposition

§FORM := Within a declared foundational context, a proposition is an expression for which the context supplies a notion of assertion/truth or inhabitation. No global identification of propositions with sets, types, or Boolean values is imposed.

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§DEP_HARD := S26-N-00000001

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000003|D|A|Proof+Compile+Meta] Predicate

§FORM := Given an ambient collection/type/domain `X`, a predicate on `X` is a proposition-valued expression `P(x)` well-formed for `x : X` (or `x in X`, according to the active foundation).

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§DEP_HARD := S26-D-00000002

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00000004|N|S|Proof+Compile+Meta] Quantifier convention

§FORM := `forall x in X, P(x)` and `exists x in X, P(x)` are binder notation. Their exact semantics are inherited from the active foundation; bounded set-theoretic and typed readings are not silently identified.

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§DEP_HARD := S26-D-00000003

§EVIDENCE := S · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000005|D|A|Proof+Compile+Meta] Equality relation in context

§FORM := An equality expression `x =_X y` denotes the equality notion supplied for objects `x,y` in ambient context `X`. Definitional equality, propositional equality, extensional equality, isomorphism, and equivalence are distinct unless an explicit bridge is present.

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§DEP_HARD := S26-N-00000001

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00000006|N|S|Proof+Compile+Meta] Equivalence is not equality

§FORM := An isomorphism, homeomorphism, diffeomorphism, equivalence of categories, logical equivalence, or computational observational equivalence must not be rewritten as literal equality without a declared quotient, univalence principle, canonical identification, or other bridge that licenses the transport.

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§DEP_HARD := S26-D-00000005

§EVIDENCE := S · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000007|D|A|Proof+Compile+Meta] Binary relation

§FORM := A binary relation from `X` to `Y` is a predicate `R(x,y)` on pairs with `x` in `X` and `y` in `Y` in the active foundational reading.

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§DEP_HARD := S26-D-00000003

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000008|D|A|Proof+Compile+Meta] Function / map

§FORM := A map `f : X -> Y` assigns to every admissible `x` of `X` a uniquely determined value `f(x)` of `Y`, interpreted in the active foundation. The graph-of-a-function encoding is one representation, not the universal definition.

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§DEP_HARD := S26-D-00000005, S26-D-00000007

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000009|D|A|Proof+Compile+Meta] Composition

§FORM := For maps `f : X -> Y` and `g : Y -> Z`, their composite `g o f : X -> Z` is defined by `(g o f)(x) = g(f(x))` whenever the expressions are well-typed/well-formed.

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§DEP_HARD := S26-D-00000008

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000010|T|A|Proof+Compile+Meta] Associativity of composition

§FORM := For composable maps `f : W -> X`, `g : X -> Y`, and `h : Y -> Z`, `h o (g o f) = (h o g) o f` extensionally, subject to the equality convention of the ambient foundation.

§ASSUME := Interpretation is relative to the declared ambient foundation/context.

§DEP_HARD := S26-D-00000005, S26-D-00000009

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000011|D|A|Proof+Compile+Meta] Identity map

§FORM := For an ambient object/domain `X`, `id_X : X -> X` is given by `id_X(x)=x`.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000008
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000012|T|A|Proof+Compile+Meta] Identity laws

§FORM := For `f : X -> Y`, `id_Y o f = f` and `f o id_X = f` extensionally.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000009, S26-D-00000011
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000013|D|S|Proof+Compile+Meta] Structure

§FORM := A structure on an ambient carrier/object consists of specified data, operations, relations, or properties together with declared laws. The word “structure” is schematic; every use in later volumes must identify its signature or canonical registry object.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-N-00000001
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000014|D|S|Proof+Compile+Meta] Structure-preserving map

§FORM := Given specified structures on `X` and `Y`, a structure-preserving map is a map whose declared preservation obligations hold. Homomorphism, linear map, continuous map, smooth map, functor, etc. are specializations and are not interchangeable.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000008, S26-D-00000013
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000015|D|A|Proof+Compile+Meta] Isomorphism relative to structure

§FORM := An isomorphism between structured objects is a structure-preserving map possessing an inverse that is structure-preserving in the same declared category/context.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000014, S26-T-00000012
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00000016|N|S|Proof+Compile+Meta] Foundational pluralism

§FORM := The Synopsis does not decree that all mathematics is intrinsically set-theoretic, type-theoretic, or categorical. A result may declare a foundation-neutral interface or a specific foundation. Transport between foundations is a typed bridge with assumptions and remainder.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-N-00000001
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000017|D|A|Proof+Compile+Meta] Set (interface-level)

§FORM := At the Synopsis interface level, a set is an object supporting membership propositions `x in X` under a declared set-theoretic or set-like semantics. Specific axiomatizations (e.g. ZF/ZFC) are separate objects.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000002, S26-N-00000016
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000018|D|A|Proof+Compile+Meta] Type (interface-level)

§FORM := At the Synopsis interface level, a type is a domain of terms equipped with the formation and equality rules of a declared type theory. No automatic identification with a set is assumed.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000005, S26-N-00000016
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00000019|B|S|Proof+Compile+Meta] Set/type transport boundary

§FORM := Statements expressed over sets and statements expressed over types may correspond under an explicit interpretation, model, coercion, or embedding. Such correspondence preserves only the properties established for that bridge; it does not license unrestricted substitution between foundations.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000017, S26-D-00000018
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000020|D|A|Proof+Compile+Meta] Preorder

§FORM := A preorder on `X` is a binary relation `<=` that is reflexive and transitive.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000007
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000021|D|A|Proof+Compile+Meta] Partial order

§FORM := A partial order is a preorder that is antisymmetric: `x <= y` and `y <= x` imply `x = y` in the declared equality.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000005, S26-D-00000020
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000022|D|A|Proof+Compile+Meta] Equivalence relation

§FORM := An equivalence relation on `X` is a binary relation that is reflexive, symmetric, and transitive.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000007
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000023|D|A|Proof+Compile+Meta] Finite set (cardinality interface)

§FORM := A set `X` is finite when it is in bijection with `{0,...,n-1}` for some natural number `n`, under a foundation where this formulation is available. Alternative constructive notions of finiteness are not silently collapsed into this definition.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000008, S26-D-00000017, S26-D-00000031, S26-D-00000040
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000024|D|S|Proof+Compile+Meta] Hard dependency

§FORM := A hard dependency edge `R -> Q` asserts that the exact statement of `R` requires `Q` as a definition, assumption, theorem, or semantic prerequisite. A release cannot contain `R` as closed while `Q` is unresolved.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-N-00000001
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000025|D|S|Proof+Compile+Meta] Soft dependency

§FORM := A soft dependency is explanatory, historical, applicative, or comparative and is not required for the logical well-formedness or support of the source entry.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000024
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000026|D|S|Proof+Compile+Meta+IG] Claim-local evidence state

§FORM := An evidence state is attached to an exact claim/version: `P` proved/definitionally closed; `A` axiomatized/assumed/conditional; `M` measured; `H` hypothesis; `S` semantic/design; `R` retired. Evidence tags do not form an automatic linear promotion ladder.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-N-00000001
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-T-00000027|T|S|Proof+Compile+Meta] No-silent-promotion invariant

§FORM := For every released entry, changing its evidence state requires an explicit promotion/demotion record naming the artifact and rule that justifies the transition. Editorial rewriting alone cannot change evidence state.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000026
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000028|D|S|Proof+Compile+Meta+IG] Typed remainder

§FORM := A remainder records information, assumptions, distinctions, or uncertainty not preserved by a translation, bridge, compression, formalization, or editorial projection.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-N-00000001
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-T-00000029|T|S|Proof+Compile+Meta] Remainder preservation invariant

§FORM := A compilation step that is known not to preserve some semantically relevant content must emit that loss as typed remainder rather than silently discarding it.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000028
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00000030|N|S|Proof+Compile+Meta] Proof-status scope

§FORM := A `P` tag means only that the exact registered claim has a proof/definitional artifact accepted by its declared verification policy. It does not certify nearby paraphrases, stronger converses, historical attribution, applications, or philosophical interpretation.
§ASSUME := Interpretation is relative to the declared ambient foundation/context.
§DEP_HARD := S26-D-00000026
§EVIDENCE := S · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00000031|D|A|Proof+Compile+Meta] Natural numbers - interface

§FORM := A natural-number object N consists of a distinguished element 0 and successor operation S, together with an induction/recursion principle sufficient to characterize the standard finite iteration generated from 0. This interface does not identify a particular set-theoretic construction with the concept itself.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-N-00000016

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000032|D|A|Proof+Compile+Meta] Zero and successor

§FORM := For the natural-number interface, 0 is the distinguished initial element and S(n) is the successor of n. In the standard natural numbers, S is injective and 0 is not in its image.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000031

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000033|T|A|Proof+Compile+Meta] Induction principle

§FORM := Let P be a predicate on N. If P(0) holds and, for every n in N, P(n) implies P(S(n)), then P(n) holds for every n in N.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000003, S26-D-00000031, S26-D-00000032

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000034|T|A|Proof+Compile+Meta] Primitive recursion principle

§FORM := Given an object a in X and an update map g : N x X -> X, there is a uniquely determined map f : N -> X satisfying f(0)=a and f(S(n))=g(n,f(n)), whenever the active foundation supplies the standard recursion principle.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000008, S26-D-00000031, S26-D-00000032

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000035|D|A|Proof+Compile+Meta+Tower] Addition on natural numbers

§FORM := Natural-number addition is the recursively determined operation + satisfying n+0=n and n+S(m)=S(n+m).

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-T-00000034

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000036|D|A|Proof+Compile+Meta+Tower] Multiplication on natural numbers

§FORM := Natural-number multiplication is the recursively determined operation * satisfying n*0=0 and n*S(m)=n*m+n.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000035, S26-T-00000034

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000037|T|A|Proof+Compile+Meta] Natural numbers form a commutative semiring

§FORM := With 0, 1=S(0), addition, and multiplication, N satisfies associativity and commutativity of + and *, additive identity 0, multiplicative identity 1, and distributivity of multiplication over addition.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000032, S26-D-00000035, S26-D-00000036

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000038|D|A|Proof+Compile+Meta] Order on natural numbers

§FORM := Define m <= n when there exists k in N such that m+k=n. This relation is the standard linear order on N.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000020, S26-D-00000035

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000039|T|A|Proof+Compile+Meta] Well-ordering of natural numbers

§FORM := Every nonempty subset of N has a least element with respect to the standard order.

§ASSUME := Classical/set-level subset formulation, or an explicitly equivalent well-founded formulation in the active foundation.

§DEP_HARD := S26-D-00000017, S26-D-00000038

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000040|D|A|Proof+Compile+Meta] Finite initial segment

§FORM := For n in N, [n] denotes the finite initial segment {0,1,...,n-1}; equivalently, the naturals k with k<n.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000031, S26-D-00000038

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000041|T|A|Proof+Compile+Meta] Closure of the finite-set definition

§FORM := The forward dependency in S26-D-00000023 is discharged by S26-D-00000031 and S26-D-00000040: the expression "in bijection with {0,...,n-1} for some natural number n" is now internally addressable.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000023, S26-D-00000031, S26-D-00000040

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000042|T|A|Proof+Compile+Meta] Uniqueness of finite cardinality

§FORM := If a set X is in bijection with [m] and also with [n], then m=n. Hence a finite set has a unique natural-number cardinality.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-D-00000023, S26-D-00000040
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000043|D|A|Proof+Compile+Meta] Countably infinite set

§FORM := A set X is countably infinite when there exists a bijection N -> X.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-D-00000008, S26-D-00000017, S26-D-00000031
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000044|D|A|Proof+Compile+Meta] Countable set

§FORM := A set X is countable when it is finite or countably infinite. Equivalent conventions using injections X -> N are admitted only through an explicit classical-choice bridge when needed.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-D-00000023, S26-D-00000043
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := Constructive settings may distinguish formulations classically treated as equivalent.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000045|D|A|Proof+Compile+Meta] Integers - interface

§FORM := The integers Z are the additive group completion of N: they contain an image of N and additive inverses, and are generated by these subject to the expected universal property. Concrete pair/quotient constructions are representations, not the interface definition.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-D-00000031, S26-D-00000022
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00000046|B|A|Proof+Compile+Meta] Natural-number embedding into integers

§FORM := There is a canonical injective map i_NZ : N -> Z preserving 0, 1, addition, multiplication, and order on nonnegative integers.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-D-00000008, S26-T-00000037, S26-D-00000045
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000047|T|A|Proof+Compile+Meta] Integers form a commutative ring

§FORM := With the induced addition and multiplication, Z is a commutative ring with identity in which every element has an additive inverse.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-D-00000045, S26-B-00000046
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000048|D|A|Proof+Compile+Meta] Order on integers

§FORM := For a,b in Z, define a <= b when b-a lies in the embedded nonnegative naturals. This is a translation-invariant linear order compatible with multiplication by nonnegative elements.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-B-00000046, S26-T-00000047, S26-D-00000021
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000049|D|A|Proof+Compile+Meta] Rational numbers - interface

§FORM := The rational numbers Q are the field of fractions of Z: elements may be represented by pairs (a,b) with a in Z and nonzero b in Z, modulo equality a/b=c/d iff ad=bc.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-T-00000047, S26-D-00000022
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00000050|B|A|Proof+Compile+Meta] Integer embedding into rationals

§FORM := There is a canonical injective ring homomorphism i_ZQ : Z -> Q given representationally by a |-> a/1.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-D-00000049, S26-T-00000047
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000051|T|A|Proof+Compile+Meta] Rationals form an ordered field

§FORM := Q is a field, and the order inherited from Z makes Q a linearly ordered field: addition preserves order and multiplication of nonnegative elements is nonnegative.
§ASSUME := Standard natural-number/number-system interface in the declared foundation.
§DEP_HARD := S26-D-00000049, S26-B-00000050, S26-D-00000048
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000052|T|A|Proof+Compile+Meta] Archimedean property of the rationals

§FORM := For every rational q there exists n in N with q<n under the canonical embeddings. Equivalently, the natural numbers are unbounded above in Q.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-B-00000046, S26-B-00000050, S26-T-00000051

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000053|D|A|Proof+Compile+Meta] Cauchy sequence in a metric space

§FORM := A sequence (x_n) in a metric space (X,d) is Cauchy when for every epsilon>0 there exists N such that m,n>=N imply d(x_m,x_n)<epsilon.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000008, S26-D-00000031

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000054|D|A|Proof+Compile+Meta] Real numbers - complete ordered field interface

§FORM := The real numbers R are represented at the interface as a complete Archimedean ordered field containing Q as an ordered subfield. Dedekind-cut and Cauchy-completion constructions are admissible realizations but are not silently identified.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-T-00000051, S26-T-00000052, S26-D-00000053

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000055|T|A|Proof+Compile+Meta] Uniqueness of the real-number interface

§FORM := Any two complete Archimedean ordered fields are isomorphic as ordered fields; once the rational embedding is fixed, the corresponding isomorphism is uniquely determined.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000054, S26-D-00000015

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00000056|B|A|Proof+Compile+Meta] Rational embedding into reals

§FORM := The real-number interface includes an injective ordered-field homomorphism i_QR : Q -> R.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000054, S26-T-00000051

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000057|T|A|Proof+Compile+Meta] Least-upper-bound completeness of the reals

§FORM := Every nonempty subset A of R that is bounded above has a least upper bound in R.

§ASSUME := R is interpreted through the complete ordered-field interface S26-D-00000054.

§DEP_HARD := S26-D-00000017, S26-D-00000021, S26-D-00000054

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000058|D|A|Proof+Compile+Meta] Complex numbers

§FORM := The complex numbers C may be represented as ordered pairs (a,b) in R^2 with addition (a,b)+(c,d)=(a+c,b+d) and multiplication (a,b)(c,d)=(ac-bd,ad+bc). Write a+bi for (a,b), where i=(0,1).

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000054

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000059|T|A|Proof+Compile+Meta] Complex numbers form a field

§FORM := With the operations of S26-D-00000058, C is a field containing an isomorphic copy of R via a |-> (a,0). Every nonzero a+bi has inverse (a-bi)/(a^2+b^2).

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000058, S26-T-00000057

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00000060|I|A|Proof+Compile+Meta] Square of the imaginary unit

§FORM := For i=(0,1) in C, i^2=-1.

§ASSUME := Standard natural-number/number-system interface in the declared foundation.

§DEP_HARD := S26-D-00000058, S26-T-00000059

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000061|D|A|Proof+Compile+Meta] Divisibility in the integers

§FORM := For integers a and b, a divides b, written a | b, when there exists an integer k such that b=ak.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-T-00000047

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000062|T|A|Proof+Compile+Meta] Elementary divisibility laws

§FORM := For integers a,b,c: a|a; if a|b and b|c then a|c; if a|b and a|c then a divides xb+yc for all integers x,y.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000061, S26-T-00000047

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000063|D|A|Proof+Compile+Meta] Common divisor and greatest common divisor

§FORM := A common divisor of integers a,b divides both. For a,b not both zero, gcd(a,b) is the unique positive common divisor divisible by every common divisor.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000061, S26-D-00000048

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000064|D|A|Proof+Compile+Meta] Coprime integers

§FORM := Integers a and b are coprime when gcd(a,b)=1.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000063

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000065|T|A|Proof+Compile+Meta] Division algorithm

§FORM := For integers a and positive b there exist unique integers q,r such that a=bq+r and 0<=r<b.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-T-00000047, S26-D-00000048

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00000066|A|A|Proof+Compile+Meta] Euclidean algorithm

§FORM := Repeatedly replace (a,b) by (b,r), where a=bq+r and 0<=r<b, until r=0. The last nonzero remainder is gcd(a,b).

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000063, S26-T-00000065

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000067|T|A|Proof+Compile+Meta] Bezout identity

§FORM := For integers a,b not both zero there exist integers x,y such that ax+by=gcd(a,b).

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-A-00000066, S26-D-00000063

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000068|T|A|Proof+Compile+Meta] Coprime Bezout criterion

§FORM := Integers a,b are coprime if and only if there exist integers x,y with ax+by=1.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000064, S26-T-00000067

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000069|D|A|Proof+Compile+Meta] Prime natural number

§FORM := A natural number p is prime when p>1 and its only positive divisors are 1 and p.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000031, S26-D-00000038, S26-D-00000061

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000070|D|A|Proof+Compile+Meta] Composite natural number

§FORM := A natural number n>1 is composite when n=ab for natural numbers a,b satisfying 1<a<n and 1<b<n.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000036, S26-D-00000038

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000071|T|A|Proof+Compile+Meta] Prime divisor lemma

§FORM := Every natural number n>1 has a prime divisor.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000069, S26-T-00000039, S26-D-00000061

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000072|T|A|Proof+Compile+Meta] Euclid lemma

§FORM := If p is prime and p divides ab, then p divides a or p divides b.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000069, S26-D-00000061, S26-T-00000067

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000073|T|A|Proof+Compile+Meta] Fundamental theorem of arithmetic

§FORM := Every natural number n>1 can be expressed as a finite product of primes, and this factorization is unique up to permutation of the prime factors.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-T-00000071, S26-T-00000072, S26-D-00000023

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000074|T|A|Proof+Compile+Meta] Infinitude of primes

§FORM := There are infinitely many prime natural numbers.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000069, S26-T-00000073
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000075|D|A|Proof+Compile+Meta] Prime exponent valuation on a positive integer

§FORM := For prime p and positive integer n, v_p(n) is the unique nonnegative exponent e such that p^e divides n and p^(e+1) does not.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000069, S26-T-00000073
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000076|T|A|Proof+Compile+Meta] Valuation of a product

§FORM := For positive integers a,b and prime p, v_p(ab)=v_p(a)+v_p(b).
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000075, S26-T-00000073
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000077|D|A|Proof+Compile+Meta] Least common multiple

§FORM := For positive integers a,b, lcm(a,b) is the least positive integer divisible by both a and b.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000061, S26-D-00000038
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00000078|I|A|Proof+Compile+Meta] GCD-LCM product identity

§FORM := For positive integers a,b, gcd(a,b) lcm(a,b)=ab.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000063, S26-D-00000077, S26-T-00000073
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000079|D|A|Proof+Compile+Meta] Congruence modulo n

§FORM := For integers a,b and positive integer n, a is congruent to b modulo n when n divides a-b.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000061, S26-D-00000048
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000080|T|A|Proof+Compile+Meta] Congruence is an equivalence relation

§FORM := For each positive n, congruence modulo n is reflexive, symmetric, and transitive on Z.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-N-00000004, S26-D-00000079
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000081|T|A|Proof+Compile+Meta] Congruence respects addition and multiplication

§FORM := If a≡b (mod n) and c≡d (mod n), then a+c≡b+d (mod n) and ac≡bd (mod n).
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000079, S26-T-00000047
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000082|D|A|Proof+Compile+Meta] Residue class ring Z/nZ

§FORM := The equivalence classes of integers under congruence modulo positive n form a commutative ring under classwise addition and multiplication.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-T-00000080, S26-T-00000081, S26-T-00000047
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000083|T|A|Proof+Compile+Meta] Invertibility criterion modulo n

§FORM := The residue class of a is invertible in Z/nZ if and only if gcd(a,n)=1.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000082, S26-T-00000067, S26-D-00000063
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000084|T|A|Proof+Compile+Meta] Prime modulus gives a field

§FORM := If p is prime, then Z/pZ is a field.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000069, S26-D-00000082, S26-T-00000083
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000085|T|A|Proof+Compile+Meta] Chinese remainder theorem - two coprime moduli

§FORM := If m,n are positive and coprime, then for any integers a,b there is an integer x satisfying x≡a (mod m) and x≡b (mod n), unique modulo mn.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000064, S26-D-00000079, S26-T-00000067

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000086|D|A|Proof+Compile+Meta] Factorial

§FORM := For n in N, define 0!=1 and (n+1)!=(n+1)n!.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-T-00000034, S26-D-00000036

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000087|D|A|Proof+Compile+Meta] Binomial coefficient

§FORM := For natural numbers 0<=k<=n, define C(n,k)=n!/(k!(n-k)!).

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000086, S26-D-00000038, S26-D-00000049

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000088|T|A|Proof+Compile+Meta] Pascal identity

§FORM := For 1<=k<=n, C(n+1,k)=C(n,k-1)+C(n,k).

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000087, S26-D-00000035

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000089|T|A|Proof+Compile+Meta] Binomial theorem

§FORM := For commuting elements x,y in a commutative ring and n in N, (x+y)^n is the sum from k=0 to n of C(n,k)x^(n-k)y^k.

§ASSUME := x and y commute in a commutative ring; exponentiation is finite natural iteration.

§DEP_HARD := S26-D-00000087, S26-T-00000037

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000090|T|A|Proof+Compile+Meta] Number of subsets of a finite set

§FORM := A finite set with n elements has exactly 2^n subsets.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-T-00000042, S26-D-00000023, S26-D-00000036

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000091|T|A|Proof+Compile+Meta] Number of k-element subsets

§FORM := A finite set with n elements has C(n,k) subsets of cardinality k for 0<=k<=n.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-T-00000042, S26-D-00000087

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000092|T|A|Proof+Compile+Meta] Pigeonhole principle

§FORM := If a finite set with more than n elements is mapped into a set with n elements, the map is not injective.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-D-00000008, S26-D-00000023, S26-T-00000042

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000093|T|A|Proof+Compile+Meta] Addition rule for disjoint finite sets

§FORM := If finite sets A and B are disjoint, then |A union B|=|A|+|B|.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-T-00000042, S26-D-00000035

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000094|T|A|Proof+Compile+Meta] Multiplication rule for finite products

§FORM := If A and B are finite, then |A x B|=|A||B|.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-T-00000042, S26-D-00000036

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000095|T|A|Proof+Compile+Meta] Inclusion-exclusion for two finite sets

§FORM := For finite sets A,B, |A union B|=|A|+|B|-|A intersection B|.

§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

§DEP_HARD := S26-T-00000093, S26-B-00000046

§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000096|D|A|Proof+Compile+Meta] Permutation of a finite set

§FORM := A permutation of a finite set X is a bijection X->X.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000008, S26-D-00000023
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000097|T|A|Proof+Compile+Meta] Number of permutations

§FORM := A finite set with n elements has n! permutations.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-D-00000096, S26-D-00000086, S26-T-00000042
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000098|T|A|Proof+Compile+Meta] Number of ordered selections without replacement

§FORM := The number of ordered k-tuples of distinct elements selected from an n-element set is n!/(n-k)! for 0<=k<=n.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-T-00000097, S26-D-00000086
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000099|T|A|Proof+Compile+Meta] Finite binomial sum

§FORM := For n in N, the sum from k=0 to n of C(n,k) equals 2^n.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-T-00000089, S26-D-00000087
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000100|T|A|Proof+Compile+Meta] Alternating finite binomial sum

§FORM := For n>0, the sum from k=0 to n of (-1)^k C(n,k) equals 0.
§ASSUME := Integers and natural numbers are interpreted through the registered interfaces of Volume 0.
§DEP_HARD := S26-T-00000089, S26-D-00000087
§EVIDENCE := A · Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000101|D|A|Proof+Compile+Meta] Commutative ring - interface

§FORM := A commutative ring is a set R with operations + and *, elements 0 and 1, such that (R,+,0) is an abelian group, multiplication is associative and commutative with identity 1, and multiplication distributes over addition.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000013
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000102|D|A|Proof+Compile+Meta] Integral domain

§FORM := An integral domain is a nonzero commutative ring in which ab=0 implies a=0 or b=0.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000101
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000103|D|A|Proof+Compile+Meta] Field - interface

§FORM := A field is a nonzero commutative ring in which every nonzero element has a multiplicative inverse.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000101
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000104|D|A|Proof+Compile+Meta] Ring homomorphism

§FORM := For commutative rings R,S, a ring homomorphism phi:R->S preserves 0, 1, addition, and multiplication.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000008, S26-D-00000101
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000105|T|A|Proof+Compile+Meta] Every field is an integral domain

§FORM := Every field, under the registered nonzero convention, is an integral domain.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000102, S26-D-00000103
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00000106|B|A|Proof+Compile+Meta] Registered number systems as abstract algebraic structures

§FORM := The integer interface instantiates a commutative ring, while the rational, real, and complex interfaces instantiate fields.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-T-00000047, S26-T-00000051, S26-D-00000054, S26-T-00000059, S26-D-00000101, S26-D-00000103
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000107|D|A|Proof+Compile+Meta] Natural scalar multiplication and powers in a commutative ring

§FORM := For a commutative ring R, n*r denotes repeated addition with 0*r=0 and (n+1)*r=n*r+r; r^n denotes repeated multiplication with r^0=1 and r^(n+1)=r^n*r.

§ASSUME := Natural numbers and the commutative-ring interface are the registered Volume 0 objects.

§DEP_HARD := S26-D-00000031, S26-D-00000101, S26-T-00000033, S26-T-00000034

§EVIDENCE := A · Exact registered recursive definitions only; no extension to negative or non-integer exponents is implied.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000108|D|A|Proof+Compile+Meta] Characteristic of a commutative ring

§FORM := For a nonzero commutative ring R, its characteristic is the least positive n with n*1=0 if such n exists, and 0 otherwise.

§ASSUME := R is a nonzero commutative ring; natural scalar multiplication is S26-D-00000107.

§DEP_HARD := S26-D-00000101, S26-D-00000107, S26-T-00000039

§EVIDENCE := A · Exact registered definition only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000109|D|A|Proof+Compile+Meta] Polynomial over a commutative ring

§FORM := For a commutative ring R, a polynomial f in R[x] is a finitely supported coefficient function a:N->R, written f(x)=sum a_k x^k.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000101, S26-D-00000031, S26-D-00000023, S26-D-00000107

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000110|D|A|Proof+Compile+Meta] Zero polynomial

§FORM := The zero polynomial is the polynomial all of whose coefficients are zero.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000109

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000111|D|A|Proof+Compile+Meta] Constant polynomial

§FORM := For c in R, the constant polynomial c in R[x] has coefficient c at degree 0 and coefficient 0 at every positive degree.

§ASSUME := R is a commutative ring and R[x] is the registered polynomial object.

§DEP_HARD := S26-D-00000109

§EVIDENCE := A · Exact registered definition only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000112|D|A|Proof+Compile+Meta] Degree of a polynomial

§FORM := For a nonzero polynomial f, deg(f) is the largest natural number k with nonzero coefficient a_k. The zero polynomial is left degree-exceptional in this interface rather than assigned an ordinary natural degree.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000109, S26-D-00000110, S26-T-00000039

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000113|D|A|Proof+Compile+Meta] Leading coefficient

§FORM := For a nonzero polynomial f, its leading coefficient is the coefficient of x^deg(f).

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000112

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000114|D|A|Proof+Compile+Meta] Monic polynomial

§FORM := A nonzero polynomial is monic when its leading coefficient equals 1.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000113

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000115|D|A|Proof+Compile+Meta] Polynomial addition and multiplication

§FORM := Polynomial addition is coefficientwise; multiplication is the Cauchy convolution of coefficient sequences.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000109, S26-D-00000035, S26-D-00000036

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000116|T|A|Proof+Compile+Meta] Polynomial ring structure

§FORM := If R is a commutative ring, then R[x] with the registered polynomial operations is a commutative ring.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000101, S26-D-00000115

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000117|T|A|Proof+Compile+Meta] Degree of a product over an integral domain

§FORM := If R is an integral domain and f,g are nonzero in R[x], then deg(fg)=deg(f)+deg(g).

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000102, S26-D-00000112, S26-D-00000115

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000118|D|A|Proof+Compile+Meta] Polynomial evaluation

§FORM := For f=sum a_k x^k in R[x] and r in R, eval_r(f)=sum a_k r^k.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000109, S26-D-00000107
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000119|T|A|Proof+Compile+Meta] Evaluation is a ring homomorphism

§FORM := For a commutative ring R and r in R, eval_r:R[x]->R is a ring homomorphism.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000104, S26-D-00000118, S26-T-00000116
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000120|D|A|Proof+Compile+Meta] Polynomial root

§FORM := For f in R[x], an element r in R is a root of f when f(r)=0.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000118
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000121|T|A|Proof+Compile+Meta] Polynomial division algorithm over a field

§FORM := If F is a field, f,g in F[x], and g is nonzero, then there exist unique q,r in F[x] such that f=qg+r and either r=0 or deg(r)<deg(g).
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000103, S26-T-00000116, S26-D-00000112
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000122|T|A|Proof+Compile+Meta] Remainder theorem

§FORM := For f in F[x] over a field F and a in F, division by x-a leaves remainder f(a).
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-T-00000121, S26-D-00000118
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000123|T|A|Proof+Compile+Meta] Factor theorem

§FORM := For f in F[x] over a field F and a in F, f(a)=0 if and only if x-a divides f.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-T-00000122, S26-D-00000120
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000124|D|A|Proof+Compile+Meta] Polynomial divisibility

§FORM := For f,g in R[x], g divides f if there exists h in R[x] with f=gh.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000109, S26-D-00000115
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000125|D|A|Proof+Compile+Meta] Multiplicity of a root

§FORM := For f in F[x], a root a has multiplicity m>=1 when (x-a)^m divides f but (x-a)^(m+1) does not.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000120, S26-D-00000124
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000126|T|A|Proof+Compile+Meta] Root bound over an integral domain

§FORM := A nonzero polynomial of degree n over an integral domain has at most n distinct roots in that domain.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000102, S26-D-00000112, S26-T-00000123
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000127|T|A|Proof+Compile+Meta] Polynomial identity principle over an infinite integral domain

§FORM := If R is an infinite integral domain and f,g in R[x] satisfy f(r)=g(r) for every r in R, then f=g.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-T-00000126, S26-D-00000118
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000128|D|A|Proof+Compile+Meta] Formal derivative of a polynomial

§FORM := For f=sum_{k=0}^n a_k x^k over a commutative ring, the formal derivative is f'=sum_{k=1}^n k a_k x^(k-1), with k interpreted via repeated addition of 1.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000109, S26-D-00000107
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000129|T|A|Proof+Compile+Meta] Linearity of the formal derivative

§FORM := For polynomials f,g and scalar c over a commutative ring, (f+g)'=f'+g' and (cf)'=c f'.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000128, S26-D-00000115
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000130|T|A|Proof+Compile+Meta] Product rule for the formal derivative

§FORM := For polynomials f,g over a commutative ring, (fg)'=f'g+fg'.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000128, S26-D-00000115
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000131|T|A|Proof+Compile+Meta] Power rule for the formal derivative

§FORM := For n>=1, (x^n)'=n x^(n-1).
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000128
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000132|D|A|Proof+Compile+Meta] Polynomial gcd over a field

§FORM := For nonzero f,g in F[x], a greatest common divisor is a common divisor d such that every common divisor of f and g divides d; a monic representative is chosen canonically.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000124, S26-D-00000114, S26-D-00000103
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00000133|A|A|Proof+Compile+Meta] Euclidean algorithm for polynomials

§FORM := Repeated polynomial division over a field terminates and yields a monic gcd of the input polynomials.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-T-00000121, S26-D-00000132
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000134|T|A|Proof+Compile+Meta] Bezout identity for polynomial gcd

§FORM := For f,g in F[x], their monic gcd d can be written d=af+bg for some a,b in F[x].
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-A-00000133, S26-D-00000132
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000135|T|A|Proof+Compile+Meta] Repeated-root derivative criterion

§FORM := For f in F[x] and a in F, a is a repeated root of f if and only if f(a)=0 and f'(a)=0.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000125, S26-D-00000128, S26-T-00000123, S26-D-00000108
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§BOUNDARY := In positive characteristic, the derivative may vanish identically for nonconstant polynomials such as x^p; the exact repeated-root statement remains valid, but derivative-based squarefreeness must respect characteristic.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000136|D|A|Proof+Compile+Meta] Squarefree polynomial over a field

§FORM := A nonzero polynomial f in F[x] is squarefree when there is no nonconstant polynomial h in F[x] such that h^2 divides f.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000124, S26-D-00000103, S26-D-00000111, S26-D-00000107
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000137|T|A|Proof+Compile+Meta] Squarefree gcd criterion

§FORM := For nonzero f in F[x], f is squarefree if and only if gcd(f,f')=1.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-D-00000136, S26-D-00000132, S26-D-00000128
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000138|T|A|Proof+Compile+Meta] Quadratic formula

§FORM := Over a field of characteristic not equal to 2, roots of ax^2+bx+c with a nonzero and with a chosen square root of the discriminant Delta=b^2-4ac are (-b +/- sqrt(Delta))/(2a).
§ASSUME := The coefficient field has characteristic not equal to 2; a is nonzero; a square root of the discriminant is available in the field or chosen extension.
§DEP_HARD := S26-D-00000103, S26-D-00000120, S26-D-00000108
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000139|D|A|Proof+Compile+Meta] Elementary symmetric polynomials

§FORM := For variables r_1,...,r_n, the k-th elementary symmetric polynomial e_k is the sum of all products r_{i1}...r_{ik} over k-element subsets of the indices.
§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.
§DEP_HARD := S26-T-00000091, S26-D-00000109
§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000140|T|A|Proof+Compile+Meta] Vieta relations for a monic split polynomial

§FORM := If f(x)=prod_{i=1}^n (x-r_i)=x^n+a_{n-1}x^(n-1)+...+a_0, then a_{n-k}=(-1)^k e_k(r_1,...,r_n).

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000139, S26-D-00000115, S26-D-00000114

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000141|T|A|Proof+Compile+Meta] Fundamental theorem of algebra - root form

§FORM := Every nonconstant polynomial with complex coefficients has at least one complex root.

§ASSUME := Complex numbers use the registered field interface; this theorem is not derived from the elementary algebraic kernel in this tranche.

§DEP_HARD := S26-D-00000058, S26-D-00000120

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000142|T|A|Proof+Compile+Meta] Complex factorization of polynomials

§FORM := Every nonzero complex polynomial f of degree n factors as c product_{i=1}^n (x-r_i), counting roots with multiplicity, for nonzero c in C.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-T-00000141, S26-T-00000123, S26-D-00000125

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000143|D|A|Proof+Compile+Meta] Primitive integer polynomial

§FORM := A polynomial in Z[x] is primitive when the gcd of all its coefficients is 1.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-B-00000106, S26-D-00000063, S26-D-00000109

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000144|D|A|Proof+Compile+Meta] Content of an integer polynomial

§FORM := For a nonzero polynomial f in Z[x], its content is the positive gcd of its coefficients.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000063, S26-D-00000109

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000145|T|A|Proof+Compile+Meta] Gauss lemma - primitive product

§FORM := The product of two primitive polynomials in Z[x] is primitive.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000143, S26-D-00000115

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000146|D|A|Proof+Compile+Meta] Irreducible polynomial over a field

§FORM := A nonconstant polynomial f in F[x] is irreducible when f=gh implies that g or h is a nonzero constant polynomial.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000103, S26-D-00000109

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000147|T|A|Proof+Compile+Meta] Rational root theorem

§FORM := If a primitive polynomial a_n x^n+...+a_0 in Z[x] has a rational root p/q in lowest terms, then p divides a_0 and q divides a_n.

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000143, S26-D-00000049, S26-D-00000061, S26-D-00000120, S26-D-00000064

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000148|T|A|Proof+Compile+Meta] Lagrange interpolation

§FORM := For distinct x_0,...,x_n in a field F and arbitrary y_0,...,y_n in F, there is a unique polynomial p of degree at most n with p(x_i)=y_i for every i; explicitly p(x)=sum_i y_i product_{j!=i}(x-x_j)/(x_i-x_j).

§ASSUME := All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

§DEP_HARD := S26-D-00000103, S26-T-00000126, S26-D-00000118

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000149|D|A|Proof+Compile+Meta] Vector space over a field

§FORM := A vector space V over a field F is an abelian group under vector addition together with scalar multiplication F x V -> V satisfying compatibility with field multiplication, identity action 1*v=v, and distributivity over both scalar and vector addition.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000103

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000150|D|A|Proof+Compile+Meta] Vector

§FORM := A vector is an element of a declared vector space V over a declared field F.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000149

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000151|D|A|Proof+Compile+Meta] Subspace

§FORM := A subspace W of a vector space V over F is a subset W of V containing 0 and closed under vector addition and scalar multiplication, with the induced vector-space operations.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000149, S26-D-00000017

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000152|T|A|Proof+Compile+Meta] Subspace criterion

§FORM := A nonempty subset W of a vector space V over F is a subspace if and only if for all u,v in W and all a,b in F, the vector a*u+b*v lies in W.

§ASSUME := V is a vector space over the field F; W is a nonempty subset of V.

§DEP_HARD := S26-D-00000151

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000153|D|A|Proof+Compile+Meta] Linear combination

§FORM := Given vectors v_1,...,v_k in V and scalars a_1,...,a_k in F, a_1*v_1+...+a_k*v_k is a linear combination of those vectors.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000149, S26-D-00000150

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000154|D|A|Proof+Compile+Meta] Span

§FORM := For S contained in a vector space V over F, span(S) is the set of all finite linear combinations of elements of S; span(emptyset)={0}.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000153, S26-D-00000031, S26-D-00000017

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000155|T|A|Proof+Compile+Meta] Span is the smallest containing subspace

§FORM := For S contained in V, span(S) is a subspace of V containing S, and every subspace of V that contains S also contains span(S).

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000154, S26-D-00000151

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000156|D|A|Proof+Compile+Meta] Linear independence

§FORM := A finite list (v_1,...,v_k) in V is linearly independent over F when a_1*v_1+...+a_k*v_k=0 implies a_1=...=a_k=0.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000153

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000157|D|A|Proof+Compile+Meta] Spanning family

§FORM := A family S in V spans V when span(S)=V.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000154

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000158|D|A|Proof+Compile+Meta] Basis

§FORM := A basis of a vector space V over F is a linearly independent family that spans V.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000156, S26-D-00000157

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000159|T|A|Proof+Compile+Meta] Unique finite coordinates in a basis

§FORM := If (b_1,...,b_n) is a finite basis of V, every v in V has a unique expression v=sum_i a_i b_i with a_i in F.

§ASSUME := (b_1,...,b_n) is a finite basis of V over F.

§DEP_HARD := S26-D-00000158, S26-D-00000153

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000160|D|A|Proof+Compile+Meta] Coordinate vector relative to an ordered basis

§FORM := For an ordered finite basis B=(b_1,...,b_n), the coordinate vector [v]_B is the column (a_1,...,a_n)^T determined by the unique expansion v=sum_i a_i b_i.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-T-00000159

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000161|D|A|Proof+Compile+Meta] Finite-dimensional vector space

§FORM := A vector space V is finite-dimensional when it has a finite basis.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000158

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000162|T|A|Proof+Compile+Meta] Basis cardinality invariance

§FORM := Any two finite bases of the same vector space V over F have the same number of vectors.

§ASSUME := V has at least one finite basis.

§DEP_HARD := S26-D-00000158, S26-T-00000091

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000163|D|A|Proof+Compile+Meta] Dimension of a finite-dimensional vector space

§FORM := For a finite-dimensional vector space V, dim_F(V) is the number of vectors in any basis of V.

§ASSUME := The basis-cardinality invariance theorem is used to make the definition well-defined.

§DEP_HARD := S26-D-00000161, S26-T-00000162

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000164|T|A|Proof+Compile+Meta] Basis extension and spanning reduction

§FORM := In a finite-dimensional vector space, every linearly independent finite set extends to a basis, and every finite spanning set contains a basis.

§ASSUME := V is finite-dimensional over F.

§DEP_HARD := S26-D-00000161, S26-D-00000156, S26-D-00000157

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000165|T|A|Proof+Compile+Meta] Dimension monotonicity for subspaces

§FORM := If W is a subspace of a finite-dimensional vector space V, then dim(W)<=dim(V), with equality if and only if W=V.

§ASSUME := V is finite-dimensional over F; W is a subspace of V.

§DEP_HARD := S26-D-00000151, S26-D-00000163

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000166|D|A|Proof+Compile+Meta] Linear map

§FORM := For vector spaces V,W over the same field F, a map T:V->W is linear when T(a*u+b*v)=a*T(u)+b*T(v) for all u,v in V and a,b in F.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000149, S26-D-00000008

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000167|T|A|Proof+Compile+Meta] Linear maps preserve zero and finite linear combinations

§FORM := If T:V->W is linear, then T(0)=0 and T(sum_i a_i v_i)=sum_i a_i T(v_i) for every finite linear combination.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000166, S26-D-00000153

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000168|D|A|Proof+Compile+Meta] Kernel of a linear map

§FORM := For a linear map T:V->W, ker(T)={v in V : T(v)=0}.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000166, S26-D-00000017

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000169|D|A|Proof+Compile+Meta] Image of a linear map

§FORM := For a linear map T:V->W, im(T)={T(v):v in V}.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000166, S26-D-00000017

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000170|T|A|Proof+Compile+Meta] Kernel and image are subspaces

§FORM := For a linear map T:V->W, ker(T) is a subspace of V and im(T) is a subspace of W.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000168, S26-D-00000169, S26-D-00000151

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000171|D|A|Proof+Compile+Meta] Rank of a linear map

§FORM := For a linear map T:V->W with finite-dimensional image, rank(T)=dim(im(T)).

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000169, S26-D-00000163

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000172|D|A|Proof+Compile+Meta] Nullity of a linear map

§FORM := For a linear map T:V->W with finite-dimensional kernel, nullity(T)=dim(ker(T)).

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000168, S26-D-00000163

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000173|T|A|Proof+Compile+Meta] Rank-nullity theorem

§FORM := If V is finite-dimensional and T:V->W is linear, then dim(V)=rank(T)+nullity(T).

§ASSUME := V is finite-dimensional over F; W may be arbitrary over the same field F.

§DEP_HARD := S26-D-00000161, S26-D-00000171, S26-D-00000172

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000174|D|A|Proof+Compile+Meta] Linear isomorphism

§FORM := A linear isomorphism T:V->W is a bijective linear map; its inverse is then linear.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000166, S26-D-00000015

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000175|T|A|Proof+Compile+Meta] Finite-dimensional isomorphism criterion

§FORM := If V and W are finite-dimensional vector spaces over F with dim(V)=dim(W), then a linear map T:V->W is injective if and only if it is surjective, and either condition is equivalent to T being a linear isomorphism.

§ASSUME := V and W are finite-dimensional over the same field F and have equal dimension.

§DEP_HARD := S26-T-00000173, S26-D-00000174, S26-D-00000163

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000176|D|A|Proof+Compile+Meta] Matrix over a commutative ring

§FORM := For positive integers m,n and a commutative ring R, an m x n matrix over R is an array A=(a_ij) with entries a_ij in R indexed by 1<=i<=m and 1<=j<=n.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000101, S26-D-00000031

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000177|D|A|Proof+Compile+Meta] Matrix addition and scalar multiplication

§FORM := Matrices of the same shape over a commutative ring are added entrywise; for r in R, scalar multiplication rA is entrywise multiplication by r.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000176, S26-D-00000101

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000178|D|A|Proof+Compile+Meta] Matrix multiplication

§FORM := For A in M_{m x n}(R) and B in M_{n x p}(R), the product AB in M_{m x p}(R) has entries (AB)_ij=sum_{k=1}^n a_ik b_kj.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000176, S26-D-00000107

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000179|T|A|Proof+Compile+Meta] Associativity and distributivity of matrix multiplication

§FORM := Whenever matrix shapes are compatible over a commutative ring R, matrix multiplication is associative and distributes over matrix addition.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000178, S26-D-00000177, S26-D-00000101

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000180|D|A|Proof+Compile+Meta] Identity matrix

§FORM := For n>=1, I_n is the n x n matrix with 1 on the diagonal and 0 off the diagonal.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000176

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000181|T|A|Proof+Compile+Meta] Matrix identity laws

§FORM := For every compatible matrix A, I_m A=A and A I_n=A.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000178, S26-D-00000180

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000182|D|A|Proof+Compile+Meta] Matrix of a linear map relative to ordered bases

§FORM := Let T:V->W be linear and let B=(b_1,...,b_n), C=(c_1,...,c_m) be ordered finite bases. The matrix [T]_{C<-B} is the m x n matrix whose j-th column is [T(b_j)]_C.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000166, S26-D-00000160, S26-D-00000176

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000183|T|A|Proof+Compile+Meta] Composition corresponds to matrix multiplication

§FORM := For linear maps T:U->V and S:V->W with compatible ordered finite bases A,B,C, [S o T]_{C<-A}=[S]_{C<-B}[T]_{B<-A}.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000182, S26-D-00000009, S26-D-00000178

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000184|D|A|Proof+Compile+Meta] Elementary row operation

§FORM := An elementary row operation on a matrix is one of: swap two rows; multiply one row by a unit scalar (over a field, any nonzero scalar); or add a scalar multiple of one row to another.

§ASSUME := For field-based Gaussian elimination, row scaling uses nonzero field scalars. Over a general commutative ring, scaling is invertible only by a unit.

§DEP_HARD := S26-D-00000176, S26-D-00000103

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000185|D|A|Proof+Compile+Meta] Row equivalence

§FORM := Two matrices over a field are row-equivalent when one can be obtained from the other by a finite sequence of elementary row operations.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000184, S26-D-00000031

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000186|D|A|Proof+Compile+Meta] Linear system represented by an augmented matrix

§FORM := For A in M_{m x n}(F), x in F^n, and b in F^m, the system Ax=b is represented by the augmented matrix [A|b].

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000176, S26-D-00000178, S26-D-00000103

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000187|T|A|Proof+Compile+Meta] Elementary row operations preserve solution sets

§FORM := Over a field F, applying an elementary row operation to an augmented matrix [A|b] does not change the solution set of Ax=b.

§ASSUME := All row operations are invertible elementary operations over the field F.

§DEP_HARD := S26-D-00000184, S26-D-00000186

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000188|D|A|Proof+Compile+Meta] Reduced row echelon form

§FORM := A matrix is in reduced row echelon form when each nonzero row has a leading 1, each leading 1 is the only nonzero entry in its column, leading positions move strictly rightward down the rows, and all zero rows occur below all nonzero rows.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000176

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000189|T|A|Proof+Compile+Meta] Existence and uniqueness of reduced row echelon form

§FORM := Every matrix over a field is row-equivalent to a unique reduced row echelon matrix.

§ASSUME := Matrix entries lie in a field.

§DEP_HARD := S26-D-00000185, S26-D-00000188

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000190|T|A|Proof+Compile+Meta] Gaussian elimination consistency criterion

§FORM := A linear system Ax=b over a field is consistent exactly when the reduced row echelon form of [A|b] contains no row of the form [0 ... 0 | c] with c nonzero; its free variables parametrize all solutions.

§ASSUME := A and b are finite matrices/vectors over a field F.

§DEP_HARD := S26-D-00000186, S26-T-00000189, S26-T-00000187

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000191|D|A|Proof+Compile+Meta] Determinant - Leibniz formula

§FORM := For an n x n matrix A over a commutative ring R, det(A)=sum_{sigma in S_n} sgn(sigma) product_{i=1}^n a_{i,sigma(i)}.

§ASSUME := R is commutative; n>=1; the sign of a permutation is defined in the usual integer-to-ring interpretation.

§DEP_HARD := S26-D-00000176, S26-D-00000086, S26-D-00000101

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000192|T|A|Proof+Compile+Meta] Determinant response to elementary row operations

§FORM := For square matrices over a commutative ring, swapping two rows negates the determinant, multiplying a row by c multiplies the determinant by c, and adding a multiple of one row to another leaves the determinant unchanged.

§ASSUME := The determinant is the registered commutative-ring determinant.

§DEP_HARD := S26-D-00000191, S26-D-00000184

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000193|T|A|Proof+Compile+Meta] Multiplicativity of determinant

§FORM := For n x n matrices A,B over a commutative ring R, det(AB)=det(A)det(B).

§ASSUME := R is a commutative ring.

§DEP_HARD := S26-D-00000191, S26-D-00000178, S26-D-00000101

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000194|T|A|Proof+Compile+Meta] Invertibility criterion by determinant over a field

§FORM := For A in M_n(F) over a field F, A is invertible if and only if det(A) is nonzero.

§ASSUME := A is square and its entries lie in a field F.

§DEP_HARD := S26-D-00000191, S26-T-00000193, S26-D-00000180, S26-D-00000103

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§BOUNDARY := Over a general commutative ring, invertibility is equivalent to det(A) being a unit, not merely nonzero.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000195|D|A|Proof+Compile+Meta] Change-of-basis matrix

§FORM := For ordered bases B and C of the same finite-dimensional vector space V, P_{C<-B} is the matrix satisfying [v]_C=P_{C<-B}[v]_B for every v in V.

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000160, S26-D-00000182, S26-D-00000174

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000196|T|A|Proof+Compile+Meta] Similarity under change of basis

§FORM := If T:V->V is linear and B,C are ordered bases of finite-dimensional V, then [T]_{C<-C}=P_{C<-B}[T]_{B<-B}P_{C<-B}^{-1}.

§ASSUME := V is finite-dimensional over F; B and C are ordered bases.

§DEP_HARD := S26-D-00000195, S26-T-00000183, S26-D-00000174

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000197|D|A|Proof+Compile+Meta] Characteristic polynomial

§FORM := For A in M_n(F), the characteristic polynomial is chi_A(lambda)=det(lambda I_n-A) in F[lambda].

§ASSUME := All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

§DEP_HARD := S26-D-00000191, S26-D-00000180, S26-D-00000109, S26-D-00000103

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000198|D|A|Proof+Compile+Meta+Spectral] Eigenvalue, eigenvector, and eigenspace

§FORM := For a linear operator T:V->V over F, lambda in F is an eigenvalue when there exists v!=0 with T(v)=lambda v; such v is an eigenvector, and E_lambda=ker(T-lambda I) is the lambda-eigenspace.

§ASSUME := T is an endomorphism of a vector space V over F.

§DEP_HARD := S26-D-00000166, S26-D-00000168, S26-D-00000011

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000199|T|A|Proof+Compile+Meta+Spectral] Eigenvalues and the characteristic polynomial

§FORM := For A in M_n(F), lambda in F is an eigenvalue of the associated linear operator if and only if chi_A(lambda)=0.

§ASSUME := A is an n x n matrix over the field F; lambda lies in F.

§DEP_HARD := S26-D-00000197, S26-D-00000198, S26-T-00000194

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000200|T|A|Proof+Compile+Meta+Spectral] Diagonalization criterion by eigenvector basis

§FORM := A linear operator T on a finite-dimensional vector space V over F is diagonalizable over F if and only if V has a basis consisting of eigenvectors of T.

§ASSUME := V is finite-dimensional over F. Diagonalizability is relative to the declared field F.

§DEP_HARD := S26-D-00000158, S26-D-00000182, S26-D-00000198

§EVIDENCE := A · Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

§BOUNDARY := A real rotation by 90 degrees on R^2 is not diagonalizable over R but becomes diagonalizable over C.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000201|D|A|Proof+Compile+Meta+Topo] Scalar conjugation on R or C

§FORM := For F equal to R or C, scalar conjugation a |-> conjugate(a) is the identity on R and ordinary complex conjugation on C; it satisfies conjugate(a+b)=conjugate(a)+conjugate(b), conjugate(ab)=conjugate(a)conjugate(b), and conjugate(conjugate(a))=a.

§ASSUME := F is exactly R or C.

§DEP_HARD := S26-T-00000059, S26-I-00000060

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000202|D|A|Proof+Compile+Meta+Topo] Inner product space over R or C

§FORM := An inner product space V over F in {R,C} is a vector space equipped with <u,v> in F that is linear in the first argument, conjugate-symmetric <u,v>=conjugate(<v,u>), and positive definite: <v,v>>=0 with equality iff v=0.

§ASSUME := F is R or C; this Synopsis uses the convention linear in the first argument and conjugate-linear in the second.

§DEP_HARD := S26-D-00000149, S26-D-00000201

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§BOUNDARY := Using an ordinary bilinear dot product on C^n can give <v,v>=0 for nonzero v.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000203|D|A|Proof+Compile+Meta+Topo] Norm induced by an inner product

§FORM := In an inner product space, ||v||=sqrt(<v,v>) is the norm induced by the inner product.

§ASSUME := The scalar <v,v> is a nonnegative real number, so the nonnegative square root is used.

§DEP_HARD := S26-D-00000202, S26-T-00000057

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000204|T|A|Proof+Compile+Meta+Topo] Cauchy-Schwarz inequality

§FORM := For all u,v in an inner product space, |<u,v>| <= ||u|| ||v||, with equality iff u and v are linearly dependent.

§ASSUME := V is an inner product space over R or C.

§DEP_HARD := S26-D-00000202, S26-D-00000203, S26-D-00000156

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000205|T|A|Proof+Compile+Meta+Topo] Triangle inequality for the induced norm

§FORM := For all u,v in an inner product space, ||u+v|| <= ||u||+||v||.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000203, S26-T-00000204

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000206|D|A|Proof+Compile+Meta+Topo] Orthogonality

§FORM := Vectors u and v in an inner product space are orthogonal, written u perp v, when <u,v>=0.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000202

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000207|T|A|Proof+Compile+Meta+Topo] Pythagorean identity

§FORM := If u and v are orthogonal, then ||u+v||^2=||u||^2+||v||^2.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000206, S26-D-00000203

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000208|D|A|Proof+Compile+Meta+Topo] Orthogonal complement

§FORM := For S contained in an inner product space V, S^perp={v in V : <v,s>=0 for every s in S}.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000206, S26-D-00000017

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000209|T|A|Proof+Compile+Meta+Topo] Orthogonal complement is a subspace

§FORM := For every subset S of an inner product space V, S^perp is a subspace of V.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000208, S26-D-00000151

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000210|D|A|Proof+Compile+Meta+Topo] Orthonormal family

§FORM := A family (e_i) in an inner product space is orthonormal when <e_i,e_j>=0 for i!=j and ||e_i||=1 for every i.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000206, S26-D-00000203

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000211|T|A|Proof+Compile+Meta+Topo] Finite orthonormal families are linearly independent

§FORM := Every finite orthonormal family is linearly independent.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000210, S26-D-00000156

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000212|T|A|Proof+Compile+Meta+Topo] Bessel inequality - finite orthonormal form

§FORM := If e_1,...,e_k is an orthonormal family, then sum_i |<v,e_i>|^2 <= ||v||^2 for every v.

§ASSUME := The family is finite.

§DEP_HARD := S26-D-00000210, S26-D-00000203

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000213|T|A|Proof+Compile+Meta+Topo] Parseval identity for a finite orthonormal basis

§FORM := If e_1,...,e_n is an orthonormal basis of V, then v=sum_i <v,e_i> e_i and ||v||^2=sum_i |<v,e_i>|^2.

§ASSUME := V is finite-dimensional and the displayed family is an orthonormal basis.

§DEP_HARD := S26-D-00000210, S26-D-00000158, S26-T-00000212

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000214|D|A|Proof+Compile+Meta+Topo] Gram matrix

§FORM := For vectors v_1,...,v_k in an inner product space, the Gram matrix G is the k x k matrix with G_ij=<v_i,v_j>.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000202, S26-D-00000176

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000215|T|A|Proof+Compile+Meta+Topo] Gram matrix positivity and independence criterion

§FORM := For every coefficient vector c, sum_{i,j} conjugate(c_i) G_ij c_j is a nonnegative real number. It is strictly positive for every nonzero c iff the family v_1,...,v_k is linearly independent.

§ASSUME := The family is finite; Hermitian and positive (semi)definite are interpreted over R or C as registered below.

§DEP_HARD := S26-D-00000214, S26-D-00000156

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000216|T|A|Proof+Compile+Meta+Topo] Gram-Schmidt orthonormalization

§FORM := Every finite linearly independent list in an inner product space can be transformed into an orthonormal list with the same successive spans.

§ASSUME := The input list is finite and linearly independent.

§DEP_HARD := S26-D-00000156, S26-D-00000210, S26-D-00000154

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000217|T|A|Proof+Compile+Meta+Topo] Existence of orthonormal bases in finite dimension

§FORM := Every finite-dimensional inner product space has an orthonormal basis.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000161, S26-T-00000164, S26-T-00000216

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000218|D|A|Proof+Compile+Meta+Topo] Orthogonal projection onto a subspace

§FORM := If W is a finite-dimensional subspace of an inner product space V with orthonormal basis e_1,...,e_k, define P_W(v)=sum_i <v,e_i> e_i.

§ASSUME := W is finite-dimensional; the definition is independent of the chosen orthonormal basis by the projection theorem.

§DEP_HARD := S26-D-00000151, S26-D-00000210, S26-T-00000213

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000219|T|A|Proof+Compile+Meta+Topo] Orthogonal decomposition theorem - finite-dimensional subspace

§FORM := If W is a finite-dimensional subspace of an inner product space V, every v in V has a unique decomposition v=w+z with w in W and z in W^perp; w=P_W(v).

§ASSUME := W is finite-dimensional; V itself need not be finite-dimensional.

§DEP_HARD := S26-D-00000218, S26-D-00000208

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000220|T|A|Proof+Compile+Meta+Topo] Best approximation property of orthogonal projection

§FORM := For W as above, P_W(v) is the unique point w in W minimizing ||v-w||.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-T-00000219, S26-T-00000207

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000221|T|A|Proof+Compile+Meta+Topo+Spectral] Projection operator identities

§FORM := For an orthogonal projection P_W onto a finite-dimensional subspace W, P_W is linear, P_W^2=P_W, im(P_W)=W, and ker(P_W)=W^perp.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000218, S26-D-00000166, S26-D-00000168, S26-D-00000169

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000222|D|A|Proof+Compile+Meta+Topo] Adjoint of a linear map

§FORM := For inner product spaces V,W and a linear map T:V->W, an adjoint T*:W->V is a linear map satisfying <Tv,w>_W=<v,T*w>_V for all v,w.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000166, S26-D-00000202

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000223|T|A|Proof+Compile+Meta+Topo] Existence and uniqueness of adjoints in finite dimension

§FORM := Every linear map between finite-dimensional inner product spaces has a unique adjoint.

§ASSUME := Both domain and codomain are finite-dimensional inner product spaces over the same F in {R,C}.

§DEP_HARD := S26-D-00000222, S26-T-00000217

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000224|D|A|Proof+Compile+Meta+Topo] Conjugate transpose of a matrix

§FORM := For a matrix A over R or C, A* is the transpose of the entrywise conjugate matrix; over R this is the ordinary transpose.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000176, S26-D-00000201

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000225|T|A|Proof+Compile+Meta+Topo] Matrix of the adjoint in orthonormal bases

§FORM := If T:V->W is linear and B,C are orthonormal bases, then [T*]_{B<-C} = ([T]_{C<-B})*.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000182, S26-T-00000223, S26-D-00000224

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000226|T|A|Proof+Compile+Meta+Topo+Spectral] Adjoint algebra

§FORM := For compatible finite-dimensional operators and scalars a,b, If pointwise operations are defined by (aS+bT)(v)=aS(v)+bT(v), then (aS+bT)*=conjugate(a)S*+conjugate(b)T*, (ST)*=T*S*, and (T*)*=T.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-T-00000223, S26-D-00000201, S26-D-00000009

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000227|D|A|Proof+Compile+Meta+Topo+Spectral] Self-adjoint operator

§FORM := An operator T on an inner product space is self-adjoint when T*=T.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000222

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000228|D|A|Proof+Compile+Meta+Topo+Spectral] Unitary and orthogonal operator

§FORM := An operator U is unitary when U*U=UU*=I. Over real inner product spaces the same condition defines an orthogonal operator.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000222, S26-D-00000011, S26-D-00000180

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000229|D|A|Proof+Compile+Meta+Topo+Spectral] Normal operator

§FORM := An operator T is normal when T*T=TT*.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000222

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000230|T|A|Proof+Compile+Meta+Topo+Spectral] Unitary operators preserve inner products and norms

§FORM := If U is unitary, then <Uv,Uw>=<v,w> and ||Uv||=||v|| for all v,w.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000228, S26-D-00000203

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000231|T|A|Proof+Compile+Meta+Topo] Finite-dimensional isometry criterion

§FORM := For an endomorphism U of a finite-dimensional inner product space, U is unitary iff it preserves the inner product; equivalently, iff it preserves the induced norm.

§ASSUME := U is a linear endomorphism of a finite-dimensional inner product space.

§DEP_HARD := S26-D-00000228, S26-T-00000230, S26-D-00000161

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000232|T|A|Proof+Compile+Meta+Topo+Spectral] Orthogonality of eigenvectors of a self-adjoint operator

§FORM := Eigenvectors of a self-adjoint operator belonging to distinct eigenvalues are orthogonal.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000227, S26-D-00000198, S26-D-00000206

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000233|T|A|Proof+Compile+Meta+Topo+Spectral] Eigenvalues of a self-adjoint operator are real

§FORM := Every eigenvalue of a self-adjoint operator on a complex inner product space is real; over a real inner product space eigenvalues are already real scalars.

§ASSUME := The eigenvalue belongs to the declared base field; in the complex case the theorem asserts it lies in R.

§DEP_HARD := S26-D-00000227, S26-D-00000198, S26-D-00000201

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000234|T|A|Proof+Compile+Meta+Topo+Spectral] Eigenvalues of a unitary operator have unit modulus

§FORM := Every eigenvalue lambda of a unitary operator satisfies |lambda|=1.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000228, S26-D-00000198, S26-D-00000203

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000235|T|A|Proof+Compile+Meta+Topo+Spectral] Finite-dimensional spectral theorem for self-adjoint operators

§FORM := Every self-adjoint operator on a finite-dimensional real or complex inner product space has an orthonormal basis of eigenvectors; equivalently, it is diagonalizable by an orthogonal/unitary change of basis with real diagonal entries.

§ASSUME := The space is finite-dimensional over R or C.

§DEP_HARD := S26-D-00000227, S26-T-00000217, S26-T-00000200, S26-T-00000232, S26-T-00000233

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000236|T|A|Proof+Compile+Meta+Topo+Spectral] Spectral decomposition of a self-adjoint operator

§FORM := If T is self-adjoint on a finite-dimensional inner product space, then T=sum_j lambda_j P_j where the lambda_j are its distinct real eigenvalues and P_j are pairwise orthogonal projections onto their eigenspaces with sum_j P_j=I.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-T-00000235, S26-D-00000218, S26-D-00000198

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000237|T|A|Proof+Compile+Meta+Topo+Spectral] Finite-dimensional complex spectral theorem for normal operators

§FORM := A linear operator on a finite-dimensional complex inner product space is normal iff it has an orthonormal basis of eigenvectors, equivalently iff it is unitarily diagonalizable.

§ASSUME := The base field is C.

§DEP_HARD := S26-D-00000229, S26-T-00000217, S26-T-00000200

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§BOUNDARY := A 90-degree rotation on R^2 is real-normal and orthogonal but has no real eigenvectors, so real normal operators need not be diagonalizable over R.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000238|D|A|Proof+Compile+Meta+Topo+Spectral] Positive semidefinite self-adjoint operator

§FORM := A self-adjoint operator A is positive semidefinite, written A>=0, when <Av,v>>=0 for every v.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000227, S26-D-00000202

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000239|D|A|Proof+Compile+Meta+Topo+Spectral] Positive definite self-adjoint operator

§FORM := A self-adjoint operator A is positive definite, written A>0, when <Av,v>>0 for every nonzero v.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000238

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000240|T|A|Proof+Compile+Meta+Topo+Spectral] Positive square root theorem - finite dimension

§FORM := Every positive semidefinite operator A on a finite-dimensional real or complex inner product space has a unique positive semidefinite operator B with B^2=A.

§ASSUME := The space is finite-dimensional.

§DEP_HARD := S26-D-00000238, S26-T-00000235, S26-D-00000011

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000241|D|A|Proof+Compile+Meta+Topo+Spectral] Rayleigh quotient

§FORM := For a self-adjoint operator A and nonzero v, the Rayleigh quotient is R_A(v)=<Av,v>/<v,v>.

§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

§DEP_HARD := S26-D-00000227, S26-D-00000202

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000242|T|A|Proof+Compile+Meta+Topo+Spectral] Extremal Rayleigh quotient characterization

§FORM := For self-adjoint A on a nonzero finite-dimensional inner product space, min_{v!=0} R_A(v)=lambda_min and max_{v!=0} R_A(v)=lambda_max, the smallest and largest eigenvalues of A.
§ASSUME := A is self-adjoint and the space is finite-dimensional and nonzero.
§DEP_HARD := S26-D-00000241, S26-T-00000235
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000243|D|A|Proof+Compile+Meta+Topo+Spectral] Induced operator norm

§FORM := For a linear map T between finite-dimensional inner product spaces, ||T||_2 = sup_{v!=0} ||Tv||/||v||.
§ASSUME := Finite dimensionality is assumed; attainment of the supremum is not part of this definition.
§DEP_HARD := S26-D-00000166, S26-D-00000203, S26-D-00000161
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000244|T|A|Proof+Compile+Meta+Topo+Spectral] Operator norm of a self-adjoint operator

§FORM := If A is self-adjoint on a finite-dimensional inner product space, then ||A||_2=max_j |lambda_j|.
§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.
§DEP_HARD := S26-D-00000243, S26-T-00000235
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000245|D|A|Proof+Compile+Meta+Topo+Spectral] Singular values

§FORM := For a linear map T between finite-dimensional inner product spaces, the singular values are the nonnegative square roots of the eigenvalues of T*T, counted with multiplicity.
§ASSUME := Domain and codomain are finite-dimensional over R or C.
§DEP_HARD := S26-T-00000223, S26-D-00000238, S26-T-00000240
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000246|T|A|Proof+Compile+Meta+Topo] Positivity of T*T

§FORM := For every linear map T between finite-dimensional inner product spaces, T*T is positive semidefinite and ker(T*T)=ker(T).
§ASSUME := All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.
§DEP_HARD := S26-T-00000223, S26-D-00000238, S26-D-00000168
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000247|T|A|Proof+Compile+Meta+Topo] Singular value decomposition

§FORM := For every linear map T:V->W between finite-dimensional real or complex inner product spaces, there exist orthonormal bases of V and W in which the matrix of T is diagonal rectangular with nonnegative diagonal entries equal to the singular values.
§ASSUME := V and W are finite-dimensional over the same F in {R,C}.
§DEP_HARD := S26-D-00000245, S26-T-00000246, S26-T-00000217
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000248|D|A|Proof+Compile+Meta+Topo] Moore-Penrose pseudoinverse via SVD

§FORM := For a matrix or finite-dimensional linear map with SVD T=U Sigma V*, the Moore-Penrose pseudoinverse is T^+=V Sigma^+ U*, where each nonzero singular value is reciprocated and each zero singular value remains zero.
§ASSUME := Finite-dimensional real or complex inner product spaces.
§DEP_HARD := S26-T-00000247, S26-D-00000224
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000249|T|A|Proof+Compile+Meta+Topo] Normal equations for least squares

§FORM := For A:F^n->F^m and b in F^m with F in {R,C}, a vector x minimizes ||Ax-b|| iff A*(Ax-b)=0, equivalently A*A x=A*b.
§ASSUME := Finite-dimensional standard inner product spaces; minimizers need not be unique when A has nontrivial kernel.
§DEP_HARD := S26-T-00000223, S26-T-00000220, S26-D-00000169
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := If A is rank deficient, the normal equations can have multiple minimizers.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000250|T|A|Proof+Compile+Meta+Topo] Pseudoinverse least-squares solution

§FORM := For a finite-dimensional least-squares problem Ax approximately b, x=A^+b is a least-squares minimizer of minimum norm; it is the unique minimum-norm minimizer.
§ASSUME := A acts between finite-dimensional real or complex inner product spaces.
§DEP_HARD := S26-D-00000248, S26-T-00000249, S26-T-00000220
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000251|D|A|Proof+Compile+Meta] Dual space

§FORM := For a vector space V over F, the dual space V* is the vector space Hom_F(V,F) of linear functionals V->F.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000166, S26-D-00000149
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000252|D|A|Proof+Compile+Meta] Dual basis

§FORM := If (e_1,...,e_n) is a basis of finite-dimensional V, its dual basis (e^1,...,e^n) in V* is defined by e^i(e_j)=delta_ij.
§ASSUME := V is finite-dimensional and the displayed basis is finite.
§DEP_HARD := S26-D-00000158, S26-D-00000251
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000253|T|A|Proof+Compile+Meta] Dual basis theorem
§FORM := Every basis of a finite-dimensional vector space has a unique dual basis, and the dual basis is a basis of V*.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000252, S26-D-00000161
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000254|T|A|Proof+Compile+Meta] Dimension of the dual
§FORM := If V is finite-dimensional, dim(V*)=dim(V).
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-T-00000253, S26-D-00000161
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000255|D|A|Proof+Compile+Meta] Evaluation map into the double dual
§FORM := For any vector space V, the canonical evaluation map J:V->V** is J(v)(phi)=phi(v).
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000251, S26-D-00000166
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000256|T|A|Proof+Compile+Meta] Finite-dimensional double-dual isomorphism
§FORM := If V is finite-dimensional, the evaluation map J:V->V** is an isomorphism.
§ASSUME := V is finite-dimensional.
§DEP_HARD := S26-D-00000255, S26-T-00000254, S26-T-00000175
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := For arbitrary infinite-dimensional algebraic vector spaces, J is injective but need not be surjective.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000257|D|A|Proof+Compile+Meta] Annihilator of a subspace
§FORM := For W<=V, the annihilator W^0 is {phi in V*: phi(w)=0 for every w in W}.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000151, S26-D-00000251
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000258|T|A|Proof+Compile+Meta] Dimension formula for annihilators
§FORM := If V is finite-dimensional and W<=V, then dim(W^0)=dim(V)-dim(W).
§ASSUME := V is finite-dimensional.
§DEP_HARD := S26-D-00000257, S26-D-00000172, S26-D-00000161
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000259|D|A|Proof+Compile+Meta] Transpose (dual) map
§FORM := For T:V->W linear, the dual map T*:W*->V* is T*(psi)=psi o T.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000166, S26-D-00000251, S26-D-00000009
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000260|T|A|Proof+Compile+Meta] Dual-map functoriality
§FORM := For composable linear maps S,T, (S o T)*=T* o S*, and the dual of an identity map is an identity map.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000259, S26-D-00000009, S26-D-00000011
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000261|D|A|Proof+Compile+Meta] Bilinear map
§FORM := A map B:V x W->U is bilinear when it is linear in each argument separately.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000166, S26-D-00000149
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000262|D|A|Proof+Compile+Meta] Multilinear map
§FORM := A map M:V_1 x ... x V_k->W is multilinear when it is linear in each argument with all other arguments fixed.
§ASSUME := k is a positive integer.
§DEP_HARD := S26-D-00000261
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000263|D|A|Proof+Compile+Meta] Alternating multilinear map
§FORM := A multilinear map is alternating when its value is zero whenever two input arguments are equal.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000262
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000264|T|A|Proof+Compile+Meta] Alternating maps change sign under a transposition

§FORM := For an alternating multilinear map over any field, swapping two arguments multiplies the value by -1.
§ASSUME := The statement uses additive inverse; in characteristic 2, -1=1, so sign change does not distinguish parity.
§DEP_HARD := S26-D-00000263
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000265|D|A|Proof+Compile+Meta] Symmetric bilinear form

§FORM := A bilinear form B:V x V->F is symmetric when B(u,v)=B(v,u).
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000261
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000266|D|A|Proof+Compile+Meta] Skew-symmetric bilinear form

§FORM := A bilinear form B is skew-symmetric when B(u,v)=-B(v,u).
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000261
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := In characteristic 2, skew-symmetry alone does not imply B(v,v)=0; alternation must be stated separately.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000267|D|A|Proof+Compile+Meta] Nondegenerate bilinear form

§FORM := A bilinear form B on V is nondegenerate when B(v,w)=0 for every w implies v=0.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000261
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000268|D|A|Proof+Compile+Meta] Tensor product - universal property

§FORM := A tensor product V tensor W is a vector space equipped with a bilinear map tau:V x W->V tensor W such that every bilinear B:V x W->U factors uniquely through a linear map V tensor W->U.
§ASSUME := Tensor products are over the declared common scalar field F.
§DEP_HARD := S26-D-00000261, S26-D-00000166
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000269|T|A|Proof+Compile+Meta] Tensor product uniqueness up to unique isomorphism

§FORM := Any two tensor products satisfying the same universal property are uniquely isomorphic by an isomorphism commuting with their canonical bilinear maps.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000268, S26-T-00000175
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000270|D|A|Proof+Compile+Meta] Pure tensor

§FORM := An element of V tensor W of the form v tensor w, the image of (v,w) under the canonical bilinear map, is called a pure tensor.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000268
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000271|T|A|Proof+Compile+Meta] Pure tensors span the tensor product

§FORM := The set of pure tensors v tensor w spans V tensor W.
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000270, S26-D-00000154
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000272|T|A|Proof+Compile+Meta] Basis of a finite tensor product

§FORM := If (e_i) is a basis of finite-dimensional V and (f_j) a basis of finite-dimensional W, then (e_i tensor f_j)_{i,j} is a basis of V tensor W.
§ASSUME := V and W are finite-dimensional.
§DEP_HARD := S26-D-00000158, S26-D-00000268, S26-T-00000271
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000273|T|A|Proof+Compile+Meta] Dimension of a tensor product

§FORM := For finite-dimensional V,W, dim(V tensor W)=dim(V) dim(W).
§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-T-00000272, S26-D-00000161
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000274|T|A|Proof+Compile+Meta] Tensor-Hom identification in finite dimension

§FORM := For finite-dimensional V,W, there is a canonical isomorphism V* tensor W ~= Hom(V,W), sending phi tensor w to the rank-one map v |-> phi(v)w.
§ASSUME := V and W are finite-dimensional.
§DEP_HARD := S26-D-00000251, S26-D-00000268, S26-D-00000166, S26-T-00000272
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000275|D|A|Proof+Compile+Meta] Tensor product of linear maps

§FORM := Given linear maps S:V->V' and T:W->W', there is a unique linear map S tensor T:V tensor W->V' tensor W' with (S tensor T)(v tensor w)=S(v) tensor T(w).

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000268, S26-D-00000166

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000276|T|A|Proof+Compile+Meta] Tensor-map functoriality

§FORM := Tensor products of linear maps satisfy (S_2 o S_1) tensor (T_2 o T_1)=(S_2 tensor T_2)o(S_1 tensor T_1) and I tensor I=I.

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000275, S26-D-00000009, S26-D-00000011

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000277|D|A|Proof+Compile+Meta] Tensor powers

§FORM := For k>=0, the k-th tensor power T^k(V) is V tensor ... tensor V with k factors, with T^0(V)=F.

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000268, S26-D-00000031

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000278|D|A|Proof+Compile+Meta] Tensor algebra

§FORM := The tensor algebra T(V)=direct_sum_{k>=0} T^k(V) is graded by tensor degree and multiplication is induced by concatenation of tensors.

§ASSUME := The direct sum is algebraic: each element has finitely many nonzero homogeneous components.

§DEP_HARD := S26-D-00000277

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000279|D|A|Proof+Compile+Meta] Covariant and contravariant tensors

§FORM := A tensor of type (r,s) on finite-dimensional V may be represented as an element of V^{tensor r} tensor (V*)^{tensor s}, equivalently as an appropriate multilinear map after the canonical finite-dimensional identifications.

§ASSUME := V is finite-dimensional; variance convention is explicitly the one stated here.

§DEP_HARD := S26-D-00000251, S26-D-00000277, S26-T-00000274

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000280|D|A|Proof+Compile+Meta] Symmetric tensor

§FORM := A covariant k-tensor is symmetric when it is invariant under every permutation of its k arguments.

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000262, S26-D-00000279

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000281|D|A|Proof+Compile+Meta] Alternating covariant tensor

§FORM := A covariant k-tensor is alternating when it is an alternating k-linear form V^k->F.

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000263, S26-D-00000279

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000282|D|A|Proof+Compile+Meta] Exterior power

§FORM := The k-th exterior power Lambda^k V is the vector space characterized by the universal alternating k-linear map V^k->Lambda^k V.

§ASSUME := k is a nonnegative integer; Lambda^0 V=F by convention.

§DEP_HARD := S26-D-00000263, S26-D-00000268

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000283|D|A|Proof+Compile+Meta] Wedge product

§FORM := The exterior algebra Lambda V=direct_sum_k Lambda^k V carries the wedge product, with decomposable elements written v_1 wedge ... wedge v_k.

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000282, S26-D-00000278

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000284|T|A|Proof+Compile+Meta] Graded commutativity of the wedge product

§FORM := For homogeneous alpha in Lambda^p V and beta in Lambda^q V, alpha wedge beta=(-1)^{pq} beta wedge alpha.

§ASSUME := Degrees p,q are nonnegative integers; in characteristic 2 the sign may equal +1.

§DEP_HARD := S26-D-00000283

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000285|T|A|Proof+Compile+Meta] Basis of an exterior power

§FORM := If e_1,...,e_n is a basis of V, then e_{i1} wedge ... wedge e_{ik} with i1<...<ik forms a basis of Lambda^k V.

§ASSUME := V is finite-dimensional of dimension n.

§DEP_HARD := S26-D-00000158, S26-D-00000282, S26-D-00000283

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000286|T|A|Proof+Compile+Meta] Dimension of an exterior power

§FORM := If dim V=n, then dim Lambda^k V=binomial(n,k); in particular Lambda^k V=0 for k>n.

§ASSUME := V is finite-dimensional of dimension n.

§DEP_HARD := S26-T-00000285, S26-T-00000095, S26-D-00000161

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000287|T|A|Proof+Compile+Meta] Top exterior power is one-dimensional

§FORM := If dim V=n, then Lambda^n V is one-dimensional.

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-T-00000286

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000288|D|A|Proof+Compile+Meta] Induced map on exterior powers

§FORM := A linear map T:V->W induces Lambda^k T:Lambda^k V->Lambda^k W by v_1 wedge ... wedge v_k |-> Tv_1 wedge ... wedge Tv_k.

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000166, S26-D-00000282, S26-D-00000283

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000289|T|A|Proof+Compile+Meta] Functoriality of exterior powers

§FORM := Lambda^k(S o T)=(Lambda^k S)o(Lambda^k T), and Lambda^k(I)=I.

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000288, S26-D-00000009, S26-D-00000011

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000290|T|A|Proof+Compile+Meta] Determinant from the top exterior power

§FORM := For an endomorphism T of an n-dimensional vector space V, Lambda^n T acts on the one-dimensional space Lambda^n V by multiplication by det(T).

§ASSUME := V is finite-dimensional of dimension n.

§DEP_HARD := S26-T-00000287, S26-D-00000288, S26-D-00000191

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000291|T|A|Proof+Compile+Meta] Multiplicativity of determinant from exterior functoriality

§FORM := For endomorphisms S,T of finite-dimensional V, det(S o T)=det(S)det(T).

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-T-00000289, S26-T-00000290, S26-T-00000193

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000292|D|A|Proof+Compile+Meta] Volume form

§FORM := On an n-dimensional real vector space, a volume form is a nonzero element of Lambda^n(V*).

§ASSUME := V is a finite-dimensional real vector space of dimension n.

§DEP_HARD := S26-D-00000251, S26-T-00000287

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000293|D|A|Proof+Compile+Meta] Orientation of a real vector space

§FORM := An orientation of an n-dimensional real vector space is a choice of one of the two equivalence classes of ordered bases, where two bases are equivalent when their change-of-basis determinant is positive.

§ASSUME := V is finite-dimensional over R.

§DEP_HARD := S26-D-00000195, S26-T-00000192

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000294|T|A|Proof+Compile+Meta] Volume forms determine orientations

§FORM := A nonzero top-degree alternating covariant form omega determines an orientation by declaring a basis (e_1,...,e_n) positive iff omega(e_1,...,e_n)>0; positive scalar multiples determine the same orientation.

§ASSUME := V is finite-dimensional over R.

§DEP_HARD := S26-D-00000292, S26-D-00000293

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000295|D|A|Proof+Compile+Meta] Pullback of covariant tensors

§FORM := For T:V->W linear and a covariant k-tensor alpha on W, the pullback T*alpha is defined by (T*alpha)(v_1,...,v_k)=alpha(Tv_1,...,Tv_k).

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000166, S26-D-00000279

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000296|T|A|Proof+Compile+Meta] Pullback preserves wedge products

§FORM := For alternating covariant tensors alpha,beta, T*(alpha wedge beta)=T*alpha wedge T*beta.

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000295, S26-D-00000283

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000297|D|A|Proof+Compile+Meta] Interior product on alternating forms

§FORM := For v in V and an alternating k-form omega, the interior product i_v omega is the (k-1)-form (i_v omega)(v_2,...,v_k)=omega(v,v_2,...,v_k).

§ASSUME := k>=1.

§DEP_HARD := S26-D-00000281

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000298|T|A|Proof+Compile+Meta] Interior product graded Leibniz rule

§FORM := For alpha of degree p, i_v(alpha wedge beta)=(i_v alpha) wedge beta+(-1)^p alpha wedge (i_v beta).

§ASSUME := All vector spaces and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000297, S26-D-00000283

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000299|D|A|Proof+Compile+Meta] Hodge star - finite-dimensional Euclidean/Hermitian exterior algebra

§FORM := Given an oriented finite-dimensional real inner product space V, the Hodge star *:Lambda^k(V*)->Lambda^{n-k}(V*) is the unique linear map satisfying alpha wedge *beta=<alpha,beta> vol for the induced inner product on exterior powers and the positively oriented unit volume form vol.

§ASSUME := V is finite-dimensional, real, oriented, and inner-product equipped; this entry does not assert the complex-Hermitian convention.

§DEP_HARD := S26-D-00000202, S26-D-00000251, S26-D-00000282, S26-T-00000294

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000300|T|A|Proof+Compile+Meta] Hodge star square in the positive-definite real case

§FORM := On an oriented n-dimensional real inner product space, the Hodge star satisfies star(star(alpha))=(-1)^{k(n-k)} alpha for alpha in Lambda^k(V*).

§ASSUME := The metric is positive definite and real; pseudo-Riemannian signatures require a modified sign.

§DEP_HARD := S26-D-00000299

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§BOUNDARY := The displayed sign is not valid unchanged for arbitrary indefinite metrics.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000301|D|A|Proof+Compile+Meta+Topo] Metric space

§FORM := A metric space is a pair (X,d) where d:X x X->[0,infinity) satisfies identity of indiscernibles, symmetry, and the triangle inequality.

§ASSUME := d is real-valued and finite.

§DEP_HARD := S26-N-00000016, S26-T-00000057

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000302|D|A|Proof+Compile+Meta+Topo] Open ball

§FORM := In a metric space (X,d), the open ball B_r(x)={y in X:d(x,y)<r} for r>0.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000301, S26-T-00000055

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000303|D|A|Proof+Compile+Meta+Topo] Open set in a metric space

§FORM := A subset U of a metric space X is open when for every x in U there exists r>0 with B_r(x) contained in U.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000302, S26-N-00000016

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000304|D|A|Proof+Compile+Meta+Topo] Topology

§FORM := A topology tau on X is a family of subsets containing emptyset and X, closed under arbitrary unions and finite intersections; (X,tau) is a topological space.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-N-00000016, S26-D-00000017

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000305|T|A|Proof+Compile+Meta+Topo] Metric topology

§FORM := The open subsets defined by a metric form a topology on X.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000301, S26-D-00000303, S26-D-00000304

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000306|D|A|Proof+Compile+Meta+Topo] Closed set

§FORM := A subset F of a topological space X is closed when X\F is open.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000304, S26-N-00000016

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000307|D|A|Proof+Compile+Meta+Topo] Interior
§FORM := The interior int(A) of A subset X is the union of all open subsets of X contained in A.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000304, S26-D-00000017
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000308|D|A|Proof+Compile+Meta+Topo] Closure
§FORM := The closure cl(A) of A subset X is the intersection of all closed subsets of X containing A.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000306, S26-D-00000017
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000309|D|A|Proof+Compile+Meta+Topo] Boundary
§FORM := The boundary of A is bd(A)=cl(A) intersect cl(X\A).
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000308, S26-N-00000016
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000310|D|A|Proof+Compile+Meta+Topo] Convergence in a metric space
§FORM := A sequence (x_n) converges to x in (X,d) when for every epsilon>0 there exists N such that n>=N implies d(x_n,x)<epsilon.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000301, S26-D-00000031, S26-T-00000055
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000311|T|A|Proof+Compile+Meta+Topo] Uniqueness of metric limits
§FORM := A convergent sequence in a metric space has a unique limit.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000310, S26-D-00000301
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000312|D|A|Proof+Compile+Meta+Topo] Continuous map - topological definition
§FORM := A map f:X->Y between topological spaces is continuous when f^{-1}(U) is open in X for every open U in Y.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000008, S26-D-00000304, S26-D-00000303
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000313|T|A|Proof+Compile+Meta+Topo] Sequential criterion for continuity in metric spaces
§FORM := For maps between metric spaces, f is continuous iff x_n->x implies f(x_n)->f(x) for every sequence.
§ASSUME := Both domain and codomain are metric spaces.
§DEP_HARD := S26-D-00000312, S26-D-00000310, S26-T-00000305
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000314|T|A|Proof+Compile+Meta+Topo] Composition of continuous maps
§FORM := The composition of continuous maps is continuous.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000312, S26-D-00000009
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000315|D|A|Proof+Compile+Meta+Topo] Homeomorphism
§FORM := A homeomorphism is a bijective continuous map whose inverse is continuous.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-T-00000010, S26-D-00000312, S26-D-00000008
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000316|D|A|Proof+Compile+Meta+Topo] Compact topological space
§FORM := A topological space X is compact when every open cover of X has a finite subcover.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000304, S26-D-00000023
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000317|T|A|Proof+Compile+Meta+Topo] Continuous image of a compact space
§FORM := If X is compact and f:X->Y is continuous, then f(X) is compact in the subspace topology.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000316, S26-D-00000312, S26-D-00000014
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000318|T|A|Proof+Compile+Meta+Topo] Heine-Borel theorem in finite-dimensional Euclidean space

§FORM := A subset K of R^n is compact iff it is closed and bounded.
§ASSUME := K is a subset of finite-dimensional Euclidean R^n with its usual metric.
§DEP_HARD := S26-D-00000316, S26-D-00000306, S26-T-00000205
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Closed and bounded need not imply compact in arbitrary metric spaces or infinite-dimensional normed spaces.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000319|T|A|Proof+Compile+Meta+Topo] Extreme value theorem

§FORM := A continuous real-valued function on a nonempty compact space attains both a maximum and a minimum.
§ASSUME := The domain is nonempty and compact; codomain is R with its usual topology.
§DEP_HARD := S26-D-00000312, S26-D-00000316, S26-T-00000055
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000320|D|A|Proof+Compile+Meta+Topo] Connected space

§FORM := A topological space is connected when it is not the union of two disjoint nonempty open subsets.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000304, S26-D-00000017
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000321|T|A|Proof+Compile+Meta+Topo] Continuous image of a connected space

§FORM := The continuous image of a connected space is connected.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000320, S26-D-00000312, S26-D-00000014
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000322|T|A|Proof+Compile+Meta+Topo] Intervals in R are connected

§FORM := Every interval in R is connected in the usual topology.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000320, S26-T-00000055
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000323|T|A|Proof+Compile+Meta+Topo] Intermediate value theorem

§FORM := If f:[a,b]->R is continuous and y lies between f(a) and f(b), then some c in [a,b] satisfies f(c)=y.
§ASSUME := a<=b and f is continuous on the closed interval [a,b].
§DEP_HARD := S26-D-00000312, S26-T-00000322
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000324|D|A|Proof+Compile+Meta+Topo] Cauchy sequence

§FORM := A sequence (x_n) in a metric space is Cauchy when for every epsilon>0 there exists N such that m,n>=N imply d(x_m,x_n)<epsilon.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000301, S26-D-00000031, S26-T-00000055
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000325|D|A|Proof+Compile+Meta+Topo] Complete metric space

§FORM := A metric space is complete when every Cauchy sequence converges to a point of the space.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000324, S26-D-00000310
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000326|T|A|Proof+Compile+Meta+Topo] Convergent sequences are Cauchy

§FORM := Every convergent sequence in a metric space is Cauchy.
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000310, S26-D-00000324
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000327|T|A|Proof+Compile+Meta+Topo] Finite-dimensional normed spaces are complete

§FORM := Every finite-dimensional normed vector space over R or C is complete.
§ASSUME := The scalar field is R or C with its usual absolute value.
§DEP_HARD := S26-D-00000203, S26-D-00000161, S26-D-00000325
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000328|T|A|Proof+Compile+Meta+Topo] Equivalence of norms in finite dimension

§FORM := Any two norms on a finite-dimensional vector space over R or C induce the same topology; quantitatively, each bounds the other up to positive constants.
§ASSUME := The vector space is finite-dimensional over R or C.
§DEP_HARD := S26-D-00000203, S26-D-00000161, S26-T-00000305
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Arbitrary norms on infinite-dimensional spaces need not be equivalent.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000329|T|A|Proof+Compile+Meta+Topo] Banach fixed-point theorem

§FORM := A contraction T on a nonempty complete metric space has a unique fixed point, and iteration x_{n+1}=T(x_n) converges to it from every starting point.

§ASSUME := There exists q with 0<=q<1 such that d(Tx,Ty)<=q d(x,y) for all x,y.

§DEP_HARD := S26-D-00000325, S26-D-00000312

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000330|D|A|Proof+Compile+Meta+Topo] Frechet derivative in finite dimensions

§FORM := For open U subset R^n and f:U->R^m, f is differentiable at a in U when there exists a linear map Df(a) such that ||f(a+h)-f(a)-Df(a)h||/||h|| -> 0 as h->0.

§ASSUME := U is open in R^n and a is in U.

§DEP_HARD := S26-D-00000166, S26-D-00000203, S26-D-00000312

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000331|T|A|Proof+Compile+Meta+Topo] Uniqueness of the derivative

§FORM := If the Frechet derivative of f at a exists, it is unique.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000330, S26-D-00000166

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000332|T|A|Proof+Compile+Meta+Topo] Differentiability implies continuity

§FORM := If f is differentiable at a, then f is continuous at a.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000330, S26-D-00000312

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000333|D|A|Proof+Compile+Meta+Topo] Directional derivative

§FORM := The directional derivative of f at a in direction v, when it exists, is D_v f(a)=lim_{t->0}(f(a+tv)-f(a))/t.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000330, S26-T-00000055

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000334|T|A|Proof+Compile+Meta+Topo] Derivative determines directional derivatives

§FORM := If f is Frechet differentiable at a, then every directional derivative exists and D_v f(a)=Df(a)v.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000330, S26-D-00000333, S26-D-00000166

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§BOUNDARY := Existence of all directional derivatives alone does not imply Frechet differentiability.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000335|D|A|Proof+Compile+Meta+Topo] Partial derivative

§FORM := For f:U subset R^n->R^m, the j-th partial derivative at a is the directional derivative in the coordinate direction e_j, when it exists.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000333, S26-D-00000158

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000336|D|A|Proof+Compile+Meta+Topo] Jacobian matrix

§FORM := For differentiable f:U subset R^n->R^m, the Jacobian J_f(a) is the matrix of Df(a) in the standard bases; its columns are the partial derivatives with respect to the input coordinates.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000330, S26-D-00000176, S26-D-00000335

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000337|T|A|Proof+Compile+Meta+Topo] Chain rule

§FORM := If f is differentiable at a and g is differentiable at f(a), then g o f is differentiable at a and D(g o f)(a)=Dg(f(a)) o Df(a).

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000330, S26-D-00000009, S26-D-00000166

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000338|D|A|Proof+Compile+Meta+Topo] Gradient

§FORM := For differentiable scalar f:U subset R^n->R, the gradient grad f(a) is the unique vector satisfying Df(a)h=<grad f(a),h> for every h.

§ASSUME := R^n has its standard Euclidean inner product.

§DEP_HARD := S26-D-00000330, S26-D-00000202, S26-D-00000251

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000339|D|A|Proof+Compile+Meta+Topo] Continuously differentiable map

§FORM := A differentiable map f:U->R^m is C^1 when a |-> Df(a) is continuous as a map into the finite-dimensional space of linear maps.

§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000330, S26-D-00000312, S26-D-00000166

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000340|T|A|Proof+Compile+Meta+Topo] Continuous partial derivatives imply C1

§FORM := If all first partial derivatives of f:U subset R^n->R^m exist on open U and are continuous, then f is C^1.
§ASSUME := U is open in R^n.
§DEP_HARD := S26-D-00000335, S26-D-00000339
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000341|T|A|Proof+Compile+Meta+Topo] Mean value theorem - scalar one-dimensional form

§FORM := If f:[a,b]->R is continuous on [a,b] and differentiable on (a,b), then some c in (a,b) satisfies f(b)-f(a)=f'(c)(b-a).
§ASSUME := a<b; differentiability is the usual one-dimensional derivative.
§DEP_HARD := S26-D-00000312, S26-D-00000330
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000342|T|A|Proof+Compile+Meta+Topo] First-order Taylor expansion

§FORM := If f:U subset R^n->R^m is differentiable at a, then f(a+h)=f(a)+Df(a)h+o(||h||) as h->0.
§ASSUME := a is an interior point of U.
§DEP_HARD := S26-D-00000330
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000343|D|A|Proof+Compile+Meta+Topo] Second derivative and Hessian

§FORM := For scalar f on open U subset R^n, when f is twice differentiable at a, the second derivative is a bilinear form D^2f(a); in standard coordinates its matrix is the Hessian H_f(a).
§ASSUME := All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000330, S26-D-00000261, S26-D-00000176
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000344|T|A|Proof+Compile+Meta+Topo] Symmetry of mixed partials - Clairaut-Schwarz

§FORM := If f has continuous second partial derivatives on a neighborhood of a, then the mixed second partials commute at a, so the Hessian is symmetric.
§ASSUME := Second partial derivatives exist and are continuous on a neighborhood of a.
§DEP_HARD := S26-D-00000343, S26-D-00000265
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Without sufficient regularity, mixed partial derivatives can exist and fail to agree.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000345|T|A|Proof+Compile+Meta+Topo] Second-order Taylor expansion

§FORM := If f:U subset R^n->R is twice differentiable at a, then f(a+h)=f(a)+Df(a)h+(1/2)D^2f(a)[h,h]+o(||h||^2) as h->0.
§ASSUME := The second Frechet derivative exists at a in the stated sense.
§DEP_HARD := S26-T-00000342, S26-D-00000343
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000346|T|A|Proof+Compile+Meta+Topo] Inverse function theorem - finite-dimensional C1 form

§FORM := If f:U subset R^n->R^n is C^1 and Df(a) is invertible, then f restricts to a C^1 diffeomorphism between neighborhoods of a and f(a), with derivative of the local inverse equal to Df(x)^{-1} at corresponding points.
§ASSUME := U is open, a in U, and Df(a) is invertible.
§DEP_HARD := S26-D-00000339, S26-T-00000175, S26-T-00000337
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000347|T|A|Proof+Compile+Meta+Topo] Implicit function theorem - finite-dimensional C1 form

§FORM := Let F:U subset R^{n+m}->R^m be C^1 near (a,b), with F(a,b)=0 and the derivative with respect to the last m variables invertible. Then near (a,b), the zero set is the graph y=g(x) of a unique C^1 function g.
§ASSUME := U is open and the indicated m x m partial Jacobian is invertible at (a,b).
§DEP_HARD := S26-D-00000339, S26-T-00000175, S26-T-00000346
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000348|T|A|Proof+Compile+Meta+Topo] First-order necessary condition for an interior extremum

§FORM := If differentiable f:U subset R^n->R has a local maximum or minimum at interior point a, then Df(a)=0, equivalently grad f(a)=0.
§ASSUME := a is an interior point of open U.
§DEP_HARD := S26-D-00000330, S26-D-00000338
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := At boundary extrema the gradient need not vanish.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000349|T|A|Proof+Compile+Meta+Topo] Second derivative test for strict local extrema

§FORM := If f is C^2 near a, grad f(a)=0, and H_f(a) is positive definite, then a is a strict local minimum; if H_f(a) is negative definite, then a is a strict local maximum.
§ASSUME := f is C^2 on a neighborhood of a.
§DEP_HARD := S26-D-00000343, S26-T-00000348, S26-T-00000242
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := A semidefinite Hessian is generally inconclusive.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000350|D|A|Proof+Compile+Meta+Topo] Convex function on a convex set

§FORM := For a convex subset C of a real vector space, f:C->R is convex when f((1-t)x+ty)<= (1-t)f(x)+t f(y) for all x,y in C and t in [0,1].
§ASSUME := C is convex and the scalar field is R.
§DEP_HARD := S26-D-00000149, S26-T-00000055
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000351|D|A|Proof+Compile+Meta] Sequence in a metric space

§FORM := A sequence in X is a map x:N->X, written (x_n); convergence is the metric convergence of entry 310.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000031, S26-D-00000301, S26-D-00000310
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000352|D|A|Proof+Compile+Meta] Subsequence

§FORM := A subsequence of (x_n) is (x_{n_k}) where n_1<n_2<... is a strictly increasing sequence of natural indices.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000351, S26-D-00000020
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000353|T|A|Proof+Compile+Meta] Subsequences preserve limits

§FORM := Every subsequence of a convergent sequence converges to the same limit.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000352, S26-D-00000310
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000354|D|A|Proof+Compile+Meta] Limit point of a sequence

§FORM := A point x is a subsequential limit of (x_n) when some subsequence converges to x.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000352, S26-D-00000310
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000355|T|A|Proof+Compile+Meta] Bolzano-Weierstrass in R^n

§FORM := Every bounded sequence in finite-dimensional Euclidean R^n has a convergent subsequence.
§ASSUME := The sequence is bounded in R^n with its usual Euclidean metric.
§DEP_HARD := S26-D-00000351, S26-T-00000318, S26-D-00000352
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := The analogous statement fails for bounded sequences in general infinite-dimensional normed spaces.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000356|D|A|Proof+Compile+Meta] Series in a normed vector space

§FORM := For a sequence (a_n) in a normed vector space, the series sum a_n converges when its partial sums s_N=sum_{n=1}^N a_n converge.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000351, S26-D-00000203, S26-D-00000310
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000357|T|A|Proof+Compile+Meta] Cauchy criterion for series

§FORM := In a complete normed space, sum a_n converges iff for every epsilon>0 there exists N such that ||sum_{k=m}^n a_k||<epsilon whenever n>=m>=N.
§ASSUME := The ambient normed space is complete.
§DEP_HARD := S26-D-00000356, S26-D-00000325, S26-D-00000324
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000358|D|A|Proof+Compile+Meta] Absolute convergence of a scalar series

§FORM := A real or complex series sum a_n is absolutely convergent when sum |a_n| converges in R.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000356, S26-T-00000055, S26-I-00000060
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000359|T|A|Proof+Compile+Meta] Absolute convergence implies convergence

§FORM := Every absolutely convergent real or complex series converges.
§ASSUME := Scalars are R or C.
§DEP_HARD := S26-D-00000358, S26-T-00000357, S26-T-00000327
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000360|T|A|Proof+Compile+Meta+Topo] Geometric series

§FORM := For scalar r with |r|<1, sum_{n=0}^infinity r^n=1/(1-r).
§ASSUME := r is real or complex and |r|<1.
§DEP_HARD := S26-D-00000356, S26-T-00000055
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := At |r|>=1 the infinite geometric series need not converge.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000361|T|A|Proof+Compile+Meta] Comparison test for nonnegative series

§FORM := If 0<=a_n<=b_n eventually and sum b_n converges, then sum a_n converges.
§ASSUME := a_n,b_n are real and nonnegative eventually.
§DEP_HARD := S26-D-00000356, S26-D-00000020
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000362|T|A|Proof+Compile+Meta] Ratio test

§FORM := For a scalar series with nonzero terms eventually, if limsup |a_{n+1}/a_n|<1 then the series converges absolutely; if liminf |a_{n+1}/a_n|>1 then terms do not tend to zero and the series diverges.
§ASSUME := The indicated ratios are defined eventually.
§DEP_HARD := S26-D-00000358, S26-T-00000360, S26-D-00000310
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000363|T|A|Proof+Compile+Meta] Root test

§FORM := For a scalar series, if limsup |a_n|^(1/n)<1 then it converges absolutely; if limsup |a_n|^(1/n)>1 then it diverges.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000358, S26-T-00000360, S26-D-00000310
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000364|D|A|Proof+Compile+Meta] Pointwise convergence of functions

§FORM := A sequence f_n:X->Y converges pointwise to f when for every x in X, f_n(x)->f(x).
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000351, S26-D-00000008, S26-D-00000310
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000365|D|A|Proof+Compile+Meta] Uniform convergence of functions

§FORM := For metric Y, f_n:X->Y converges uniformly to f when for every epsilon>0 there exists N such that n>=N implies d(f_n(x),f(x))<epsilon for every x in X.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000364, S26-D-00000301
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000366|T|A|Proof+Compile+Meta] Uniform convergence implies pointwise convergence

§FORM := Uniform convergence of functions implies pointwise convergence.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000365, S26-D-00000364
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000367|T|A|Proof+Compile+Meta] Uniform limit theorem

§FORM := If f_n:X->Y are continuous, Y is metric, and f_n converges uniformly to f, then f is continuous.
§ASSUME := Continuity is with respect to the topology on X and metric topology on Y.
§DEP_HARD := S26-D-00000365, S26-D-00000312
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := A pointwise limit of continuous functions need not be continuous.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000368|D|A|Proof+Compile+Meta] Supremum norm

§FORM := For bounded scalar-valued f on X, ||f||_infinity=sup_{x in X}|f(x)|.
§ASSUME := f is bounded so the supremum is finite.
§DEP_HARD := S26-D-00000203, S26-D-00000021, S26-T-00000055
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000369|T|A|Proof+Compile+Meta] Uniform convergence and supremum norm

§FORM := For bounded scalar functions, f_n->f uniformly iff ||f_n-f||_infinity->0.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000365, S26-D-00000368
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000370|T|A|Proof+Compile+Meta] Weierstrass M-test

§FORM := If |f_n(x)|<=M_n for every x and sum M_n converges, then sum f_n converges uniformly and absolutely pointwise.
§ASSUME := M_n>=0 and the scalar codomain is R or C.
§DEP_HARD := S26-D-00000356, S26-D-00000365, S26-T-00000361
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000371|D|A|Proof+Compile+Meta] Power series

§FORM := A power series centered at a is sum_{n=0}^infinity c_n(x-a)^n over R or C.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000356, S26-T-00000126, S26-T-00000055
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000372|D|A|Proof+Compile+Meta] Radius of convergence

§FORM := The radius R in [0,infinity] of a power series is the threshold such that the series converges absolutely for |x-a|<R and diverges for |x-a|>R.

§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000371, S26-D-00000358

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000373|T|A|Proof+Compile+Meta] Cauchy-Hadamard formula

§FORM := For a power series sum c_n(x-a)^n, its radius satisfies 1/R=limsup |c_n|^(1/n), with the usual conventions for 0 and infinity.

§ASSUME := Coefficients are real or complex.

§DEP_HARD := S26-D-00000372, S26-T-00000363

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000374|T|A|Proof+Compile+Meta] Uniform convergence inside the radius

§FORM := A power series converges uniformly on every closed disk |x-a|<=r with r<R.

§ASSUME := 0<=r<R.

§DEP_HARD := S26-T-00000370, S26-D-00000372

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000375|T|A|Proof+Compile+Meta] Termwise differentiation of a power series

§FORM := Inside its radius of convergence, a power series is differentiable and its derivative is obtained term by term; the derived series has the same radius.

§ASSUME := Real one-variable or complex power series, evaluated strictly inside the radius of convergence.

§DEP_HARD := S26-D-00000371, S26-T-00000374, S26-D-00000330

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000376|T|A|Proof+Compile+Meta] Termwise integration of a power series

§FORM := Inside its radius of convergence, a real power series may be integrated term by term on compact subintervals, yielding an antiderivative series with the same radius.

§ASSUME := Real one-variable power series on an interval strictly inside the radius.

§DEP_HARD := S26-D-00000371, S26-T-00000374

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000377|D|A|Proof+Compile+Meta] Partition of a compact interval

§FORM := A partition P of [a,b] is a finite ordered set a=x_0<x_1<...<x_n=b.

§ASSUME := a<b are real.

§DEP_HARD := S26-D-00000023, S26-D-00000020, S26-T-00000055

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000378|D|A|Proof+Compile+Meta] Mesh of a partition

§FORM := The mesh ||P|| of P is max_i(x_i-x_{i-1}).

§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000377, S26-D-00000021

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000379|D|A|Proof+Compile+Meta] Tagged partition

§FORM := A tagged partition is a partition P together with tags t_i in [x_{i-1},x_i].

§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000377

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000380|D|A|Proof+Compile+Meta] Riemann sum

§FORM := For bounded f:[a,b]->R and tagged partition, the Riemann sum is sum_i f(t_i)(x_i-x_{i-1}).

§ASSUME := f is bounded on [a,b].

§DEP_HARD := S26-D-00000379, S26-D-00000356

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000381|D|A|Proof+Compile+Meta] Riemann integrability - tagged-sum definition

§FORM := A bounded f:[a,b]->R is Riemann integrable with integral I if for every epsilon>0 there exists delta>0 such that every tagged partition with mesh<delta has Riemann sum within epsilon of I.

§ASSUME := f is bounded on the compact interval [a,b].

§DEP_HARD := S26-D-00000380, S26-D-00000378

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000382|T|A|Proof+Compile+Meta] Uniqueness of the Riemann integral

§FORM := A Riemann-integrable function has a unique integral value.

§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000381, S26-T-00000055

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000383|T|A|Proof+Compile+Meta] Continuous functions are Riemann integrable

§FORM := Every continuous real-valued function on [a,b] is Riemann integrable.
§ASSUME := a<b and f:[a,b]->R is continuous.
§DEP_HARD := S26-D-00000312, S26-T-00000318, S26-D-00000381
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000384|T|A|Proof+Compile+Meta] Monotone functions are Riemann integrable

§FORM := Every bounded monotone real-valued function on [a,b] is Riemann integrable.
§ASSUME := a<b and f is monotone.
§DEP_HARD := S26-D-00000381, S26-D-00000020
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000385|T|A|Proof+Compile+Meta] Linearity of the Riemann integral

§FORM := If f and g are Riemann integrable and alpha,beta are real, then integral(alpha f+beta g)=alpha integral f+beta integral g.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000381, S26-D-00000149
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000386|T|A|Proof+Compile+Meta] Order preservation of the Riemann integral

§FORM := If f and g are Riemann integrable and f(x)<=g(x) on [a,b], then integral f<=integral g.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000381, S26-D-00000020
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000387|T|A|Proof+Compile+Meta] Integral absolute-value bound

§FORM := If f is Riemann integrable, then |integral_a^b f|<=integral_a^b |f|.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000381, S26-T-00000386
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000388|T|A|Proof+Compile+Meta] Additivity over adjacent intervals

§FORM := If f is Riemann integrable on [a,b] and c in [a,b], then integral_a^b f=integral_a^c f+integral_c^b f.
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000381, S26-T-00000385
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000389|T|A|Proof+Compile+Meta] Uniform convergence preserves Riemann integrals

§FORM := If Riemann-integrable f_n on [a,b] converge uniformly to f, then f is Riemann integrable and integral f_n -> integral f.
§ASSUME := All functions are real-valued on the same compact interval.
§DEP_HARD := S26-D-00000365, S26-D-00000381, S26-T-00000387
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000390|D|A|Proof+Compile+Meta] Indefinite integral function

§FORM := For Riemann-integrable f on [a,b], define F(x)=integral_a^x f(t)dt for x in [a,b].
§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000381, S26-T-00000388
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000391|T|A|Proof+Compile+Meta] Fundamental theorem of calculus - part I

§FORM := If f:[a,b]->R is continuous and F(x)=integral_a^x f(t)dt, then F is differentiable on (a,b) and F'(x)=f(x).
§ASSUME := f is continuous on [a,b].
§DEP_HARD := S26-T-00000383, S26-D-00000390, S26-D-00000330
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000392|T|A|Proof+Compile+Meta] Fundamental theorem of calculus - part II

§FORM := If F:[a,b]->R is C^1, then integral_a^b F'(x)dx=F(b)-F(a).
§ASSUME := F is continuously differentiable on [a,b].
§DEP_HARD := S26-D-00000339, S26-T-00000383, S26-T-00000391
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000393|T|A|Proof+Compile+Meta] Integration by parts

§FORM := If u,v are C^1 on [a,b], then integral_a^b u v'=u(b)v(b)-u(a)v(a)-integral_a^b u' v.
§ASSUME := u and v are continuously differentiable.
§DEP_HARD := S26-T-00000392, S26-T-00000337
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000394|T|A|Proof+Compile+Meta] Change of variables - one-dimensional Riemann form

§FORM := If phi:[alpha,beta]->[a,b] is C^1 and f is continuous on [a,b], then integral_alpha^beta f(phi(t))phi'(t)dt=integral_{phi(alpha)}^{phi(beta)} f(x)dx.

§ASSUME := phi is C^1; f is continuous on an interval containing phi([alpha,beta]).

§DEP_HARD := S26-T-00000392, S26-T-00000337

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000395|T|A|Proof+Compile+Meta] Mean value theorem for integrals

§FORM := If f is continuous on [a,b] and a<b, then some c in [a,b] satisfies integral_a^b f=f(c)(b-a).

§ASSUME := f is continuous and real-valued.

§DEP_HARD := S26-T-00000383, S26-T-00000319, S26-T-00000323

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000396|D|A|Proof+Compile+Meta] Improper integral on an unbounded interval

§FORM := For f on [a,infinity), the improper integral integral_a^infinity f is defined as lim_{b->infinity} integral_a^b f when this limit exists.

§ASSUME := f is Riemann integrable on every finite interval [a,b].

§DEP_HARD := S26-D-00000381, S26-D-00000310

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000397|T|A|Proof+Compile+Meta] Absolute convergence criterion for improper integrals

§FORM := If integral_a^infinity |f(x)|dx converges, then integral_a^infinity f(x)dx converges.

§ASSUME := f is Riemann integrable on every compact subinterval.

§DEP_HARD := S26-D-00000396, S26-T-00000387

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000398|T|A|Proof+Compile+Meta] p-integral criterion

§FORM := For real p, integral_1^infinity x^{-p}dx converges iff p>1.

§ASSUME := All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000396, S26-T-00000055

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000399|T|A|Proof+Compile+Meta] Integral test for positive decreasing series

§FORM := If f:[1,infinity)->[0,infinity) is continuous and decreasing and a_n=f(n), then sum_{n=1}^infinity a_n converges iff integral_1^infinity f(x)dx converges.

§ASSUME := f is continuous, nonnegative, and decreasing.

§DEP_HARD := S26-D-00000356, S26-D-00000396, S26-D-00000020

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000400|T|A|Proof+Compile+Meta] Uniform convergence permits termwise integration

§FORM := If continuous f_n:[a,b]->R and sum f_n converges uniformly to S, then S is continuous and integral_a^b S=sum_n integral_a^b f_n.

§ASSUME := The function series converges uniformly on the compact interval [a,b].

§DEP_HARD := S26-T-00000370, S26-T-00000367, S26-T-00000389

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§BOUNDARY := Pointwise convergence alone does not justify interchange of infinite summation and integration.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000401|D|A|Proof+Compile+Meta+IG] Sigma-algebra

§FORM := A sigma-algebra Sigma on a set X is a family of subsets containing X and closed under complements in X and countable unions.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-N-00000016, S26-D-00000023, S26-D-00000031

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000402|D|A|Proof+Compile+Meta+IG] Measurable space

§FORM := A measurable space is a pair (X,Sigma) consisting of a set X and a sigma-algebra Sigma on X.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000401

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000403|D|A|Proof+Compile+Meta+IG] Generated sigma-algebra

§FORM := For a family C of subsets of X, sigma(C) is the intersection of all sigma-algebras on X containing C.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000401, S26-D-00000018

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000404|D|A|Proof+Compile+Meta+Topo+IG] Borel sigma-algebra

§FORM := For a topological space X, the Borel sigma-algebra B(X) is the sigma-algebra generated by the open subsets of X.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000403, S26-T-00000305

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000405|D|A|Proof+Compile+Meta+IG] Measure

§FORM := A measure on (X,Sigma) is a map mu:Sigma->[0,infinity] with mu(emptyset)=0 and countable additivity on pairwise disjoint measurable sets.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000402, S26-D-00000031

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000406|D|A|Proof+Compile+Meta+IG] Measure space

§FORM := A measure space is a triple (X,Sigma,mu) where (X,Sigma) is measurable and mu is a measure on it.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000405

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000407|T|A|Proof+Compile+Meta+IG] Continuity from below

§FORM := If E_1 subset E_2 subset ... are measurable, then mu(union_n E_n)=lim_n mu(E_n).

§ASSUME := (E_n) is an increasing sequence of measurable sets.

§DEP_HARD := S26-D-00000405, S26-D-00000310

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000408|T|A|Proof+Compile+Meta+IG] Continuity from above

§FORM := If E_1 superset E_2 superset ... are measurable and mu(E_1)<infinity, then mu(intersection_n E_n)=lim_n mu(E_n).

§ASSUME := (E_n) is decreasing and mu(E_1) is finite.

§DEP_HARD := S26-D-00000405, S26-D-00000310

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§BOUNDARY := The finite-measure hypothesis cannot in general be dropped.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000409|D|A|Proof+Compile+Meta+IG] Null set

§FORM := A measurable set N is null when mu(N)=0.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000405

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000410|D|A|Proof+Compile+Meta+IG] Almost everywhere

§FORM := A property holds mu-almost everywhere when the measurable set on which it fails is null.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000409

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000411|D|A|Proof+Compile+Meta+IG] Complete measure

§FORM := A measure space is complete when every subset of every measurable null set is measurable.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000406, S26-D-00000409

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000412|D|A|Proof+Compile+Meta+IG] Measurable function

§FORM := A function f:(X,Sigma)->(Y,Tau) is measurable when f^{-1}(A) is in Sigma for every A in Tau.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000008, S26-D-00000402

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000413|T|A|Proof+Compile+Meta+Topo+IG] Continuous maps are Borel measurable

§FORM := Every continuous map between topological spaces is measurable with respect to their Borel sigma-algebras.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000312, S26-D-00000404, S26-D-00000412

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000414|T|A|Proof+Compile+Meta+IG] Composition of measurable maps

§FORM := The composition of measurable maps is measurable.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000412, S26-D-00000011

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000415|T|A|Proof+Compile+Meta+IG] Pointwise limits preserve measurability

§FORM := A pointwise limit of measurable real- or extended-real-valued functions is measurable whenever the pointwise limit is defined.

§ASSUME := The codomain carries its Borel sigma-algebra; extended-real conventions are used where needed.

§DEP_HARD := S26-D-00000412, S26-D-00000364

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000416|D|A|Proof+Compile+Meta+IG] Indicator function

§FORM := For E subset X, the indicator 1_E:X->{0,1} equals 1 on E and 0 outside E.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-N-00000016, S26-D-00000008
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000417|D|A|Proof+Compile+Meta+IG] Nonnegative simple function

§FORM := A nonnegative simple measurable function is a finite sum sum_i a_i 1_{E_i} with a_i>=0 and measurable E_i.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000416, S26-D-00000412, S26-D-00000020
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000418|D|A|Proof+Compile+Meta+IG] Integral of a nonnegative simple function

§FORM := For a representation s=sum_i a_i 1_{E_i} with disjoint measurable E_i, define integral s dmu=sum_i a_i mu(E_i).
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000417, S26-D-00000405
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000419|D|A|Proof+Compile+Meta+IG] Lebesgue integral of a nonnegative function

§FORM := For measurable f:X->[0,infinity], define integral f dmu as the supremum of integrals of nonnegative simple measurable s with 0<=s<=f.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000418, S26-D-00000021, S26-D-00000412
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000420|T|A|Proof+Compile+Meta+IG] Monotone convergence theorem

§FORM := If 0<=f_n<=f_{n+1} are measurable and f_n->f pointwise, then integral f_n dmu -> integral f dmu, allowing infinity.
§ASSUME := All f_n are nonnegative measurable functions.
§DEP_HARD := S26-D-00000419, S26-D-00000364
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000421|T|A|Proof+Compile+Meta+IG] Fatou lemma

§FORM := For nonnegative measurable f_n, integral liminf_n f_n dmu <= liminf_n integral f_n dmu.
§ASSUME := All f_n are nonnegative measurable functions.
§DEP_HARD := S26-D-00000419, S26-T-00000420
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000422|D|A|Proof+Compile+Meta+IG] Positive and negative parts

§FORM := For a real-valued measurable f, define f^+=max(f,0), f^-=max(-f,0), so f=f^+-f^- and |f|=f^++f^-.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000412, S26-D-00000020, S26-T-00000055
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000423|D|A|Proof+Compile+Meta+IG] Lebesgue integrable function

§FORM := A measurable real- or complex-valued function f is integrable when integral |f| dmu<infinity; its integral is defined from positive/negative parts (or real/imaginary parts).
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000419, S26-D-00000422, S26-I-00000060
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000424|T|A|Proof+Compile+Meta+IG] Linearity of the Lebesgue integral

§FORM := For integrable f,g and scalars alpha,beta, integral(alpha f+beta g)=alpha integral f+beta integral g.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000423, S26-D-00000149
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000425|T|A|Proof+Compile+Meta+IG] Integral absolute-value inequality

§FORM := For integrable f, |integral f dmu|<=integral |f| dmu.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000423, S26-D-00000020
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000426|T|A|Proof+Compile+Meta+IG] Dominated convergence theorem

§FORM := If measurable f_n->f almost everywhere and |f_n|<=g almost everywhere for an integrable g, then f is integrable and integral |f_n-f| dmu->0; hence integral f_n dmu->integral f dmu.
§ASSUME := g is integrable and dominates every |f_n| almost everywhere.
§DEP_HARD := S26-D-00000410, S26-D-00000423, S26-T-00000421
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Pointwise or almost-everywhere convergence without an integrable dominator does not in general permit exchanging limit and integral.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000427|T|A|Proof+Compile+Meta+IG] Bounded convergence on finite measure spaces

§FORM := If mu(X)<infinity, f_n are uniformly bounded by a finite constant, and f_n->f almost everywhere, then integral |f_n-f| dmu->0.

§ASSUME := mu(X)<infinity and a common finite essential bound exists.

§DEP_HARD := S26-T-00000426, S26-D-00000405

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000428|T|A|Proof+Compile+Meta+IG] Riemann integrable implies Lebesgue integrable

§FORM := A bounded Riemann-integrable function on [a,b] is Lebesgue measurable and Lebesgue integrable, and its Lebesgue integral equals its Riemann integral.

§ASSUME := Lebesgue measure on a compact real interval is used.

§DEP_HARD := S26-D-00000381, S26-D-00000423, S26-D-00000404

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000429|D|A|Proof+Compile+Meta+IG] Product sigma-algebra

§FORM := For measurable spaces (X,Sigma),(Y,Tau), Sigma tensor Tau is the sigma-algebra on X x Y generated by measurable rectangles A x B.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000403, S26-D-00000402, S26-D-00000017

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000430|D|A|Proof+Compile+Meta+IG] Product measure

§FORM := For sigma-finite measure spaces, the product measure mu x nu is the measure on the product sigma-algebra satisfying (mu x nu)(A x B)=mu(A)nu(B) on measurable rectangles.

§ASSUME := The factors are sigma-finite measure spaces.

§DEP_HARD := S26-D-00000429, S26-D-00000405

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000431|T|A|Proof+Compile+Meta+IG] Tonelli theorem

§FORM := For a nonnegative measurable f on a product of sigma-finite measure spaces, the product integral equals either iterated integral, allowing infinity.

§ASSUME := The product measure spaces are sigma-finite and f is nonnegative measurable.

§DEP_HARD := S26-D-00000419, S26-D-00000430

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000432|T|A|Proof+Compile+Meta+IG] Fubini theorem

§FORM := For integrable f on a product of sigma-finite measure spaces, the iterated integrals exist almost everywhere and equal the product integral in either order.

§ASSUME := f is integrable with respect to the product measure and the factors are sigma-finite.

§DEP_HARD := S26-D-00000423, S26-D-00000430, S26-T-00000431

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§BOUNDARY := Without absolute integrability, changing the order of integration can fail or become undefined.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000433|D|A|Proof+Compile+Meta+IG] L^p seminorm

§FORM := For 1<=p<infinity define ||f||_p=(integral |f|^p dmu)^(1/p); define ||f||_infinity as the essential supremum.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000423, S26-D-00000021, S26-D-00000368

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000434|D|A|Proof+Compile+Meta+IG] L^p space

§FORM := L^p(X,mu) is the vector space of measurable functions with finite L^p seminorm modulo equality almost everywhere.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000433, S26-D-00000410, S26-D-00000149

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000435|T|A|Proof+Compile+Meta+IG] Holder inequality

§FORM := If 1<=p,q<=infinity with 1/p+1/q=1, then ||fg||_1<=||f||_p ||g||_q.

§ASSUME := p and q are conjugate exponents and f in L^p, g in L^q.

§DEP_HARD := S26-D-00000433, S26-D-00000434

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000436|T|A|Proof+Compile+Meta+IG] Minkowski inequality

§FORM := For 1<=p<=infinity, ||f+g||_p<=||f||_p+||g||_p.

§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

§DEP_HARD := S26-D-00000433, S26-T-00000435

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000437|T|A|Proof+Compile+Meta+IG] L^p completeness

§FORM := For 1<=p<=infinity, L^p(X,mu) is a Banach space under ||.||_p.

§ASSUME := Functions are identified modulo equality almost everywhere.

§DEP_HARD := S26-D-00000434, S26-T-00000436, S26-D-00000325

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000438|D|A|Proof+Compile+Meta+IG] L^2 inner product

§FORM := For complex L^2 define <f,g>=integral conjugate(f)g dmu; for real L^2 omit conjugation.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000434, S26-D-00000201, S26-I-00000060
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000439|T|A|Proof+Compile+Meta+IG] Cauchy-Schwarz in L^2

§FORM := For f,g in L^2, |<f,g>|<=||f||_2||g||_2.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000438, S26-T-00000435
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000440|T|A|Proof+Compile+Meta+IG+Spectral] L^2 is a Hilbert space

§FORM := L^2(X,mu), modulo equality almost everywhere, is complete under the norm induced by its inner product.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-T-00000437, S26-D-00000438, S26-T-00000209
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000441|D|A|Proof+Compile+Meta+IG] Convergence in measure

§FORM := A sequence f_n converges to f in measure when for every epsilon>0, mu({|f_n-f|>epsilon})->0.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000405, S26-D-00000310, S26-D-00000412
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000442|T|A|Proof+Compile+Meta+IG] L^p convergence implies convergence in measure

§FORM := If 1<=p<infinity and ||f_n-f||_p->0, then f_n->f in measure.
§ASSUME := 1<=p<infinity.
§DEP_HARD := S26-D-00000433, S26-D-00000441
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000443|T|A|Proof+Compile+Meta+IG] Almost-everywhere convergence need not imply L^1 convergence

§FORM := There exist integrable f_n converging to 0 almost everywhere while ||f_n||_1 does not tend to 0.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000410, S26-D-00000434
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := On [0,1], f_n=n 1_(0,1/n) converges to 0 almost everywhere but has L^1 norm 1.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000444|D|A|Proof+Compile+Meta+IG] Pushforward measure

§FORM := For measurable T:(X,Sigma)->(Y,Tau), the pushforward T_*mu is defined by (T_*mu)(A)=mu(T^{-1}(A)).
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000412, S26-D-00000405
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000445|T|A|Proof+Compile+Meta+IG] Integration against a pushforward

§FORM := For nonnegative measurable or integrable g on Y, integral_Y g d(T_*mu)=integral_X g(T(x)) dmu.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000444, S26-D-00000419, S26-D-00000423
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000446|D|A|Proof+Compile+Meta+IG] Probability space

§FORM := A probability space is a measure space (Omega,F,P) with P(Omega)=1.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000406
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000447|D|A|Proof+Compile+Meta+IG] Random variable

§FORM := A random variable with values in a measurable space S is a measurable map X:(Omega,F)->S.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000446, S26-D-00000412
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000448|D|A|Proof+Compile+Meta+IG] Distribution of a random variable

§FORM := The distribution or law of X is the pushforward probability measure X_*P.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000447, S26-D-00000444
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000449|D|A|Proof+Compile+Meta+IG] Expectation

§FORM := For an integrable real- or complex-valued random variable X, E[X]=integral_Omega X dP.
§ASSUME := All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.
§DEP_HARD := S26-D-00000447, S26-D-00000423
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000450|T|A|Proof+Compile+Meta+IG] Markov inequality

§FORM := If X>=0 is an integrable random variable and a>0, then P(X>=a)<=E[X]/a.
§ASSUME := X is nonnegative and integrable; a>0.
§DEP_HARD := S26-D-00000449, S26-D-00000020
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000451|D|A|Proof+Compile+Meta+IG] Event

§FORM := An event in a probability space (Omega,F,P) is an element A of the sigma-algebra F.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000446, S26-D-00000401
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000452|D|A|Proof+Compile+Meta+IG] Conditional probability

§FORM := For events A,B with P(B)>0, define P(A|B)=P(A intersection B)/P(B).
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000451, S26-D-00000020
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000453|T|A|Proof+Compile+Meta+IG] Multiplication rule

§FORM := If P(B)>0, then P(A intersection B)=P(A|B)P(B).
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000452
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000454|T|A|Proof+Compile+Meta+IG] Law of total probability

§FORM := If (B_i) is a finite or countable measurable partition with P(B_i)>0, then P(A)=sum_i P(A|B_i)P(B_i).
§ASSUME := (B_i) is a finite or countable partition of Omega and each used conditional probability is defined.
§DEP_HARD := S26-D-00000452, S26-D-00000031
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000455|T|A|Proof+Compile+Meta+IG] Bayes theorem

§FORM := If P(A)>0 and P(B)>0, then P(A|B)=P(B|A)P(A)/P(B).
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000452, S26-T-00000453
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000456|D|A|Proof+Compile+Meta+IG] Independence of two events

§FORM := Events A and B are independent when P(A intersection B)=P(A)P(B).
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000451
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000457|D|A|Proof+Compile+Meta+IG] Mutual independence

§FORM := A family of events is mutually independent when every finite subfamily has intersection probability equal to the product of its individual probabilities.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000456, S26-D-00000031
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000458|D|A|Proof+Compile+Meta+IG] Independence of random variables

§FORM := Random variables X_i are independent when their generated sigma-algebras are mutually independent; equivalently, finite inverse-image events factorize.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000447, S26-D-00000403, S26-D-00000457
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000459|D|A|Proof+Compile+Meta+IG] Identically distributed random variables

§FORM := Random variables X and Y are identically distributed when their pushforward laws are equal.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000448
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000460|D|A|Proof+Compile+Meta+IG] IID family

§FORM := A family of random variables is IID when its members are mutually independent and identically distributed.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000458, S26-D-00000459
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000461|D|A|Proof+Compile+Meta+IG] Distribution function

§FORM := For a real-valued random variable X, F_X(x)=P(X<=x).
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000447, S26-D-00000448
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000462|T|A|Proof+Compile+Meta+IG] CDF characterization

§FORM := A real distribution function is nondecreasing, right-continuous, tends to 0 at -infinity and 1 at +infinity; conversely every such function is the CDF of a unique Borel probability measure on R.
§ASSUME := The converse concerns Borel probability measures on R.
§DEP_HARD := S26-D-00000461, S26-D-00000404, S26-D-00000310
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000463|D|A|Proof+Compile+Meta+IG] Discrete random variable

§FORM := A random variable is discrete when its law is supported on a finite or countable set.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000448, S26-D-00000031
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000464|D|A|Proof+Compile+Meta+IG] Probability mass function

§FORM := For a discrete random variable X, p_X(x)=P(X=x); its nonnegative masses sum to 1 over the support.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000463
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000465|D|A|Proof+Compile+Meta+IG] Absolutely continuous distribution

§FORM := A real random variable has an absolutely continuous law when there exists an integrable f>=0 such that P(X in A)=integral_A f dx for every Borel A.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000448, S26-D-00000423, S26-D-00000404
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000466|D|A|Proof+Compile+Meta+IG] Probability density function

§FORM := A density f of an absolutely continuous law is a nonnegative measurable function with integral_R f dx=1 representing probabilities by integration.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000465
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000467|T|A|Proof+Compile+Meta+IG] Expectation via the law

§FORM := For integrable measurable g(X), E[g(X)]=integral g d mu_X, where mu_X is the law of X.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000445, S26-D-00000448, S26-D-00000449
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000468|D|A|Proof+Compile+Meta+IG] Moment

§FORM := When integrable, the k-th raw moment of X is E[X^k].
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000449, S26-D-00000021
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000469|D|A|Proof+Compile+Meta+IG] Variance

§FORM := For square-integrable real X, Var(X)=E[(X-E[X])^2]=E[X^2]-E[X]^2.
§ASSUME := X is real-valued and square-integrable.
§DEP_HARD := S26-D-00000449, S26-D-00000468
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000470|D|A|Proof+Compile+Meta+IG] Covariance

§FORM := For square-integrable real X,Y, Cov(X,Y)=E[(X-E[X])(Y-E[Y])].
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000469, S26-D-00000449
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000471|D|A|Proof+Compile+Meta+IG] Correlation coefficient

§FORM := If Var(X),Var(Y)>0, rho(X,Y)=Cov(X,Y)/(sqrt(Var(X))sqrt(Var(Y))).

§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000470, S26-D-00000469

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000472|T|A|Proof+Compile+Meta+IG] Cauchy-Schwarz covariance bound

§FORM := For square-integrable real X,Y, |Cov(X,Y)|<=sqrt(Var(X)Var(Y)); hence |rho|<=1 when correlation is defined.

§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000439, S26-D-00000470, S26-D-00000471

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000473|T|A|Proof+Compile+Meta+IG] Variance of a sum

§FORM := For square-integrable X_1,...,X_n, Var(sum_i X_i)=sum_i Var(X_i)+2 sum_{i<j} Cov(X_i,X_j).

§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000469, S26-D-00000470, S26-D-00000020

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000474|T|A|Proof+Compile+Meta+IG] Variance additivity under independence

§FORM := For pairwise independent square-integrable X_i, Var(sum_i X_i)=sum_i Var(X_i).

§ASSUME := The variables are pairwise independent and square-integrable.

§DEP_HARD := S26-D-00000458, S26-T-00000473

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000475|T|A|Proof+Compile+Meta+IG] Chebyshev inequality

§FORM := If X has finite variance, then for a>0, P(|X-E[X]|>=a)<=Var(X)/a^2.

§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000450, S26-D-00000469

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000476|D|A|Proof+Compile+Meta+IG] Conditional expectation

§FORM := For integrable X and sub-sigma-algebra G subset F, E[X|G] is any G-measurable integrable random variable Y satisfying integral_G Y dP=integral_G X dP for every G in G; it is unique almost surely.

§ASSUME := X is integrable and G is a sub-sigma-algebra of F.

§DEP_HARD := S26-D-00000449, S26-D-00000401, S26-D-00000410

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000477|T|A|Proof+Compile+Meta+IG+Tower] Tower property

§FORM := If H subset G subset F are sub-sigma-algebras and X is integrable, then E[E[X|G]|H]=E[X|H] almost surely.

§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000476

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000478|T|A|Proof+Compile+Meta+IG] Taking out what is known

§FORM := If X is integrable, Y is G-measurable, and YX is integrable under suitable boundedness/integrability conditions, then E[YX|G]=Y E[X|G] almost surely.

§ASSUME := Products and conditional expectations appearing in the statement are integrable; a standard sufficient case is bounded G-measurable Y and integrable X.

§DEP_HARD := S26-D-00000476, S26-T-00000424

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000479|T|A|Proof+Compile+Meta+IG] Conditional expectation under independence

§FORM := If integrable X is independent of G, then E[X|G]=E[X] almost surely.

§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000458, S26-D-00000476

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000480|D|A|Proof+Compile+Meta+IG] Moment generating function

§FORM := When finite on a neighborhood of 0, the moment generating function is M_X(t)=E[e^{tX}].

§ASSUME := The expectation is finite for the values of t under consideration.

§DEP_HARD := S26-D-00000449

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000481|D|A|Proof+Compile+Meta+IG] Characteristic function

§FORM := For real X, phi_X(t)=E[e^{itX}], defined for every real t.

§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000449, S26-I-00000060

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000482|T|A|Proof+Compile+Meta+IG] Characteristic function of an independent sum

§FORM := If X and Y are independent, phi_{X+Y}(t)=phi_X(t)phi_Y(t).
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000458, S26-D-00000481
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000483|T|A|Proof+Compile+Meta+IG] Uniqueness of characteristic functions

§FORM := Two Borel probability measures on R with the same characteristic function are equal.
§ASSUME := The measures are Borel probability measures on R.
§DEP_HARD := S26-D-00000481, S26-D-00000404
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000484|D|A|Proof+Compile+Meta+IG] Convergence almost surely

§FORM := X_n converges almost surely to X when P({omega: X_n(omega)->X(omega)})=1.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000410, S26-D-00000447, S26-D-00000364
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000485|D|A|Proof+Compile+Meta+IG] Convergence in probability

§FORM := X_n converges in probability to X when for every epsilon>0, P(|X_n-X|>epsilon)->0.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000441, S26-D-00000446
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000486|D|A|Proof+Compile+Meta+IG] Convergence in L^p

§FORM := For 1<=p<infinity, X_n converges to X in L^p when E[|X_n-X|^p]->0.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000433, S26-D-00000449
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000487|D|A|Proof+Compile+Meta+IG] Convergence in distribution

§FORM := Real random variables X_n converge in distribution to X when their laws converge weakly; equivalently, F_{X_n}(x)->F_X(x) at every continuity point of F_X.
§ASSUME := For the CDF criterion the variables are real-valued.
§DEP_HARD := S26-D-00000448, S26-D-00000461
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000488|T|A|Proof+Compile+Meta+IG] Almost-sure convergence implies convergence in probability

§FORM := If X_n->X almost surely, then X_n->X in probability.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000484, S26-D-00000485
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000489|T|A|Proof+Compile+Meta+IG] L^p convergence implies convergence in probability

§FORM := For 1<=p<infinity, X_n->X in L^p implies X_n->X in probability.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000442, S26-D-00000486, S26-D-00000485
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000490|T|A|Proof+Compile+Meta+IG] Convergence in probability implies convergence in distribution

§FORM := If X_n->X in probability, then X_n->X in distribution.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000485, S26-D-00000487
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000491|T|A|Proof+Compile+Meta+IG] Convergence in distribution need not imply convergence in probability

§FORM := Equal or converging laws alone do not force random variables on a common probability space to become close in probability.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000487, S26-D-00000485
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Independent Bernoulli(1/2) variables X_n and X on one product space have identical laws but P(|X_n-X|>1/2)=1/2.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000492|T|A|Proof+Compile+Meta+IG] Borel-Cantelli lemma I

§FORM := For events A_n, if sum_n P(A_n)<infinity then P(A_n infinitely often)=0.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000451, S26-D-00000358, S26-T-00000407
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000493|T|A|Proof+Compile+Meta+IG] Borel-Cantelli lemma II

§FORM := If events A_n are independent and sum_n P(A_n)=infinity, then P(A_n infinitely often)=1.
§ASSUME := The events are mutually independent.
§DEP_HARD := S26-D-00000457, S26-T-00000492
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000494|D|A|Proof+Compile+Meta+IG] Sample mean

§FORM := For X_1,...,X_n, the sample mean is bar X_n=(1/n)sum_{k=1}^n X_k.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000020, S26-D-00000447
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000495|T|A|Proof+Compile+Meta+IG] Weak law of large numbers

§FORM := For IID random variables with finite mean mu and finite variance, the sample mean converges in probability to mu.
§ASSUME := The variables are IID with finite mean and finite variance.
§DEP_HARD := S26-D-00000460, S26-T-00000475, S26-D-00000494, S26-D-00000485
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000496|T|A|Proof+Compile+Meta+IG] Strong law of large numbers

§FORM := For IID integrable real random variables with mean mu, the sample mean converges almost surely to mu.
§ASSUME := The variables are IID and E[|X_1|]<infinity.
§DEP_HARD := S26-D-00000460, S26-D-00000494, S26-D-00000484
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000497|D|A|Proof+Compile+Meta+IG] Standardized partial sum

§FORM := For IID real X_k with mean mu and variance sigma^2 in (0,infinity), define Z_n=(sum_{k=1}^n X_k-n mu)/(sigma sqrt(n)).
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000460, S26-D-00000469
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000498|T|A|Proof+Compile+Meta+IG] Lindeberg-Levy central limit theorem

§FORM := For IID real X_k with finite mean mu and finite positive variance sigma^2, the standardized partial sums Z_n converge in distribution to N(0,1).
§ASSUME := The variables are IID with finite mean and variance 0<sigma^2<infinity.
§DEP_HARD := S26-D-00000497, S26-D-00000487
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000499|D|A|Proof+Compile+Meta+IG] Standard normal distribution

§FORM := The standard normal law N(0,1) is the Borel probability measure on R with density (2 pi)^(-1/2) exp(-x^2/2).
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000466, S26-D-00000404
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000500|T|A|Proof+Compile+Meta+IG] Central limit theorem CDF form

§FORM := Under the Lindeberg-Levy hypotheses, P(Z_n<=x)->Phi(x) for every real x, where Phi is the standard normal CDF.
§ASSUME := All probability spaces, random variables, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000498, S26-D-00000499, S26-D-00000461
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000501|D|A|Proof+Compile+Meta+IG] Stochastic process

§FORM := A stochastic process indexed by T is a family (X_t)_{t in T} of random variables on one probability space.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000447
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000502|D|A|Proof+Compile+Meta+IG] Sample path

§FORM := For fixed omega, the map t |-> X_t(omega) is the sample path (trajectory) of a stochastic process.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000501
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000503|D|A|Proof+Compile+Meta+IG] Finite-dimensional distribution

§FORM := For t_1,...,t_n, the joint law of (X_{t_1},...,X_{t_n}) is a finite-dimensional distribution of the process.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000501, S26-D-00000448
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000504|T|A|Proof+Compile+Meta+IG] Process law determined by finite-dimensional distributions on product cylinders

§FORM := On the canonical product measurable space, equality of all finite-dimensional distributions implies equality of process laws.
§ASSUME := Processes are viewed on the canonical product measurable space with the cylinder sigma-algebra.
§DEP_HARD := S26-D-00000503, S26-D-00000403
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000505|D|A|Proof+Compile+Meta+IG] Filtration

§FORM := A filtration (F_t) is an increasing family of sub-sigma-algebras: s<=t implies F_s subset F_t.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000401, S26-D-00000309
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000506|D|A|Proof+Compile+Meta+IG] Filtered probability space

§FORM := A filtered probability space is (Omega,F,(F_t),P) with (F_t) a filtration contained in F.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000446, S26-D-00000505
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000507|D|A|Proof+Compile+Meta+IG] Natural filtration

§FORM := The natural filtration of X is F_t^X=sigma(X_s: s<=t).
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000501, S26-D-00000505
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000508|D|A|Proof+Compile+Meta+IG] Adapted process

§FORM := X is adapted to (F_t) when X_t is F_t-measurable for every t.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000501, S26-D-00000505, S26-D-00000405
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000509|D|A|Proof+Compile+Meta+IG] Stopping time

§FORM := A random time tau is a stopping time when {tau<=t} belongs to F_t for every t.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000505, S26-D-00000451
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000510|D|A|Proof+Compile+Meta+IG] Stopped process

§FORM := For a process X and stopping time tau, the stopped process is X^tau_t=X_{t wedge tau}.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000501, S26-D-00000509
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000511|D|A|Proof+Compile+Meta+IG] Sigma-algebra at a stopping time

§FORM := F_tau consists of events A such that A intersection {tau<=t} belongs to F_t for every t.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000509, S26-D-00000401
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000512|D|A|Proof+Compile+Meta+IG] Discrete-time martingale

§FORM := An adapted integrable process (M_n) is a martingale when E[M_{n+1}|F_n]=M_n almost surely for every n.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000508, S26-D-00000476
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000513|D|A|Proof+Compile+Meta+IG] Submartingale

§FORM := An adapted integrable process is a submartingale when E[X_{n+1}|F_n]>=X_n almost surely.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000512
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000514|D|A|Proof+Compile+Meta+IG] Supermartingale

§FORM := An adapted integrable process is a supermartingale when E[X_{n+1}|F_n]<=X_n almost surely.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000512
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000515|T|A|Proof+Compile+Meta+IG] Martingale expectation conservation

§FORM := For an integrable martingale, E[M_n]=E[M_0] for every n.

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000512, S26-T-00000477

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000516|T|A|Proof+Compile+Meta+IG] Conditional-expectation martingale

§FORM := If X is integrable and F_n is a filtration, M_n=E[X|F_n] is a martingale.

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000476, S26-D-00000505, S26-D-00000512

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000517|T|A|Proof+Compile+Meta+IG] Martingale transform by bounded predictable sequence

§FORM := If M is a discrete-time martingale and H_n is bounded and F_{n-1}-measurable, then the stochastic sum sum_{k=1}^n H_k(M_k-M_{k-1}) is a martingale.

§ASSUME := H_n is bounded and predictable; all displayed terms are integrable.

§DEP_HARD := S26-D-00000512, S26-T-00000478

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000518|T|A|Proof+Compile+Meta+IG] Optional stopping: bounded stopping times

§FORM := If M_n is an integrable martingale and tau is a bounded stopping time, then E[M_tau]=E[M_0].

§ASSUME := tau is bounded.

§DEP_HARD := S26-D-00000509, S26-D-00000512, S26-T-00000515

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000519|T|A|Proof+Compile+Meta+IG] Optional stopping requires hypotheses

§FORM := Optional-stopping identities are not valid for arbitrary unbounded stopping times without additional hypotheses.

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000518

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§BOUNDARY := A fair random walk stopped at the first hit of +1 has an unbounded stopping time and naive unrestricted optional-stopping reasoning can fail integrability/uniform-integrability conditions.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000520|D|A|Proof+Compile+Meta+IG] Transition kernel

§FORM := A Markov transition kernel K from (E,E) to (F,F) assigns to x a probability measure K(x,.) and to measurable B a measurable function x |-> K(x,B).

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000403, S26-D-00000446

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000521|D|A|Proof+Compile+Meta+IG] Markov chain

§FORM := A discrete-time process is Markov with kernel K when P(X_{n+1} in A | F_n)=K(X_n,A) almost surely.

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000501, S26-D-00000505, S26-D-00000520, S26-D-00000476

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000522|D|A|Proof+Compile+Meta+IG] Transition matrix

§FORM := For a countable-state Markov chain, P=(p_ij) with p_ij=P(X_{n+1}=j|X_n=i) is a row-stochastic transition matrix.

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000521, S26-D-00000464, S26-D-00000176

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000523|T|A|Proof+Compile+Meta+IG] Chapman-Kolmogorov equation

§FORM := For a time-homogeneous Markov chain, n+m step transition kernels compose: P^{n+m}(i,j)=sum_k P^n(i,k)P^m(k,j).

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000521, S26-D-00000522, S26-D-00000031

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000524|D|A|Proof+Compile+Meta+IG] Stationary distribution

§FORM := A probability distribution pi is stationary for transition matrix P when pi P=pi.

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000522, S26-D-00000448

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000525|T|A|Proof+Compile+Meta+IG] Stationary initialization

§FORM := If X_0 has stationary law pi and the chain has kernel P, then every X_n has law pi.

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000521, S26-D-00000524

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000526|D|A|Proof+Compile+Meta+IG] Communicating states
§FORM := States i,j communicate when each is reachable from the other with positive probability in some number of steps.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000523
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000527|D|A|Proof+Compile+Meta+IG] Irreducible Markov chain
§FORM := A Markov chain is irreducible when every pair of states communicates.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000526
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000528|D|A|Proof+Compile+Meta+IG] Period of a state
§FORM := The period of state i is gcd{n>=1: P^n(i,i)>0}.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000523, S26-T-00000119
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000529|D|A|Proof+Compile+Meta+IG] Aperiodic chain
§FORM := An irreducible chain is aperiodic when its states have period one.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000527, S26-D-00000528
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000530|D|A|Proof+Compile+Meta+IG] Return time
§FORM := The first return time to state i is T_i^+=inf{n>=1:X_n=i}.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000509, S26-D-00000521
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000531|D|A|Proof+Compile+Meta+IG] Recurrent state
§FORM := State i is recurrent when P_i(T_i^+<infinity)=1.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000530
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000532|D|A|Proof+Compile+Meta+IG] Transient state
§FORM := State i is transient when P_i(T_i^+<infinity)<1.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000530
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000533|D|A|Proof+Compile+Meta+IG] Positive recurrent state
§FORM := A recurrent state i is positive recurrent when E_i[T_i^+]<infinity.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000531, S26-D-00000449
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000534|T|A|Proof+Compile+Meta+IG] Finite irreducible chains are positive recurrent
§FORM := Every state of a finite irreducible Markov chain is positive recurrent.
§ASSUME := The state space is finite.
§DEP_HARD := S26-D-00000527, S26-D-00000533
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000535|T|A|Proof+Compile+Meta+IG] Finite irreducible stationary distribution
§FORM := A finite irreducible Markov chain has a unique stationary distribution with strictly positive coordinates.
§ASSUME := The state space is finite.
§DEP_HARD := S26-D-00000524, S26-D-00000527, S26-T-00000534
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000536|T|A|Proof+Compile+Meta+IG] Finite ergodic-chain convergence
§FORM := For a finite irreducible aperiodic Markov chain with stationary distribution pi, P^n(i,j)->pi_j for all states i,j.
§ASSUME := The state space is finite, irreducible, and aperiodic.
§DEP_HARD := S26-D-00000529, S26-T-00000535
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000537|D|A|Proof+Compile+Meta+IG] Strict stationarity
§FORM := A process is strictly stationary when every finite-dimensional distribution is invariant under admissible time shifts.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000503
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000538|D|A|Proof+Compile+Meta+IG] Weak stationarity

§FORM := A second-order process is weakly stationary when its mean is constant and covariance depends only on time lag.
§ASSUME := The process has finite second moments.
§DEP_HARD := S26-D-00000470, S26-D-00000501
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000539|T|A|Proof+Compile+Meta+IG] Strict stationarity plus second moments implies weak stationarity

§FORM := A strictly stationary process with finite second moments is weakly stationary.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000537, S26-D-00000538
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000540|T|A|Proof+Compile+Meta+IG] Weak stationarity need not imply strict stationarity

§FORM := Second-order stationarity alone does not determine all finite-dimensional distributions.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000537, S26-D-00000538
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Processes can share constant mean and lag covariance while having time-varying higher-order joint laws.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000541|D|A|Proof+Compile+Meta+IG] Measure-preserving transformation

§FORM := A measurable T on a probability space is measure-preserving when P(T^{-1}A)=P(A) for every measurable A.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000446, S26-D-00000405
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000542|D|A|Proof+Compile+Meta+IG] Invariant event

§FORM := An event A is invariant under T when T^{-1}A=A modulo null sets.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000541, S26-D-00000410
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000543|D|A|Proof+Compile+Meta+IG] Ergodic transformation

§FORM := A measure-preserving transformation is ergodic when every invariant event has probability 0 or 1.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000541, S26-D-00000542
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000544|D|A|Proof+Compile+Meta+IG] Stationary process from a measure-preserving system

§FORM := Given measure-preserving T and measurable f, X_n=f composed with T^n defines a stationary process.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000541, S26-D-00000537, S26-D-00000405
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000545|T|A|Proof+Compile+Meta+IG] Birkhoff ergodic theorem interface

§FORM := For measure-preserving T and f in L^1, time averages n^{-1}sum_{k=0}^{n-1} f(T^k omega) converge almost surely to E[f|I], where I is the invariant sigma-algebra.
§ASSUME := T is measure-preserving and f is integrable.
§DEP_HARD := S26-D-00000476, S26-D-00000541, S26-D-00000542, S26-D-00000433
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000546|T|A|Proof+Compile+Meta+IG] Ergodic specialization of Birkhoff

§FORM := If T is ergodic and f in L^1, the Birkhoff time average converges almost surely to E[f].
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000543, S26-T-00000545
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000547|D|A|Proof+Compile+Meta+IG] Hitting time

§FORM := For measurable A, tau_A=inf{n>=0:X_n in A} is the hitting time of A.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000509, S26-D-00000501
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000548|D|A|Proof+Compile+Meta+IG] First-passage probability

§FORM := For states or measurable sets A,B, first-passage probabilities quantify reaching B before A or before a specified time.
§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000547, S26-D-00000451
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000549|T|A|Proof+Compile+Meta+IG] Strong Markov property interface

§FORM := For a Markov chain and an almost surely finite stopping time tau, the post-tau process conditionally restarts from X_tau with the same transition kernel.

§ASSUME := tau is a stopping time satisfying the standard finiteness/measurability hypotheses.

§DEP_HARD := S26-D-00000509, S26-D-00000521

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000550|T|A|Proof+Compile+Meta+IG] Stochastic-process foundation frontier

§FORM := Filtrations, martingales, stopping times, Markov kernels/chains, stationarity, and ergodic interfaces form the finite/discrete stochastic-process substrate before continuous-time Brownian motion and stochastic integration.

§ASSUME := All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000512, S26-D-00000521, S26-D-00000537, S26-T-00000545

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000551|D|A|Proof+Compile+Meta+IG] Brownian motion

§FORM := A standard Brownian motion W has W_0=0 a.s., independent stationary Gaussian increments, and almost surely continuous paths.

§ASSUME := Continuous time t>=0; standard real Brownian motion.

§DEP_HARD := S26-D-00000501, S26-D-00000466

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000552|T|A|Proof+Compile+Meta+IG] Brownian increment law

§FORM := For 0<=s<t, W_t-W_s is N(0,t-s).

§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000551

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000553|T|A|Proof+Compile+Meta+IG] Brownian covariance

§FORM := E[W_s W_t]=min(s,t).

§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000551, S26-D-00000470

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000554|D|A|Proof+Compile+Meta+IG] Gaussian process

§FORM := A process is Gaussian when every finite-dimensional distribution is multivariate Gaussian.

§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000503, S26-D-00000466

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000555|T|A|Proof+Compile+Meta+IG] Brownian motion is Gaussian

§FORM := Standard Brownian motion is a centered Gaussian process with covariance min(s,t).

§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000551, S26-D-00000554

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000556|T|A|Proof+Compile+Meta+IG] Brownian scaling

§FORM := For c>0, (c^{-1/2}W_{ct}) has the same finite-dimensional laws as Brownian motion.

§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000551, S26-T-00000504

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000557|T|A|Proof+Compile+Meta+IG] Brownian Markov property

§FORM := Brownian motion is a time-homogeneous Markov process with Gaussian transition kernel.

§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000551, S26-D-00000521

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000558|T|A|Proof+Compile+Meta+IG] Brownian martingale property

§FORM := Brownian motion adapted to its natural filtration is a martingale.

§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000551, S26-D-00000507, S26-D-00000512

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000559|D|A|Proof+Compile+Meta+IG] Continuous local martingale interface

§FORM := A continuous local martingale is an adapted continuous process that becomes a martingale after a localizing sequence of stopping times.

§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000508, S26-D-00000509, S26-D-00000512

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000560|D|A|Proof+Compile+Meta+IG] Partition quadratic variation

§FORM := For a partition Pi of [0,t], the quadratic-variation sum is sum (X_{t_{i+1}}-X_{t_i})^2.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000501, S26-T-00000357
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000561|T|A|Proof+Compile+Meta+IG] Brownian quadratic variation

§FORM := Along refining partitions with mesh tending to zero, Brownian quadratic-variation sums converge in probability to t.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000551, S26-D-00000560, S26-D-00000486
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000562|T|A|Proof+Compile+Meta+IG] Brownian infinite variation

§FORM := Brownian paths have infinite total variation on every nontrivial compact interval almost surely.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000551, S26-D-00000410
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Continuity does not imply bounded variation.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000563|D|A|Proof+Compile+Meta+IG] Predictable process interface

§FORM := A predictable process is measurable with respect to the predictable sigma-algebra generated by adapted left-continuous processes.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000508, S26-D-00000401
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000564|D|A|Proof+Compile+Meta+IG] Simple adapted integrand

§FORM := A simple adapted integrand has form H_t=sum H_k 1_(t_k,t_{k+1}](t), with H_k measurable at time t_k.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000508, S26-D-00000405
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000565|D|A|Proof+Compile+Meta+IG] Ito integral for simple processes

§FORM := For simple adapted H, integral H dW is sum H_k(W_{t_{k+1}}-W_{t_k}).
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000551, S26-D-00000564
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000566|T|A|Proof+Compile+Meta+IG] Ito isometry for simple processes

§FORM := E[(integral_0^T H dW)^2]=E[integral_0^T H_t^2 dt] for square-integrable simple adapted H.
§ASSUME := H is simple, adapted, and square-integrable.
§DEP_HARD := S26-D-00000565, S26-T-00000439
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000567|D|A|Proof+Compile+Meta+IG] L2 Ito integral

§FORM := The Ito integral for square-integrable predictable H is the L2 limit of integrals of simple predictable approximants.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000563, S26-T-00000566, S26-D-00000438
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000568|T|A|Proof+Compile+Meta+IG] Ito integral has mean zero

§FORM := For square-integrable predictable H, E[integral_0^T H dW]=0.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000567, S26-D-00000449
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000569|T|A|Proof+Compile+Meta+IG] Ito integral martingale

§FORM := M_t=integral_0^t H_s dW_s is a continuous martingale under square-integrability.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000567, S26-D-00000512
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000570|T|A|Proof+Compile+Meta+IG] Ito isometry

§FORM := For square-integrable predictable H, E[(integral_0^T H dW)^2]=E[integral_0^T H_t^2 dt].
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000567, S26-T-00000566
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000571|D|A|Proof+Compile+Meta+IG] Stochastic differential notation

§FORM := dX_t=b_t dt+sigma_t dW_t abbreviates the corresponding integral equation, not an ordinary pointwise differential identity.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000567, S26-T-00000398
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Brownian paths are almost surely nowhere classically differentiable.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000572|T|A|Proof+Compile+Meta+IG] Ito formula one-dimensional

§FORM := For dX_t=b_t dt+sigma_t dW_t and sufficiently smooth f, df(X_t)=f'(X_t)dX_t+(1/2)f''(X_t)sigma_t^2 dt.
§ASSUME := f is C^2 and coefficients satisfy conditions making all terms well-defined.
§DEP_HARD := S26-D-00000571, S26-T-00000344
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000573|T|A|Proof+Compile+Meta+IG] Ito correction term

§FORM := The second-order term in Ito formula arises from nonzero quadratic variation: heuristically (dW_t)^2=dt.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000561, S26-T-00000572
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000574|T|A|Proof+Compile+Meta+IG] Ito product rule

§FORM := d(X_tY_t)=X_t dY_t+Y_t dX_t+d[X,Y]_t for continuous semimartingales.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000572, S26-D-00000560
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000575|D|A|Proof+Compile+Meta+IG] Covariation interface

§FORM := Quadratic covariation [X,Y] is obtained from limits of sums of paired increments along refining partitions.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000560
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000576|T|A|Proof+Compile+Meta+IG] Polarization of covariation

§FORM := When quadratic variations exist, [X,Y]=([X+Y]-[X-Y])/4.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000575, S26-T-00000216
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000577|D|A|Proof+Compile+Meta+IG] Ito process

§FORM := An Ito process has X_t=X_0+integral b_s ds+integral sigma_s dW_s under suitable integrability and adaptedness.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000567, S26-D-00000571
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000578|D|A|Proof+Compile+Meta+IG] Stochastic differential equation

§FORM := An SDE specifies an adapted process through an integral equation driven by a stochastic process, typically Brownian motion.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000577
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000579|D|A|Proof+Compile+Meta+IG] Strong solution

§FORM := A strong solution is adapted to the given driving filtration and is constructed on the given probability space with the given Brownian motion.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000578, S26-D-00000506
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000580|D|A|Proof+Compile+Meta+IG] Weak solution

§FORM := A weak solution allows the probability space and driving Brownian motion to be part of the solution.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000578, S26-D-00000446
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000581|D|A|Proof+Compile+Meta+IG] Pathwise uniqueness

§FORM := Pathwise uniqueness means two solutions on the same space with the same Brownian motion and initial condition agree almost surely.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000579, S26-D-00000410
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000582|T|A|Proof+Compile+Meta+IG] Lipschitz SDE existence and uniqueness

§FORM := Globally Lipschitz coefficients with linear growth give a unique strong solution to the standard Ito SDE.
§ASSUME := Coefficients satisfy standard global Lipschitz and linear-growth bounds.
§DEP_HARD := S26-D-00000578, S26-D-00000579, S26-D-00000581
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000583|D|A|Proof+Compile+Meta+IG] Generator of an Ito diffusion

§FORM := For dX=b(X)dt+sigma(X)dW, the generator acts as Lf=b·grad f+(1/2)Tr(sigma sigma^T Hess f).
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000577, S26-T-00000341, S26-D-00000343
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000584|T|A|Proof+Compile+Meta+IG] Dynkin formula interface

§FORM := Under suitable regularity, E[f(X_tau)] = f(X_0)+E[integral_0^tau Lf(X_s)ds] for admissible stopping times.
§ASSUME := Regularity, integrability, and stopping hypotheses required by Dynkin formula hold.
§DEP_HARD := S26-D-00000583, S26-D-00000509
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000585|T|A|Proof+Compile+Meta+IG] Kolmogorov backward equation interface

§FORM := For a regular diffusion semigroup, u(t,x)=E_x[g(X_t)] formally satisfies partial_t u=L u with initial data g.
§ASSUME := Sufficient regularity for the classical PDE statement.
§DEP_HARD := S26-D-00000583, S26-D-00000449
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000586|T|A|Proof+Compile+Meta+IG] Fokker-Planck interface

§FORM := When a diffusion admits a sufficiently regular density, its density evolves under the adjoint generator equation.
§ASSUME := A sufficiently regular transition density exists.
§DEP_HARD := S26-D-00000583, S26-T-00000467
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000587|D|A|Proof+Compile+Meta+IG] Ornstein-Uhlenbeck process

§FORM := The Ornstein-Uhlenbeck process solves dX_t=-theta X_t dt+sigma dW_t.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000578
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000588|T|A|Proof+Compile+Meta+IG] Ornstein-Uhlenbeck explicit solution

§FORM := For constant theta,sigma, X_t=e^{-theta t}X_0+sigma integral_0^t e^{-theta(t-s)}dW_s.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000587, S26-D-00000567
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000589|D|A|Proof+Compile+Meta+Topo+IG] Geometric Brownian motion

§FORM := Geometric Brownian motion solves dS_t=mu S_t dt+sigma S_t dW_t.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000578
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000590|T|A|Proof+Compile+Meta+Topo+IG] Geometric Brownian explicit solution

§FORM := S_t=S_0 exp((mu-sigma^2/2)t+sigma W_t).
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000589, S26-T-00000572
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000591|D|A|Proof+Compile+Meta+IG] Exponential martingale

§FORM := For constant lambda, exp(lambda W_t-lambda^2 t/2) is the canonical Brownian exponential martingale.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000551, S26-T-00000572
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000592|T|A|Proof+Compile+Meta+IG] Exponential martingale property

§FORM := The Brownian exponential process is a martingale with expectation one.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000591, S26-T-00000569
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000593|D|A|Proof+Compile+Meta+IG] Stochastic integral versus pathwise integral

§FORM := The Ito integral is an L2/probabilistic construction and generally is not a Riemann-Stieltjes integral against Brownian paths.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000567, S26-T-00000389
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Brownian paths have unbounded variation almost surely.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000594|T|A|Proof+Compile+Meta+IG] Adaptedness boundary for Ito integration

§FORM := The elementary Ito construction requires non-anticipating integrands; future-dependent integrands are outside this definition.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000563, S26-D-00000567
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := An integrand depending on W_T at times t<T is generally anticipative.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000595|T|A|Proof+Compile+Meta+IG] Ito-Stratonovich distinction interface

§FORM := Ito and Stratonovich stochastic integrals use different limiting conventions and differ by a quadratic-covariation correction.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000573, S26-D-00000575
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000596|D|A|Proof+Compile+Meta+IG] Multidimensional Brownian motion

§FORM := A d-dimensional Brownian motion has independent standard Brownian coordinate processes.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000551, S26-D-00000465
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000597|T|A|Proof+Compile+Meta+IG] Multidimensional Ito formula interface

§FORM := For a sufficiently smooth scalar f and an Ito diffusion in R^d, Ito formula uses the gradient and Hessian with covariance matrix sigma sigma^T.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000596, S26-D-00000583, S26-D-00000343
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000598|D|A|Proof+Compile+Meta+IG] Stochastic flow interface

§FORM := Under stronger regularity, solutions indexed by initial state may define a random flow of maps.
§ASSUME := Coefficient regularity sufficient for the stated flow property.
§DEP_HARD := S26-T-00000582, S26-D-00000178
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000599|T|A|Proof+Compile+Meta+IG] Stochastic-calculus hypothesis fence

§FORM := Stochastic differential notation does not license ordinary calculus: adaptedness, integrability, quadratic variation, and the chosen integral convention remain load-bearing.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000571, S26-T-00000572, S26-T-00000594, S26-T-00000595
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000600|T|A|Proof+Compile+Meta+IG] Stochastic-calculus frontier

§FORM := Brownian motion, quadratic variation, Ito integration, Ito formula, and basic SDEs form the continuous stochastic-calculus substrate before advanced martingale theory, stochastic control, SPDEs, and mathematical finance.
§ASSUME := All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000551, S26-D-00000567, S26-T-00000572, S26-T-00000582
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000601|D|A|Proof+Compile+Meta] Normed vector space

§FORM := A normed vector space is a vector space X over R or C equipped with a norm ||.|| satisfying definiteness, absolute homogeneity, and the triangle inequality.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000149, S26-D-00000202
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000602|T|A|Proof+Compile+Meta] Norm metric

§FORM := Every norm induces the metric d(x,y)=||x-y||.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000601, S26-D-00000301
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000603|D|A|Proof+Compile+Meta+Topo] Equivalent norms

§FORM := Two norms on the same vector space are equivalent when each bounds the other up to positive constants; equivalent norms induce the same topology.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000601, S26-D-00000315
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000604|T|A|Proof+Compile+Meta] Finite-dimensional norm equivalence
§FORM := All norms on a finite-dimensional vector space over R or C are equivalent.
§ASSUME := The vector space is finite-dimensional over R or C.
§DEP_HARD := S26-T-00000164, S26-D-00000603
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Infinite-dimensional normed spaces can carry inequivalent norms.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000605|D|A|Proof+Compile+Meta] Banach space
§FORM := A Banach space is a complete normed vector space.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000601, S26-T-00000327
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000606|D|A|Proof+Compile+Meta+Spectral] Bounded linear operator
§FORM := A linear map T:X->Y between normed spaces is bounded when ||Tx||<=C||x|| for some C>=0 and all x.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000166, S26-D-00000601
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000607|T|A|Proof+Compile+Meta] Bounded iff continuous linear
§FORM := A linear map between normed spaces is continuous iff it is bounded.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000606, S26-T-00000317
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000608|D|A|Proof+Compile+Meta+Spectral] Operator norm
§FORM := For bounded linear T:X->Y, ||T||=sup_{||x||<=1}||Tx||.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000606, S26-D-00000210
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000609|T|A|Proof+Compile+Meta+Spectral] Operator norm inequality
§FORM := For bounded linear T, ||Tx||<=||T|| ||x||.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000608
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000610|T|A|Proof+Compile+Meta+Spectral] Submultiplicativity of operator norm
§FORM := For bounded composable operators S,T, ||ST||<=||S|| ||T||.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000608, S26-D-00000171
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000611|T|A|Proof+Compile+Meta+Spectral] Bounded operators form a normed space
§FORM := B(X,Y), the bounded linear maps X->Y, is a normed vector space under the operator norm.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000608, S26-D-00000601
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000612|T|A|Proof+Compile+Meta+Spectral] Completeness of operator space
§FORM := If Y is Banach, then B(X,Y) is Banach for every normed X.
§ASSUME := Y is Banach.
§DEP_HARD := S26-T-00000611, S26-D-00000605
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000613|D|A|Proof+Compile+Meta] Continuous dual
§FORM := The continuous dual X* is B(X,F), the bounded linear functionals from X to its scalar field F.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000611, S26-D-00000251
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000614|T|A|Proof+Compile+Meta] Dual is Banach
§FORM := The continuous dual X* of every normed space is Banach.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000612, S26-D-00000613
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000615|D|A|Proof+Compile+Meta] Annihilator

§FORM := For M subset X, M^perp={f in X*: f(m)=0 for all m in M}.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000613, S26-D-00000156
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000616|T|A|Proof+Compile+Meta] Hahn-Banach extension theorem

§FORM := A bounded linear functional on a subspace of a real or complex normed space extends to the whole space without increasing its norm.
§ASSUME := Standard real or complex Hahn-Banach hypotheses.
§DEP_HARD := S26-D-00000613, S26-D-00000156
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000617|T|A|Proof+Compile+Meta] Dual separates points

§FORM := For nonzero x in a normed space, there exists f in X* with ||f||=1 and f(x)=||x||.
§ASSUME := X is a normed space over R or C.
§DEP_HARD := S26-T-00000616
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000618|T|A|Proof+Compile+Meta] Norm recovered from dual

§FORM := For x in a normed space, ||x||=sup{|f(x)|: f in X*, ||f||<=1}.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000617, S26-D-00000608
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000619|D|A|Proof+Compile+Meta+Spectral] Pointwise bounded operator family

§FORM := A family F subset B(X,Y) is pointwise bounded when sup_{T in F}||Tx||<infinity for every x in X.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000611
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000620|T|A|Proof+Compile+Meta] Uniform boundedness principle

§FORM := If X is Banach and a family F subset B(X,Y) is pointwise bounded, then sup_{T in F}||T||<infinity.
§ASSUME := X is Banach.
§DEP_HARD := S26-D-00000605, S26-D-00000619
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Completeness of the domain is load-bearing in the standard theorem.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000621|T|A|Proof+Compile+Meta] Open mapping theorem

§FORM := A surjective bounded linear map between Banach spaces is an open map.
§ASSUME := Domain and codomain are Banach; the operator is bounded, linear, and surjective.
§DEP_HARD := S26-D-00000605, S26-D-00000606, S26-T-00000322
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000622|T|A|Proof+Compile+Meta] Bounded inverse theorem

§FORM := A bijective bounded linear map between Banach spaces has bounded inverse.
§ASSUME := Domain and codomain are Banach.
§DEP_HARD := S26-T-00000621, S26-D-00000172
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000623|D|A|Proof+Compile+Meta+Spectral] Closed operator graph

§FORM := The graph of T:X->Y is {(x,Tx):x in X}; T is closed when this graph is closed in X x Y.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000166, S26-D-00000308, S26-T-00000305
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000624|T|A|Proof+Compile+Meta+Spectral] Closed graph theorem

§FORM := A linear operator T:X->Y between Banach spaces is bounded iff its graph is closed.
§ASSUME := X and Y are Banach and T is defined on all of X.
§DEP_HARD := S26-D-00000623, S26-D-00000605, S26-D-00000606
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000625|D|A|Proof+Compile+Meta+Topo] Weak topology

§FORM := The weak topology sigma(X,X*) is the coarsest topology making every continuous linear functional f in X* continuous.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000613, S26-D-00000306
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000626|D|A|Proof+Compile+Meta] Weak convergence
§FORM := x_n converges weakly to x when f(x_n)->f(x) for every f in X*.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000625, S26-D-00000351
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000627|T|A|Proof+Compile+Meta] Norm convergence implies weak convergence
§FORM := If x_n->x in norm, then x_n->x weakly.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000607, S26-D-00000626
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000628|T|A|Proof+Compile+Meta] Weak convergence need not imply norm convergence
§FORM := In infinite-dimensional normed spaces, weak convergence can occur without norm convergence.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000626
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := The standard basis e_n in l2 converges weakly to 0 but ||e_n||=1.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000629|D|A|Proof+Compile+Meta+Topo] Weak-star topology
§FORM := The weak-star topology sigma(X*,X) is pointwise convergence of functionals on X.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000613, S26-D-00000306
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000630|T|A|Proof+Compile+Meta+Topo] Banach-Alaoglu theorem
§FORM := The closed unit ball of X* is compact in the weak-star topology.
§ASSUME := Compactness is with respect to the weak-star topology, not generally the norm topology.
§DEP_HARD := S26-D-00000629, S26-T-00000323
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000631|D|A|Proof+Compile+Meta+Spectral] Hilbert space
§FORM := A Hilbert space is a complete inner-product space with norm induced by its inner product.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000201, S26-D-00000605
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000632|T|A|Proof+Compile+Meta] Parallelogram law
§FORM := A norm induced by an inner product satisfies ||x+y||^2+||x-y||^2=2||x||^2+2||y||^2.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000631, S26-T-00000216
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000633|T|A|Proof+Compile+Meta] Jordan-von Neumann interface
§FORM := A norm satisfying the parallelogram law arises from a unique compatible inner product via polarization.
§ASSUME := Real or complex normed space with the appropriate polarization identity.
§DEP_HARD := S26-T-00000632, S26-T-00000216
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000634|D|A|Proof+Compile+Meta+Spectral] Orthogonal complement in Hilbert space
§FORM := For M subset H, M^perp={x in H:<x,m>=0 for every m in M}.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000631, S26-T-00000211
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000635|T|A|Proof+Compile+Meta] Closedness of orthogonal complement
§FORM := M^perp is a closed linear subspace of H.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000634, S26-D-00000308
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000636|T|A|Proof+Compile+Meta+Spectral] Hilbert projection theorem
§FORM := For a nonempty closed convex subset C of a Hilbert space, every x has a unique nearest point in C.
§ASSUME := C is nonempty, closed, and convex.
§DEP_HARD := S26-D-00000631, S26-T-00000349, S26-D-00000308
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000637|T|A|Proof+Compile+Meta] Orthogonal projection onto closed subspace

§FORM := For a closed subspace M of H, each x uniquely decomposes x=P_M x+(x-P_Mx) with P_Mx in M and x-P_Mx in M^perp.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000636, S26-D-00000634
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000638|T|A|Proof+Compile+Meta] Orthogonal decomposition

§FORM := For a closed subspace M of H, H=M direct-sum M^perp.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000637, S26-D-00000158
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000639|T|A|Proof+Compile+Meta+Spectral] Projection operator properties

§FORM := The orthogonal projection P_M onto a closed subspace is bounded, self-adjoint, idempotent, and has norm <=1.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000637, S26-T-00000226, S26-D-00000608
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000640|T|A|Proof+Compile+Meta+Spectral] Riesz representation theorem for Hilbert spaces

§FORM := Every bounded linear functional f on a Hilbert space H has a unique representing vector y with f(x)=<x,y> for all x, up to the chosen inner-product convention.
§ASSUME := H is a real or complex Hilbert space; conjugate-linearity convention is fixed.
§DEP_HARD := S26-D-00000631, S26-D-00000613
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000641|T|A|Proof+Compile+Meta+Spectral] Hilbert space self-duality

§FORM := The Riesz map identifies a Hilbert space isometrically with its continuous dual, conjugate-linearly in the standard complex convention.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000640, S26-T-00000173
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000642|D|A|Proof+Compile+Meta+Spectral] Adjoint of bounded Hilbert operator

§FORM := For bounded T:H->K between Hilbert spaces, T* is the unique bounded operator satisfying <Tx,y>=<x,T*y>.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000640, S26-D-00000222, S26-D-00000606
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000643|T|A|Proof+Compile+Meta+Spectral] Adjoint norm equality

§FORM := For bounded Hilbert-space operators, ||T*||=||T|| and ||T*T||=||T||^2.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000642, S26-D-00000608
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000644|D|A|Proof+Compile+Meta+Spectral] Compact operator

§FORM := A bounded operator T:X->Y is compact when it maps bounded sets to relatively compact sets.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000606, S26-T-00000323
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000645|T|A|Proof+Compile+Meta+Spectral] Finite-rank operators are compact

§FORM := Every bounded finite-rank operator between normed spaces is compact.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000644, S26-T-00000170, S26-T-00000164
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000646|T|A|Proof+Compile+Meta+Spectral] Compact operators form an operator-norm closed ideal

§FORM := Compact operators are closed under addition and operator-norm limits, and composition with bounded operators preserves compactness.
§ASSUME := Operators are between compatible normed spaces.
§DEP_HARD := S26-D-00000644, S26-T-00000610
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000647|T|A|Proof+Compile+Meta+Spectral] Compact self-adjoint spectral theorem interface

§FORM := A compact self-adjoint operator on a Hilbert space admits an orthonormal eigenvector decomposition on the closure of its range, with nonzero eigenvalues of finite multiplicity accumulating only at zero.
§ASSUME := Operator is compact and self-adjoint on a Hilbert space.
§DEP_HARD := S26-D-00000644, S26-T-00000225, S26-D-00000631
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000648|T|A|Proof+Compile+Meta] Infinite-dimensional compactness fence

§FORM := Closed and bounded subsets of infinite-dimensional normed spaces need not be compact.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000323, S26-D-00000601
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := The closed unit ball of an infinite-dimensional Hilbert space is not norm-compact.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000649|T|A|Proof+Compile+Meta+Topo] Functional-analysis hypothesis fence

§FORM := Finite-dimensional intuitions do not automatically extend: completeness, topology choice, closedness, boundedness, and compactness hypotheses remain load-bearing.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000604, S26-T-00000620, S26-T-00000621, S26-T-00000630, S26-T-00000648
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000650|T|A|Proof+Compile+Meta+Topo+Spectral] Functional-analysis frontier

§FORM := Normed and Banach spaces, bounded operators, duality principles, weak topologies, Hilbert projection/Riesz representation, and compact operators form the substrate for infinite-dimensional spectral theory, distributions, PDE, and operator theory.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000605, S26-T-00000616, S26-T-00000620, S26-D-00000631, S26-T-00000640, S26-D-00000644
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000651|D|A|Proof+Compile+Meta+Spectral] Spectrum of a bounded operator

§FORM := For T in B(X) on a complex Banach space, the spectrum sigma(T) is the set of lambda in C for which T-lambda I is not boundedly invertible.
§ASSUME := X is a complex Banach space and T is bounded.
§DEP_HARD := S26-T-00000622, S26-D-00000606
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000652|D|A|Proof+Compile+Meta+Spectral] Resolvent set and resolvent operator

§FORM := The resolvent set rho(T)=C\sigma(T); for lambda in rho(T), R(lambda,T)=(lambda I-T)^(-1).
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000651
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000653|T|A|Proof+Compile+Meta+Spectral] Resolvent identity

§FORM := For lambda,mu in rho(T), R(lambda,T)-R(mu,T)=(mu-lambda)R(lambda,T)R(mu,T).
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000652
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000654|T|A|Proof+Compile+Meta+Spectral] Spectrum nonempty compact

§FORM := The spectrum of a bounded operator on a nonzero complex Banach space is a nonempty compact subset of C.
§ASSUME := X is a nonzero complex Banach space; T is bounded.
§DEP_HARD := S26-D-00000651, S26-T-00000323
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000655|D|A|Proof+Compile+Meta+Spectral] Spectral radius

§FORM := The spectral radius is r(T)=sup{|lambda|:lambda in sigma(T)}.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000651
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000656|T|A|Proof+Compile+Meta+Spectral] Spectral radius formula

§FORM := For T in B(X), r(T)=lim_{n->infinity} ||T^n||^(1/n)=inf_n ||T^n||^(1/n).
§ASSUME := X is a complex Banach space and T is bounded.
§DEP_HARD := S26-D-00000655, S26-T-00000610
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000657|T|A|Proof+Compile+Meta+Spectral] Neumann series inverse

§FORM := If ||T||<1 on a Banach space, I-T is invertible with (I-T)^(-1)=sum_{n>=0}T^n in operator norm.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000605, S26-D-00000608, S26-D-00000356
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000658|T|A|Proof+Compile+Meta+Spectral] Resolvent outside norm disk

§FORM := If |lambda|>||T||, then lambda belongs to rho(T), obtained by a Neumann-series expansion.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000657, S26-D-00000652
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000659|D|A|Proof+Compile+Meta+Spectral] Point spectrum

§FORM := The point spectrum consists of eigenvalues: lambda for which T-lambda I is not injective.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000651, S26-D-00000197
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000660|T|A|Proof+Compile+Meta+Spectral] Compact operator nonzero spectrum

§FORM := For a compact operator on an infinite-dimensional complex Banach space, every nonzero spectral value is an eigenvalue of finite algebraic multiplicity; zero is the only possible accumulation point.
§ASSUME := T is compact on an infinite-dimensional complex Banach space.
§DEP_HARD := S26-D-00000644, S26-D-00000651
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000661|D|A|Proof+Compile+Meta+Spectral] Unbounded linear operator

§FORM := An unbounded-operator datum is a linear map A:D(A) subset X -> Y together with its specified domain D(A).
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000166, S26-D-00000156
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000662|D|A|Proof+Compile+Meta+Spectral] Densely defined operator

§FORM := A:D(A) subset X->Y is densely defined when the closure of D(A) equals X.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000661, S26-D-00000308
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000663|D|A|Proof+Compile+Meta+Spectral] Closed unbounded operator

§FORM := A is closed when its graph is closed in X x Y.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000661, S26-D-00000623
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000664|D|A|Proof+Compile+Meta+Spectral] Closable operator

§FORM := A is closable when the closure of its graph is the graph of an operator, called its closure.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000663, S26-D-00000308
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000665|D|A|Proof+Compile+Meta+Spectral] Adjoint of a densely defined operator

§FORM := For densely defined A:D(A) subset H->K, y lies in D(A*) when x -> <Ax,y> is bounded on D(A) in the H norm; A*y is its Riesz representing vector.
§ASSUME := A is densely defined between Hilbert spaces.
§DEP_HARD := S26-D-00000662, S26-T-00000640
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000666|T|A|Proof+Compile+Meta+Spectral] Adjoint is closed

§FORM := The adjoint A* of a densely defined operator is closed.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000665, S26-D-00000663
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000667|D|A|Proof+Compile+Meta+Spectral] Symmetric operator

§FORM := A densely defined operator A on H is symmetric when <Ax,y>=<x,Ay> for all x,y in D(A), equivalently A is contained in A*.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000665
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000668|D|A|Proof+Compile+Meta+Spectral] Self-adjoint unbounded operator

§FORM := A is self-adjoint when A=A*, including equality of domains.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000665
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Equality of the differential expressions without equality of domains does not establish self-adjointness.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000669|T|A|Proof+Compile+Meta+Spectral] Self-adjoint implies symmetric

§FORM := Every self-adjoint operator is symmetric.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000667, S26-D-00000668
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000670|T|A|Proof+Compile+Meta+Spectral] Symmetric need not be self-adjoint

§FORM := A symmetric densely defined operator need not be self-adjoint; domain and deficiency data can obstruct equality with its adjoint.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000667, S26-D-00000668
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§BOUNDARY := Minimal differential operators can be symmetric but require boundary conditions for self-adjoint extensions.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000671|D|A|Proof+Compile+Meta+Spectral] Positive self-adjoint operator

§FORM := A self-adjoint operator A is positive when <Ax,x>>=0 for every x in D(A).
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000668, S26-T-00000231
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000672|T|A|Proof+Compile+Meta+Spectral] Spectral theorem for bounded self-adjoint operators

§FORM := A bounded self-adjoint operator on a complex Hilbert space admits a projection-valued spectral representation T=integral lambda dE(lambda).
§ASSUME := T is bounded and self-adjoint on a complex Hilbert space.
§DEP_HARD := S26-T-00000225, S26-D-00000631
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000673|D|A|Proof+Compile+Meta+Topo+IG+Spectral] Projection-valued measure

§FORM := A projection-valued measure assigns orthogonal projections E(B) to measurable sets B with countable additivity in the strong operator topology and the usual measure identities.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000434, S26-T-00000639
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000674|T|A|Proof+Compile+Meta+Spectral] Borel functional calculus interface

§FORM := For bounded self-adjoint T, bounded Borel functions f define operators f(T)=integral f(lambda)dE(lambda), compatible with algebraic operations.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000672, S26-D-00000673
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000675|T|A|Proof+Compile+Meta] Continuous functional calculus interface

§FORM := For bounded normal T, continuous functions on sigma(T) define f(T) through the continuous functional calculus.
§ASSUME := T is bounded normal on a complex Hilbert space.
§DEP_HARD := S26-D-00000227, S26-D-00000651
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000676|T|A|Proof+Compile+Meta+IG+Spectral] Spectral theorem for unbounded self-adjoint operators

§FORM := A self-adjoint operator admits a projection-valued spectral representation A=integral lambda dE(lambda), with its domain determined by square-integrability of lambda against the spectral measures.
§ASSUME := A is self-adjoint on a complex Hilbert space.
§DEP_HARD := S26-D-00000668, S26-D-00000673
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000677|D|A|Proof+Compile+Meta] Test-function space

§FORM := D(Omega)=C_c^infinity(Omega) is the space of smooth compactly supported test functions on an open set Omega subset R^n.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000335, S26-T-00000323
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000678|D|A|Proof+Compile+Meta+Topo] Distribution

§FORM := A distribution on Omega is a continuous linear functional on D(Omega) with its standard locally convex topology.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000677, S26-D-00000613
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000679|D|A|Proof+Compile+Meta] Regular distribution

§FORM := A locally integrable function f defines the distribution phi -> integral_Omega f phi.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000678, S26-D-00000429
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000680|D|A|Proof+Compile+Meta] Distributional derivative

§FORM := For a distribution T, its derivative is defined by <partial_i T,phi>=-<T,partial_i phi>.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000678, S26-T-00000337
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000681|T|A|Proof+Compile+Meta] Distributional integration by parts

§FORM := Distributional differentiation transfers derivatives to test functions with the sign dictated by integration by parts.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000680, S26-T-00000399
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000682|D|A|Proof+Compile+Meta] Dirac distribution

§FORM := At x0 in Omega, delta_x0(phi)=phi(x0) defines a distribution.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000678
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000683|T|A|Proof+Compile+Meta] Heaviside derivative

§FORM := The distributional derivative of the Heaviside step function is the Dirac distribution at zero.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000679, S26-D-00000680, S26-D-00000682
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000684|D|A|Proof+Compile+Meta] Weak derivative

§FORM := A locally integrable v is the weak derivative partial_i u when integral u partial_i phi = - integral v phi for all test functions phi.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000677, S26-D-00000429, S26-D-00000680
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000685|D|A|Proof+Compile+Meta] Sobolev space W^{k,p}

§FORM := W^{k,p}(Omega) consists of L^p functions whose weak derivatives through order k belong to L^p.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000440, S26-D-00000684
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000686|D|A|Proof+Compile+Meta+Spectral] H^k Sobolev space

§FORM := H^k(Omega)=W^{k,2}(Omega) is the Hilbertian Sobolev scale.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000685, S26-T-00000442
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000687|T|A|Proof+Compile+Meta] W^{k,p} completeness

§FORM := Sobolev spaces W^{k,p}(Omega) are Banach for 1<=p<=infinity under the standard Sobolev norm.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000685, S26-D-00000605
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000688|T|A|Proof+Compile+Meta+Spectral] H^k Hilbert structure

§FORM := H^k(Omega) is a Hilbert space with the standard derivative inner product.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000686, S26-D-00000631
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000689|D|A|Proof+Compile+Meta] Weak formulation of Poisson equation

§FORM := A weak Dirichlet solution of -Delta u=f seeks u in H_0^1(Omega) satisfying integral grad u dot grad v = integral f v for all admissible test v.
§ASSUME := The domain and data are such that the displayed pairings are well-defined.
§DEP_HARD := S26-D-00000686, S26-T-00000341, S26-T-00000399
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000690|D|A|Proof+Compile+Meta+Spectral] Bounded bilinear form on Hilbert space

§FORM := A bilinear or sesquilinear form a on H is bounded when |a(u,v)|<=C||u|||v||.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000254, S26-D-00000631
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000691|D|A|Proof+Compile+Meta] Coercive bilinear form

§FORM := A form a on H is coercive when Re a(u,u)>=alpha||u||^2 for some alpha>0.
§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000690
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000692|T|A|Proof+Compile+Meta+Spectral] Lax-Milgram theorem

§FORM := A bounded coercive sesquilinear form on a Hilbert space represents each bounded anti/linear functional by a unique solution u, with the standard norm bound.

§ASSUME := H is Hilbert; the form is bounded and coercive; scalar convention is fixed.

§DEP_HARD := S26-D-00000690, S26-D-00000691, S26-T-00000640

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000693|T|A|Proof+Compile+Meta] Poisson weak-solution interface

§FORM := When the Dirichlet energy form is bounded and coercive on H_0^1(Omega), Lax-Milgram yields existence and uniqueness of the weak Poisson solution for admissible forcing.

§ASSUME := All domain regularity, Poincare/coercivity, and forcing-duality hypotheses needed for the stated weak problem hold.

§DEP_HARD := S26-D-00000689, S26-T-00000692

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000694|D|A|Proof+Compile+Meta+Spectral] Operator domain fence

§FORM := For unbounded operators, the domain is part of the operator; changing boundary conditions can change symmetry, adjoints, self-adjointness, and spectrum.

§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000661, S26-D-00000665, S26-D-00000668

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000695|D|A|Proof+Compile+Meta+Spectral] Spectrum-type fence

§FORM := Point spectrum, continuous spectrum, residual spectrum, and resolvent behavior must not be collapsed into the finite-dimensional notion of eigenvalues.

§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000651, S26-D-00000659

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000696|D|A|Proof+Compile+Meta] Weak-versus-classical derivative fence

§FORM := A weak or distributional derivative need not arise from pointwise classical differentiability.

§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000680, S26-D-00000684, S26-D-00000335

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§BOUNDARY := The absolute-value function has a weak first derivative despite failure of classical differentiability at zero.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000697|D|A|Proof+Compile+Meta+Spectral] Functional-calculus hypothesis fence

§FORM := Functional calculi depend on operator class and function class; polynomial, continuous, Borel, holomorphic, bounded and unbounded calculi are not interchangeable.

§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000674, S26-T-00000675, S26-T-00000676

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000698|T|A|Proof+Compile+Meta+Spectral] Spectral-PDE bridge

§FORM := Self-adjoint operator theory and Sobolev weak formulations provide complementary spectral and variational languages for differential operators.

§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000676, S26-D-00000685, S26-T-00000692

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000699|T|A|Proof+Compile+Meta+Topo+Spectral] Unbounded-operator epistemic fence

§FORM := Finite-dimensional matrix identities do not by themselves validate claims about unbounded operators: domains, closure, density, topology, and spectral type are load-bearing.

§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000661, S26-D-00000663, S26-D-00000668, S26-D-00000695

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000700|T|A|Proof+Compile+Meta+Topo+Spectral] Operator-analysis frontier

§FORM := Spectrum/resolvent theory, unbounded self-adjoint operators, distributions, Sobolev spaces, and variational existence principles form the substrate for PDE, harmonic analysis, quantum mechanics, and geometric analysis.

§ASSUME := All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000656, S26-T-00000676, S26-D-00000678, S26-D-00000685, S26-T-00000692

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000701|D|A|Proof+Compile+Meta+Spectral] Fourier coefficient

§FORM := For f in L1(T), the nth Fourier coefficient is fhat(n)=(1/2pi) integral_0^{2pi} f(x)e^{-inx} dx.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000429

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000702|D|A|Proof+Compile+Meta+Spectral] Fourier series
§FORM := The formal Fourier series of f is sum_{n in Z} fhat(n)e^{inx}.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000701
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000703|T|A|Proof+Compile+Meta] Bessel inequality
§FORM := For f in L2(T), sum_n |fhat(n)|^2 <= ||f||_2^2 under the normalized convention.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000442, S26-D-00000701
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000704|T|A|Proof+Compile+Meta+Spectral] Parseval identity on the circle
§FORM := For f in L2(T), the Fourier coefficients satisfy Parseval equality with the chosen normalization.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000703
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000705|T|A|Proof+Compile+Meta+Spectral] L2 Fourier convergence
§FORM := Partial Fourier sums converge to f in L2(T).
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000704
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000706|D|A|Proof+Compile+Meta+Spectral] Dirichlet kernel
§FORM := The Nth Dirichlet kernel D_N is sum_{|n|<=N}e^{inx}, and Fourier partial sums are convolution with D_N.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000702, S26-D-00000711
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000707|D|A|Proof+Compile+Meta] Fejer kernel
§FORM := The Fejer kernel is the Cesaro average of Dirichlet kernels and is nonnegative with normalized integral one.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000706
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000708|T|A|Proof+Compile+Meta+Spectral] Fejer theorem
§FORM := For continuous periodic f, its Cesaro Fourier means converge uniformly to f.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000707, S26-T-00000366
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000709|D|A|Proof+Compile+Meta+Spectral] Fourier transform on R^n
§FORM := For f in L1(R^n), define Ff(xi)=integral f(x)e^{-2pi i x dot xi} dx under the fixed convention.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000429, S26-I-00000060
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000710|D|A|Proof+Compile+Meta+Spectral] Inverse Fourier transform convention
§FORM := The inverse transform uses the dual exponential sign and normalization paired with entry 709.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000709
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000711|D|A|Proof+Compile+Meta] Convolution
§FORM := For suitable f,g on R^n, (f*g)(x)=integral f(x-y)g(y)dy.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000429
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000712|T|A|Proof+Compile+Meta] Young L1 convolution bound
§FORM := If f,g in L1, then f*g in L1 and ||f*g||_1<=||f||_1||g||_1.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000711, S26-T-00000440
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000713|T|A|Proof+Compile+Meta+Spectral] Convolution theorem
§FORM := For f,g in L1, F(f*g)=F(f)F(g) under the fixed Fourier convention.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000709, S26-D-00000711
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000714|T|A|Proof+Compile+Meta+Spectral] Product-transform interface

§FORM := Under hypotheses making both sides meaningful, Fourier transform converts products into convolution with convention-dependent constants.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000709, S26-D-00000711

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000715|T|A|Proof+Compile+Meta] Riemann-Lebesgue lemma

§FORM := If f in L1(R^n), then Ff is continuous and vanishes at infinity.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000709

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000716|D|A|Proof+Compile+Meta] Schwartz space

§FORM := S(R^n) consists of smooth functions whose derivatives decay faster than every polynomial power.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000335

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000717|T|A|Proof+Compile+Meta+Spectral] Fourier invariance of Schwartz space

§FORM := The Fourier transform is an automorphism of S(R^n).

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000709, S26-D-00000716

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000718|T|A|Proof+Compile+Meta+Spectral] Fourier inversion on Schwartz space

§FORM := For f in S(R^n), inverse Fourier transform recovers f pointwise under the paired convention.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000710, S26-T-00000717

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000719|T|A|Proof+Compile+Meta+Spectral] Plancherel theorem

§FORM := The Fourier transform extends uniquely from S cap L2 to a unitary operator on L2(R^n), under unitary normalization.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000442, S26-T-00000717

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000720|T|A|Proof+Compile+Meta+Spectral] L2 Fourier inversion

§FORM := For f in L2, Fourier inversion holds in the L2 sense through the Plancherel extension.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000719

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000721|D|A|Proof+Compile+Meta] Approximate identity

§FORM := A family phi_epsilon is an approximate identity when mass is normalized and concentrates near the identity while remaining uniformly L1-controlled.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000711, S26-T-00000440

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000722|T|A|Proof+Compile+Meta] Approximate-identity convergence

§FORM := For 1<=p<infinity, convolution with a standard approximate identity converges to f in Lp.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000721, S26-T-00000440

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000723|D|A|Proof+Compile+Meta] Mollifier

§FORM := A mollifier is a smooth compactly supported approximate identity obtained by rescaling a unit-mass test function.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000677, S26-D-00000721

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000724|T|A|Proof+Compile+Meta] Mollification regularizes

§FORM := Convolution with a mollifier produces smooth functions and approximates Lp functions in the appropriate norm.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000723, S26-T-00000722

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000725|D|A|Proof+Compile+Meta] Tempered distribution

§FORM := A tempered distribution is a continuous linear functional on Schwartz space S(R^n).
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000716, S26-D-00000678
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000726|T|A|Proof+Compile+Meta+Spectral] Fourier transform of tempered distributions

§FORM := Fourier transform extends to S-prime by duality: <FT,phi>=<T,Fphi> up to the fixed convention.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000725, S26-T-00000717
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000727|D|A|Proof+Compile+Meta+Spectral] Fourier transform of Dirac

§FORM := The Fourier transform of delta_0 is the constant distribution 1 under the standard convention.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000682, S26-T-00000726
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000728|T|A|Proof+Compile+Meta+Spectral] Differentiation-multiplication duality

§FORM := Fourier transformation converts distributional differentiation into multiplication by the frequency coordinate, with convention-dependent factors.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000680, S26-T-00000726
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000729|T|A|Proof+Compile+Meta+Spectral] Multiplication-differentiation duality

§FORM := Multiplication by x_j corresponds under Fourier transform to differentiation in xi_j, with convention-dependent factors.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000726
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000730|D|A|Proof+Compile+Meta+Spectral] Translation operator

§FORM := Translation tau_a f(x)=f(x-a) acts compatibly with Fourier modulation.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000709
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000731|D|A|Proof+Compile+Meta+Spectral] Modulation operator

§FORM := Modulation M_eta f(x)=e^{2pi i eta dot x}f(x) acts compatibly with Fourier translation.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000709
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000732|T|A|Proof+Compile+Meta+Spectral+Swirl] Translation-modulation covariance

§FORM := Fourier transform exchanges translations and modulations according to the fixed phase convention.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000730, S26-D-00000731
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000733|D|A|Proof+Compile+Meta+Spectral] Dilation operator

§FORM := Dilation rescales spatial variables and induces reciprocal frequency scaling with the Jacobian factor.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000709
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000734|T|A|Proof+Compile+Meta+Spectral] Fourier scaling law

§FORM := The Fourier transform of f(ax) is the reciprocally rescaled transform with determinant/Jacobian factor.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000733
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000735|D|A|Proof+Compile+Meta+Spectral] Gaussian Fourier interface

§FORM := Gaussians transform to Gaussians under compatible normalization.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000709
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000736|T|A|Proof+Compile+Meta] Heisenberg uncertainty inequality

§FORM := For suitably normalized f with x f and grad f square-integrable, spatial and frequency dispersions obey a positive lower bound.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000719, S26-T-00000728
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000737|D|A|Proof+Compile+Meta+Spectral] Fourier multiplier

§FORM := A Fourier multiplier T_m acts formally by F(T_m f)=m Ff on a specified function space.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000709
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000738|T|A|Proof+Compile+Meta+Spectral] L2 multiplier bound

§FORM := If m is essentially bounded, the associated Fourier multiplier is bounded on L2 with norm at most ||m||_infinity.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000719, S26-D-00000737
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000739|D|A|Proof+Compile+Meta] Fractional differentiation multiplier

§FORM := Fractional powers of |D| are represented formally by multipliers |xi|^s on suitable domains.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000737, S26-D-00000685
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000740|T|A|Proof+Compile+Meta+Spectral] Sobolev Fourier characterization

§FORM := On R^n, H^s can be characterized by weighted square-integrability of the Fourier transform with weight comparable to (1+|xi|^2)^s.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000686, S26-T-00000719
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000741|D|A|Proof+Compile+Meta+Spectral] Fundamental solution interface

§FORM := A fundamental solution E of a constant-coefficient operator P(D) satisfies P(D)E=delta in distributions.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000680, S26-D-00000682
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000742|T|A|Proof+Compile+Meta+Spectral] Fourier-symbol PDE reduction

§FORM := For constant-coefficient differential operators, Fourier transform converts P(D)u=f into multiplication by its symbol P(2pi i xi), subject to distributional/domain hypotheses.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000726, S26-T-00000728, S26-D-00000741
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000743|D|A|Proof+Compile+Meta+Spectral] Laplacian Fourier symbol

§FORM := Under the 2pi convention, F(-Delta f)(xi)=4pi^2|xi|^2 Ff(xi) for Schwartz f.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000341, S26-T-00000728, S26-D-00000716
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000744|T|A|Proof+Compile+Meta+Spectral] Heat semigroup Fourier solution

§FORM := For suitable initial data, the heat equation Fourier transform evolves by multiplication with exp(-4pi^2 t|xi|^2).
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000743, S26-T-00000719
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000745|T|A|Proof+Compile+Meta+Spectral] Poisson equation Fourier interface

§FORM := On R^n, solving -Delta u=f formally corresponds to division by 4pi^2|xi|^2 away from zero, with low-frequency and distributional issues handled explicitly.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000743, S26-T-00000742
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000746|D|A|Proof+Compile+Meta+Spectral] Fourier normalization fence

§FORM := All constants, inverse formulas, differentiation factors, and Plancherel identities depend on the declared Fourier normalization.
§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000709, S26-D-00000710
§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000747|D|A|Proof+Compile+Meta+Spectral] Convergence-mode fence for Fourier series

§FORM := L2, pointwise, uniform, Cesaro, and almost-everywhere convergence of Fourier expansions are distinct claims and require distinct hypotheses.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000705, S26-T-00000708, S26-D-00000365

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000748|D|A|Proof+Compile+Meta] Distribution-versus-function fence

§FORM := A tempered distribution need not be represented by an ordinary function; algebraic operations require their own domains of definition.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000725, S26-D-00000679

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000749|T|A|Proof+Compile+Meta+Spectral] Harmonic-analysis epistemic fence

§FORM := Finite FFT experiments or smooth examples do not prove general Fourier inversion, multiplier, uncertainty, or PDE theorems.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000719, S26-T-00000736, S26-T-00000742

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000750|T|A|Proof+Compile+Meta+Spectral] Harmonic-analysis frontier

§FORM := Fourier series, transforms, convolution, Schwartz/tempered distributions, multipliers, Sobolev frequency descriptions, and PDE symbols form the substrate for systematic PDE and harmonic analysis.

§ASSUME := Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

§DEP_HARD := S26-T-00000708, S26-T-00000719, S26-T-00000726, S26-T-00000740, S26-T-00000742

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000751|D|A|Proof+Compile+Meta] Partial differential equation

§FORM := A PDE is a relation involving an unknown function of several variables and some of its partial derivatives.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000336

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000752|D|A|Proof+Compile+Meta+Spectral] Linear PDE

§FORM := A PDE is linear when the unknown function and its derivatives enter through a linear differential operator.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000751, S26-T-00000226

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000753|D|A|Proof+Compile+Meta] Semilinear PDE

§FORM := A semilinear PDE has a linear highest-order part while lower-order terms may depend nonlinearly on the unknown.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000752

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000754|D|A|Proof+Compile+Meta] Quasilinear PDE

§FORM := A quasilinear PDE is linear in its highest-order derivatives but may have coefficients depending on lower-order derivatives and the unknown.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000751

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000755|D|A|Proof+Compile+Meta] Fully nonlinear PDE

§FORM := A fully nonlinear PDE may depend nonlinearly on its highest-order derivatives.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000751

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000756|D|A|Proof+Compile+Meta+Spectral] Principal symbol

§FORM := The principal symbol records the highest-order part of a differential operator as a homogeneous polynomial in the cotangent/frequency variable.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000742

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000757|D|A|Proof+Compile+Meta+Spectral] Second-order principal matrix

§FORM := For a scalar second-order operator, the symmetric coefficient matrix of second derivatives determines the principal quadratic form.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000756, S26-D-00000239

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000758|D|A|Proof+Compile+Meta+Spectral] Elliptic operator

§FORM := A second-order scalar operator is elliptic at a point when its principal quadratic form is definite there; uniform ellipticity requires quantitative bounds.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000757

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000759|D|A|Proof+Compile+Meta+Spectral] Parabolic prototype

§FORM := The heat operator partial_t-Delta is the canonical parabolic prototype; general parabolicity requires a time-space structure and quantitative ellipticity in spatial variables.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000744, S26-D-00000758

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000760|D|A|Proof+Compile+Meta+Spectral] Hyperbolic prototype

§FORM := The wave operator partial_t^2-Delta is the canonical hyperbolic prototype, with finite-speed propagation under the standard constant-coefficient model.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000743, S26-D-00000751

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000761|D|A|Proof+Compile+Meta] Boundary value problem

§FORM := A boundary value problem couples a PDE in a domain with conditions prescribed on its boundary.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000751, S26-D-00000306

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000762|D|A|Proof+Compile+Meta] Initial value problem

§FORM := An initial value problem couples an evolution PDE with data prescribed at an initial time.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000751, S26-T-00000346

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000763|D|A|Proof+Compile+Meta] Dirichlet boundary condition

§FORM := A Dirichlet condition prescribes the trace/value of the solution on the boundary where that trace is defined.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000761, S26-T-00000687

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000764|D|A|Proof+Compile+Meta] Neumann boundary condition

§FORM := A Neumann condition prescribes an appropriate normal derivative or flux on the boundary, requiring sufficient boundary and trace structure.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000761, S26-T-00000688

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000765|D|A|Proof+Compile+Meta] Classical PDE solution

§FORM := A classical solution has enough ordinary differentiability for the PDE and prescribed data to hold pointwise in their declared sense.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000751, S26-D-00000335

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000766|D|A|Proof+Compile+Meta] Weak PDE solution

§FORM := A weak solution satisfies an integrated variational identity in a specified function space rather than necessarily satisfying the PDE pointwise.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000692, S26-D-00000765

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000767|D|A|Proof+Compile+Meta] Distributional PDE solution

§FORM := A distributional solution satisfies the PDE after interpreting derivatives in the sense of distributions.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000679, S26-D-00000680, S26-D-00000751

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000768|T|A|Proof+Compile+Meta] Classical implies distributional

§FORM := A sufficiently regular classical solution induces a distributional solution of the same PDE.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000765, S26-D-00000767

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000769|D|A|Proof+Compile+Meta+Topo] Well-posedness

§FORM := A problem is well posed in a declared class when existence, uniqueness, and continuous dependence on data all hold in the specified topologies.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000306, S26-T-00000319, S26-D-00000751

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000770|D|A|Proof+Compile+Meta] A priori estimate

§FORM := An a priori estimate bounds a solution quantity by prescribed data independently of a particular construction of the solution.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000769, S26-T-00000305

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000771|T|A|Proof+Compile+Meta] Energy method interface

§FORM := Multiplying an evolution PDE by a suitable solution/test quantity and integrating can yield an energy identity or inequality under justified regularity and boundary conditions.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000430, S26-D-00000766

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000772|D|A|Proof+Compile+Meta+Spectral] Wave energy

§FORM := For the standard homogeneous wave equation, the energy combines squared time derivative and squared spatial gradient integrals.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000760, S26-T-00000771

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000773|T|A|Proof+Compile+Meta+Spectral] Wave energy conservation

§FORM := For sufficiently regular solutions of the homogeneous wave equation with compatible boundary/decay conditions, the standard energy is conserved.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000772

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000774|D|A|Proof+Compile+Meta] Heat energy dissipation

§FORM := For the homogeneous heat equation under compatible conditions, an L2 energy decreases and its derivative is controlled by the gradient norm.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000744, S26-T-00000771, S26-T-00000442

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000775|D|A|Proof+Compile+Meta+Spectral] Maximum principle interface

§FORM := Maximum principles constrain extrema of solutions or subsolutions of elliptic/parabolic equations under operator, domain, and boundary hypotheses.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000758, S26-D-00000759, S26-T-00000348

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000776|T|A|Proof+Compile+Meta] Weak maximum principle for harmonic functions

§FORM := A continuous harmonic function on a bounded domain, sufficiently regular inside, attains its maximum and minimum on the boundary.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000341, S26-D-00000775, S26-D-00000310

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000777|T|A|Proof+Compile+Meta+Spectral] Uniqueness from maximum principle

§FORM := For the Laplace Dirichlet problem, the maximum principle yields uniqueness in the corresponding classical solution class.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000763, S26-T-00000776

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000778|D|A|Proof+Compile+Meta] Green identity interface

§FORM := Integration by parts relates Laplacians, gradients, and boundary flux terms under sufficient regularity.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000430, S26-T-00000341, S26-D-00000764

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000779|D|A|Proof+Compile+Meta] Green function

§FORM := A Green function represents the response of a linear boundary value problem to a point source and encodes the boundary condition.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000741, S26-D-00000761, S26-D-00000682

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000780|T|A|Proof+Compile+Meta] Green representation interface

§FORM := Under suitable regularity, domain, and boundary hypotheses, solutions of linear elliptic problems admit integral representations using an associated Green function plus boundary terms.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000779, S26-D-00000778

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000781|D|A|Proof+Compile+Meta+Spectral] Variational elliptic problem

§FORM := A second-order elliptic boundary problem can often be recast as finding u in a Hilbert/Sobolev space such that a(u,v)=F(v) for all test functions v.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000692, S26-T-00000699

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000782|T|A|Proof+Compile+Meta+Spectral] Lax-Milgram elliptic existence interface

§FORM := A bounded coercive bilinear form on a Hilbert space yields a unique weak solution with an a priori norm bound.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000699, S26-D-00000781

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000783|D|A|Proof+Compile+Meta] Coercivity fence

§FORM := Coercivity is a quantitative lower bound on the bilinear form relative to the chosen norm; positivity without a uniform lower bound is not enough for Lax-Milgram.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000782

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000784|D|A|Proof+Compile+Meta+Topo] Poincare inequality interface

§FORM := On appropriate domains and subspaces, an Lp norm of a function can be controlled by a norm of its gradient, with constants depending on geometry and normalization.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000686, S26-T-00000440

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000785|T|A|Proof+Compile+Meta] Dirichlet energy coercivity interface

§FORM := For H_0^1-type spaces on suitable bounded domains, Poincare inequality makes the gradient energy coercive in the H1 norm.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000784, S26-D-00000781

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000786|D|A|Proof+Compile+Meta] C0 semigroup

§FORM := A strongly continuous one-parameter semigroup on a Banach space satisfies T(0)=I, T(t+s)=T(t)T(s), and strong continuity in time.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000604, S26-T-00000617

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000787|D|A|Proof+Compile+Meta+Spectral] Semigroup generator

§FORM := The generator A of a C0 semigroup is the generally unbounded operator defined by the strong derivative at zero on the domain where that limit exists.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000786, S26-D-00000661

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000788|T|A|Proof+Compile+Meta] Abstract evolution equation interface

§FORM := When A generates a suitable C0 semigroup, u(t)=T(t)u0 provides the mild solution of u prime=Au with initial datum u0.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000787, S26-D-00000762

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000789|D|A|Proof+Compile+Meta] Mild solution

§FORM := A mild solution of an evolution equation is defined through a semigroup/integral formulation and need not possess the derivatives required of a classical solution.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-T-00000788, S26-D-00000765

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000790|D|A|Proof+Compile+Meta] Duhamel principle

§FORM := For an inhomogeneous linear evolution equation, the solution is represented by homogeneous semigroup evolution plus a time convolution with the forcing, under appropriate hypotheses.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000786, S26-D-00000789, S26-D-00000711

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000791|D|A|Proof+Compile+Meta] Elliptic regularity interface

§FORM := Elliptic regularity asks when improved regularity of coefficients, data, and domain forces improved regularity of a weak/distributional solution.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000758, S26-D-00000766, S26-D-00000686

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000792|T|A|Proof+Compile+Meta+Spectral] Interior elliptic regularity schematic

§FORM := For uniformly elliptic operators with sufficiently regular coefficients, solutions gain interior regularity relative to the forcing in theorem-specific Sobolev or Holder scales.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000791, S26-D-00000758

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000793|D|A|Proof+Compile+Meta] Parabolic smoothing interface

§FORM := Parabolic equations such as the heat equation can improve spatial regularity for positive time, subject to the data and solution framework.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000759, S26-T-00000744, S26-D-00000791

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000794|D|A|Proof+Compile+Meta+Spectral] Finite propagation interface

§FORM := Hyperbolic equations such as the wave equation exhibit domain-of-dependence and finite-speed phenomena under suitable principal-part hypotheses.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000760, S26-D-00000769

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000795|D|A|Proof+Compile+Meta] Characteristic surface interface

§FORM := For a PDE, characteristic covectors/surfaces are determined by vanishing of the principal symbol and govern propagation behavior.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000756, S26-D-00000760

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000796|D|A|Proof+Compile+Meta] Solution-concept fence

§FORM := Classical, strong, weak, mild, viscosity, and distributional solutions are distinct typed notions; implications require explicit hypotheses.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000765, S26-D-00000766, S26-D-00000767, S26-D-00000789

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000797|D|A|Proof+Compile+Meta+Topo] Boundary-regularity fence

§FORM := Boundary traces, normal derivatives, Green formulas, and boundary regularity require geometric and function-space hypotheses and cannot be inferred from interior regularity alone.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000763, S26-D-00000764, S26-D-00000778, S26-D-00000791

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000798|D|A|Proof+Compile+Meta] PDE classification fence

§FORM := Elliptic, parabolic, and hyperbolic labels refer to principal-symbol structure and may depend on point, coefficients, formulation, and degeneracy; prototype equations do not define every case.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000758, S26-D-00000759, S26-D-00000760, S26-D-00000756

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000799|T|A|Proof+Compile+Meta] PDE epistemic fence

§FORM := Grid experiments, symbolic residuals, or smooth manufactured solutions can test implementations but do not prove general existence, uniqueness, regularity, stability, or convergence theorems.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000769, S26-D-00000770, S26-D-00000791

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000800|T|A|Proof+Compile+Meta+Topo] PDE foundations frontier

§FORM := Classification, boundary/initial data, weak formulations, energy and maximum methods, variational existence, semigroups, and regularity interfaces form the substrate for nonlinear PDE, numerical PDE, and geometric analysis.

§ASSUME := Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

§DEP_HARD := S26-D-00000769, S26-T-00000782, S26-T-00000788, S26-T-00000792, S26-D-00000794

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000801|D|A|Proof+Compile+Meta] Functional

§FORM := A functional is a map from a space of admissible functions or states into the scalar field or extended real line.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-T-00000226, S26-T-00000604

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000802|D|A|Proof+Compile+Meta] Admissible class

§FORM := An admissible class specifies the function space together with boundary, integral, inequality, or other constraints over which a variational problem is posed.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000801, S26-D-00000761

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000803|D|A|Proof+Compile+Meta] Variational minimizer

§FORM := A minimizer of J over an admissible class A is u in A with J(u)<=J(v) for every v in A.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000801, S26-D-00000802

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000804|D|A|Proof+Compile+Meta+Topo] Local variational minimizer

§FORM := A local minimizer minimizes a functional only relative to an explicitly chosen topology or norm in a neighborhood of the candidate.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000803, S26-D-00000303

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000805|D|A|Proof+Compile+Meta] Variation

§FORM := A variation of u is a curve u+t phi through the admissible space, with phi restricted when constraints or boundary data require it.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000802, S26-T-00000331

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000806|D|A|Proof+Compile+Meta] First variation

§FORM := The first variation delta J(u)[phi] is the directional derivative at t=0 of J(u+t phi), when this derivative exists.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000805, S26-T-00000329

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000807|D|A|Proof+Compile+Meta] Stationary point of a functional

§FORM := A stationary point has vanishing first variation for every admissible test direction.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000806

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000808|T|A|Proof+Compile+Meta] Variational necessary condition

§FORM := An interior local minimizer at which the first variation exists in every admissible direction is stationary.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000804, S26-D-00000806, S26-D-00000807

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000809|D|A|Proof+Compile+Meta] Integral functional

§FORM := A basic variational functional has the form J(u)=integral_Omega L(x,u,grad u) dx on a declared Sobolev or smooth admissible class.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000801, S26-D-00000430, S26-D-00000686

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000810|D|A|Proof+Compile+Meta] Lagrangian density

§FORM := The integrand L(x,z,p) of an integral functional is its Lagrangian density; regularity and growth assumptions determine which manipulations are legitimate.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000809

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000811|T|A|Proof+Compile+Meta] Euler-Lagrange equation interface

§FORM := For sufficiently smooth L and admissible u, vanishing first variation yields the Euler-Lagrange equation in weak form and, with additional regularity, pointwise form.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000806, S26-D-00000809, S26-D-00000766

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000812|D|A|Proof+Compile+Meta] Weak Euler-Lagrange equation

§FORM := The weak Euler-Lagrange equation is the test-function identity obtained after differentiating the functional and transferring derivatives by integration by parts.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-T-00000811, S26-D-00000778

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000813|D|A|Proof+Compile+Meta] Second variation

§FORM := The second variation is the second directional derivative of a functional along admissible variations when it exists.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000806, S26-T-00000337

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000814|T|A|Proof+Compile+Meta] Second-variation necessary condition

§FORM := At a sufficiently regular local minimizer, the second variation is nonnegative in admissible directions, under hypotheses ensuring the second derivative test applies.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000804, S26-D-00000813, S26-T-00000349

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000815|D|A|Proof+Compile+Meta] Convex functional

§FORM := A functional J on a convex set is convex when J((1-t)u+tv)<=(1-t)J(u)+tJ(v) for 0<=t<=1.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-T-00000355, S26-D-00000801

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000816|D|A|Proof+Compile+Meta] Strictly convex functional

§FORM := A functional is strictly convex when the convexity inequality is strict for distinct points and 0<t<1.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000815

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000817|T|A|Proof+Compile+Meta] Strict-convexity uniqueness

§FORM := A strictly convex functional has at most one minimizer on a convex admissible set.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000816, S26-D-00000803

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000818|D|A|Proof+Compile+Meta+Topo] Coercive functional

§FORM := A functional is coercive relative to a norm or topology when J(u) tends to +infinity as the declared size of u tends to infinity.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000801, S26-T-00000604

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000819|D|A|Proof+Compile+Meta] Weak lower semicontinuity

§FORM := J is sequentially weakly lower semicontinuous when u_n converges weakly to u implies J(u)<=liminf J(u_n).

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-T-00000632, S26-D-00000801, S26-T-00000373

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000820|D|A|Proof+Compile+Meta] Minimizing sequence

§FORM := A minimizing sequence satisfies J(u_n)->inf_A J without itself being assumed to converge.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000803, S26-T-00000362

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000821|D|A|Proof+Compile+Meta] Direct method of calculus of variations

§FORM := The direct method seeks a bounded minimizing sequence, extracts a convergent subsequence using compactness, proves admissibility of the limit, and uses lower semicontinuity to obtain a minimizer.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000818, S26-D-00000819, S26-D-00000820, S26-T-00000311

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000822|T|A|Proof+Compile+Meta] Direct-method existence interface

§FORM := On a reflexive Banach space, coercivity and sequential weak lower semicontinuity, together with weakly closed nonempty admissibility, give a minimizer under the theorem-specific hypotheses.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000821, S26-D-00000634, S26-T-00000633

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000823|D|A|Proof+Compile+Meta+Topo] Compactness fence for direct methods

§FORM := Boundedness of a minimizing sequence is not itself strong compactness in an infinite-dimensional space; the topology and compactness theorem used must be named.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000820, S26-D-00000623, S26-T-00000633

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000824|D|A|Proof+Compile+Meta] Lower-semicontinuity fence

§FORM := Weak convergence does not permit passing arbitrary nonlinear energies to the limit; weak lower semicontinuity or another justified limit theorem is required.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000819, S26-T-00000632

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000825|D|A|Proof+Compile+Meta] Constraint functional

§FORM := A constraint is represented by an equation or inequality on the admissible state, often through one or more auxiliary functionals.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000802, S26-D-00000801

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000826|D|A|Proof+Compile+Meta] Lagrange multiplier interface in function spaces

§FORM := Under a suitable constraint qualification, a constrained stationary point may satisfy a multiplier equation involving derivatives of the objective and constraint functionals.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000825, S26-D-00000806

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000827|D|A|Proof+Compile+Meta] Obstacle problem interface

§FORM := An obstacle problem minimizes an energy over functions constrained to lie above or below a prescribed obstacle, leading naturally to a variational inequality.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000803, S26-D-00000825, S26-D-00000781

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000828|D|A|Proof+Compile+Meta] Variational inequality

§FORM := A variational inequality replaces equality against every test direction by an inequality over a convex admissible set.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000827, S26-D-00000815

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000829|D|A|Proof+Compile+Meta+Spectral] Nonlinear operator equation

§FORM := A nonlinear PDE may be encoded abstractly as A(u)=f between suitable function spaces.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000755, S26-T-00000604

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000830|D|A|Proof+Compile+Meta+Spectral] Monotone operator

§FORM := An operator A from a real Banach space into its dual is monotone when <A(u)-A(v),u-v>>=0 for all u,v in its domain.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000829, S26-T-00000621

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000831|D|A|Proof+Compile+Meta+Spectral] Strictly monotone operator

§FORM := A monotone operator is strictly monotone when the monotonicity pairing is positive for distinct admissible points.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000830

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000832|D|A|Proof+Compile+Meta+Spectral] Strongly monotone operator

§FORM := A is strongly monotone when the monotonicity pairing controls a positive power, typically the squared norm, of u-v by a uniform constant.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000830, S26-T-00000604

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000833|D|A|Proof+Compile+Meta+Spectral] Hemicontinuity interface

§FORM := Hemicontinuity requires suitable scalar continuity of t -> <A(u+tv),w> and is one standard hypothesis in monotone-operator existence theorems.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000830, S26-T-00000319

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000834|D|A|Proof+Compile+Meta+Spectral] Browder-Minty interface

§FORM := A coercive, hemicontinuous monotone operator on an appropriate reflexive Banach space is surjective under the theorem-specific hypotheses; strict monotonicity yields uniqueness.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000830, S26-D-00000833, S26-D-00000634

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000835|D|A|Proof+Compile+Meta] p-Laplacian

§FORM := The p-Laplacian is Delta_p u=div(|grad u|^(p-2) grad u), naturally interpreted weakly on W^{1,p}-type spaces.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-T-00000688, S26-D-00000751, S26-D-00000830

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000836|D|A|Proof+Compile+Meta+Spectral] Weak p-Laplace equation

§FORM := A weak solution of -Delta_p u=f satisfies the corresponding integral identity against test functions, with integrability exponents and boundary conditions explicitly specified.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000835, S26-D-00000766, S26-T-00000443

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000837|D|A|Proof+Compile+Meta+Spectral] Nonlinear elliptic equation

§FORM := A nonlinear elliptic equation has a nonlinear operator whose principal behavior satisfies an appropriate ellipticity condition; definitions vary by divergence/nondivergence and solution framework.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000755, S26-D-00000758, S26-D-00000829

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000838|D|A|Proof+Compile+Meta] Quasilinear divergence-form equation

§FORM := A typical quasilinear divergence-form equation is -div A(x,u,grad u)=f, interpreted under growth, coercivity, and monotonicity assumptions.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000754, S26-T-00000688, S26-D-00000830

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000839|D|A|Proof+Compile+Meta] Weak compactness interface for nonlinear PDE

§FORM := Uniform bounds in reflexive Sobolev spaces can yield weakly convergent subsequences, but identification of nonlinear limits requires additional structure.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000634, S26-D-00000686, S26-T-00000632

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000840|D|A|Proof+Compile+Meta] Strong compactness interface

§FORM := Compact embeddings can upgrade bounded sequences to strongly convergent subsequences in weaker norms under domain and exponent hypotheses.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-T-00000649, S26-D-00000686, S26-T-00000311

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000841|D|A|Proof+Compile+Meta] Rellich-Kondrachov interface

§FORM := On suitable bounded domains, specified Sobolev embeddings are compact when the exponents and regularity lie in the theorem-specific compact range.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000840, S26-D-00000686

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000842|D|A|Proof+Compile+Meta] Nonlinear limit-identification fence

§FORM := Weak convergence u_n -> u generally does not imply F(u_n)->F(u) for a nonlinear F; monotonicity, compactness, convexity, or another identification argument is needed.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000839, S26-D-00000829

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000843|D|A|Proof+Compile+Meta] Schauder fixed-point interface

§FORM := A continuous map from a nonempty closed convex set into a relatively compact subset of itself has a fixed point under the standard Banach-space formulation.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000354, S26-T-00000649, S26-D-00000315

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000844|D|A|Proof+Compile+Meta+Topo] Leray-Schauder continuation interface

§FORM := Compact fixed-point problems can be treated by continuation/degree arguments when the homotopy fixed points satisfy suitable a priori bounds.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000843, S26-D-00000770

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000845|D|A|Proof+Compile+Meta] Banach fixed-point nonlinear PDE interface

§FORM := If a nonlinear PDE reformulation is a contraction on a complete invariant set, Banach fixed point gives a unique local or global solution in that set.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000315, S26-D-00000829

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000846|D|A|Proof+Compile+Meta] Galerkin approximation interface

§FORM := Galerkin methods approximate an infinite-dimensional variational or evolution problem in finite-dimensional subspaces and require stability plus compactness/consistency to pass to a limit.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000781, S26-D-00000840, S26-D-00000214

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000847|D|A|Proof+Compile+Meta] A priori bounds before compactness

§FORM := In nonlinear existence proofs, uniform estimates are established before extracting subsequences; compactness cannot substitute for missing bounds.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000770, S26-D-00000839, S26-D-00000840

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000848|D|A|Proof+Compile+Meta] Nonlinear uniqueness fence

§FORM := Existence by compactness or Schauder fixed point does not imply uniqueness; uniqueness requires an independent mechanism such as contraction or strict monotonicity.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000843, S26-D-00000845, S26-D-00000831

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000849|T|A|Proof+Compile+Meta] Variational and nonlinear-PDE epistemic fence

§FORM := Discrete minimization, symbolic Euler-Lagrange residuals, and finite Galerkin experiments can test constructions but do not prove general lower semicontinuity, compactness, existence, uniqueness, or regularity theorems.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-D-00000821, S26-D-00000842, S26-D-00000846

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000850|T|A|Proof+Compile+Meta+Topo+Spectral] Calculus-of-variations and nonlinear-PDE frontier

§FORM := First variation, Euler-Lagrange structure, direct methods, variational inequalities, monotone operators, compactness, and fixed-point interfaces form a controlled substrate for nonlinear PDE, geometric variational problems, and numerical approximation.

§ASSUME := Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

§DEP_HARD := S26-T-00000811, S26-T-00000822, S26-D-00000828, S26-D-00000834, S26-D-00000841, S26-D-00000843, S26-D-00000846

§EVIDENCE := A · Exact registered statement only; bounded checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000851|D|A|Proof+Compile+Meta] Numerical approximation

§FORM := Numerical analysis studies computable approximations together with explicit error, stability, conditioning, and complexity models.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-T-00000850

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000852|D|A|Proof+Compile+Meta] Absolute error

§FORM := Absolute error is the norm or magnitude of the difference between an approximation and the declared exact target.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-T-00000604, S26-D-00000851
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000853|D|A|Proof+Compile+Meta] Relative error

§FORM := Relative error scales absolute error by a nonzero reference magnitude; it is undefined or unsuitable when that reference vanishes or is negligibly small.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000852
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000854|D|A|Proof+Compile+Meta+IG] Forward error

§FORM := Forward error measures error in the computed output relative to the exact output for the original input.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000852
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000855|D|A|Proof+Compile+Meta] Backward error

§FORM := Backward error asks for the smallest perturbation of the input for which the computed result is exact.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000854
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000856|D|A|Proof+Compile+Meta] Conditioning

§FORM := Conditioning describes sensitivity of the mathematical problem itself to perturbations of its data, independently of a particular algorithm.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-T-00000319, S26-D-00000855
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000857|D|A|Proof+Compile+Meta] Absolute condition number

§FORM := A local absolute condition number bounds first-order output perturbation per unit input perturbation in declared norms.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000856, S26-T-00000329
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000858|D|A|Proof+Compile+Meta] Relative condition number

§FORM := A relative condition number compares relative output sensitivity with relative input sensitivity where the required scales are nonzero.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000853, S26-D-00000857
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000859|T|A|Proof+Compile+Meta] Conditioning-algorithm fence

§FORM := An ill-conditioned problem can defeat a stable algorithm, while an unstable algorithm can fail on a well-conditioned problem; conditioning and stability are distinct.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000856
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000860|D|A|Proof+Compile+Meta] Floating-point model

§FORM := A standard floating-point model represents an elementary rounded operation as fl(x op y)=(x op y)(1+delta), with |delta| bounded by unit roundoff when model hypotheses hold.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000851
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000861|D|A|Proof+Compile+Meta] Machine epsilon and unit roundoff

§FORM := Machine epsilon and unit roundoff quantify floating-point spacing/rounding under a specified format and rounding convention; conventions must not be silently conflated.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000860
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000862|D|A|Proof+Compile+Meta] Roundoff error

§FORM := Roundoff error is error introduced by finite-precision representation and arithmetic, distinct from discretization and modeling error.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000860, S26-D-00000852
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000863|D|A|Proof+Compile+Meta] Catastrophic cancellation

§FORM := Subtraction of nearby floating-point quantities can destroy relative accuracy when significant leading digits cancel.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000862
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000864|D|A|Proof+Compile+Meta] Backward stable algorithm

§FORM := An algorithm is backward stable when its computed answer is the exact answer to a nearby problem under a declared perturbation metric.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000855, S26-D-00000862
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000865|T|A|Proof+Compile+Meta] Forward-error conditioning interface

§FORM := Backward stability plus favorable conditioning can yield a small forward error, with the precise bound depending on the problem and perturbation model.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000856, S26-D-00000864
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000866|D|A|Proof+Compile+Meta+Spectral] Consistency of a discretization

§FORM := A discretization is consistent when its discrete operator approximates the continuous operator with truncation defect tending to zero in the specified limit.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000851, S26-D-00000775
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000867|D|A|Proof+Compile+Meta+Spectral] Stability of a discretization

§FORM := Stability controls amplification of perturbations or data by the discrete solution operator in specified norms.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000866, S26-T-00000859
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000868|D|A|Proof+Compile+Meta] Convergence of a discretization

§FORM := A discretization converges when discrete solutions approach the exact continuous solution in the declared norm as resolution is refined.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000866, S26-D-00000867
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000869|T|A|Proof+Compile+Meta] Lax equivalence interface

§FORM := For a properly posed linear initial-value problem and a consistent finite-difference approximation, stability is equivalent to convergence under the theorem-specific framework.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000766, S26-D-00000866, S26-D-00000867, S26-D-00000868
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000870|D|A|Proof+Compile+Meta] Order of convergence

§FORM := A method has order p when its error admits the stated asymptotic bound C h^p, with norm, regime, regularity, and constants declared.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000868
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000871|D|A|Proof+Compile+Meta] Interpolation problem

§FORM := Interpolation constructs an approximant matching prescribed data values or functionals at selected nodes.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-T-00000147, S26-D-00000851
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000872|D|A|Proof+Compile+Meta] Polynomial interpolation

§FORM := At n+1 distinct scalar nodes there is a unique polynomial of degree at most n matching n+1 prescribed scalar values.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000146, S26-D-00000871
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000873|D|A|Proof+Compile+Meta] Lagrange interpolation formula

§FORM := The Lagrange basis gives an explicit representation of the unique polynomial interpolant at distinct nodes.
§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.
§DEP_HARD := S26-D-00000872
§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000874|D|A|Proof+Compile+Meta] Interpolation remainder

§FORM := For sufficiently smooth functions, polynomial interpolation error has a derivative-based remainder involving the nodal polynomial under theorem-specific hypotheses.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000872, S26-D-00000338

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000875|D|A|Proof+Compile+Meta+Spectral] Lebesgue constant

§FORM := The Lebesgue constant is the operator norm controlling worst-case amplification from sampled data to the interpolation polynomial in the chosen sup norm.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000619, S26-D-00000872

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000876|D|A|Proof+Compile+Meta] Runge phenomenon

§FORM := High-degree interpolation at poorly distributed nodes can diverge or oscillate strongly even for smooth functions; increasing degree alone is not a convergence guarantee.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000872, S26-D-00000875

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000877|D|A|Proof+Compile+Meta] Chebyshev-node interface

§FORM := Chebyshev-type nodes reduce interpolation instability relative to equally spaced nodes and give near-optimal logarithmic Lebesgue-constant growth in the classical setting.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000875

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000878|D|A|Proof+Compile+Meta] Numerical differentiation

§FORM := Numerical differentiation approximates derivatives from discrete data and is typically sensitive to noise and roundoff.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-T-00000329, S26-D-00000862

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000879|D|A|Proof+Compile+Meta+Spectral] Finite-difference stencil

§FORM := A finite-difference stencil is a weighted combination of nearby sampled values designed to approximate a derivative or differential operator.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000878, S26-D-00000775

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000880|D|A|Proof+Compile+Meta] Local truncation error

§FORM := Local truncation error is the residual produced by inserting the exact sufficiently smooth solution into the discrete formula.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000866, S26-D-00000879

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000881|D|A|Proof+Compile+Meta] Numerical quadrature

§FORM := Quadrature approximates a definite integral by a weighted sum of function values or other accessible data.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-T-00000395, S26-D-00000851

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000882|D|A|Proof+Compile+Meta] Newton-Cotes interface

§FORM := Newton-Cotes rules integrate polynomial interpolants on equally spaced nodes; their accuracy and stability depend on degree and rule choice.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000872, S26-D-00000881

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000883|D|A|Proof+Compile+Meta] Gaussian quadrature

§FORM := An n-node Gaussian quadrature rule associated with a positive weight is exact for polynomials through degree 2n-1 under the standard orthogonality hypotheses.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000881, S26-D-00000224

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000884|D|A|Proof+Compile+Meta] Adaptive quadrature

§FORM := Adaptive quadrature refines sampling according to an error indicator or estimator and requires a stopping tolerance tied to a declared error model.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000881, S26-D-00000852

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000885|D|A|Proof+Compile+Meta+Tower] Initial-value ODE discretization

§FORM := A time-stepping method approximates an ODE trajectory by recursively advancing discrete states from initial data.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-T-00000331, S26-D-00000851

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000886|D|A|Proof+Compile+Meta] Explicit Euler method

§FORM := Explicit Euler advances y_{n+1}=y_n+h f(t_n,y_n) and is first-order under standard smoothness and stability assumptions.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000885, S26-D-00000870

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000887|D|A|Proof+Compile+Meta] Implicit Euler method

§FORM := Implicit Euler advances y_{n+1}=y_n+h f(t_{n+1},y_{n+1}) and requires solving an implicit equation at each step.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000885, S26-D-00000829

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000888|D|A|Proof+Compile+Meta] Runge-Kutta method

§FORM := A Runge-Kutta method combines staged evaluations of the vector field to construct a one-step update with order determined by its coefficients.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000885

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000889|D|A|Proof+Compile+Meta] Absolute stability region

§FORM := For a time-stepping method, the absolute stability region is the set of scaled test-equation parameters for which the numerical amplification factor does not grow.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000867, S26-D-00000885

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000890|D|A|Proof+Compile+Meta] A-stability

§FORM := A method is A-stable when its absolute stability region contains the entire open left half-plane.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000889

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000891|D|A|Proof+Compile+Meta] Stiffness interface

§FORM := Stiff problems contain separated dynamical scales that can impose severe stability restrictions on explicit methods; stiffness is problem-and-method dependent.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000889, S26-D-00000890

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000892|D|A|Proof+Compile+Meta+Spectral] Finite-difference PDE method

§FORM := A finite-difference PDE method replaces differential operators on a grid by discrete stencils and requires consistency, stability, boundary treatment, and convergence analysis.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000879, S26-D-00000866, S26-D-00000867, S26-D-00000775

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000893|D|A|Proof+Compile+Meta] Mesh and mesh size

§FORM := A computational mesh partitions or samples a domain; mesh size and shape-regularity parameters must be declared when error constants depend on them.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000761, S26-D-00000851

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000894|D|A|Proof+Compile+Meta] Finite element space

§FORM := A finite element space is a finite-dimensional trial/test space assembled from local polynomial-like functions over a mesh with specified continuity and boundary constraints.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000781, S26-D-00000893, S26-D-00000214

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000895|D|A|Proof+Compile+Meta] Galerkin orthogonality

§FORM := For a conforming linear Galerkin approximation, the error is orthogonal to the discrete test space with respect to the governing bilinear form.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000781, S26-D-00000846, S26-D-00000894

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000896|T|A|Proof+Compile+Meta] Cea lemma interface

§FORM := For a continuous coercive bilinear variational problem, conforming Galerkin error is bounded by a stability constant times the best approximation error in the energy-space norm.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-T-00000782, S26-D-00000895

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000897|D|A|Proof+Compile+Meta] A priori error estimate

§FORM := An a priori estimate bounds discretization error using known regularity, approximation properties, mesh parameters, and stability constants before computing the solution.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000870, S26-T-00000896

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000898|D|A|Proof+Compile+Meta+IG] A posteriori error estimator

§FORM := An a posteriori estimator uses the computed solution and residual/jump information to estimate or bound error after computation, with reliability and efficiency requiring separate proof.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000897, S26-D-00000884

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000899|T|A|Proof+Compile+Meta] Numerical epistemic error decomposition

§FORM := Modeling error, discretization error, algebraic-solver error, roundoff error, and implementation error are distinct objects and must not be collapsed into a single unexplained numerical discrepancy.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000862, S26-D-00000868, S26-D-00000897, S26-D-00000898

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000900|T|A|Proof+Compile+Meta] Numerical-analysis frontier

§FORM := Conditioning, floating-point stability, approximation, interpolation, quadrature, time stepping, finite differences, and finite elements form a controlled computational layer connecting exact mathematics to reproducible numerical evidence.

§ASSUME := Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

§DEP_HARD := S26-D-00000856, S26-D-00000864, S26-D-00000868, S26-D-00000872, S26-D-00000881, S26-D-00000885, S26-D-00000892, S26-T-00000896, S26-T-00000899

§EVIDENCE := A · Exact registered statement only; computational checks do not promote the general theorem to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000901|D|A|Proof+Compile+Meta+Topo] Topological manifold

§FORM := An n-dimensional topological manifold is a Hausdorff, second-countable space locally homeomorphic to open subsets of R^n.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000309, S26-D-00000312

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000902|D|A|Proof+Compile+Meta+Topo] Coordinate chart

§FORM := A chart is a homeomorphism from an open subset of a manifold onto an open subset of R^n.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000901

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000903|D|A|Proof+Compile+Meta+Topo] Atlas

§FORM := An atlas is a collection of compatible charts covering the manifold.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000902

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000904|D|A|Proof+Compile+Meta+Topo] Smooth compatibility

§FORM := Two charts are smoothly compatible when their transition maps are smooth wherever defined.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000903, S26-T-00000329

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000905|D|A|Proof+Compile+Meta+Topo] Smooth manifold

§FORM := A smooth manifold is a topological manifold equipped with a maximal smooth atlas.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000901, S26-D-00000904

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000906|D|A|Proof+Compile+Meta+Topo] Smooth map of manifolds

§FORM := A map between smooth manifolds is smooth when its coordinate representatives are smooth; this property is chart-independent.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000905, S26-T-00000329
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000907|D|A|Proof+Compile+Meta+Topo] Diffeomorphism

§FORM := A diffeomorphism is a bijective smooth map with smooth inverse.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000906, S26-D-00000048
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000908|D|A|Proof+Compile+Meta+Topo] Smooth curve

§FORM := A smooth curve in M is a smooth map from an interval into M.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000906
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000909|D|A|Proof+Compile+Meta+Topo] Tangent vector by derivations

§FORM := A tangent vector at p may be defined as a derivation on germs of smooth functions at p satisfying the Leibniz rule.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000905, S26-D-00000128
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000910|D|A|Proof+Compile+Meta+Topo] Tangent space

§FORM := The tangent space T_pM is the vector space of tangent vectors at p.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000909, S26-D-00000149
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000911|D|A|Proof+Compile+Meta+Topo] Velocity of a curve

§FORM := A smooth curve through p determines a tangent vector at p by differentiating smooth functions along the curve.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000908, S26-D-00000909
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000912|D|A|Proof+Compile+Meta+Topo] Differential of a smooth map

§FORM := For smooth F:M->N, the differential dF_p:T_pM->T_{F(p)}N is the induced linear map on tangent vectors.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000906, S26-D-00000910, S26-D-00000169
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000913|T|A|Proof+Compile+Meta+Topo] Manifold chain rule

§FORM := Differentials compose: d(G o F)_p=dG_{F(p)} o dF_p.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000912, S26-T-00000334
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000914|D|A|Proof+Compile+Meta+Topo] Tangent bundle

§FORM := TM is the disjoint union of tangent spaces with its canonical smooth-manifold structure.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000910, S26-D-00000905
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000915|D|A|Proof+Compile+Meta+Topo] Vector field

§FORM := A smooth vector field is a smooth section X:M->TM of the tangent bundle.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000914
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000916|D|A|Proof+Compile+Meta+Topo] Integral curve

§FORM := An integral curve of X satisfies gamma prime(t)=X(gamma(t)) in the tangent bundle.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000915, S26-T-00000331
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000917|D|A|Proof+Compile+Meta+Topo] Flow

§FORM := A local flow is the local one-parameter family of diffeomorphisms generated by a vector field where the ODE exists.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000916, S26-D-00000907
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000918|D|A|Proof+Compile+Meta+Topo] Lie bracket

§FORM := The Lie bracket [X,Y] is the vector field acting on smooth functions by X(Yf)-Y(Xf).
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000915
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000919|T|A|Proof+Compile+Meta+Topo] Lie bracket properties

§FORM := The vector-field bracket is bilinear, antisymmetric, and satisfies the Jacobi identity.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000918
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000920|D|A|Proof+Compile+Meta+Topo] Cotangent space

§FORM := The cotangent space T_p^*M is the algebraic dual of T_pM.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000910, S26-D-00000251
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000921|D|A|Proof+Compile+Meta+Topo] Cotangent bundle

§FORM := T^*M is the smooth vector bundle formed from the cotangent spaces.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000920, S26-D-00000905
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000922|D|A|Proof+Compile+Meta+Topo] Differential of a function

§FORM := For smooth f:M->R, df is the 1-form defined by df_p(v)=v(f).
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000912, S26-D-00000920
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000923|D|A|Proof+Compile+Meta+Topo] Differential k-form

§FORM := A differential k-form is a smooth section of the bundle of alternating k-covectors Lambda^k T^*M.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000921, S26-T-00000276
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000924|D|A|Proof+Compile+Meta+Topo] Wedge product of forms

§FORM := Differential forms carry a graded wedge product induced pointwise by exterior algebra.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000923, S26-D-00000277
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000925|D|A|Proof+Compile+Meta+Topo] Exterior derivative

§FORM := The exterior derivative d maps k-forms to (k+1)-forms, is local and graded-Leibniz, agrees with df on functions, and satisfies d^2=0.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000923, S26-D-00000924
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000926|D|A|Proof+Compile+Meta+Topo] Pullback of differential forms

§FORM := A smooth map F:M->N pulls differential forms on N back to forms on M via its differential.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000912, S26-D-00000923, S26-T-00000289
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000927|T|A|Proof+Compile+Meta+Topo] Naturality of exterior derivative

§FORM := Pullback commutes with exterior differentiation: F^*(d omega)=d(F^* omega).
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000925, S26-D-00000926
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000928|D|A|Proof+Compile+Meta+Topo] Closed form

§FORM := A differential form omega is closed when d omega=0.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000925
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000929|D|A|Proof+Compile+Meta+Topo] Exact form

§FORM := A differential form is exact when omega=d eta for some form eta.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000925
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000930|T|A|Proof+Compile+Meta+Topo] Exact implies closed

§FORM := Every exact differential form is closed because d^2=0.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000928, S26-D-00000929
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000931|D|A|Proof+Compile+Meta+Topo] de Rham cohomology

§FORM := H^k_dR(M) is the quotient of closed k-forms by exact k-forms.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000928, S26-D-00000929, S26-D-00000113
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000932|D|A|Proof+Compile+Meta+Topo] Orientation of a manifold

§FORM := An orientation is a consistent choice of orientation on tangent spaces, equivalently an oriented-atlas or positive top-form structure under standard hypotheses.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-T-00000290, S26-D-00000910
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000933|D|A|Proof+Compile+Meta+Topo] Integration of top forms

§FORM := Compactly supported top-degree forms on an oriented manifold admit a coordinate-independent integral assembled using oriented charts and partitions of unity.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000932, S26-D-00000923, S26-T-00000445
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000934|D|A|Proof+Compile+Meta+Topo] Manifold with boundary

§FORM := A manifold with boundary is locally modeled on the closed half-space, with boundary points characterized by half-space charts.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000901
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000935|T|A|Proof+Compile+Meta+Topo] General Stokes theorem

§FORM := For a compact oriented smooth manifold with boundary and a smooth (n-1)-form omega, integral_M d omega = integral_boundaryM omega with induced boundary orientation.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000925, S26-D-00000933, S26-D-00000934
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000936|D|A|Proof+Compile+Meta+Topo] Riemannian metric

§FORM := A Riemannian metric is a smooth assignment of positive-definite inner products g_p on tangent spaces.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000910, S26-D-00000201
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000937|D|A|Proof+Compile+Meta+Topo] Riemannian length

§FORM := The length of a piecewise smooth curve is the integral of the norm of its velocity induced by the Riemannian metric.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000936, S26-D-00000908, S26-T-00000395
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000938|D|A|Proof+Compile+Meta+Topo] Riemannian distance

§FORM := The Riemannian distance is the infimum of lengths of piecewise smooth curves joining two points in the same connected component.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000937, S26-D-00000304
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000939|D|A|Proof+Compile+Meta+Topo] Musical isomorphisms

§FORM := A Riemannian metric identifies tangent and cotangent vectors via flat and sharp bundle maps.
§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.
§DEP_HARD := S26-D-00000936, S26-D-00000920, S26-D-00000208
§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000940|D|A|Proof+Compile+Meta+Topo] Gradient on a Riemannian manifold

§FORM := The gradient grad f is defined by g(grad f,X)=df(X) for every vector field X.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000922, S26-D-00000936, S26-D-00000939

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000941|D|A|Proof+Compile+Meta+Topo] Affine connection

§FORM := A connection nabla differentiates vector fields along vector fields, linearly in the direction and satisfying a Leibniz rule in the differentiated field.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000915

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000942|D|A|Proof+Compile+Meta+Topo] Covariant derivative along a curve

§FORM := A connection induces differentiation of vector fields along smooth curves.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000941, S26-D-00000908

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000943|D|A|Proof+Compile+Meta+Topo] Torsion

§FORM := The torsion of a connection is T(X,Y)=nabla_XY-nabla_YX-[X,Y].

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000941, S26-D-00000918

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000944|T|A|Proof+Compile+Meta+Topo] Levi-Civita theorem

§FORM := Every Riemannian manifold has a unique torsion-free connection compatible with the metric.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000936, S26-D-00000941, S26-D-00000943

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000945|D|A|Proof+Compile+Meta+Topo] Geodesic

§FORM := A geodesic is a curve whose velocity is parallel along itself: nabla_{gamma prime} gamma prime=0.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000942, S26-T-00000944

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000946|D|A|Proof+Compile+Meta+Topo] Exponential map

§FORM := The exponential map exp_p sends an initial tangent vector to the endpoint at unit time of its geodesic when defined.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000945, S26-D-00000910

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000947|D|A|Proof+Compile+Meta+Topo+Spectral] Riemann curvature tensor

§FORM := The curvature operator is R(X,Y)Z=nabla_Xnabla_YZ-nabla_Ynabla_XZ-nabla_[X,Y]Z.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000941, S26-D-00000918

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000948|D|A|Proof+Compile+Meta+Topo] Sectional curvature

§FORM := Sectional curvature assigns to each tangent 2-plane a scalar curvature determined by the Riemann tensor and metric.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000947, S26-D-00000936

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000949|D|A|Proof+Compile+Meta+Topo] Ricci and scalar curvature

§FORM := Ricci curvature is the metric trace of the Riemann tensor, and scalar curvature is the trace of Ricci curvature.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000947, S26-D-00000936, S26-D-00000218

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000950|T|A|Proof+Compile+Meta+Topo] Differential-geometry frontier

§FORM := Smooth manifolds, tangent/cotangent bundles, forms, Stokes theory, Riemannian metrics, connections, geodesics, and curvature form the coordinate-invariant geometric layer joining topology, calculus, exterior algebra, analysis, and mathematical physics.

§ASSUME := Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

§DEP_HARD := S26-D-00000905, S26-D-00000910, S26-D-00000923, S26-T-00000935, S26-D-00000936, S26-T-00000944, S26-D-00000945, S26-D-00000947, S26-D-00000949

§EVIDENCE := A · Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

§BOUNDARY := A coordinate formula is not itself an invariant definition; chart-dependent expressions must transform correctly.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000951|D|A|Proof+Compile+Meta+Topo] Lie group

§FORM := A Lie group is a smooth manifold G equipped with a group structure whose multiplication GxG->G and inversion G->G are smooth.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000905, S26-D-00000102

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000952|D|A|Proof+Compile+Meta+Topo] Lie-group homomorphism

§FORM := A Lie-group homomorphism is a group homomorphism that is also smooth.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000951, S26-D-00000906

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000953|D|A|Proof+Compile+Meta+Topo] Left and right translations

§FORM := For g in a Lie group G, L_g(h)=gh and R_g(h)=hg are diffeomorphisms of G.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000951, S26-D-00000907

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000954|D|A|Proof+Compile+Meta+Topo] Lie algebra of a Lie group

§FORM := The Lie algebra g of a Lie group G is T_eG equipped with the bracket induced by left-invariant vector fields.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000951, S26-D-00000910, S26-D-00000918

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000955|D|A|Proof+Compile+Meta+Topo] Left-invariant vector field

§FORM := A vector field X on G is left-invariant when (L_g)_*X_h=X_{gh} for all g,h.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000953, S26-D-00000915, S26-D-00000912

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000956|T|A|Proof+Compile+Meta+Topo] Left-invariant fields identify with T_eG

§FORM := Evaluation at the identity gives a vector-space isomorphism between left-invariant vector fields on G and T_eG.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000954, S26-D-00000955, S26-T-00000175

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000957|T|A|Proof+Compile+Meta+Topo] Lie-algebra bracket properties

§FORM := The bracket on a Lie algebra is bilinear, alternating (hence antisymmetric in characteristic not 2), and satisfies the Jacobi identity.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000954, S26-T-00000919

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§BOUNDARY := Alternating and antisymmetric formulations require characteristic care in abstract algebraic generalizations.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000958|D|A|Proof+Compile+Meta+Topo] One-parameter subgroup

§FORM := A one-parameter subgroup is a smooth homomorphism gamma:(R,+)->G.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000952, S26-D-00000054

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000959|D|A|Proof+Compile+Meta+Topo] Lie-group exponential map

§FORM := For X in g, exp(X) is the value at time 1 of the one-parameter subgroup whose tangent at 0 is X.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000954, S26-D-00000958, S26-D-00000917

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000960|T|A|Proof+Compile+Meta+Topo] Exponential one-parameter law

§FORM := For fixed X in g, exp((s+t)X)=exp(sX)exp(tX) whenever the exponential is defined by the global one-parameter subgroup.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000959, S26-D-00000958

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000961|D|A|Proof+Compile+Meta+Topo] Adjoint representation

§FORM := Conjugation C_g(h)=ghg^{-1} differentiates at the identity to a linear automorphism Ad_g:g->g.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000951, S26-D-00000912, S26-D-00000954

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000962|D|A|Proof+Compile+Meta+Topo+Tower] Infinitesimal adjoint map

§FORM := For X in g, ad_X is the linear map Y |-> [X,Y].

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000954, S26-T-00000957

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000963|T|A|Proof+Compile+Meta+Topo] Adjoint intertwines the bracket

§FORM := Ad_g([X,Y])=[Ad_gX,Ad_gY].

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000961, S26-T-00000957

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000964|T|A|Proof+Compile+Meta+Topo] Baker-Campbell-Hausdorff local interface

§FORM := Near the identity, log(exp X exp Y) is expressed by the BCH Lie series X+Y+1/2[X,Y]+higher nested commutators under its applicable convergence/formal hypotheses.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000959, S26-T-00000957

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§BOUNDARY := The BCH series is a local/formal statement; convergence and global logarithms require additional hypotheses.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000965|D|A|Proof+Compile+Meta+Topo] Smooth Lie-group action

§FORM := A smooth left action of G on M is a smooth map GxM->M satisfying e.p=p and g.(h.p)=(gh).p.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000951, S26-D-00000905

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000966|D|A|Proof+Compile+Meta+Topo] Orbit

§FORM := The orbit G.p is the set of points reachable from p under the group action.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000965, S26-D-00000017

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000967|D|A|Proof+Compile+Meta+Topo] Stabilizer subgroup

§FORM := The stabilizer G_p consists of group elements fixing p.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000965, S26-D-00000101

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000968|D|A|Proof+Compile+Meta+Topo] Fundamental vector field

§FORM := An element X of g induces a vector field X_M by differentiating t |-> exp(tX).p at t=0.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000959, S26-D-00000965, S26-D-00000915

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000969|D|A|Proof+Compile+Meta+Topo] Equivariant map

§FORM := For G-spaces M and N, a map F:M->N is equivariant when F(g.p)=g.F(p).

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000965, S26-D-00000906

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000970|D|A|Proof+Compile+Meta+Topo] Free action

§FORM := An action is free when every stabilizer is trivial.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000967

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000971|D|A|Proof+Compile+Meta+Topo] Proper action

§FORM := A smooth action is proper when the map GxM->MxM, (g,p)|->(g.p,p), is proper.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000965, S26-T-00000322

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000972|T|A|Proof+Compile+Meta+Topo] Quotient manifold interface

§FORM := A free proper smooth action has a smooth quotient M/G for which the quotient map is a submersion and M->M/G is a principal G-bundle.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000970, S26-D-00000971, S26-D-00000905

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§BOUNDARY := Without freeness or properness the orbit space can be singular or non-Hausdorff.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000973|D|A|Proof+Compile+Meta+Topo] Homogeneous space

§FORM := A homogeneous G-space is a smooth G-space on which G acts transitively.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000965

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000974|T|A|Proof+Compile+Meta+Topo] Orbit-stabilizer geometry interface

§FORM := Under standard embeddedness/closedness hypotheses, an orbit is modeled by a homogeneous space G/G_p and its tangent space is generated by fundamental vector fields.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000966, S26-D-00000967, S26-D-00000968, S26-D-00000973

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§BOUNDARY := Topological or smooth quotient conclusions require closedness/properness hypotheses.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000975|T|A|Proof+Compile+Meta+Topo+Tower] Lie-symmetry frontier

§FORM := Lie groups and Lie algebras encode continuous symmetries through global transformations, infinitesimal generators, actions, and homogeneous spaces.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000951, S26-D-00000954, S26-D-00000959, S26-D-00000965, S26-D-00000968

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000976|D|A|Proof+Compile+Meta+Topo] Symplectic vector space

§FORM := A symplectic vector space is an even-dimensional real vector space V with a nondegenerate alternating bilinear form omega.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000149, S26-T-00000254, S26-T-00000264

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000977|D|A|Proof+Compile+Meta+Topo] Symplectic orthogonal complement

§FORM := For W subseteq V, W^omega consists of vectors v such that omega(v,w)=0 for every w in W.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000976, S26-T-00000155

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000978|D|A|Proof+Compile+Meta+Topo] Lagrangian subspace

§FORM := A subspace L of a 2n-dimensional symplectic vector space is Lagrangian when L=L^omega, equivalently when omega|_L=0 and dim L=n.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000976, S26-D-00000977, S26-T-00000165

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000979|D|A|Proof+Compile+Meta+Topo] Symplectic manifold

§FORM := A symplectic manifold is a smooth even-dimensional manifold M with a closed nondegenerate differential 2-form omega.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000905, S26-D-00000923, S26-D-00000928

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§BOUNDARY := Nondegeneracy without d omega=0 does not define a symplectic form.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000980|D|A|Proof+Compile+Meta+Topo] Symplectomorphism

§FORM := A diffeomorphism F:(M,omega_M)->(N,omega_N) is symplectic when F^*omega_N=omega_M.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000979, S26-D-00000907, S26-D-00000926

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000981|D|A|Proof+Compile+Meta+Topo] Canonical one-form on a cotangent bundle

§FORM := The tautological one-form theta on T^*Q is defined by theta_alpha(v)=alpha(d pi(v)), where pi:T^*Q->Q.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000921, S26-D-00000912

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000982|D|A|Proof+Compile+Meta+Topo] Canonical symplectic form on T^*Q

§FORM := The cotangent bundle carries the canonical symplectic form omega=-d theta (up to the explicitly declared sign convention).

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000981, S26-D-00000925, S26-D-00000979

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§BOUNDARY := The sign convention omega=d theta versus omega=-d theta varies across sources and must be declared.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000983|T|A|Proof+Compile+Meta+Topo] Darboux theorem

§FORM := Every symplectic manifold is locally symplectomorphic to the standard symplectic vector space; symplectic forms have no local scalar curvature-type invariants.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000979, S26-D-00000980, S26-D-00000976

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000984|D|A|Proof+Compile+Meta+Spectral] Hamiltonian vector field

§FORM := For H in C^infty(M), the Hamiltonian vector field X_H is uniquely defined by i_{X_H}omega=dH under the chosen sign convention.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000979, S26-D-00000288, S26-D-00000922, S26-D-00000915

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000985|D|A|Proof+Compile+Meta+Spectral] Poisson bracket

§FORM := The Poisson bracket of f and g is defined from the symplectic form and Hamiltonian vector fields, with sign convention declared consistently, e.g. {f,g}=omega(X_f,X_g).

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000984, S26-D-00000979

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000986|T|A|Proof+Compile+Meta+Spectral] Poisson algebra identities

§FORM := The Poisson bracket is bilinear, antisymmetric, satisfies the Jacobi identity, and is a derivation in each argument.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000985, S26-T-00000957

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000987|T|A|Proof+Compile+Meta+Spectral] Hamiltonian conservation of H

§FORM := Along the Hamiltonian flow of H, the Hamiltonian H is constant because dH(X_H)=omega(X_H,X_H)=0.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000984, S26-D-00000976

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000988|T|A|Proof+Compile+Meta+Spectral] Hamiltonian flow is symplectic

§FORM := The local flow generated by X_H preserves omega: phi_t^*omega=omega.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000984, S26-D-00000917, S26-D-00000925, S26-D-00000980

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000989|D|A|Proof+Compile+Meta+Spectral] Hamilton equations in canonical coordinates

§FORM := In canonical coordinates (q,p) with the standard symplectic convention, Hamiltonian trajectories satisfy qdot=partial H/partial p and pdot=-partial H/partial q.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000982, S26-D-00000984, S26-T-00000331

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000990|D|A|Proof+Compile+Meta+Spectral] Symplectic gradient matrix

§FORM := In R^{2n} with standard matrix J, Hamiltonian dynamics may be written zdot=J grad H(z), with J^T=-J and J^2=-I under the declared coordinate convention.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000976, S26-T-00000213, S26-D-00000343

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000991|D|A|Proof+Compile+Meta+Topo] Liouville volume

§FORM := On a 2n-dimensional symplectic manifold, omega^n/n! is a nowhere-vanishing volume form.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000979, S26-D-00000924, S26-T-00000287

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000992|T|A|Proof+Compile+Meta+Spectral+Swirl] Liouville theorem for Hamiltonian flow

§FORM := Hamiltonian flow preserves the symplectic volume, and hence phase-space volume, wherever the flow is defined.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-T-00000988, S26-D-00000991

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000993|D|A|Proof+Compile+Meta+Spectral] Momentum map

§FORM := For a symplectic G-action, a momentum map J:M->g^* packages Hamiltonians for fundamental vector fields when the action is Hamiltonian, subject to the chosen sign and equivariance conventions.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000965, S26-D-00000968, S26-D-00000979, S26-D-00000251, S26-D-00000984

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§BOUNDARY := Not every symplectic group action admits a globally defined equivariant momentum map.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000994|T|A|Proof+Compile+Meta+Spectral] Hamiltonian Noether interface

§FORM := For a Hamiltonian symmetry generated by a momentum-map component that Poisson-commutes with H, that component is conserved along the H-flow.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000993, S26-D-00000985, S26-T-00000987

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000995|D|A|Proof+Compile+Meta+Topo] Lagrangian submanifold

§FORM := A submanifold L of a symplectic 2n-manifold is Lagrangian when each tangent space T_pL is a Lagrangian subspace of T_pM.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000978, S26-D-00000979, S26-D-00000910

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000996|D|A|Proof+Compile+Meta+Topo] Exact Lagrangian interface

§FORM := In an exact symplectic manifold omega=-d theta, a Lagrangian L is exact when theta|_L is an exact 1-form.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000982, S26-D-00000995, S26-D-00000929

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00000997|D|A|Proof+Compile+Meta+Spectral+Swirl] Canonical transformation

§FORM := A canonical transformation in Hamiltonian mechanics is a symplectomorphism of phase space; coordinate formulas must preserve the symplectic form rather than merely be invertible.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000980, S26-D-00000989

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§BOUNDARY := An arbitrary invertible coordinate change need not be canonical/symplectic.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00000998|B|A|Proof+Compile+Meta+Spectral] Lagrangian-Hamiltonian bridge

§FORM := Under regularity/nondegeneracy hypotheses on the Lagrangian, the Legendre transform transports Euler-Lagrange dynamics on TQ to Hamiltonian dynamics on T^*Q.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-D-00000803, S26-D-00000982, S26-D-00000989, S26-D-00000180

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§BOUNDARY := Degenerate Lagrangians require constraint theory; the ordinary Legendre transform need not be invertible.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00000999|T|A|Proof+Compile+Meta+Topo+Spectral+Swirl] Symmetry-geometry-mechanics synthesis

§FORM := Lie actions, momentum maps, symplectic forms, Poisson brackets, and variational dynamics form a common structure linking continuous symmetry to conservation and phase-space evolution.

§ASSUME := Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

§DEP_HARD := S26-T-00000975, S26-D-00000979, S26-D-00000985, S26-D-00000993, S26-T-00000994, S26-B-00000998

§EVIDENCE := A · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00001000|N|S|Proof+Compile+Meta] Volume 0 closure invariant

§FORM := Volume 0 closes the foundational substrate at 1000 addressable objects. Later volumes must reuse these IDs, preserve evidence states and correction history, and introduce new primitives only when an explicit obstruction cannot be represented by the existing substrate.

§ASSUME := This is a project-governance invariant, not a mathematical theorem.

§DEP_HARD := S26-T-00000027, S26-T-00000029, S26-N-00000030, S26-T-00000950, S26-T-00000999

§EVIDENCE := S · Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§PACK Synopsis_2026_Volume_I_sLANG_S1

volume := I · Number, Algebra & Discrete Structure

canonical_range := 1001-2000

§CLAIM[S26-D-00001001|D|A|Proof+Compile+Meta] Arithmetic function

§FORM := An arithmetic function is a function f:N_{>0}->C (or another declared codomain) studied through the multiplicative structure of positive integers.

§DEP_HARD := S26-T-00000076

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001002|D|A|Proof+Compile+Meta] Multiplicative arithmetic function

§FORM := An arithmetic function f is multiplicative when f(1)=1 and f(mn)=f(m)f(n) whenever gcd(m,n)=1.

§DEP_HARD := S26-D-00001001

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001003|D|A|Proof+Compile+Meta] Completely multiplicative function

§FORM := An arithmetic function f is completely multiplicative when f(mn)=f(m)f(n) for all positive integers m,n.

§DEP_HARD := S26-D-00001002

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001004|D|A|Proof+Compile+Meta] Dirichlet convolution

§FORM := For arithmetic functions f and g, (f*g)(n)=sum_{d|n} f(d)g(n/d).

§DEP_HARD := S26-D-00001003

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001005|T|A|Proof+Compile+Meta] Dirichlet convolution is commutative and associative

§FORM := Dirichlet convolution on arithmetic functions is commutative and associative; the delta function epsilon with epsilon(1)=1 and epsilon(n)=0 for n>1 is its identity.

§DEP_HARD := S26-D-00001004

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001006|T|A|Proof+Compile+Meta] Multiplicative functions are closed under Dirichlet convolution

§FORM := If f and g are multiplicative arithmetic functions, then f*g is multiplicative.

§DEP_HARD := S26-T-00001005

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001007|D|A|Proof+Compile+Meta] Moebius function

§FORM := The Moebius function mu satisfies mu(1)=1; mu(n)=0 when a square of a prime divides n; otherwise mu(n)=(-1)^r when n is a product of r distinct primes.

§DEP_HARD := S26-T-00001006

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001008|T|A|Proof+Compile+Meta] Moebius inversion

§FORM := If F(n)=sum_{d|n} f(d), then f(n)=sum_{d|n} mu(d)F(n/d). Equivalently, mu is the Dirichlet-convolution inverse of the constant-one function.

§DEP_HARD := S26-D-00001007

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001009|D|A|Proof+Compile+Meta] Euler totient function

§FORM := phi(n) is the number of residue classes modulo n that are coprime to n.

§DEP_HARD := S26-T-00001008

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001010|I|A|Proof+Compile+Meta] Euler totient product formula

§FORM := For n=product p_i^{a_i}, phi(n)=n product_{p|n}(1-1/p).

§DEP_HARD := S26-D-00001009

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001011|D|A|Proof+Compile+Meta] Divisor-counting function

§FORM := tau(n)=sum_{d|n}1 counts the positive divisors of n.

§DEP_HARD := S26-I-00001010

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001012|I|A|Proof+Compile+Meta] Divisor-counting prime-power formula

§FORM := If n=product p_i^{a_i}, then tau(n)=product_i(a_i+1).

§DEP_HARD := S26-D-00001011

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001013|D|A|Proof+Compile+Meta] Sum-of-divisors function

§FORM := sigma_k(n)=sum_{d|n} d^k; sigma_1 is the ordinary sum-of-divisors function.
§DEP_HARD := S26-I-00001012
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001014|I|A|Proof+Compile+Meta] Sum-of-divisors prime-power formula

§FORM := For k not equal to 0 and n=product p_i^{a_i}, sigma_k(n)=product_i (p_i^{k(a_i+1)}-1)/(p_i^k-1).
§DEP_HARD := S26-D-00001013
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001015|D|A|Proof+Compile+Meta] Liouville function

§FORM := lambda(n)=(-1)^{Omega(n)}, where Omega counts prime factors with multiplicity.
§DEP_HARD := S26-I-00001014
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001016|D|A|Proof+Compile+Meta] von Mangoldt function

§FORM := Lambda(n)=log p if n=p^k for a prime p and integer k>=1, and Lambda(n)=0 otherwise.
§DEP_HARD := S26-D-00001015
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001017|I|A|Proof+Compile+Meta] Logarithm as divisor sum of von Mangoldt

§FORM := For n>=1, log n=sum_{d|n} Lambda(d).
§DEP_HARD := S26-D-00001016
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001018|D|A|Proof+Compile+Meta] Jordan totient function

§FORM := J_k(n)=n^k product_{p|n}(1-p^{-k}); it counts ordered k-tuples modulo n that together with n are coprime in the standard gcd sense.
§DEP_HARD := S26-I-00001017
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001019|D|A|Proof+Compile+Meta] Ramanujan sum

§FORM := c_q(n)=sum_{1<=a<=q, gcd(a,q)=1} exp(2 pi i a n/q).
§DEP_HARD := S26-D-00001018
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001020|I|A|Proof+Compile+Meta] Ramanujan sum divisor formula

§FORM := c_q(n)=sum_{d|gcd(q,n)} d mu(q/d).
§DEP_HARD := S26-D-00001019
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001021|E|M|Proof+Compile+Meta] Arithmetic profile of 1

§FORM := 1 has prime factorization {}; phi(1)=1, tau(1)=1, sigma_1(1)=1, and mu(1)=1.
§DEP_HARD := S26-I-00001020
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001022|E|M|Proof+Compile+Meta] Arithmetic profile of 2

§FORM := 2 has prime factorization {2: 1}; phi(2)=1, tau(2)=2, sigma_1(2)=3, and mu(2)=-1.
§DEP_HARD := S26-E-00001021
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001023|E|M|Proof+Compile+Meta] Arithmetic profile of 3

§FORM := 3 has prime factorization {3: 1}; phi(3)=2, tau(3)=2, sigma_1(3)=4, and mu(3)=-1.
§DEP_HARD := S26-E-00001022
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001024|E|M|Proof+Compile+Meta] Arithmetic profile of 4

§FORM := 4 has prime factorization {2: 2}; phi(4)=2, tau(4)=3, sigma_1(4)=7, and mu(4)=0.
§DEP_HARD := S26-E-00001023
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001025|E|M|Proof+Compile+Meta] Arithmetic profile of 5

§FORM := 5 has prime factorization {5: 1}; phi(5)=4, tau(5)=2, sigma_1(5)=6, and mu(5)=-1.
§DEP_HARD := S26-E-00001024
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001026|E|M|Proof+Compile+Meta] Arithmetic profile of 6

§FORM := 6 has prime factorization {2: 1, 3: 1}; phi(6)=2, tau(6)=4, sigma_1(6)=12, and mu(6)=1.
§DEP_HARD := S26-E-00001025
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001027|E|M|Proof+Compile+Meta] Arithmetic profile of 7

§FORM := 7 has prime factorization {7: 1}; phi(7)=6, tau(7)=2, sigma_1(7)=8, and mu(7)=-1.
§DEP_HARD := S26-E-00001026
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001028|E|M|Proof+Compile+Meta] Arithmetic profile of 8

§FORM := 8 has prime factorization {2: 3}; phi(8)=4, tau(8)=4, sigma_1(8)=15, and mu(8)=0.
§DEP_HARD := S26-E-00001027
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001029|E|M|Proof+Compile+Meta] Arithmetic profile of 9

§FORM := 9 has prime factorization {3: 2}; phi(9)=6, tau(9)=3, sigma_1(9)=13, and mu(9)=0.
§DEP_HARD := S26-E-00001028
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001030|E|M|Proof+Compile+Meta] Arithmetic profile of 10

§FORM := 10 has prime factorization {2: 1, 5: 1}; phi(10)=4, tau(10)=4, sigma_1(10)=18, and mu(10)=1.
§DEP_HARD := S26-E-00001029
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001031|E|M|Proof+Compile+Meta] Arithmetic profile of 11

§FORM := 11 has prime factorization {11: 1}; phi(11)=10, tau(11)=2, sigma_1(11)=12, and mu(11)=-1.
§DEP_HARD := S26-E-00001030
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001032|E|M|Proof+Compile+Meta] Arithmetic profile of 12

§FORM := 12 has prime factorization {2: 2, 3: 1}; phi(12)=4, tau(12)=6, sigma_1(12)=28, and mu(12)=0.
§DEP_HARD := S26-E-00001031
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001033|E|M|Proof+Compile+Meta] Arithmetic profile of 13

§FORM := 13 has prime factorization {13: 1}; phi(13)=12, tau(13)=2, sigma_1(13)=14, and mu(13)=-1.
§DEP_HARD := S26-E-00001032
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001034|E|M|Proof+Compile+Meta] Arithmetic profile of 14

§FORM := 14 has prime factorization {2: 1, 7: 1}; phi(14)=6, tau(14)=4, sigma_1(14)=24, and mu(14)=1.
§DEP_HARD := S26-E-00001033
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001035|E|M|Proof+Compile+Meta] Arithmetic profile of 15

§FORM := 15 has prime factorization {3: 1, 5: 1}; phi(15)=8, tau(15)=4, sigma_1(15)=24, and mu(15)=1.
§DEP_HARD := S26-E-00001034
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001036|E|M|Proof+Compile+Meta] Arithmetic profile of 16

§FORM := 16 has prime factorization {2: 4}; phi(16)=8, tau(16)=5, sigma_1(16)=31, and mu(16)=0.
§DEP_HARD := S26-E-00001035
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001037|E|M|Proof+Compile+Meta] Arithmetic profile of 17

§FORM := 17 has prime factorization {17: 1}; phi(17)=16, tau(17)=2, sigma_1(17)=18, and mu(17)=-1.
§DEP_HARD := S26-E-00001036
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001038|E|M|Proof+Compile+Meta] Arithmetic profile of 18

§FORM := 18 has prime factorization {2: 1, 3: 2}; phi(18)=6, tau(18)=6, sigma_1(18)=39, and mu(18)=0.
§DEP_HARD := S26-E-00001037
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001039|E|M|Proof+Compile+Meta] Arithmetic profile of 19

§FORM := 19 has prime factorization {19: 1}; phi(19)=18, tau(19)=2, sigma_1(19)=20, and mu(19)=-1.
§DEP_HARD := S26-E-00001038
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001040|E|M|Proof+Compile+Meta] Arithmetic profile of 20

§FORM := 20 has prime factorization {2: 2, 5: 1}; phi(20)=8, tau(20)=6, sigma_1(20)=42, and mu(20)=0.
§DEP_HARD := S26-E-00001039
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001041|E|M|Proof+Compile+Meta] Arithmetic profile of 21

§FORM := 21 has prime factorization {3: 1, 7: 1}; phi(21)=12, tau(21)=4, sigma_1(21)=32, and mu(21)=1.
§DEP_HARD := S26-E-00001040
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001042|E|M|Proof+Compile+Meta] Arithmetic profile of 22

§FORM := 22 has prime factorization {2: 1, 11: 1}; phi(22)=10, tau(22)=4, sigma_1(22)=36, and mu(22)=1.
§DEP_HARD := S26-E-00001041
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001043|E|M|Proof+Compile+Meta] Arithmetic profile of 23

§FORM := 23 has prime factorization {23: 1}; phi(23)=22, tau(23)=2, sigma_1(23)=24, and mu(23)=-1.
§DEP_HARD := S26-E-00001042
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001044|E|M|Proof+Compile+Meta] Arithmetic profile of 24

§FORM := 24 has prime factorization {2: 3, 3: 1}; phi(24)=8, tau(24)=8, sigma_1(24)=60, and mu(24)=0.
§DEP_HARD := S26-E-00001043
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001045|E|M|Proof+Compile+Meta] Arithmetic profile of 25

§FORM := 25 has prime factorization {5: 2}; phi(25)=20, tau(25)=3, sigma_1(25)=31, and mu(25)=0.
§DEP_HARD := S26-E-00001044
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001046|E|M|Proof+Compile+Meta] Arithmetic profile of 26

§FORM := 26 has prime factorization {2: 1, 13: 1}; phi(26)=12, tau(26)=4, sigma_1(26)=42, and mu(26)=1.
§DEP_HARD := S26-E-00001045
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001047|E|M|Proof+Compile+Meta] Arithmetic profile of 27

§FORM := 27 has prime factorization {3: 3}; phi(27)=18, tau(27)=4, sigma_1(27)=40, and mu(27)=0.
§DEP_HARD := S26-E-00001046
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001048|E|M|Proof+Compile+Meta] Arithmetic profile of 28

§FORM := 28 has prime factorization {2: 2, 7: 1}; phi(28)=12, tau(28)=6, sigma_1(28)=56, and mu(28)=0.
§DEP_HARD := S26-E-00001047
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001049|E|M|Proof+Compile+Meta] Arithmetic profile of 29

§FORM := 29 has prime factorization {29: 1}; phi(29)=28, tau(29)=2, sigma_1(29)=30, and mu(29)=-1.
§DEP_HARD := S26-E-00001048
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001050|E|M|Proof+Compile+Meta] Arithmetic profile of 30

§FORM := 30 has prime factorization {2: 1, 3: 1, 5: 1}; phi(30)=8, tau(30)=8, sigma_1(30)=72, and mu(30)=-1.
§DEP_HARD := S26-E-00001049
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-B-00001051|B|A|Proof+Compile+Meta] Congruence relation reuse in Volume I

§FORM := Volume I reuses the frozen congruence relation S26-D-00000079 and treats it as the ambient equivalence relation for the modular-arithmetic developments that follow.
§DEP_HARD := S26-D-00000079
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001052|T|A|Proof+Compile+Meta] Congruence respects ring operations

§FORM := If a congruent b and c congruent d modulo n, then a+c congruent b+d and ac congruent bd modulo n.
§DEP_HARD := S26-B-00001051
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001053|T|A|Proof+Compile+Meta] Linear congruence solvability

§FORM := The congruence ax congruent b mod n is solvable iff gcd(a,n) divides b; when solvable it has exactly gcd(a,n) residue-class solutions modulo n.
§DEP_HARD := S26-T-00001052
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001054|T|A|Proof+Compile+Meta] Chinese remainder theorem
§FORM := For pairwise coprime moduli n_1,...,n_r, the natural map Z/(product n_i) -> product Z/n_i is a ring isomorphism.
§DEP_HARD := S26-T-00001053
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001055|D|A|Proof+Compile+Meta] Multiplicative order modulo n
§FORM := For gcd(a,n)=1, ord_n(a) is the least positive k with a^k congruent 1 mod n.
§DEP_HARD := S26-T-00001054
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001056|T|A|Proof+Compile+Meta] Order divides group exponent
§FORM := ord_n(a) divides phi(n); more sharply it divides the exponent lambda_C(n) of (Z/nZ)^x.
§DEP_HARD := S26-D-00001055
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001057|D|A|Proof+Compile+Meta] Carmichael function
§FORM := The Carmichael function lambda_C(n) is the exponent of the finite group (Z/nZ)^x.
§DEP_HARD := S26-T-00001056
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001058|T|A|Proof+Compile+Meta] Euler theorem
§FORM := If gcd(a,n)=1, then a^{phi(n)} congruent 1 mod n.
§DEP_HARD := S26-D-00001057
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001059|T|A|Proof+Compile+Meta] Fermat little theorem
§FORM := For prime p, a^p congruent a mod p; if p does not divide a then a^{p-1} congruent 1 mod p.
§DEP_HARD := S26-T-00001058
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001060|D|A|Proof+Compile+Meta] Primitive root modulo n
§FORM := A primitive root modulo n is a generator of the cyclic group (Z/nZ)^x when that group is cyclic.
§DEP_HARD := S26-T-00001059
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001061|T|A|Proof+Compile+Meta] Primitive-root classification
§FORM := Primitive roots exist modulo n exactly for n in {1,2,4,p^k,2p^k} with p an odd prime.
§DEP_HARD := S26-D-00001060
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001062|D|A|Proof+Compile+Meta] Quadratic residue
§FORM := For odd prime p and p not dividing a, a is a quadratic residue mod p when x^2 congruent a mod p is solvable.
§DEP_HARD := S26-T-00001061
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001063|D|A|Proof+Compile+Meta] Legendre symbol
§FORM := For odd prime p, (a/p) is 0 if p|a, 1 if a is a nonzero quadratic residue mod p, and -1 otherwise.
§DEP_HARD := S26-D-00001062
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001064|T|A|Proof+Compile+Meta] Euler criterion
§FORM := For odd prime p, (a/p) congruent a^{(p-1)/2} mod p, interpreted as the values 0,+1,-1.
§DEP_HARD := S26-D-00001063
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001065|T|A|Proof+Compile+Meta] Quadratic reciprocity
§FORM := For distinct odd primes p,q, (p/q)(q/p)=(-1)^{((p-1)/2)((q-1)/2)}.
§DEP_HARD := S26-T-00001064
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001066|I|A|Proof+Compile+Meta] First supplementary law
§FORM := For odd prime p, (-1/p)=(-1)^{(p-1)/2}.
§DEP_HARD := S26-T-00001065
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001067|I|A|Proof+Compile+Meta] Second supplementary law
§FORM := For odd prime p, (2/p)=(-1)^{(p^2-1)/8}.
§DEP_HARD := S26-I-00001066
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001068|D|A|Proof+Compile+Meta] Jacobi symbol

§FORM := For odd positive n with prime factorization n=product p_i^{e_i}, (a/n)=product (a/p_i)^{e_i}; value 1 does not by itself imply a is a square mod composite n.

§DEP_HARD := S26-I-00001067

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001069|T|A|Proof+Compile+Meta] Hensel lifting for simple roots

§FORM := If f(a) congruent 0 mod p and f'(a) is nonzero mod p, then a lifts uniquely to a root modulo p^k for every k>=1.

§DEP_HARD := S26-D-00001068

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001070|B|A|Proof+Compile+Meta+Tower] Local modular viewpoint

§FORM := Congruence problems modulo composite n decompose into prime-power problems by the Chinese remainder theorem, providing the finite local model used later in p-adic and algebraic-number-theoretic analysis.

§DEP_HARD := S26-T-00001069

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001071|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 3

§FORM := Modulo the prime 3, a primitive root is 2; the nonzero quadratic residues are [1].

§DEP_HARD := S26-B-00001070

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001072|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 5

§FORM := Modulo the prime 5, a primitive root is 2; the nonzero quadratic residues are [1, 4].

§DEP_HARD := S26-E-00001071

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001073|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 7

§FORM := Modulo the prime 7, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4].

§DEP_HARD := S26-E-00001072

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001074|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 11

§FORM := Modulo the prime 11, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 5, 9].

§DEP_HARD := S26-E-00001073

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001075|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 13

§FORM := Modulo the prime 13, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 9, 10, 12].

§DEP_HARD := S26-E-00001074

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001076|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 17

§FORM := Modulo the prime 17, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 8, 9, 13, 15, 16].

§DEP_HARD := S26-E-00001075

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001077|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 19

§FORM := Modulo the prime 19, a primitive root is 2; the nonzero quadratic residues are [1, 4, 5, 6, 7, 9, 11, 16, 17].

§DEP_HARD := S26-E-00001076

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001078|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 23

§FORM := Modulo the prime 23, a primitive root is 5; the nonzero quadratic residues are [1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18].

§DEP_HARD := S26-E-00001077

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001079|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 29

§FORM := Modulo the prime 29, a primitive root is 2; the nonzero quadratic residues are [1, 4, 5, 6, 7, 9, 13, 16, 20, 22, 23, 24, 25, 28].

§DEP_HARD := S26-E-00001078

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001080|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 31

§FORM := Modulo the prime 31, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 5, 7, 8, 9, 10, 14, 16, 18, 19, 20, 25, 28].

§DEP_HARD := S26-E-00001079

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001081|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 37

§FORM := Modulo the prime 37, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 7, 9, 10, 11, 12, 16, 21, 25, 26, 27, 28, 30, 33, 34, 36].

§DEP_HARD := S26-E-00001080

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001082|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 41

§FORM := Modulo the prime 41, a primitive root is 6; the nonzero quadratic residues are [1, 2, 4, 5, 8, 9, 10, 16, 18, 20, 21, 23, 25, 31, 32, 33, 36, 37, 39, 40].

§DEP_HARD := S26-E-00001081

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001083|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 43

§FORM := Modulo the prime 43, a primitive root is 3; the nonzero quadratic residues are [1, 4, 6, 9, 10, 11, 13, 14, 15, 16, 17, 21, 23, 24, 25, 31, 35, 36, 38, 40, 41].

§DEP_HARD := S26-E-00001082

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001084|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 47

§FORM := Modulo the prime 47, a primitive root is 5; the nonzero quadratic residues are [1, 2, 3, 4, 6, 7, 8, 9, 12, 14, 16, 17, 18, 21, 24, 25, 27, 28, 32, 34, 36, 37, 42].

§DEP_HARD := S26-E-00001083

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001085|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 53

§FORM := Modulo the prime 53, a primitive root is 2; the nonzero quadratic residues are [1, 4, 6, 7, 9, 10, 11, 13, 15, 16, 17, 24, 25, 28, 29, 36, 37, 38, 40, 42, 43, 44, 46, 47, 49, 52].

§DEP_HARD := S26-E-00001084

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001086|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 59

§FORM := Modulo the prime 59, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 5, 7, 9, 12, 15, 16, 17, 19, 20, 21, 22, 25, 26, 27, 28, 29, 35, 36, 41, 45, 46, 48, 49, 51, 53, 57].

§DEP_HARD := S26-E-00001085

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001087|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 61

§FORM := Modulo the prime 61, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 5, 9, 12, 13, 14, 15, 16, 19, 20, 22, 25, 27, 34, 36, 39, 41, 42, 45, 46, 47, 48, 49, 52, 56, 57, 58, 60].

§DEP_HARD := S26-E-00001086

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001088|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 67

§FORM := Modulo the prime 67, a primitive root is 2; the nonzero quadratic residues are [1, 4, 6, 9, 10, 14, 15, 16, 17, 19, 21, 22, 23, 24, 25, 26, 29, 33, 35, 36, 37, 39, 40, 47, 49, 54, 55, 56, 59, 60, 62, 64, 65].

§DEP_HARD := S26-E-00001087

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001089|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 71

§FORM := Modulo the prime 71, a primitive root is 7; the nonzero quadratic residues are [1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 16, 18, 19, 20, 24, 25, 27, 29, 30, 32, 36, 37, 38, 40, 43, 45, 48, 49, 50, 54, 57, 58, 60, 64].

§DEP_HARD := S26-E-00001088

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001090|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 73

§FORM := Modulo the prime 73, a primitive root is 5; the nonzero quadratic residues are [1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 19, 23, 24, 25, 27, 32, 35, 36, 37, 38, 41, 46, 48, 49, 50, 54, 55, 57, 61, 64, 65, 67, 69, 70, 71, 72].

§DEP_HARD := S26-E-00001089

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001091|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 79

§FORM := Modulo the prime 79, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 5, 8, 9, 10, 11, 13, 16, 18, 19, 20, 21, 22, 23, 25, 26, 31, 32, 36, 38, 40, 42, 44, 45, 46, 49, 50, 51, 52, 55, 62, 64, 65, 67, 72, 73, 76].

§DEP_HARD := S26-E-00001090

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001092|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 83

§FORM := Modulo the prime 83, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 7, 9, 10, 11, 12, 16, 17, 21, 23, 25, 26, 27, 28, 29, 30, 31, 33, 36, 37, 38, 40, 41, 44, 48, 49, 51, 59, 61, 63, 64, 65, 68, 69, 70, 75, 77, 78, 81].

§DEP_HARD := S26-E-00001091

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001093|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 89

§FORM := Modulo the prime 89, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 5, 8, 9, 10, 11, 16, 17, 18, 20, 21, 22, 25, 32, 34, 36, 39, 40, 42, 44, 45, 47, 49, 50, 53, 55, 57, 64, 67, 68, 69, 71, 72, 73, 78, 79, 80, 81, 84, 85, 87, 88].

§DEP_HARD := S26-E-00001092

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001094|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 97

§FORM := Modulo the prime 97, a primitive root is 5; the nonzero quadratic residues are [1, 2, 3, 4, 6, 8, 9, 11, 12, 16, 18, 22, 24, 25, 27, 31, 32, 33, 35, 36, 43, 44, 47, 48, 49, 50, 53, 54, 61, 62, 64, 65, 66, 70, 72, 73, 75, 79, 81, 85, 86, 88, 89, 91, 93, 94, 95, 96].

§DEP_HARD := S26-E-00001093

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001095|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 101

§FORM := Modulo the prime 101, a primitive root is 2; the nonzero quadratic residues are [1, 4, 5, 6, 9, 13, 14, 16, 17, 19, 20, 21, 22, 23, 24, 25, 30, 31, 33, 36, 37, 43, 45, 47, 49, 52, 54, 56, 58, 64, 65, 68, 70, 71, 76, 77, 78, 79, 80, 81, 82, 84, 85, 87, 88, 92, 95, 96, 97, 100].

§DEP_HARD := S26-E-00001094

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001096|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 103

§FORM := Modulo the prime 103, a primitive root is 5; the nonzero quadratic residues are [1, 2, 4, 7, 8, 9, 13, 14, 15, 16, 17, 18, 19, 23, 25, 26, 28, 29, 30, 32, 33, 34, 36, 38, 41, 46, 49, 50, 52, 55, 56, 58, 59, 60, 61, 63, 64, 66, 68, 72, 76, 79, 81, 82, 83, 91, 92, 93, 97, 98, 100].

§DEP_HARD := S26-E-00001095

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001097|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 107

§FORM := Modulo the prime 107, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 9, 10, 11, 12, 13, 14, 16, 19, 23, 25, 27, 29, 30, 33, 34, 35, 36, 37, 39, 40, 41, 42, 44, 47, 48, 49, 52, 53, 56, 57, 61, 62, 64, 69, 75, 76, 79, 81, 83, 85, 86, 87, 89, 90, 92, 99, 100, 101, 102, 105].

§DEP_HARD := S26-E-00001096

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001098|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 109

§FORM := Modulo the prime 109, a primitive root is 6; the nonzero quadratic residues are [1, 3, 4, 5, 7, 9, 12, 15, 16, 20, 21, 22, 25, 26, 27, 28, 29, 31, 34, 35, 36, 38, 43, 45, 46, 48, 49, 60, 61, 63, 64, 66, 71, 73, 74, 75, 78, 80, 81, 82, 83, 84, 87, 88, 89, 93, 94, 97, 100, 102, 104, 105, 106, 108].

§DEP_HARD := S26-E-00001097

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001099|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 113

§FORM := Modulo the prime 113, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 7, 8, 9, 11, 13, 14, 15, 16, 18, 22, 25, 26, 28, 30, 31, 32, 36, 41, 44, 49, 50, 51, 52, 53, 56, 57, 60, 61, 62, 63, 64, 69, 72, 77, 81, 82, 83, 85, 87, 88, 91, 95, 97, 98, 99, 100, 102, 104, 105, 106, 109, 111, 112].

§DEP_HARD := S26-E-00001098

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001100|E|M|Proof+Compile+Meta] Quadratic-residue profile modulo 127

§FORM := Modulo the prime 127, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 8, 9, 11, 13, 15, 16, 17, 18, 19, 21, 22, 25, 26, 30, 31, 32, 34, 35, 36, 37, 38, 41, 42, 44, 47, 49, 50, 52, 60, 61, 62, 64, 68, 69, 70, 71, 72, 73, 74, 76, 79, 81, 82, 84, 87, 88, 94, 98, 99, 100, 103, 104, 107, 113, 115, 117, 120, 121, 122, 124].

§DEP_HARD := S26-E-00001099

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001101|D|A|Proof+Compile+Meta] Diophantine equation

§FORM := A Diophantine equation is a polynomial equation with integer coefficients for which integer or rational solutions are sought.

§DEP_HARD := S26-T-00000072

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001102|T|A|Proof+Compile+Meta] Linear Diophantine equation criterion

§FORM := ax+by=c has integer solutions iff gcd(a,b) divides c.

§DEP_HARD := S26-D-00001101

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001103|I|A|Proof+Compile+Meta] Parametrization of linear Diophantine solutions

§FORM := If ax_0+by_0=c and g=gcd(a,b), all integer solutions are x=x_0+(b/g)t, y=y_0-(a/g)t with t in Z.

§DEP_HARD := S26-T-00001102

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001104|D|A|Proof+Compile+Meta] Pythagorean triple

§FORM := A Pythagorean triple is an integer solution x^2+y^2=z^2; it is primitive when gcd(x,y,z)=1.

§DEP_HARD := S26-I-00001103

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001105|T|A|Proof+Compile+Meta] Primitive Pythagorean parametrization

§FORM := Up to swapping legs, every primitive positive Pythagorean triple is x=m^2-n^2, y=2mn, z=m^2+n^2 for coprime m>n of opposite parity.

§DEP_HARD := S26-D-00001104

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001106|D|A|Proof+Compile+Meta] Pell equation

§FORM := For nonsquare positive D, Pell's equation is x^2-Dy^2=1.

§DEP_HARD := S26-T-00001105

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001107|T|A|Proof+Compile+Meta] Pell solvability

§FORM := For every positive nonsquare D, Pell's equation has infinitely many integer solutions generated from powers of a fundamental unit of norm 1 in Q(sqrt(D)).

§DEP_HARD := S26-D-00001106

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001108|D|A|Proof+Compile+Meta] Generalized Pell equation

§FORM := x^2-Dy^2=N is the norm equation N_{Q(sqrt D)/Q}(x+y sqrt D)=N.

§DEP_HARD := S26-T-00001107

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001109|T|A|Proof+Compile+Meta] Fermat theorem on sums of two squares

§FORM := A positive integer n is a sum of two integer squares iff every prime congruent 3 mod 4 occurs to an even exponent in the factorization of n.

§DEP_HARD := S26-D-00001108

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001110|I|A|Proof+Compile+Meta] Two-square representation count

§FORM := The number r_2(n) of ordered signed representations n=x^2+y^2 satisfies r_2(n)=4(d_1(n)-d_3(n)), where d_j counts divisors congruent j mod 4.

§DEP_HARD := S26-T-00001109

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001111|T|A|Proof+Compile+Meta] Lagrange four-square theorem

§FORM := Every nonnegative integer is a sum of four integer squares.

§DEP_HARD := S26-I-00001110

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001112|T|A|Proof+Compile+Meta] Legendre three-square theorem

§FORM := A nonnegative integer n is a sum of three integer squares iff n is not of the form 4^a(8b+7).

§DEP_HARD := S26-T-00001111

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001113|D|A|Proof+Compile+Meta] Mordell equation

§FORM := A Mordell equation has the form y^2=x^3+k with fixed nonzero integer k.

§DEP_HARD := S26-T-00001112

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001114|D|A|Proof+Compile+Meta] Thue equation

§FORM := A Thue equation F(x,y)=m uses an irreducible binary form F of degree at least 3 and fixed nonzero integer m.

§DEP_HARD := S26-D-00001113

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001115|T|A|Proof+Compile+Meta] Thue finiteness theorem

§FORM := For irreducible binary form F of degree at least 3 and nonzero m, F(x,y)=m has finitely many integer solutions.

§DEP_HARD := S26-D-00001114

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001116|D|A|Proof+Compile+Meta] S-unit equation

§FORM := An S-unit equation has form x+y=1 with x,y units outside a finite set S of places.

§DEP_HARD := S26-T-00001115

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001117|T|A|Proof+Compile+Meta] S-unit finiteness

§FORM := For a fixed number field and finite S, the S-unit equation x+y=1 has finitely many solutions.

§DEP_HARD := S26-D-00001116

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001118|D|A|Proof+Compile+Meta] Local obstruction

§FORM := A Diophantine equation has a local obstruction if it has no solution over some completion such as R or Q_p; then it has no rational solution.

§DEP_HARD := S26-T-00001117

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00001119|N|A|Proof+Compile+Meta] Hasse-principle warning

§FORM := Solvability over every completion need not imply rational solvability for arbitrary varieties; local-to-global principles require theorem-specific hypotheses.

§DEP_HARD := S26-D-00001118

§EVIDENCE := A · Exact registered statement only.

§BOUNDARY := Local solvability is not universally sufficient for global solvability.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001120|B|A|Proof+Compile+Meta+Topo] Diophantine-to-geometry bridge

§FORM := Polynomial Diophantine problems can be recast as rational/integral-point problems on algebraic varieties, opening access to geometry, Galois theory, and height methods.

§DEP_HARD := S26-N-00001119

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001121|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=2

§FORM := The least positive solution returned for x^2-2y^2=1 is x=3, y=2; direct substitution gives 1.

§DEP_HARD := S26-B-00001120

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001122|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=3

§FORM := The least positive solution returned for x^2-3y^2=1 is x=2, y=1; direct substitution gives 1.

§DEP_HARD := S26-E-00001121

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001123|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=5

§FORM := The least positive solution returned for x^2-5y^2=1 is x=9, y=4; direct substitution gives 1.

§DEP_HARD := S26-E-00001122

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001124|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=6

§FORM := The least positive solution returned for x^2-6y^2=1 is x=5, y=2; direct substitution gives 1.

§DEP_HARD := S26-E-00001123

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001125|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=7

§FORM := The least positive solution returned for x^2-7y^2=1 is x=8, y=3; direct substitution gives 1.

§DEP_HARD := S26-E-00001124

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001126|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=8

§FORM := The least positive solution returned for x^2-8y^2=1 is x=3, y=1; direct substitution gives 1.

§DEP_HARD := S26-E-00001125

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001127|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=10

§FORM := The least positive solution returned for x^2-10y^2=1 is x=19, y=6; direct substitution gives 1.

§DEP_HARD := S26-E-00001126

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001128|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=11

§FORM := The least positive solution returned for x^2-11y^2=1 is x=10, y=3; direct substitution gives 1.

§DEP_HARD := S26-E-00001127

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001129|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=12

§FORM := The least positive solution returned for x^2-12y^2=1 is x=7, y=2; direct substitution gives 1.

§DEP_HARD := S26-E-00001128

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001130|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=13

§FORM := The least positive solution returned for x^2-13y^2=1 is x=649, y=180; direct substitution gives 1.

§DEP_HARD := S26-E-00001129

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001131|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=14

§FORM := The least positive solution returned for x^2-14y^2=1 is x=15, y=4; direct substitution gives 1.
§DEP_HARD := S26-E-00001130
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001132|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=15

§FORM := The least positive solution returned for x^2-15y^2=1 is x=4, y=1; direct substitution gives 1.
§DEP_HARD := S26-E-00001131
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001133|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=17

§FORM := The least positive solution returned for x^2-17y^2=1 is x=33, y=8; direct substitution gives 1.
§DEP_HARD := S26-E-00001132
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001134|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=18

§FORM := The least positive solution returned for x^2-18y^2=1 is x=17, y=4; direct substitution gives 1.
§DEP_HARD := S26-E-00001133
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001135|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=19

§FORM := The least positive solution returned for x^2-19y^2=1 is x=170, y=39; direct substitution gives 1.
§DEP_HARD := S26-E-00001134
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001136|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=20

§FORM := The least positive solution returned for x^2-20y^2=1 is x=9, y=2; direct substitution gives 1.
§DEP_HARD := S26-E-00001135
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001137|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=21

§FORM := The least positive solution returned for x^2-21y^2=1 is x=55, y=12; direct substitution gives 1.
§DEP_HARD := S26-E-00001136
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001138|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=22

§FORM := The least positive solution returned for x^2-22y^2=1 is x=197, y=42; direct substitution gives 1.
§DEP_HARD := S26-E-00001137
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001139|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=23

§FORM := The least positive solution returned for x^2-23y^2=1 is x=24, y=5; direct substitution gives 1.
§DEP_HARD := S26-E-00001138
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001140|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=24

§FORM := The least positive solution returned for x^2-24y^2=1 is x=5, y=1; direct substitution gives 1.
§DEP_HARD := S26-E-00001139
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001141|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=26

§FORM := The least positive solution returned for x^2-26y^2=1 is x=51, y=10; direct substitution gives 1.
§DEP_HARD := S26-E-00001140
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001142|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=27

§FORM := The least positive solution returned for x^2-27y^2=1 is x=26, y=5; direct substitution gives 1.
§DEP_HARD := S26-E-00001141
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001143|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=28

§FORM := The least positive solution returned for x^2-28y^2=1 is x=127, y=24; direct substitution gives 1.
§DEP_HARD := S26-E-00001142
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001144|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=29

§FORM := The least positive solution returned for x^2-29y^2=1 is x=9801, y=1820; direct substitution gives 1.
§DEP_HARD := S26-E-00001143
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001145|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=30
§FORM := The least positive solution returned for x^2-30y^2=1 is x=11, y=2; direct substitution gives 1.
§DEP_HARD := S26-E-00001144
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001146|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=31
§FORM := The least positive solution returned for x^2-31y^2=1 is x=1520, y=273; direct substitution gives 1.
§DEP_HARD := S26-E-00001145
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001147|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=32
§FORM := The least positive solution returned for x^2-32y^2=1 is x=17, y=3; direct substitution gives 1.
§DEP_HARD := S26-E-00001146
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001148|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=33
§FORM := The least positive solution returned for x^2-33y^2=1 is x=23, y=4; direct substitution gives 1.
§DEP_HARD := S26-E-00001147
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001149|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=34
§FORM := The least positive solution returned for x^2-34y^2=1 is x=35, y=6; direct substitution gives 1.
§DEP_HARD := S26-E-00001148
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001150|E|M|Proof+Compile+Meta] Fundamental Pell solution for D=35
§FORM := The least positive solution returned for x^2-35y^2=1 is x=6, y=1; direct substitution gives 1.
§DEP_HARD := S26-E-00001149
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001151|D|A|Proof+Compile+Meta] Prime-counting function
§FORM := pi(x) is the number of primes p<=x.
§DEP_HARD := S26-D-00000069
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001152|D|A|Proof+Compile+Meta] Chebyshev theta function
§FORM := theta(x)=sum_{p<=x} log p.
§DEP_HARD := S26-D-00001151
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001153|D|A|Proof+Compile+Meta] Chebyshev psi function
§FORM := psi(x)=sum_{n<=x} Lambda(n)=sum_{p^k<=x} log p.
§DEP_HARD := S26-D-00001152
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001154|T|A|Proof+Compile+Meta] Prime number theorem
§FORM := pi(x) is asymptotic to x/log x as x->infinity; equivalently theta(x)~x and psi(x)~x.
§DEP_HARD := S26-D-00001153
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001155|D|A|Proof+Compile+Meta] Riemann zeta function
§FORM := For Re(s)>1, zeta(s)=sum_{n>=1} n^{-s}=product_p(1-p^{-s})^{-1}.
§DEP_HARD := S26-T-00001154
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001156|I|A|Proof+Compile+Meta] Euler product for zeta
§FORM := Absolute convergence for Re(s)>1 permits the identity zeta(s)=product_p(1-p^{-s})^{-1}.
§DEP_HARD := S26-D-00001155
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001157|T|A|Proof+Compile+Meta] Meromorphic continuation of zeta
§FORM := zeta(s) extends meromorphically to C with a single simple pole at s=1 of residue 1.
§DEP_HARD := S26-I-00001156
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001158|I|A|Proof+Compile+Meta] Zeta functional equation
§FORM := The completed function xi(s)=1/2 s(s-1) pi^{-s/2} Gamma(s/2) zeta(s) satisfies xi(s)=xi(1-s).
§DEP_HARD := S26-T-00001157
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001159|D|A|Proof+Compile+Meta] Dirichlet character

§FORM := A Dirichlet character modulo q is a completely multiplicative periodic function chi:Z->C that vanishes on integers not coprime to q and restricts to a group character on (Z/qZ)^x.
§DEP_HARD := S26-I-00001158
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001160|D|A|Proof+Compile+Meta] Dirichlet L-function

§FORM := For Re(s)>1, L(s,chi)=sum_{n>=1} chi(n)n^{-s}, with an Euler product over primes.
§DEP_HARD := S26-D-00001159
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001161|T|A|Proof+Compile+Meta] Dirichlet theorem on arithmetic progressions

§FORM := If gcd(a,q)=1, infinitely many primes are congruent to a modulo q.
§DEP_HARD := S26-D-00001160
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001162|D|A|Proof+Compile+Meta] Abscissa of convergence

§FORM := For a Dirichlet series sum a_n n^{-s}, the abscissa of convergence is the boundary between its half-plane of convergence and divergence.
§DEP_HARD := S26-T-00001161
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001163|D|A|Proof+Compile+Meta] Summatory function

§FORM := For an arithmetic function f, F(x)=sum_{n<=x} f(n) is its summatory function.
§DEP_HARD := S26-D-00001162
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001164|T|A|Proof+Compile+Meta] Partial summation

§FORM := If A(x)=sum_{n<=x} a_n and f is continuously differentiable, then sum_{n<=x}a_n f(n)=A(x)f(x)-integral_1^x A(t)f'(t)dt, with endpoint conventions as appropriate.
§DEP_HARD := S26-D-00001163
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001165|D|A|Proof+Compile+Meta] Average order

§FORM := A function g is an average order of f when sum_{n<=x}f(n) is asymptotic to integral_1^x g(t)dt in the declared sense.
§DEP_HARD := S26-T-00001164
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001166|T|A|Proof+Compile+Meta] Average order of divisor function

§FORM := sum_{n<=x} tau(n)=x log x+(2 gamma-1)x+O(sqrt(x)) in a classical elementary estimate.
§DEP_HARD := S26-D-00001165
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001167|T|A|Proof+Compile+Meta] Mertens product asymptotic

§FORM := product_{p<=x}(1-1/p)^{-1} ~ e^gamma log x.
§DEP_HARD := S26-T-00001166
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-C-00001168|C|H|Proof+Compile+Meta] Riemann hypothesis

§FORM := The Riemann hypothesis asserts that every nontrivial zero of zeta(s) has real part 1/2.
§DEP_HARD := S26-T-00001167
§EVIDENCE := H · Exact registered conjecture only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | hypothesis_remains_open

§CLAIM[S26-N-00001169|N|A|Proof+Compile+Meta] Unproved-status fence for RH

§FORM := The Riemann hypothesis remains a conjecture; numerical zero verification or consequences under RH do not constitute proof.
§DEP_HARD := S26-C-00001168
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := Finite computations cannot establish an infinite zero-location claim.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001170|B|A|Proof+Compile+Meta+Spectral] Spectral/arithmetic interface

§FORM := Zeros and poles of zeta- and L-functions control explicit formulas relating primes to analytic spectral data; later volumes separate proved explicit formulas from conjectural spectral interpretations.
§DEP_HARD := S26-N-00001169
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001171|E|M|Proof+Compile+Meta] Prime-counting profile at x=10

§FORM := pi(10)=4; the finite ratio pi(x)/(x/log x) is approximately 0.92103404.
§DEP_HARD := S26-B-00001170
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001172|E|M|Proof+Compile+Meta] Prime-counting profile at x=20

§FORM := pi(20)=8; the finite ratio pi(x)/(x/log x) is approximately 1.19829291.
§DEP_HARD := S26-E-00001171
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001173|E|M|Proof+Compile+Meta] Prime-counting profile at x=30

§FORM := pi(30)=10; the finite ratio pi(x)/(x/log x) is approximately 1.13373246.
§DEP_HARD := S26-E-00001172
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001174|E|M|Proof+Compile+Meta] Prime-counting profile at x=40

§FORM := pi(40)=12; the finite ratio pi(x)/(x/log x) is approximately 1.10666384.
§DEP_HARD := S26-E-00001173
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001175|E|M|Proof+Compile+Meta] Prime-counting profile at x=50

§FORM := pi(50)=15; the finite ratio pi(x)/(x/log x) is approximately 1.17360690.
§DEP_HARD := S26-E-00001174
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001176|E|M|Proof+Compile+Meta] Prime-counting profile at x=60

§FORM := pi(60)=17; the finite ratio pi(x)/(x/log x) is approximately 1.16006429.
§DEP_HARD := S26-E-00001175
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001177|E|M|Proof+Compile+Meta] Prime-counting profile at x=70

§FORM := pi(70)=19; the finite ratio pi(x)/(x/log x) is approximately 1.15316299.
§DEP_HARD := S26-E-00001176
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001178|E|M|Proof+Compile+Meta] Prime-counting profile at x=80

§FORM := pi(80)=22; the finite ratio pi(x)/(x/log x) is approximately 1.20505732.
§DEP_HARD := S26-E-00001177
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001179|E|M|Proof+Compile+Meta] Prime-counting profile at x=90

§FORM := pi(90)=24; the finite ratio pi(x)/(x/log x) is approximately 1.19994925.
§DEP_HARD := S26-E-00001178
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001180|E|M|Proof+Compile+Meta] Prime-counting profile at x=100

§FORM := pi(100)=25; the finite ratio pi(x)/(x/log x) is approximately 1.15129255.
§DEP_HARD := S26-E-00001179
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001181|E|M|Proof+Compile+Meta] Prime-counting profile at x=110

§FORM := pi(110)=29; the finite ratio pi(x)/(x/log x) is approximately 1.23921755.
§DEP_HARD := S26-E-00001180
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001182|E|M|Proof+Compile+Meta] Prime-counting profile at x=120

§FORM := pi(120)=30; the finite ratio pi(x)/(x/log x) is approximately 1.19687294.
§DEP_HARD := S26-E-00001181
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001183|E|M|Proof+Compile+Meta] Prime-counting profile at x=130

§FORM := pi(130)=31; the finite ratio pi(x)/(x/log x) is approximately 1.16071975.
§DEP_HARD := S26-E-00001182
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001184|E|M|Proof+Compile+Meta] Prime-counting profile at x=140

§FORM := pi(140)=34; the finite ratio pi(x)/(x/log x) is approximately 1.20011316.
§DEP_HARD := S26-E-00001183
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001185|E|M|Proof+Compile+Meta] Prime-counting profile at x=150

§FORM := pi(150)=35; the finite ratio pi(x)/(x/log x) is approximately 1.16914824.
§DEP_HARD := S26-E-00001184
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001186|E|M|Proof+Compile+Meta] Prime-counting profile at x=160
§FORM := pi(160)=37; the finite ratio pi(x)/(x/log x) is approximately 1.17363394.
§DEP_HARD := S26-E-00001185
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001187|E|M|Proof+Compile+Meta] Prime-counting profile at x=170
§FORM := pi(170)=39; the finite ratio pi(x)/(x/log x) is approximately 1.17821258.
§DEP_HARD := S26-E-00001186
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001188|E|M|Proof+Compile+Meta] Prime-counting profile at x=180
§FORM := pi(180)=41; the finite ratio pi(x)/(x/log x) is approximately 1.18284017.
§DEP_HARD := S26-E-00001187
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001189|E|M|Proof+Compile+Meta] Prime-counting profile at x=190
§FORM := pi(190)=42; the finite ratio pi(x)/(x/log x) is approximately 1.15986848.
§DEP_HARD := S26-E-00001188
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001190|E|M|Proof+Compile+Meta] Prime-counting profile at x=200
§FORM := pi(200)=46; the finite ratio pi(x)/(x/log x) is approximately 1.21861299.
§DEP_HARD := S26-E-00001189
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001191|E|M|Proof+Compile+Meta] Prime-counting profile at x=1000
§FORM := pi(1000)=168; the finite ratio pi(x)/(x/log x) is approximately 1.16050289.
§DEP_HARD := S26-E-00001190
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001192|E|M|Proof+Compile+Meta] Prime-counting profile at x=10000
§FORM := pi(10000)=1229; the finite ratio pi(x)/(x/log x) is approximately 1.13195083.
§DEP_HARD := S26-E-00001191
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001193|E|M|Proof+Compile+Meta] Prime-counting profile at x=100000
§FORM := pi(100000)=9592; the finite ratio pi(x)/(x/log x) is approximately 1.10431981.
§DEP_HARD := S26-E-00001192
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001194|E|M|Proof+Compile+Meta] Prime-counting profile at x=1000000
§FORM := pi(1000000)=78498; the finite ratio pi(x)/(x/log x) is approximately 1.08448995.
§DEP_HARD := S26-E-00001193
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001195|E|M|Proof+Compile+Meta] Prime-counting profile at x=10000000
§FORM := pi(10000000)=664579; the finite ratio pi(x)/(x/log x) is approximately 1.07117479.
§DEP_HARD := S26-E-00001194
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001196|E|M|Proof+Compile+Meta] Prime-counting profile at x=100000000
§FORM := pi(100000000)=5761455; the finite ratio pi(x)/(x/log x) is approximately 1.06129923.
§DEP_HARD := S26-E-00001195
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001197|E|M|Proof+Compile+Meta] Prime-counting profile at x=1000000000
§FORM := pi(1000000000)=50847534; the finite ratio pi(x)/(x/log x) is approximately 1.05372696.
§DEP_HARD := S26-E-00001196
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001198|E|M|Proof+Compile+Meta] Prime-counting profile at x=10000000000
§FORM := pi(10000000000)=455052511; the finite ratio pi(x)/(x/log x) is approximately 1.04779713.
§DEP_HARD := S26-E-00001197
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001199|E|M|Proof+Compile+Meta] Prime-counting profile at x=100000000000
§FORM := pi(100000000000)=4118054813; the finite ratio pi(x)/(x/log x) is approximately 1.04303888.
§DEP_HARD := S26-E-00001198
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001200|E|M|Proof+Compile+Meta] Prime-counting profile at x=1000000000000

§FORM := pi(1000000000000)=37607912018; the finite ratio pi(x)/(x/log x) is approximately 1.03914501.
§DEP_HARD := S26-E-00001199
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001201|D|A|Proof+Compile+Meta] Number field

§FORM := A number field is a finite extension K/Q.
§DEP_HARD := S26-D-00000101
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001202|D|A|Proof+Compile+Meta] Algebraic integer

§FORM := An element alpha of a number field is an algebraic integer when it satisfies a monic polynomial with coefficients in Z.
§DEP_HARD := S26-D-00001201
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001203|D|A|Proof+Compile+Meta] Ring of integers

§FORM := O_K is the ring of all algebraic integers in K.
§DEP_HARD := S26-D-00001202
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001204|T|A|Proof+Compile+Meta] Integral closure of Z in a number field

§FORM := O_K is the integral closure of Z in K and is a free Z-module of rank [K:Q].
§DEP_HARD := S26-D-00001203
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001205|D|A|Proof+Compile+Meta] Field norm

§FORM := For alpha in K, N_{K/Q}(alpha) is the determinant of multiplication by alpha as a Q-linear map.
§DEP_HARD := S26-T-00001204
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001206|D|A|Proof+Compile+Meta] Field trace

§FORM := For alpha in K, Tr_{K/Q}(alpha) is the trace of multiplication by alpha as a Q-linear map.
§DEP_HARD := S26-D-00001205
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001207|I|A|Proof+Compile+Meta] Norm and trace from embeddings

§FORM := For separable K/Q, the norm is the product and the trace the sum of the images of alpha over embeddings K->C, counted appropriately.
§DEP_HARD := S26-D-00001206
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001208|D|A|Proof+Compile+Meta] Discriminant of a basis

§FORM := For a Q-basis alpha_1,...,alpha_n of K, disc(alpha_i)=det(Tr(alpha_i alpha_j)).
§DEP_HARD := S26-I-00001207
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001209|D|A|Proof+Compile+Meta] Field discriminant

§FORM := The discriminant d_K is the discriminant of an integral basis of O_K and is independent of the chosen integral basis.
§DEP_HARD := S26-D-00001208
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001210|D|A|Proof+Compile+Meta] Fractional ideal

§FORM := A fractional ideal of O_K is a nonzero O_K-submodule I of K such that dI is an integral ideal for some nonzero d in O_K.
§DEP_HARD := S26-D-00001209
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001211|T|A|Proof+Compile+Meta] Dedekind ideal factorization

§FORM := Every nonzero proper ideal of O_K factors uniquely as a product of nonzero prime ideals.
§DEP_HARD := S26-D-00001210
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001212|D|A|Proof+Compile+Meta] Ideal norm

§FORM := For nonzero integral ideal I, N(I)=|O_K/I|.
§DEP_HARD := S26-T-00001211
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001213|I|A|Proof+Compile+Meta] Multiplicativity of ideal norm

§FORM := For nonzero ideals I,J in O_K, N(IJ)=N(I)N(J).
§DEP_HARD := S26-D-00001212
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001214|D|A|Proof+Compile+Meta] Ideal class group
§FORM := Cl(K) is the quotient of nonzero fractional ideals by principal fractional ideals.
§DEP_HARD := S26-I-00001213
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001215|T|A|Proof+Compile+Meta] Finiteness of class group
§FORM := The ideal class group of a number field is finite.
§DEP_HARD := S26-D-00001214
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001216|D|A|Proof+Compile+Meta] Unit group
§FORM := O_K^x is the multiplicative group of units in the ring of integers.
§DEP_HARD := S26-T-00001215
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001217|T|A|Proof+Compile+Meta] Dirichlet unit theorem
§FORM := If K has r_1 real embeddings and r_2 conjugate pairs of complex embeddings, O_K^x is isomorphic to mu(K) x Z^{r_1+r_2-1}.
§DEP_HARD := S26-D-00001216
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001218|D|A|Proof+Compile+Meta+IG] Ramification index and residue degree
§FORM := For prime ideal P over p, e(P/p) measures the exponent of P in pO_K and f(P/p)=[O_K/P:F_p].
§DEP_HARD := S26-T-00001217
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001219|I|A|Proof+Compile+Meta] Prime decomposition degree formula
§FORM := For pO_K=product P_i^{e_i}, sum_i e_i f_i=[K:Q].
§DEP_HARD := S26-D-00001218
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001220|B|A|Proof+Compile+Meta+Topo] Arithmetic geometry gateway
§FORM := Ideal classes, units, valuations, and ramification provide the local/global algebraic infrastructure used by class field theory and arithmetic geometry.
§DEP_HARD := S26-I-00001219
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001221|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-1
§FORM := For K=Q(sqrt(-1)) with squarefree d=-1, the standard quadratic-field discriminant is -4.
§DEP_HARD := S26-B-00001220
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001222|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-2
§FORM := For K=Q(sqrt(-2)) with squarefree d=-2, the standard quadratic-field discriminant is -8.
§DEP_HARD := S26-E-00001221
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001223|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-3
§FORM := For K=Q(sqrt(-3)) with squarefree d=-3, the standard quadratic-field discriminant is -3.
§DEP_HARD := S26-E-00001222
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001224|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-5
§FORM := For K=Q(sqrt(-5)) with squarefree d=-5, the standard quadratic-field discriminant is -20.
§DEP_HARD := S26-E-00001223
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001225|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-6
§FORM := For K=Q(sqrt(-6)) with squarefree d=-6, the standard quadratic-field discriminant is -24.
§DEP_HARD := S26-E-00001224
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001226|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-7
§FORM := For K=Q(sqrt(-7)) with squarefree d=-7, the standard quadratic-field discriminant is -7.
§DEP_HARD := S26-E-00001225
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001227|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-10
§FORM := For K=Q(sqrt(-10)) with squarefree d=-10, the standard quadratic-field discriminant is -40.
§DEP_HARD := S26-E-00001226
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001228|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-11
§FORM := For K=Q(sqrt(-11)) with squarefree d=-11, the standard quadratic-field discriminant is -11.
§DEP_HARD := S26-E-00001227
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001229|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-13
§FORM := For K=Q(sqrt(-13)) with squarefree d=-13, the standard quadratic-field discriminant is -52.
§DEP_HARD := S26-E-00001228
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001230|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=-15
§FORM := For K=Q(sqrt(-15)) with squarefree d=-15, the standard quadratic-field discriminant is -15.
§DEP_HARD := S26-E-00001229
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001231|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=2
§FORM := For K=Q(sqrt(2)) with squarefree d=2, the standard quadratic-field discriminant is 8.
§DEP_HARD := S26-E-00001230
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001232|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=3
§FORM := For K=Q(sqrt(3)) with squarefree d=3, the standard quadratic-field discriminant is 12.
§DEP_HARD := S26-E-00001231
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001233|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=5
§FORM := For K=Q(sqrt(5)) with squarefree d=5, the standard quadratic-field discriminant is 5.
§DEP_HARD := S26-E-00001232
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001234|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=6
§FORM := For K=Q(sqrt(6)) with squarefree d=6, the standard quadratic-field discriminant is 24.
§DEP_HARD := S26-E-00001233
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001235|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=7
§FORM := For K=Q(sqrt(7)) with squarefree d=7, the standard quadratic-field discriminant is 28.
§DEP_HARD := S26-E-00001234
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001236|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=10
§FORM := For K=Q(sqrt(10)) with squarefree d=10, the standard quadratic-field discriminant is 40.
§DEP_HARD := S26-E-00001235
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001237|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=11
§FORM := For K=Q(sqrt(11)) with squarefree d=11, the standard quadratic-field discriminant is 44.
§DEP_HARD := S26-E-00001236
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001238|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=13
§FORM := For K=Q(sqrt(13)) with squarefree d=13, the standard quadratic-field discriminant is 13.
§DEP_HARD := S26-E-00001237
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001239|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=14
§FORM := For K=Q(sqrt(14)) with squarefree d=14, the standard quadratic-field discriminant is 56.
§DEP_HARD := S26-E-00001238
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001240|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=17
§FORM := For K=Q(sqrt(17)) with squarefree d=17, the standard quadratic-field discriminant is 17.
§DEP_HARD := S26-E-00001239
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001241|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=19
§FORM := For K=Q(sqrt(19)) with squarefree d=19, the standard quadratic-field discriminant is 76.
§DEP_HARD := S26-E-00001240
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001242|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=21
§FORM := For K=Q(sqrt(21)) with squarefree d=21, the standard quadratic-field discriminant is 21.
§DEP_HARD := S26-E-00001241
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001243|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=22
§FORM := For K=Q(sqrt(22)) with squarefree d=22, the standard quadratic-field discriminant is 88.
§DEP_HARD := S26-E-00001242
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001244|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=23
§FORM := For K=Q(sqrt(23)) with squarefree d=23, the standard quadratic-field discriminant is 92.
§DEP_HARD := S26-E-00001243
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001245|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=26
§FORM := For K=Q(sqrt(26)) with squarefree d=26, the standard quadratic-field discriminant is 104.
§DEP_HARD := S26-E-00001244
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001246|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=29
§FORM := For K=Q(sqrt(29)) with squarefree d=29, the standard quadratic-field discriminant is 29.
§DEP_HARD := S26-E-00001245
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001247|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=30
§FORM := For K=Q(sqrt(30)) with squarefree d=30, the standard quadratic-field discriminant is 120.
§DEP_HARD := S26-E-00001246
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001248|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=31
§FORM := For K=Q(sqrt(31)) with squarefree d=31, the standard quadratic-field discriminant is 124.
§DEP_HARD := S26-E-00001247
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001249|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=33
§FORM := For K=Q(sqrt(33)) with squarefree d=33, the standard quadratic-field discriminant is 33.
§DEP_HARD := S26-E-00001248
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001250|E|M|Proof+Compile+Meta] Quadratic-field discriminant profile d=37
§FORM := For K=Q(sqrt(37)) with squarefree d=37, the standard quadratic-field discriminant is 37.
§DEP_HARD := S26-E-00001249
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001251|D|A|Proof+Compile+Meta] Finite field
§FORM := A finite field is a field with finitely many elements.
§DEP_HARD := S26-D-00000101
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001252|T|A|Proof+Compile+Meta] Finite-field order theorem
§FORM := Every finite field has order p^n for a prime p and n>=1; for every p^n there exists a finite field of that order, unique up to isomorphism.
§DEP_HARD := S26-D-00001251
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001253|T|A|Proof+Compile+Meta] Finite-field multiplicative group
§FORM := The multiplicative group F_q^x is cyclic of order q-1.
§DEP_HARD := S26-T-00001252
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001254|I|A|Proof+Compile+Meta] Frobenius identity in F_q
§FORM := Every x in F_q satisfies x^q=x.
§DEP_HARD := S26-T-00001253
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001255|D|A|Proof+Compile+Meta] Frobenius automorphism
§FORM := For F_{p^n}, Frobenius is F(x)=x^p; it generates Gal(F_{p^n}/F_p), a cyclic group of order n.
§DEP_HARD := S26-I-00001254
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001256|T|A|Proof+Compile+Meta] Subfield classification of finite fields
§FORM := F_{p^m} embeds as a subfield of F_{p^n} iff m divides n.
§DEP_HARD := S26-D-00001255
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001257|D|A|Proof+Compile+Meta] Trace over finite fields
§FORM := Tr_{F_{q^n}/F_q}(x)=x+x^q+...+x^{q^{n-1}}.
§DEP_HARD := S26-T-00001256
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001258|D|A|Proof+Compile+Meta] Norm over finite fields
§FORM := N_{F_{q^n}/F_q}(x)=x^{(q^n-1)/(q-1)}.
§DEP_HARD := S26-D-00001257
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001259|T|A|Proof+Compile+Meta] Irreducible-polynomial construction
§FORM := If f in F_q[x] is irreducible of degree n, then F_q[x]/(f) is a field with q^n elements.
§DEP_HARD := S26-D-00001258
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001260|D|A|Proof+Compile+Meta] Cyclotomic polynomial
§FORM := Phi_n(x) is the monic polynomial whose roots are the primitive n-th roots of unity.
§DEP_HARD := S26-T-00001259
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001261|I|A|Proof+Compile+Meta] Cyclotomic factorization
§FORM := x^n-1=product_{d|n} Phi_d(x).
§DEP_HARD := S26-D-00001260
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001262|T|A|Proof+Compile+Meta] Degree of cyclotomic polynomial
§FORM := deg Phi_n=phi(n).
§DEP_HARD := S26-I-00001261
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001263|T|A|Proof+Compile+Meta] Cyclotomic irreducibility
§FORM := Phi_n(x) is irreducible over Q.
§DEP_HARD := S26-T-00001262
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001264|D|A|Proof+Compile+Meta] Cyclotomic field
§FORM := Q(zeta_n) is the field obtained by adjoining a primitive n-th root of unity.
§DEP_HARD := S26-T-00001263
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001265|T|A|Proof+Compile+Meta] Cyclotomic Galois group
§FORM := Gal(Q(zeta_n)/Q) is canonically isomorphic to (Z/nZ)^x.
§DEP_HARD := S26-D-00001264
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001266|D|A|Proof+Compile+Meta] Gaussian period
§FORM := A Gaussian period is a sum of roots of unity over a coset of a multiplicative subgroup of a finite cyclic group, used in explicit cyclotomic constructions.
§DEP_HARD := S26-T-00001265
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001267|D|A|Proof+Compile+Meta] Discrete logarithm in F_q
§FORM := For a generator g of F_q^x and nonzero h, the discrete logarithm asks for k modulo q-1 with g^k=h.
§DEP_HARD := S26-D-00001266
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00001268|N|A|Proof+Compile+Meta] Finite-field representation fence
§FORM := A finite field is unique only up to isomorphism; concrete polynomial-basis or normal-basis encodings are representations, not the abstract field itself.
§DEP_HARD := S26-D-00001267
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := Distinct irreducible-polynomial quotients can represent isomorphic fields.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001269|B|A|Proof+Compile+Meta+Topo] Finite fields to coding theory

§FORM := Finite fields supply alphabets and linear-algebra structure for Reed-Solomon, BCH, cyclic, and algebraic-geometric codes.
§DEP_HARD := S26-N-00001268
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001270|B|A|Proof+Compile+Meta] Finite fields to cryptography

§FORM := Finite-field groups and elliptic curves over finite fields provide finite algebraic groups used in public-key cryptography; security depends on computational hardness assumptions, not field axioms alone.
§DEP_HARD := S26-B-00001269
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001271|E|M|Proof+Compile+Meta] Finite-field order profile q=2

§FORM := q=2=2^1 is a prime power; F_2 exists up to isomorphism and F_2^x has cyclic order 1.
§DEP_HARD := S26-B-00001270
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001272|E|M|Proof+Compile+Meta] Finite-field order profile q=3

§FORM := q=3=3^1 is a prime power; F_3 exists up to isomorphism and F_3^x has cyclic order 2.
§DEP_HARD := S26-E-00001271
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001273|E|M|Proof+Compile+Meta] Finite-field order profile q=4

§FORM := q=4=2^2 is a prime power; F_4 exists up to isomorphism and F_4^x has cyclic order 3.
§DEP_HARD := S26-E-00001272
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001274|E|M|Proof+Compile+Meta] Finite-field order profile q=5

§FORM := q=5=5^1 is a prime power; F_5 exists up to isomorphism and F_5^x has cyclic order 4.
§DEP_HARD := S26-E-00001273
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001275|E|M|Proof+Compile+Meta] Finite-field order profile q=7

§FORM := q=7=7^1 is a prime power; F_7 exists up to isomorphism and F_7^x has cyclic order 6.
§DEP_HARD := S26-E-00001274
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001276|E|M|Proof+Compile+Meta] Finite-field order profile q=8

§FORM := q=8=2^3 is a prime power; F_8 exists up to isomorphism and F_8^x has cyclic order 7.
§DEP_HARD := S26-E-00001275
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001277|E|M|Proof+Compile+Meta] Finite-field order profile q=9

§FORM := q=9=3^2 is a prime power; F_9 exists up to isomorphism and F_9^x has cyclic order 8.
§DEP_HARD := S26-E-00001276
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001278|E|M|Proof+Compile+Meta] Finite-field order profile q=11

§FORM := q=11=11^1 is a prime power; F_11 exists up to isomorphism and F_11^x has cyclic order 10.
§DEP_HARD := S26-E-00001277
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001279|E|M|Proof+Compile+Meta] Finite-field order profile q=13

§FORM := q=13=13^1 is a prime power; F_13 exists up to isomorphism and F_13^x has cyclic order 12.
§DEP_HARD := S26-E-00001278
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001280|E|M|Proof+Compile+Meta] Finite-field order profile q=16

§FORM := q=16=2^4 is a prime power; F_16 exists up to isomorphism and F_16^x has cyclic order 15.
§DEP_HARD := S26-E-00001279
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001281|E|M|Proof+Compile+Meta] Finite-field order profile q=17

§FORM := q=17=17^1 is a prime power; F_17 exists up to isomorphism and F_17^x has cyclic order 16.
§DEP_HARD := S26-E-00001280
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001282|E|M|Proof+Compile+Meta] Finite-field order profile q=19

§FORM := q=19=19^1 is a prime power; F_19 exists up to isomorphism and F_19^x has cyclic order 18.
§DEP_HARD := S26-E-00001281
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001283|E|M|Proof+Compile+Meta] Finite-field order profile q=23

§FORM := q=23=23^1 is a prime power; F_23 exists up to isomorphism and F_23^x has cyclic order 22.
§DEP_HARD := S26-E-00001282
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001284|E|M|Proof+Compile+Meta] Finite-field order profile q=25

§FORM := q=25=5^2 is a prime power; F_25 exists up to isomorphism and F_25^x has cyclic order 24.
§DEP_HARD := S26-E-00001283
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001285|E|M|Proof+Compile+Meta] Finite-field order profile q=27

§FORM := q=27=3^3 is a prime power; F_27 exists up to isomorphism and F_27^x has cyclic order 26.
§DEP_HARD := S26-E-00001284
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001286|E|M|Proof+Compile+Meta] Finite-field order profile q=29

§FORM := q=29=29^1 is a prime power; F_29 exists up to isomorphism and F_29^x has cyclic order 28.
§DEP_HARD := S26-E-00001285
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001287|E|M|Proof+Compile+Meta] Finite-field order profile q=31

§FORM := q=31=31^1 is a prime power; F_31 exists up to isomorphism and F_31^x has cyclic order 30.
§DEP_HARD := S26-E-00001286
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001288|E|M|Proof+Compile+Meta] Finite-field order profile q=32

§FORM := q=32=2^5 is a prime power; F_32 exists up to isomorphism and F_32^x has cyclic order 31.
§DEP_HARD := S26-E-00001287
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001289|E|M|Proof+Compile+Meta] Finite-field order profile q=37

§FORM := q=37=37^1 is a prime power; F_37 exists up to isomorphism and F_37^x has cyclic order 36.
§DEP_HARD := S26-E-00001288
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001290|E|M|Proof+Compile+Meta] Finite-field order profile q=41

§FORM := q=41=41^1 is a prime power; F_41 exists up to isomorphism and F_41^x has cyclic order 40.
§DEP_HARD := S26-E-00001289
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001291|E|M|Proof+Compile+Meta] Finite-field order profile q=43

§FORM := q=43=43^1 is a prime power; F_43 exists up to isomorphism and F_43^x has cyclic order 42.
§DEP_HARD := S26-E-00001290
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001292|E|M|Proof+Compile+Meta] Finite-field order profile q=47

§FORM := q=47=47^1 is a prime power; F_47 exists up to isomorphism and F_47^x has cyclic order 46.
§DEP_HARD := S26-E-00001291
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001293|E|M|Proof+Compile+Meta] Finite-field order profile q=49

§FORM := q=49=7^2 is a prime power; F_49 exists up to isomorphism and F_49^x has cyclic order 48.
§DEP_HARD := S26-E-00001292
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001294|E|M|Proof+Compile+Meta] Finite-field order profile q=53

§FORM := q=53=53^1 is a prime power; F_53 exists up to isomorphism and F_53^x has cyclic order 52.
§DEP_HARD := S26-E-00001293
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001295|E|M|Proof+Compile+Meta] Finite-field order profile q=59

§FORM := q=59=59^1 is a prime power; F_59 exists up to isomorphism and F_59^x has cyclic order 58.
§DEP_HARD := S26-E-00001294
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001296|E|M|Proof+Compile+Meta] Finite-field order profile q=81

§FORM := q=81=3^4 is a prime power; F_81 exists up to isomorphism and F_81^x has cyclic order 80.
§DEP_HARD := S26-E-00001295
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001297|E|M|Proof+Compile+Meta] Finite-field order profile q=121

§FORM := q=121=11^2 is a prime power; F_121 exists up to isomorphism and F_121^x has cyclic order 120.
§DEP_HARD := S26-E-00001296
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001298|E|M|Proof+Compile+Meta] Finite-field order profile q=125

§FORM := q=125=5^3 is a prime power; F_125 exists up to isomorphism and F_125^x has cyclic order 124.
§DEP_HARD := S26-E-00001297
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001299|E|M|Proof+Compile+Meta] Finite-field order profile q=169

§FORM := q=169=13^2 is a prime power; F_169 exists up to isomorphism and F_169^x has cyclic order 168.
§DEP_HARD := S26-E-00001298
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001300|E|M|Proof+Compile+Meta] Finite-field order profile q=243

§FORM := q=243=3^5 is a prime power; F_243 exists up to isomorphism and F_243^x has cyclic order 242.
§DEP_HARD := S26-E-00001299
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001301|D|A|Proof+Compile+Meta] Group action

§FORM := A left action of G on X is a map GxX->X with e.x=x and (gh).x=g.(h.x).
§DEP_HARD := S26-D-00000103
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001302|T|A|Proof+Compile+Meta] Orbit as a homogeneous quotient

§FORM := For a group action and x in X, the orbit G.x is naturally identified with the coset space G/G_x as a G-set.
§DEP_HARD := S26-D-00001301
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001303|D|A|Proof+Compile+Meta] Stabilizer

§FORM := G_x={g in G:g.x=x} is the stabilizer subgroup of x.
§DEP_HARD := S26-T-00001302
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001304|T|A|Proof+Compile+Meta] Orbit-stabilizer theorem

§FORM := For finite G, |G.x|=[G:G_x], hence |G|=|G.x||G_x|.
§DEP_HARD := S26-D-00001303
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001305|D|A|Proof+Compile+Meta] Kernel of an action

§FORM := The kernel of a G-action is the intersection of all point stabilizers; the action is faithful iff this kernel is trivial.
§DEP_HARD := S26-T-00001304
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001306|T|A|Proof+Compile+Meta+Tower] Cayley theorem

§FORM := Every group G is isomorphic to a subgroup of the symmetric group on the underlying set of G.
§DEP_HARD := S26-D-00001305
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001307|D|A|Proof+Compile+Meta] Conjugacy class

§FORM := The conjugacy class of g is {xgx^{-1}:x in G}; its stabilizer under conjugation is the centralizer C_G(g).
§DEP_HARD := S26-T-00001306
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001308|I|A|Proof+Compile+Meta] Class equation

§FORM := For finite G, |G|=|Z(G)|+sum [G:C_G(x_i)] over representatives of noncentral conjugacy classes.
§DEP_HARD := S26-D-00001307
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001309|T|A|Proof+Compile+Meta] Cauchy theorem for finite groups

§FORM := If prime p divides |G|, then G contains an element of order p.
§DEP_HARD := S26-I-00001308
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001310|T|A|Proof+Compile+Meta] Sylow existence theorem

§FORM := If |G|=p^n m with p not dividing m, G has a subgroup of order p^n.
§DEP_HARD := S26-T-00001309
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001311|T|A|Proof+Compile+Meta] Sylow conjugacy theorem
§FORM := All Sylow p-subgroups of a finite group are conjugate.
§DEP_HARD := S26-T-00001310
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001312|T|A|Proof+Compile+Meta] Sylow counting theorem
§FORM := The number n_p of Sylow p-subgroups satisfies n_p congruent 1 mod p and n_p divides m when |G|=p^n m.
§DEP_HARD := S26-T-00001311
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001313|D|A|Proof+Compile+Meta] Normal subgroup
§FORM := N<=G is normal when gNg^{-1}=N for all g in G; then the coset set G/N inherits a group structure.
§DEP_HARD := S26-T-00001312
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001314|T|A|Proof+Compile+Meta] First isomorphism theorem for groups
§FORM := For homomorphism f:G->H, G/ker f is isomorphic to im f.
§DEP_HARD := S26-D-00001313
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001315|T|A|Proof+Compile+Meta] Second isomorphism theorem for groups
§FORM := For A<=G and N normal in G, AN/N is isomorphic to A/(A intersect N).
§DEP_HARD := S26-T-00001314
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001316|T|A|Proof+Compile+Meta] Third isomorphism theorem for groups
§FORM := If N normal H normal G, then (G/N)/(H/N) is isomorphic to G/H.
§DEP_HARD := S26-T-00001315
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001317|D|A|Proof+Compile+Meta] Semidirect product
§FORM := Given N,H and an action H->Aut(N), the semidirect product N rtimes H has multiplication (n,h)(n',h')=(n h.n',hh').
§DEP_HARD := S26-T-00001316
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001318|D|A|Proof+Compile+Meta] Direct product of groups
§FORM := GxH has componentwise multiplication; both factors embed as commuting normal subgroups with trivial intersection.
§DEP_HARD := S26-D-00001317
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001319|D|A|Proof+Compile+Meta] Free group
§FORM := The free group F(S) on a set S is characterized by the universal property that every map S->G extends uniquely to a homomorphism F(S)->G.
§DEP_HARD := S26-D-00001318
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001320|B|A|Proof+Compile+Meta+Topo] Groups as symmetry objects
§FORM := Group actions turn abstract multiplication into symmetry on sets, vector spaces, manifolds, graphs, and fields; stabilizers and quotients encode loss of symmetry.
§DEP_HARD := S26-D-00001319
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001321|E|M|Proof+Compile+Meta] Cyclic group C_1 profile
§FORM := The cyclic group C_1 has order 1; it has phi(1)=1 generators. Representative generator exponents relatively prime to 1 begin [1].
§DEP_HARD := S26-B-00001320
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001322|E|M|Proof+Compile+Meta] Cyclic group C_2 profile
§FORM := The cyclic group C_2 has order 2; it has phi(2)=1 generators. Representative generator exponents relatively prime to 2 begin [1].
§DEP_HARD := S26-E-00001321
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001323|E|M|Proof+Compile+Meta] Cyclic group C_3 profile
§FORM := The cyclic group C_3 has order 3; it has phi(3)=2 generators. Representative generator exponents relatively prime to 3 begin [1, 2].
§DEP_HARD := S26-E-00001322
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001324|E|M|Proof+Compile+Meta] Cyclic group C_4 profile

§FORM := The cyclic group C_4 has order 4; it has phi(4)=2 generators. Representative generator exponents relatively prime to 4 begin [1, 3].

§DEP_HARD := S26-E-00001323

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001325|E|M|Proof+Compile+Meta] Cyclic group C_5 profile

§FORM := The cyclic group C_5 has order 5; it has phi(5)=4 generators. Representative generator exponents relatively prime to 5 begin [1, 2, 3, 4].

§DEP_HARD := S26-E-00001324

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001326|E|M|Proof+Compile+Meta] Cyclic group C_6 profile

§FORM := The cyclic group C_6 has order 6; it has phi(6)=2 generators. Representative generator exponents relatively prime to 6 begin [1, 5].

§DEP_HARD := S26-E-00001325

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001327|E|M|Proof+Compile+Meta] Cyclic group C_7 profile

§FORM := The cyclic group C_7 has order 7; it has phi(7)=6 generators. Representative generator exponents relatively prime to 7 begin [1, 2, 3, 4, 5, 6].

§DEP_HARD := S26-E-00001326

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001328|E|M|Proof+Compile+Meta] Cyclic group C_8 profile

§FORM := The cyclic group C_8 has order 8; it has phi(8)=4 generators. Representative generator exponents relatively prime to 8 begin [1, 3, 5, 7].

§DEP_HARD := S26-E-00001327

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001329|E|M|Proof+Compile+Meta] Cyclic group C_9 profile

§FORM := The cyclic group C_9 has order 9; it has phi(9)=6 generators. Representative generator exponents relatively prime to 9 begin [1, 2, 4, 5, 7, 8].

§DEP_HARD := S26-E-00001328

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001330|E|M|Proof+Compile+Meta] Cyclic group C_10 profile

§FORM := The cyclic group C_10 has order 10; it has phi(10)=4 generators. Representative generator exponents relatively prime to 10 begin [1, 3, 7, 9].

§DEP_HARD := S26-E-00001329

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001331|E|M|Proof+Compile+Meta] Cyclic group C_11 profile

§FORM := The cyclic group C_11 has order 11; it has phi(11)=10 generators. Representative generator exponents relatively prime to 11 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

§DEP_HARD := S26-E-00001330

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001332|E|M|Proof+Compile+Meta] Cyclic group C_12 profile

§FORM := The cyclic group C_12 has order 12; it has phi(12)=4 generators. Representative generator exponents relatively prime to 12 begin [1, 5, 7, 11].

§DEP_HARD := S26-E-00001331

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001333|E|M|Proof+Compile+Meta] Cyclic group C_13 profile

§FORM := The cyclic group C_13 has order 13; it has phi(13)=12 generators. Representative generator exponents relatively prime to 13 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

§DEP_HARD := S26-E-00001332

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001334|E|M|Proof+Compile+Meta] Cyclic group C_14 profile

§FORM := The cyclic group C_14 has order 14; it has phi(14)=6 generators. Representative generator exponents relatively prime to 14 begin [1, 3, 5, 9, 11, 13].

§DEP_HARD := S26-E-00001333

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001335|E|M|Proof+Compile+Meta] Cyclic group C_15 profile

§FORM := The cyclic group C_15 has order 15; it has phi(15)=8 generators. Representative generator exponents relatively prime to 15 begin [1, 2, 4, 7, 8, 11, 13, 14].

§DEP_HARD := S26-E-00001334

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001336|E|M|Proof+Compile+Meta] Cyclic group C_16 profile

§FORM := The cyclic group C_16 has order 16; it has phi(16)=8 generators. Representative generator exponents relatively prime to 16 begin [1, 3, 5, 7, 9, 11, 13, 15].

§DEP_HARD := S26-E-00001335

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001337|E|M|Proof+Compile+Meta] Cyclic group C_17 profile

§FORM := The cyclic group C_17 has order 17; it has phi(17)=16 generators. Representative generator exponents relatively prime to 17 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

§DEP_HARD := S26-E-00001336

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001338|E|M|Proof+Compile+Meta] Cyclic group C_18 profile

§FORM := The cyclic group C_18 has order 18; it has phi(18)=6 generators. Representative generator exponents relatively prime to 18 begin [1, 5, 7, 11, 13, 17].

§DEP_HARD := S26-E-00001337

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001339|E|M|Proof+Compile+Meta] Cyclic group C_19 profile

§FORM := The cyclic group C_19 has order 19; it has phi(19)=18 generators. Representative generator exponents relatively prime to 19 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

§DEP_HARD := S26-E-00001338

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001340|E|M|Proof+Compile+Meta] Cyclic group C_20 profile

§FORM := The cyclic group C_20 has order 20; it has phi(20)=8 generators. Representative generator exponents relatively prime to 20 begin [1, 3, 7, 9, 11, 13, 17, 19].

§DEP_HARD := S26-E-00001339

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001341|E|M|Proof+Compile+Meta] Cyclic group C_21 profile

§FORM := The cyclic group C_21 has order 21; it has phi(21)=12 generators. Representative generator exponents relatively prime to 21 begin [1, 2, 4, 5, 8, 10, 11, 13, 16, 17].

§DEP_HARD := S26-E-00001340

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001342|E|M|Proof+Compile+Meta] Cyclic group C_22 profile

§FORM := The cyclic group C_22 has order 22; it has phi(22)=10 generators. Representative generator exponents relatively prime to 22 begin [1, 3, 5, 7, 9, 13, 15, 17, 19, 21].

§DEP_HARD := S26-E-00001341

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001343|E|M|Proof+Compile+Meta] Cyclic group C_23 profile

§FORM := The cyclic group C_23 has order 23; it has phi(23)=22 generators. Representative generator exponents relatively prime to 23 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

§DEP_HARD := S26-E-00001342

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001344|E|M|Proof+Compile+Meta] Cyclic group C_24 profile

§FORM := The cyclic group C_24 has order 24; it has phi(24)=8 generators. Representative generator exponents relatively prime to 24 begin [1, 5, 7, 11, 13, 17, 19, 23].

§DEP_HARD := S26-E-00001343

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001345|E|M|Proof+Compile+Meta] Cyclic group C_25 profile

§FORM := The cyclic group C_25 has order 25; it has phi(25)=20 generators. Representative generator exponents relatively prime to 25 begin [1, 2, 3, 4, 6, 7, 8, 9, 11, 12].

§DEP_HARD := S26-E-00001344

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001346|E|M|Proof+Compile+Meta] Cyclic group C_26 profile

§FORM := The cyclic group C_26 has order 26; it has phi(26)=12 generators. Representative generator exponents relatively prime to 26 begin [1, 3, 5, 7, 9, 11, 15, 17, 19, 21].

§DEP_HARD := S26-E-00001345

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001347|E|M|Proof+Compile+Meta] Cyclic group C_27 profile

§FORM := The cyclic group C_27 has order 27; it has phi(27)=18 generators. Representative generator exponents relatively prime to 27 begin [1, 2, 4, 5, 7, 8, 10, 11, 13, 14].

§DEP_HARD := S26-E-00001346

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001348|E|M|Proof+Compile+Meta] Cyclic group C_28 profile

§FORM := The cyclic group C_28 has order 28; it has phi(28)=12 generators. Representative generator exponents relatively prime to 28 begin [1, 3, 5, 9, 11, 13, 15, 17, 19, 23].

§DEP_HARD := S26-E-00001347

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001349|E|M|Proof+Compile+Meta] Cyclic group C_29 profile

§FORM := The cyclic group C_29 has order 29; it has phi(29)=28 generators. Representative generator exponents relatively prime to 29 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

§DEP_HARD := S26-E-00001348

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001350|E|M|Proof+Compile+Meta] Cyclic group C_30 profile

§FORM := The cyclic group C_30 has order 30; it has phi(30)=8 generators. Representative generator exponents relatively prime to 30 begin [1, 7, 11, 13, 17, 19, 23, 29].

§DEP_HARD := S26-E-00001349

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001351|D|A|Proof+Compile+Meta+IG] Commutator of group elements

§FORM := [g,h]=g^{-1}h^{-1}gh measures the failure of g and h to commute.

§DEP_HARD := S26-D-00000103

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001352|D|A|Proof+Compile+Meta] Derived subgroup

§FORM := G'=[G,G] is the subgroup generated by all commutators; G/G' is the abelianization of G.

§DEP_HARD := S26-D-00001351

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001353|D|A|Proof+Compile+Meta] Derived series

§FORM := G^{(0)}=G and G^{(n+1)}=[G^{(n)},G^{(n)}].

§DEP_HARD := S26-D-00001352

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001354|D|A|Proof+Compile+Meta] Solvable group

§FORM := G is solvable when its derived series reaches the identity after finitely many steps.

§DEP_HARD := S26-D-00001353

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001355|D|A|Proof+Compile+Meta] Lower central series

§FORM := gamma_1(G)=G and gamma_{n+1}(G)=[gamma_n(G),G].

§DEP_HARD := S26-D-00001354

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001356|D|A|Proof+Compile+Meta] Nilpotent group

§FORM := G is nilpotent when its lower central series reaches the identity after finitely many steps.

§DEP_HARD := S26-D-00001355

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001357|T|A|Proof+Compile+Meta] Finite p-groups have nontrivial center

§FORM := Every nontrivial finite group of prime-power order has a nontrivial center.

§DEP_HARD := S26-D-00001356

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001358|T|A|Proof+Compile+Meta] Finite p-groups are nilpotent

§FORM := Every finite p-group is nilpotent.

§DEP_HARD := S26-T-00001357

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001359|D|A|Proof+Compile+Meta] Simple group

§FORM := A nontrivial group is simple when its only normal subgroups are the identity and the whole group.

§DEP_HARD := S26-T-00001358

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001360|T|A|Proof+Compile+Meta] Simplicity of alternating groups

§FORM := A_n is simple for n>=5.

§DEP_HARD := S26-D-00001359

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001361|D|A|Proof+Compile+Meta] Composition series

§FORM := A composition series is a finite subnormal chain whose factors are simple.
§DEP_HARD := S26-T-00001360
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001362|T|A|Proof+Compile+Meta] Jordan-Holder theorem

§FORM := Any two composition series of a finite-length group have the same multiset of simple factors up to isomorphism and permutation.
§DEP_HARD := S26-D-00001361
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001363|D|A|Proof+Compile+Meta] Characteristic subgroup

§FORM := H<=G is characteristic when every automorphism of G preserves H; characteristic subgroups are normal.
§DEP_HARD := S26-T-00001362
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001364|D|A|Proof+Compile+Meta] Frattini subgroup

§FORM := Phi(G) is the intersection of all maximal subgroups of G, with Phi(G)=G if there are none.
§DEP_HARD := S26-D-00001363
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001365|D|A|Proof+Compile+Meta] Center of a group

§FORM := Z(G)={z in G:zg=gz for all g in G}.
§DEP_HARD := S26-D-00001364
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001366|D|A|Proof+Compile+Meta] Central product

§FORM := A central product glues groups along central subgroups; unlike direct products, the factors may intersect nontrivially in the center.
§DEP_HARD := S26-D-00001365
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001367|T|A|Proof+Compile+Meta] Structure theorem for finitely generated abelian groups

§FORM := Every finitely generated abelian group is isomorphic to Z^r direct-sum a finite direct sum of cyclic p-power groups; the invariants are unique up to order.
§DEP_HARD := S26-D-00001366
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001368|T|A|Proof+Compile+Meta] Burnside p^a q^b theorem

§FORM := Every finite group whose order is divisible by at most two distinct primes is solvable.
§DEP_HARD := S26-T-00001367
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00001369|N|A|Proof+Compile+Meta] Classification boundary

§FORM := The classification of finite simple groups is a deep theorem family; this Synopsis records its consequence-level interface separately from proof infrastructure.
§DEP_HARD := S26-T-00001368
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001370|B|A|Proof+Compile+Meta] Group extensions

§FORM := An exact sequence 1->N->G->Q->1 presents G as an extension of Q by N; extension theory links group structure to cohomological invariants.
§DEP_HARD := S26-N-00001369
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001371|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=3

§FORM := |S_3|=6 and |A_3|=3. A_3 is simple for n>=5.
§DEP_HARD := S26-B-00001370
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001372|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=4

§FORM := |S_4|=24 and |A_4|=12. A_4 is simple for n>=5.
§DEP_HARD := S26-E-00001371
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001373|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=5

§FORM := |S_5|=120 and |A_5|=60. A_5 is simple for n>=5.
§DEP_HARD := S26-E-00001372
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001374|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=6
§FORM := |S_6|=720 and |A_6|=360. A_6 is simple for n>=5.
§DEP_HARD := S26-E-00001373
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001375|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=7
§FORM := |S_7|=5040 and |A_7|=2520. A_7 is simple for n>=5.
§DEP_HARD := S26-E-00001374
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001376|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=8
§FORM := |S_8|=40320 and |A_8|=20160. A_8 is simple for n>=5.
§DEP_HARD := S26-E-00001375
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001377|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=9
§FORM := |S_9|=362880 and |A_9|=181440. A_9 is simple for n>=5.
§DEP_HARD := S26-E-00001376
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001378|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=10
§FORM := |S_10|=3628800 and |A_10|=1814400. A_10 is simple for n>=5.
§DEP_HARD := S26-E-00001377
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001379|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=11
§FORM := |S_11|=39916800 and |A_11|=19958400. A_11 is simple for n>=5.
§DEP_HARD := S26-E-00001378
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001380|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=12
§FORM := |S_12|=479001600 and |A_12|=239500800. A_12 is simple for n>=5.
§DEP_HARD := S26-E-00001379
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001381|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=13
§FORM := |S_13|=6227020800 and |A_13|=3113510400. A_13 is simple for n>=5.
§DEP_HARD := S26-E-00001380
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001382|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=14
§FORM := |S_14|=87178291200 and |A_14|=43589145600. A_14 is simple for n>=5.
§DEP_HARD := S26-E-00001381
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001383|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=15
§FORM := |S_15|=1307674368000 and |A_15|=653837184000. A_15 is simple for n>=5.
§DEP_HARD := S26-E-00001382
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001384|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=16
§FORM := |S_16|=20922789888000 and |A_16|=10461394944000. A_16 is simple for n>=5.
§DEP_HARD := S26-E-00001383
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001385|E|M|Proof+Compile+Meta] Symmetric and alternating group orders at n=17
§FORM := |S_17|=355687428096000 and |A_17|=177843714048000. A_17 is simple for n>=5.
§DEP_HARD := S26-E-00001384
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001386|E|M|Proof+Compile+Meta] Dihedral group D_2 order profile
§FORM := With the convention D_2=<r,s | r^2=s^2=1, srs=r^(-1)>, the group has order 4.
§DEP_HARD := S26-E-00001385
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001387|E|M|Proof+Compile+Meta] Dihedral group D_3 order profile
§FORM := With the convention D_3=<r,s | r^3=s^2=1, srs=r^(-1)>, the group has order 6.
§DEP_HARD := S26-E-00001386
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001388|E|M|Proof+Compile+Meta] Dihedral group D_4 order profile
§FORM := With the convention D_4=<r,s | r^4=s^2=1, srs=r^(-1)>, the group has order 8.
§DEP_HARD := S26-E-00001387
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001389|E|M|Proof+Compile+Meta] Dihedral group D_5 order profile
§FORM := With the convention D_5=<r,s | r^5=s^2=1, srs=r^(-1)>, the group has order 10.
§DEP_HARD := S26-E-00001388
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001390|E|M|Proof+Compile+Meta] Dihedral group D_6 order profile
§FORM := With the convention D_6=<r,s | r^6=s^2=1, srs=r^(-1)>, the group has order 12.
§DEP_HARD := S26-E-00001389
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001391|E|M|Proof+Compile+Meta] Dihedral group D_7 order profile
§FORM := With the convention D_7=<r,s | r^7=s^2=1, srs=r^(-1)>, the group has order 14.
§DEP_HARD := S26-E-00001390
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001392|E|M|Proof+Compile+Meta] Dihedral group D_8 order profile
§FORM := With the convention D_8=<r,s | r^8=s^2=1, srs=r^(-1)>, the group has order 16.
§DEP_HARD := S26-E-00001391
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001393|E|M|Proof+Compile+Meta] Dihedral group D_9 order profile
§FORM := With the convention D_9=<r,s | r^9=s^2=1, srs=r^(-1)>, the group has order 18.
§DEP_HARD := S26-E-00001392
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001394|E|M|Proof+Compile+Meta] Dihedral group D_10 order profile
§FORM := With the convention D_10=<r,s | r^10=s^2=1, srs=r^(-1)>, the group has order 20.
§DEP_HARD := S26-E-00001393
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001395|E|M|Proof+Compile+Meta] Dihedral group D_11 order profile
§FORM := With the convention D_11=<r,s | r^11=s^2=1, srs=r^(-1)>, the group has order 22.
§DEP_HARD := S26-E-00001394
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001396|E|M|Proof+Compile+Meta] Dihedral group D_12 order profile
§FORM := With the convention D_12=<r,s | r^12=s^2=1, srs=r^(-1)>, the group has order 24.
§DEP_HARD := S26-E-00001395
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001397|E|M|Proof+Compile+Meta] Dihedral group D_13 order profile
§FORM := With the convention D_13=<r,s | r^13=s^2=1, srs=r^(-1)>, the group has order 26.
§DEP_HARD := S26-E-00001396
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001398|E|M|Proof+Compile+Meta] Dihedral group D_14 order profile
§FORM := With the convention D_14=<r,s | r^14=s^2=1, srs=r^(-1)>, the group has order 28.
§DEP_HARD := S26-E-00001397
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001399|E|M|Proof+Compile+Meta] Dihedral group D_15 order profile
§FORM := With the convention D_15=<r,s | r^15=s^2=1, srs=r^(-1)>, the group has order 30.
§DEP_HARD := S26-E-00001398
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001400|E|M|Proof+Compile+Meta] Dihedral group D_16 order profile
§FORM := With the convention D_16=<r,s | r^16=s^2=1, srs=r^(-1)>, the group has order 32.
§DEP_HARD := S26-E-00001399
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001401|D|A|Proof+Compile+Meta] Linear representation
§FORM := A representation of G on vector space V over field F is a homomorphism rho:G->GL(V).
§DEP_HARD := S26-D-00000149
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001402|D|A|Proof+Compile+Meta] G-module

§FORM := A G-module over F is an F-vector space V with an F-linear G-action; equivalently it is a module over the group algebra F[G].
§DEP_HARD := S26-D-00001401
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001403|D|A|Proof+Compile+Meta] Invariant subspace

§FORM := W<=V is G-invariant when rho(g)W subseteq W for every g in G.
§DEP_HARD := S26-D-00001402
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001404|D|A|Proof+Compile+Meta] Irreducible representation

§FORM := A nonzero representation is irreducible when it has no invariant subspaces other than 0 and V.
§DEP_HARD := S26-D-00001403
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001405|D|A|Proof+Compile+Meta] Completely reducible representation

§FORM := A representation is completely reducible when it is a direct sum of irreducible representations.
§DEP_HARD := S26-D-00001404
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001406|T|A|Proof+Compile+Meta] Maschke theorem

§FORM := If G is finite and char(F) does not divide |G|, every finite-dimensional F-representation of G is completely reducible.
§DEP_HARD := S26-D-00001405
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001407|D|A|Proof+Compile+Meta] Character of a representation

§FORM := For finite-dimensional rho, chi(g)=Tr(rho(g)); characters are constant on conjugacy classes.
§DEP_HARD := S26-T-00001406
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001408|D|A|Proof+Compile+Meta] Character inner product

§FORM := For complex class functions on finite G, <chi,psi>=(1/|G|)sum_{g in G} chi(g) overline{psi(g)}.
§DEP_HARD := S26-D-00001407
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001409|T|A|Proof+Compile+Meta] Orthogonality of irreducible characters

§FORM := Distinct irreducible complex characters are orthogonal and each irreducible character has norm 1 under the character inner product.
§DEP_HARD := S26-D-00001408
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001410|T|A|Proof+Compile+Meta] Number of irreducible characters

§FORM := For a finite group over C, the number of irreducible characters equals the number of conjugacy classes.
§DEP_HARD := S26-T-00001409
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001411|I|A|Proof+Compile+Meta] Sum of squared irreducible dimensions

§FORM := For finite G over C, sum_{rho irreducible}(dim rho)^2=|G|.
§DEP_HARD := S26-T-00001410
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001412|D|A|Proof+Compile+Meta] Regular representation

§FORM := The left regular representation acts on F[G] by left multiplication.
§DEP_HARD := S26-I-00001411
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001413|I|A|Proof+Compile+Meta] Regular-character decomposition

§FORM := Over C, the regular representation contains each irreducible representation with multiplicity equal to its dimension.
§DEP_HARD := S26-D-00001412
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001414|D|A|Proof+Compile+Meta] Induced representation

§FORM := For H<=G and an H-representation W, Ind_H^G W is obtained from F[G] tensor_{F[H]} W.
§DEP_HARD := S26-I-00001413
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001415|T|A|Proof+Compile+Meta] Frobenius reciprocity

§FORM := For suitable finite-dimensional representations, Hom_G(Ind_H^G W,V) is naturally isomorphic to Hom_H(W,Res_H^G V).
§DEP_HARD := S26-D-00001414
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001416|D|A|Proof+Compile+Meta] Tensor product representation

§FORM := rho tensor sigma acts by (rho tensor sigma)(g)=rho(g) tensor sigma(g).

§DEP_HARD := S26-T-00001415

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001417|I|A|Proof+Compile+Meta] Character of a tensor product

§FORM := chi_{V tensor W}(g)=chi_V(g) chi_W(g).

§DEP_HARD := S26-D-00001416

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001418|D|A|Proof+Compile+Meta] Dual representation

§FORM := The dual representation on V* is rho*(g)=rho(g^{-1})^T in a chosen basis.

§DEP_HARD := S26-I-00001417

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001419|I|A|Proof+Compile+Meta] Character of the dual

§FORM := For complex unitary realizations of finite-group representations, chi_{V*}(g)=overline{chi_V(g)}.

§DEP_HARD := S26-D-00001418

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001420|B|A|Proof+Compile+Meta+Spectral] Representation-theoretic Fourier transform

§FORM := For finite groups, decomposing functions on G into irreducible representations generalizes the ordinary Fourier transform from abelian groups to noncommutative symmetry.

§DEP_HARD := S26-I-00001419

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001421|E|M|Proof+Compile+Meta] Complex irreducibles of C_1

§FORM := The cyclic group C_1 has exactly 1 one-dimensional irreducible complex representations, indexed by the 1-th roots of unity. The sum of squared dimensions is 1.

§DEP_HARD := S26-B-00001420

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001422|E|M|Proof+Compile+Meta] Complex irreducibles of C_2

§FORM := The cyclic group C_2 has exactly 2 one-dimensional irreducible complex representations, indexed by the 2-th roots of unity. The sum of squared dimensions is 2.

§DEP_HARD := S26-E-00001421

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001423|E|M|Proof+Compile+Meta] Complex irreducibles of C_3

§FORM := The cyclic group C_3 has exactly 3 one-dimensional irreducible complex representations, indexed by the 3-th roots of unity. The sum of squared dimensions is 3.

§DEP_HARD := S26-E-00001422

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001424|E|M|Proof+Compile+Meta] Complex irreducibles of C_4

§FORM := The cyclic group C_4 has exactly 4 one-dimensional irreducible complex representations, indexed by the 4-th roots of unity. The sum of squared dimensions is 4.

§DEP_HARD := S26-E-00001423

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001425|E|M|Proof+Compile+Meta] Complex irreducibles of C_5

§FORM := The cyclic group C_5 has exactly 5 one-dimensional irreducible complex representations, indexed by the 5-th roots of unity. The sum of squared dimensions is 5.

§DEP_HARD := S26-E-00001424

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001426|E|M|Proof+Compile+Meta] Complex irreducibles of C_6

§FORM := The cyclic group C_6 has exactly 6 one-dimensional irreducible complex representations, indexed by the 6-th roots of unity. The sum of squared dimensions is 6.

§DEP_HARD := S26-E-00001425

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001427|E|M|Proof+Compile+Meta] Complex irreducibles of C_7

§FORM := The cyclic group C_7 has exactly 7 one-dimensional irreducible complex representations, indexed by the 7-th roots of unity. The sum of squared dimensions is 7.

§DEP_HARD := S26-E-00001426

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001428|E|M|Proof+Compile+Meta] Complex irreducibles of C_8

§FORM := The cyclic group C_8 has exactly 8 one-dimensional irreducible complex representations, indexed by the 8-th roots of unity. The sum of squared dimensions is 8.

§DEP_HARD := S26-E-00001427

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001429|E|M|Proof+Compile+Meta] Complex irreducibles of C_9

§FORM := The cyclic group C_9 has exactly 9 one-dimensional irreducible complex representations, indexed by the 9-th roots of unity. The sum of squared dimensions is 9.

§DEP_HARD := S26-E-00001428

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001430|E|M|Proof+Compile+Meta] Complex irreducibles of C_10

§FORM := The cyclic group C_10 has exactly 10 one-dimensional irreducible complex representations, indexed by the 10-th roots of unity. The sum of squared dimensions is 10.

§DEP_HARD := S26-E-00001429

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001431|E|M|Proof+Compile+Meta] Complex irreducibles of C_11

§FORM := The cyclic group C_11 has exactly 11 one-dimensional irreducible complex representations, indexed by the 11-th roots of unity. The sum of squared dimensions is 11.

§DEP_HARD := S26-E-00001430

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001432|E|M|Proof+Compile+Meta] Complex irreducibles of C_12

§FORM := The cyclic group C_12 has exactly 12 one-dimensional irreducible complex representations, indexed by the 12-th roots of unity. The sum of squared dimensions is 12.

§DEP_HARD := S26-E-00001431

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001433|E|M|Proof+Compile+Meta] Complex irreducibles of C_13

§FORM := The cyclic group C_13 has exactly 13 one-dimensional irreducible complex representations, indexed by the 13-th roots of unity. The sum of squared dimensions is 13.

§DEP_HARD := S26-E-00001432

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001434|E|M|Proof+Compile+Meta] Complex irreducibles of C_14

§FORM := The cyclic group C_14 has exactly 14 one-dimensional irreducible complex representations, indexed by the 14-th roots of unity. The sum of squared dimensions is 14.

§DEP_HARD := S26-E-00001433

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001435|E|M|Proof+Compile+Meta] Complex irreducibles of C_15

§FORM := The cyclic group C_15 has exactly 15 one-dimensional irreducible complex representations, indexed by the 15-th roots of unity. The sum of squared dimensions is 15.

§DEP_HARD := S26-E-00001434

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001436|E|M|Proof+Compile+Meta] Complex irreducibles of C_16

§FORM := The cyclic group C_16 has exactly 16 one-dimensional irreducible complex representations, indexed by the 16-th roots of unity. The sum of squared dimensions is 16.

§DEP_HARD := S26-E-00001435

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001437|E|M|Proof+Compile+Meta] Complex irreducibles of C_17

§FORM := The cyclic group C_17 has exactly 17 one-dimensional irreducible complex representations, indexed by the 17-th roots of unity. The sum of squared dimensions is 17.

§DEP_HARD := S26-E-00001436

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001438|E|M|Proof+Compile+Meta] Complex irreducibles of C_18

§FORM := The cyclic group C_18 has exactly 18 one-dimensional irreducible complex representations, indexed by the 18-th roots of unity. The sum of squared dimensions is 18.

§DEP_HARD := S26-E-00001437

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001439|E|M|Proof+Compile+Meta] Complex irreducibles of C_19

§FORM := The cyclic group C_19 has exactly 19 one-dimensional irreducible complex representations, indexed by the 19-th roots of unity. The sum of squared dimensions is 19.

§DEP_HARD := S26-E-00001438

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001440|E|M|Proof+Compile+Meta] Complex irreducibles of C_20

§FORM := The cyclic group C_20 has exactly 20 one-dimensional irreducible complex representations, indexed by the 20-th roots of unity. The sum of squared dimensions is 20.

§DEP_HARD := S26-E-00001439

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001441|E|M|Proof+Compile+Meta] Complex irreducibles of C_21

§FORM := The cyclic group C_21 has exactly 21 one-dimensional irreducible complex representations, indexed by the 21-th roots of unity. The sum of squared dimensions is 21.

§DEP_HARD := S26-E-00001440

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001442|E|M|Proof+Compile+Meta] Complex irreducibles of C_22

§FORM := The cyclic group C_22 has exactly 22 one-dimensional irreducible complex representations, indexed by the 22-th roots of unity. The sum of squared dimensions is 22.

§DEP_HARD := S26-E-00001441

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001443|E|M|Proof+Compile+Meta] Complex irreducibles of C_23

§FORM := The cyclic group C_23 has exactly 23 one-dimensional irreducible complex representations, indexed by the 23-th roots of unity. The sum of squared dimensions is 23.

§DEP_HARD := S26-E-00001442

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001444|E|M|Proof+Compile+Meta] Complex irreducibles of C_24

§FORM := The cyclic group C_24 has exactly 24 one-dimensional irreducible complex representations, indexed by the 24-th roots of unity. The sum of squared dimensions is 24.

§DEP_HARD := S26-E-00001443

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001445|E|M|Proof+Compile+Meta] Complex irreducibles of C_25

§FORM := The cyclic group C_25 has exactly 25 one-dimensional irreducible complex representations, indexed by the 25-th roots of unity. The sum of squared dimensions is 25.

§DEP_HARD := S26-E-00001444

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001446|E|M|Proof+Compile+Meta] Irreducible-count profile for S_1

§FORM := The symmetric group S_1 has 1 complex irreducible representations, equal to the number of partitions of 1.

§DEP_HARD := S26-E-00001445

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001447|E|M|Proof+Compile+Meta] Irreducible-count profile for S_2

§FORM := The symmetric group S_2 has 2 complex irreducible representations, equal to the number of partitions of 2.

§DEP_HARD := S26-E-00001446

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001448|E|M|Proof+Compile+Meta] Irreducible-count profile for S_3

§FORM := The symmetric group S_3 has 3 complex irreducible representations, equal to the number of partitions of 3.

§DEP_HARD := S26-E-00001447

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001449|E|M|Proof+Compile+Meta] Irreducible-count profile for S_4

§FORM := The symmetric group S_4 has 5 complex irreducible representations, equal to the number of partitions of 4.

§DEP_HARD := S26-E-00001448

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001450|E|M|Proof+Compile+Meta] Irreducible-count profile for S_5

§FORM := The symmetric group S_5 has 7 complex irreducible representations, equal to the number of partitions of 5.

§DEP_HARD := S26-E-00001449

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001451|D|A|Proof+Compile+Meta] Ring ideal

§FORM := An ideal I of a ring R is an additive subgroup stable under multiplication by arbitrary ring elements on the required side; in commutative rings left/right distinctions disappear.

§DEP_HARD := S26-D-00000101

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001452|D|A|Proof+Compile+Meta] Prime ideal

§FORM := In a commutative ring R, a proper ideal p is prime when ab in p implies a in p or b in p.

§DEP_HARD := S26-D-00001451

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001453|D|A|Proof+Compile+Meta] Maximal ideal
§FORM := A proper ideal m is maximal when there is no proper ideal strictly between m and R.
§DEP_HARD := S26-D-00001452
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001454|T|A|Proof+Compile+Meta] Maximal ideals are prime
§FORM := Every maximal ideal in a commutative ring with identity is prime.
§DEP_HARD := S26-D-00001453
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001455|T|A|Proof+Compile+Meta] Prime quotient criterion
§FORM := For commutative R, p is prime iff R/p is an integral domain.
§DEP_HARD := S26-T-00001454
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001456|T|A|Proof+Compile+Meta] Maximal quotient criterion
§FORM := For commutative R, m is maximal iff R/m is a field.
§DEP_HARD := S26-T-00001455
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001457|D|A|Proof+Compile+Meta] Principal ideal domain
§FORM := A PID is an integral domain in which every ideal is generated by one element.
§DEP_HARD := S26-T-00001456
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001458|D|A|Proof+Compile+Meta] Euclidean domain
§FORM := A Euclidean domain is an integral domain with a Euclidean function supporting division with remainder; every Euclidean domain is a PID.
§DEP_HARD := S26-D-00001457
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001459|T|A|Proof+Compile+Meta] PID implies UFD
§FORM := Every principal ideal domain is a unique factorization domain.
§DEP_HARD := S26-D-00001458
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001460|D|A|Proof+Compile+Meta] Module over a ring
§FORM := An R-module generalizes a vector space by allowing scalars from a ring R rather than necessarily a field.
§DEP_HARD := S26-T-00001459
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001461|D|A|Proof+Compile+Meta] Submodule and quotient module
§FORM := A submodule N<=M is closed under module operations; M/N inherits an R-module structure.
§DEP_HARD := S26-D-00001460
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001462|T|A|Proof+Compile+Meta] First isomorphism theorem for modules
§FORM := For an R-linear map f:M->N, M/ker f is isomorphic to im f.
§DEP_HARD := S26-D-00001461
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001463|D|A|Proof+Compile+Meta] Free module
§FORM := A free R-module has a basis in the module sense and is isomorphic to a direct sum of copies of R.
§DEP_HARD := S26-T-00001462
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001464|D|A|Proof+Compile+Meta] Torsion element of a module
§FORM := For module M over an integral domain R, m is torsion when rm=0 for some nonzero r in R.
§DEP_HARD := S26-D-00001463
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001465|D|A|Proof+Compile+Meta] Finitely generated module
§FORM := M is finitely generated when a finite subset spans M as an R-module.
§DEP_HARD := S26-D-00001464
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001466|T|A|Proof+Compile+Meta] Structure theorem over a PID

§FORM := Every finitely generated module over a PID decomposes into a free part plus cyclic torsion modules determined by invariant factors or elementary divisors.

§DEP_HARD := S26-D-00001465

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001467|D|A|Proof+Compile+Meta] Exact sequence of modules

§FORM := A sequence of module maps is exact at a term when the image of the incoming map equals the kernel of the outgoing map.

§DEP_HARD := S26-T-00001466

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001468|D|A|Proof+Compile+Meta] Projective module

§FORM := P is projective when Hom(P,-) preserves epimorphism lifting; equivalently every short exact sequence ending in P splits.

§DEP_HARD := S26-D-00001467

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001469|D|A|Proof+Compile+Meta] Injective module

§FORM := I is injective when Hom(-,I) sends monomorphisms to surjections on Hom sets, equivalently extensions across submodules exist.

§DEP_HARD := S26-D-00001468

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001470|D|A|Proof+Compile+Meta] Flat module

§FORM := An R-module M is flat when tensoring with M preserves exactness of short exact sequences in the appropriate module category; equivalently it preserves injections in the standard one-sided formulation.

§DEP_HARD := S26-D-00001469

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001471|E|M|Proof+Compile+Meta] Residue ring Z/2Z profile

§FORM := Z/2Z has 2 elements and 1 units. It is a field iff 2 is prime; its unit residues are [1].

§DEP_HARD := S26-D-00001470

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001472|E|M|Proof+Compile+Meta] Residue ring Z/3Z profile

§FORM := Z/3Z has 3 elements and 2 units. It is a field iff 3 is prime; its unit residues are [1, 2].

§DEP_HARD := S26-E-00001471

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001473|E|M|Proof+Compile+Meta] Residue ring Z/4Z profile

§FORM := Z/4Z has 4 elements and 2 units. It is a field iff 4 is prime; its unit residues are [1, 3].

§DEP_HARD := S26-E-00001472

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001474|E|M|Proof+Compile+Meta] Residue ring Z/5Z profile

§FORM := Z/5Z has 5 elements and 4 units. It is a field iff 5 is prime; its unit residues are [1, 2, 3, 4].

§DEP_HARD := S26-E-00001473

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001475|E|M|Proof+Compile+Meta] Residue ring Z/6Z profile

§FORM := Z/6Z has 6 elements and 2 units. It is a field iff 6 is prime; its unit residues are [1, 5].

§DEP_HARD := S26-E-00001474

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001476|E|M|Proof+Compile+Meta] Residue ring Z/7Z profile

§FORM := Z/7Z has 7 elements and 6 units. It is a field iff 7 is prime; its unit residues are [1, 2, 3, 4, 5, 6].

§DEP_HARD := S26-E-00001475

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001477|E|M|Proof+Compile+Meta] Residue ring Z/8Z profile

§FORM := Z/8Z has 8 elements and 4 units. It is a field iff 8 is prime; its unit residues are [1, 3, 5, 7].

§DEP_HARD := S26-E-00001476

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001478|E|M|Proof+Compile+Meta] Residue ring Z/9Z profile

§FORM := Z/9Z has 9 elements and 6 units. It is a field iff 9 is prime; its unit residues are [1, 2, 4, 5, 7, 8].

§DEP_HARD := S26-E-00001477

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001479|E|M|Proof+Compile+Meta] Residue ring Z/10Z profile

§FORM := Z/10Z has 10 elements and 4 units. It is a field iff 10 is prime; its unit residues are [1, 3, 7, 9].
§DEP_HARD := S26-E-00001478
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001480|E|M|Proof+Compile+Meta] Residue ring Z/11Z profile

§FORM := Z/11Z has 11 elements and 10 units. It is a field iff 11 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].
§DEP_HARD := S26-E-00001479
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001481|E|M|Proof+Compile+Meta] Residue ring Z/12Z profile

§FORM := Z/12Z has 12 elements and 4 units. It is a field iff 12 is prime; its unit residues are [1, 5, 7, 11].
§DEP_HARD := S26-E-00001480
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001482|E|M|Proof+Compile+Meta] Residue ring Z/13Z profile

§FORM := Z/13Z has 13 elements and 12 units. It is a field iff 13 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12].
§DEP_HARD := S26-E-00001481
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001483|E|M|Proof+Compile+Meta] Residue ring Z/14Z profile

§FORM := Z/14Z has 14 elements and 6 units. It is a field iff 14 is prime; its unit residues are [1, 3, 5, 9, 11, 13].
§DEP_HARD := S26-E-00001482
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001484|E|M|Proof+Compile+Meta] Residue ring Z/15Z profile

§FORM := Z/15Z has 15 elements and 8 units. It is a field iff 15 is prime; its unit residues are [1, 2, 4, 7, 8, 11, 13, 14].
§DEP_HARD := S26-E-00001483
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001485|E|M|Proof+Compile+Meta] Residue ring Z/16Z profile

§FORM := Z/16Z has 16 elements and 8 units. It is a field iff 16 is prime; its unit residues are [1, 3, 5, 7, 9, 11, 13, 15].
§DEP_HARD := S26-E-00001484
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001486|E|M|Proof+Compile+Meta] Residue ring Z/17Z profile

§FORM := Z/17Z has 17 elements and 16 units. It is a field iff 17 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16].
§DEP_HARD := S26-E-00001485
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001487|E|M|Proof+Compile+Meta] Residue ring Z/18Z profile

§FORM := Z/18Z has 18 elements and 6 units. It is a field iff 18 is prime; its unit residues are [1, 5, 7, 11, 13, 17].
§DEP_HARD := S26-E-00001486
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001488|E|M|Proof+Compile+Meta] Residue ring Z/19Z profile

§FORM := Z/19Z has 19 elements and 18 units. It is a field iff 19 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18].
§DEP_HARD := S26-E-00001487
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001489|E|M|Proof+Compile+Meta] Residue ring Z/20Z profile

§FORM := Z/20Z has 20 elements and 8 units. It is a field iff 20 is prime; its unit residues are [1, 3, 7, 9, 11, 13, 17, 19].
§DEP_HARD := S26-E-00001488
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001490|E|M|Proof+Compile+Meta] Residue ring Z/21Z profile

§FORM := Z/21Z has 21 elements and 12 units. It is a field iff 21 is prime; its unit residues are [1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20].
§DEP_HARD := S26-E-00001489
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001491|E|M|Proof+Compile+Meta] Residue ring Z/22Z profile

§FORM := Z/22Z has 22 elements and 10 units. It is a field iff 22 is prime; its unit residues are [1, 3, 5, 7, 9, 13, 15, 17, 19, 21].
§DEP_HARD := S26-E-00001490
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001492|E|M|Proof+Compile+Meta] Residue ring Z/23Z profile

§FORM := Z/23Z has 23 elements and 22 units. It is a field iff 23 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] ....

§DEP_HARD := S26-E-00001491

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001493|E|M|Proof+Compile+Meta] Residue ring Z/24Z profile

§FORM := Z/24Z has 24 elements and 8 units. It is a field iff 24 is prime; its unit residues are [1, 5, 7, 11, 13, 17, 19, 23].

§DEP_HARD := S26-E-00001492

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001494|E|M|Proof+Compile+Meta] Residue ring Z/25Z profile

§FORM := Z/25Z has 25 elements and 20 units. It is a field iff 25 is prime; its unit residues are [1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17, 18, 19, 21, 22, 23, 24].

§DEP_HARD := S26-E-00001493

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001495|E|M|Proof+Compile+Meta] Residue ring Z/26Z profile

§FORM := Z/26Z has 26 elements and 12 units. It is a field iff 26 is prime; its unit residues are [1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25].

§DEP_HARD := S26-E-00001494

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001496|E|M|Proof+Compile+Meta] Residue ring Z/27Z profile

§FORM := Z/27Z has 27 elements and 18 units. It is a field iff 27 is prime; its unit residues are [1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26].

§DEP_HARD := S26-E-00001495

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001497|E|M|Proof+Compile+Meta] Residue ring Z/28Z profile

§FORM := Z/28Z has 28 elements and 12 units. It is a field iff 28 is prime; its unit residues are [1, 3, 5, 9, 11, 13, 15, 17, 19, 23, 25, 27].

§DEP_HARD := S26-E-00001496

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001498|E|M|Proof+Compile+Meta] Residue ring Z/29Z profile

§FORM := Z/29Z has 29 elements and 28 units. It is a field iff 29 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] ....

§DEP_HARD := S26-E-00001497

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001499|E|M|Proof+Compile+Meta] Residue ring Z/30Z profile

§FORM := Z/30Z has 30 elements and 8 units. It is a field iff 30 is prime; its unit residues are [1, 7, 11, 13, 17, 19, 23, 29].

§DEP_HARD := S26-E-00001498

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001500|E|M|Proof+Compile+Meta] Residue ring Z/31Z profile

§FORM := Z/31Z has 31 elements and 30 units. It is a field iff 31 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] ....

§DEP_HARD := S26-E-00001499

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001501|D|A|Proof+Compile+Meta] Noetherian ring

§FORM := A commutative ring R is Noetherian when every ascending chain of ideals stabilizes, equivalently every ideal is finitely generated.

§DEP_HARD := S26-D-00000101

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001502|T|A|Proof+Compile+Meta+Spectral] Hilbert basis theorem

§FORM := If R is Noetherian, then R[x] is Noetherian; iterating gives R[x_1,...,x_n] Noetherian.

§DEP_HARD := S26-D-00001501

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001503|D|A|Proof+Compile+Meta] Localization of a ring

§FORM := For multiplicative S subset R, S^{-1}R formally inverts all elements of S.

§DEP_HARD := S26-T-00001502

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001504|D|A|Proof+Compile+Meta] Localization at a prime

§FORM := R_p=(R minus p)^(-1)R is the local ring obtained by inverting all elements outside prime p.

§DEP_HARD := S26-D-00001503

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001505|D|A|Proof+Compile+Meta] Local ring
§FORM := A commutative ring is local when it has a unique maximal ideal.
§DEP_HARD := S26-D-00001504
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001506|T|A|Proof+Compile+Meta] Unit criterion in a local ring
§FORM := In local ring (R,m), the units are exactly R minus m.
§DEP_HARD := S26-D-00001505
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001507|D|A|Proof+Compile+Meta+Tower] Radical of an ideal
§FORM := sqrt(I)={r in R:r^n in I for some n>=1}.
§DEP_HARD := S26-T-00001506
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001508|T|A|Proof+Compile+Meta+Tower] Radical as intersection of primes
§FORM := sqrt(I) is the intersection of all prime ideals containing I.
§DEP_HARD := S26-D-00001507
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001509|D|A|Proof+Compile+Meta] Primary ideal
§FORM := A proper ideal Q is primary when ab in Q and a notin Q imply b^n in Q for some n>=1.
§DEP_HARD := S26-T-00001508
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001510|T|A|Proof+Compile+Meta] Lasker-Noether primary decomposition
§FORM := In a Noetherian commutative ring, every proper ideal admits a finite primary decomposition.
§DEP_HARD := S26-D-00001509
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001511|D|A|Proof+Compile+Meta] Krull dimension
§FORM := dim R is the supremum of lengths of strict chains of prime ideals.
§DEP_HARD := S26-T-00001510
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001512|D|A|Proof+Compile+Meta] Integral dependence
§FORM := s in an R-algebra S is integral over R when it satisfies a monic polynomial with coefficients in R.
§DEP_HARD := S26-D-00001511
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001513|T|A|Proof+Compile+Meta] Going-up theorem
§FORM := For an integral extension R subset S, chains of prime ideals in R can be lifted upward to chains in S under the standard lying-over setup.
§DEP_HARD := S26-D-00001512
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001514|D|A|Proof+Compile+Meta] Tensor product of modules
§FORM := M tensor_R N represents R-balanced bilinear maps out of MxN by a universal property.
§DEP_HARD := S26-T-00001513
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001515|D|A|Proof+Compile+Meta] Tor functor
§FORM := Tor_n^R(M,N) is the n-th left-derived functor of tensor product, measuring failure of flatness/exactness.
§DEP_HARD := S26-D-00001514
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001516|D|A|Proof+Compile+Meta] Ext functor
§FORM := Ext_R^n(M,N) is the n-th right-derived functor of Hom_R(M,-) or Hom_R(-,N) in the corresponding resolved variable.
§DEP_HARD := S26-D-00001515
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001517|D|A|Proof+Compile+Meta] Regular sequence
§FORM := x_1,...,x_r is an M-regular sequence when each x_i is a non-zero-divisor on M/(x_1,...,x_{i-1})M and the final quotient is nonzero under the chosen convention.
§DEP_HARD := S26-D-00001516
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001518|D|A|Proof+Compile+Meta] Cohen-Macaulay ring
§FORM := A Noetherian local ring is Cohen-Macaulay when its depth equals its Krull dimension.
§DEP_HARD := S26-D-00001517
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001519|D|A|Proof+Compile+Meta] Nakayama lemma
§FORM := If M is a finitely generated module over local ring (R,m) and mM=M, then M=0; more generally generators modulo m lift to generators of M.
§DEP_HARD := S26-D-00001518
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001520|B|A|Proof+Compile+Meta+Topo] Commutative algebra to geometry
§FORM := Prime ideals become points of Spec R, localization becomes passage to neighborhoods/stalks, and modules become algebraic data for sheaves.
§DEP_HARD := S26-D-00001519
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001521|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**2 + 1
§FORM := Over F_2, x**2 + 1 factors as (x + 1)^2.
§DEP_HARD := S26-B-00001520
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001522|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**2 - 2
§FORM := Over F_2, x**2 - 2 factors as (x)^2.
§DEP_HARD := S26-E-00001521
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001523|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**2 + x + 1
§FORM := Over F_2, x**2 + x + 1 factors as (x**2 + x + 1)^1.
§DEP_HARD := S26-E-00001522
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001524|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**3 - 2
§FORM := Over F_2, x**3 - 2 factors as (x)^3.
§DEP_HARD := S26-E-00001523
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001525|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**3 - x - 1
§FORM := Over F_2, x**3 - x - 1 factors as (x**3 + x + 1)^1.
§DEP_HARD := S26-E-00001524
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001526|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**4 + 1
§FORM := Over F_2, x**4 + 1 factors as (x + 1)^4.
§DEP_HARD := S26-E-00001525
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001527|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**4 - x + 1
§FORM := Over F_2, x**4 - x + 1 factors as (x**4 + x + 1)^1.
§DEP_HARD := S26-E-00001526
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001528|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**4 + 2
§FORM := Over F_2, x**4 + 2 factors as (x)^4.
§DEP_HARD := S26-E-00001527
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001529|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**5 - 2
§FORM := Over F_2, x**5 - 2 factors as (x)^5.
§DEP_HARD := S26-E-00001528
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001530|E|M|Proof+Compile+Meta] Polynomial factorization over F_2: x**5 - x - 1
§FORM := Over F_2, x**5 - x - 1 factors as (x**2 + x + 1)^1 * (x**3 + x**2 + 1)^1.
§DEP_HARD := S26-E-00001529
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001531|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x2 + 1**
§FORM := Over F_3, x**2 + 1 factors as (x**2 + 1)^1.
§DEP_HARD := S26-E-00001530
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001532|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x2 - 2**
§FORM := Over F_3, x**2 - 2 factors as (x**2 + 1)^1.
§DEP_HARD := S26-E-00001531
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001533|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x2 + x + 1**
§FORM := Over F_3, x**2 + x + 1 factors as (x - 1)^2.
§DEP_HARD := S26-E-00001532
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001534|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x3 - 2**
§FORM := Over F_3, x**3 - 2 factors as (x + 1)^3.
§DEP_HARD := S26-E-00001533
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001535|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x3 - x - 1**
§FORM := Over F_3, x**3 - x - 1 factors as (x**3 - x - 1)^1.
§DEP_HARD := S26-E-00001534
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001536|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x4 + 1**
§FORM := Over F_3, x**4 + 1 factors as (x**2 + x - 1)^1 * (x**2 - x - 1)^1.
§DEP_HARD := S26-E-00001535
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001537|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x4 - x + 1**
§FORM := Over F_3, x**4 - x + 1 factors as (x + 1)^1 * (x**3 - x**2 + x + 1)^1.
§DEP_HARD := S26-E-00001536
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001538|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x4 + 2**
§FORM := Over F_3, x**4 + 2 factors as (x + 1)^1 * (x - 1)^1 * (x**2 + 1)^1.
§DEP_HARD := S26-E-00001537
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001539|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x5 - 2**
§FORM := Over F_3, x**5 - 2 factors as (x + 1)^1 * (x**4 - x**3 + x**2 - x + 1)^1.
§DEP_HARD := S26-E-00001538
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001540|E|M|Proof+Compile+Meta] Polynomial factorization over F_3: x5 - x - 1**
§FORM := Over F_3, x**5 - x - 1 factors as (x**5 - x - 1)^1.
§DEP_HARD := S26-E-00001539
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001541|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x2 + 1**
§FORM := Over F_5, x**2 + 1 factors as (x + 2)^1 * (x - 2)^1.
§DEP_HARD := S26-E-00001540
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001542|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x2 - 2**
§FORM := Over F_5, x**2 - 2 factors as (x**2 - 2)^1.
§DEP_HARD := S26-E-00001541
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001543|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x2 + x + 1**
§FORM := Over F_5, x**2 + x + 1 factors as (x**2 + x + 1)^1.
§DEP_HARD := S26-E-00001542
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001544|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x3 - 2**
§FORM := Over F_5, x**3 - 2 factors as (x + 2)^1 * (x**2 - 2*x - 1)^1.
§DEP_HARD := S26-E-00001543
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001545|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x**3 - x - 1

§FORM := Over F_5, x**3 - x - 1 factors as (x - 2)^1 * (x**2 + 2*x - 2)^1.
§DEP_HARD := S26-E-00001544
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001546|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x**4 + 1

§FORM := Over F_5, x**4 + 1 factors as (x**2 + 2)^1 * (x**2 - 2)^1.
§DEP_HARD := S26-E-00001545
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001547|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x**4 - x + 1

§FORM := Over F_5, x**4 - x + 1 factors as (x - 2)^1 * (x**3 + 2*x**2 - x + 2)^1.
§DEP_HARD := S26-E-00001546
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001548|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x**4 + 2

§FORM := Over F_5, x**4 + 2 factors as (x**4 + 2)^1.
§DEP_HARD := S26-E-00001547
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001549|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x**5 - 2

§FORM := Over F_5, x**5 - 2 factors as (x - 2)^5.
§DEP_HARD := S26-E-00001548
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001550|E|M|Proof+Compile+Meta] Polynomial factorization over F_5: x**5 - x - 1

§FORM := Over F_5, x**5 - x - 1 factors as (x**5 - x - 1)^1.
§DEP_HARD := S26-E-00001549
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001551|D|A|Proof+Compile+Meta] Field extension degree

§FORM := For extension L/K, [L:K] is the dimension of L as a K-vector space.
§DEP_HARD := S26-D-00000101
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001552|T|A|Proof+Compile+Meta+Tower] Tower law

§FORM := For K subset L subset M with finite degrees, [M:K]=[M:L][L:K].
§DEP_HARD := S26-D-00001551
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001553|D|A|Proof+Compile+Meta] Algebraic extension

§FORM := L/K is algebraic when every element of L is algebraic over K.
§DEP_HARD := S26-T-00001552
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001554|D|A|Proof+Compile+Meta] Transcendence degree

§FORM := A transcendence basis is a maximal algebraically independent subset; its cardinality is trdeg_K L.
§DEP_HARD := S26-D-00001553
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001555|D|A|Proof+Compile+Meta] Splitting field

§FORM := A splitting field of f in K[x] is a minimal extension in which f factors into linear factors.
§DEP_HARD := S26-D-00001554
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001556|D|A|Proof+Compile+Meta] Separable extension

§FORM := An algebraic extension is separable when every element has a separable minimal polynomial over the base field.
§DEP_HARD := S26-D-00001555
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001557|D|A|Proof+Compile+Meta] Normal extension

§FORM := An algebraic extension L/K is normal when every irreducible polynomial over K having one root in L splits in L.
§DEP_HARD := S26-D-00001556
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001558|D|A|Proof+Compile+Meta] Galois extension

§FORM := A finite extension is Galois iff it is normal and separable.
§DEP_HARD := S26-D-00001557
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001559|D|A|Proof+Compile+Meta] Galois group

§FORM := Gal(L/K) is the group of field automorphisms of L fixing K pointwise.
§DEP_HARD := S26-D-00001558
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001560|T|A|Proof+Compile+Meta] Fundamental theorem of Galois theory

§FORM := For finite Galois L/K, intermediate fields E correspond inclusion-reversingly to subgroups H<=Gal(L/K), with E=L^H and H=Gal(L/E).
§DEP_HARD := S26-D-00001559
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001561|I|A|Proof+Compile+Meta] Degree-index relation in Galois theory

§FORM := For finite Galois L/K and H<=G, [L:L^H]=|H| and [L^H:K]=[G:H].
§DEP_HARD := S26-T-00001560
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001562|D|A|Proof+Compile+Meta] Fixed field

§FORM := For H<=Aut(L), L^H={x in L:sigma(x)=x for all sigma in H}.
§DEP_HARD := S26-I-00001561
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001563|D|A|Proof+Compile+Meta+Tower] Solvable polynomial by radicals

§FORM := A polynomial over characteristic-zero field is solvable by radicals when its splitting field lies in a tower obtained by adjoining radicals and roots of unity as needed.
§DEP_HARD := S26-D-00001562
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001564|T|A|Proof+Compile+Meta+Tower] Galois criterion for solvability by radicals

§FORM := Over characteristic zero, a polynomial is solvable by radicals iff its Galois group is solvable.
§DEP_HARD := S26-D-00001563
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001565|D|A|Proof+Compile+Meta] Artin-Schreier extension

§FORM := In characteristic p, an extension generated by a root of x^p-x-a is an Artin-Schreier extension when the polynomial is irreducible.
§DEP_HARD := S26-T-00001564
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001566|D|A|Proof+Compile+Meta] Kummer extension

§FORM := When the base contains relevant roots of unity and characteristic does not divide n, adjoining n-th roots gives Kummer extensions controlled by multiplicative classes modulo n-th powers.
§DEP_HARD := S26-D-00001565
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001567|T|A|Proof+Compile+Meta] Primitive element theorem

§FORM := Every finite separable extension L/K is simple: L=K(alpha) for some alpha.
§DEP_HARD := S26-D-00001566
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001568|D|A|Proof+Compile+Meta] Algebraic closure

§FORM := An algebraic closure of K is an algebraic extension that is algebraically closed; it is unique up to noncanonical K-isomorphism.
§DEP_HARD := S26-T-00001567
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00001569|N|A|Proof+Compile+Meta] Galois correspondence direction fence

§FORM := Larger subgroups correspond to smaller fixed fields; the correspondence reverses inclusions.
§DEP_HARD := S26-D-00001568
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := Treating subgroup inclusion as field inclusion in the same direction reverses the theorem.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001570|B|A|Proof+Compile+Meta] Galois theory as symmetry of equations

§FORM := Galois groups encode permutation symmetry of roots while preserving base-field algebraic relations, turning solvability questions into group-theoretic structure.
§DEP_HARD := S26-N-00001569
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001571|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=3

§FORM := [Q(zeta_3):Q]=phi(3)=2; its Galois group has order 2 and is isomorphic to (Z/3Z)^x.
§DEP_HARD := S26-B-00001570
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001572|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=4
§FORM := [Q(zeta_4):Q]=phi(4)=2; its Galois group has order 2 and is isomorphic to (Z/4Z)^x.
§DEP_HARD := S26-E-00001571
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001573|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=5
§FORM := [Q(zeta_5):Q]=phi(5)=4; its Galois group has order 4 and is isomorphic to (Z/5Z)^x.
§DEP_HARD := S26-E-00001572
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001574|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=6
§FORM := [Q(zeta_6):Q]=phi(6)=2; its Galois group has order 2 and is isomorphic to (Z/6Z)^x.
§DEP_HARD := S26-E-00001573
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001575|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=7
§FORM := [Q(zeta_7):Q]=phi(7)=6; its Galois group has order 6 and is isomorphic to (Z/7Z)^x.
§DEP_HARD := S26-E-00001574
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001576|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=8
§FORM := [Q(zeta_8):Q]=phi(8)=4; its Galois group has order 4 and is isomorphic to (Z/8Z)^x.
§DEP_HARD := S26-E-00001575
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001577|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=9
§FORM := [Q(zeta_9):Q]=phi(9)=6; its Galois group has order 6 and is isomorphic to (Z/9Z)^x.
§DEP_HARD := S26-E-00001576
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001578|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=10
§FORM := [Q(zeta_10):Q]=phi(10)=4; its Galois group has order 4 and is isomorphic to (Z/10Z)^x.
§DEP_HARD := S26-E-00001577
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001579|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=11
§FORM := [Q(zeta_11):Q]=phi(11)=10; its Galois group has order 10 and is isomorphic to (Z/11Z)^x.
§DEP_HARD := S26-E-00001578
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001580|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=12
§FORM := [Q(zeta_12):Q]=phi(12)=4; its Galois group has order 4 and is isomorphic to (Z/12Z)^x.
§DEP_HARD := S26-E-00001579
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001581|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=13
§FORM := [Q(zeta_13):Q]=phi(13)=12; its Galois group has order 12 and is isomorphic to (Z/13Z)^x.
§DEP_HARD := S26-E-00001580
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001582|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=14
§FORM := [Q(zeta_14):Q]=phi(14)=6; its Galois group has order 6 and is isomorphic to (Z/14Z)^x.
§DEP_HARD := S26-E-00001581
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001583|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=15
§FORM := [Q(zeta_15):Q]=phi(15)=8; its Galois group has order 8 and is isomorphic to (Z/15Z)^x.
§DEP_HARD := S26-E-00001582
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001584|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=16
§FORM := [Q(zeta_16):Q]=phi(16)=8; its Galois group has order 8 and is isomorphic to (Z/16Z)^x.
§DEP_HARD := S26-E-00001583
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001585|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=17
§FORM := [Q(zeta_17):Q]=phi(17)=16; its Galois group has order 16 and is isomorphic to (Z/17Z)^x.
§DEP_HARD := S26-E-00001584
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001586|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=18
§FORM := [Q(zeta_18):Q]=phi(18)=6; its Galois group has order 6 and is isomorphic to (Z/18Z)^x.
§DEP_HARD := S26-E-00001585
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001587|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=19
§FORM := [Q(zeta_19):Q]=phi(19)=18; its Galois group has order 18 and is isomorphic to (Z/19Z)^x.
§DEP_HARD := S26-E-00001586
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001588|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=20
§FORM := [Q(zeta_20):Q]=phi(20)=8; its Galois group has order 8 and is isomorphic to (Z/20Z)^x.
§DEP_HARD := S26-E-00001587
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001589|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=21
§FORM := [Q(zeta_21):Q]=phi(21)=12; its Galois group has order 12 and is isomorphic to (Z/21Z)^x.
§DEP_HARD := S26-E-00001588
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001590|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=22
§FORM := [Q(zeta_22):Q]=phi(22)=10; its Galois group has order 10 and is isomorphic to (Z/22Z)^x.
§DEP_HARD := S26-E-00001589
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001591|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=23
§FORM := [Q(zeta_23):Q]=phi(23)=22; its Galois group has order 22 and is isomorphic to (Z/23Z)^x.
§DEP_HARD := S26-E-00001590
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001592|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=24
§FORM := [Q(zeta_24):Q]=phi(24)=8; its Galois group has order 8 and is isomorphic to (Z/24Z)^x.
§DEP_HARD := S26-E-00001591
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001593|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=25
§FORM := [Q(zeta_25):Q]=phi(25)=20; its Galois group has order 20 and is isomorphic to (Z/25Z)^x.
§DEP_HARD := S26-E-00001592
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001594|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=26
§FORM := [Q(zeta_26):Q]=phi(26)=12; its Galois group has order 12 and is isomorphic to (Z/26Z)^x.
§DEP_HARD := S26-E-00001593
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001595|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=27
§FORM := [Q(zeta_27):Q]=phi(27)=18; its Galois group has order 18 and is isomorphic to (Z/27Z)^x.
§DEP_HARD := S26-E-00001594
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001596|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=28
§FORM := [Q(zeta_28):Q]=phi(28)=12; its Galois group has order 12 and is isomorphic to (Z/28Z)^x.
§DEP_HARD := S26-E-00001595
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001597|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=29
§FORM := [Q(zeta_29):Q]=phi(29)=28; its Galois group has order 28 and is isomorphic to (Z/29Z)^x.
§DEP_HARD := S26-E-00001596
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001598|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=30
§FORM := [Q(zeta_30):Q]=phi(30)=8; its Galois group has order 8 and is isomorphic to (Z/30Z)^x.
§DEP_HARD := S26-E-00001597
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001599|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=31
§FORM := [Q(zeta_31):Q]=phi(31)=30; its Galois group has order 30 and is isomorphic to (Z/31Z)^x.
§DEP_HARD := S26-E-00001598
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001600|E|M|Proof+Compile+Meta] Cyclotomic degree profile n=32

§FORM := [Q(zeta_32):Q]=phi(32)=16; its Galois group has order 16 and is isomorphic to (Z/32Z)^x.
§DEP_HARD := S26-E-00001599
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001601|D|A|Proof+Compile+Meta] Chain complex

§FORM := A chain complex C consists of objects C_n and differentials d_n:C_n->C_{n-1} with d_{n-1}d_n=0.
§DEP_HARD := S26-D-00000267
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001602|D|A|Proof+Compile+Meta] Cycle and boundary

§FORM := Z_n=ker d_n are n-cycles and B_n=im d_{n+1} are n-boundaries; B_n subseteq Z_n.
§DEP_HARD := S26-D-00001601
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001603|D|A|Proof+Compile+Meta] Homology group

§FORM := H_n(C)=Z_n/B_n.
§DEP_HARD := S26-D-00001602
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001604|D|A|Proof+Compile+Meta] Cochain complex

§FORM := A cochain complex has maps d^n:C^n->C^{n+1} with d^{n+1}d^n=0; its cohomology is H^n=ker d^n/im d^{n-1}.
§DEP_HARD := S26-D-00001603
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001605|D|A|Proof+Compile+Meta] Chain map

§FORM := A chain map f:C->D is a family f_n commuting with differentials.
§DEP_HARD := S26-D-00001604
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001606|D|A|Proof+Compile+Meta+Topo] Chain homotopy

§FORM := Chain maps f,g:C->D are chain-homotopic when f-g=dh+hd for a graded family h of degree +1.
§DEP_HARD := S26-D-00001605
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001607|T|A|Proof+Compile+Meta+Topo] Chain-homotopic maps induce equal homology maps

§FORM := If f and g are chain homotopic, H_n(f)=H_n(g) for all n.
§DEP_HARD := S26-D-00001606
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001608|D|A|Proof+Compile+Meta] Exact complex

§FORM := A chain complex is exact when all homology groups vanish.
§DEP_HARD := S26-T-00001607
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001609|D|A|Proof+Compile+Meta] Projective resolution

§FORM := A projective resolution of M is an exact complex of projective modules ending in M.
§DEP_HARD := S26-D-00001608
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001610|D|A|Proof+Compile+Meta] Injective resolution

§FORM := An injective resolution embeds M into an exact cochain complex of injective modules.
§DEP_HARD := S26-D-00001609
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001611|D|A|Proof+Compile+Meta] Group algebra

§FORM := For group G and field/ring F, F[G] is the free F-module on G with multiplication extended linearly from the group law.
§DEP_HARD := S26-D-00001610
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001612|D|A|Proof+Compile+Meta] Associative algebra

§FORM := An associative algebra over commutative ring k is simultaneously a k-module and an associative ring with compatible scalar multiplication.
§DEP_HARD := S26-D-00001611
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001613|D|A|Proof+Compile+Meta] Lie algebra

§FORM := A Lie algebra is a vector space with bilinear bracket [x,y] that is alternating/skew under suitable characteristic assumptions and satisfies the Jacobi identity.
§DEP_HARD := S26-D-00001612
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001614|D|A|Proof+Compile+Meta] Universal enveloping algebra

§FORM := U(g) is the associative algebra universal for Lie-algebra maps g->A^- into commutator Lie algebras.
§DEP_HARD := S26-D-00001613
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001615|T|A|Proof+Compile+Meta] Poincare-Birkhoff-Witt theorem

§FORM := For a Lie algebra over a field under standard hypotheses, ordered monomials in a basis give a vector-space basis of its universal enveloping algebra.
§DEP_HARD := S26-D-00001614
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001616|D|A|Proof+Compile+Meta] Center of an algebra

§FORM := Z(A)={a in A:ab=ba for all b in A}.
§DEP_HARD := S26-T-00001615
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001617|D|A|Proof+Compile+Meta] Simple algebra

§FORM := An associative algebra A is simple when it has no nontrivial two-sided ideals and A^2 is nonzero under the usual convention.
§DEP_HARD := S26-D-00001616
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001618|T|A|Proof+Compile+Meta] Artin-Wedderburn theorem

§FORM := A semisimple Artinian ring is a finite product of matrix rings over division rings; over an algebraically closed field, finite-dimensional semisimple algebras are products of full matrix algebras.
§DEP_HARD := S26-D-00001617
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001619|D|A|Proof+Compile+Meta] Morita equivalence

§FORM := Rings R and S are Morita equivalent when their module categories are equivalent.
§DEP_HARD := S26-T-00001618
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001620|B|A|Proof+Compile+Meta+Topo] Homology as obstruction extractor

§FORM := Passing from cycles to cycles modulo boundaries converts failure of exactness into invariant algebraic data and recurs across topology, geometry, algebra, and mathematical physics.
§DEP_HARD := S26-D-00001619
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001621|E|M|Proof+Compile+Meta] Two-step chain-complex profile 1

§FORM := For d1=[2,3] and d2=[3,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-B-00001620
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001622|E|M|Proof+Compile+Meta] Two-step chain-complex profile 2

§FORM := For d1=[3,5] and d2=[5,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001621
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001623|E|M|Proof+Compile+Meta] Two-step chain-complex profile 3

§FORM := For d1=[4,7] and d2=[7,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001622
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001624|E|M|Proof+Compile+Meta] Two-step chain-complex profile 4

§FORM := For d1=[5,2] and d2=[2,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001623
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001625|E|M|Proof+Compile+Meta] Two-step chain-complex profile 5

§FORM := For d1=[1,4] and d2=[4,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001624
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001626|E|M|Proof+Compile+Meta] Two-step chain-complex profile 6
§FORM := For d1=[2,6] and d2=[6,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001625
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001627|E|M|Proof+Compile+Meta] Two-step chain-complex profile 7
§FORM := For d1=[3,1] and d2=[1,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001626
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001628|E|M|Proof+Compile+Meta] Two-step chain-complex profile 8
§FORM := For d1=[4,3] and d2=[3,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001627
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001629|E|M|Proof+Compile+Meta] Two-step chain-complex profile 9
§FORM := For d1=[5,5] and d2=[5,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001628
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001630|E|M|Proof+Compile+Meta] Two-step chain-complex profile 10
§FORM := For d1=[1,7] and d2=[7,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001629
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001631|E|M|Proof+Compile+Meta] Two-step chain-complex profile 11
§FORM := For d1=[2,2] and d2=[2,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001630
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001632|E|M|Proof+Compile+Meta] Two-step chain-complex profile 12
§FORM := For d1=[3,4] and d2=[4,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001631
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001633|E|M|Proof+Compile+Meta] Two-step chain-complex profile 13
§FORM := For d1=[4,6] and d2=[6,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001632
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001634|E|M|Proof+Compile+Meta] Two-step chain-complex profile 14
§FORM := For d1=[5,1] and d2=[1,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001633
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001635|E|M|Proof+Compile+Meta] Two-step chain-complex profile 15
§FORM := For d1=[1,3] and d2=[3,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001634
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001636|E|M|Proof+Compile+Meta] Two-step chain-complex profile 16
§FORM := For d1=[2,5] and d2=[5,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001635
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001637|E|M|Proof+Compile+Meta] Two-step chain-complex profile 17
§FORM := For d1=[3,7] and d2=[7,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001636
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001638|E|M|Proof+Compile+Meta] Two-step chain-complex profile 18
§FORM := For d1=[4,2] and d2=[2,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001637
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001639|E|M|Proof+Compile+Meta] Two-step chain-complex profile 19
§FORM := For d1=[5,4] and d2=[4,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001638
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001640|E|M|Proof+Compile+Meta] Two-step chain-complex profile 20
§FORM := For d1=[1,6] and d2=[6,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001639
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001641|E|M|Proof+Compile+Meta] Two-step chain-complex profile 21
§FORM := For d1=[2,1] and d2=[1,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001640
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001642|E|M|Proof+Compile+Meta] Two-step chain-complex profile 22
§FORM := For d1=[3,3] and d2=[3,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001641
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001643|E|M|Proof+Compile+Meta] Two-step chain-complex profile 23
§FORM := For d1=[4,5] and d2=[5,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001642
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001644|E|M|Proof+Compile+Meta] Two-step chain-complex profile 24
§FORM := For d1=[5,7] and d2=[7,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001643
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001645|E|M|Proof+Compile+Meta] Two-step chain-complex profile 25
§FORM := For d1=[1,2] and d2=[2,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001644
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001646|E|M|Proof+Compile+Meta] Two-step chain-complex profile 26
§FORM := For d1=[2,4] and d2=[4,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001645
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001647|E|M|Proof+Compile+Meta] Two-step chain-complex profile 27
§FORM := For d1=[3,6] and d2=[6,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001646
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001648|E|M|Proof+Compile+Meta] Two-step chain-complex profile 28
§FORM := For d1=[4,1] and d2=[1,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001647
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001649|E|M|Proof+Compile+Meta] Two-step chain-complex profile 29
§FORM := For d1=[5,3] and d2=[3,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001648
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001650|E|M|Proof+Compile+Meta] Two-step chain-complex profile 30
§FORM := For d1=[1,5] and d2=[5,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.
§DEP_HARD := S26-E-00001649
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001651|D|A|Proof+Compile+Meta] Ordinary generating function
§FORM := For sequence a_n, A(x)=sum_{n>=0}a_n x^n is its ordinary generating function as a formal power series or analytic function where convergence is invoked.
§DEP_HARD := S26-T-00000088
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001652|D|A|Proof+Compile+Meta] Exponential generating function
§FORM := For sequence a_n, E(x)=sum_{n>=0} a_n x^n/n! is its exponential generating function.
§DEP_HARD := S26-D-00001651
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001653|T|A|Proof+Compile+Meta] Product rule for ordinary generating functions
§FORM := The coefficient of x^n in A(x)B(x) is the convolution sum_{k=0}^n a_k b_{n-k}.
§DEP_HARD := S26-D-00001652
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001654|I|A|Proof+Compile+Meta] Factorial formula for binomial coefficients

§FORM := For integers 0<=k<=n, C(n,k)=n!/(k!(n-k)!), refining the frozen combinatorial coefficient interface by its factorial identity.
§DEP_HARD := S26-T-00001653
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001655|B|A|Proof+Compile+Meta+Tower] Pascal recurrence as a combinatorial bridge

§FORM := The frozen Pascal identity S26-T-00000088 gives the recursive decomposition underlying lattice-path and subset-counting interpretations of binomial coefficients.
§DEP_HARD := S26-I-00001654
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001656|B|A|Proof+Compile+Meta] Binomial theorem coefficient bridge

§FORM := The frozen binomial theorem S26-T-00000089 identifies C(n,k) as the coefficient of x^{n-k}y^k in (x+y)^n, linking algebraic expansion to subset enumeration.
§DEP_HARD := S26-B-00001655
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001657|D|A|Proof+Compile+Meta] Stirling numbers of the second kind

§FORM := S(n,k) counts partitions of an n-element set into k nonempty unlabeled blocks.
§DEP_HARD := S26-B-00001656
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001658|I|A|Proof+Compile+Meta] Stirling second-kind recurrence

§FORM := S(n,k)=k S(n-1,k)+S(n-1,k-1).
§DEP_HARD := S26-D-00001657
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001659|D|A|Proof+Compile+Meta] Stirling numbers of the first kind

§FORM := Unsigned c(n,k) counts permutations of n elements with exactly k cycles.
§DEP_HARD := S26-I-00001658
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001660|D|A|Proof+Compile+Meta] Bell number

§FORM := B_n=sum_k S(n,k) counts set partitions of an n-element set.
§DEP_HARD := S26-D-00001659
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001661|D|A|Proof+Compile+Meta] Catalan number

§FORM := C_n=(1/(n+1)) C(2n,n), counting many equivalent families such as Dyck paths of semilength n.
§DEP_HARD := S26-D-00001660
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001662|I|A|Proof+Compile+Meta] Catalan recurrence

§FORM := C_{n+1}=sum_{k=0}^n C_k C_{n-k}.
§DEP_HARD := S26-D-00001661
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001663|D|A|Proof+Compile+Meta] Derangement number

§FORM := !n counts permutations of n elements with no fixed points.
§DEP_HARD := S26-I-00001662
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001664|I|A|Proof+Compile+Meta] Derangement formula

§FORM := !n=n! sum_{k=0}^n (-1)^k/k!, hence !n is the nearest integer to n!/e for n>=1.
§DEP_HARD := S26-D-00001663
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001665|T|A|Proof+Compile+Meta] Inclusion-exclusion principle

§FORM := For finite sets A_1,...,A_m, |union_i A_i|=sum_{empty!=J subseteq [m]}(-1)^{|J|+1}|intersection_{j in J}A_j|.
§DEP_HARD := S26-I-00001664
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001666|T|A|Proof+Compile+Meta] Burnside counting lemma

§FORM := The number of orbits of a finite group G acting on finite X is (1/|G|)sum_{g in G}|Fix(g)|.
§DEP_HARD := S26-T-00001665
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001667|T|A|Proof+Compile+Meta] Polya enumeration theorem

§FORM := For a finite permutation group acting on positions, the cycle index substituted with color inventory counts colorings up to symmetry.

§DEP_HARD := S26-T-00001666

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001668|D|A|Proof+Compile+Meta] Species of structures

§FORM := A combinatorial species assigns sets of structures functorially to finite label sets; its exponential generating function records labeled counts.

§DEP_HARD := S26-T-00001667

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001669|D|A|Proof+Compile+Meta] Recurrence relation

§FORM := A recurrence defines sequence terms from earlier terms; linear recurrences with constant coefficients are governed by the roots of a characteristic polynomial.

§DEP_HARD := S26-D-00001668

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001670|B|A|Proof+Compile+Meta+IG] Enumeration-probability bridge

§FORM := Generating functions encode both combinatorial counts and probability distributions; normalization and analytic domain determine which interpretation is valid.

§DEP_HARD := S26-D-00001669

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001671|E|M|Proof+Compile+Meta] Enumerative profile n=0

§FORM := At n=0: Catalan C_n=1, Bell B_n=1, and derangements !n=1.

§DEP_HARD := S26-B-00001670

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001672|E|M|Proof+Compile+Meta] Enumerative profile n=1

§FORM := At n=1: Catalan C_n=1, Bell B_n=1, and derangements !n=0.

§DEP_HARD := S26-E-00001671

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001673|E|M|Proof+Compile+Meta] Enumerative profile n=2

§FORM := At n=2: Catalan C_n=2, Bell B_n=2, and derangements !n=1.

§DEP_HARD := S26-E-00001672

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001674|E|M|Proof+Compile+Meta] Enumerative profile n=3

§FORM := At n=3: Catalan C_n=5, Bell B_n=5, and derangements !n=2.

§DEP_HARD := S26-E-00001673

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001675|E|M|Proof+Compile+Meta] Enumerative profile n=4

§FORM := At n=4: Catalan C_n=14, Bell B_n=15, and derangements !n=9.

§DEP_HARD := S26-E-00001674

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001676|E|M|Proof+Compile+Meta] Enumerative profile n=5

§FORM := At n=5: Catalan C_n=42, Bell B_n=52, and derangements !n=44.

§DEP_HARD := S26-E-00001675

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001677|E|M|Proof+Compile+Meta] Enumerative profile n=6

§FORM := At n=6: Catalan C_n=132, Bell B_n=203, and derangements !n=265.

§DEP_HARD := S26-E-00001676

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001678|E|M|Proof+Compile+Meta] Enumerative profile n=7

§FORM := At n=7: Catalan C_n=429, Bell B_n=877, and derangements !n=1854.

§DEP_HARD := S26-E-00001677

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001679|E|M|Proof+Compile+Meta] Enumerative profile n=8

§FORM := At n=8: Catalan C_n=1430, Bell B_n=4140, and derangements !n=14833.

§DEP_HARD := S26-E-00001678

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001680|E|M|Proof+Compile+Meta] Enumerative profile n=9

§FORM := At n=9: Catalan C_n=4862, Bell B_n=21147, and derangements !n=133496.
§DEP_HARD := S26-E-00001679
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001681|E|M|Proof+Compile+Meta] Enumerative profile n=10

§FORM := At n=10: Catalan C_n=16796, Bell B_n=115975, and derangements !n=1334961.
§DEP_HARD := S26-E-00001680
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001682|E|M|Proof+Compile+Meta] Enumerative profile n=11

§FORM := At n=11: Catalan C_n=58786, Bell B_n=678570, and derangements !n=14684570.
§DEP_HARD := S26-E-00001681
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001683|E|M|Proof+Compile+Meta] Enumerative profile n=12

§FORM := At n=12: Catalan C_n=208012, Bell B_n=4213597, and derangements !n=176214841.
§DEP_HARD := S26-E-00001682
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001684|E|M|Proof+Compile+Meta] Enumerative profile n=13

§FORM := At n=13: Catalan C_n=742900, Bell B_n=27644437, and derangements !n=2290792932.
§DEP_HARD := S26-E-00001683
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001685|E|M|Proof+Compile+Meta] Enumerative profile n=14

§FORM := At n=14: Catalan C_n=2674440, Bell B_n=190899322, and derangements !n=32071101049.
§DEP_HARD := S26-E-00001684
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001686|E|M|Proof+Compile+Meta] Enumerative profile n=15

§FORM := At n=15: Catalan C_n=9694845, Bell B_n=1382958545, and derangements !n=481066515734.
§DEP_HARD := S26-E-00001685
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001687|E|M|Proof+Compile+Meta] Enumerative profile n=16

§FORM := At n=16: Catalan C_n=35357670, Bell B_n=10480142147, and derangements !n=7697064251745.
§DEP_HARD := S26-E-00001686
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001688|E|M|Proof+Compile+Meta] Enumerative profile n=17

§FORM := At n=17: Catalan C_n=129644790, Bell B_n=82864869804, and derangements !n=130850092279664.
§DEP_HARD := S26-E-00001687
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001689|E|M|Proof+Compile+Meta] Enumerative profile n=18

§FORM := At n=18: Catalan C_n=477638700, Bell B_n=682076806159, and derangements !n=2355301661033953.
§DEP_HARD := S26-E-00001688
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001690|E|M|Proof+Compile+Meta] Enumerative profile n=19

§FORM := At n=19: Catalan C_n=1767263190, Bell B_n=5832742205057, and derangements !n=44750731559645106.
§DEP_HARD := S26-E-00001689
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001691|E|M|Proof+Compile+Meta] Enumerative profile n=20

§FORM := At n=20: Catalan C_n=6564120420, Bell B_n=51724158235372, and derangements !n=895014631192902121.
§DEP_HARD := S26-E-00001690
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001692|E|M|Proof+Compile+Meta] Enumerative profile n=21

§FORM := At n=21: Catalan C_n=24466267020, Bell B_n=474869816156751, and derangements !n=18795307255050944540.
§DEP_HARD := S26-E-00001691
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001693|E|M|Proof+Compile+Meta] Enumerative profile n=22

§FORM := At n=22: Catalan C_n=91482563640, Bell B_n=4506715738447323, and derangements !n=413496759611120779881.
§DEP_HARD := S26-E-00001692
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001694|E|M|Proof+Compile+Meta] Enumerative profile n=23

§FORM := At n=23: Catalan C_n=343059613650, Bell B_n=44152005855084346, and derangements !n=951042547105577937262.
§DEP_HARD := S26-E-00001693
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001695|E|M|Proof+Compile+Meta] Enumerative profile n=24

§FORM := At n=24: Catalan C_n=1289904147324, Bell B_n=445958869294805289, and derangements !n=228250211305338670494289.
§DEP_HARD := S26-E-00001694
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001696|E|M|Proof+Compile+Meta] Enumerative profile n=25

§FORM := At n=25: Catalan C_n=4861946401452, Bell B_n=4638590332229999353, and derangements !n=5706255282633466762357224.
§DEP_HARD := S26-E-00001695
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001697|E|M|Proof+Compile+Meta] Enumerative profile n=26

§FORM := At n=26: Catalan C_n=18367353072152, Bell B_n=49631246523618756274, and derangements !n=1483626373484701358212878 25.
§DEP_HARD := S26-E-00001696
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001698|E|M|Proof+Compile+Meta] Enumerative profile n=27

§FORM := At n=27: Catalan C_n=69533550916004, Bell B_n=545717047936059989389, and derangements !n=400579120840869367174771274.
§DEP_HARD := S26-E-00001697
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001699|E|M|Proof+Compile+Meta] Enumerative profile n=28

§FORM := At n=28: Catalan C_n=263747951750360, Bell B_n=6160539404599934652455, and derangements !n=11216215383544342268089359 5673.
§DEP_HARD := S26-E-00001698
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001700|E|M|Proof+Compile+Meta] Enumerative profile n=29

§FORM := At n=29: Catalan C_n=1002242216651368, Bell B_n=71339801938860275191172, and derangements !n=3252702461227859257745914274516.
§DEP_HARD := S26-E-00001699
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001701|D|A|Proof+Compile+Meta] Integer partition

§FORM := A partition of n is a weakly decreasing finite sequence of positive integers summing to n.
§DEP_HARD := S26-D-00000086
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001702|D|A|Proof+Compile+Meta] Partition function

§FORM := p(n) is the number of integer partitions of n, with p(0)=1.
§DEP_HARD := S26-D-00001701
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001703|I|A|Proof+Compile+Meta] Euler partition generating function

§FORM := sum_{n>=0}p(n)q^n=product_{m>=1}(1-q^m)^{-1} as a formal power series.
§DEP_HARD := S26-D-00001702
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001704|T|A|Proof+Compile+Meta] Euler pentagonal recurrence

§FORM := p(n)=sum_{k!=0}(-1)^{k-1}p(n-k(3k-1)/2), with p(0)=1 and p(m)=0 for m<0.
§DEP_HARD := S26-I-00001703
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001705|T|A|Proof+Compile+Meta] Euler distinct-odd partition theorem

§FORM := The number of partitions of n into distinct parts equals the number of partitions of n into odd parts.
§DEP_HARD := S26-T-00001704
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001706|D|A|Proof+Compile+Meta] Ferrers diagram

§FORM := A Ferrers/Young diagram represents a partition by left-justified rows whose lengths are the parts.
§DEP_HARD := S26-T-00001705
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001707|D|A|Proof+Compile+Meta] Conjugate partition
§FORM := The conjugate partition transposes the Ferrers diagram; row lengths become column lengths.
§DEP_HARD := S26-D-00001706
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001708|D|A|Proof+Compile+Meta] Hook length
§FORM := For a cell of a Young diagram, its hook length is one plus the number of cells strictly right in its row plus strictly below in its column.
§DEP_HARD := S26-D-00001707
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001709|T|A|Proof+Compile+Meta] Hook-length formula
§FORM := The number f^lambda of standard Young tableaux of shape lambda partitioning n is n! divided by the product of hook lengths of lambda.
§DEP_HARD := S26-D-00001708
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001710|D|A|Proof+Compile+Meta] q-binomial coefficient
§FORM := The Gaussian binomial [n choose k]_q is product_{i=0}^{k-1}(1-q^{n-i})/(1-q^{k-i}).
§DEP_HARD := S26-T-00001709
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001711|T|A|Proof+Compile+Meta] Finite-field subspace count
§FORM := [n choose k]_q counts k-dimensional subspaces of F_q^n.
§DEP_HARD := S26-D-00001710
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001712|D|A|Proof+Compile+Meta] q-Pochhammer symbol
§FORM := (a;q)_n=product_{j=0}^{n-1}(1-aq^j), with an infinite version when convergence/formal context is declared.
§DEP_HARD := S26-T-00001711
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001713|D|A|Proof+Compile+Meta] Theta series
§FORM := A theta series is a generating function sum q^{Q(x)} attached to a quadratic form/lattice, encoding representation counts.
§DEP_HARD := S26-D-00001712
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001714|D|A|Proof+Compile+Meta] Modular form interface
§FORM := A modular form is a holomorphic function on the upper half-plane satisfying a specified modular transformation law and cusp regularity; weight and group are part of the data.
§DEP_HARD := S26-D-00001713
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00001715|N|A|Proof+Compile+Meta] q-series convergence fence
§FORM := Formal-power-series identities and analytic q-series identities are distinct claims; analytic manipulations require a declared domain such as |q|<1.
§DEP_HARD := S26-D-00001714
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := Formal coefficient equality alone does not justify analytic limit interchange.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001716|T|A|Proof+Compile+Meta] Ramanujan congruence mod 5
§FORM := For n>=0, p(5n+4) is divisible by 5.
§DEP_HARD := S26-N-00001715
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001717|T|A|Proof+Compile+Meta] Ramanujan congruence mod 7
§FORM := For n>=0, p(7n+5) is divisible by 7.
§DEP_HARD := S26-T-00001716
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001718|T|A|Proof+Compile+Meta] Ramanujan congruence mod 11
§FORM := For n>=0, p(11n+6) is divisible by 11.
§DEP_HARD := S26-T-00001717
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001719|D|A|Proof+Compile+Meta] Partition rank
§FORM := The rank of a partition is its largest part minus its number of parts.
§DEP_HARD := S26-T-00001718
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001720|B|A|Proof+Compile+Meta] Partitions to representation theory

§FORM := Young diagrams index irreducible representations of symmetric groups in characteristic zero and organize Schur functors, symmetric functions, and tableaux combinatorics.

§DEP_HARD := S26-D-00001719

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001721|E|M|Proof+Compile+Meta] Partition number p(0)

§FORM := The exact partition count is p(0)=1.

§DEP_HARD := S26-B-00001720

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001722|E|M|Proof+Compile+Meta] Partition number p(1)

§FORM := The exact partition count is p(1)=1.

§DEP_HARD := S26-E-00001721

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001723|E|M|Proof+Compile+Meta] Partition number p(2)

§FORM := The exact partition count is p(2)=2.

§DEP_HARD := S26-E-00001722

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001724|E|M|Proof+Compile+Meta] Partition number p(3)

§FORM := The exact partition count is p(3)=3.

§DEP_HARD := S26-E-00001723

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001725|E|M|Proof+Compile+Meta] Partition number p(4)

§FORM := The exact partition count is p(4)=5.

§DEP_HARD := S26-E-00001724

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001726|E|M|Proof+Compile+Meta] Partition number p(5)

§FORM := The exact partition count is p(5)=7.

§DEP_HARD := S26-E-00001725

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001727|E|M|Proof+Compile+Meta] Partition number p(6)

§FORM := The exact partition count is p(6)=11.

§DEP_HARD := S26-E-00001726

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001728|E|M|Proof+Compile+Meta] Partition number p(7)

§FORM := The exact partition count is p(7)=15.

§DEP_HARD := S26-E-00001727

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001729|E|M|Proof+Compile+Meta] Partition number p(8)

§FORM := The exact partition count is p(8)=22.

§DEP_HARD := S26-E-00001728

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001730|E|M|Proof+Compile+Meta] Partition number p(9)

§FORM := The exact partition count is p(9)=30.

§DEP_HARD := S26-E-00001729

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001731|E|M|Proof+Compile+Meta] Partition number p(10)

§FORM := The exact partition count is p(10)=42.

§DEP_HARD := S26-E-00001730

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001732|E|M|Proof+Compile+Meta] Partition number p(11)

§FORM := The exact partition count is p(11)=56.

§DEP_HARD := S26-E-00001731

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001733|E|M|Proof+Compile+Meta] Partition number p(12)

§FORM := The exact partition count is p(12)=77.

§DEP_HARD := S26-E-00001732

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001734|E|M|Proof+Compile+Meta] Partition number p(13)

§FORM := The exact partition count is p(13)=101.
§DEP_HARD := S26-E-00001733
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001735|E|M|Proof+Compile+Meta] Partition number p(14)

§FORM := The exact partition count is p(14)=135.
§DEP_HARD := S26-E-00001734
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001736|E|M|Proof+Compile+Meta] Partition number p(15)

§FORM := The exact partition count is p(15)=176.
§DEP_HARD := S26-E-00001735
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001737|E|M|Proof+Compile+Meta] Partition number p(16)

§FORM := The exact partition count is p(16)=231.
§DEP_HARD := S26-E-00001736
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001738|E|M|Proof+Compile+Meta] Partition number p(17)

§FORM := The exact partition count is p(17)=297.
§DEP_HARD := S26-E-00001737
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001739|E|M|Proof+Compile+Meta] Partition number p(18)

§FORM := The exact partition count is p(18)=385.
§DEP_HARD := S26-E-00001738
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001740|E|M|Proof+Compile+Meta] Partition number p(19)

§FORM := The exact partition count is p(19)=490.
§DEP_HARD := S26-E-00001739
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001741|E|M|Proof+Compile+Meta] Partition number p(20)

§FORM := The exact partition count is p(20)=627.
§DEP_HARD := S26-E-00001740
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001742|E|M|Proof+Compile+Meta] Partition number p(21)

§FORM := The exact partition count is p(21)=792.
§DEP_HARD := S26-E-00001741
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001743|E|M|Proof+Compile+Meta] Partition number p(22)

§FORM := The exact partition count is p(22)=1002.
§DEP_HARD := S26-E-00001742
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001744|E|M|Proof+Compile+Meta] Partition number p(23)

§FORM := The exact partition count is p(23)=1255.
§DEP_HARD := S26-E-00001743
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001745|E|M|Proof+Compile+Meta] Partition number p(24)

§FORM := The exact partition count is p(24)=1575.
§DEP_HARD := S26-E-00001744
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001746|E|M|Proof+Compile+Meta] Partition number p(25)

§FORM := The exact partition count is p(25)=1958.
§DEP_HARD := S26-E-00001745
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001747|E|M|Proof+Compile+Meta] Partition number p(26)

§FORM := The exact partition count is p(26)=2436.
§DEP_HARD := S26-E-00001746
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001748|E|M|Proof+Compile+Meta] Partition number p(27)
§FORM := The exact partition count is p(27)=3010.
§DEP_HARD := S26-E-00001747
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001749|E|M|Proof+Compile+Meta] Partition number p(28)
§FORM := The exact partition count is p(28)=3718.
§DEP_HARD := S26-E-00001748
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001750|E|M|Proof+Compile+Meta] Partition number p(29)
§FORM := The exact partition count is p(29)=4565.
§DEP_HARD := S26-E-00001749
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001751|D|A|Proof+Compile+Meta] Simple graph
§FORM := A simple graph G=(V,E) has a vertex set V and a set E of unordered two-element subsets of V, with no loops or parallel edges.
§DEP_HARD := S26-D-00000149
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001752|D|A|Proof+Compile+Meta] Degree of a vertex
§FORM := deg(v) is the number of edges incident to v in a finite simple graph.
§DEP_HARD := S26-D-00001751
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001753|T|A|Proof+Compile+Meta] Handshake lemma
§FORM := For a finite graph, sum_{v in V}deg(v)=2|E|.
§DEP_HARD := S26-D-00001752
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001754|D|A|Proof+Compile+Meta] Path and cycle
§FORM := A path is a sequence of adjacent vertices without repeated vertices under the simple-path convention; a cycle is a closed path with distinct interior vertices.
§DEP_HARD := S26-T-00001753
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001755|D|A|Proof+Compile+Meta] Connected graph
§FORM := A graph is connected when every pair of vertices is joined by a path.
§DEP_HARD := S26-D-00001754
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001756|D|A|Proof+Compile+Meta] Tree
§FORM := A tree is a connected acyclic graph.
§DEP_HARD := S26-D-00001755
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001757|T|A|Proof+Compile+Meta] Tree edge count
§FORM := A finite tree with n vertices has n-1 edges; conversely a connected n-vertex graph with n-1 edges is a tree.
§DEP_HARD := S26-D-00001756
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001758|T|A|Proof+Compile+Meta+Tower] Cayley formula
§FORM := There are n^{n-2} labeled trees on n labeled vertices for n>=2.
§DEP_HARD := S26-T-00001757
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001759|D|A|Proof+Compile+Meta] Spanning tree
§FORM := A spanning tree of connected G is a tree containing all vertices of G.
§DEP_HARD := S26-T-00001758
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001760|T|A|Proof+Compile+Meta] Matrix-tree theorem
§FORM := The number of spanning trees of a finite graph equals any cofactor of its Laplacian matrix.
§DEP_HARD := S26-D-00001759
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001761|D|A|Proof+Compile+Meta] Graph Laplacian
§FORM := For simple graph G, L=D-A where D is the diagonal degree matrix and A the adjacency matrix.
§DEP_HARD := S26-T-00001760
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001762|T|A|Proof+Compile+Meta+Spectral] Laplacian kernel and components
§FORM := The multiplicity of eigenvalue 0 of the graph Laplacian equals the number of connected components.
§DEP_HARD := S26-D-00001761
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001763|D|A|Proof+Compile+Meta] Eulerian circuit
§FORM := An Eulerian circuit traverses every edge exactly once and returns to its start.
§DEP_HARD := S26-T-00001762
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001764|T|A|Proof+Compile+Meta] Euler criterion for connected graphs
§FORM := A finite connected graph has an Eulerian circuit iff every vertex has even degree.
§DEP_HARD := S26-D-00001763
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001765|D|A|Proof+Compile+Meta+Spectral] Hamiltonian cycle
§FORM := A Hamiltonian cycle visits every vertex exactly once before returning to its start.
§DEP_HARD := S26-T-00001764
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00001766|N|A|Proof+Compile+Meta+Spectral] Eulerian/Hamiltonian fence
§FORM := Eulerian conditions concern edges and have a complete degree criterion; Hamiltonicity concerns vertices and has no analogous simple necessary-and-sufficient degree criterion.
§DEP_HARD := S26-D-00001765
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := Do not transfer Euler degree parity to Hamiltonicity.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001767|D|A|Proof+Compile+Meta] Planar graph
§FORM := A graph is planar when it admits an embedding in the sphere/plane with edges meeting only at common endpoints.
§DEP_HARD := S26-N-00001766
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001768|I|A|Proof+Compile+Meta] Euler formula for connected planar graphs
§FORM := For a connected planar embedding, |V|-|E|+|F|=2.
§DEP_HARD := S26-D-00001767
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001769|T|A|Proof+Compile+Meta] Kuratowski theorem
§FORM := A finite graph is planar iff it contains no subdivision of K_5 or K_{3,3}.
§DEP_HARD := S26-I-00001768
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001770|B|A|Proof+Compile+Meta+Spectral] Graphs to linear algebra
§FORM := Adjacency, incidence, and Laplacian matrices turn connectivity, cuts, walks, and expansion into spectral linear-algebraic questions.
§DEP_HARD := S26-T-00001769
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001771|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=3
§FORM := On n=3 labeled vertices, K_n has 3 edges, C_n has 3 edges, and there are 3 labeled trees by Cayley's formula.
§DEP_HARD := S26-B-00001770
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001772|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=4
§FORM := On n=4 labeled vertices, K_n has 6 edges, C_n has 4 edges, and there are 16 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001771
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001773|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=5
§FORM := On n=5 labeled vertices, K_n has 10 edges, C_n has 5 edges, and there are 125 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001772
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001774|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=6
§FORM := On n=6 labeled vertices, K_n has 15 edges, C_n has 6 edges, and there are 1296 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001773
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001775|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=7

§FORM := On n=7 labeled vertices, K_n has 21 edges, C_n has 7 edges, and there are 16807 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001774
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001776|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=8

§FORM := On n=8 labeled vertices, K_n has 28 edges, C_n has 8 edges, and there are 262144 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001775
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001777|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=9

§FORM := On n=9 labeled vertices, K_n has 36 edges, C_n has 9 edges, and there are 4782969 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001776
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001778|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=10

§FORM := On n=10 labeled vertices, K_n has 45 edges, C_n has 10 edges, and there are 100000000 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001777
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001779|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=11

§FORM := On n=11 labeled vertices, K_n has 55 edges, C_n has 11 edges, and there are 2357947691 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001778
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001780|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=12

§FORM := On n=12 labeled vertices, K_n has 66 edges, C_n has 12 edges, and there are 61917364224 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001779
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001781|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=13

§FORM := On n=13 labeled vertices, K_n has 78 edges, C_n has 13 edges, and there are 1792160394037 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001780
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001782|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=14

§FORM := On n=14 labeled vertices, K_n has 91 edges, C_n has 14 edges, and there are 56693912375296 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001781
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001783|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=15

§FORM := On n=15 labeled vertices, K_n has 105 edges, C_n has 15 edges, and there are 1946195068359375 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001782
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001784|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=16

§FORM := On n=16 labeled vertices, K_n has 120 edges, C_n has 16 edges, and there are 72057594037927936 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001783
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001785|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=17

§FORM := On n=17 labeled vertices, K_n has 136 edges, C_n has 17 edges, and there are 2862423051509815793 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001784
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001786|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=18

§FORM := On n=18 labeled vertices, K_n has 153 edges, C_n has 18 edges, and there are 121439531096594251776 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001785
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001787|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=19

§FORM := On n=19 labeled vertices, K_n has 171 edges, C_n has 19 edges, and there are 5480386857784802185939 labeled trees by Cayley's formula.
§DEP_HARD := S26-E-00001786
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001788|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=20

§FORM := On n=20 labeled vertices, K_n has 190 edges, C_n has 20 edges, and there are 262144000000000000000000 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001787

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001789|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=21

§FORM := On n=21 labeled vertices, K_n has 210 edges, C_n has 21 edges, and there are 1324849664033102612558078 1 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001788

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001790|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=22

§FORM := On n=22 labeled vertices, K_n has 231 edges, C_n has 22 edges, and there are 705429498686404044207947776 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001789

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001791|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=23

§FORM := On n=23 labeled vertices, K_n has 253 edges, C_n has 23 edges, and there are 394715841206954858872495896 23 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001790

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001792|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=24

§FORM := On n=24 labeled vertices, K_n has 276 edges, C_n has 24 edges, and there are 2315513501476187716057433112576 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001791

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001793|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=25

§FORM := On n=25 labeled vertices, K_n has 300 edges, C_n has 25 edges, and there are 142108547152020037174224853515625 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001792

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001794|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=26

§FORM := On n=26 labeled vertices, K_n has 325 edges, C_n has 26 edges, and there are 910668576953721495679981403609497 6 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001793

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001795|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=27

§FORM := On n=27 labeled vertices, K_n has 351 edges, C_n has 27 edges, and there are 608266787713357709119683992618861307 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001794

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001796|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=28

§FORM := On n=28 labeled vertices, K_n has 378 edges, C_n has 28 edges, and there are 422774529505782842634856227221487319 04 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001795

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001797|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=29

§FORM := On n=29 labeled vertices, K_n has 406 edges, C_n has 29 edges, and there are 30531345459705245357453367594899121595 09 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001796

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001798|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=30

§FORM := On n=30 labeled vertices, K_n has 435 edges, C_n has 30 edges, and there are 22876792454961000000000000000000000000000000 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001797

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001799|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=31

§FORM := On n=31 labeled vertices, K_n has 465 edges, C_n has 31 edges, and there are 17761887753093897979823770061456102763834271 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001798

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001800|E|M|Proof+Compile+Meta+Tower] Standard graph counts at n=32

§FORM := On n=32 labeled vertices, K_n has 496 edges, C_n has 32 edges, and there are 142724769270595988105828596944949513638274662 4 labeled trees by Cayley's formula.

§DEP_HARD := S26-E-00001799

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001801|D|A|Proof+Compile+Meta] Partially ordered set

§FORM := A poset (P,<=) has a reflexive, antisymmetric, transitive relation.

§DEP_HARD := S26-D-00000020

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001802|D|A|Proof+Compile+Meta] Chain and antichain

§FORM := A chain is a subset of pairwise comparable elements; an antichain is a subset of pairwise incomparable elements.

§DEP_HARD := S26-D-00001801

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001803|D|A|Proof+Compile+Meta] Width of a poset

§FORM := The width is the maximum size of an antichain.

§DEP_HARD := S26-D-00001802

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001804|T|A|Proof+Compile+Meta] Dilworth theorem

§FORM := In a finite poset, the maximum antichain size equals the minimum number of chains in a partition of the poset.

§DEP_HARD := S26-D-00001803

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001805|T|A|Proof+Compile+Meta] Mirsky theorem

§FORM := In a finite poset, the maximum chain size equals the minimum number of antichains in a partition of the poset.

§DEP_HARD := S26-T-00001804

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001806|D|A|Proof+Compile+Meta] Lattice

§FORM := A lattice is a poset in which every pair has a meet and join.

§DEP_HARD := S26-T-00001805

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001807|D|A|Proof+Compile+Meta] Distributive lattice

§FORM := A lattice is distributive when meet and join distribute over one another.

§DEP_HARD := S26-D-00001806

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001808|T|A|Proof+Compile+Meta] Birkhoff representation theorem

§FORM := Every finite distributive lattice is isomorphic to the lattice of order ideals of its poset of join-irreducible elements.

§DEP_HARD := S26-D-00001807

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001809|D|A|Proof+Compile+Meta] Moebius function of a poset

§FORM := For locally finite poset P, mu(x,x)=1 and sum_{x<=z<=y}mu(x,z)=0 for x<y.

§DEP_HARD := S26-T-00001808

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001810|T|A|Proof+Compile+Meta] Poset Moebius inversion

§FORM := If g(y)=sum_{x<=y} f(x), then f(y)=sum_{x<=y}mu(x,y)g(x) for locally finite posets.

§DEP_HARD := S26-D-00001809

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001811|D|A|Proof+Compile+Meta] Matroid

§FORM := A matroid M=(E,I) consists of a finite ground set E and independent subsets I satisfying heredity and augmentation axioms.

§DEP_HARD := S26-T-00001810

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001812|D|A|Proof+Compile+Meta] Rank function of a matroid

§FORM := r(A) is the maximum size of an independent subset of A.

§DEP_HARD := S26-D-00001811

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001813|D|A|Proof+Compile+Meta] Basis of a matroid

§FORM := A basis is a maximal independent subset; all bases have the same size r(E).

§DEP_HARD := S26-D-00001812

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001814|D|A|Proof+Compile+Meta] Circuit of a matroid
§FORM := A circuit is a minimal dependent subset.
§DEP_HARD := S26-D-00001813
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001815|D|A|Proof+Compile+Meta] Closure in a matroid
§FORM := cl(A)={e:r(A union {e})=r(A)}.
§DEP_HARD := S26-D-00001814
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001816|D|A|Proof+Compile+Meta] Dual matroid
§FORM := The dual M* has bases E minus B as B ranges over bases of M.
§DEP_HARD := S26-D-00001815
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001817|D|A|Proof+Compile+Meta] Graphic matroid
§FORM := For graph G, the graphic matroid has E(G) as ground set and forests as independent sets.
§DEP_HARD := S26-D-00001816
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001818|D|A|Proof+Compile+Meta] Uniform matroid
§FORM := U_{r,n} has ground set of size n and independent sets exactly those of size at most r.
§DEP_HARD := S26-D-00001817
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001819|T|A|Proof+Compile+Meta] Greedy characterization of matroids
§FORM := A finite independence system is a matroid iff the greedy algorithm produces a maximum-weight basis for every assignment of weights under the standard ordering convention.
§DEP_HARD := S26-D-00001818
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001820|B|A|Proof+Compile+Meta] Matroids as independence abstraction
§FORM := Matroids unify linear independence, forests, transversal systems, and greedy optimization while forgetting structure not captured by independence axioms.
§DEP_HARD := S26-T-00001819
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001821|E|M|Proof+Compile+Meta] Boolean lattice B_1 profile
§FORM := The Boolean lattice of subsets of an 1-element set has 2 elements, rank sizes [1, 1], and width 1.
§DEP_HARD := S26-B-00001820
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001822|E|M|Proof+Compile+Meta] Boolean lattice B_2 profile
§FORM := The Boolean lattice of subsets of an 2-element set has 4 elements, rank sizes [1, 2, 1], and width 2.
§DEP_HARD := S26-E-00001821
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001823|E|M|Proof+Compile+Meta] Boolean lattice B_3 profile
§FORM := The Boolean lattice of subsets of an 3-element set has 8 elements, rank sizes [1, 3, 3, 1], and width 3.
§DEP_HARD := S26-E-00001822
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001824|E|M|Proof+Compile+Meta] Boolean lattice B_4 profile
§FORM := The Boolean lattice of subsets of an 4-element set has 16 elements, rank sizes [1, 4, 6, 4, 1], and width 6.
§DEP_HARD := S26-E-00001823
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001825|E|M|Proof+Compile+Meta] Boolean lattice B_5 profile
§FORM := The Boolean lattice of subsets of an 5-element set has 32 elements, rank sizes [1, 5, 10, 10, 5, 1], and width 10.
§DEP_HARD := S26-E-00001824
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001826|E|M|Proof+Compile+Meta] Boolean lattice B_6 profile
§FORM := The Boolean lattice of subsets of an 6-element set has 64 elements, rank sizes [1, 6, 15, 20, 15, 6, 1], and width 20.
§DEP_HARD := S26-E-00001825
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001827|E|M|Proof+Compile+Meta] Boolean lattice B_7 profile

§FORM := The Boolean lattice of subsets of an 7-element set has 128 elements, rank sizes [1, 7, 21, 35, 35, 21, 7, 1], and width 35.

§DEP_HARD := S26-E-00001826

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001828|E|M|Proof+Compile+Meta] Boolean lattice B_8 profile

§FORM := The Boolean lattice of subsets of an 8-element set has 256 elements, rank sizes [1, 8, 28, 56, 70, 56, 28, 8, 1], and width 70.

§DEP_HARD := S26-E-00001827

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001829|E|M|Proof+Compile+Meta] Boolean lattice B_9 profile

§FORM := The Boolean lattice of subsets of an 9-element set has 512 elements, rank sizes [1, 9, 36, 84, 126, 126, 84, 36, 9, 1], and width 126.

§DEP_HARD := S26-E-00001828

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001830|E|M|Proof+Compile+Meta] Boolean lattice B_10 profile

§FORM := The Boolean lattice of subsets of an 10-element set has 1024 elements, rank sizes [1, 10, 45, 120, 210, 252, 210, 120, 45, 10, 1], and width 252.

§DEP_HARD := S26-E-00001829

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001831|E|M|Proof+Compile+Meta] Boolean lattice B_11 profile

§FORM := The Boolean lattice of subsets of an 11-element set has 2048 elements, rank sizes [1, 11, 55, 165, 330, 462, 462, 330, 165, 55, 11, 1], and width 462.

§DEP_HARD := S26-E-00001830

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001832|E|M|Proof+Compile+Meta] Boolean lattice B_12 profile

§FORM := The Boolean lattice of subsets of an 12-element set has 4096 elements, rank sizes [1, 12, 66, 220, 495, 792, 924, 792, 495, 220, 66, 12, 1], and width 924.

§DEP_HARD := S26-E-00001831

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001833|E|M|Proof+Compile+Meta] Boolean lattice B_13 profile

§FORM := The Boolean lattice of subsets of an 13-element set has 8192 elements, rank sizes [1, 13, 78, 286, 715, 1287, 1716, 1716, 1287, 715, 286, 78, 13, 1], and width 1716.

§DEP_HARD := S26-E-00001832

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001834|E|M|Proof+Compile+Meta] Boolean lattice B_14 profile

§FORM := The Boolean lattice of subsets of an 14-element set has 16384 elements, rank sizes [1, 14, 91, 364, 1001, 2002, 3003, 3432, 3003, 2002, 1001, 364, 91, 14, 1], and width 3432.

§DEP_HARD := S26-E-00001833

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001835|E|M|Proof+Compile+Meta] Boolean lattice B_15 profile

§FORM := The Boolean lattice of subsets of an 15-element set has 32768 elements, rank sizes [1, 15, 105, 455, 1365, 3003, 5005, 6435, 6435, 5005, 3003, 1365, 455, 105, 15, 1], and width 6435.

§DEP_HARD := S26-E-00001834

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001836|E|M|Proof+Compile+Meta] Boolean lattice B_16 profile

§FORM := The Boolean lattice of subsets of an 16-element set has 65536 elements, rank sizes [1, 16, 120, 560, 1820, 4368, 8008, 11440, 12870, 11440, 8008, 4368, 1820, 560, 120, 16, 1], and width 12870.

§DEP_HARD := S26-E-00001835

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001837|E|M|Proof+Compile+Meta] Boolean lattice B_17 profile

§FORM := The Boolean lattice of subsets of an 17-element set has 131072 elements, rank sizes [1, 17, 136, 680, 2380, 6188, 12376, 19448, 24310, 24310, 19448, 12376, 6188, 2380, 680, 136, 17, 1], and width 24310.

§DEP_HARD := S26-E-00001836

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001838|E|M|Proof+Compile+Meta] Boolean lattice B_18 profile

§FORM := The Boolean lattice of subsets of an 18-element set has 262144 elements, rank sizes [1, 18, 153, 816, 3060, 8568, 18564, 31824, 43758, 48620, 43758, 31824, 18564, 8568, 3060, 816, 153, 18, 1], and width 48620.

§DEP_HARD := S26-E-00001837

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001839|E|M|Proof+Compile+Meta] Boolean lattice B_19 profile

§FORM := The Boolean lattice of subsets of an 19-element set has 524288 elements, rank sizes [1, 19, 171, 969, 3876, 11628, 27132, 50388, 75582, 92378, 92378, 75582, 50388, 27132, 11628, 3876, 969, 171, 19, 1], and width 92378.

§DEP_HARD := S26-E-00001838

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001840|E|M|Proof+Compile+Meta] Boolean lattice B_20 profile

§FORM := The Boolean lattice of subsets of an 20-element set has 1048576 elements, rank sizes [1, 20, 190, 1140, 4845, 15504, 38760, 77520, 125970, 167960, 184756, 167960, 125970, 77520, 38760, 15504, 4845, 1140, 190, 20, 1], and width 184756.

§DEP_HARD := S26-E-00001839

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001841|E|M|Proof+Compile+Meta] Boolean lattice B_21 profile

§FORM := The Boolean lattice of subsets of an 21-element set has 2097152 elements, rank sizes [1, 21, 210, 1330, 5985, 20349, 54264, 116280, 203490, 293930, 352716, 352716, 293930, 203490, 116280, 54264, 20349, 5985, 1330, 210, 21, 1], and width 352716.

§DEP_HARD := S26-E-00001840

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001842|E|M|Proof+Compile+Meta] Boolean lattice B_22 profile

§FORM := The Boolean lattice of subsets of an 22-element set has 4194304 elements, rank sizes [1, 22, 231, 1540, 7315, 26334, 74613, 170544, 319770, 497420, 646646, 705432, 646646, 497420, 319770, 170544, 74613, 26334, 7315, 1540, 231, 22, 1], and width 705432.

§DEP_HARD := S26-E-00001841

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001843|E|M|Proof+Compile+Meta] Boolean lattice B_23 profile

§FORM := The Boolean lattice of subsets of an 23-element set has 8388608 elements, rank sizes [1, 23, 253, 1771, 8855, 33649, 100947, 245157, 490314, 817190, 1144066, 1352078, 1352078, 1144066, 817190, 490314, 245157, 100947, 33649, 8855, 1771, 253, 23, 1], and width 1352078.

§DEP_HARD := S26-E-00001842

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001844|E|M|Proof+Compile+Meta] Boolean lattice B_24 profile

§FORM := The Boolean lattice of subsets of an 24-element set has 16777216 elements, rank sizes [1, 24, 276, 2024, 10626, 42504, 134596, 346104, 735471, 1307504, 1961256, 2496144, 2704156, 2496144, 1961256, 1307504, 735471, 346104, 134596, 42504, 10626, 2024, 276, 24, 1], and width 2704156.

§DEP_HARD := S26-E-00001843

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001845|E|M|Proof+Compile+Meta] Boolean lattice B_25 profile

§FORM := The Boolean lattice of subsets of an 25-element set has 33554432 elements, rank sizes [1, 25, 300, 2300, 12650, 53130, 177100, 480700, 1081575, 2042975, 3268760, 4457400, 5200300, 5200300, 4457400, 3268760, 2042975, 1081575, 480700, 177100, 53130, 12650, 2300, 300, 25, 1], and width 5200300.

§DEP_HARD := S26-E-00001844

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001846|E|M|Proof+Compile+Meta] Boolean lattice B_26 profile

§FORM := The Boolean lattice of subsets of an 26-element set has 67108864 elements, rank sizes [1, 26, 325, 2600, 14950, 65780, 230230, 657800, 1562275, 3124550, 5311735, 7726160, 9657700, 10400600, 9657700, 7726160, 5311735, 3124550, 1562275, 657800, 230230, 65780, 14950, 2600, 325, 26, 1], and width 10400600.

§DEP_HARD := S26-E-00001845

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001847|E|M|Proof+Compile+Meta] Boolean lattice B_27 profile

§FORM := The Boolean lattice of subsets of an 27-element set has 134217728 elements, rank sizes [1, 27, 351, 2925, 17550, 80730, 296010, 888030, 2220075, 4686825, 8436285, 13037895, 17383860, 20058300, 20058300, 17383860, 13037895, 8436285, 4686825, 2220075, 888030, 296010, 80730, 17550, 2925, 351, 27, 1], and width 20058300.

§DEP_HARD := S26-E-00001846

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001848|E|M|Proof+Compile+Meta] Boolean lattice B_28 profile

§FORM := The Boolean lattice of subsets of an 28-element set has 268435456 elements, rank sizes [1, 28, 378, 3276, 20475, 98280, 376740, 1184040, 3108105, 6906900, 13123110, 21474180, 30421755, 37442160, 40116600, 37442160, 30421755, 21474180, 13123110, 6906900, 3108105, 1184040, 376740, 98280, 20475, 3276, 378, 28, 1], and width 40116600.

§DEP_HARD := S26-E-00001847

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001849|E|M|Proof+Compile+Meta] Boolean lattice B_29 profile

§FORM := The Boolean lattice of subsets of an 29-element set has 536870912 elements, rank sizes [1, 29, 406, 3654, 23751, 118755, 475020, 1560780, 4292145, 10015005, 20030010, 34597290, 51895935, 67863915, 77558760, 77558760, 67863915, 51895935, 34597290, 20030010, 10015005, 4292145, 1560780, 475020, 118755, 23751, 3654, 406, 29, 1], and width 77558760.

§DEP_HARD := S26-E-00001848

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001850|E|M|Proof+Compile+Meta] Boolean lattice B_30 profile

§FORM := The Boolean lattice of subsets of an 30-element set has 1073741824 elements, rank sizes [1, 30, 435, 4060, 27405, 142506, 593775, 2035800, 5852925, 14307150, 30045015, 54627300, 86493225, 119759850, 145422675, 155117520, 145422675, 119759850, 86493225, 54627300, 30045015, 14307150, 5852925, 2035800, 593775, 142506, 27405, 4060, 435, 30, 1], and width 155117520.

§DEP_HARD := S26-E-00001849

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001851|D|A|Proof+Compile+Meta] q-ary block code

§FORM := A q-ary block code of length n is a subset C of A^n for an alphabet A with |A|=q.

§DEP_HARD := S26-D-00001251

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001852|D|A|Proof+Compile+Meta] Hamming distance

§FORM := d_H(x,y) is the number of coordinates in which words x and y differ.

§DEP_HARD := S26-D-00001851

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001853|D|A|Proof+Compile+Meta] Minimum distance of a code

§FORM := d(C)=min_{x!=y in C} d_H(x,y).

§DEP_HARD := S26-D-00001852

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001854|T|A|Proof+Compile+Meta] Error correction from minimum distance

§FORM := A code of minimum distance d corrects every pattern of at most floor((d-1)/2) symbol errors by nearest-neighbor decoding when the nearest codeword is unique.

§DEP_HARD := S26-D-00001853

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001855|D|A|Proof+Compile+Meta] Linear code

§FORM := A linear [n,k]_q code is a k-dimensional subspace of F_q^n.

§DEP_HARD := S26-T-00001854

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001856|D|A|Proof+Compile+Meta] Generator matrix

§FORM := A generator matrix G for a linear [n,k]_q code has row space C.

§DEP_HARD := S26-D-00001855

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001857|D|A|Proof+Compile+Meta] Parity-check matrix

§FORM := A parity-check matrix H satisfies C=ker H; for an [n,k]_q code, H may be taken with n-k independent rows.

§DEP_HARD := S26-D-00001856

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001858|D|A|Proof+Compile+Meta] Dual code

§FORM := C^perp={y:<x,y>=0 for every x in C}.

§DEP_HARD := S26-D-00001857

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001859|I|A|Proof+Compile+Meta] Dimension of dual code

§FORM := For linear C<=F_q^n, dim C + dim C^perp=n.

§DEP_HARD := S26-D-00001858

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001860|T|A|Proof+Compile+Meta] Singleton bound

§FORM := For a q-ary code of size M and minimum distance d, M<=q^{n-d+1}; for linear [n,k,d]_q, d<=n-k+1.

§DEP_HARD := S26-I-00001859

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001861|D|A|Proof+Compile+Meta] MDS code

§FORM := A linear code meeting the Singleton bound with equality is maximum-distance separable.

§DEP_HARD := S26-T-00001860

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001862|T|A|Proof+Compile+Meta] Hamming sphere-packing bound

§FORM := For a q-ary code correcting t errors, |C| sum_{i=0}^t C(n,i)(q-1)^i <= q^n.

§DEP_HARD := S26-D-00001861

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001863|T|A|Proof+Compile+Meta] Gilbert-Varshamov bound interface

§FORM := A greedy packing argument gives lower bounds on the size/existence of codes with prescribed length and minimum distance; exact form depends on linear vs nonlinear setting.

§DEP_HARD := S26-T-00001862

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001864|D|A|Proof+Compile+Meta] Hamming code

§FORM := A q-ary Hamming code is a perfect single-error-correcting linear code with parity-check columns representing projective points under the standard construction.

§DEP_HARD := S26-T-00001863

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001865|D|A|Proof+Compile+Meta] Reed-Solomon code

§FORM := A Reed-Solomon code evaluates low-degree polynomials at distinct field points; it is an MDS code.

§DEP_HARD := S26-D-00001864

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001866|D|A|Proof+Compile+Meta] Cyclic code

§FORM := A linear code C<=F_q^n is cyclic when cyclic coordinate shift preserves C; equivalently it corresponds to an ideal of F_q[x]/(x^n-1).

§DEP_HARD := S26-D-00001865

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001867|D|A|Proof+Compile+Meta] Block design

§FORM := A t-(v,k,lambda) design is a family of k-subsets (blocks) of a v-set such that every t-subset occurs in exactly lambda blocks.

§DEP_HARD := S26-D-00001866

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001868|D|A|Proof+Compile+Meta] Steiner system

§FORM := A Steiner system S(t,k,v) is a t-(v,k,1) design.

§DEP_HARD := S26-D-00001867

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001869|I|A|Proof+Compile+Meta] BIBD parameter relation

§FORM := For a 2-(v,k,lambda) design with b blocks and replication number r, vr=bk and r(k-1)=lambda(v-1).

§DEP_HARD := S26-D-00001868

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001870|B|A|Proof+Compile+Meta+Topo] Codes and finite geometry

§FORM := Linear codes, projective point sets, finite-field polynomials, and block designs share incidence structures; translations preserve some but not all parameters.

§DEP_HARD := S26-I-00001869

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001871|E|M|Proof+Compile+Meta] Binary Hamming code profile m=2

§FORM := The standard binary Hamming code has parameters [3,1,3]_2 and corrects one bit error.

§DEP_HARD := S26-B-00001870

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001872|E|M|Proof+Compile+Meta] Binary Hamming code profile m=3

§FORM := The standard binary Hamming code has parameters [7,4,3]_2 and corrects one bit error.

§DEP_HARD := S26-E-00001871

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001873|E|M|Proof+Compile+Meta] Binary Hamming code profile m=4

§FORM := The standard binary Hamming code has parameters [15,11,3]_2 and corrects one bit error.

§DEP_HARD := S26-E-00001872

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001874|E|M|Proof+Compile+Meta] Binary Hamming code profile m=5

§FORM := The standard binary Hamming code has parameters [31,26,3]_2 and corrects one bit error.

§DEP_HARD := S26-E-00001873

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001875|E|M|Proof+Compile+Meta] Binary Hamming code profile m=6

§FORM := The standard binary Hamming code has parameters [63,57,3]_2 and corrects one bit error.

§DEP_HARD := S26-E-00001874

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001876|E|M|Proof+Compile+Meta] Binary Hamming code profile m=7

§FORM := The standard binary Hamming code has parameters [127,120,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001875
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001877|E|M|Proof+Compile+Meta] Binary Hamming code profile m=8

§FORM := The standard binary Hamming code has parameters [255,247,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001876
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001878|E|M|Proof+Compile+Meta] Binary Hamming code profile m=9

§FORM := The standard binary Hamming code has parameters [511,502,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001877
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001879|E|M|Proof+Compile+Meta] Binary Hamming code profile m=10

§FORM := The standard binary Hamming code has parameters [1023,1013,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001878
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001880|E|M|Proof+Compile+Meta] Binary Hamming code profile m=11

§FORM := The standard binary Hamming code has parameters [2047,2036,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001879
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001881|E|M|Proof+Compile+Meta] Binary Hamming code profile m=12

§FORM := The standard binary Hamming code has parameters [4095,4083,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001880
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001882|E|M|Proof+Compile+Meta] Binary Hamming code profile m=13

§FORM := The standard binary Hamming code has parameters [8191,8178,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001881
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001883|E|M|Proof+Compile+Meta] Binary Hamming code profile m=14

§FORM := The standard binary Hamming code has parameters [16383,16369,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001882
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001884|E|M|Proof+Compile+Meta] Binary Hamming code profile m=15

§FORM := The standard binary Hamming code has parameters [32767,32752,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001883
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001885|E|M|Proof+Compile+Meta] Binary Hamming code profile m=16

§FORM := The standard binary Hamming code has parameters [65535,65519,3]_2 and corrects one bit error.
§DEP_HARD := S26-E-00001884
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001886|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=3

§FORM := Over a field F_3, evaluating polynomials of degree <1 at 2 distinct nonzero field points gives an [2,1,2]_3 Reed-Solomon MDS code.
§DEP_HARD := S26-E-00001885
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001887|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=5

§FORM := Over a field F_5, evaluating polynomials of degree <2 at 4 distinct nonzero field points gives an [4,2,3]_5 Reed-Solomon MDS code.
§DEP_HARD := S26-E-00001886
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001888|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=7

§FORM := Over a field F_7, evaluating polynomials of degree <3 at 6 distinct nonzero field points gives an [6,3,4]_7 Reed-Solomon MDS code.
§DEP_HARD := S26-E-00001887
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001889|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=8

§FORM := Over a field F_8, evaluating polynomials of degree <4 at 7 distinct nonzero field points gives an [7,4,4]_8 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001888

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001890|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=9

§FORM := Over a field F_9, evaluating polynomials of degree <4 at 8 distinct nonzero field points gives an [8,4,5]_9 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001889

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001891|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=11

§FORM := Over a field F_11, evaluating polynomials of degree <5 at 10 distinct nonzero field points gives an [10,5,6]_11 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001890

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001892|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=13

§FORM := Over a field F_13, evaluating polynomials of degree <6 at 12 distinct nonzero field points gives an [12,6,7]_13 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001891

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001893|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=16

§FORM := Over a field F_16, evaluating polynomials of degree <8 at 15 distinct nonzero field points gives an [15,8,8]_16 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001892

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001894|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=17

§FORM := Over a field F_17, evaluating polynomials of degree <8 at 16 distinct nonzero field points gives an [16,8,9]_17 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001893

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001895|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=19

§FORM := Over a field F_19, evaluating polynomials of degree <9 at 18 distinct nonzero field points gives an [18,9,10]_19 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001894

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001896|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=23

§FORM := Over a field F_23, evaluating polynomials of degree <11 at 22 distinct nonzero field points gives an [22,11,12]_23 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001895

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001897|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=25

§FORM := Over a field F_25, evaluating polynomials of degree <12 at 24 distinct nonzero field points gives an [24,12,13]_25 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001896

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001898|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=27

§FORM := Over a field F_27, evaluating polynomials of degree <13 at 26 distinct nonzero field points gives an [26,13,14]_27 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001897

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001899|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=29

§FORM := Over a field F_29, evaluating polynomials of degree <14 at 28 distinct nonzero field points gives an [28,14,15]_29 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001898

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001900|E|M|Proof+Compile+Meta] Reed-Solomon parameter profile q=31

§FORM := Over a field F_31, evaluating polynomials of degree <15 at 30 distinct nonzero field points gives an [30,15,16]_31 Reed-Solomon MDS code.

§DEP_HARD := S26-E-00001899

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001901|D|A|Proof+Compile+Meta] Sumset

§FORM := For subsets A,B of an abelian group, A+B={a+b:a in A,b in B}.

§DEP_HARD := S26-D-00000709

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001902|D|A|Proof+Compile+Meta] Difference set

§FORM := A-B={a-b:a in A,b in B}.

§DEP_HARD := S26-D-00001901

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001903|D|A|Proof+Compile+Meta] Additive energy

§FORM := For finite A,B in an abelian group, E(A,B) is the number of quadruples a_1+b_1=a_2+b_2 with a_i in A,b_i in B.

§DEP_HARD := S26-D-00001902

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001904|I|A|Proof+Compile+Meta] Energy via representation function

§FORM := If r_{A+B}(x)=|{(a,b):a+b=x}|, then E(A,B)=sum_x r_{A+B}(x)^2.

§DEP_HARD := S26-D-00001903

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001905|T|A|Proof+Compile+Meta] Cauchy-Schwarz sumset bound

§FORM := For finite nonempty A,B, E(A,B)>=|A|^2|B|^2/|A+B|.

§DEP_HARD := S26-I-00001904

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001906|T|A|Proof+Compile+Meta] Cauchy-Davenport theorem

§FORM := For nonempty A,B subset F_p, |A+B|>=min(p,|A|+|B|-1).

§DEP_HARD := S26-T-00001905

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001907|T|A|Proof+Compile+Meta] Vosper theorem interface

§FORM := Under the nontrivial equality case of Cauchy-Davenport with sufficiently small sumset, subsets of F_p are arithmetic progressions up to the standard exceptional cases.

§DEP_HARD := S26-T-00001906

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001908|D|A|Proof+Compile+Meta] Arithmetic progression

§FORM := A finite arithmetic progression in an abelian group is {a+jd:0<=j<k} with declared nondegeneracy conditions.

§DEP_HARD := S26-T-00001907

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001909|D|A|Proof+Compile+Meta] Generalized arithmetic progression

§FORM := A rank-r generalized progression is {a+n_1d_1+...+n_rd_r:0<=n_i<L_i}; properness means the parameterization is injective.

§DEP_HARD := S26-D-00001908

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001910|T|A|Proof+Compile+Meta] Freiman theorem interface

§FORM := Finite subsets of torsion-free abelian groups with small doubling are controlled by generalized arithmetic progressions; constants depend on the doubling parameter and chosen formulation.

§DEP_HARD := S26-D-00001909

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001911|D|A|Proof+Compile+Meta] Sidon set

§FORM := A Sidon/B_2 set has essentially unique pairwise sums under a declared ordered/unordered convention.

§DEP_HARD := S26-T-00001910

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001912|D|A|Proof+Compile+Meta] Convex lattice polytope

§FORM := A lattice polytope is the convex hull of finitely many points in Z^d.

§DEP_HARD := S26-D-00001911

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001913|D|A|Proof+Compile+Meta] Ehrhart polynomial

§FORM := For a d-dimensional lattice polytope P, L_P(n)=|nP intersect Z^d| is a polynomial in n of degree d for n>=0.

§DEP_HARD := S26-D-00001912

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001914|T|A|Proof+Compile+Meta] Ehrhart leading coefficient

§FORM := The leading coefficient of the Ehrhart polynomial is the Euclidean d-volume of P under the lattice-normalized coordinate convention.

§DEP_HARD := S26-D-00001913

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001915|T|A|Proof+Compile+Meta] Pick theorem

§FORM := For a lattice polygon in R^2, area=I+B/2-1, where I and B are the numbers of interior and boundary lattice points.

§DEP_HARD := S26-T-00001914

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001916|D|A|Proof+Compile+Meta] Hypercube

§FORM := The d-dimensional unit hypercube is [0,1]^d with 2^d vertices.

§DEP_HARD := S26-T-00001915

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00001917|I|A|Proof+Compile+Meta] Hypercube face numbers

§FORM := The number of k-dimensional faces of the d-cube is 2^{d-k} C(d,k).

§DEP_HARD := S26-D-00001916

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001918|D|A|Proof+Compile+Meta] Cross-polytope

§FORM := The d-dimensional cross-polytope is conv{+/- e_1,...,+/- e_d}; it is dual to the hypercube.

§DEP_HARD := S26-I-00001917

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00001919|N|A|Proof+Compile+Meta] Additive-structure fence

§FORM := Small doubling is a structural hypothesis, not a synonym for randomness or pseudorandomness; the relevant inverse theorem depends on the ambient group and scale.

§DEP_HARD := S26-D-00001918

§EVIDENCE := A · Exact registered statement only.

§BOUNDARY := A set can have small doubling because it is highly structured.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001920|B|A|Proof+Compile+Meta+Spectral] Additive combinatorics to harmonic analysis

§FORM := Fourier coefficients, convolution, energy, and sumsets translate additive structure into analytic concentration and back.

§DEP_HARD := S26-N-00001919

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001921|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=1

§FORM := The 1-cube has face vector f_k=2^(d-k)C(d,k): [2, 1]. In particular it has 2 vertices and 1 edges.

§DEP_HARD := S26-B-00001920

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001922|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=2

§FORM := The 2-cube has face vector f_k=2^(d-k)C(d,k): [4, 4, 1]. In particular it has 4 vertices and 4 edges.

§DEP_HARD := S26-E-00001921

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001923|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=3

§FORM := The 3-cube has face vector f_k=2^(d-k)C(d,k): [8, 12, 6, 1]. In particular it has 8 vertices and 12 edges.

§DEP_HARD := S26-E-00001922

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001924|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=4

§FORM := The 4-cube has face vector f_k=2^(d-k)C(d,k): [16, 32, 24, 8, 1]. In particular it has 16 vertices and 32 edges.

§DEP_HARD := S26-E-00001923

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001925|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=5

§FORM := The 5-cube has face vector f_k=2^(d-k)C(d,k): [32, 80, 80, 40, 10, 1]. In particular it has 32 vertices and 80 edges.

§DEP_HARD := S26-E-00001924

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001926|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=6

§FORM := The 6-cube has face vector f_k=2^(d-k)C(d,k): [64, 192, 240, 160, 60, 12, 1]. In particular it has 64 vertices and 192 edges.

§DEP_HARD := S26-E-00001925

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001927|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=7

§FORM := The 7-cube has face vector f_k=2^(d-k)C(d,k): [128, 448, 672, 560, 280, 84, 14, 1]. In particular it has 128 vertices and 448 edges.

§DEP_HARD := S26-E-00001926

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001928|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=8

§FORM := The 8-cube has face vector f_k=2^(d-k)C(d,k): [256, 1024, 1792, 1792, 1120, 448, 112, 16, 1]. In particular it has 256 vertices and 1024 edges.

§DEP_HARD := S26-E-00001927

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001929|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=9

§FORM := The 9-cube has face vector f_k=2^(d-k)C(d,k): [512, 2304, 4608, 5376, 4032, 2016, 672, 144, 18, 1]. In particular it has 512 vertices and 2304 edges.

§DEP_HARD := S26-E-00001928

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001930|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=10

§FORM := The 10-cube has face vector f_k=2^(d-k)C(d,k): [1024, 5120, 11520, 15360, 13440, 8064, 3360, 960, 180, 20, 1]. In particular it has 1024 vertices and 5120 edges.

§DEP_HARD := S26-E-00001929

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001931|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=11

§FORM := The 11-cube has face vector f_k=2^(d-k)C(d,k): [2048, 11264, 28160, 42240, 42240, 29568, 14784, 5280, 1320, 220, 22, 1]. In particular it has 2048 vertices and 11264 edges.

§DEP_HARD := S26-E-00001930

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001932|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=12

§FORM := The 12-cube has face vector f_k=2^(d-k)C(d,k): [4096, 24576, 67584, 112640, 126720, 101376, 59136, 25344, 7920, 1760, 264, 24, 1]. In particular it has 4096 vertices and 24576 edges.

§DEP_HARD := S26-E-00001931

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001933|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=13

§FORM := The 13-cube has face vector f_k=2^(d-k)C(d,k): [8192, 53248, 159744, 292864, 366080, 329472, 219648, 109824, 41184, 11440, 2288, 312, 26, 1]. In particular it has 8192 vertices and 53248 edges.

§DEP_HARD := S26-E-00001932

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001934|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=14

§FORM := The 14-cube has face vector f_k=2^(d-k)C(d,k): [16384, 114688, 372736, 745472, 1025024, 1025024, 768768, 439296, 192192, 64064, 16016, 2912, 364, 28, 1]. In particular it has 16384 vertices and 114688 edges.

§DEP_HARD := S26-E-00001933

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001935|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=15

§FORM := The 15-cube has face vector f_k=2^(d-k)C(d,k): [32768, 245760, 860160, 1863680, 2795520, 3075072, 2562560, 1647360, 823680, 320320, 96096, 21840, 3640, 420, 30, 1]. In particular it has 32768 vertices and 245760 edges.

§DEP_HARD := S26-E-00001934

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001936|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=16

§FORM := The 16-cube has face vector f_k=2^(d-k)C(d,k): [65536, 524288, 1966080, 4587520, 7454720, 8945664, 8200192, 5857280, 3294720, 1464320, 512512, 139776, 29120, 4480, 480, 32, 1]. In particular it has 65536 vertices and 524288 edges.

§DEP_HARD := S26-E-00001935

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001937|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=17

§FORM := The 17-cube has face vector f_k=2^(d-k)C(d,k): [131072, 1114112, 4456448, 11141120, 19496960, 25346048, 25346048, 19914752, 12446720, 6223360, 2489344, 792064, 198016, 38080, 5440, 544, 34, 1]. In particular it has 131072 vertices and 1114112 edges.

§DEP_HARD := S26-E-00001936

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001938|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=18

§FORM := The 18-cube has face vector f_k=2^(d-k)C(d,k): [262144, 2359296, 10027008, 26738688, 50135040, 70189056, 76038144, 65175552, 44808192, 24893440, 11202048, 4073472, 1188096, 274176, 48960, 6528, 612, 36, 1]. In particular it has 262144 vertices and 2359296 edges.

§DEP_HARD := S26-E-00001937

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001939|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=19

§FORM := The 19-cube has face vector f_k=2^(d-k)C(d,k): [524288, 4980736, 22413312, 63504384, 127008768, 190513152, 222265344, 206389248, 154791936, 94595072, 47297536, 19348992, 6449664, 1736448, 372096, 62016, 7752, 684, 38, 1]. In particular it has 524288 vertices and 4980736 edges.

§DEP_HARD := S26-E-00001938

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001940|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=20

§FORM := The 20-cube has face vector f_k=2^(d-k)C(d,k): [1048576, 10485760, 49807360, 149422080, 317521920, 508035072, 635043840, 635043840, 515973120, 343982080, 189190144, 85995520, 32248320, 9922560, 2480640, 496128, 77520, 9120, 760, 40, 1]. In particular it has 1048576 vertices and 10485760 edges.

§DEP_HARD := S26-E-00001939

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001941|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=21

§FORM := The 21-cube has face vector f_k=2^(d-k)C(d,k): [2097152, 22020096, 110100480, 348651520, 784465920, 1333592064, 1778122752, 1905131520, 1666990080, 1203937280, 722362368, 361181184, 150492160, 52093440, 14883840, 3472896, 651168, 95760, 10640, 840, 42, 1]. In particular it has 2097152 vertices and 22020096 edges.

§DEP_HARD := S26-E-00001940

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001942|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=22

§FORM := The 22-cube has face vector f_k=2^(d-k)C(d,k): [4194304, 46137344, 242221056, 807403520, 1917583360, 3451650048, 4889837568, 5588385792, 5239111680, 4074864640, 2648662016, 1444724736, 662165504, 254679040, 81861120, 21829632, 4775232, 842688, 117040, 12320, 924, 44, 1]. In particular it has 4194304 vertices and 46137344 edges.

§DEP_HARD := S26-E-00001941

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001943|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=23

§FORM := The 23-cube has face vector f_k=2^(d-k)C(d,k): [8388608, 96468992, 530579456, 1857028096, 4642570240, 8820883456, 13231325184, 16066609152, 16066609152, 13388840960, 9372188672, 5538111488, 2769055744, 1171523584, 418401280, 125520384, 31380096, 6460608, 1076768, 141680, 14168, 1012, 46, 1]. In particular it has 8388608 vertices and 96468992 edges.

§DEP_HARD := S26-E-00001942

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001944|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=24

§FORM := The 24-cube has face vector f_k=2^(d-k)C(d,k): [16777216, 201326592, 1157627904, 4244635648, 11142168576, 22284337152, 35283533824, 45364543488, 48199827456, 42844291072, 32133218304, 20448411648, 11076222976, 5112102912, 2008326144, 669442048, 188280576, 44301312, 8614144, 1360128, 170016, 16192, 1104, 48, 1]. In particular it has 16777216 vertices and 201326592 edges.

§DEP_HARD := S26-E-00001943

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001945|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=25

§FORM := The 25-cube has face vector f_k=2^(d-k)C(d,k): [33554432, 419430400, 2516582400, 9646899200, 26528972800, 55710842880, 92851404800, 126012620800, 141764198400, 133888409600, 107110727680, 73030041600, 42600857600, 21300428800, 9128755200, 3347210240, 1046003200, 276883200, 61529600, 11334400, 1700160, 202400, 18400, 1200, 50, 1]. In particular it has 33554432 vertices and 419430400 edges.

§DEP_HARD := S26-E-00001944

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001946|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=26

§FORM := The 26-cube has face vector f_k=2^(d-k)C(d,k): [67108864, 872415232, 5452595200, 21810380800, 62704844800, 137950658560, 241413652480, 344876646400, 409541017600, 409541017600, 348109864960, 253170810880, 158231756800, 85201715200, 39557939200, 15823175680, 5439216640, 1599769600, 399942400, 84198400, 14734720, 2104960, 239200, 20800, 1300, 52, 1]. In particular it has 67108864 vertices and 872415232 edges.

§DEP_HARD := S26-E-00001945

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001947|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=27

§FORM := The 27-cube has face vector f_k=2^(d-k)C(d,k): [134217728, 1811939328, 11777605632, 49073356800, 147220070400, 338606161920, 620777963520, 931166945280, 1163958681600, 1228623052800, 1105760747520, 854451486720, 569634324480, 328635187200, 164317593600, 71204290560, 26701608960, 8638755840, 2399654400, 568339200, 113667840, 18944640, 2583360, 280800, 23400, 1404, 54, 1]. In particular it has 134217728 vertices and 1811939328 edges.

§DEP_HARD := S26-E-00001946

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001948|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=28

§FORM := The 28-cube has face vector f_k=2^(d-k)C(d,k): [268435456, 3758096384, 25367150592, 109924319232, 343513497600, 824432394240, 1580162088960, 2483111854080, 3259084308480, 3621204787200, 3440144547840, 2814663720960, 1993720135680, 1226904698880, 657270374400, 306726174720, 124607508480, 43979120640, 13438064640, 3536332800, 795674880, 151557120, 24111360, 3144960, 327600, 26208, 1512, 56, 1]. In particular it has 268435456 vertices and 3758096384 edges.

§DEP_HARD := S26-E-00001947

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001949|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=29

§FORM := The 29-cube has face vector f_k=2^(d-k)C(d,k): [536870912, 7784628224, 54492397568, 245215789056, 796951314432, 1992378286080, 3984756572160, 6546385797120, 9001280471040, 10501493882880, 10501493882880, 9069471989760, 6802103992320, 4447529533440, 2541445447680, 1270722723840, 555941191680, 212565749760, 70855249920, 20510730240, 5127682560, 1098789120, 199779840, 30401280, 3800160, 380016, 29232, 1624, 58, 1]. In particular it has 536870912 vertices and 7784628224 edges.

§DEP_HARD := S26-E-00001948

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001950|E|M|Proof+Compile+Meta+Topo] Hypercube face profile d=30

§FORM := The 30-cube has face vector f_k=2^(d-k)C(d,k): [1073741824, 16106127360, 116769423360, 544923975680, 1839118417920, 4781707886592, 9961891430400, 17077528166400, 24548946739200, 30004268236800, 31504481648640, 28640437862400, 22673679974400, 15697163059200, 9530420428800, 5082890895360, 2382605107200, 981072691200, 354276249600, 111876710400, 30766095360, 7325260800, 1498348800, 260582400, 38001600, 4560192, 438480, 32480, 1740, 60, 1]. In particular it has 1073741824 vertices and 16106127360 edges.

§DEP_HARD := S26-E-00001949

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00001951|D|A|Proof+Compile+Meta] Asymptotic time complexity

§FORM := A time bound T(n)=O(f(n)) means there exist constants C,n_0 such that T(n)<=C f(n) for n>=n_0 under the declared computational model.

§DEP_HARD := S26-A-00000066

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001952|D|A|Proof+Compile+Meta] Polynomial-time algorithm

§FORM := An algorithm runs in polynomial time when its worst-case running time is bounded by n^{O(1)} in the input length under a specified reasonable machine model.

§DEP_HARD := S26-D-00001951

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001953|D|A|Proof+Compile+Meta] Complexity class P

§FORM := P is the class of decision problems solvable by deterministic Turing machines in polynomial time.

§DEP_HARD := S26-D-00001952

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001954|D|A|Proof+Compile+Meta] Complexity class NP

§FORM := NP is the class of decision problems whose yes-instances have polynomial-size certificates verifiable in deterministic polynomial time; equivalently solvable by nondeterministic polynomial time.

§DEP_HARD := S26-D-00001953

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001955|D|A|Proof+Compile+Meta] Polynomial-time many-one reduction

§FORM := A<=_p B when a polynomial-time computable map f satisfies x in A iff f(x) in B.

§DEP_HARD := S26-D-00001954

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001956|D|A|Proof+Compile+Meta] NP-hardness

§FORM := A problem B is NP-hard when every problem in NP reduces to B by the declared polynomial-time reduction.

§DEP_HARD := S26-D-00001955

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00001957|D|A|Proof+Compile+Meta] NP-completeness

§FORM := A decision problem is NP-complete when it lies in NP and is NP-hard.

§DEP_HARD := S26-D-00001956

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001958|T|A|Proof+Compile+Meta] Cook-Levin theorem

§FORM := Boolean satisfiability is NP-complete under polynomial-time many-one reductions.

§DEP_HARD := S26-D-00001957

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00001959|A|A|Proof+Compile+Meta] Euclidean algorithm complexity interface

§FORM := The Euclidean algorithm computes gcd(a,b) by repeated division with remainder; bit-complexity analysis depends on integer-arithmetic cost, while the number of divisions is logarithmic in the smaller positive input up to classical worst-case constants.

§DEP_HARD := S26-T-00001958

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00001960|A|A|Proof+Compile+Meta] Fast exponentiation

§FORM := Binary exponentiation computes a^n using O(log n) multiplications by repeated squaring and multiplication according to the binary digits of n.

§DEP_HARD := S26-A-00001959

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00001961|A|A|Proof+Compile+Meta] Modular exponentiation

§FORM := Binary exponentiation combined with modular reduction computes a^n mod m using O(log n) modular multiplications.

§DEP_HARD := S26-A-00001960

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00001962|A|A|Proof+Compile+Meta] Sieve of Eratosthenes

§FORM := The sieve marks composites up to N by prime multiples and enumerates all primes <=N; standard implementations use O(N log log N) arithmetic marking operations up to model-dependent refinements.

§DEP_HARD := S26-A-00001961

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00001963|A|A|Proof+Compile+Meta] Breadth-first search

§FORM := BFS explores an unweighted graph by distance layers and runs in O(|V|+|E|) with adjacency-list representation.

§DEP_HARD := S26-A-00001962

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00001964|A|A|Proof+Compile+Meta+Topo] Depth-first search

§FORM := DFS explores along depth-first branches and runs in O(|V|+|E|) with adjacency lists; it supports component, cycle, and topological-order routines.

§DEP_HARD := S26-A-00001963

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00001965|A|A|Proof+Compile+Meta] Dijkstra algorithm

§FORM := With nonnegative edge weights, Dijkstra's algorithm computes single-source shortest-path distances; complexity depends on the priority-queue implementation.

§DEP_HARD := S26-A-00001964

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00001966|A|A|Proof+Compile+Meta] Kruskal algorithm

§FORM := Kruskal's algorithm sorts edges by weight and greedily adds edges that do not create a cycle, yielding a minimum spanning forest.

§DEP_HARD := S26-A-00001965

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00001967|A|A|Proof+Compile+Meta] Prim algorithm

§FORM := Prim's algorithm grows a minimum spanning tree by repeatedly adding the cheapest edge crossing the current cut in a connected weighted graph.

§DEP_HARD := S26-A-00001966

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00001968|T|A|Proof+Compile+Meta] Greedy correctness for matroid bases

§FORM := Sorting elements by weight and repeatedly adding an element that preserves independence yields a maximum-weight basis of a matroid.

§DEP_HARD := S26-A-00001967

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00001969|N|A|Proof+Compile+Meta] Complexity-model fence

§FORM := Asymptotic complexity claims require an input encoding and cost model; arithmetic-operation counts and bit complexity are not interchangeable.

§DEP_HARD := S26-T-00001968

§EVIDENCE := A · Exact registered statement only.

§BOUNDARY := An O(log n) multiplication count does not imply O(log n) bit operations for arbitrarily large operands.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00001970|B|A|Proof+Compile+Meta] Discrete mathematics to computation

§FORM := Graphs, matroids, finite fields, groups, and combinatorial structures become algorithmic objects once encodings and cost models are declared; theorem-level structure and implementation complexity remain distinct evidence layers.

§DEP_HARD := S26-N-00001969

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00001971|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 1

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 1 (0b1) uses 0 squarings and 0 extra multiplies, for 0 multiplications total.

§DEP_HARD := S26-B-00001970

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001972|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 2

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 2 (0b10) uses 1 squarings and 0 extra multiplies, for 1 multiplications total.

§DEP_HARD := S26-E-00001971

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001973|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 3

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 3 (0b11) uses 1 squarings and 1 extra multiplies, for 2 multiplications total.

§DEP_HARD := S26-E-00001972

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001974|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 4

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 4 (0b100) uses 2 squarings and 0 extra multiplies, for 2 multiplications total.

§DEP_HARD := S26-E-00001973

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001975|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 5

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 5 (0b101) uses 2 squarings and 1 extra multiplies, for 3 multiplications total.

§DEP_HARD := S26-E-00001974

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001976|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 6

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 6 (0b110) uses 2 squarings and 1 extra multiplies, for 3 multiplications total.

§DEP_HARD := S26-E-00001975

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001977|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 7

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 7 (0b111) uses 2 squarings and 2 extra multiplies, for 4 multiplications total.

§DEP_HARD := S26-E-00001976

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001978|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 8

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 8 (0b1000) uses 3 squarings and 0 extra multiplies, for 3 multiplications total.

§DEP_HARD := S26-E-00001977

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001979|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 9

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 9 (0b1001) uses 3 squarings and 1 extra multiplies, for 4 multiplications total.

§DEP_HARD := S26-E-00001978

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001980|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 10

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 10 (0b1010) uses 3 squarings and 1 extra multiplies, for 4 multiplications total.

§DEP_HARD := S26-E-00001979

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001981|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 11

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 11 (0b1011) uses 3 squarings and 2 extra multiplies, for 5 multiplications total.

§DEP_HARD := S26-E-00001980

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001982|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 12

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 12 (0b1100) uses 3 squarings and 1 extra multiplies, for 4 multiplications total.

§DEP_HARD := S26-E-00001981

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001983|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 13

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 13 (0b1101) uses 3 squarings and 2 extra multiplies, for 5 multiplications total.

§DEP_HARD := S26-E-00001982

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001984|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 14

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 14 (0b1110) uses 3 squarings and 2 extra multiplies, for 5 multiplications total.

§DEP_HARD := S26-E-00001983

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001985|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 15

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 15 (0b1111) uses 3 squarings and 3 extra multiplies, for 6 multiplications total.

§DEP_HARD := S26-E-00001984

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001986|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 16

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 16 (0b10000) uses 4 squarings and 0 extra multiplies, for 4 multiplications total.

§DEP_HARD := S26-E-00001985

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001987|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 17

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 17 (0b10001) uses 4 squarings and 1 extra multiplies, for 5 multiplications total.

§DEP_HARD := S26-E-00001986

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001988|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 18

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 18 (0b10010) uses 4 squarings and 1 extra multiplies, for 5 multiplications total.

§DEP_HARD := S26-E-00001987

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001989|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 19

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 19 (0b10011) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

§DEP_HARD := S26-E-00001988

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001990|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 20

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 20 (0b10100) uses 4 squarings and 1 extra multiplies, for 5 multiplications total.

§DEP_HARD := S26-E-00001989

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001991|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 21

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 21 (0b10101) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

§DEP_HARD := S26-E-00001990

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001992|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 22

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 22 (0b10110) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

§DEP_HARD := S26-E-00001991

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001993|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 23

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 23 (0b10111) uses 4 squarings and 3 extra multiplies, for 7 multiplications total.

§DEP_HARD := S26-E-00001992

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001994|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 24

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 24 (0b11000) uses 4 squarings and 1 extra multiplies, for 5 multiplications total.

§DEP_HARD := S26-E-00001993

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001995|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 25

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 25 (0b11001) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

§DEP_HARD := S26-E-00001994

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001996|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 26

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 26 (0b11010) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

§DEP_HARD := S26-E-00001995

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001997|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 27

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 27 (0b11011) uses 4 squarings and 3 extra multiplies, for 7 multiplications total.

§DEP_HARD := S26-E-00001996

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001998|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 28

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 28 (0b11100) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

§DEP_HARD := S26-E-00001997

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00001999|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 29

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 29 (0b11101) uses 4 squarings and 3 extra multiplies, for 7 multiplications total.

§DEP_HARD := S26-E-00001998

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002000|E|M|Proof+Compile+Meta] Binary-exponentiation count for exponent 30

§FORM := Using left-to-right binary exponentiation initialized at the leading 1, exponent 30 (0b11110) uses 4 squarings and 3 extra multiplies, for 7 multiplications total.

§DEP_HARD := S26-E-00001999

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§PACK Synopsis_2026_Volume_II_sLANG_S1

volume := II · Analysis

canonical_range := 2001-3000

§CLAIM[S26-D-00002001|D|A|Proof+Compile+Meta] Asymptotic equivalence of sequences

§FORM := For eventually nonzero b_n, a_n~b_n means a_n/b_n -> 1.

§DEP_HARD := S26-D-00000351

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002002|D|A|Proof+Compile+Meta] Little-o relation for sequences

§FORM := a_n=o(b_n) means a_n/b_n -> 0 wherever the ratio is eventually defined.

§DEP_HARD := S26-D-00002001

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002003|D|A|Proof+Compile+Meta] Big-O relation for sequences

§FORM := a_n=O(b_n) means |a_n|<=C|b_n| for all sufficiently large n and some finite C.

§DEP_HARD := S26-D-00002002

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002004|D|A|Proof+Compile+Meta] Theta relation for positive scales

§FORM := a_n=Theta(b_n) means both a_n=O(b_n) and b_n=O(a_n), with positivity/nonvanishing declared as needed.

§DEP_HARD := S26-D-00002003

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002005|D|A|Proof+Compile+Meta] Asymptotic scale

§FORM := An asymptotic scale phi_0,phi_1,... satisfies phi_{j+1}=o(phi_j) in the declared limit regime.

§DEP_HARD := S26-D-00002004

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002006|D|A|Proof+Compile+Meta] Poincare asymptotic expansion

§FORM := f(x)~sum_{j>=0} a_j phi_j(x) means the remainder after N terms is o(phi_N(x)) for every fixed N.

§DEP_HARD := S26-D-00002005

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002007|T|A|Proof+Compile+Meta] Cesaro convergence preserves ordinary limits

§FORM := If a_n->L, then the arithmetic means (1/n)sum_{k<=n}a_k also converge to L.

§DEP_HARD := S26-D-00002006

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002008|T|A|Proof+Compile+Meta] Stolz-Cesaro theorem

§FORM := Under the standard monotonicity/unboundedness hypotheses on b_n, limits of (a_{n+1}-a_n)/(b_{n+1}-b_n) control the limit of a_n/b_n.

§DEP_HARD := S26-T-00002007

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002009|T|A|Proof+Compile+Meta] Cauchy condensation criterion

§FORM := For a nonincreasing nonnegative sequence a_n, sum a_n converges iff sum 2^k a_{2^k} converges.

§DEP_HARD := S26-T-00002008

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002010|I|A|Proof+Compile+Meta] Abel summation formula

§FORM := For A(x)=sum_{n<=x}a_n, sum_{n<=N}a_n b_n = A(N)b_N + sum_{n<N}A(n)(b_n-b_{n+1}).

§DEP_HARD := S26-T-00002009

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002011|D|A|Proof+Compile+Meta] Regularly varying sequence

§FORM := A positive sequence a_n is regularly varying with index rho when a_{floor(tn)}/a_n -> t^rho for every t>0.

§DEP_HARD := S26-I-00002010

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002012|D|A|Proof+Compile+Meta] Slowly varying sequence

§FORM := Regular variation with index 0 is called slow variation.

§DEP_HARD := S26-D-00002011

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002013|D|A|Proof+Compile+Meta] Limit superior of a real sequence

§FORM := limsup a_n is the limit of the decreasing sequence sup_{k>=n}a_k in the extended reals.

§DEP_HARD := S26-D-00002012

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002014|D|A|Proof+Compile+Meta] Limit inferior of a real sequence

§FORM := liminf a_n is the limit of the increasing sequence inf_{k>=n}a_k in the extended reals.
§DEP_HARD := S26-D-00002013
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002015|T|A|Proof+Compile+Meta] Subsequence characterization of limsup

§FORM := For a bounded real sequence, limsup is the largest subsequential limit and is attained by some subsequence.
§DEP_HARD := S26-D-00002014
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002016|T|A|Proof+Compile+Meta] Ratio test in limsup form

§FORM := For nonzero terms eventually, limsup |a_{n+1}/a_n|<1 implies absolute convergence of sum a_n.
§DEP_HARD := S26-T-00002015
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002017|T|A|Proof+Compile+Meta] Root test in limsup form

§FORM := limsup |a_n|^{1/n}<1 implies absolute convergence; liminf |a_n|^{1/n}>1 implies divergence of terms from zero.
§DEP_HARD := S26-T-00002016
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002018|D|A|Proof+Compile+Meta] Tauberian principle

§FORM := A Tauberian theorem supplements a one-way summability implication with side conditions that recover ordinary convergence or sharper asymptotics.
§DEP_HARD := S26-T-00002017
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002019|N|A|Proof+Compile+Meta] Asymptotic nonuniformity fence

§FORM := An asymptotic statement in one parameter cannot be treated as uniform in another parameter without an explicit uniform remainder bound.
§DEP_HARD := S26-D-00002018
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := Pointwise asymptotics can fail after exchanging limits or optimizing over the second parameter.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002020|B|A|Proof+Compile+Meta] Asymptotics to numerical analysis bridge

§FORM := Rates expressed by O, o, and asymptotic expansions become computational error models only after constants, norms, and finite parameter ranges are specified.
§DEP_HARD := S26-N-00002019
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002021|E|M|Proof+Compile+Meta] Exponential-limit profile n=10

§FORM := For n=10, (1+1/n)^n=2.593742460100, lying 1.245e-01 below e in floating evaluation.
§DEP_HARD := S26-B-00002020
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002022|E|M|Proof+Compile+Meta] Exponential-limit profile n=20

§FORM := For n=20, (1+1/n)^n=2.653297705144, lying 6.498e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002021
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002023|E|M|Proof+Compile+Meta] Exponential-limit profile n=30

§FORM := For n=30, (1+1/n)^n=2.674318775870, lying 4.396e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002022
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002024|E|M|Proof+Compile+Meta] Exponential-limit profile n=40

§FORM := For n=40, (1+1/n)^n=2.685063838390, lying 3.322e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002023
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002025|E|M|Proof+Compile+Meta] Exponential-limit profile n=50

§FORM := For n=50, (1+1/n)^n=2.691588029074, lying 2.669e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002024
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002026|E|M|Proof+Compile+Meta] Exponential-limit profile n=60

§FORM := For n=60, (1+1/n)^n=2.695970139330, lying 2.231e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002025
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002027|E|M|Proof+Compile+Meta] Exponential-limit profile n=70

§FORM := For n=70, (1+1/n)^n=2.699116370976, lying 1.917e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002026
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002028|E|M|Proof+Compile+Meta] Exponential-limit profile n=80

§FORM := For n=80, (1+1/n)^n=2.701484940753, lying 1.680e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002027
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002029|E|M|Proof+Compile+Meta] Exponential-limit profile n=90

§FORM := For n=90, (1+1/n)^n=2.703332461058, lying 1.495e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002028
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002030|E|M|Proof+Compile+Meta] Exponential-limit profile n=100

§FORM := For n=100, (1+1/n)^n=2.704813829422, lying 1.347e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002029
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002031|E|M|Proof+Compile+Meta] Exponential-limit profile n=110

§FORM := For n=110, (1+1/n)^n=2.706028081505, lying 1.225e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002030
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002032|E|M|Proof+Compile+Meta] Exponential-limit profile n=120

§FORM := For n=120, (1+1/n)^n=2.707041490862, lying 1.124e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002031
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002033|E|M|Proof+Compile+Meta] Exponential-limit profile n=130

§FORM := For n=130, (1+1/n)^n=2.707900081718, lying 1.038e-02 below e in floating evaluation.
§DEP_HARD := S26-E-00002032
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002034|E|M|Proof+Compile+Meta] Exponential-limit profile n=140

§FORM := For n=140, (1+1/n)^n=2.708636813921, lying 9.645e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002033
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002035|E|M|Proof+Compile+Meta] Exponential-limit profile n=150

§FORM := For n=150, (1+1/n)^n=2.709275911335, lying 9.006e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002034
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002036|E|M|Proof+Compile+Meta] Exponential-limit profile n=160

§FORM := For n=160, (1+1/n)^n=2.709835576308, lying 8.446e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002035
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002037|E|M|Proof+Compile+Meta] Exponential-limit profile n=170

§FORM := For n=170, (1+1/n)^n=2.710329751224, lying 7.952e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002036
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002038|E|M|Proof+Compile+Meta] Exponential-limit profile n=180

§FORM := For n=180, (1+1/n)^n=2.710769295839, lying 7.513e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002037
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002039|E|M|Proof+Compile+Meta] Exponential-limit profile n=190

§FORM := For n=190, (1+1/n)^n=2.711162794611, lying 7.119e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002038
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002040|E|M|Proof+Compile+Meta] Exponential-limit profile n=200

§FORM := For n=200, (1+1/n)^n=2.711517122929, lying 6.765e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002039
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002041|E|M|Proof+Compile+Meta] Exponential-limit profile n=210
§FORM := For n=210, (1+1/n)^n=2.711837852272, lying 6.444e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002040
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002042|E|M|Proof+Compile+Meta] Exponential-limit profile n=220
§FORM := For n=220, (1+1/n)^n=2.712129545321, lying 6.152e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002041
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002043|E|M|Proof+Compile+Meta] Exponential-limit profile n=230
§FORM := For n=230, (1+1/n)^n=2.712395974395, lying 5.886e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002042
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002044|E|M|Proof+Compile+Meta] Exponential-limit profile n=240
§FORM := For n=240, (1+1/n)^n=2.712640285482, lying 5.642e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002043
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002045|E|M|Proof+Compile+Meta] Exponential-limit profile n=250
§FORM := For n=250, (1+1/n)^n=2.712865123051, lying 5.417e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002044
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002046|E|M|Proof+Compile+Meta] Exponential-limit profile n=260
§FORM := For n=260, (1+1/n)^n=2.713072726158, lying 5.209e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002045
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002047|E|M|Proof+Compile+Meta] Exponential-limit profile n=270
§FORM := For n=270, (1+1/n)^n=2.713265003267, lying 5.017e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002046
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002048|E|M|Proof+Compile+Meta] Exponential-limit profile n=280
§FORM := For n=280, (1+1/n)^n=2.713443591094, lying 4.838e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002047
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002049|E|M|Proof+Compile+Meta] Exponential-limit profile n=290
§FORM := For n=290, (1+1/n)^n=2.713609901311, lying 4.672e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002048
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002050|E|M|Proof+Compile+Meta] Exponential-limit profile n=300
§FORM := For n=300, (1+1/n)^n=2.713765157943, lying 4.517e-03 below e in floating evaluation.
§DEP_HARD := S26-E-00002049
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002051|D|A|Proof+Compile+Meta] Upper right Dini derivative
§FORM := D^+f(x)=limsup_{h downarrow 0}(f(x+h)-f(x))/h.
§DEP_HARD := S26-E-00002050
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002052|D|A|Proof+Compile+Meta] Lower right Dini derivative
§FORM := D_+f(x)=liminf_{h downarrow 0}(f(x+h)-f(x))/h.
§DEP_HARD := S26-D-00002051
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002053|D|A|Proof+Compile+Meta] Function of bounded variation on an interval
§FORM := f has bounded variation on [a,b] when the supremum of sum |f(x_i)-f(x_{i-1})| over partitions is finite.
§DEP_HARD := S26-D-00002052
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002054|D|A|Proof+Compile+Meta] Total variation function
§FORM := V_f(x) is the variation of f on [a,x], giving a nondecreasing control function for BV maps.
§DEP_HARD := S26-D-00002053
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002055|T|A|Proof+Compile+Meta] Jordan decomposition for BV functions

§FORM := A real BV function on an interval is the difference of two bounded nondecreasing functions.
§DEP_HARD := S26-D-00002054
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002056|D|A|Proof+Compile+Meta] Absolutely continuous function on an interval

§FORM := f is absolutely continuous when disjoint intervals of sufficiently small total length force the sum of corresponding oscillations of f to be small.
§DEP_HARD := S26-T-00002055
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002057|T|A|Proof+Compile+Meta] Absolute continuity implies bounded variation

§FORM := Every absolutely continuous function on a compact interval is of bounded variation.
§DEP_HARD := S26-D-00002056
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002058|T|A|Proof+Compile+Meta] Fundamental theorem for absolutely continuous functions

§FORM := f is absolutely continuous iff f has an L1 derivative almost everywhere and f(x)=f(a)+integral_a^x f'(t)dt.
§DEP_HARD := S26-T-00002057
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002059|D|A|Proof+Compile+Meta] Lipschitz function on a metric domain

§FORM := f is Lipschitz with constant L when d_Y(f(x),f(y))<=L d_X(x,y).
§DEP_HARD := S26-T-00002058
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002060|T|A|Proof+Compile+Meta] Rademacher theorem in Euclidean space

§FORM := Every Lipschitz map R^n->R^m is Frechet differentiable almost everywhere.
§DEP_HARD := S26-D-00002059
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002061|D|A|Proof+Compile+Meta] Approximate derivative

§FORM := The approximate derivative uses density-one approach sets rather than all nearby points.
§DEP_HARD := S26-T-00002060
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002062|D|A|Proof+Compile+Meta] Symmetric derivative

§FORM := The symmetric derivative is lim_{h->0}(f(x+h)-f(x-h))/(2h) when the limit exists.
§DEP_HARD := S26-D-00002061
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002063|T|A|Proof+Compile+Meta] Lebesgue differentiation theorem

§FORM := For f in L1_loc(R^n), averages of f over shrinking balls converge to f(x) for almost every x.
§DEP_HARD := S26-D-00002062
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002064|D|A|Proof+Compile+Meta+Spectral] Hardy-Littlewood local averaging operator

§FORM := A_r f(x)=|B(x,r)|^{-1} integral_{B(x,r)} f(y)dy records the scale-r local average.
§DEP_HARD := S26-T-00002063
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002065|T|A|Proof+Compile+Meta] Monotone functions are differentiable almost everywhere

§FORM := A real monotone function on an interval has a finite derivative almost everywhere.
§DEP_HARD := S26-D-00002064
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002066|D|A|Proof+Compile+Meta] Singular function

§FORM := A singular increasing function is continuous and nonconstant while having derivative zero almost everywhere.
§DEP_HARD := S26-T-00002065
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002067|N|A|Proof+Compile+Meta] Derivative-zero fence

§FORM := Derivative zero almost everywhere does not imply constancy without additional regularity such as absolute continuity.
§DEP_HARD := S26-D-00002066
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := The Cantor function is nonconstant with derivative zero almost everywhere.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002068|D|A|Proof+Compile+Meta+IG] One-dimensional coarea identity for BV

§FORM := For suitable BV functions on an interval, total variation can be represented by integrating the counting measure of level crossings.

§DEP_HARD := S26-N-00002067

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002069|B|A|Proof+Compile+Meta+IG] BV to measure bridge

§FORM := Distributional derivatives of BV functions are finite signed Radon measures, connecting real-variable regularity to measure theory.

§DEP_HARD := S26-D-00002068

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002070|B|A|Proof+Compile+Meta+IG] Differentiation to optimization bridge

§FORM := Generalized one-sided derivatives and BV/AC distinctions control when local slope information legitimately supports optimization or dynamical conclusions.

§DEP_HARD := S26-B-00002069

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002071|E|M|Proof+Compile+Meta] Central-difference profile h=6.813e-02

§FORM := For f(x)=sin x at x=1 and h=6.813e-02, the centered quotient is 0.539884426003; absolute error versus cos(1) is 4.179e-04.

§DEP_HARD := S26-B-00002070

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002072|E|M|Proof+Compile+Meta] Central-difference profile h=4.642e-02

§FORM := For f(x)=sin x at x=1 and h=4.642e-02, the centered quotient is 0.540108319094; absolute error versus cos(1) is 1.940e-04.

§DEP_HARD := S26-E-00002071

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002073|E|M|Proof+Compile+Meta] Central-difference profile h=3.162e-02

§FORM := For f(x)=sin x at x=1 and h=3.162e-02, the centered quotient is 0.540212259986; absolute error versus cos(1) is 9.005e-05.

§DEP_HARD := S26-E-00002072

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002074|E|M|Proof+Compile+Meta] Central-difference profile h=2.154e-02

§FORM := For f(x)=sin x at x=1 and h=2.154e-02, the centered quotient is 0.540260509152; absolute error versus cos(1) is 4.180e-05.

§DEP_HARD := S26-E-00002073

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002075|E|M|Proof+Compile+Meta] Central-difference profile h=1.468e-02

§FORM := For f(x)=sin x at x=1 and h=1.468e-02, the centered quotient is 0.540282905310; absolute error versus cos(1) is 1.940e-05.

§DEP_HARD := S26-E-00002074

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002076|E|M|Proof+Compile+Meta] Central-difference profile h=1.000e-02

§FORM := For f(x)=sin x at x=1 and h=1.000e-02, the centered quotient is 0.540293300875; absolute error versus cos(1) is 9.005e-06.

§DEP_HARD := S26-E-00002075

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002077|E|M|Proof+Compile+Meta] Central-difference profile h=6.813e-03

§FORM := For f(x)=sin x at x=1 and h=6.813e-03, the centered quotient is 0.540298126109; absolute error versus cos(1) is 4.180e-06.

§DEP_HARD := S26-E-00002076

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002078|E|M|Proof+Compile+Meta] Central-difference profile h=4.642e-03

§FORM := For f(x)=sin x at x=1 and h=4.642e-03, the centered quotient is 0.540300365794; absolute error versus cos(1) is 1.940e-06.

§DEP_HARD := S26-E-00002077

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002079|E|M|Proof+Compile+Meta] Central-difference profile h=3.162e-03

§FORM := For f(x)=sin x at x=1 and h=3.162e-03, the centered quotient is 0.540301405365; absolute error versus cos(1) is 9.005e-07.

§DEP_HARD := S26-E-00002078

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002080|E|M|Proof+Compile+Meta] Central-difference profile h=2.154e-03

§FORM := For f(x)=sin x at x=1 and h=2.154e-03, the centered quotient is 0.540301887891; absolute error versus cos(1) is 4.180e-07.

§DEP_HARD := S26-E-00002079

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002081|E|M|Proof+Compile+Meta] Central-difference profile h=1.468e-03

§FORM := For f(x)=sin x at x=1 and h=1.468e-03, the centered quotient is 0.540302111860; absolute error versus cos(1) is 1.940e-07.
§DEP_HARD := S26-E-00002080
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002082|E|M|Proof+Compile+Meta] Central-difference profile h=1.000e-03

§FORM := For f(x)=sin x at x=1 and h=1.000e-03, the centered quotient is 0.540302215818; absolute error versus cos(1) is 9.005e-08.
§DEP_HARD := S26-E-00002081
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002083|E|M|Proof+Compile+Meta] Central-difference profile h=6.813e-04

§FORM := For f(x)=sin x at x=1 and h=6.813e-04, the centered quotient is 0.540302264070; absolute error versus cos(1) is 4.180e-08.
§DEP_HARD := S26-E-00002082
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002084|E|M|Proof+Compile+Meta] Central-difference profile h=4.642e-04

§FORM := For f(x)=sin x at x=1 and h=4.642e-04, the centered quotient is 0.540302286467; absolute error versus cos(1) is 1.940e-08.
§DEP_HARD := S26-E-00002083
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002085|E|M|Proof+Compile+Meta] Central-difference profile h=3.162e-04

§FORM := For f(x)=sin x at x=1 and h=3.162e-04, the centered quotient is 0.540302296863; absolute error versus cos(1) is 9.005e-09.
§DEP_HARD := S26-E-00002084
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002086|E|M|Proof+Compile+Meta] Central-difference profile h=2.154e-04

§FORM := For f(x)=sin x at x=1 and h=2.154e-04, the centered quotient is 0.540302301688; absolute error versus cos(1) is 4.180e-09.
§DEP_HARD := S26-E-00002085
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002087|E|M|Proof+Compile+Meta] Central-difference profile h=1.468e-04

§FORM := For f(x)=sin x at x=1 and h=1.468e-04, the centered quotient is 0.540302303928; absolute error versus cos(1) is 1.940e-09.
§DEP_HARD := S26-E-00002086
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002088|E|M|Proof+Compile+Meta] Central-difference profile h=1.000e-04

§FORM := For f(x)=sin x at x=1 and h=1.000e-04, the centered quotient is 0.540302304968; absolute error versus cos(1) is 9.004e-10.
§DEP_HARD := S26-E-00002087
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002089|E|M|Proof+Compile+Meta] Central-difference profile h=6.813e-05

§FORM := For f(x)=sin x at x=1 and h=6.813e-05, the centered quotient is 0.540302305450; absolute error versus cos(1) is 4.178e-10.
§DEP_HARD := S26-E-00002088
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002090|E|M|Proof+Compile+Meta] Central-difference profile h=4.642e-05

§FORM := For f(x)=sin x at x=1 and h=4.642e-05, the centered quotient is 0.540302305674; absolute error versus cos(1) is 1.939e-10.
§DEP_HARD := S26-E-00002089
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002091|E|M|Proof+Compile+Meta] Central-difference profile h=3.162e-05

§FORM := For f(x)=sin x at x=1 and h=3.162e-05, the centered quotient is 0.540302305779; absolute error versus cos(1) is 8.893e-11.
§DEP_HARD := S26-E-00002090
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002092|E|M|Proof+Compile+Meta] Central-difference profile h=2.154e-05

§FORM := For f(x)=sin x at x=1 and h=2.154e-05, the centered quotient is 0.540302305827; absolute error versus cos(1) is 4.150e-11.
§DEP_HARD := S26-E-00002091
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002093|E|M|Proof+Compile+Meta] Central-difference profile h=1.468e-05

§FORM := For f(x)=sin x at x=1 and h=1.468e-05, the centered quotient is 0.540302305848; absolute error versus cos(1) is 2.012e-11.
§DEP_HARD := S26-E-00002092
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002094|E|M|Proof+Compile+Meta] Central-difference profile h=1.000e-05

§FORM := For f(x)=sin x at x=1 and h=1.000e-05, the centered quotient is 0.540302305857; absolute error versus cos(1) is 1.114e-11.
§DEP_HARD := S26-E-00002093
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002095|E|M|Proof+Compile+Meta] Central-difference profile h=6.813e-06
§FORM := For f(x)=sin x at x=1 and h=6.813e-06, the centered quotient is 0.540302305857; absolute error versus cos(1) is 1.124e-11.
§DEP_HARD := S26-E-00002094
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002096|E|M|Proof+Compile+Meta] Central-difference profile h=4.642e-06
§FORM := For f(x)=sin x at x=1 and h=4.642e-06, the centered quotient is 0.540302305856; absolute error versus cos(1) is 1.212e-11.
§DEP_HARD := S26-E-00002095
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002097|E|M|Proof+Compile+Meta] Central-difference profile h=3.162e-06
§FORM := For f(x)=sin x at x=1 and h=3.162e-06, the centered quotient is 0.540302305869; absolute error versus cos(1) is 6.005e-13.
§DEP_HARD := S26-E-00002096
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002098|E|M|Proof+Compile+Meta] Central-difference profile h=2.154e-06
§FORM := For f(x)=sin x at x=1 and h=2.154e-06, the centered quotient is 0.540302305832; absolute error versus cos(1) is 3.634e-11.
§DEP_HARD := S26-E-00002097
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002099|E|M|Proof+Compile+Meta] Central-difference profile h=1.468e-06
§FORM := For f(x)=sin x at x=1 and h=1.468e-06, the centered quotient is 0.540302305871; absolute error versus cos(1) is 2.570e-12.
§DEP_HARD := S26-E-00002098
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002100|E|M|Proof+Compile+Meta] Central-difference profile h=1.000e-06
§FORM := For f(x)=sin x at x=1 and h=1.000e-06, the centered quotient is 0.540302305896; absolute error versus cos(1) is 2.772e-11.
§DEP_HARD := S26-E-00002099
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002101|D|A|Proof+Compile+Meta] Strictly convex function
§FORM := A convex function is strictly convex when the convexity inequality is strict for distinct endpoints and weights in (0,1).
§DEP_HARD := S26-E-00002100
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002102|D|A|Proof+Compile+Meta] Uniformly convex function
§FORM := A convex function is uniformly convex when midpoint deficits admit a positive modulus depending only on separation.
§DEP_HARD := S26-D-00002101
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002103|D|A|Proof+Compile+Meta] Strongly convex differentiable function
§FORM := f is m-strongly convex when f(y)>=f(x)+<grad f(x),y-x>+(m/2)||y-x||^2.
§DEP_HARD := S26-D-00002102
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002104|D|A|Proof+Compile+Meta] Convex conjugate
§FORM := f*(y)=sup_x(<x,y>-f(x)) is the Legendre-Fenchel conjugate.
§DEP_HARD := S26-D-00002103
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002105|T|A|Proof+Compile+Meta] Fenchel-Young inequality
§FORM := For proper convex f, <x,y><=f(x)+f*(y).
§DEP_HARD := S26-D-00002104
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002106|T|A|Proof+Compile+Meta] Fenchel-Moreau biconjugation
§FORM := A proper lower-semicontinuous convex function on a suitable locally convex space equals its biconjugate.
§DEP_HARD := S26-T-00002105
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002107|D|A|Proof+Compile+Meta] Subgradient of a convex function
§FORM := g is a subgradient of f at x when f(y)>=f(x)+<g,y-x> for all y.
§DEP_HARD := S26-T-00002106
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002108|D|A|Proof+Compile+Meta] Subdifferential
§FORM := partial f(x) is the set of all subgradients at x.
§DEP_HARD := S26-D-00002107
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002109|T|A|Proof+Compile+Meta] Subgradient optimality criterion

§FORM := For proper convex f, 0 in partial f(x) is equivalent to x being a global minimizer.

§DEP_HARD := S26-D-00002108

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002110|D|A|Proof+Compile+Meta] Bregman divergence

§FORM := D_f(x,y)=f(x)-f(y)-<grad f(y),x-y> for differentiable convex f; it is generally not symmetric and need not satisfy the triangle inequality.

§DEP_HARD := S26-T-00002109

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002111|T|A|Proof+Compile+Meta] Young inequality with conjugate exponents

§FORM := For a,b>=0 and p,q>1 with 1/p+1/q=1, ab<=a^p/p+b^q/q.

§DEP_HARD := S26-D-00002110

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002112|T|A|Proof+Compile+Meta] Weighted AM-GM inequality

§FORM := For positive x_i and weights lambda_i>=0 summing to one, product x_i^{lambda_i}<=sum lambda_i x_i.

§DEP_HARD := S26-T-00002111

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002113|T|A|Proof+Compile+Meta] Log-sum inequality

§FORM := For positive a_i,b_i, sum a_i log(a_i/b_i)>=(sum a_i)log((sum a_i)/(sum b_i)).

§DEP_HARD := S26-T-00002112

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002114|D|A|Proof+Compile+Meta] Log-convex function

§FORM := A positive function f is log-convex when log f is convex.

§DEP_HARD := S26-T-00002113

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002115|T|A|Proof+Compile+Meta] Three-lines theorem

§FORM := A bounded holomorphic function on a vertical strip has logarithmically convex sup norms across intermediate vertical lines under the standard boundary hypotheses.

§DEP_HARD := S26-D-00002114

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002116|D|A|Proof+Compile+Meta] Modulus of convexity of a normed space

§FORM := delta_X(epsilon)=inf{1-||(x+y)/2||:||x||,||y||<=1, ||x-y||>=epsilon}.

§DEP_HARD := S26-T-00002115

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002117|D|A|Proof+Compile+Meta] Uniform convexity of a Banach space

§FORM := X is uniformly convex when delta_X(epsilon)>0 for every epsilon>0.

§DEP_HARD := S26-D-00002116

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002118|T|A|Proof+Compile+Meta] Uniform convexity implies reflexivity

§FORM := Every uniformly convex Banach space is reflexive.

§DEP_HARD := S26-D-00002117

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002119|N|A|Proof+Compile+Meta] Convexity smoothness fence

§FORM := Strict convexity, strong convexity, smoothness, and uniform convexity are distinct properties and cannot be substituted without explicit hypotheses.

§DEP_HARD := S26-T-00002118

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002120|B|A|Proof+Compile+Meta+Spectral] Convex duality bridge

§FORM := Conjugates and subdifferentials connect inequalities, variational problems, optimization duality, monotone operators and PDE energy methods.

§DEP_HARD := S26-N-00002119

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002121|E|M|Proof+Compile+Meta] Young-inequality profile 1

§FORM := With p=1.200, q=6.000, a=0.370, b=1.680: ab=0.621600000 and a^p/p+b^q/q=3.999911346; gap=3.378e+00.

§DEP_HARD := S26-B-00002120

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002122|E|M|Proof+Compile+Meta] Young-inequality profile 2

§FORM := With p=1.300, q=4.333, a=0.440, b=1.660: ab=0.730400000 and a^p/p+b^q/q=2.339388897; gap=1.609e+00.
§DEP_HARD := S26-E-00002121
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002123|E|M|Proof+Compile+Meta] Young-inequality profile 3

§FORM := With p=1.400, q=3.500, a=0.510, b=1.640: ab=0.836400000 and a^p/p+b^q/q=1.892205160; gap=1.056e+00.
§DEP_HARD := S26-E-00002122
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002124|E|M|Proof+Compile+Meta] Young-inequality profile 4

§FORM := With p=1.500, q=3.000, a=0.580, b=1.620: ab=0.939600000 and a^p/p+b^q/q=1.711652560; gap=7.721e-01.
§DEP_HARD := S26-E-00002123
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002125|E|M|Proof+Compile+Meta] Young-inequality profile 5

§FORM := With p=1.600, q=2.667, a=0.650, b=1.600: ab=1.040000000 and a^p/p+b^q/q=1.626980936; gap=5.870e-01.
§DEP_HARD := S26-E-00002124
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002126|E|M|Proof+Compile+Meta] Young-inequality profile 6

§FORM := With p=1.700, q=2.429, a=0.720, b=1.580: ab=1.137600000 and a^p/p+b^q/q=1.587077049; gap=4.495e-01.
§DEP_HARD := S26-E-00002125
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002127|E|M|Proof+Compile+Meta] Young-inequality profile 7

§FORM := With p=1.800, q=2.250, a=0.790, b=1.560: ab=1.232400000 and a^p/p+b^q/q=1.572241262; gap=3.398e-01.
§DEP_HARD := S26-E-00002126
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002128|E|M|Proof+Compile+Meta] Young-inequality profile 8

§FORM := With p=1.900, q=2.111, a=0.860, b=1.540: ab=1.324400000 and a^p/p+b^q/q=1.573777411; gap=2.494e-01.
§DEP_HARD := S26-E-00002127
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002129|E|M|Proof+Compile+Meta] Young-inequality profile 9

§FORM := With p=2.000, q=2.000, a=0.930, b=1.520: ab=1.413600000 and a^p/p+b^q/q=1.587650000; gap=1.741e-01.
§DEP_HARD := S26-E-00002128
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002130|E|M|Proof+Compile+Meta] Young-inequality profile 10

§FORM := With p=2.100, q=1.909, a=1.000, b=1.500: ab=1.500000000 and a^p/p+b^q/q=1.612110126; gap=1.121e-01.
§DEP_HARD := S26-E-00002129
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002131|E|M|Proof+Compile+Meta] Young-inequality profile 11

§FORM := With p=2.200, q=1.833, a=1.070, b=1.480: ab=1.583600000 and a^p/p+b^q/q=1.646692137; gap=6.309e-02.
§DEP_HARD := S26-E-00002130
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002132|E|M|Proof+Compile+Meta] Young-inequality profile 12

§FORM := With p=2.300, q=1.769, a=1.140, b=1.460: ab=1.664400000 and a^p/p+b^q/q=1.691759261; gap=2.736e-02.
§DEP_HARD := S26-E-00002131
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002133|E|M|Proof+Compile+Meta] Young-inequality profile 13

§FORM := With p=2.400, q=1.714, a=1.210, b=1.440: ab=1.742400000 and a^p/p+b^q/q=1.748297310; gap=5.897e-03.
§DEP_HARD := S26-E-00002132
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002134|E|M|Proof+Compile+Meta] Young-inequality profile 14

§FORM := With p=2.500, q=1.667, a=1.280, b=1.420: ab=1.817600000 and a^p/p+b^q/q=1.817834055; gap=2.341e-04.
§DEP_HARD := S26-E-00002133
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002135|E|M|Proof+Compile+Meta] Young-inequality profile 15

§FORM := With p=2.600, q=1.625, a=1.350, b=1.400: ab=1.890000000 and a^p/p+b^q/q=1.902430836; gap=1.243e-02.
§DEP_HARD := S26-E-00002134
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002136|E|M|Proof+Compile+Meta] Young-inequality profile 16

§FORM := With p=2.700, q=1.588, a=1.420, b=1.380: ab=1.959600000 and a^p/p+b^q/q=2.004722590; gap=4.512e-02.
§DEP_HARD := S26-E-00002135
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002137|E|M|Proof+Compile+Meta] Young-inequality profile 17

§FORM := With p=2.800, q=1.556, a=1.490, b=1.360: ab=2.026400000 and a^p/p+b^q/q=2.127996415; gap=1.016e-01.
§DEP_HARD := S26-E-00002136
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002138|E|M|Proof+Compile+Meta] Young-inequality profile 18

§FORM := With p=2.900, q=1.526, a=1.560, b=1.340: ab=2.090400000 and a^p/p+b^q/q=2.276306197; gap=1.859e-01.
§DEP_HARD := S26-E-00002137
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002139|E|M|Proof+Compile+Meta] Young-inequality profile 19

§FORM := With p=3.000, q=1.500, a=1.630, b=1.320: ab=2.151600000 and a^p/p+b^q/q=2.454625359; gap=3.030e-01.
§DEP_HARD := S26-E-00002138
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002140|E|M|Proof+Compile+Meta] Young-inequality profile 20

§FORM := With p=3.100, q=1.476, a=1.700, b=1.300: ab=2.210000000 and a^p/p+b^q/q=2.669043113; gap=4.590e-01.
§DEP_HARD := S26-E-00002139
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002141|E|M|Proof+Compile+Meta] Young-inequality profile 21

§FORM := With p=3.200, q=1.455, a=1.770, b=1.280: ab=2.265600000 and a^p/p+b^q/q=2.927012596; gap=6.614e-01.
§DEP_HARD := S26-E-00002140
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002142|E|M|Proof+Compile+Meta] Young-inequality profile 22

§FORM := With p=3.300, q=1.435, a=1.840, b=1.260: ab=2.318400000 and a^p/p+b^q/q=3.237662385; gap=9.193e-01.
§DEP_HARD := S26-E-00002141
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002143|E|M|Proof+Compile+Meta] Young-inequality profile 23

§FORM := With p=3.400, q=1.417, a=1.910, b=1.240: ab=2.368400000 and a^p/p+b^q/q=3.612186418; gap=1.244e+00.
§DEP_HARD := S26-E-00002142
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002144|E|M|Proof+Compile+Meta] Young-inequality profile 24

§FORM := With p=3.500, q=1.400, a=1.980, b=1.220: ab=2.415600000 and a^p/p+b^q/q=4.064331691; gap=1.649e+00.
§DEP_HARD := S26-E-00002143
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002145|E|M|Proof+Compile+Meta] Young-inequality profile 25

§FORM := With p=3.600, q=1.385, a=2.050, b=1.200: ab=2.460000000 and a^p/p+b^q/q=4.611008416; gap=2.151e+00.
§DEP_HARD := S26-E-00002144
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002146|E|M|Proof+Compile+Meta] Young-inequality profile 26

§FORM := With p=3.700, q=1.370, a=2.120, b=1.180: ab=2.501600000 and a^p/p+b^q/q=5.273054040; gap=2.771e+00.
§DEP_HARD := S26-E-00002145
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002147|E|M|Proof+Compile+Meta] Young-inequality profile 27

§FORM := With p=3.800, q=1.357, a=2.190, b=1.160: ab=2.540400000 and a^p/p+b^q/q=6.076191064; gap=3.536e+00.
§DEP_HARD := S26-E-00002146
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002148|E|M|Proof+Compile+Meta] Young-inequality profile 28

§FORM := With p=3.900, q=1.345, a=2.260, b=1.140: ab=2.576400000 and a^p/p+b^q/q=7.052229394; gap=4.476e+00.
§DEP_HARD := S26-E-00002147
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002149|E|M|Proof+Compile+Meta] Young-inequality profile 29

§FORM := With p=4.000, q=1.333, a=2.330, b=1.120: ab=2.609600000 and a^p/p+b^q/q=8.240577812; gap=5.631e+00.
§DEP_HARD := S26-E-00002148
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002150|E|M|Proof+Compile+Meta] Young-inequality profile 30

§FORM := With p=4.100, q=1.323, a=2.400, b=1.100: ab=2.640000000 and a^p/p+b^q/q=9.690146781; gap=7.050e+00.
§DEP_HARD := S26-E-00002149
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002151|D|A|Proof+Compile+Meta+IG] Signed measure

§FORM := A signed measure is a countably additive map into the extended reals that does not take both +infinity and -infinity.
§DEP_HARD := S26-E-00002150
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002152|D|A|Proof+Compile+Meta+IG] Positive set for a signed measure

§FORM := A measurable set P is positive for nu when every measurable E subset P has nu(E)>=0.
§DEP_HARD := S26-D-00002151
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002153|T|A|Proof+Compile+Meta+IG] Hahn decomposition theorem

§FORM := Every signed measure space decomposes into disjoint positive and negative measurable sets, unique up to nu-null sets.
§DEP_HARD := S26-D-00002152
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002154|T|A|Proof+Compile+Meta+IG] Jordan decomposition theorem for signed measures

§FORM := Every signed measure nu can be written uniquely as nu=nu^+-nu^- with mutually singular positive measures.
§DEP_HARD := S26-T-00002153
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002155|D|A|Proof+Compile+Meta+IG] Total variation measure

§FORM := |nu|=nu^++nu^- is the total variation measure of a signed measure.
§DEP_HARD := S26-T-00002154
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002156|D|A|Proof+Compile+Meta+IG] Absolute continuity of measures

§FORM := nu<<mu means mu(E)=0 implies nu(E)=0 for every measurable E.
§DEP_HARD := S26-D-00002155
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002157|D|A|Proof+Compile+Meta+IG] Mutually singular measures

§FORM := nu perpendicular mu means the measures are supported on disjoint measurable pieces up to null sets.
§DEP_HARD := S26-D-00002156
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002158|T|A|Proof+Compile+Meta+IG] Radon-Nikodym theorem

§FORM := For sigma-finite positive measures mu,nu with nu << mu, there exists a nonnegative measurable density f, unique mu-a.e., with nu(E)=integral_E f dmu for every measurable E; if nu is finite, then f is in L^1(mu).
§ASSUME := mu and nu are sigma-finite positive measures; nu << mu. | Global L^1(mu) integrability is asserted only when nu is finite.
§DEP_HARD := S26-D-00002157
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002159|T|A|Proof+Compile+Meta+IG] Lebesgue decomposition of measures

§FORM := Under standard sigma-finiteness hypotheses, nu decomposes uniquely into a part absolutely continuous with respect to mu and a part singular to mu.
§DEP_HARD := S26-T-00002158
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002160|D|A|Proof+Compile+Meta+Topo+IG] Radon measure on a locally compact space

§FORM := A Radon measure is locally finite and inner regular, with the precise regularity convention declared for the ambient topological setting.
§DEP_HARD := S26-T-00002159
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002161|D|A|Proof+Compile+Meta+IG] Support of a measure

§FORM := supp(mu) is the complement of the largest open set of mu-measure zero.
§DEP_HARD := S26-D-00002160
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002162|D|A|Proof+Compile+Meta+IG] Pushforward density transformation

§FORM := When a smooth injective change of variables is valid, densities transform by the absolute Jacobian determinant.
§DEP_HARD := S26-D-00002161
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002163|T|A|Proof+Compile+Meta+IG] Change-of-variables theorem for Lebesgue integrals

§FORM := For an appropriate C1 diffeomorphism Phi, integral_{Phi(U)}g(y)dy=integral_U g(Phi(x))|det D Phi(x)|dx.
§DEP_HARD := S26-D-00002162
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002164|D|A|Proof+Compile+Meta+IG] Weak convergence of finite measures

§FORM := mu_n converges weakly to mu when integrals against bounded continuous test functions converge.
§DEP_HARD := S26-T-00002163
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002165|D|A|Proof+Compile+Meta+IG] Vague convergence of Radon measures

§FORM := Vague convergence tests against compactly supported continuous functions.
§DEP_HARD := S26-D-00002164
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002166|D|A|Proof+Compile+Meta+IG] Tight family of probability measures

§FORM := A family is tight when for every epsilon>0 there is a compact K carrying mass at least 1-epsilon for every measure in the family.
§DEP_HARD := S26-D-00002165
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002167|T|A|Proof+Compile+Meta+IG] Prokhorov compactness interface

§FORM := On Polish spaces, tightness of probability measures is equivalent to relative compactness for weak convergence.
§DEP_HARD := S26-D-00002166
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002168|N|A|Proof+Compile+Meta+IG] Density representation fence

§FORM := Absolute continuity and a density are relative to a reference measure; a singular measure need not possess a Lebesgue density.
§DEP_HARD := S26-T-00002167
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002169|B|A|Proof+Compile+Meta+IG] Measure duality bridge

§FORM := Radon measures act as continuous linear functionals on spaces of continuous compactly supported functions under the appropriate representation theorem.
§DEP_HARD := S26-N-00002168
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002170|B|A|Proof+Compile+Meta+IG] Weak-measure convergence bridge

§FORM := Weak convergence of measures links analysis to probability, empirical distributions, PDE compactness and variational limits.
§DEP_HARD := S26-B-00002169
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002171|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=3

§FORM := On atoms 0,...,2 with weights proportional to 1,...,3, total mass is 1.000000000000 and the exact floating mean is 1.333333333333.
§DEP_HARD := S26-B-00002170
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002172|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=4

§FORM := On atoms 0,...,3 with weights proportional to 1,...,4, total mass is 1.000000000000 and the exact floating mean is 2.000000000000.
§DEP_HARD := S26-E-00002171
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002173|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=5

§FORM := On atoms 0,...,4 with weights proportional to 1,...,5, total mass is 1.000000000000 and the exact floating mean is 2.666666666667.
§DEP_HARD := S26-E-00002172
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002174|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=6

§FORM := On atoms 0,...,5 with weights proportional to 1,...,6, total mass is 1.000000000000 and the exact floating mean is 3.333333333333.
§DEP_HARD := S26-E-00002173
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002175|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=7

§FORM := On atoms 0,...,6 with weights proportional to 1,...,7, total mass is 1.000000000000 and the exact floating mean is 4.000000000000.
§DEP_HARD := S26-E-00002174
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002176|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=8

§FORM := On atoms 0,...,7 with weights proportional to 1,...,8, total mass is 1.000000000000 and the exact floating mean is 4.666666666667.

§DEP_HARD := S26-E-00002175

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002177|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=9

§FORM := On atoms 0,...,8 with weights proportional to 1,...,9, total mass is 1.000000000000 and the exact floating mean is 5.333333333333.

§DEP_HARD := S26-E-00002176

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002178|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=10

§FORM := On atoms 0,...,9 with weights proportional to 1,...,10, total mass is 1.000000000000 and the exact floating mean is 6.000000000000.

§DEP_HARD := S26-E-00002177

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002179|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=11

§FORM := On atoms 0,...,10 with weights proportional to 1,...,11, total mass is 1.000000000000 and the exact floating mean is 6.666666666667.

§DEP_HARD := S26-E-00002178

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002180|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=12

§FORM := On atoms 0,...,11 with weights proportional to 1,...,12, total mass is 1.000000000000 and the exact floating mean is 7.333333333333.

§DEP_HARD := S26-E-00002179

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002181|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=13

§FORM := On atoms 0,...,12 with weights proportional to 1,...,13, total mass is 1.000000000000 and the exact floating mean is 8.000000000000.

§DEP_HARD := S26-E-00002180

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002182|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=14

§FORM := On atoms 0,...,13 with weights proportional to 1,...,14, total mass is 1.000000000000 and the exact floating mean is 8.666666666667.

§DEP_HARD := S26-E-00002181

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002183|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=15

§FORM := On atoms 0,...,14 with weights proportional to 1,...,15, total mass is 1.000000000000 and the exact floating mean is 9.333333333333.

§DEP_HARD := S26-E-00002182

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002184|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=16

§FORM := On atoms 0,...,15 with weights proportional to 1,...,16, total mass is 1.000000000000 and the exact floating mean is 10.000000000000.

§DEP_HARD := S26-E-00002183

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002185|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=17

§FORM := On atoms 0,...,16 with weights proportional to 1,...,17, total mass is 1.000000000000 and the exact floating mean is 10.666666666667.

§DEP_HARD := S26-E-00002184

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002186|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=18

§FORM := On atoms 0,...,17 with weights proportional to 1,...,18, total mass is 1.000000000000 and the exact floating mean is 11.333333333333.

§DEP_HARD := S26-E-00002185

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002187|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=19

§FORM := On atoms 0,...,18 with weights proportional to 1,...,19, total mass is 1.000000000000 and the exact floating mean is 12.000000000000.

§DEP_HARD := S26-E-00002186

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002188|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=20

§FORM := On atoms 0,...,19 with weights proportional to 1,...,20, total mass is 1.000000000000 and the exact floating mean is 12.666666666667.

§DEP_HARD := S26-E-00002187

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002189|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=21

§FORM := On atoms 0,...,20 with weights proportional to 1,...,21, total mass is 1.000000000000 and the exact floating mean is 13.333333333333.

§DEP_HARD := S26-E-00002188

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002190|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=22

§FORM := On atoms 0,...,21 with weights proportional to 1,...,22, total mass is 1.000000000000 and the exact floating mean is 14.000000000000.

§DEP_HARD := S26-E-00002189

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002191|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=23

§FORM := On atoms 0,...,22 with weights proportional to 1,...,23, total mass is 1.000000000000 and the exact floating mean is 14.666666666667.

§DEP_HARD := S26-E-00002190

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002192|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=24

§FORM := On atoms 0,...,23 with weights proportional to 1,...,24, total mass is 1.000000000000 and the exact floating mean is 15.333333333333.

§DEP_HARD := S26-E-00002191

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002193|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=25

§FORM := On atoms 0,...,24 with weights proportional to 1,...,25, total mass is 1.000000000000 and the exact floating mean is 16.000000000000.

§DEP_HARD := S26-E-00002192

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002194|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=26

§FORM := On atoms 0,...,25 with weights proportional to 1,...,26, total mass is 1.000000000000 and the exact floating mean is 16.666666666667.

§DEP_HARD := S26-E-00002193

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002195|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=27

§FORM := On atoms 0,...,26 with weights proportional to 1,...,27, total mass is 1.000000000000 and the exact floating mean is 17.333333333333.

§DEP_HARD := S26-E-00002194

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002196|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=28

§FORM := On atoms 0,...,27 with weights proportional to 1,...,28, total mass is 1.000000000000 and the exact floating mean is 18.000000000000.

§DEP_HARD := S26-E-00002195

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002197|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=29

§FORM := On atoms 0,...,28 with weights proportional to 1,...,29, total mass is 1.000000000000 and the exact floating mean is 18.666666666667.

§DEP_HARD := S26-E-00002196

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002198|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=30

§FORM := On atoms 0,...,29 with weights proportional to 1,...,30, total mass is 1.000000000000 and the exact floating mean is 19.333333333333.

§DEP_HARD := S26-E-00002197

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002199|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=31

§FORM := On atoms 0,...,30 with weights proportional to 1,...,31, total mass is 1.000000000000 and the exact floating mean is 20.000000000000.

§DEP_HARD := S26-E-00002198

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002200|E|M|Proof+Compile+Meta+IG] Finite probability-measure profile n=32

§FORM := On atoms 0,...,31 with weights proportional to 1,...,32, total mass is 1.000000000000 and the exact floating mean is 20.666666666667.

§DEP_HARD := S26-E-00002199

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002201|D|A|Proof+Compile+Meta] Essential supremum norm

§FORM := ||f||_infinity is the infimum of M such that |f|<=M almost everywhere.

§DEP_HARD := S26-E-00002200

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002202|D|A|Proof+Compile+Meta] Weak Lp space

§FORM := L^{p,infinity} consists of measurable f with sup_{lambda>0} lambda mu(|f|>lambda)^{1/p}<infinity.

§DEP_HARD := S26-D-00002201

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002203|D|A|Proof+Compile+Meta] Lorentz space interface

§FORM := Lorentz spaces L^{p,q} refine Lp by integrating the decreasing rearrangement with a second exponent q.

§DEP_HARD := S26-D-00002202

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002204|D|A|Proof+Compile+Meta] Decreasing rearrangement

§FORM := f*(t) is the right-continuous decreasing rearrangement determined by the distribution function of |f|.

§DEP_HARD := S26-D-00002203

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002205|T|A|Proof+Compile+Meta+IG] Hardy-Littlewood rearrangement inequality

§FORM := For nonnegative measurable f,g, integral fg is bounded above by integral f* g* under the standard measure-space formulation.

§DEP_HARD := S26-D-00002204

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002206|T|A|Proof+Compile+Meta] Generalized Holder inequality

§FORM := If sum_i 1/p_i=1, then ||product_i f_i||_1<=product_i ||f_i||_{p_i}.

§DEP_HARD := S26-T-00002205

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002207|T|A|Proof+Compile+Meta] Minkowski integral inequality

§FORM := Under standard measurability and integrability hypotheses, an Lp norm may be moved outside an integral with the Minkowski bound for p>=1.

§DEP_HARD := S26-T-00002206

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002208|D|A|Proof+Compile+Meta+Topo] Interpolation couple

§FORM := An interpolation couple is a pair of compatible Banach spaces continuously embedded in a common Hausdorff topological vector space.

§DEP_HARD := S26-T-00002207

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002209|T|A|Proof+Compile+Meta+Spectral] Riesz-Thorin interpolation theorem

§FORM := A linear operator bounded on two Lp endpoint pairs is bounded on the interpolated exponent pairs with logarithmically interpolated norm bound.

§DEP_HARD := S26-D-00002208

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002210|T|A|Proof+Compile+Meta] Marcinkiewicz interpolation theorem

§FORM := Sublinear weak-type endpoint bounds interpolate to strong-type interior Lp bounds under the standard exponent hypotheses.

§DEP_HARD := S26-T-00002209

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002211|D|A|Proof+Compile+Meta] Uniform integrability

§FORM := A family in L1 is uniformly integrable when large values contribute uniformly small L1 mass.

§DEP_HARD := S26-T-00002210

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002212|T|A|Proof+Compile+Meta+IG] Vitali convergence theorem

§FORM := Convergence in measure plus uniform integrability yields L1 convergence on finite-measure spaces under standard hypotheses.

§DEP_HARD := S26-D-00002211

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002213|D|A|Proof+Compile+Meta] Reflexive Lp regime
§FORM := For 1<p<infinity, Lp is reflexive.
§DEP_HARD := S26-T-00002212
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002214|D|A|Proof+Compile+Meta] Dual exponent
§FORM := For 1<p<infinity, p'=p/(p-1) is the Holder conjugate exponent.
§DEP_HARD := S26-D-00002213
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002215|T|A|Proof+Compile+Meta+IG] Lp duality on sigma-finite spaces
§FORM := For 1<p<infinity under standard measure hypotheses, every continuous linear functional on Lp is integration against a unique L^{p'} function.
§DEP_HARD := S26-D-00002214
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002216|D|A|Proof+Compile+Meta] Orlicz function
§FORM := A Young function is convex, increasing from zero and diverges to infinity, generating an Orlicz integrability scale.
§DEP_HARD := S26-T-00002215
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002217|D|A|Proof+Compile+Meta] Orlicz space interface
§FORM := An Orlicz space replaces power growth |f|^p by a Young function Phi(|f|) in the integrability criterion.
§DEP_HARD := S26-D-00002216
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002218|N|A|Proof+Compile+Meta] Endpoint interpolation fence
§FORM := Interpolation conclusions at p=1 or p=infinity often require separate weak-type or endpoint arguments and cannot be inferred mechanically from interior formulas.
§DEP_HARD := S26-D-00002217
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002219|B|A|Proof+Compile+Meta+Spectral] Lp to harmonic-analysis bridge
§FORM := Weak Lp, Lorentz and interpolation spaces encode endpoint behavior of maximal, singular-integral and Fourier operators.
§DEP_HARD := S26-N-00002218
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002220|B|A|Proof+Compile+Meta+IG] Uniform-integrability bridge
§FORM := Uniform integrability converts convergence in measure into norm convergence and is a central compactness mechanism in probability and PDE.
§DEP_HARD := S26-B-00002219
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002221|E|M|Proof+Compile+Meta] Finite lp-norm profile n=3, p=1.15
§FORM := For v=(1,...,3), ||v||_1.15=5.266253703961 and ||v||_infinity=3.
§DEP_HARD := S26-B-00002220
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002222|E|M|Proof+Compile+Meta] Finite lp-norm profile n=4, p=1.30
§FORM := For v=(1,...,4), ||v||_1.30=7.487050309871 and ||v||_infinity=4.
§DEP_HARD := S26-E-00002221
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002223|E|M|Proof+Compile+Meta] Finite lp-norm profile n=5, p=1.45
§FORM := For v=(1,...,5), ||v||_1.45=9.566741833882 and ||v||_infinity=5.
§DEP_HARD := S26-E-00002222
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002224|E|M|Proof+Compile+Meta] Finite lp-norm profile n=6, p=1.60
§FORM := For v=(1,...,6), ||v||_1.60=11.492357851814 and ||v||_infinity=6.
§DEP_HARD := S26-E-00002223
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002225|E|M|Proof+Compile+Meta] Finite lp-norm profile n=7, p=1.75
§FORM := For v=(1,...,7), ||v||_1.75=13.278104237489 and ||v||_infinity=7.
§DEP_HARD := S26-E-00002224
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002226|E|M|Proof+Compile+Meta] Finite lp-norm profile n=8, p=1.90
§FORM := For v=(1,...,8), ||v||_1.90=14.944705402164 and ||v||_infinity=8.
§DEP_HARD := S26-E-00002225
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002227|E|M|Proof+Compile+Meta] Finite lp-norm profile n=9, p=2.05
§FORM := For v=(1,...,9), ||v||_2.05=16.512266295892 and ||v||_infinity=9.
§DEP_HARD := S26-E-00002226
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002228|E|M|Proof+Compile+Meta] Finite lp-norm profile n=10, p=2.20
§FORM := For v=(1,...,10), ||v||_2.20=17.998172269504 and ||v||_infinity=10.
§DEP_HARD := S26-E-00002227
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002229|E|M|Proof+Compile+Meta] Finite lp-norm profile n=11, p=2.35
§FORM := For v=(1,...,11), ||v||_2.35=19.416799575465 and ||v||_infinity=11.
§DEP_HARD := S26-E-00002228
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002230|E|M|Proof+Compile+Meta] Finite lp-norm profile n=12, p=2.50
§FORM := For v=(1,...,12), ||v||_2.50=20.779820602243 and ||v||_infinity=12.
§DEP_HARD := S26-E-00002229
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002231|E|M|Proof+Compile+Meta] Finite lp-norm profile n=13, p=2.65
§FORM := For v=(1,...,13), ||v||_2.65=22.096652593901 and ||v||_infinity=13.
§DEP_HARD := S26-E-00002230
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002232|E|M|Proof+Compile+Meta] Finite lp-norm profile n=14, p=2.80
§FORM := For v=(1,...,14), ||v||_2.80=23.374890814190 and ||v||_infinity=14.
§DEP_HARD := S26-E-00002231
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002233|E|M|Proof+Compile+Meta] Finite lp-norm profile n=15, p=2.95
§FORM := For v=(1,...,15), ||v||_2.95=24.620678492252 and ||v||_infinity=15.
§DEP_HARD := S26-E-00002232
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002234|E|M|Proof+Compile+Meta] Finite lp-norm profile n=16, p=3.10
§FORM := For v=(1,...,16), ||v||_3.10=25.839007140489 and ||v||_infinity=16.
§DEP_HARD := S26-E-00002233
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002235|E|M|Proof+Compile+Meta] Finite lp-norm profile n=17, p=3.25
§FORM := For v=(1,...,17), ||v||_3.25=27.033954750223 and ||v||_infinity=17.
§DEP_HARD := S26-E-00002234
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002236|E|M|Proof+Compile+Meta] Finite lp-norm profile n=18, p=3.40
§FORM := For v=(1,...,18), ||v||_3.40=28.208872775623 and ||v||_infinity=18.
§DEP_HARD := S26-E-00002235
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002237|E|M|Proof+Compile+Meta] Finite lp-norm profile n=19, p=3.55
§FORM := For v=(1,...,19), ||v||_3.55=29.366532399636 and ||v||_infinity=19.
§DEP_HARD := S26-E-00002236
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002238|E|M|Proof+Compile+Meta] Finite lp-norm profile n=20, p=3.70
§FORM := For v=(1,...,20), ||v||_3.70=30.509238984748 and ||v||_infinity=20.
§DEP_HARD := S26-E-00002237
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002239|E|M|Proof+Compile+Meta] Finite lp-norm profile n=21, p=3.85
§FORM := For v=(1,...,21), ||v||_3.85=31.638921860664 and ||v||_infinity=21.
§DEP_HARD := S26-E-00002238
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002240|E|M|Proof+Compile+Meta] Finite lp-norm profile n=22, p=4.00
§FORM := For v=(1,...,22), ||v||_4.00=32.757205042625 and ||v||_infinity=22.
§DEP_HARD := S26-E-00002239
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002241|E|M|Proof+Compile+Meta] Finite lp-norm profile n=23, p=4.15
§FORM := For v=(1,...,23), ||v||_4.15=33.865463197083 and ||v||_infinity=23.
§DEP_HARD := S26-E-00002240
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002242|E|M|Proof+Compile+Meta] Finite lp-norm profile n=24, p=4.30
§FORM := For v=(1,...,24), ||v||_4.30=34.964866165474 and ||v||_infinity=24.
§DEP_HARD := S26-E-00002241
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002243|E|M|Proof+Compile+Meta] Finite lp-norm profile n=25, p=4.45
§FORM := For v=(1,...,25), ||v||_4.45=36.056414580626 and ||v||_infinity=25.
§DEP_HARD := S26-E-00002242
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002244|E|M|Proof+Compile+Meta] Finite lp-norm profile n=26, p=4.60
§FORM := For v=(1,...,26), ||v||_4.60=37.140968517555 and ||v||_infinity=26.
§DEP_HARD := S26-E-00002243
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002245|E|M|Proof+Compile+Meta] Finite lp-norm profile n=27, p=4.75
§FORM := For v=(1,...,27), ||v||_4.75=38.219270669982 and ||v||_infinity=27.
§DEP_HARD := S26-E-00002244
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002246|E|M|Proof+Compile+Meta] Finite lp-norm profile n=28, p=4.90
§FORM := For v=(1,...,28), ||v||_4.90=39.291965202042 and ||v||_infinity=28.
§DEP_HARD := S26-E-00002245
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002247|E|M|Proof+Compile+Meta] Finite lp-norm profile n=29, p=5.05
§FORM := For v=(1,...,29), ||v||_5.05=40.359613164958 and ||v||_infinity=29.
§DEP_HARD := S26-E-00002246
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002248|E|M|Proof+Compile+Meta] Finite lp-norm profile n=30, p=5.20
§FORM := For v=(1,...,30), ||v||_5.20=41.422705170734 and ||v||_infinity=30.
§DEP_HARD := S26-E-00002247
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002249|E|M|Proof+Compile+Meta] Finite lp-norm profile n=31, p=5.35
§FORM := For v=(1,...,31), ||v||_5.35=42.481671863774 and ||v||_infinity=31.
§DEP_HARD := S26-E-00002248
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002250|E|M|Proof+Compile+Meta] Finite lp-norm profile n=32, p=5.50
§FORM := For v=(1,...,32), ||v||_5.50=43.536892615411 and ||v||_infinity=32.
§DEP_HARD := S26-E-00002249
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002251|D|A|Proof+Compile+Meta] Complex differentiability at a point
§FORM := f is complex differentiable at z when lim_{h->0}(f(z+h)-f(z))/h exists with h complex.
§DEP_HARD := S26-E-00002250
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002252|D|A|Proof+Compile+Meta] Holomorphic function
§FORM := A function is holomorphic on an open set when it is complex differentiable at every point of that set.
§DEP_HARD := S26-D-00002251
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002253|D|A|Proof+Compile+Meta] Entire function
§FORM := An entire function is holomorphic on all of C.
§DEP_HARD := S26-D-00002252
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002254|T|A|Proof+Compile+Meta] Cauchy-Riemann equations
§FORM := If f=u+iv is complex differentiable with suitable real differentiability, then u_x=v_y and u_y=-v_x.
§DEP_HARD := S26-D-00002253
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002255|T|A|Proof+Compile+Meta] Holomorphic functions are analytic
§FORM := Every holomorphic function has a locally convergent complex power-series expansion.
§DEP_HARD := S26-T-00002254
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002256|T|A|Proof+Compile+Meta] Cauchy integral formula
§FORM := If f is holomorphic on and inside a positively oriented simple contour, f(z0)=(1/(2pi i)) integral f(z)/(z-z0) dz for interior z0.
§DEP_HARD := S26-T-00002255
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002257|I|A|Proof+Compile+Meta] Cauchy derivative formula
§FORM := f^{(n)}(z0)=n!/(2pi i) integral f(z)/(z-z0)^{n+1} dz under the Cauchy hypotheses.
§DEP_HARD := S26-T-00002256
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002258|T|A|Proof+Compile+Meta] Cauchy estimates
§FORM := If |f|<=M on |z-z0|=R, then |f^{(n)}(z0)|<=n! M/R^n.
§DEP_HARD := S26-I-00002257
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002259|T|A|Proof+Compile+Meta] Liouville theorem
§FORM := Every bounded entire function is constant.
§DEP_HARD := S26-T-00002258
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002260|T|A|Proof+Compile+Meta] Fundamental theorem of algebra via complex analysis
§FORM := Every nonconstant complex polynomial has a complex zero; Liouville gives one standard analytic proof.
§DEP_HARD := S26-T-00002259
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002261|T|A|Proof+Compile+Meta] Identity theorem for holomorphic functions
§FORM := If two holomorphic functions agree on a set with an accumulation point inside a connected domain, they agree everywhere on that domain.
§DEP_HARD := S26-T-00002260
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002262|T|A|Proof+Compile+Meta] Open mapping theorem for holomorphic functions
§FORM := If U is a connected open subset of C and f:U->C is holomorphic and nonconstant, then f is an open map. On a disconnected domain, apply the conclusion componentwise on components where f is nonconstant.
§ASSUME := U is connected and open in C; f is holomorphic and nonconstant.
§DEP_HARD := S26-T-00002261
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002263|T|A|Proof+Compile+Meta] Maximum modulus principle
§FORM := A nonconstant holomorphic function cannot attain a strict interior maximum of its modulus on a connected domain.
§DEP_HARD := S26-T-00002262
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002264|T|A|Proof+Compile+Meta] Schwarz lemma
§FORM := A holomorphic self-map of the unit disk fixing 0 satisfies |f(z)|<=|z| and |f'(0)|<=1, with rigidity in the equality case.
§DEP_HARD := S26-T-00002263
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002265|D|A|Proof+Compile+Meta] Order of a zero
§FORM := A zero z0 of a holomorphic function has order m when f(z)=(z-z0)^m g(z) with g(z0) nonzero.
§DEP_HARD := S26-T-00002264
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002266|D|A|Proof+Compile+Meta] Isolated singularity
§FORM := A point z0 is an isolated singularity when f is holomorphic on a punctured neighborhood of z0.
§DEP_HARD := S26-D-00002265
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002267|D|A|Proof+Compile+Meta] Laurent series
§FORM := On an annulus, a holomorphic function admits a unique doubly infinite Laurent expansion.
§DEP_HARD := S26-D-00002266
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002268|D|A|Proof+Compile+Meta] Removable, pole and essential singularities
§FORM := An isolated singularity is removable, a pole, or essential according to the negative part of its Laurent series.
§DEP_HARD := S26-D-00002267
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002269|N|A|Proof+Compile+Meta] Complex-real differentiability fence
§FORM := Real differentiability of f:R^2->R^2 does not imply complex differentiability; the Cauchy-Riemann structure is additional.
§DEP_HARD := S26-D-00002268
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002270|B|A|Proof+Compile+Meta] Holomorphy rigidity bridge
§FORM := Complex differentiability forces analytic continuation, strong maximum principles and coefficient bounds unavailable to generic smooth real functions.
§DEP_HARD := S26-N-00002269
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002271|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 1
§FORM := At z=0.030-0.020i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.221e-16.
§DEP_HARD := S26-B-00002270
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002272|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 2
§FORM := At z=0.060-0.040i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.180e-16.
§DEP_HARD := S26-E-00002271
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002273|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 3
§FORM := At z=0.090-0.060i, the degree-8 Taylor polynomial for exp(z) has absolute error 5.764e-15.
§DEP_HARD := S26-E-00002272
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002274|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 4
§FORM := At z=0.120-0.080i, the degree-8 Taylor polynomial for exp(z) has absolute error 7.527e-14.
§DEP_HARD := S26-E-00002273
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002275|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 5
§FORM := At z=0.150-0.100i, the degree-8 Taylor polynomial for exp(z) has absolute error 5.628e-13.
§DEP_HARD := S26-E-00002274
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002276|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 6
§FORM := At z=0.180-0.120i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.912e-12.
§DEP_HARD := S26-E-00002275
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002277|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 7
§FORM := At z=0.210-0.140i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.170e-11.
§DEP_HARD := S26-E-00002276
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002278|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 8
§FORM := At z=0.240-0.160i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.902e-11.
§DEP_HARD := S26-E-00002277
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002279|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 9
§FORM := At z=0.270-0.180i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.130e-10.
§DEP_HARD := S26-E-00002278
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002280|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 10
§FORM := At z=0.300-0.200i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.925e-10.
§DEP_HARD := S26-E-00002279
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002281|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 11

§FORM := At z=0.330-0.220i, the degree-8 Taylor polynomial for exp(z) has absolute error 6.918e-10.
§DEP_HARD := S26-E-00002280
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002282|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 12

§FORM := At z=0.360-0.240i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.518e-09.
§DEP_HARD := S26-E-00002281
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002283|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 13

§FORM := At z=0.390-0.260i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.130e-09.
§DEP_HARD := S26-E-00002282
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002284|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 14

§FORM := At z=0.420-0.280i, the degree-8 Taylor polynomial for exp(z) has absolute error 6.117e-09.
§DEP_HARD := S26-E-00002283
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002285|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 15

§FORM := At z=0.450-0.300i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.142e-08.
§DEP_HARD := S26-E-00002284
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002286|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 16

§FORM := At z=0.480-0.320i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.047e-08.
§DEP_HARD := S26-E-00002285
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002287|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 17

§FORM := At z=0.510-0.340i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.543e-08.
§DEP_HARD := S26-E-00002286
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002288|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 18

§FORM := At z=0.540-0.360i, the degree-8 Taylor polynomial for exp(z) has absolute error 5.945e-08.
§DEP_HARD := S26-E-00002287
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002289|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 19

§FORM := At z=0.570-0.380i, the degree-8 Taylor polynomial for exp(z) has absolute error 9.701e-08.
§DEP_HARD := S26-E-00002288
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002290|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 20

§FORM := At z=0.600-0.400i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.544e-07.
§DEP_HARD := S26-E-00002289
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002291|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 21

§FORM := At z=0.630-0.420i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.403e-07.
§DEP_HARD := S26-E-00002290
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002292|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 22

§FORM := At z=0.660-0.440i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.663e-07.
§DEP_HARD := S26-E-00002291
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002293|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 23

§FORM := At z=0.690-0.460i, the degree-8 Taylor polynomial for exp(z) has absolute error 5.482e-07.
§DEP_HARD := S26-E-00002292
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002294|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 24

§FORM := At z=0.720-0.480i, the degree-8 Taylor polynomial for exp(z) has absolute error 8.066e-07.
§DEP_HARD := S26-E-00002293
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002295|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 25
§FORM := At z=0.750-0.500i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.168e-06.
§DEP_HARD := S26-E-00002294
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002296|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 26
§FORM := At z=0.780-0.520i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.668e-06.
§DEP_HARD := S26-E-00002295
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002297|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 27
§FORM := At z=0.810-0.540i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.350e-06.
§DEP_HARD := S26-E-00002296
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002298|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 28
§FORM := At z=0.840-0.560i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.270e-06.
§DEP_HARD := S26-E-00002297
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002299|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 29
§FORM := At z=0.870-0.580i, the degree-8 Taylor polynomial for exp(z) has absolute error 4.499e-06.
§DEP_HARD := S26-E-00002298
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002300|E|M|Proof+Compile+Meta] Entire exponential Taylor profile 30
§FORM := At z=0.900-0.600i, the degree-8 Taylor polynomial for exp(z) has absolute error 6.123e-06.
§DEP_HARD := S26-E-00002299
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002301|D|A|Proof+Compile+Meta] Residue at an isolated singularity
§FORM := Res(f,z0) is the coefficient of (z-z0)^{-1} in the Laurent expansion.
§DEP_HARD := S26-E-00002300
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002302|I|A|Proof+Compile+Meta] Residue at a simple pole
§FORM := If f=g/h with g,h holomorphic, h(z0)=0 and h'(z0) nonzero, then Res(f,z0)=g(z0)/h'(z0).
§DEP_HARD := S26-D-00002301
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002303|T|A|Proof+Compile+Meta] Residue theorem
§FORM := For a meromorphic function with finitely many poles inside a closed contour, the contour integral equals 2pi i times the sum of enclosed residues, counted with winding numbers in the general form.
§DEP_HARD := S26-I-00002302
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002304|D|A|Proof+Compile+Meta] Winding number
§FORM := Ind(gamma,z0)=(1/(2pi i)) integral_gamma dz/(z-z0) for z0 off the closed curve.
§DEP_HARD := S26-T-00002303
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002305|T|A|Proof+Compile+Meta] Argument principle
§FORM := The change of argument of a meromorphic f around a contour equals 2pi times the number of zeros minus poles inside, counted with multiplicity.
§DEP_HARD := S26-D-00002304
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002306|T|A|Proof+Compile+Meta] Rouche theorem
§FORM := If |g|<|f| on a simple closed contour, then f and f+g have the same number of zeros inside, counted with multiplicity.
§DEP_HARD := S26-T-00002305
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002307|D|A|Proof+Compile+Meta] Principal part of a Laurent series
§FORM := The principal part is the sum of negative-power terms and determines the isolated singular behavior.
§DEP_HARD := S26-T-00002306
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002308|D|A|Proof+Compile+Meta] Meromorphic function
§FORM := A meromorphic function is holomorphic except at isolated poles.
§DEP_HARD := S26-D-00002307
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002309|T|A|Proof+Compile+Meta] Mittag-Leffler interface
§FORM := Prescribed principal parts at a discrete set can be realized by a meromorphic function under the classical Mittag-Leffler construction.
§DEP_HARD := S26-D-00002308
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002310|T|A|Proof+Compile+Meta] Weierstrass product interface
§FORM := Entire functions with prescribed zeros can be constructed by convergent canonical products subject to convergence factors.
§DEP_HARD := S26-T-00002309
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002311|D|A|Proof+Compile+Meta+Topo] Contour deformation principle
§FORM := Holomorphic integrals are invariant under admissible homotopies that do not cross singularities, expressed precisely through Cauchy theory/homology.
§DEP_HARD := S26-T-00002310
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002312|T|A|Proof+Compile+Meta+Spectral] Jordan lemma interface
§FORM := For suitable rational-decay factors, semicircular contour contributions vanish, enabling Fourier-type real integral evaluation.
§DEP_HARD := S26-D-00002311
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002313|D|A|Proof+Compile+Meta] Cauchy principal value
§FORM := PV integral is a symmetric limiting prescription for singular integrals and is not ordinary absolute integration.
§DEP_HARD := S26-T-00002312
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002314|D|A|Proof+Compile+Meta+Topo] Indentation contour
§FORM := A small contour detour around a pole records half-residue or related contributions only under the declared orientation and limiting geometry.
§DEP_HARD := S26-D-00002313
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002315|T|A|Proof+Compile+Meta] Residue evaluation of rational trigonometric integrals
§FORM := The substitution z=e^{it} converts many periodic rational trigonometric integrals into contour integrals on the unit circle.
§DEP_HARD := S26-D-00002314
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002316|T|A|Proof+Compile+Meta] Residue evaluation of real rational integrals
§FORM := Suitable rational functions decaying in a half-plane admit real-line integral evaluations from enclosed residues when arc contributions vanish.
§DEP_HARD := S26-T-00002315
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002317|D|A|Proof+Compile+Meta] Pole at infinity
§FORM := The residue at infinity is minus the coefficient of z^{-1} in the Laurent expansion at infinity and equals minus the sum of finite residues on the Riemann sphere.
§DEP_HARD := S26-T-00002316
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002318|N|A|Proof+Compile+Meta] Contour crossing fence
§FORM := Deforming a contour across a singularity changes the integral by the corresponding residue contribution; deformation invariance is not unconditional.
§DEP_HARD := S26-D-00002317
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002319|B|A|Proof+Compile+Meta+Spectral] Residues to spectral inversion bridge
§FORM := Residue calculus extracts modal contributions from inverse Laplace/Fourier transforms and resolvents when poles represent isolated spectral data.
§DEP_HARD := S26-N-00002318
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002320|B|A|Proof+Compile+Meta] Argument principle to root counting bridge
§FORM := Contour integrals can convert analytic zero counting into winding-number computation, motivating robust root-location algorithms.
§DEP_HARD := S26-B-00002319
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002321|E|M|Proof+Compile+Meta] Simple-pole residue profile 1
§FORM := For f(z)=1/((z-0.230)(z--0.380)), the residue at z=0.230 is 1.639344262295.
§DEP_HARD := S26-B-00002320
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002322|E|M|Proof+Compile+Meta] Simple-pole residue profile 2
§FORM := For f(z)=1/((z-0.260)(z--0.360)), the residue at z=0.260 is 1.612903225806.
§DEP_HARD := S26-E-00002321
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002323|E|M|Proof+Compile+Meta] Simple-pole residue profile 3
§FORM := For f(z)=1/((z-0.290)(z--0.340)), the residue at z=0.290 is 1.587301587302.
§DEP_HARD := S26-E-00002322
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002324|E|M|Proof+Compile+Meta] Simple-pole residue profile 4
§FORM := For f(z)=1/((z-0.320)(z--0.320)), the residue at z=0.320 is 1.562500000000.
§DEP_HARD := S26-E-00002323
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002325|E|M|Proof+Compile+Meta] Simple-pole residue profile 5
§FORM := For f(z)=1/((z-0.350)(z--0.300)), the residue at z=0.350 is 1.538461538462.
§DEP_HARD := S26-E-00002324
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002326|E|M|Proof+Compile+Meta] Simple-pole residue profile 6
§FORM := For f(z)=1/((z-0.380)(z--0.280)), the residue at z=0.380 is 1.515151515152.
§DEP_HARD := S26-E-00002325
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002327|E|M|Proof+Compile+Meta] Simple-pole residue profile 7
§FORM := For f(z)=1/((z-0.410)(z--0.260)), the residue at z=0.410 is 1.492537313433.
§DEP_HARD := S26-E-00002326
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002328|E|M|Proof+Compile+Meta] Simple-pole residue profile 8
§FORM := For f(z)=1/((z-0.440)(z--0.240)), the residue at z=0.440 is 1.470588235294.
§DEP_HARD := S26-E-00002327
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002329|E|M|Proof+Compile+Meta] Simple-pole residue profile 9
§FORM := For f(z)=1/((z-0.470)(z--0.220)), the residue at z=0.470 is 1.449275362319.
§DEP_HARD := S26-E-00002328
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002330|E|M|Proof+Compile+Meta] Simple-pole residue profile 10
§FORM := For f(z)=1/((z-0.500)(z--0.200)), the residue at z=0.500 is 1.428571428571.
§DEP_HARD := S26-E-00002329
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002331|E|M|Proof+Compile+Meta] Simple-pole residue profile 11
§FORM := For f(z)=1/((z-0.530)(z--0.180)), the residue at z=0.530 is 1.408450704225.
§DEP_HARD := S26-E-00002330
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002332|E|M|Proof+Compile+Meta] Simple-pole residue profile 12
§FORM := For f(z)=1/((z-0.560)(z--0.160)), the residue at z=0.560 is 1.388888888889.
§DEP_HARD := S26-E-00002331
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002333|E|M|Proof+Compile+Meta] Simple-pole residue profile 13
§FORM := For f(z)=1/((z-0.590)(z--0.140)), the residue at z=0.590 is 1.369863013699.
§DEP_HARD := S26-E-00002332
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002334|E|M|Proof+Compile+Meta] Simple-pole residue profile 14
§FORM := For f(z)=1/((z-0.620)(z--0.120)), the residue at z=0.620 is 1.351351351351.
§DEP_HARD := S26-E-00002333
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002335|E|M|Proof+Compile+Meta] Simple-pole residue profile 15
§FORM := For f(z)=1/((z-0.650)(z--0.100)), the residue at z=0.650 is 1.333333333333.
§DEP_HARD := S26-E-00002334
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002336|E|M|Proof+Compile+Meta] Simple-pole residue profile 16
§FORM := For f(z)=1/((z-0.680)(z--0.080)), the residue at z=0.680 is 1.315789473684.
§DEP_HARD := S26-E-00002335
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002337|E|M|Proof+Compile+Meta] Simple-pole residue profile 17
§FORM := For f(z)=1/((z-0.710)(z--0.060)), the residue at z=0.710 is 1.298701298701.
§DEP_HARD := S26-E-00002336
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002338|E|M|Proof+Compile+Meta] Simple-pole residue profile 18
§FORM := For f(z)=1/((z-0.740)(z--0.040)), the residue at z=0.740 is 1.282051282051.
§DEP_HARD := S26-E-00002337
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002339|E|M|Proof+Compile+Meta] Simple-pole residue profile 19
§FORM := For f(z)=1/((z-0.770)(z--0.020)), the residue at z=0.770 is 1.265822784810.
§DEP_HARD := S26-E-00002338
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002340|E|M|Proof+Compile+Meta] Simple-pole residue profile 20
§FORM := For f(z)=1/((z-0.800)(z-0.000)), the residue at z=0.800 is 1.250000000000.
§DEP_HARD := S26-E-00002339
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002341|E|M|Proof+Compile+Meta] Simple-pole residue profile 21
§FORM := For f(z)=1/((z-0.830)(z-0.020)), the residue at z=0.830 is 1.234567901235.
§DEP_HARD := S26-E-00002340
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002342|E|M|Proof+Compile+Meta] Simple-pole residue profile 22
§FORM := For f(z)=1/((z-0.860)(z-0.040)), the residue at z=0.860 is 1.219512195122.
§DEP_HARD := S26-E-00002341
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002343|E|M|Proof+Compile+Meta] Simple-pole residue profile 23
§FORM := For f(z)=1/((z-0.890)(z-0.060)), the residue at z=0.890 is 1.204819277108.
§DEP_HARD := S26-E-00002342
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002344|E|M|Proof+Compile+Meta] Simple-pole residue profile 24
§FORM := For f(z)=1/((z-0.920)(z-0.080)), the residue at z=0.920 is 1.190476190476.
§DEP_HARD := S26-E-00002343
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002345|E|M|Proof+Compile+Meta] Simple-pole residue profile 25
§FORM := For f(z)=1/((z-0.950)(z-0.100)), the residue at z=0.950 is 1.176470588235.
§DEP_HARD := S26-E-00002344
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002346|E|M|Proof+Compile+Meta] Simple-pole residue profile 26
§FORM := For f(z)=1/((z-0.980)(z-0.120)), the residue at z=0.980 is 1.162790697674.
§DEP_HARD := S26-E-00002345
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002347|E|M|Proof+Compile+Meta] Simple-pole residue profile 27
§FORM := For f(z)=1/((z-1.010)(z-0.140)), the residue at z=1.010 is 1.149425287356.
§DEP_HARD := S26-E-00002346
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002348|E|M|Proof+Compile+Meta] Simple-pole residue profile 28

§FORM := For f(z)=1/((z-1.040)(z-0.160)), the residue at z=1.040 is 1.136363636364.
§DEP_HARD := S26-E-00002347
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002349|E|M|Proof+Compile+Meta] Simple-pole residue profile 29

§FORM := For f(z)=1/((z-1.070)(z-0.180)), the residue at z=1.070 is 1.123595505618.
§DEP_HARD := S26-E-00002348
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002350|E|M|Proof+Compile+Meta] Simple-pole residue profile 30

§FORM := For f(z)=1/((z-1.100)(z-0.200)), the residue at z=1.100 is 1.111111111111.
§DEP_HARD := S26-E-00002349
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002351|D|A|Proof+Compile+Meta] Conformal map

§FORM := A holomorphic map with nonzero derivative is locally angle preserving in the oriented sense.
§DEP_HARD := S26-E-00002350
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002352|D|A|Proof+Compile+Meta] Biholomorphism

§FORM := A biholomorphism is a bijective holomorphic map whose inverse is holomorphic.
§DEP_HARD := S26-D-00002351
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002353|T|A|Proof+Compile+Meta] Riemann mapping theorem

§FORM := Every nonempty simply connected proper open subset of C is biholomorphic to the unit disk.
§DEP_HARD := S26-D-00002352
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002354|D|A|Proof+Compile+Meta] Mobius transformation

§FORM := z |-> (az+b)/(cz+d) with ad-bc nonzero acts conformally on the Riemann sphere.
§DEP_HARD := S26-T-00002353
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002355|T|A|Proof+Compile+Meta] Mobius transformations preserve generalized circles

§FORM := Mobius maps send circles and lines on the Riemann sphere to circles or lines.
§DEP_HARD := S26-D-00002354
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002356|D|A|Proof+Compile+Meta] Harmonic function

§FORM := A C2 real-valued function u is harmonic when Delta u=0.
§DEP_HARD := S26-T-00002355
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002357|T|A|Proof+Compile+Meta] Real and imaginary parts of holomorphic functions are harmonic

§FORM := If f=u+iv is holomorphic, then Delta u=Delta v=0.
§DEP_HARD := S26-D-00002356
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002358|D|A|Proof+Compile+Meta+Topo] Harmonic conjugate

§FORM := v is a harmonic conjugate of u when u+iv is holomorphic; global existence can be obstructed by topology.
§DEP_HARD := S26-T-00002357
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002359|T|A|Proof+Compile+Meta] Mean-value property for harmonic functions

§FORM := A harmonic function equals its average over sufficiently small circles or balls centered at the point.
§DEP_HARD := S26-D-00002358
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002360|T|A|Proof+Compile+Meta] Maximum principle for harmonic functions

§FORM := A nonconstant harmonic function on a connected domain cannot attain an interior maximum or minimum.
§DEP_HARD := S26-T-00002359
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002361|D|A|Proof+Compile+Meta] Poisson kernel for the unit disk

§FORM := P_r(theta)=(1-r^2)/(1-2r cos theta+r^2) is the boundary reproducing kernel for harmonic extension in the disk.
§DEP_HARD := S26-T-00002360
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002362|I|A|Proof+Compile+Meta] Poisson integral formula

§FORM := Boundary data on the circle is harmonically extended to the disk by convolution with the Poisson kernel under suitable integrability/regularity assumptions.
§DEP_HARD := S26-D-00002361
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002363|D|A|Proof+Compile+Meta] Green function in planar potential theory

§FORM := A Green function is harmonic away from its pole, has logarithmic singularity at the pole, and satisfies the declared boundary normalization.
§DEP_HARD := S26-I-00002362
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002364|D|A|Proof+Compile+Meta] Subharmonic function

§FORM := An upper-semicontinuous function is subharmonic when it lies below its local harmonic averages in the appropriate sense.
§DEP_HARD := S26-D-00002363
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002365|T|A|Proof+Compile+Meta] Maximum principle for subharmonic functions

§FORM := A subharmonic function cannot possess a strict interior maximum unless locally constant under standard connectedness assumptions.
§DEP_HARD := S26-D-00002364
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002366|D|A|Proof+Compile+Meta] Hyperbolic metric of the disk

§FORM := The Poincare disk metric has line element proportional to |dz|/(1-|z|^2), with normalization convention declared.
§DEP_HARD := S26-T-00002365
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002367|T|A|Proof+Compile+Meta] Schwarz-Pick theorem

§FORM := Holomorphic self-maps of the disk are nonexpansive for the Poincare metric, with equality rigidity for automorphisms.
§DEP_HARD := S26-D-00002366
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002368|N|A|Proof+Compile+Meta] Global conjugate fence

§FORM := A harmonic function need not possess a single-valued global harmonic conjugate on a nonsimply connected domain.
§DEP_HARD := S26-T-00002367
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002369|B|A|Proof+Compile+Meta+Topo] Conformal geometry bridge

§FORM := Planar conformal maps translate boundary-value problems between domains while preserving harmonic structure up to the appropriate transformation law.
§DEP_HARD := S26-N-00002368
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002370|B|A|Proof+Compile+Meta+IG] Potential theory bridge

§FORM := Harmonic, subharmonic, Green and Poisson objects connect complex analysis, elliptic PDE, probability and capacity theory.
§DEP_HARD := S26-B-00002369
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002371|E|M|Proof+Compile+Meta] Poisson-kernel profile 1

§FORM := For disk radius parameter r=0.020 and angle theta=0.100, P_r(theta)=1.040599805721.
§DEP_HARD := S26-B-00002370
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002372|E|M|Proof+Compile+Meta] Poisson-kernel profile 2

§FORM := For disk radius parameter r=0.040 and angle theta=0.200, P_r(theta)=1.081462045183.
§DEP_HARD := S26-E-00002371
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002373|E|M|Proof+Compile+Meta] Poisson-kernel profile 3

§FORM := For disk radius parameter r=0.060 and angle theta=0.300, P_r(theta)=1.120860808658.
§DEP_HARD := S26-E-00002372
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002374|E|M|Proof+Compile+Meta] Poisson-kernel profile 4

§FORM := For disk radius parameter r=0.080 and angle theta=0.400, P_r(theta)=1.156653110245.
§DEP_HARD := S26-E-00002373
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002375|E|M|Proof+Compile+Meta] Poisson-kernel profile 5

§FORM := For disk radius parameter r=0.100 and angle theta=0.500, P_r(theta)=1.186362599961.
§DEP_HARD := S26-E-00002374
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002376|E|M|Proof+Compile+Meta] Poisson-kernel profile 6

§FORM := For disk radius parameter r=0.120 and angle theta=0.600, P_r(theta)=1.207370468848.
§DEP_HARD := S26-E-00002375
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002377|E|M|Proof+Compile+Meta] Poisson-kernel profile 7

§FORM := For disk radius parameter r=0.140 and angle theta=0.700, P_r(theta)=1.217216555960.
§DEP_HARD := S26-E-00002376
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002378|E|M|Proof+Compile+Meta] Poisson-kernel profile 8

§FORM := For disk radius parameter r=0.160 and angle theta=0.800, P_r(theta)=1.213972868064.
§DEP_HARD := S26-E-00002377
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002379|E|M|Proof+Compile+Meta] Poisson-kernel profile 9

§FORM := For disk radius parameter r=0.180 and angle theta=0.900, P_r(theta)=1.196605955442.
§DEP_HARD := S26-E-00002378
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002380|E|M|Proof+Compile+Meta] Poisson-kernel profile 10

§FORM := For disk radius parameter r=0.200 and angle theta=1.000, P_r(theta)=1.165219540148.
§DEP_HARD := S26-E-00002379
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002381|E|M|Proof+Compile+Meta] Poisson-kernel profile 11

§FORM := For disk radius parameter r=0.220 and angle theta=1.100, P_r(theta)=1.121088771631.
§DEP_HARD := S26-E-00002380
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002382|E|M|Proof+Compile+Meta] Poisson-kernel profile 12

§FORM := For disk radius parameter r=0.240 and angle theta=1.200, P_r(theta)=1.066463540246.
§DEP_HARD := S26-E-00002381
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002383|E|M|Proof+Compile+Meta] Poisson-kernel profile 13

§FORM := For disk radius parameter r=0.260 and angle theta=1.300, P_r(theta)=1.004199664326.
§DEP_HARD := S26-E-00002382
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002384|E|M|Proof+Compile+Meta] Poisson-kernel profile 14

§FORM := For disk radius parameter r=0.280 and angle theta=1.400, P_r(theta)=0.937329895395.
§DEP_HARD := S26-E-00002383
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002385|E|M|Proof+Compile+Meta] Poisson-kernel profile 15

§FORM := For disk radius parameter r=0.300 and angle theta=1.500, P_r(theta)=0.868687250586.
§DEP_HARD := S26-E-00002384
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002386|E|M|Proof+Compile+Meta] Poisson-kernel profile 16

§FORM := For disk radius parameter r=0.320 and angle theta=1.600, P_r(theta)=0.800651014711.
§DEP_HARD := S26-E-00002385
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002387|E|M|Proof+Compile+Meta] Poisson-kernel profile 17

§FORM := For disk radius parameter r=0.340 and angle theta=1.700, P_r(theta)=0.735031184597.
§DEP_HARD := S26-E-00002386
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002388|E|M|Proof+Compile+Meta] Poisson-kernel profile 18

§FORM := For disk radius parameter r=0.360 and angle theta=1.800, P_r(theta)=0.673066620755.
§DEP_HARD := S26-E-00002387
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002389|E|M|Proof+Compile+Meta] Poisson-kernel profile 19

§FORM := For disk radius parameter r=0.380 and angle theta=1.900, P_r(theta)=0.615495256753.
§DEP_HARD := S26-E-00002388
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002390|E|M|Proof+Compile+Meta] Poisson-kernel profile 20

§FORM := For disk radius parameter r=0.400 and angle theta=2.000, P_r(theta)=0.562656688872.
§DEP_HARD := S26-E-00002389
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002391|E|M|Proof+Compile+Meta] Poisson-kernel profile 21

§FORM := For disk radius parameter r=0.420 and angle theta=2.100, P_r(theta)=0.514598602562.
§DEP_HARD := S26-E-00002390
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002392|E|M|Proof+Compile+Meta] Poisson-kernel profile 22

§FORM := For disk radius parameter r=0.440 and angle theta=2.200, P_r(theta)=0.471170879445.
§DEP_HARD := S26-E-00002391
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002393|E|M|Proof+Compile+Meta] Poisson-kernel profile 23

§FORM := For disk radius parameter r=0.460 and angle theta=2.300, P_r(theta)=0.432100877305.
§DEP_HARD := S26-E-00002392
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002394|E|M|Proof+Compile+Meta] Poisson-kernel profile 24

§FORM := For disk radius parameter r=0.480 and angle theta=2.400, P_r(theta)=0.397049376894.
§DEP_HARD := S26-E-00002393
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002395|E|M|Proof+Compile+Meta] Poisson-kernel profile 25

§FORM := For disk radius parameter r=0.500 and angle theta=2.500, P_r(theta)=0.365649676754.
§DEP_HARD := S26-E-00002394
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002396|E|M|Proof+Compile+Meta] Poisson-kernel profile 26

§FORM := For disk radius parameter r=0.520 and angle theta=2.600, P_r(theta)=0.337533331216.
§DEP_HARD := S26-E-00002395
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002397|E|M|Proof+Compile+Meta] Poisson-kernel profile 27

§FORM := For disk radius parameter r=0.540 and angle theta=2.700, P_r(theta)=0.312345966379.
§DEP_HARD := S26-E-00002396
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002398|E|M|Proof+Compile+Meta] Poisson-kernel profile 28

§FORM := For disk radius parameter r=0.560 and angle theta=2.800, P_r(theta)=0.289756081334.
§DEP_HARD := S26-E-00002397
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002399|E|M|Proof+Compile+Meta] Poisson-kernel profile 29

§FORM := For disk radius parameter r=0.580 and angle theta=2.900, P_r(theta)=0.269459093223.
§DEP_HARD := S26-E-00002398
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002400|E|M|Proof+Compile+Meta] Poisson-kernel profile 30

§FORM := For disk radius parameter r=0.600 and angle theta=3.000, P_r(theta)=0.251178281644.
§DEP_HARD := S26-E-00002399
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002401|D|A|Proof+Compile+Meta] Gamma function

§FORM := For Re(s)>0, Gamma(s)=integral_0^infinity t^{s-1}e^{-t}dt, extended meromorphically to C.
§DEP_HARD := S26-E-00002400
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002402|I|A|Proof+Compile+Meta] Gamma recurrence

§FORM := Gamma(s+1)=s Gamma(s).
§DEP_HARD := S26-D-00002401
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002403|I|A|Proof+Compile+Meta] Gamma factorial identity
§FORM := Gamma(n+1)=n! for nonnegative integers n.
§DEP_HARD := S26-I-00002402
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002404|T|A|Proof+Compile+Meta] Euler reflection formula
§FORM := Gamma(z)Gamma(1-z)=pi/sin(pi z) away from poles.
§DEP_HARD := S26-I-00002403
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002405|T|A|Proof+Compile+Meta] Legendre duplication formula
§FORM := Gamma(z)Gamma(z+1/2)=2^{1-2z}sqrt(pi)Gamma(2z).
§DEP_HARD := S26-T-00002404
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002406|D|A|Proof+Compile+Meta] Beta function
§FORM := B(x,y)=integral_0^1 t^{x-1}(1-t)^{y-1}dt for Re x,Re y>0.
§DEP_HARD := S26-T-00002405
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002407|I|A|Proof+Compile+Meta] Beta-Gamma identity
§FORM := B(x,y)=Gamma(x)Gamma(y)/Gamma(x+y).
§DEP_HARD := S26-D-00002406
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002408|D|A|Proof+Compile+Meta] Digamma function
§FORM := psi(z)=Gamma'(z)/Gamma(z) is the logarithmic derivative of Gamma.
§DEP_HARD := S26-I-00002407
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002409|D|A|Proof+Compile+Meta] Polygamma functions
§FORM := psi^{(m)} are derivatives of the digamma function.
§DEP_HARD := S26-D-00002408
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002410|D|A|Proof+Compile+Meta] Error function
§FORM := erf(x)=2/sqrt(pi) integral_0^x e^{-t^2}dt.
§DEP_HARD := S26-D-00002409
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002411|D|A|Proof+Compile+Meta] Complementary error function
§FORM := erfc(x)=1-erf(x).
§DEP_HARD := S26-D-00002410
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002412|D|A|Proof+Compile+Meta] Bessel equation
§FORM := x^2 y''+x y'+(x^2-nu^2)y=0 defines the classical Bessel-function solution family.
§DEP_HARD := S26-D-00002411
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002413|D|A|Proof+Compile+Meta] Bessel function of the first kind
§FORM := J_nu is the solution of Bessel equation specified by its standard power series near zero.
§DEP_HARD := S26-D-00002412
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002414|D|A|Proof+Compile+Meta] Modified Bessel equation
§FORM := x^2 y''+x y'-(x^2+nu^2)y=0 defines I_nu and K_nu solution families.
§DEP_HARD := S26-D-00002413
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002415|D|A|Proof+Compile+Meta] Airy equation
§FORM := y''-x y=0 has canonical solutions Ai and Bi.
§DEP_HARD := S26-D-00002414
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002416|D|A|Proof+Compile+Meta+Topo] Hypergeometric differential equation
§FORM := z(1-z)y''+[c-(a+b+1)z]y'-ab y=0 defines the Gauss hypergeometric family.
§DEP_HARD := S26-D-00002415
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002417|D|A|Proof+Compile+Meta+Topo] Gauss hypergeometric series

§FORM := _2F_1(a,b;c;z)=sum_{n>=0}(a)_n(b)_n/((c)_n n!) z^n in its disk of convergence, with analytic continuation beyond.
§DEP_HARD := S26-D-00002416
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002418|T|A|Proof+Compile+Meta] Stirling asymptotic for Gamma

§FORM := Gamma(z) has the classical Stirling expansion in sectors excluding the negative real axis, with the sector and remainder convention declared.
§DEP_HARD := S26-D-00002417
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002419|N|A|Proof+Compile+Meta] Special-function branch fence

§FORM := Analytic continuation of special functions can involve branch cuts and convention-dependent branches; numerical values require the branch convention.
§DEP_HARD := S26-T-00002418
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002420|B|A|Proof+Compile+Meta+Topo] Special-functions bridge

§FORM := Gamma, Bessel, Airy and hypergeometric functions encode canonical solutions of differential equations, transforms and asymptotic transition regimes.
§DEP_HARD := S26-N-00002419
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002421|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=0.15

§FORM := At x=0.15: Gamma(x+1)=0.933040931107, B(x+1,3/2)=0.556753322025, erf(x)=0.167995971427.
§DEP_HARD := S26-B-00002420
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002422|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=0.30

§FORM := At x=0.30: Gamma(x+1)=0.897470696306, B(x+1,3/2)=0.474421154996, erf(x)=0.328626759459.
§DEP_HARD := S26-E-00002421
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002423|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=0.45

§FORM := At x=0.45: Gamma(x+1)=0.885661380271, B(x+1,3/2)=0.410775804234, erf(x)=0.475481719787.
§DEP_HARD := S26-E-00002422
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002424|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=0.60

§FORM := At x=0.60: Gamma(x+1)=0.893515349288, B(x+1,3/2)=0.360324924478, erf(x)=0.603856090848.
§DEP_HARD := S26-E-00002423
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002425|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=0.75

§FORM := At x=0.75: Gamma(x+1)=0.919062526849, B(x+1,3/2)=0.319504062596, erf(x)=0.711155633654.
§DEP_HARD := S26-E-00002424
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002426|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=0.90

§FORM := At x=0.90: Gamma(x+1)=0.961765831907, B(x+1,3/2)=0.285905319589, erf(x)=0.796908212423.
§DEP_HARD := S26-E-00002425
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002427|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=1.05

§FORM := At x=1.05: Gamma(x+1)=1.02217947884, B(x+1,3/2)=0.257847413019, erf(x)=0.86243610609.
§DEP_HARD := S26-E-00002426
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002428|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=1.20

§FORM := At x=1.20: Gamma(x+1)=1.10180249088, B(x+1,3/2)=0.234123365979, erf(x)=0.91031397823.
§DEP_HARD := S26-E-00002427
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002429|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=1.35

§FORM := At x=1.35: Gamma(x+1)=1.20305444673, B(x+1,3/2)=0.2138459543, erf(x)=0.943762196123.
§DEP_HARD := S26-E-00002428
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002430|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=1.50

§FORM := At x=1.50: Gamma(x+1)=1.32934038818, B(x+1,3/2)=0.196349540849, erf(x)=0.966105146475.
§DEP_HARD := S26-E-00002429
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002431|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=1.65

§FORM := At x=1.65: Gamma(x+1)=1.48519274692, B(x+1,3/2)=0.181125755612, erf(x)=0.980375585023.
§DEP_HARD := S26-E-00002430
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002432|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=1.80

§FORM := At x=1.80: Gamma(x+1)=1.67649078776, B(x+1,3/2)=0.167780199583, erf(x)=0.989090501636.
§DEP_HARD := S26-E-00002431
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002433|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=1.95

§FORM := At x=1.95: Gamma(x+1)=1.91076726998, B(x+1,3/2)=0.156002603074, erf(x)=0.994179333592.
§DEP_HARD := S26-E-00002432
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002434|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=2.10

§FORM := At x=2.10: Gamma(x+1)=2.19762027839, B(x+1,3/2)=0.145545822832, erf(x)=0.997020533344.
§DEP_HARD := S26-E-00002433
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002435|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=2.25

§FORM := At x=2.25: Gamma(x+1)=2.54925696672, B(x+1,3/2)=0.136210782041, erf(x)=0.998537283413.
§DEP_HARD := S26-E-00002434
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002436|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=2.40

§FORM := At x=2.40: Gamma(x+1)=2.98120642681, B(x+1,3/2)=0.127835489725, erf(x)=0.999311486103.
§DEP_HARD := S26-E-00002435
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002437|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=2.55

§FORM := At x=2.55: Gamma(x+1)=3.51325214469, B(x+1,3/2)=0.120286912946, erf(x)=0.999689339657.
§DEP_HARD := S26-E-00002436
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002438|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=2.70

§FORM := At x=2.70: Gamma(x+1)=4.1706517838, B(x+1,3/2)=0.113454877835, erf(x)=0.99986566726.
§DEP_HARD := S26-E-00002437
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002439|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=2.85

§FORM := At x=2.85: Gamma(x+1)=4.9857349276, B(x+1,3/2)=0.107247435651, erf(x)=0.99994434372.
§DEP_HARD := S26-E-00002438
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002440|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=3.00

§FORM := At x=3.00: Gamma(x+1)=6, B(x+1,3/2)=0.101587301587, erf(x)=0.999977909503.
§DEP_HARD := S26-E-00002439
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002441|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=3.15

§FORM := At x=3.15: Gamma(x+1)=7.26687266181, B(x+1,3/2)=0.0964090891631, erf(x)=0.999991601789.
§DEP_HARD := S26-E-00002440
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002442|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=3.30

§FORM := At x=3.30: Gamma(x+1)=8.85534336045, B(x+1,3/2)=0.0916571416108, erf(x)=0.99999694229.
§DEP_HARD := S26-E-00002441
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002443|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=3.45

§FORM := At x=3.45: Gamma(x+1)=10.8547765843, B(x+1,3/2)=0.0872838161259, erf(x)=0.999998933948.
§DEP_HARD := S26-E-00002442
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002444|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=3.60

§FORM := At x=3.60: Gamma(x+1)=13.3812858709, B(x+1,3/2)=0.0832481151014, erf(x)=0.999999644137.
§DEP_HARD := S26-E-00002443
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002445|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=3.75

§FORM := At x=3.75: Gamma(x+1)=16.5862065392, B(x+1,3/2)=0.079514585714, erf(x)=0.999999886273.
§DEP_HARD := S26-E-00002444
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002446|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=3.90

§FORM := At x=3.90: Gamma(x+1)=20.6673859619, B(x+1,3/2)=0.0760524288552, erf(x)=0.999999965208.
§DEP_HARD := S26-E-00002445
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002447|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=4.05

§FORM := At x=4.05: Gamma(x+1)=25.8842676254, B(x+1,3/2)=0.0728347727068, erf(x)=0.999999989812.
§DEP_HARD := S26-E-00002446
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002448|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=4.20

§FORM := At x=4.20: Gamma(x+1)=32.5780960503, B(x+1,3/2)=0.0698380767861, erf(x)=0.999999997145.
§DEP_HARD := S26-E-00002447
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002449|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=4.35

§FORM := At x=4.35: Gamma(x+1)=41.1990506738, B(x+1,3/2)=0.0670416401215, erf(x)=0.999999999234.
§DEP_HARD := S26-E-00002448
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002450|E|M|Proof+Compile+Meta] Gamma-Beta-erf profile x=4.50

§FORM := At x=4.50: Gamma(x+1)=52.3427777846, B(x+1,3/2)=0.0644271930912, erf(x)=0.999999999803.
§DEP_HARD := S26-E-00002449
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002451|D|A|Proof+Compile+Meta] Best approximation error

§FORM := For f in a normed space and approximation set V, E(f,V)=inf_{v in V}||f-v||.
§DEP_HARD := S26-E-00002450
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002452|D|A|Proof+Compile+Meta] Polynomial approximation space

§FORM := P_n is the space of polynomials of degree at most n on the declared interval/domain.
§DEP_HARD := S26-D-00002451
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002453|T|A|Proof+Compile+Meta] Weierstrass approximation theorem

§FORM := Every continuous real-valued function on a compact interval can be uniformly approximated by polynomials.
§DEP_HARD := S26-D-00002452
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002454|T|A|Proof+Compile+Meta] Stone-Weierstrass theorem

§FORM := A separating subalgebra containing constants is dense in C(K) under the standard real form; the complex form requires conjugation closure.
§DEP_HARD := S26-T-00002453
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002455|D|A|Proof+Compile+Meta] Modulus of continuity

§FORM := omega(f,delta)=sup{|f(x)-f(y)|:|x-y|<=delta} quantifies uniform continuity.
§DEP_HARD := S26-T-00002454
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002456|T|A|Proof+Compile+Meta] Jackson approximation interface

§FORM := Polynomial/trigonometric approximation errors admit upper bounds in terms of moduli of smoothness under the corresponding Jackson theorem.
§DEP_HARD := S26-D-00002455
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002457|D|A|Proof+Compile+Meta] Bernstein polynomial

§FORM := B_n f(x)=sum_{k=0}^n f(k/n) binom(n,k)x^k(1-x)^{n-k} on [0,1].
§DEP_HARD := S26-T-00002456
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002458|T|A|Proof+Compile+Meta] Bernstein uniform convergence

§FORM := For f continuous on [0,1], B_n f converges uniformly to f.
§DEP_HARD := S26-D-00002457
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002459|D|A|Proof+Compile+Meta] Chebyshev alternation property

§FORM := A best uniform polynomial approximation to a continuous function is characterized by an alternating extremal error pattern under the classical degree conditions.
§DEP_HARD := S26-T-00002458
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002460|D|A|Proof+Compile+Meta+Spectral] Lebesgue constant for interpolation

§FORM := The Lebesgue constant is the sup norm of the interpolation operator and quantifies amplification of nodal data errors.
§DEP_HARD := S26-D-00002459
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002461|T|A|Proof+Compile+Meta] Interpolation error formula

§FORM := For f sufficiently differentiable, the polynomial interpolation error at x equals f^{(n+1)}(xi)/(n+1)! times the nodal polynomial for some xi in the interpolation interval.
§DEP_HARD := S26-D-00002460
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002462|D|A|Proof+Compile+Meta] Chebyshev nodes

§FORM := Chebyshev interpolation nodes cluster near interval endpoints and reduce worst-case polynomial interpolation amplification relative to equally spaced nodes.
§DEP_HARD := S26-T-00002461
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002463|D|A|Proof+Compile+Meta] Minimax polynomial approximation

§FORM := A minimax approximant minimizes the maximum absolute error on a compact set.
§DEP_HARD := S26-D-00002462
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002464|D|A|Proof+Compile+Meta] Rational approximation

§FORM := Rational approximation uses quotients p/q and can represent poles or boundary layers more efficiently than fixed-degree polynomials.
§DEP_HARD := S26-D-00002463
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002465|D|A|Proof+Compile+Meta] Pade approximant

§FORM := A [m/n] Pade approximant matches as many initial power-series coefficients as possible with a rational function having numerator degree m and denominator degree n.
§DEP_HARD := S26-D-00002464
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002466|D|A|Proof+Compile+Meta] Spline approximation

§FORM := A spline is a piecewise polynomial with prescribed smoothness across knots.
§DEP_HARD := S26-D-00002465
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002467|T|A|Proof+Compile+Meta] Cubic spline interpolation interface

§FORM := Natural or clamped cubic splines give unique C2 piecewise-cubic interpolants after the corresponding boundary conditions are specified.
§DEP_HARD := S26-D-00002466
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002468|N|A|Proof+Compile+Meta+Topo] Interpolation-convergence fence

§FORM := Exact interpolation at more nodes does not by itself imply uniform convergence; node geometry and Lebesgue growth matter.
§DEP_HARD := S26-T-00002467
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := Equally spaced polynomial interpolation can exhibit the Runge phenomenon.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002469|B|A|Proof+Compile+Meta+Spectral] Approximation to numerics bridge

§FORM := Approximation error, conditioning and discretization error must be tracked separately in interpolation and spectral algorithms.
§DEP_HARD := S26-N-00002468
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002470|B|A|Proof+Compile+Meta] Approximation to learning bridge

§FORM := Finite-dimensional approximation spaces and regularized best approximation provide analytic prototypes for regression and function learning.

§DEP_HARD := S26-B-00002469

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002471|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 2

§FORM := Using 3 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 5.647e-02.

§DEP_HARD := S26-B-00002470

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002472|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 3

§FORM := Using 4 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.657e-03.

§DEP_HARD := S26-E-00002471

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002473|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 4

§FORM := Using 5 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.397e-04.

§DEP_HARD := S26-E-00002472

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002474|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 5

§FORM := Using 6 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 5.180e-05.

§DEP_HARD := S26-E-00002473

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002475|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 6

§FORM := Using 7 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.620e-06.

§DEP_HARD := S26-E-00002474

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002476|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 7

§FORM := Using 8 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 2.224e-07.

§DEP_HARD := S26-E-00002475

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002477|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 8

§FORM := Using 9 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.219e-08.

§DEP_HARD := S26-E-00002476

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002478|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 9

§FORM := Using 10 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.027e-10.

§DEP_HARD := S26-E-00002477

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002479|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 10

§FORM := Using 11 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 2.714e-11.

§DEP_HARD := S26-E-00002478

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002480|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 11

§FORM := Using 12 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.123e-12.

§DEP_HARD := S26-E-00002479

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002481|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 12

§FORM := Using 13 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 4.530e-14.

§DEP_HARD := S26-E-00002480

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002482|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 13

§FORM := Using 14 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.109e-15.

§DEP_HARD := S26-E-00002481

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002483|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 14

§FORM := Using 15 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.109e-15.

§DEP_HARD := S26-E-00002482

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002484|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 15
§FORM := Using 16 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 8.438e-15.
§DEP_HARD := S26-E-00002483
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002485|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 16
§FORM := Using 17 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.465e-14.
§DEP_HARD := S26-E-00002484
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002486|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 17
§FORM := Using 18 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 2.665e-15.
§DEP_HARD := S26-E-00002485
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002487|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 18
§FORM := Using 19 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.109e-15.
§DEP_HARD := S26-E-00002486
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002488|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 19
§FORM := Using 20 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 2.220e-15.
§DEP_HARD := S26-E-00002487
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002489|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 20
§FORM := Using 21 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.997e-15.
§DEP_HARD := S26-E-00002488
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002490|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 21
§FORM := Using 22 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.110e-14.
§DEP_HARD := S26-E-00002489
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002491|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 22
§FORM := Using 23 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.465e-14.
§DEP_HARD := S26-E-00002490
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002492|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 23
§FORM := Using 24 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.661e-15.
§DEP_HARD := S26-E-00002491
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002493|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 24
§FORM := Using 25 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.997e-15.
§DEP_HARD := S26-E-00002492
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002494|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 25
§FORM := Using 26 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.217e-15.
§DEP_HARD := S26-E-00002493
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002495|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 26
§FORM := Using 27 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.217e-15.
§DEP_HARD := S26-E-00002494
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002496|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 27
§FORM := Using 28 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.510e-14.
§DEP_HARD := S26-E-00002495
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002497|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 28
§FORM := Using 29 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 4.885e-15.
§DEP_HARD := S26-E-00002496
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002498|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 29

§FORM := Using 30 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 8.438e-15.
§DEP_HARD := S26-E-00002497
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002499|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 30

§FORM := Using 31 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 8.882e-15.
§DEP_HARD := S26-E-00002498
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002500|E|M|Proof+Compile+Meta] Chebyshev interpolation profile degree 31

§FORM := Using 32 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 7.994e-15.
§DEP_HARD := S26-E-00002499
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002501|D|A|Proof+Compile+Meta] Orthogonal polynomial sequence

§FORM := Polynomials p_n are orthogonal for weight w on an interval when integral p_n p_m w=0 for n!=m.
§DEP_HARD := S26-E-00002500
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002502|D|A|Proof+Compile+Meta] Monic orthogonal polynomial

§FORM := A monic orthogonal polynomial has leading coefficient one and is orthogonal to all lower-degree polynomials.
§DEP_HARD := S26-D-00002501
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002503|T|A|Proof+Compile+Meta+IG] Three-term recurrence for orthogonal polynomials

§FORM := Under a positive measure with infinite support and finite moments, monic orthogonal polynomials satisfy a three-term recurrence.
§DEP_HARD := S26-D-00002502
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002504|D|A|Proof+Compile+Meta] Legendre polynomial

§FORM := P_n is the degree-n polynomial orthogonal on [-1,1] with weight 1 and standard normalization P_n(1)=1.
§DEP_HARD := S26-T-00002503
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002505|I|A|Proof+Compile+Meta] Rodrigues formula for Legendre polynomials

§FORM := P_n(x)=1/(2^n n!) d^n/dx^n (x^2-1)^n.
§DEP_HARD := S26-D-00002504
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002506|D|A|Proof+Compile+Meta] Chebyshev polynomial of first kind

§FORM := T_n(cos theta)=cos(n theta).
§DEP_HARD := S26-I-00002505
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002507|D|A|Proof+Compile+Meta] Chebyshev polynomial of second kind

§FORM := U_n(cos theta)=sin((n+1)theta)/sin theta.
§DEP_HARD := S26-D-00002506
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002508|D|A|Proof+Compile+Meta+IG] Hermite polynomial

§FORM := Hermite polynomials form an orthogonal family for Gaussian weight, with physicists/probabilists normalizations distinguished by convention.
§DEP_HARD := S26-D-00002507
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002509|D|A|Proof+Compile+Meta] Laguerre polynomial

§FORM := Laguerre polynomials are orthogonal on [0,infinity) for exponential weights, with generalized parameter alpha when declared.
§DEP_HARD := S26-D-00002508
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002510|D|A|Proof+Compile+Meta] Jacobi polynomial

§FORM := P_n^{(alpha,beta)} is orthogonal on [-1,1] for weight (1-x)^alpha(1+x)^beta with alpha,beta>-1.
§DEP_HARD := S26-D-00002509
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002511|T|A|Proof+Compile+Meta] Zeros of orthogonal polynomials

§FORM := For a positive weight on an interval, the degree-n orthogonal polynomial has n simple zeros in the interior of the convex hull of the support.

§DEP_HARD := S26-D-00002510

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002512|D|A|Proof+Compile+Meta] Christoffel-Darboux kernel

§FORM := K_n(x,y)=sum_{k=0}^n p_k(x)p_k(y)/h_k is the finite orthogonal-projection kernel.

§DEP_HARD := S26-T-00002511

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002513|T|A|Proof+Compile+Meta] Christoffel-Darboux formula

§FORM := The orthogonal-polynomial kernel admits a quotient representation using consecutive polynomials and recurrence coefficients.

§DEP_HARD := S26-D-00002512

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002514|D|A|Proof+Compile+Meta] Gaussian quadrature rule

§FORM := An n-node Gaussian quadrature rule uses the zeros of the degree-n orthogonal polynomial as nodes.

§DEP_HARD := S26-T-00002513

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002515|T|A|Proof+Compile+Meta] Gaussian quadrature exactness

§FORM := An n-node Gaussian quadrature rule is exact for polynomials of degree at most 2n-1 under the associated weight.

§DEP_HARD := S26-D-00002514

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002516|D|A|Proof+Compile+Meta] Moment sequence

§FORM := m_k=integral x^k dmu(x) is the kth moment when finite.

§DEP_HARD := S26-T-00002515

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002517|D|A|Proof+Compile+Meta+IG] Hankel moment matrix

§FORM := H_n=(m_{i+j})_{0<=i,j<=n} is positive semidefinite for every positive measure with the relevant finite moments.

§DEP_HARD := S26-D-00002516

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002518|N|A|Proof+Compile+Meta] Normalization fence for classical polynomials

§FORM := Legendre, Hermite, Laguerre and Jacobi polynomials have multiple normalization conventions; recurrence and norm constants depend on the declared convention.

§DEP_HARD := S26-D-00002517

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002519|B|A|Proof+Compile+Meta+Spectral] Orthogonal-polynomial spectral bridge

§FORM := Jacobi matrices encode three-term recurrences and connect orthogonal polynomials to self-adjoint spectral theory.

§DEP_HARD := S26-N-00002518

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002520|B|A|Proof+Compile+Meta+Spectral] Orthogonal-polynomial numerical bridge

§FORM := Gaussian quadrature and spectral approximation turn orthogonality into high-order numerical integration and discretization schemes.

§DEP_HARD := S26-B-00002519

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002521|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=1

§FORM := At x=0.37, P_1(x)=0.37 and T_1(x)=0.37 under standard normalizations.

§DEP_HARD := S26-B-00002520

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002522|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=2

§FORM := At x=0.37, P_2(x)=-0.29465 and T_2(x)=-0.7262 under standard normalizations.

§DEP_HARD := S26-E-00002521

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002523|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=3

§FORM := At x=0.37, P_3(x)=-0.4283675 and T_3(x)=-0.907388 under standard normalizations.

§DEP_HARD := S26-E-00002522

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002524|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=4
§FORM := At x=0.37, P_4(x)=-0.05638045625 and T_4(x)=0.05473288 under standard normalizations.
§DEP_HARD := S26-E-00002523
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002525|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=5
§FORM := At x=0.37, P_5(x)=0.305144616137 and T_5(x)=0.9478903312 under standard normalizations.
§DEP_HARD := S26-E-00002524
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002526|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=6
§FORM := At x=0.37, P_6(x)=0.253973478155 and T_6(x)=0.646705965088 under standard normalizations.
§DEP_HARD := S26-E-00002525
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002527|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=7
§FORM := At x=0.37, P_7(x)=-0.0870364667 and T_7(x)=-0.469327917035 under standard normalizations.
§DEP_HARD := S26-E-00002526
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002528|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=8
§FORM := At x=0.37, P_8(x)=-0.282608342159 and T_8(x)=-0.994008623694 under standard normalizations.
§DEP_HARD := S26-E-00002527
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002529|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=9
§FORM := At x=0.37, P_9(x)=-0.120146082064 and T_9(x)=-0.266238464499 under standard normalizations.
§DEP_HARD := S26-E-00002528
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002530|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=10
§FORM := At x=0.37, P_10(x)=0.169884812252 and T_10(x)=0.796992159965 under standard normalizations.
§DEP_HARD := S26-E-00002529
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002531|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=11
§FORM := At x=0.37, P_11(x)=0.229224164713 and T_11(x)=0.856012662873 under standard normalizations.
§DEP_HARD := S26-E-00002530
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002532|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=12
§FORM := At x=0.37, P_12(x)=0.00683039224464 and T_12(x)=-0.163542789439 under standard normalizations.
§DEP_HARD := S26-E-00002531
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002533|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=13
§FORM := At x=0.37, P_13(x)=-0.206731449868 and T_13(x)=-0.977034327058 under standard normalizations.
§DEP_HARD := S26-E-00002532
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002534|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=14
§FORM := At x=0.37, P_14(x)=-0.153860163097 and T_14(x)=-0.559462612583 under standard normalizations.
§DEP_HARD := S26-E-00002533
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002535|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=15
§FORM := At x=0.37, P_15(x)=0.0828880498747 and T_15(x)=0.563031993746 under standard normalizations.
§DEP_HARD := S26-E-00002534
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002536|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=16
§FORM := At x=0.37, P_16(x)=0.203664273658 and T_16(x)=0.976106287955 under standard normalizations.
§DEP_HARD := S26-E-00002535
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002537|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=17
§FORM := At x=0.37, P_17(x)=0.0682665872569 and T_17(x)=0.159286659341 under standard normalizations.
§DEP_HARD := S26-E-00002536
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002538|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=18
§FORM := At x=0.37, P_18(x)=-0.143235574845 and T_18(x)=-0.858234160043 under standard normalizations.
§DEP_HARD := S26-E-00002537
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002539|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=19
§FORM := At x=0.37, P_19(x)=-0.167878610013 and T_19(x)=-0.794379937773 under standard normalizations.
§DEP_HARD := S26-E-00002538
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002540|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=20
§FORM := At x=0.37, P_20(x)=0.0149493789781 and T_20(x)=0.270393006091 under standard normalizations.
§DEP_HARD := S26-E-00002539
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002541|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=21
§FORM := At x=0.37, P_21(x)=0.170683537112 and T_21(x)=0.99447076228 under standard normalizations.
§DEP_HARD := S26-E-00002540
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002542|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=22
§FORM := At x=0.37, P_22(x)=0.109165368951 and T_22(x)=0.465515357996 under standard normalizations.
§DEP_HARD := S26-E-00002541
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002543|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=23
§FORM := At x=0.37, P_23(x)=-0.08423627928 and T_23(x)=-0.649989397363 under standard normalizations.
§DEP_HARD := S26-E-00002542
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002544|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=24
§FORM := At x=0.37, P_24(x)=-0.165653015939 and T_24(x)=-0.946507512045 under standard normalizations.
§DEP_HARD := S26-E-00002543
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002545|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=25
§FORM := At x=0.37, P_25(x)=-0.0392647390505 and T_25(x)=-0.0504261615503 under standard normalizations.
§DEP_HARD := S26-E-00002544
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002546|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=26
§FORM := At x=0.37, P_26(x)=0.130784606639 and T_26(x)=0.909192152498 under standard normalizations.
§DEP_HARD := S26-E-00002545
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002547|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=27
§FORM := At x=0.37, P_27(x)=0.13279886487 and T_27(x)=0.723228354399 under standard normalizations.
§DEP_HARD := S26-E-00002546
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002548|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=28
§FORM := At x=0.37, P_28(x)=-0.0295974099691 and T_28(x)=-0.374003170243 under standard normalizations.
§DEP_HARD := S26-E-00002547
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002549|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=29
§FORM := At x=0.37, P_29(x)=-0.149744054918 and T_29(x)=-0.999990700378 under standard normalizations.
§DEP_HARD := S26-E-00002548
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002550|E|M|Proof+Compile+Meta] Legendre-Chebyshev profile n=30
§FORM := At x=0.37, P_30(x)=-0.0803529276584 and T_30(x)=-0.365989948037 under standard normalizations.
§DEP_HARD := S26-E-00002549
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002551|D|A|Proof+Compile+Meta+Spectral] Fourier coefficient on the circle
§FORM := For integrable 2pi-periodic f, fhat(k)=(1/(2pi)) integral_{-pi}^{pi}f(x)e^{-ikx}dx.
§DEP_HARD := S26-E-00002550
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002552|D|A|Proof+Compile+Meta+Spectral] Symmetric Fourier partial sum

§FORM := S_N f=sum_{|k|<=N} fhat(k)e^{ikx}.
§DEP_HARD := S26-D-00002551
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002553|D|A|Proof+Compile+Meta+Spectral] Dirichlet kernel explicit formula refinement

§FORM := D_N(x)=sum_{|k|<=N}e^{ikx}=sin((N+1/2)x)/sin(x/2).
§DEP_HARD := S26-D-00002552
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002554|D|A|Proof+Compile+Meta+Spectral] Fejer mean

§FORM := sigma_N f=(1/(N+1))sum_{n=0}^N S_n f is the Cesaro mean of Fourier partial sums.
§DEP_HARD := S26-D-00002553
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002555|D|A|Proof+Compile+Meta+Spectral] Fejer kernel positivity refinement

§FORM := K_N=(1/(N+1))sum_{n=0}^N D_n is nonnegative and integrates to 2pi under the standard circle normalization.
§DEP_HARD := S26-D-00002554
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002556|T|A|Proof+Compile+Meta+Spectral] Fejer convergence theorem refinement

§FORM := For continuous periodic f, Fejer means converge uniformly to f; for L1 functions they converge at Lebesgue points in the standard sense.
§DEP_HARD := S26-D-00002555
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002557|T|A|Proof+Compile+Meta+Spectral] Riemann-Lebesgue lemma on the circle

§FORM := If f in L1(T), then fhat(k)->0 as |k|->infinity.
§DEP_HARD := S26-T-00002556
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002558|T|A|Proof+Compile+Meta+Spectral] Bessel inequality for Fourier coefficients

§FORM := For f in L2(T), sum_k |fhat(k)|^2 <= ||f||_2^2 under matching normalization.
§DEP_HARD := S26-T-00002557
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002559|T|A|Proof+Compile+Meta+Spectral] Parseval identity with circle normalization

§FORM := For f in L2(T), the squared L2 norm equals the sum of squared Fourier coefficients under the chosen normalization.
§DEP_HARD := S26-T-00002558
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002560|D|A|Proof+Compile+Meta+Spectral] Sobolev regularity from Fourier coefficients on T

§FORM := Periodic H^s regularity is characterized by square summability of (1+|k|^2)^{s/2} fhat(k).
§DEP_HARD := S26-T-00002559
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002561|T|A|Proof+Compile+Meta+Spectral] Fourier differentiation rule on the circle

§FORM := When justified by regularity/distribution theory, the kth Fourier coefficient of f' is ik fhat(k).
§DEP_HARD := S26-D-00002560
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002562|D|A|Proof+Compile+Meta+Spectral] Gibbs phenomenon

§FORM := Fourier partial sums near a jump exhibit a persistent relative overshoot even as the transition region narrows.
§DEP_HARD := S26-T-00002561
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002563|D|A|Proof+Compile+Meta+Spectral] Fourier multiplier on the torus

§FORM := A multiplier operator acts by (Tf)^hat(k)=m(k)fhat(k) on Fourier modes.
§DEP_HARD := S26-D-00002562
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002564|T|A|Proof+Compile+Meta+Spectral] L2 multiplier bound on the torus

§FORM := If m is bounded, the torus Fourier multiplier is bounded on L2 with norm at most sup|m| under Parseval.
§DEP_HARD := S26-D-00002563
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002565|D|A|Proof+Compile+Meta+Spectral] Periodic convolution

§FORM := (f*g)(x)=(1/(2pi))integral f(x-y)g(y)dy under the normalized circle convention.
§DEP_HARD := S26-T-00002564
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002566|I|A|Proof+Compile+Meta+Spectral] Fourier coefficient of periodic convolution

§FORM := With normalized convolution, (f*g)^hat(k)=fhat(k) ghat(k).
§DEP_HARD := S26-D-00002565
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002567|D|A|Proof+Compile+Meta+Spectral] Lacunary Fourier series

§FORM := A lacunary series uses frequencies whose successive ratios are bounded below by a constant greater than one.
§DEP_HARD := S26-I-00002566
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002568|N|A|Proof+Compile+Meta+Spectral] Fourier convergence fence

§FORM := L2 convergence of Fourier series does not imply pointwise or uniform convergence; each mode of convergence requires separate hypotheses.
§DEP_HARD := S26-D-00002567
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002569|B|A|Proof+Compile+Meta+Spectral] Fourier series to PDE bridge

§FORM := Periodic differential operators diagonalize on Fourier modes, reducing linear constant-coefficient PDE on tori to algebraic mode equations.
§DEP_HARD := S26-N-00002568
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002570|B|A|Proof+Compile+Meta+Spectral] Fourier series to signal bridge

§FORM := Truncation, windowing and Cesaro summation correspond to different spectral filters and have different localization/convergence behavior.
§DEP_HARD := S26-B-00002569
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002571|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=1

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=1 is 4/(pi n)=1.27323954474; even harmonics vanish.
§DEP_HARD := S26-B-00002570
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002572|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=3

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=3 is 4/(pi n)=0.424413181578; even harmonics vanish.
§DEP_HARD := S26-E-00002571
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002573|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=5

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=5 is 4/(pi n)=0.254647908947; even harmonics vanish.
§DEP_HARD := S26-E-00002572
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002574|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=7

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=7 is 4/(pi n)=0.181891363534; even harmonics vanish.
§DEP_HARD := S26-E-00002573
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002575|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=9

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=9 is 4/(pi n)=0.141471060526; even harmonics vanish.
§DEP_HARD := S26-E-00002574
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002576|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=11

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=11 is 4/(pi n)=0.115749049521; even harmonics vanish.
§DEP_HARD := S26-E-00002575
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002577|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=13

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=13 is 4/(pi n)=0.0979415034412; even harmonics vanish.

§DEP_HARD := S26-E-00002576

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002578|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=15

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=15 is 4/(pi n)=0.0848826363157; even harmonics vanish.

§DEP_HARD := S26-E-00002577

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002579|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=17

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=17 is 4/(pi n)=0.074896443808; even harmonics vanish.

§DEP_HARD := S26-E-00002578

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002580|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=19

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=19 is 4/(pi n)=0.0670126076176; even harmonics vanish.

§DEP_HARD := S26-E-00002579

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002581|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=21

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=21 is 4/(pi n)=0.0606304545112; even harmonics vanish.

§DEP_HARD := S26-E-00002580

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002582|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=23

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=23 is 4/(pi n)=0.0553582410754; even harmonics vanish.

§DEP_HARD := S26-E-00002581

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002583|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=25

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=25 is 4/(pi n)=0.0509295817894; even harmonics vanish.

§DEP_HARD := S26-E-00002582

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002584|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=27

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=27 is 4/(pi n)=0.0471570201754; even harmonics vanish.

§DEP_HARD := S26-E-00002583

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002585|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=29

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=29 is 4/(pi n)=0.0439048118874; even harmonics vanish.

§DEP_HARD := S26-E-00002584

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002586|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=31

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=31 is 4/(pi n)=0.0410722433786; even harmonics vanish.

§DEP_HARD := S26-E-00002585

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002587|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=33

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=33 is 4/(pi n)=0.0385830165071; even harmonics vanish.

§DEP_HARD := S26-E-00002586

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002588|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=35

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=35 is 4/(pi n)=0.0363782727067; even harmonics vanish.

§DEP_HARD := S26-E-00002587

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002589|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=37

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=37 is 4/(pi n)=0.0344118795874; even harmonics vanish.

§DEP_HARD := S26-E-00002588

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002590|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=39

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=39 is 4/(pi n)=0.0326471678137; even harmonics vanish.

§DEP_HARD := S26-E-00002589

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002591|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=41

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=41 is 4/(pi n)=0.0310546230423; even harmonics vanish.

§DEP_HARD := S26-E-00002590

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002592|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=43

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=43 is 4/(pi n)=0.0296102219706; even harmonics vanish.

§DEP_HARD := S26-E-00002591

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002593|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=45

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=45 is 4/(pi n)=0.0282942121052; even harmonics vanish.

§DEP_HARD := S26-E-00002592

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002594|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=47

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=47 is 4/(pi n)=0.0270902030795; even harmonics vanish.

§DEP_HARD := S26-E-00002593

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002595|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=49

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=49 is 4/(pi n)=0.0259844805048; even harmonics vanish.

§DEP_HARD := S26-E-00002594

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002596|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=51

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=51 is 4/(pi n)=0.0249654812693; even harmonics vanish.

§DEP_HARD := S26-E-00002595

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002597|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=53

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=53 is 4/(pi n)=0.0240233876365; even harmonics vanish.

§DEP_HARD := S26-E-00002596

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002598|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=55

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=55 is 4/(pi n)=0.0231498099043; even harmonics vanish.

§DEP_HARD := S26-E-00002597

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002599|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=57

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=57 is 4/(pi n)=0.0223375358725; even harmonics vanish.

§DEP_HARD := S26-E-00002598

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002600|E|M|Proof+Compile+Meta+Spectral] Square-wave Fourier sine profile n=59

§FORM := For the standard +/-1 square wave, the sine coefficient at odd harmonic n=59 is 4/(pi n)=0.0215803312667; even harmonics vanish.

§DEP_HARD := S26-E-00002599

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002601|D|A|Proof+Compile+Meta+Spectral] Fourier transform normalization pair

§FORM := A Fourier transform convention must declare both exponent sign and constants; all multiplier and Plancherel formulas depend on that choice.

§DEP_HARD := S26-E-00002600

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002602|T|A|Proof+Compile+Meta+Spectral] Riemann-Lebesgue lemma on Rn

§FORM := For f in L1(R^n), its Fourier transform is continuous and vanishes at infinity.

§DEP_HARD := S26-D-00002601

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002603|T|A|Proof+Compile+Meta+Spectral+Swirl] Fourier transform of translation

§FORM := Translation in physical space multiplies the Fourier transform by a phase factor under the declared convention.

§DEP_HARD := S26-T-00002602

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002604|T|A|Proof+Compile+Meta+Spectral] Fourier transform of modulation

§FORM := Modulation in physical space translates the Fourier transform in frequency.

§DEP_HARD := S26-T-00002603

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002605|T|A|Proof+Compile+Meta+Spectral] Fourier scaling law under declared convention

§FORM := Dilation of f rescales the Fourier transform with the reciprocal spatial scale and the Jacobian factor.

§DEP_HARD := S26-T-00002604

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002606|T|A|Proof+Compile+Meta+Spectral] Plancherel extension on Rn

§FORM := The Fourier transform extends from Schwartz functions to a unitary operator on L2 after normalization.

§DEP_HARD := S26-T-00002605

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002607|T|A|Proof+Compile+Meta+Spectral] Hausdorff-Young inequality

§FORM := For 1<=p<=2, the Fourier transform maps Lp to L^{p'} with the appropriate normalization-dependent bound.

§DEP_HARD := S26-T-00002606

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002608|D|A|Proof+Compile+Meta+Spectral] Fourier multiplier on Rn

§FORM := A multiplier T_m is defined formally by Fourier(T_m f)(xi)=m(xi) Fourier(f)(xi).

§DEP_HARD := S26-T-00002607

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002609|T|A|Proof+Compile+Meta+Spectral] L2 Fourier multiplier theorem

§FORM := Every essentially bounded multiplier m defines a bounded L2 multiplier with operator norm ||m||_infinity.

§DEP_HARD := S26-D-00002608

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002610|D|A|Proof+Compile+Meta+Spectral] Fractional Laplacian via Fourier symbol

§FORM := (-Delta)^{s/2} is defined on suitable functions/distributions by multiplier |xi|^s.

§DEP_HARD := S26-T-00002609

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002611|D|A|Proof+Compile+Meta+Spectral] Bessel potential operator

§FORM := (I-Delta)^{-s/2} is the Fourier multiplier (1+|xi|^2)^{-s/2}.

§DEP_HARD := S26-D-00002610

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002612|D|A|Proof+Compile+Meta+Spectral] Band-limited function

§FORM := f is band-limited when its Fourier transform is supported in a bounded frequency set.

§DEP_HARD := S26-D-00002611

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002613|T|A|Proof+Compile+Meta+Spectral] Paley-Wiener interface

§FORM := Compact support on one side of Fourier duality corresponds to analytic/exponential-type constraints on the other, with several precise Paley-Wiener formulations.

§DEP_HARD := S26-D-00002612

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002614|D|A|Proof+Compile+Meta+Spectral] Approximate identity by rescaling

§FORM := phi_epsilon(x)=epsilon^{-n}phi(x/epsilon) with integral phi=1 forms an approximate identity under standard integrability assumptions.
§DEP_HARD := S26-T-00002613
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002615|T|A|Proof+Compile+Meta+Spectral] Approximate-identity convergence in Lp

§FORM := For 1<=p<infinity, phi_epsilon*f -> f in Lp under the standard approximate-identity hypotheses.
§DEP_HARD := S26-D-00002614
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002616|D|A|Proof+Compile+Meta+Spectral] Autocorrelation

§FORM := The autocorrelation of f is the convolution of f with the reflected-conjugate function where defined.
§DEP_HARD := S26-T-00002615
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002617|T|A|Proof+Compile+Meta+IG+Spectral] Wiener-Khinchin interface

§FORM := Under suitable hypotheses, the Fourier transform of autocorrelation is the power spectral density |fhat|^2, with probabilistic variants for stationary processes.
§DEP_HARD := S26-D-00002616
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002618|N|A|Proof+Compile+Meta+Spectral] Transform-normalization fence

§FORM := A formula imported from another Fourier convention can differ by powers of 2pi and sign changes; convention transport must be explicit.
§DEP_HARD := S26-T-00002617
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002619|B|A|Proof+Compile+Meta+Spectral] Fourier transform to pseudodifferential bridge

§FORM := Variable-coefficient symbol calculus generalizes constant Fourier multipliers and underlies microlocal analysis.
§DEP_HARD := S26-N-00002618
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002620|B|A|Proof+Compile+Meta+Spectral] Fourier transform to uncertainty bridge

§FORM := Spatial localization and frequency localization compete quantitatively; exact uncertainty statements depend on norm and normalization.
§DEP_HARD := S26-B-00002619
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002621|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.30, xi=0.1

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.7516501637.
§DEP_HARD := S26-B-00002620
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002622|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.35, xi=0.2

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.875173093283.
§DEP_HARD := S26-E-00002621
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002623|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.40, xi=0.3

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.995458146883.
§DEP_HARD := S26-E-00002622
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002624|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.45, xi=0.4

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.10985662131.
§DEP_HARD := S26-E-00002623
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002625|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.50, xi=0.5

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.21475371513.
§DEP_HARD := S26-E-00002624
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002626|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.55, xi=0.6

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.3055854129.
§DEP_HARD := S26-E-00002625
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002627|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.60, xi=0.7
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.37700783473.
§DEP_HARD := S26-E-00002626
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002628|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.65, xi=0.8
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.42326797329.
§DEP_HARD := S26-E-00002627
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002629|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.70, xi=0.9
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.43880598028.
§DEP_HARD := S26-E-00002628
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002630|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.75, xi=1.0
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.41907671687.
§DEP_HARD := S26-E-00002629
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002631|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.80, xi=1.1
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.36151082228.
§DEP_HARD := S26-E-00002630
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002632|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.85, xi=1.2
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.26645239727.
§DEP_HARD := S26-E-00002631
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002633|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.90, xi=1.3
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.13783577779.
§DEP_HARD := S26-E-00002632
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002634|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=0.95, xi=1.4
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.983335651184.
§DEP_HARD := S26-E-00002633
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002635|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.00, xi=1.5
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.813783054109.
§DEP_HARD := S26-E-00002634
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002636|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.05, xi=1.6
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.641804651578.
§DEP_HARD := S26-E-00002635
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002637|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.10, xi=1.7
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.479888599212.
§DEP_HARD := S26-E-00002636
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002638|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.15, xi=1.8
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.338324588123.
§DEP_HARD := S26-E-00002637
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002639|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.20, xi=1.9
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.223590302164.
§DEP_HARD := S26-E-00002638
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002640|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.25, xi=2.0
§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.137666950151.
§DEP_HARD := S26-E-00002639
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002641|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.30, xi=2.1

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.078461501415.
§DEP_HARD := S26-E-00002640
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002642|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.35, xi=2.2

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.0411139880167.
§DEP_HARD := S26-E-00002641
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002643|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.40, xi=2.3

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.0196674709233.
§DEP_HARD := S26-E-00002642
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002644|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.45, xi=2.4

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.00852546330755.
§DEP_HARD := S26-E-00002643
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002645|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.50, xi=2.5

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.00332313601624.
§DEP_HARD := S26-E-00002644
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002646|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.55, xi=2.6

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.00115546024269.
§DEP_HARD := S26-E-00002645
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002647|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.60, xi=2.7

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.000355403129202.
§DEP_HARD := S26-E-00002646
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002648|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.65, xi=2.8

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 9.58729980928e-05.
§DEP_HARD := S26-E-00002647
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002649|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.70, xi=2.9

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 2.24800296108e-05.
§DEP_HARD := S26-E-00002648
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002650|E|M|Proof+Compile+Meta+Spectral] Gaussian Fourier profile sigma=1.75, xi=3.0

§FORM := For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 4.53949094495e-06.
§DEP_HARD := S26-E-00002649
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002651|D|A|Proof+Compile+Meta] Hardy-Littlewood maximal function

§FORM := Mf(x)=sup_{r>0}|B(x,r)|^{-1} integral_{B(x,r)}|f(y)|dy.
§DEP_HARD := S26-E-00002650
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002652|T|A|Proof+Compile+Meta+Spectral] Weak (1,1) maximal inequality

§FORM := The Hardy-Littlewood maximal operator satisfies mu({Mf>lambda})<=C_n ||f||_1/lambda on R^n.
§DEP_HARD := S26-D-00002651
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002653|T|A|Proof+Compile+Meta+Spectral] Strong Lp maximal inequality

§FORM := For 1<p<=infinity, the maximal operator is bounded on Lp(R^n).
§DEP_HARD := S26-T-00002652
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002654|D|A|Proof+Compile+Meta+Spectral] Hilbert transform

§FORM := Hf(x)=PV (1/pi) integral f(y)/(x-y)dy under the standard real-line normalization.
§DEP_HARD := S26-T-00002653
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002655|T|A|Proof+Compile+Meta+Spectral] L2 Hilbert-transform multiplier
§FORM := The Hilbert transform has Fourier multiplier -i sign(xi) and is bounded on L2.
§DEP_HARD := S26-D-00002654
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002656|T|A|Proof+Compile+Meta+Spectral] Lp boundedness of Hilbert transform
§FORM := For 1<p<infinity, H extends boundedly on Lp(R).
§DEP_HARD := S26-T-00002655
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002657|D|A|Proof+Compile+Meta+Spectral] Singular integral kernel
§FORM := A singular kernel has nonintegrable point singularity together with cancellation and size/smoothness structure sufficient for a principal-value operator.
§DEP_HARD := S26-T-00002656
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002658|D|A|Proof+Compile+Meta+Spectral] Calderon-Zygmund operator interface
§FORM := A Calderon-Zygmund operator is L2 bounded and represented off the diagonal by a standard singular kernel satisfying size and smoothness estimates.
§DEP_HARD := S26-D-00002657
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002659|T|A|Proof+Compile+Meta+Spectral] Calderon-Zygmund Lp theorem
§FORM := A Calderon-Zygmund operator extends boundedly to Lp for 1<p<infinity and satisfies weak-type endpoint control under standard assumptions.
§DEP_HARD := S26-D-00002658
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002660|D|A|Proof+Compile+Meta+Spectral] Riesz transform
§FORM := R_j has Fourier multiplier proportional to -i xi_j/|xi| under the declared Fourier convention.
§DEP_HARD := S26-T-00002659
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002661|D|A|Proof+Compile+Meta] Maximal truncation
§FORM := A maximal truncation takes the supremum over singular-integral cutoffs and controls almost-everywhere convergence.
§DEP_HARD := S26-D-00002660
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002662|D|A|Proof+Compile+Meta] Hardy space H1 interface
§FORM := Real-variable H1 replaces L1 at certain singular-integral endpoints and admits atomic/maximal characterizations.
§DEP_HARD := S26-D-00002661
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002663|D|A|Proof+Compile+Meta] BMO space
§FORM := A locally integrable function is in BMO when its mean oscillation over balls/cubes is uniformly bounded.
§DEP_HARD := S26-D-00002662
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002664|T|A|Proof+Compile+Meta] John-Nirenberg inequality
§FORM := BMO oscillations have exponentially decaying distribution tails on each cube, up to dimension-dependent constants.
§DEP_HARD := S26-D-00002663
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002665|T|A|Proof+Compile+Meta] H1-BMO duality interface
§FORM := The dual of real Hardy space H1 can be identified with BMO modulo constants under the classical pairing.
§DEP_HARD := S26-T-00002664
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002666|D|A|Proof+Compile+Meta] Muckenhoupt Ap weight
§FORM := A weight w belongs to Ap when its local average and reciprocal-power average satisfy the Ap characteristic bound.
§DEP_HARD := S26-T-00002665
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002667|T|A|Proof+Compile+Meta+Spectral] Weighted maximal theorem
§FORM := The maximal operator is bounded on Lp(w) exactly for Muckenhoupt Ap weights when 1<p<infinity.
§DEP_HARD := S26-D-00002666
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002668|N|A|Proof+Compile+Meta] Principal-value fence

§FORM := Existence of a principal value is weaker than absolute integrability and can depend on symmetric cancellation.
§DEP_HARD := S26-T-00002667
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002669|B|A|Proof+Compile+Meta+Spectral] Singular integrals to PDE bridge

§FORM := Riesz transforms and Calderon-Zygmund estimates convert elliptic PDE derivatives into bounded singular-integral operators.
§DEP_HARD := S26-N-00002668
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002670|B|A|Proof+Compile+Meta+Spectral] Maximal operators to differentiation bridge

§FORM := Maximal inequalities provide the quantitative domination behind almost-everywhere differentiation theorems.
§DEP_HARD := S26-B-00002669
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002671|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=5

§FORM := For v=(1,...,5) at its central index, centered finite interval averages are [3.0, 3.0, 3.0]; their maximum is 3.000000.
§DEP_HARD := S26-B-00002670
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002672|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=7

§FORM := For v=(1,...,7) at its central index, centered finite interval averages are [4.0, 4.0, 4.0, 4.0]; their maximum is 4.000000.
§DEP_HARD := S26-E-00002671
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002673|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=9

§FORM := For v=(1,...,9) at its central index, centered finite interval averages are [5.0, 5.0, 5.0, 5.0, 5.0]; their maximum is 5.000000.
§DEP_HARD := S26-E-00002672
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002674|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=11

§FORM := For v=(1,...,11) at its central index, centered finite interval averages are [6.0, 6.0, 6.0, 6.0, 6.0, 6.0]; their maximum is 6.000000.
§DEP_HARD := S26-E-00002673
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002675|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=13

§FORM := For v=(1,...,13) at its central index, centered finite interval averages are [7.0, 7.0, 7.0, 7.0, 7.0, 7.0, 7.0]; their maximum is 7.000000.
§DEP_HARD := S26-E-00002674
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002676|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=15

§FORM := For v=(1,...,15) at its central index, centered finite interval averages are [8.0, 8.0, 8.0, 8.0, 8.0, 8.0, 8.0, 8.0]; their maximum is 8.000000.
§DEP_HARD := S26-E-00002675
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002677|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=17

§FORM := For v=(1,...,17) at its central index, centered finite interval averages are [9.0, 9.0, 9.0, 9.0, 9.0, 9.0, 9.0, 9.0, 9.0]; their maximum is 9.000000.
§DEP_HARD := S26-E-00002676
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002678|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=19

§FORM := For v=(1,...,19) at its central index, centered finite interval averages are [10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0]; their maximum is 10.000000.
§DEP_HARD := S26-E-00002677
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002679|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=21

§FORM := For v=(1,...,21) at its central index, centered finite interval averages are [11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0]; their maximum is 11.000000.
§DEP_HARD := S26-E-00002678
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002680|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=23

§FORM := For v=(1,...,23) at its central index, centered finite interval averages are [12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0]; their maximum is 12.000000.
§DEP_HARD := S26-E-00002679
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002681|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=25

§FORM := For v=(1,...,25) at its central index, centered finite interval averages are [13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0]; their maximum is 13.000000.

§DEP_HARD := S26-E-00002680

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002682|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=27

§FORM := For v=(1,...,27) at its central index, centered finite interval averages are [14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0]; their maximum is 14.000000.

§DEP_HARD := S26-E-00002681

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002683|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=29

§FORM := For v=(1,...,29) at its central index, centered finite interval averages are [15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0]; their maximum is 15.000000.

§DEP_HARD := S26-E-00002682

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002684|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=31

§FORM := For v=(1,...,31) at its central index, centered finite interval averages are [16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0]; their maximum is 16.000000.

§DEP_HARD := S26-E-00002683

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002685|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=33

§FORM := For v=(1,...,33) at its central index, centered finite interval averages are [17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0]; their maximum is 17.000000.

§DEP_HARD := S26-E-00002684

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002686|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=35

§FORM := For v=(1,...,35) at its central index, centered finite interval averages are [18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0]; their maximum is 18.000000.

§DEP_HARD := S26-E-00002685

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002687|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=37

§FORM := For v=(1,...,37) at its central index, centered finite interval averages are [19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0]; their maximum is 19.000000.

§DEP_HARD := S26-E-00002686

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002688|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=39

§FORM := For v=(1,...,39) at its central index, centered finite interval averages are [20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0]; their maximum is 20.000000.

§DEP_HARD := S26-E-00002687

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002689|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=41

§FORM := For v=(1,...,41) at its central index, centered finite interval averages are [21.0, 21.0]; their maximum is 21.000000.

§DEP_HARD := S26-E-00002688

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002690|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=43

§FORM := For v=(1,...,43) at its central index, centered finite interval averages are [22.0, 22.0]; their maximum is 22.000000.

§DEP_HARD := S26-E-00002689

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002691|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=45

§FORM := For v=(1,...,45) at its central index, centered finite interval averages are [23.0, 23.0]; their maximum is 23.000000.

§DEP_HARD := S26-E-00002690

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002692|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=47

§FORM := For v=(1,...,47) at its central index, centered finite interval averages are [24.0, 24.0]; their maximum is 24.000000.

§DEP_HARD := S26-E-00002691

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002693|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=49

§FORM := For v=(1,...,49) at its central index, centered finite interval averages are [25.0, 25.0]; their maximum is 25.000000.

§DEP_HARD := S26-E-00002692

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002694|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=51

§FORM := For v=(1,...,51) at its central index, centered finite interval averages are [26.0, 26.0]; their maximum is 26.000000.

§DEP_HARD := S26-E-00002693

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002695|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=53

§FORM := For v=(1,...,53) at its central index, centered finite interval averages are [27.0, 27.0]; their maximum is 27.000000.

§DEP_HARD := S26-E-00002694

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002696|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=55

§FORM := For v=(1,...,55) at its central index, centered finite interval averages are [28.0, 28.0]; their maximum is 28.000000.

§DEP_HARD := S26-E-00002695

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002697|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=57

§FORM := For v=(1,...,57) at its central index, centered finite interval averages are [29.0, 29.0]; their maximum is 29.000000.

§DEP_HARD := S26-E-00002696

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002698|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=59

§FORM := For v=(1,...,59) at its central index, centered finite interval averages are [30.0, 30.0]; their maximum is 30.000000.

§DEP_HARD := S26-E-00002697

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002699|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=61

§FORM := For v=(1,...,61) at its central index, centered finite interval averages are [31.0, 31.0]; their maximum is 31.000000.

§DEP_HARD := S26-E-00002698

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002700|E|M|Proof+Compile+Meta] Discrete maximal-average profile n=63

§FORM := For v=(1,...,63) at its central index, centered finite interval averages are [32.0, 32.0]; their maximum is 32.000000.

§DEP_HARD := S26-E-00002699

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002701|D|A|Proof+Compile+Meta] Holder space C^{0,alpha}

§FORM := For 0<alpha<=1, C^{0,alpha} consists of bounded functions with finite seminorm sup_{x!=y}|f(x)-f(y)|/|x-y|^alpha.

§DEP_HARD := S26-E-00002700

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002702|D|A|Proof+Compile+Meta] Holder space C^{k,alpha}

§FORM := C^{k,alpha} consists of C^k functions whose kth derivatives are alpha-Holder continuous.

§DEP_HARD := S26-D-00002701

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002703|D|A|Proof+Compile+Meta] Lipschitz space as Holder endpoint

§FORM := C^{0,1} is the Lipschitz class under the standard bounded-domain convention.

§DEP_HARD := S26-D-00002702

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002704|D|A|Proof+Compile+Meta] Morrey space interface

§FORM := Morrey spaces control local Lp mass with an additional scaling exponent, interpolating between Lp and uniform regularity phenomena.

§DEP_HARD := S26-D-00002703

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002705|D|A|Proof+Compile+Meta] Besov space via differences

§FORM := Besov B^s_{p,q} can be defined using Lp norms of finite differences integrated over scale with q-summability.
§DEP_HARD := S26-D-00002704
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002706|D|A|Proof+Compile+Meta+Tower] Besov space via dyadic decomposition

§FORM := Equivalent Besov norms can be built from Littlewood-Paley dyadic frequency pieces under standard decomposition choices.
§DEP_HARD := S26-D-00002705
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002707|D|A|Proof+Compile+Meta+Tower] Triebel-Lizorkin space interface

§FORM := Triebel-Lizorkin F^s_{p,q} places the dyadic q-aggregation inside the Lp norm and includes many Sobolev/Hardy scales as special cases.
§DEP_HARD := S26-D-00002706
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002708|D|A|Proof+Compile+Meta+Tower] Littlewood-Paley partition of unity

§FORM := A dyadic smooth frequency partition decomposes a distribution into low-frequency and annular pieces.
§DEP_HARD := S26-D-00002707
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002709|T|A|Proof+Compile+Meta+Tower] Littlewood-Paley Lp equivalence

§FORM := For 1<p<infinity, square-function norms built from dyadic pieces are equivalent to Lp norms under standard constructions.
§DEP_HARD := S26-D-00002708
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002710|D|A|Proof+Compile+Meta] Sobolev-Slobodeckij seminorm

§FORM := For 0<s<1, [f]_{W^{s,p}}^p=double integral |f(x)-f(y)|^p/|x-y|^{n+sp} dxdy.
§DEP_HARD := S26-T-00002709
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002711|D|A|Proof+Compile+Meta] Fractional Sobolev space W^{s,p}

§FORM := W^{s,p} combines Lp integrability with the fractional difference seminorm for noninteger s in the standard regime.
§DEP_HARD := S26-D-00002710
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002712|T|A|Proof+Compile+Meta] Morrey embedding interface

§FORM := If p>n, W^{1,p} functions on suitable domains have Holder-continuous representatives with exponent 1-n/p.
§DEP_HARD := S26-D-00002711
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002713|T|A|Proof+Compile+Meta] Sobolev trace theorem interface

§FORM := Sobolev functions on sufficiently regular domains admit boundary traces in lower-order fractional spaces when the regularity index crosses the trace threshold.
§DEP_HARD := S26-T-00002712
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002714|D|A|Proof+Compile+Meta+Spectral] Extension operator for Sobolev spaces

§FORM := An extension operator maps Sobolev functions on a domain to the ambient space while preserving the function on the domain and controlling the Sobolev norm.
§DEP_HARD := S26-T-00002713
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002715|T|A|Proof+Compile+Meta+Spectral] Sobolev extension theorem for Lipschitz domains

§FORM := Bounded Lipschitz domains admit bounded linear extension operators for standard W^{k,p} ranges.
§DEP_HARD := S26-D-00002714
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002716|D|A|Proof+Compile+Meta+IG] Campanato space interface

§FORM := Campanato seminorms measure polynomial/mean oscillation with scale weights and identify with Holder spaces in supercritical regimes.
§DEP_HARD := S26-T-00002715
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002717|D|A|Proof+Compile+Meta] Function-space scaling dimension

§FORM := A function-space norm has a scaling law under f(x)->f(lambda x); critical exponents occur when the target estimate is scale invariant.
§DEP_HARD := S26-D-00002716
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002718|N|A|Proof+Compile+Meta] Embedding-domain fence

§FORM := Sobolev and trace embeddings depend on dimension, domain regularity, exponent and boundary conditions; Euclidean formulas are not domain-free.

§DEP_HARD := S26-D-00002717

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002719|B|A|Proof+Compile+Meta] Function spaces to regularity bridge

§FORM := Holder, Sobolev, Besov and Triebel-Lizorkin scales provide a vocabulary for PDE regularity, approximation rates and harmonic-analysis bounds.

§DEP_HARD := S26-N-00002718

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002720|B|A|Proof+Compile+Meta+IG+Spectral] Function spaces to numerics bridge

§FORM := Smoothness measured in Besov/Sobolev scales predicts approximation sparsity and convergence rates for wavelet, finite-element and spectral methods.

§DEP_HARD := S26-B-00002719

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002721|E|M|Proof+Compile+Meta] Holder power profile alpha=0.125

§FORM := For f(x)=x^0.125 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-B-00002720

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002722|E|M|Proof+Compile+Meta] Holder power profile alpha=0.150

§FORM := For f(x)=x^0.150 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-E-00002721

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002723|E|M|Proof+Compile+Meta] Holder power profile alpha=0.175

§FORM := For f(x)=x^0.175 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-E-00002722

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002724|E|M|Proof+Compile+Meta] Holder power profile alpha=0.200

§FORM := For f(x)=x^0.200 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-E-00002723

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002725|E|M|Proof+Compile+Meta] Holder power profile alpha=0.225

§FORM := For f(x)=x^0.225 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-E-00002724

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002726|E|M|Proof+Compile+Meta] Holder power profile alpha=0.250

§FORM := For f(x)=x^0.250 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-E-00002725

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002727|E|M|Proof+Compile+Meta] Holder power profile alpha=0.275

§FORM := For f(x)=x^0.275 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-E-00002726

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002728|E|M|Proof+Compile+Meta] Holder power profile alpha=0.300

§FORM := For f(x)=x^0.300 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-E-00002727

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002729|E|M|Proof+Compile+Meta] Holder power profile alpha=0.325

§FORM := For f(x)=x^0.325 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-E-00002728

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002730|E|M|Proof+Compile+Meta] Holder power profile alpha=0.350

§FORM := For f(x)=x^0.350 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

§DEP_HARD := S26-E-00002729

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002731|E|M|Proof+Compile+Meta] Holder power profile alpha=0.375

§FORM := For f(x)=x^0.375 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002730
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002732|E|M|Proof+Compile+Meta] Holder power profile alpha=0.400

§FORM := For f(x)=x^0.400 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002731
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002733|E|M|Proof+Compile+Meta] Holder power profile alpha=0.425

§FORM := For f(x)=x^0.425 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002732
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002734|E|M|Proof+Compile+Meta] Holder power profile alpha=0.450

§FORM := For f(x)=x^0.450 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002733
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002735|E|M|Proof+Compile+Meta] Holder power profile alpha=0.475

§FORM := For f(x)=x^0.475 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002734
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002736|E|M|Proof+Compile+Meta] Holder power profile alpha=0.500

§FORM := For f(x)=x^0.500 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002735
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002737|E|M|Proof+Compile+Meta] Holder power profile alpha=0.525

§FORM := For f(x)=x^0.525 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002736
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002738|E|M|Proof+Compile+Meta] Holder power profile alpha=0.550

§FORM := For f(x)=x^0.550 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002737
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002739|E|M|Proof+Compile+Meta] Holder power profile alpha=0.575

§FORM := For f(x)=x^0.575 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002738
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002740|E|M|Proof+Compile+Meta] Holder power profile alpha=0.600

§FORM := For f(x)=x^0.600 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002739
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002741|E|M|Proof+Compile+Meta] Holder power profile alpha=0.625

§FORM := For f(x)=x^0.625 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002740
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002742|E|M|Proof+Compile+Meta] Holder power profile alpha=0.650

§FORM := For f(x)=x^0.650 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002741
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002743|E|M|Proof+Compile+Meta] Holder power profile alpha=0.675

§FORM := For f(x)=x^0.675 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002742
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002744|E|M|Proof+Compile+Meta] Holder power profile alpha=0.700

§FORM := For f(x)=x^0.700 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002743
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002745|E|M|Proof+Compile+Meta] Holder power profile alpha=0.725

§FORM := For f(x)=x^0.725 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002744
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002746|E|M|Proof+Compile+Meta] Holder power profile alpha=0.750

§FORM := For f(x)=x^0.750 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002745
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002747|E|M|Proof+Compile+Meta] Holder power profile alpha=0.775

§FORM := For f(x)=x^0.775 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002746
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002748|E|M|Proof+Compile+Meta] Holder power profile alpha=0.800

§FORM := For f(x)=x^0.800 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002747
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002749|E|M|Proof+Compile+Meta] Holder power profile alpha=0.825

§FORM := For f(x)=x^0.825 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002748
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002750|E|M|Proof+Compile+Meta] Holder power profile alpha=0.850

§FORM := For f(x)=x^0.850 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.
§DEP_HARD := S26-E-00002749
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002751|D|A|Proof+Compile+Meta] Reflexive Banach space

§FORM := X is reflexive when the canonical embedding X->X** is onto.
§DEP_HARD := S26-E-00002750
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002752|D|A|Proof+Compile+Meta] Strictly convex norm

§FORM := A norm is strictly convex when the unit sphere contains no nontrivial line segment.
§DEP_HARD := S26-D-00002751
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002753|D|A|Proof+Compile+Meta] Smooth norm

§FORM := A norm is smooth when it has a unique supporting functional at every nonzero point, with equivalent differentiability formulations under standard conditions.
§DEP_HARD := S26-D-00002752
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002754|D|A|Proof+Compile+Meta] Uniformly smooth Banach space

§FORM := Uniform smoothness is quantified by a modulus rho_X(t) satisfying rho_X(t)/t->0 as t->0.
§DEP_HARD := S26-D-00002753
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002755|D|A|Proof+Compile+Meta] Modulus of smoothness

§FORM := rho_X(t)=sup_{||x||=||y||=1}((||x+ty||+||x-ty||)/2-1).
§DEP_HARD := S26-D-00002754
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002756|T|A|Proof+Compile+Meta] Duality of uniform convexity and smoothness

§FORM := Uniform convexity of X is dual to uniform smoothness of X* in the standard quantitative duality framework.
§DEP_HARD := S26-D-00002755
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002757|D|A|Proof+Compile+Meta] Schauder basis

§FORM := A sequence (e_n) is a Schauder basis if every x has a unique norm-convergent expansion x=sum a_n e_n.
§DEP_HARD := S26-T-00002756
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002758|D|A|Proof+Compile+Meta] Unconditional basis

§FORM := A Schauder basis is unconditional when the expansion converges regardless of signs/permutations in the equivalent standard formulations.

§DEP_HARD := S26-D-00002757

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002759|D|A|Proof+Compile+Meta] Basic sequence

§FORM := A basic sequence is a Schauder basis for its closed linear span.

§DEP_HARD := S26-D-00002758

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002760|D|A|Proof+Compile+Meta] Finite representability

§FORM := X is finitely representable in Y when every finite-dimensional subspace of X embeds into Y with arbitrarily small distortion above one.

§DEP_HARD := S26-D-00002759

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002761|D|A|Proof+Compile+Meta] Banach-Mazur distance

§FORM := For isomorphic finite-dimensional normed spaces X,Y, d(X,Y)=inf ||T||||T^{-1}|| over linear isomorphisms T:X->Y.

§DEP_HARD := S26-D-00002760

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002762|D|A|Proof+Compile+Meta] Type of a Banach space

§FORM := Rademacher type p controls expected norms of signed sums by the l^p sum of vector norms.

§DEP_HARD := S26-D-00002761

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002763|D|A|Proof+Compile+Meta] Cotype of a Banach space

§FORM := Rademacher cotype q controls the reverse comparison between vector norms and expected signed sums.

§DEP_HARD := S26-D-00002762

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002764|T|A|Proof+Compile+Meta] Kadec-Klee property interface

§FORM := In spaces with the Kadec-Klee property, weak convergence plus convergence of norms implies norm convergence.

§DEP_HARD := S26-D-00002763

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002765|D|A|Proof+Compile+Meta] Weakly Cauchy sequence

§FORM := A sequence x_n is weakly Cauchy when phi(x_n) is Cauchy for every phi in X*.

§DEP_HARD := S26-T-00002764

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002766|T|A|Proof+Compile+Meta] Eberlein-Smulian interface

§FORM := For Banach spaces, weak compactness is equivalent to weak sequential compactness.

§DEP_HARD := S26-D-00002765

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002767|D|A|Proof+Compile+Meta+IG] Radon-Nikodym property interface

§FORM := A Banach space has the Radon-Nikodym property when suitable vector measures absolutely continuous with respect to scalar measures possess Bochner-integrable densities.

§DEP_HARD := S26-T-00002766

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002768|N|A|Proof+Compile+Meta] Basis fence

§FORM := A separable Banach space need not have a Schauder basis, and a Schauder basis need not be unconditional.

§DEP_HARD := S26-D-00002767

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002769|B|A|Proof+Compile+Meta+Topo+IG] Banach geometry to probability bridge

§FORM := Type, cotype and uniform convexity govern concentration and vector-valued random-series inequalities.

§DEP_HARD := S26-N-00002768

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002770|B|A|Proof+Compile+Meta+Topo] Banach geometry to algorithms bridge

§FORM := Norm geometry and Banach-Mazur distortion quantify conditioning and dimension-dependent losses when transporting algorithms between normed models.

§DEP_HARD := S26-B-00002769

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002771|E|M|Proof+Compile+Meta] lp midpoint profile p=1.10, n=5

§FORM := For normalized reversed positive vectors in l^1.10_5, the midpoint norm is 0.988310066232; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-B-00002770

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002772|E|M|Proof+Compile+Meta] lp midpoint profile p=1.20, n=6

§FORM := For normalized reversed positive vectors in l^1.20_6, the midpoint norm is 0.975426882711; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002771

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002773|E|M|Proof+Compile+Meta] lp midpoint profile p=1.30, n=7

§FORM := For normalized reversed positive vectors in l^1.30_7, the midpoint norm is 0.962206636680; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002772

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002774|E|M|Proof+Compile+Meta] lp midpoint profile p=1.40, n=8

§FORM := For normalized reversed positive vectors in l^1.40_8, the midpoint norm is 0.949109987475; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002773

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002775|E|M|Proof+Compile+Meta] lp midpoint profile p=1.50, n=9

§FORM := For normalized reversed positive vectors in l^1.50_9, the midpoint norm is 0.936383710782; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002774

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002776|E|M|Proof+Compile+Meta] lp midpoint profile p=1.60, n=10

§FORM := For normalized reversed positive vectors in l^1.60_10, the midpoint norm is 0.924154076451; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002775

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002777|E|M|Proof+Compile+Meta] lp midpoint profile p=1.70, n=4

§FORM := For normalized reversed positive vectors in l^1.70_4, the midpoint norm is 0.935311358159; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002776

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002778|E|M|Proof+Compile+Meta] lp midpoint profile p=1.80, n=5

§FORM := For normalized reversed positive vectors in l^1.80_5, the midpoint norm is 0.920254786622; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002777

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002779|E|M|Proof+Compile+Meta] lp midpoint profile p=1.90, n=6

§FORM := For normalized reversed positive vectors in l^1.90_6, the midpoint norm is 0.906737660524; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002778

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002780|E|M|Proof+Compile+Meta] lp midpoint profile p=2.00, n=7

§FORM := For normalized reversed positive vectors in l^2.00_7, the midpoint norm is 0.894427191000; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002779

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002781|E|M|Proof+Compile+Meta] lp midpoint profile p=2.10, n=8

§FORM := For normalized reversed positive vectors in l^2.10_8, the midpoint norm is 0.883106934357; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002780

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002782|E|M|Proof+Compile+Meta] lp midpoint profile p=2.20, n=9

§FORM := For normalized reversed positive vectors in l^2.20_9, the midpoint norm is 0.872623919168; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002781

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002783|E|M|Proof+Compile+Meta] lp midpoint profile p=2.30, n=10

§FORM := For normalized reversed positive vectors in l^2.30_10, the midpoint norm is 0.862863535879; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002782

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002784|E|M|Proof+Compile+Meta] lp midpoint profile p=2.40, n=4

§FORM := For normalized reversed positive vectors in l^2.40_4, the midpoint norm is 0.886943703761; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002783

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002785|E|M|Proof+Compile+Meta] lp midpoint profile p=2.50, n=5

§FORM := For normalized reversed positive vectors in l^2.50_5, the midpoint norm is 0.870803508081; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002784

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002786|E|M|Proof+Compile+Meta] lp midpoint profile p=2.60, n=6

§FORM := For normalized reversed positive vectors in l^2.60_6, the midpoint norm is 0.857607742895; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002785

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002787|E|M|Proof+Compile+Meta] lp midpoint profile p=2.70, n=7

§FORM := For normalized reversed positive vectors in l^2.70_7, the midpoint norm is 0.846324902449; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002786

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002788|E|M|Proof+Compile+Meta] lp midpoint profile p=2.80, n=8

§FORM := For normalized reversed positive vectors in l^2.80_8, the midpoint norm is 0.836389360421; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002787

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002789|E|M|Proof+Compile+Meta] lp midpoint profile p=2.90, n=9

§FORM := For normalized reversed positive vectors in l^2.90_9, the midpoint norm is 0.827461206652; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002788

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002790|E|M|Proof+Compile+Meta] lp midpoint profile p=3.00, n=10

§FORM := For normalized reversed positive vectors in l^3.00_10, the midpoint norm is 0.819321270601; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002789

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002791|E|M|Proof+Compile+Meta] lp midpoint profile p=3.10, n=4

§FORM := For normalized reversed positive vectors in l^3.10_4, the midpoint norm is 0.850333663578; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002790

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002792|E|M|Proof+Compile+Meta] lp midpoint profile p=3.20, n=5

§FORM := For normalized reversed positive vectors in l^3.20_5, the midpoint norm is 0.833902099906; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002791

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002793|E|M|Proof+Compile+Meta] lp midpoint profile p=3.30, n=6

§FORM := For normalized reversed positive vectors in l^3.30_6, the midpoint norm is 0.821165226896; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002792

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002794|E|M|Proof+Compile+Meta] lp midpoint profile p=3.40, n=7

§FORM := For normalized reversed positive vectors in l^3.40_7, the midpoint norm is 0.810694560788; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002793

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002795|E|M|Proof+Compile+Meta] lp midpoint profile p=3.50, n=8

§FORM := For normalized reversed positive vectors in l^3.50_8, the midpoint norm is 0.801739915782; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002794

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002796|E|M|Proof+Compile+Meta] lp midpoint profile p=3.60, n=9

§FORM := For normalized reversed positive vectors in l^3.60_9, the midpoint norm is 0.793867140500; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002795

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002797|E|M|Proof+Compile+Meta] lp midpoint profile p=3.70, n=10

§FORM := For normalized reversed positive vectors in l^3.70_10, the midpoint norm is 0.786805931648; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002796

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002798|E|M|Proof+Compile+Meta] lp midpoint profile p=3.80, n=4

§FORM := For normalized reversed positive vectors in l^3.80_4, the midpoint norm is 0.822013869190; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002797

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002799|E|M|Proof+Compile+Meta] lp midpoint profile p=3.90, n=5

§FORM := For normalized reversed positive vectors in l^3.90_5, the midpoint norm is 0.805513391165; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002798

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002800|E|M|Proof+Compile+Meta] lp midpoint profile p=4.00, n=6

§FORM := For normalized reversed positive vectors in l^4.00_6, the midpoint norm is 0.793159456850; a value below 1 records this finite convexity instance.

§DEP_HARD := S26-E-00002799

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002801|D|A|Proof+Compile+Meta+Spectral] Resolvent set of a closed operator

§FORM := rho(A) consists of lambda for which lambda I-A is bijective with bounded inverse on the ambient Banach space.

§DEP_HARD := S26-E-00002800

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002802|D|A|Proof+Compile+Meta+Spectral] Resolvent operator

§FORM := R(lambda,A)=(lambda I-A)^{-1} for lambda in rho(A).

§DEP_HARD := S26-D-00002801

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002803|I|A|Proof+Compile+Meta+Spectral] Resolvent identity for closed operators

§FORM := R(lambda,A)-R(mu,A)=(mu-lambda)R(lambda,A)R(mu,A).

§DEP_HARD := S26-D-00002802

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002804|T|A|Proof+Compile+Meta+Spectral] Resolvent analyticity

§FORM := The resolvent is operator-valued holomorphic on the resolvent set.

§DEP_HARD := S26-I-00002803

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002805|D|A|Proof+Compile+Meta+Spectral] Spectral projection by contour integral

§FORM := For isolated spectral components, P=(1/(2pi i)) integral_Gamma (zI-A)^{-1}dz defines the corresponding Riesz projection.

§DEP_HARD := S26-T-00002804

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002806|D|A|Proof+Compile+Meta+Spectral] Fredholm operator

§FORM := T is Fredholm when ker T and cokernel are finite-dimensional and range T is closed.

§DEP_HARD := S26-D-00002805

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002807|D|A|Proof+Compile+Meta+Spectral] Fredholm index

§FORM := ind(T)=dim ker T-dim coker T.

§DEP_HARD := S26-D-00002806

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002808|T|A|Proof+Compile+Meta+Spectral] Index stability under compact perturbations

§FORM := The Fredholm index is invariant under compact perturbations that preserve Fredholmness; in fact T+K is Fredholm with the same index when T is Fredholm and K compact.

§DEP_HARD := S26-D-00002807

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002809|T|A|Proof+Compile+Meta+Spectral] Fredholm alternative for compact operators

§FORM := For compact K, I-K is invertible iff the homogeneous equation (I-K)x=0 has only the zero solution, with the dual finite-dimensional obstruction formulation.

§DEP_HARD := S26-T-00002808

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002810|D|A|Proof+Compile+Meta+Spectral] Essential spectrum interface

§FORM := The essential spectrum removes isolated finite-multiplicity eigenvalues according to a declared equivalent/Fredholm definition appropriate to the operator class.

§DEP_HARD := S26-T-00002809

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002811|D|A|Proof+Compile+Meta+Spectral] Numerical range

§FORM := W(A)={<Ax,x>:||x||=1} for a bounded operator on a complex Hilbert space.

§DEP_HARD := S26-D-00002810

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002812|T|A|Proof+Compile+Meta+Spectral] Toeplitz-Hausdorff theorem

§FORM := The numerical range of a bounded operator on a complex Hilbert space is convex.

§DEP_HARD := S26-D-00002811

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002813|D|A|Proof+Compile+Meta+Spectral] Compact resolvent

§FORM := A closed operator has compact resolvent when R(lambda,A) is compact for one, hence every, lambda in the resolvent set.

§DEP_HARD := S26-T-00002812

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002814|T|A|Proof+Compile+Meta+Spectral] Compact-resolvent spectral discreteness interface

§FORM := A closed operator with compact resolvent has discrete spectrum with finite algebraic multiplicities under standard Banach/Hilbert hypotheses, accumulating only at infinity.

§DEP_HARD := S26-D-00002813

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002815|D|A|Proof+Compile+Meta+Spectral] Sectorial operator

§FORM := A sectorial operator has spectrum in a sector and resolvent bounds outside wider sectors, with angle and constants declared.

§DEP_HARD := S26-T-00002814

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002816|D|A|Proof+Compile+Meta+Spectral] Fractional power of a sectorial operator

§FORM := Fractional powers A^alpha can be defined through holomorphic functional calculus or equivalent integral formulas under sectorial hypotheses.

§DEP_HARD := S26-D-00002815

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002817|D|A|Proof+Compile+Meta+Spectral] Functional calculus algebra homomorphism

§FORM := A holomorphic functional calculus maps admissible scalar functions to operators while preserving sums, products and the identity on its domain of definition.

§DEP_HARD := S26-D-00002816

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002818|N|A|Proof+Compile+Meta+Spectral] Spectrum-eigenvalue fence for infinite dimensions

§FORM := Spectrum can contain continuous or residual parts with no eigenvectors; finite-matrix intuition does not exhaust operator spectrum.

§DEP_HARD := S26-D-00002817

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002819|B|A|Proof+Compile+Meta+Spectral] Resolvent to PDE bridge

§FORM := Elliptic and evolution estimates are often resolvent bounds for differential operators on function spaces.

§DEP_HARD := S26-N-00002818

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002820|B|A|Proof+Compile+Meta+Spectral] Operator calculus bridge
§FORM := Contour, holomorphic and spectral functional calculi turn scalar analytic functions into operator constructions such as semigroups, fractional powers and projections.
§DEP_HARD := S26-B-00002819
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002821|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=3, lambda=3.60
§FORM := For A=diag(1,...,3) and lambda=3.60, ||(lambda I-A)^(-1)||_2=1.66666666667.
§DEP_HARD := S26-B-00002820
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002822|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=4, lambda=4.70
§FORM := For A=diag(1,...,4) and lambda=4.70, ||(lambda I-A)^(-1)||_2=1.42857142857.
§DEP_HARD := S26-E-00002821
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002823|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=5, lambda=5.80
§FORM := For A=diag(1,...,5) and lambda=5.80, ||(lambda I-A)^(-1)||_2=1.25.
§DEP_HARD := S26-E-00002822
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002824|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=6, lambda=6.90
§FORM := For A=diag(1,...,6) and lambda=6.90, ||(lambda I-A)^(-1)||_2=1.11111111111.
§DEP_HARD := S26-E-00002823
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002825|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=2, lambda=3.00
§FORM := For A=diag(1,...,2) and lambda=3.00, ||(lambda I-A)^(-1)||_2=1.
§DEP_HARD := S26-E-00002824
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002826|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=3, lambda=4.10
§FORM := For A=diag(1,...,3) and lambda=4.10, ||(lambda I-A)^(-1)||_2=0.909090909091.
§DEP_HARD := S26-E-00002825
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002827|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=4, lambda=5.20
§FORM := For A=diag(1,...,4) and lambda=5.20, ||(lambda I-A)^(-1)||_2=0.833333333333.
§DEP_HARD := S26-E-00002826
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002828|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=5, lambda=6.30
§FORM := For A=diag(1,...,5) and lambda=6.30, ||(lambda I-A)^(-1)||_2=0.769230769231.
§DEP_HARD := S26-E-00002827
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002829|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=6, lambda=7.40
§FORM := For A=diag(1,...,6) and lambda=7.40, ||(lambda I-A)^(-1)||_2=0.714285714286.
§DEP_HARD := S26-E-00002828
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002830|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=2, lambda=3.50
§FORM := For A=diag(1,...,2) and lambda=3.50, ||(lambda I-A)^(-1)||_2=0.666666666667.
§DEP_HARD := S26-E-00002829
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002831|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=3, lambda=4.60
§FORM := For A=diag(1,...,3) and lambda=4.60, ||(lambda I-A)^(-1)||_2=0.625.
§DEP_HARD := S26-E-00002830
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002832|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=4, lambda=5.70
§FORM := For A=diag(1,...,4) and lambda=5.70, ||(lambda I-A)^(-1)||_2=0.588235294118.
§DEP_HARD := S26-E-00002831
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002833|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=5, lambda=6.80
§FORM := For A=diag(1,...,5) and lambda=6.80, ||(lambda I-A)^(-1)||_2=0.555555555556.
§DEP_HARD := S26-E-00002832
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002834|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=6, lambda=7.90
§FORM := For A=diag(1,...,6) and lambda=7.90, ||(lambda I-A)^(-1)||_2=0.526315789474.
§DEP_HARD := S26-E-00002833
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002835|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=2, lambda=4.00
§FORM := For A=diag(1,...,2) and lambda=4.00, ||(lambda I-A)^(-1)||_2=0.5.
§DEP_HARD := S26-E-00002834
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002836|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=3, lambda=5.10
§FORM := For A=diag(1,...,3) and lambda=5.10, ||(lambda I-A)^(-1)||_2=0.47619047619.
§DEP_HARD := S26-E-00002835
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002837|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=4, lambda=6.20
§FORM := For A=diag(1,...,4) and lambda=6.20, ||(lambda I-A)^(-1)||_2=0.454545454545.
§DEP_HARD := S26-E-00002836
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002838|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=5, lambda=7.30
§FORM := For A=diag(1,...,5) and lambda=7.30, ||(lambda I-A)^(-1)||_2=0.434782608696.
§DEP_HARD := S26-E-00002837
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002839|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=6, lambda=8.40
§FORM := For A=diag(1,...,6) and lambda=8.40, ||(lambda I-A)^(-1)||_2=0.416666666667.
§DEP_HARD := S26-E-00002838
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002840|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=2, lambda=4.50
§FORM := For A=diag(1,...,2) and lambda=4.50, ||(lambda I-A)^(-1)||_2=0.4.
§DEP_HARD := S26-E-00002839
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002841|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=3, lambda=5.60
§FORM := For A=diag(1,...,3) and lambda=5.60, ||(lambda I-A)^(-1)||_2=0.384615384615.
§DEP_HARD := S26-E-00002840
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002842|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=4, lambda=6.70
§FORM := For A=diag(1,...,4) and lambda=6.70, ||(lambda I-A)^(-1)||_2=0.37037037037.
§DEP_HARD := S26-E-00002841
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002843|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=5, lambda=7.80
§FORM := For A=diag(1,...,5) and lambda=7.80, ||(lambda I-A)^(-1)||_2=0.357142857143.
§DEP_HARD := S26-E-00002842
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002844|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=6, lambda=8.90
§FORM := For A=diag(1,...,6) and lambda=8.90, ||(lambda I-A)^(-1)||_2=0.344827586207.
§DEP_HARD := S26-E-00002843
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002845|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=2, lambda=5.00
§FORM := For A=diag(1,...,2) and lambda=5.00, ||(lambda I-A)^(-1)||_2=0.333333333333.
§DEP_HARD := S26-E-00002844
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002846|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=3, lambda=6.10
§FORM := For A=diag(1,...,3) and lambda=6.10, ||(lambda I-A)^(-1)||_2=0.322580645161.
§DEP_HARD := S26-E-00002845
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002847|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=4, lambda=7.20
§FORM := For A=diag(1,...,4) and lambda=7.20, ||(lambda I-A)^(-1)||_2=0.3125.
§DEP_HARD := S26-E-00002846
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002848|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=5, lambda=8.30

§FORM := For A=diag(1,...,5) and lambda=8.30, ||(lambda I-A)^(-1)||_2=0.30303030303.
§DEP_HARD := S26-E-00002847
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002849|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=6, lambda=9.40

§FORM := For A=diag(1,...,6) and lambda=9.40, ||(lambda I-A)^(-1)||_2=0.294117647059.
§DEP_HARD := S26-E-00002848
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002850|E|M|Proof+Compile+Meta+Spectral] Diagonal resolvent profile n=2, lambda=5.50

§FORM := For A=diag(1,...,2) and lambda=5.50, ||(lambda I-A)^(-1)||_2=0.285714285714.
§DEP_HARD := S26-E-00002849
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002851|D|A|Proof+Compile+Meta] Distributional support

§FORM := supp(T) is the complement of the largest open set on which the distribution T vanishes.
§DEP_HARD := S26-E-00002850
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002852|D|A|Proof+Compile+Meta] Distribution of finite order

§FORM := T has finite order m on a compact set when its action is bounded by derivatives of test functions through order m with a compact-dependent constant.
§DEP_HARD := S26-D-00002851
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002853|D|A|Proof+Compile+Meta] Convolution of a distribution with a test function

§FORM := For T in D' and phi in C_c^infinity, T*phi is smooth and defined by translating the test function against T.
§DEP_HARD := S26-D-00002852
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002854|T|A|Proof+Compile+Meta] Distributional convolution regularizes

§FORM := Convolution of a distribution with a compactly supported smooth test function produces a smooth function.
§DEP_HARD := S26-D-00002853
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002855|D|A|Proof+Compile+Meta] Compactly supported mollifier interface

§FORM := A mollifier rho_epsilon is a rescaled smooth compactly supported approximate identity.
§DEP_HARD := S26-T-00002854
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002856|T|A|Proof+Compile+Meta+Topo] Mollification convergence for distributions

§FORM := T*rho_epsilon -> T in the distribution topology as epsilon->0 under the standard mollifier construction.
§DEP_HARD := S26-D-00002855
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002857|D|A|Proof+Compile+Meta] Distributional fundamental solution

§FORM := E is a fundamental solution of P(D) when P(D)E=delta in distributions.
§DEP_HARD := S26-T-00002856
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002858|T|A|Proof+Compile+Meta+Spectral] Malgrange-Ehrenpreis theorem

§FORM := Every nonzero constant-coefficient linear differential operator on R^n has a distributional fundamental solution.
§DEP_HARD := S26-D-00002857
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002859|D|A|Proof+Compile+Meta] Singular support

§FORM := sing supp(T) is the complement of the largest open set on which T is represented by a smooth function.
§DEP_HARD := S26-T-00002858
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002860|D|A|Proof+Compile+Meta+Spectral] Wavefront set interface

§FORM := WF(T) refines singular support by recording both singular positions and cotangent directions in which localized Fourier decay fails.
§DEP_HARD := S26-D-00002859
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002861|D|A|Proof+Compile+Meta+Spectral] Microlocal smoothness

§FORM := T is microlocally smooth near (x0,xi0) when a localization near x0 has rapidly decaying Fourier transform in a cone around xi0.
§DEP_HARD := S26-D-00002860
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002862|T|A|Proof+Compile+Meta+Spectral] Wavefront projection to singular support

§FORM := The projection of WF(T) to the base space equals sing supp(T).
§DEP_HARD := S26-D-00002861
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002863|D|A|Proof+Compile+Meta] Pseudodifferential symbol class S^m

§FORM := A symbol a(x,xi) is of order m when derivatives obey polynomial growth bounds in xi with order reduced by xi-derivatives.
§DEP_HARD := S26-T-00002862
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002864|D|A|Proof+Compile+Meta+Spectral] Pseudodifferential operator interface

§FORM := Op(a) acts by oscillatory Fourier synthesis with symbol a(x,xi), with quantization convention declared.
§DEP_HARD := S26-D-00002863
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002865|D|A|Proof+Compile+Meta+Spectral] Principal symbol of a pseudodifferential operator

§FORM := The top-order homogeneous/equivalence class of a symbol determines its principal microlocal behavior.
§DEP_HARD := S26-D-00002864
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002866|D|A|Proof+Compile+Meta] Elliptic symbol

§FORM := A symbol is elliptic when its magnitude has the expected lower bound of its order for large frequencies in the region of interest.
§DEP_HARD := S26-D-00002865
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002867|T|A|Proof+Compile+Meta] Elliptic regularity microlocal interface

§FORM := For elliptic P, singularities of u and Pu coincide microlocally up to the characteristic set; ellipticity prevents new singularities.
§DEP_HARD := S26-D-00002866
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002868|N|A|Proof+Compile+Meta+Spectral] Wavefront-support fence

§FORM := Support records where a distribution is nonzero; singular support records where it is nonsmooth; wavefront set additionally records singular directions. These are different objects.
§DEP_HARD := S26-T-00002867
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002869|B|A|Proof+Compile+Meta+Spectral] Microlocal PDE bridge

§FORM := Wavefront sets and symbols track propagation and removal of singularities beyond what global Sobolev norms can express.
§DEP_HARD := S26-N-00002868
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002870|B|A|Proof+Compile+Meta+Spectral] Microlocal spectral bridge

§FORM := Pseudodifferential parametrices connect local symbol inversion to resolvents, elliptic spectra and heat-kernel asymptotics.
§DEP_HARD := S26-B-00002869
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002871|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.020

§FORM := For the normalized Gaussian mollifier at epsilon=0.020, rho_epsilon(0.3)=2.76535477492e-48. This is a smooth approximate-identity instance, not compactly supported.
§DEP_HARD := S26-B-00002870
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002872|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.040

§FORM := For the normalized Gaussian mollifier at epsilon=0.040, rho_epsilon(0.3)=6.08580133257e-12. This is a smooth approximate-identity instance, not compactly supported.
§DEP_HARD := S26-E-00002871
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002873|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.060

§FORM := For the normalized Gaussian mollifier at epsilon=0.060, rho_epsilon(0.3)=2.47786585789e-05. This is a smooth approximate-identity instance, not compactly supported.
§DEP_HARD := S26-E-00002872
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002874|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.080

§FORM := For the normalized Gaussian mollifier at epsilon=0.080, rho_epsilon(0.3)=0.00440744602959. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002873

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002875|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.100

§FORM := For the normalized Gaussian mollifier at epsilon=0.100, rho_epsilon(0.3)=0.0443184841194. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002874

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002876|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.120

§FORM := For the normalized Gaussian mollifier at epsilon=0.120, rho_epsilon(0.3)=0.14606917078. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002875

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002877|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.140

§FORM := For the normalized Gaussian mollifier at epsilon=0.140, rho_epsilon(0.3)=0.286864860292. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002876

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002878|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.160

§FORM := For the normalized Gaussian mollifier at epsilon=0.160, rho_epsilon(0.3)=0.429914223917. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002877

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002879|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.180

§FORM := For the normalized Gaussian mollifier at epsilon=0.180, rho_epsilon(0.3)=0.552650771071. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002878

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002880|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.200

§FORM := For the normalized Gaussian mollifier at epsilon=0.200, rho_epsilon(0.3)=0.647587978329. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002879

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002881|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.220

§FORM := For the normalized Gaussian mollifier at epsilon=0.220, rho_epsilon(0.3)=0.715650852813. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002880

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002882|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.240

§FORM := For the normalized Gaussian mollifier at epsilon=0.240, rho_epsilon(0.3)=0.761037855788. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002881

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002883|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.260

§FORM := For the normalized Gaussian mollifier at epsilon=0.260, rho_epsilon(0.3)=0.788561122133. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002882

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002884|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.280

§FORM := For the normalized Gaussian mollifier at epsilon=0.280, rho_epsilon(0.3)=0.802556959258. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002883

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002885|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.300

§FORM := For the normalized Gaussian mollifier at epsilon=0.300, rho_epsilon(0.3)=0.80656908173. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002884

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002886|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.320

§FORM := For the normalized Gaussian mollifier at epsilon=0.320, rho_epsilon(0.3)=0.803355960459. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002885

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002887|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.340

§FORM := For the normalized Gaussian mollifier at epsilon=0.340, rho_epsilon(0.3)=0.795008694242. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002886

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002888|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.360

§FORM := For the normalized Gaussian mollifier at epsilon=0.360, rho_epsilon(0.3)=0.783088542806. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002887

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002889|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.380

§FORM := For the normalized Gaussian mollifier at epsilon=0.380, rho_epsilon(0.3)=0.768750464239. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002888

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002890|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.400

§FORM := For the normalized Gaussian mollifier at epsilon=0.400, rho_epsilon(0.3)=0.752843580387. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002889

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002891|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.420

§FORM := For the normalized Gaussian mollifier at epsilon=0.420, rho_epsilon(0.3)=0.735989073379. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002890

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002892|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.440

§FORM := For the normalized Gaussian mollifier at epsilon=0.440, rho_epsilon(0.3)=0.718639097674. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002891

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002893|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.460

§FORM := For the normalized Gaussian mollifier at epsilon=0.460, rho_epsilon(0.3)=0.701120751911. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002892

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002894|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.480

§FORM := For the normalized Gaussian mollifier at epsilon=0.480, rho_epsilon(0.3)=0.68366868448. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002893

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002895|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.500

§FORM := For the normalized Gaussian mollifier at epsilon=0.500, rho_epsilon(0.3)=0.666449205784. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002894

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002896|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.520

§FORM := For the normalized Gaussian mollifier at epsilon=0.520, rho_epsilon(0.3)=0.649578112742. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002895

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002897|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.540

§FORM := For the normalized Gaussian mollifier at epsilon=0.540, rho_epsilon(0.3)=0.633133878085. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002896

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002898|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.560

§FORM := For the normalized Gaussian mollifier at epsilon=0.560, rho_epsilon(0.3)=0.617167426637. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002897

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002899|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.580

§FORM := For the normalized Gaussian mollifier at epsilon=0.580, rho_epsilon(0.3)=0.601709396676. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002898

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002900|E|M|Proof+Compile+Meta] Gaussian mollifier profile epsilon=0.600

§FORM := For the normalized Gaussian mollifier at epsilon=0.600, rho_epsilon(0.3)=0.586775544607. This is a smooth approximate-identity instance, not compactly supported.

§DEP_HARD := S26-E-00002899

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002901|D|A|Proof+Compile+Meta+Spectral] Laplace-type integral

§FORM := A Laplace-type integral has the form I(lambda)=integral e^{-lambda phi(x)}a(x)dx for large positive lambda.

§DEP_HARD := S26-E-00002900

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002902|T|A|Proof+Compile+Meta+Spectral] Laplace method at a nondegenerate minimum

§FORM := If phi has a unique nondegenerate interior minimum x0 and a is smooth, I(lambda) has leading term a(x0)e^{-lambda phi(x0)}sqrt(2pi/(lambda phi''(x0))) in one dimension.

§DEP_HARD := S26-D-00002901

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002903|D|A|Proof+Compile+Meta+Spectral] Watson lemma

§FORM := Watson-type results turn a local expansion of an integrand near an endpoint into an asymptotic expansion of its Laplace transform.

§DEP_HARD := S26-T-00002902

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002904|D|A|Proof+Compile+Meta+Swirl] Oscillatory integral

§FORM := An oscillatory integral has rapidly varying phase e^{i lambda phi(x)} and is interpreted by ordinary or regularized integration depending on decay.

§DEP_HARD := S26-D-00002903

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002905|T|A|Proof+Compile+Meta+Swirl] Nonstationary phase principle

§FORM := If grad phi has no zeros on the support of a smooth compactly supported amplitude, repeated integration by parts gives decay faster than any prescribed power when the data are smooth enough.

§DEP_HARD := S26-D-00002904

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002906|T|A|Proof+Compile+Meta+Swirl] Stationary phase at a nondegenerate critical point

§FORM := A nondegenerate critical point contributes a lambda^{-n/2} leading term with determinant/Hessian phase factor under the standard stationary-phase theorem.

§DEP_HARD := S26-T-00002905

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002907|D|A|Proof+Compile+Meta+Swirl] Steepest descent contour

§FORM := A steepest-descent contour follows directions on which the real part of the complex phase decreases most rapidly away from a saddle.

§DEP_HARD := S26-T-00002906

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002908|D|A|Proof+Compile+Meta+Swirl] Saddle point

§FORM := A saddle point is a critical point of a complex phase used to determine dominant asymptotic contributions after contour deformation.

§DEP_HARD := S26-D-00002907

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002909|D|A|Proof+Compile+Meta] Mellin transform

§FORM := M[f](s)=integral_0^infinity f(x)x^{s-1}dx on its fundamental strip.

§DEP_HARD := S26-D-00002908

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00002910|I|A|Proof+Compile+Meta] Mellin scaling law

§FORM := M[f(ax)](s)=a^{-s}M[f](s) for a>0 where both sides converge.

§DEP_HARD := S26-D-00002909

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002911|D|A|Proof+Compile+Meta] Fundamental strip of a Mellin transform

§FORM := The fundamental strip is the vertical region in which the defining Mellin integral converges absolutely.

§DEP_HARD := S26-I-00002910

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002912|D|A|Proof+Compile+Meta] Mellin inversion interface

§FORM := Under standard regularity/decay assumptions, f is recovered by a vertical-line inverse Mellin integral inside the fundamental strip.

§DEP_HARD := S26-D-00002911

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002913|D|A|Proof+Compile+Meta] Asymptotic matching

§FORM := Matched asymptotics constructs local expansions in distinct regions and matches their common overlap behavior.

§DEP_HARD := S26-D-00002912

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002914|D|A|Proof+Compile+Meta] Boundary layer

§FORM := A boundary layer is a thin region in which a solution varies on a different scale from the outer solution.

§DEP_HARD := S26-D-00002913

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002915|D|A|Proof+Compile+Meta] Multiple-scales expansion

§FORM := The method of multiple scales introduces independent slow/fast variables to remove secular growth in perturbative approximations.

§DEP_HARD := S26-D-00002914

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002916|D|A|Proof+Compile+Meta] Poincare-Lindstedt method

§FORM := Frequency is expanded along with the solution to eliminate secular terms in weakly nonlinear periodic problems.

§DEP_HARD := S26-D-00002915

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002917|D|A|Proof+Compile+Meta] Borel transform interface

§FORM := A formal power series can be mapped to a factorially rescaled Borel transform to study summability beyond ordinary convergence.

§DEP_HARD := S26-D-00002916

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002918|N|A|Proof+Compile+Meta] Asymptotic-series fence

§FORM := An asymptotic expansion can diverge for every fixed nonzero parameter; usefulness comes from truncated remainder behavior, not convergence of the infinite series.

§DEP_HARD := S26-D-00002917

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002919|B|A|Proof+Compile+Meta] Asymptotics to special-functions bridge

§FORM := Airy, Bessel and related functions arise as canonical uniform approximants near turning points and coalescing stationary points.

§DEP_HARD := S26-N-00002918

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002920|B|A|Proof+Compile+Meta] Asymptotics to computation bridge

§FORM := Optimal truncation, exponentially small remainders and uniformity regions must be separated from mere agreement at sampled parameter values.

§DEP_HARD := S26-B-00002919

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002921|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=2

§FORM := Integral_R exp(-lambda x^2)dx at lambda=2 equals sqrt(pi/lambda)=1.25331413732; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-B-00002920

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002922|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=4

§FORM := Integral_R exp(-lambda x^2)dx at lambda=4 equals sqrt(pi/lambda)=0.886226925453; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002921

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002923|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=6

§FORM := Integral_R exp(-lambda x^2)dx at lambda=6 equals sqrt(pi/lambda)=0.723601254558; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002922

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002924|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=8

§FORM := Integral_R exp(-lambda x^2)dx at lambda=8 equals sqrt(pi/lambda)=0.626657068658; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002923

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002925|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=10

§FORM := Integral_R exp(-lambda x^2)dx at lambda=10 equals sqrt(pi/lambda)=0.56049912164; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002924

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002926|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=12

§FORM := Integral_R exp(-lambda x^2)dx at lambda=12 equals sqrt(pi/lambda)=0.511663353973; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002925

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002927|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=14

§FORM := Integral_R exp(-lambda x^2)dx at lambda=14 equals sqrt(pi/lambda)=0.473708217425; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002926

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002928|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=16

§FORM := Integral_R exp(-lambda x^2)dx at lambda=16 equals sqrt(pi/lambda)=0.443113462726; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002927

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002929|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=18

§FORM := Integral_R exp(-lambda x^2)dx at lambda=18 equals sqrt(pi/lambda)=0.417771379105; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002928

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002930|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=20

§FORM := Integral_R exp(-lambda x^2)dx at lambda=20 equals sqrt(pi/lambda)=0.396332729761; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002929

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002931|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=22

§FORM := Integral_R exp(-lambda x^2)dx at lambda=22 equals sqrt(pi/lambda)=0.37788843072; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002930

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002932|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=24

§FORM := Integral_R exp(-lambda x^2)dx at lambda=24 equals sqrt(pi/lambda)=0.361800627279; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002931

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002933|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=26

§FORM := Integral_R exp(-lambda x^2)dx at lambda=26 equals sqrt(pi/lambda)=0.34760679895; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002932

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002934|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=28

§FORM := Integral_R exp(-lambda x^2)dx at lambda=28 equals sqrt(pi/lambda)=0.334962292845; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002933

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002935|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=30

§FORM := Integral_R exp(-lambda x^2)dx at lambda=30 equals sqrt(pi/lambda)=0.323604318759; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002934

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002936|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=32

§FORM := Integral_R exp(-lambda x^2)dx at lambda=32 equals sqrt(pi/lambda)=0.313328534329; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002935

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002937|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=34

§FORM := Integral_R exp(-lambda x^2)dx at lambda=34 equals sqrt(pi/lambda)=0.303973327661; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002936

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002938|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=36

§FORM := Integral_R exp(-lambda x^2)dx at lambda=36 equals sqrt(pi/lambda)=0.295408975151; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002937

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002939|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=38

§FORM := Integral_R exp(-lambda x^2)dx at lambda=38 equals sqrt(pi/lambda)=0.287529982583; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002938

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002940|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=40

§FORM := Integral_R exp(-lambda x^2)dx at lambda=40 equals sqrt(pi/lambda)=0.28024956082; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002939

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002941|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=42

§FORM := Integral_R exp(-lambda x^2)dx at lambda=42 equals sqrt(pi/lambda)=0.273495566848; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002940

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002942|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=44

§FORM := Integral_R exp(-lambda x^2)dx at lambda=44 equals sqrt(pi/lambda)=0.267207471894; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002941

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002943|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=46

§FORM := Integral_R exp(-lambda x^2)dx at lambda=46 equals sqrt(pi/lambda)=0.261334062972; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002942

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002944|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=48

§FORM := Integral_R exp(-lambda x^2)dx at lambda=48 equals sqrt(pi/lambda)=0.255831676987; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002943

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002945|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=50

§FORM := Integral_R exp(-lambda x^2)dx at lambda=50 equals sqrt(pi/lambda)=0.250662827463; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002944

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002946|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=52

§FORM := Integral_R exp(-lambda x^2)dx at lambda=52 equals sqrt(pi/lambda)=0.245795124724; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002945

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002947|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=54

§FORM := Integral_R exp(-lambda x^2)dx at lambda=54 equals sqrt(pi/lambda)=0.241200418186; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002946

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002948|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=56

§FORM := Integral_R exp(-lambda x^2)dx at lambda=56 equals sqrt(pi/lambda)=0.236854108713; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002947

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002949|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=58

§FORM := Integral_R exp(-lambda x^2)dx at lambda=58 equals sqrt(pi/lambda)=0.232734592571; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002948

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002950|E|M|Proof+Compile+Meta+Spectral] Gaussian Laplace profile lambda=60

§FORM := Integral_R exp(-lambda x^2)dx at lambda=60 equals sqrt(pi/lambda)=0.228822808216; the quadratic Laplace leading term is exact in this case.

§DEP_HARD := S26-E-00002949

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00002951|D|A|Proof+Compile+Meta+Spectral] Strongly continuous operator semigroup

§FORM := A family T(t), t>=0, is a C0 semigroup when T(0)=I, T(t+s)=T(t)T(s), and T(t)x->x as t downarrow 0 for every x.

§DEP_HARD := S26-E-00002950

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002952|D|A|Proof+Compile+Meta+Tower] Infinitesimal generator of a C0 semigroup

§FORM := Ax=lim_{t downarrow0}(T(t)x-x)/t on the set of x for which the strong limit exists.

§DEP_HARD := S26-D-00002951

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002953|T|A|Proof+Compile+Meta+Spectral] Generator is closed and densely defined

§FORM := The generator of a C0 semigroup on a Banach space is a closed densely defined operator.

§DEP_HARD := S26-D-00002952

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002954|D|A|Proof+Compile+Meta] Growth bound of a semigroup

§FORM := omega_0 is the infimum of omega for which ||T(t)||<=M e^{omega t} holds for some M and all t>=0.

§DEP_HARD := S26-T-00002953

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002955|T|A|Proof+Compile+Meta+Spectral] Hille-Yosida theorem interface

§FORM := Closed densely defined operators satisfying the Hille-Yosida resolvent bounds are exactly generators of contraction or bounded-growth C0 semigroups in the corresponding form.

§DEP_HARD := S26-D-00002954

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002956|D|A|Proof+Compile+Meta] Analytic semigroup

§FORM := A C0 semigroup is analytic when it extends holomorphically to a sector and satisfies the compatible semigroup law there.

§DEP_HARD := S26-T-00002955

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002957|D|A|Proof+Compile+Meta] Mild solution of abstract Cauchy problem

§FORM := For u'=Au+f, a mild solution satisfies u(t)=T(t)u0+integral_0^t T(t-s)f(s)ds.

§DEP_HARD := S26-D-00002956

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002958|D|A|Proof+Compile+Meta] Classical solution of abstract Cauchy problem

§FORM := A classical solution is differentiable in time, lies in D(A), and satisfies u'=Au+f pointwise in the Banach space.

§DEP_HARD := S26-D-00002957

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00002959|T|A|Proof+Compile+Meta] Variation-of-constants formula

§FORM := The mild-solution formula is the semigroup form of Duhamel variation of constants under the standard integrability hypotheses.

§DEP_HARD := S26-D-00002958

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002960|D|A|Proof+Compile+Meta] Exponential stability
§FORM := A semigroup is exponentially stable when ||T(t)||<=M e^{-omega t} for some M,omega>0.
§DEP_HARD := S26-T-00002959
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002961|D|A|Proof+Compile+Meta+Spectral] Spectral bound
§FORM := s(A)=sup{Re lambda:lambda in sigma(A)}.
§DEP_HARD := S26-D-00002960
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00002962|N|A|Proof+Compile+Meta+Spectral] Growth-spectral bound fence
§FORM := In general Banach spaces the semigroup growth bound need not equal the spectral bound without additional hypotheses.
§DEP_HARD := S26-D-00002961
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002963|D|A|Proof+Compile+Meta+Spectral] Resolvent Laplace representation
§FORM := For Re lambda above the growth bound, R(lambda,A)x=integral_0^infinity e^{-lambda t}T(t)x dt.
§DEP_HARD := S26-N-00002962
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002964|D|A|Proof+Compile+Meta] Evolution family
§FORM := For nonautonomous problems, U(t,s) generalizes a semigroup with U(t,r)U(r,s)=U(t,s).
§DEP_HARD := S26-D-00002963
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002965|D|A|Proof+Compile+Meta] Maximal regularity interface
§FORM := Maximal Lp regularity asks whether f in Lp forces u' and Au to belong to Lp for the abstract evolution equation.
§DEP_HARD := S26-D-00002964
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002966|D|A|Proof+Compile+Meta] Trotter product interface
§FORM := Product formulas approximate a semigroup generated by A+B through alternating short-time factors generated by A and B under appropriate hypotheses.
§DEP_HARD := S26-D-00002965
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002967|D|A|Proof+Compile+Meta+Spectral] Chernoff approximation interface
§FORM := Chernoff-type theorems derive semigroup limits from consistent families of approximate evolution operators.
§DEP_HARD := S26-D-00002966
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00002968|D|A|Proof+Compile+Meta+Spectral] Fractional evolution equation interface
§FORM := Replacing the first time derivative by a fractional derivative leads to solution operators built from sectorial calculus/subordination rather than an ordinary semigroup in time.
§DEP_HARD := S26-D-00002967
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002969|B|A|Proof+Compile+Meta+Spectral] Semigroup to PDE bridge
§FORM := Heat, diffusion and many linear evolution PDE are abstract Cauchy problems generated by differential operators on Banach/Hilbert spaces.
§DEP_HARD := S26-D-00002968
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00002970|B|A|Proof+Compile+Meta+IG] Semigroup to probability bridge
§FORM := Markov semigroups encode transition evolution of stochastic processes; positivity and mass preservation are additional structures beyond the C0 semigroup axioms.
§DEP_HARD := S26-B-00002969
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00002971|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=0.1
§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.904837418036, 0.818730753078, 0.740818220682] at t=0.1; its 2-norm is 0.904837418036.
§DEP_HARD := S26-B-00002970
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002972|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=0.2
§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.818730753078, 0.670320046036, 0.548811636094] at t=0.2; its 2-norm is 0.818730753078.
§DEP_HARD := S26-E-00002971
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002973|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=0.3

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.740818220682, 0.548811636094, 0.406569659741] at t=0.3; its 2-norm is 0.740818220682.

§DEP_HARD := S26-E-00002972

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002974|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=0.4

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.670320046036, 0.449328964117, 0.301194211912] at t=0.4; its 2-norm is 0.670320046036.

§DEP_HARD := S26-E-00002973

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002975|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=0.5

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.606530659713, 0.367879441171, 0.223130160148] at t=0.5; its 2-norm is 0.606530659713.

§DEP_HARD := S26-E-00002974

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002976|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=0.6

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.548811636094, 0.301194211912, 0.165298888222] at t=0.6; its 2-norm is 0.548811636094.

§DEP_HARD := S26-E-00002975

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002977|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=0.7

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.496585303791, 0.246596963942, 0.122456428253] at t=0.7; its 2-norm is 0.496585303791.

§DEP_HARD := S26-E-00002976

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002978|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=0.8

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.449328964117, 0.201896517995, 0.090717953289] at t=0.8; its 2-norm is 0.449328964117.

§DEP_HARD := S26-E-00002977

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002979|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=0.9

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.406569659741, 0.165298888222, 0.06720551274] at t=0.9; its 2-norm is 0.406569659741.

§DEP_HARD := S26-E-00002978

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002980|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.0

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.367879441171, 0.135335283237, 0.049787068368] at t=1.0; its 2-norm is 0.367879441171.

§DEP_HARD := S26-E-00002979

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002981|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.1

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.332871083698, 0.110803158362, 0.036883167401] at t=1.1; its 2-norm is 0.332871083698.

§DEP_HARD := S26-E-00002980

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002982|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.2

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.301194211912, 0.090717953289, 0.027323722447] at t=1.2; its 2-norm is 0.301194211912.

§DEP_HARD := S26-E-00002981

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002983|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.3

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.272531793034, 0.074273578214, 0.020241911446] at t=1.3; its 2-norm is 0.272531793034.

§DEP_HARD := S26-E-00002982

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002984|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.4

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.246596963942, 0.060810062625, 0.01499557682] at t=1.4; its 2-norm is 0.246596963942.

§DEP_HARD := S26-E-00002983

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002985|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.5

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.223130160148, 0.049787068368, 0.011108996538] at t=1.5; its 2-norm is 0.223130160148.

§DEP_HARD := S26-E-00002984

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002986|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.6

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.201896517995, 0.040762203978, 0.008229747049] at t=1.6; its 2-norm is 0.201896517995.

§DEP_HARD := S26-E-00002985

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002987|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.7

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.182683524053, 0.03337326996, 0.006096746566] at t=1.7; its 2-norm is 0.182683524053.

§DEP_HARD := S26-E-00002986

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002988|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.8

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.165298888222, 0.027323722447, 0.004516580943] at t=1.8; its 2-norm is 0.165298888222.

§DEP_HARD := S26-E-00002987

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002989|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=1.9

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.149568619223, 0.022370771856, 0.003345965457] at t=1.9; its 2-norm is 0.149568619223.

§DEP_HARD := S26-E-00002988

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002990|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.0

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.135335283237, 0.018315638889, 0.002478752177] at t=2.0; its 2-norm is 0.135335283237.

§DEP_HARD := S26-E-00002989

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002991|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.1

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.122456428253, 0.01499557682, 0.001836304777] at t=2.1; its 2-norm is 0.122456428253.

§DEP_HARD := S26-E-00002990

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002992|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.2

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.110803158362, 0.012277339903, 0.001360368038] at t=2.2; its 2-norm is 0.110803158362.

§DEP_HARD := S26-E-00002991

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002993|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.3

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.100258843723, 0.010051835745, 0.001007785429] at t=2.3; its 2-norm is 0.100258843723.

§DEP_HARD := S26-E-00002992

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002994|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.4

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.090717953289, 0.008229747049, 0.000746585808] at t=2.4; its 2-norm is 0.0907179532894.

§DEP_HARD := S26-E-00002993

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002995|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.5

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.082084998624, 0.006737946999, 0.00055308437] at t=2.5; its 2-norm is 0.0820849986239.

§DEP_HARD := S26-E-00002994

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002996|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.6

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.074273578214, 0.005516564421, 0.000409734979] at t=2.6; its 2-norm is 0.0742735782143.

§DEP_HARD := S26-E-00002995

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002997|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.7

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.06720551274, 0.004516580943, 0.000303539138] at t=2.7; its 2-norm is 0.0672055127397.

§DEP_HARD := S26-E-00002996

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002998|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.8

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.060810062625, 0.003697863716, 0.000224867324] at t=2.8; its 2-norm is 0.0608100626252.

§DEP_HARD := S26-E-00002997

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00002999|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=2.9

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.055023220056, 0.003027554745, 0.000166585811] at t=2.9; its 2-norm is 0.0550232200564.

§DEP_HARD := S26-E-00002998

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003000|E|M|Proof+Compile+Meta] Contractive diagonal semigroup profile t=3.0

§FORM := For A=diag(-1,-2,-3), e^(tA) has diagonal [0.049787068368, 0.002478752177, 0.000123409804] at t=3.0; its 2-norm is 0.0497870683679.

§DEP_HARD := S26-E-00002999

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§PACK Synopsis_2026_Volume_III_sLANG_S1

volume := III · Geometry & Topology
canonical_range := 3001-4000

§CLAIM[S26-D-00003001|D|A|Proof+Compile+Meta+Topo] Topology generated by a subbasis
§FORM := A subbasis S on X generates the smallest topology containing S; finite intersections of subbasic sets form a basis.
§DEP_HARD := S26-D-00000901
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003002|D|A|Proof+Compile+Meta+Topo] Initial topology of a family of maps
§FORM := The initial topology on X induced by maps f_i:X->Y_i is the coarsest topology making every f_i continuous.
§DEP_HARD := S26-D-00003001
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003003|D|A|Proof+Compile+Meta+Topo] Final topology of a family of maps
§FORM := The final topology on Y induced by maps g_i:X_i->Y is the finest topology making every g_i continuous.
§DEP_HARD := S26-D-00003002
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003004|D|A|Proof+Compile+Meta+Topo] Nets in a topological space
§FORM := A net is a function from a directed set into a space and generalizes sequences for detecting topology.
§DEP_HARD := S26-D-00003003
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003005|T|A|Proof+Compile+Meta+Topo] Net characterization of closure
§FORM := x lies in the closure of A iff some net in A converges to x.
§DEP_HARD := S26-D-00003004
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003006|T|A|Proof+Compile+Meta+Topo] Net characterization of continuity
§FORM := A map is continuous iff it preserves convergence of all nets.
§DEP_HARD := S26-T-00003005
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003007|D|A|Proof+Compile+Meta+Topo] Filters and filter convergence
§FORM := A filter is an upward-closed finite-intersection family of subsets; it converges to x when every neighborhood of x belongs to the filter.
§DEP_HARD := S26-T-00003006
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003008|D|A|Proof+Compile+Meta+Topo] Ultrafilter
§FORM := An ultrafilter is a maximal proper filter; equivalently, for every subset A, exactly one of A or its complement belongs to it.
§DEP_HARD := S26-D-00003007
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003009|T|A|Proof+Compile+Meta+Topo] Compactness by ultrafilters
§FORM := A topological space is compact iff every ultrafilter on it converges.
§DEP_HARD := S26-D-00003008
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003010|D|A|Proof+Compile+Meta+Topo] First-countable space
§FORM := A space is first countable when each point has a countable neighborhood basis.
§DEP_HARD := S26-T-00003009
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003011|D|A|Proof+Compile+Meta+Topo] Second-countable space
§FORM := A space is second countable when its topology has a countable basis.
§DEP_HARD := S26-D-00003010
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003012|T|A|Proof+Compile+Meta+Topo] Second countable implies separable
§FORM := Every second-countable space is separable.
§DEP_HARD := S26-D-00003011
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003013|T|A|Proof+Compile+Meta+Topo] Second countable implies Lindelof
§FORM := Every second-countable space is Lindelof.
§DEP_HARD := S26-T-00003012
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003014|D|A|Proof+Compile+Meta+Topo] Regular topological space

§FORM := A T1 space is regular when points and disjoint closed sets can be separated by neighborhoods.
§DEP_HARD := S26-T-00003013
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003015|D|A|Proof+Compile+Meta+Topo] Normal topological space

§FORM := A T1 space is normal when disjoint closed sets have disjoint neighborhoods.
§DEP_HARD := S26-D-00003014
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003016|T|A|Proof+Compile+Meta+Topo] Urysohn lemma

§FORM := In a normal space, disjoint closed sets can be separated by a continuous map to [0,1].
§DEP_HARD := S26-D-00003015
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003017|T|A|Proof+Compile+Meta+Topo] Tietze extension theorem

§FORM := A continuous real-valued function on a closed subset of a normal space extends continuously to the whole space.
§DEP_HARD := S26-T-00003016
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003018|N|A|Proof+Compile+Meta+Topo] Sequence-detection fence

§FORM := Sequential tests characterize closure and continuity only in suitable spaces; arbitrary topological spaces may require nets or filters.
§DEP_HARD := S26-T-00003017
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := There are nonsequential spaces where sequential closure is strictly smaller than closure.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003019|B|A|Proof+Compile+Meta+Topo+IG] Topology to analysis bridge

§FORM := Initial/final topologies, nets and compactness principles supply the language behind weak topologies, convergence of measures, and function-space constructions.
§DEP_HARD := S26-N-00003018
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003020|B|A|Proof+Compile+Meta+Topo] Topology to geometry bridge

§FORM := Countability and separation axioms are structural hypotheses used in manifold definitions, partitions of unity, embeddings and metrization.
§DEP_HARD := S26-B-00003019
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003021|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=3

§FORM := On a labeled 3-point set, the indiscrete topology has 2 open sets and the discrete topology has 8 open sets; these are exact finite counts.
§DEP_HARD := S26-B-00003020
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003022|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=4

§FORM := On a labeled 4-point set, the indiscrete topology has 2 open sets and the discrete topology has 16 open sets; these are exact finite counts.
§DEP_HARD := S26-E-00003021
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003023|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=5

§FORM := On a labeled 5-point set, the indiscrete topology has 2 open sets and the discrete topology has 32 open sets; these are exact finite counts.
§DEP_HARD := S26-E-00003022
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003024|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=6

§FORM := On a labeled 6-point set, the indiscrete topology has 2 open sets and the discrete topology has 64 open sets; these are exact finite counts.
§DEP_HARD := S26-E-00003023
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003025|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=7

§FORM := On a labeled 7-point set, the indiscrete topology has 2 open sets and the discrete topology has 128 open sets; these are exact finite counts.
§DEP_HARD := S26-E-00003024
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003026|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=8

§FORM := On a labeled 8-point set, the indiscrete topology has 2 open sets and the discrete topology has 256 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003025

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003027|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=9

§FORM := On a labeled 9-point set, the indiscrete topology has 2 open sets and the discrete topology has 512 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003026

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003028|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=10

§FORM := On a labeled 10-point set, the indiscrete topology has 2 open sets and the discrete topology has 1024 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003027

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003029|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=11

§FORM := On a labeled 11-point set, the indiscrete topology has 2 open sets and the discrete topology has 2048 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003028

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003030|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=12

§FORM := On a labeled 12-point set, the indiscrete topology has 2 open sets and the discrete topology has 4096 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003029

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003031|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=13

§FORM := On a labeled 13-point set, the indiscrete topology has 2 open sets and the discrete topology has 8192 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003030

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003032|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=14

§FORM := On a labeled 14-point set, the indiscrete topology has 2 open sets and the discrete topology has 16384 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003031

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003033|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=15

§FORM := On a labeled 15-point set, the indiscrete topology has 2 open sets and the discrete topology has 32768 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003032

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003034|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=16

§FORM := On a labeled 16-point set, the indiscrete topology has 2 open sets and the discrete topology has 65536 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003033

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003035|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=17

§FORM := On a labeled 17-point set, the indiscrete topology has 2 open sets and the discrete topology has 131072 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003034

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003036|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=18

§FORM := On a labeled 18-point set, the indiscrete topology has 2 open sets and the discrete topology has 262144 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003035

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003037|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=19

§FORM := On a labeled 19-point set, the indiscrete topology has 2 open sets and the discrete topology has 524288 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003036

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003038|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=20

§FORM := On a labeled 20-point set, the indiscrete topology has 2 open sets and the discrete topology has 1048576 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003037

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003039|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=21

§FORM := On a labeled 21-point set, the indiscrete topology has 2 open sets and the discrete topology has 2097152 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003038

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003040|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=22

§FORM := On a labeled 22-point set, the indiscrete topology has 2 open sets and the discrete topology has 4194304 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003039

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003041|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=23

§FORM := On a labeled 23-point set, the indiscrete topology has 2 open sets and the discrete topology has 8388608 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003040

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003042|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=24

§FORM := On a labeled 24-point set, the indiscrete topology has 2 open sets and the discrete topology has 16777216 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003041

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003043|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=25

§FORM := On a labeled 25-point set, the indiscrete topology has 2 open sets and the discrete topology has 33554432 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003042

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003044|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=26

§FORM := On a labeled 26-point set, the indiscrete topology has 2 open sets and the discrete topology has 67108864 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003043

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003045|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=27

§FORM := On a labeled 27-point set, the indiscrete topology has 2 open sets and the discrete topology has 134217728 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003044

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003046|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=28

§FORM := On a labeled 28-point set, the indiscrete topology has 2 open sets and the discrete topology has 268435456 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003045

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003047|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=29

§FORM := On a labeled 29-point set, the indiscrete topology has 2 open sets and the discrete topology has 536870912 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003046

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003048|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=30

§FORM := On a labeled 30-point set, the indiscrete topology has 2 open sets and the discrete topology has 1073741824 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003047

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003049|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=31

§FORM := On a labeled 31-point set, the indiscrete topology has 2 open sets and the discrete topology has 2147483648 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003048

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003050|E|M|Proof+Compile+Meta+Topo] Finite topology cardinality bounds n=32

§FORM := On a labeled 32-point set, the indiscrete topology has 2 open sets and the discrete topology has 4294967296 open sets; these are exact finite counts.

§DEP_HARD := S26-E-00003049

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003051|D|A|Proof+Compile+Meta+Topo] Product topology on an arbitrary family

§FORM := The product topology on product_i X_i is generated by inverse images of open sets under coordinate projections, with finitely many constrained coordinates in a basic open set.

§DEP_HARD := S26-E-00003050

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003052|T|A|Proof+Compile+Meta+Topo] Universal property of product topology

§FORM := A map f:Y->product_i X_i is continuous iff every coordinate projection pi_i∘f is continuous.

§DEP_HARD := S26-D-00003051

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003053|T|A|Proof+Compile+Meta+Topo] Tychonoff theorem

§FORM := In the classical ZFC reading used for this theorem, an arbitrary product of compact spaces is compact in the product topology.

§ASSUME := Classical ZFC/choice foundation is assumed for the arbitrary-product theorem.

§DEP_HARD := S26-T-00003052

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003054|D|A|Proof+Compile+Meta+Topo] Box topology

§FORM := The box topology permits arbitrary products of open coordinate sets and is generally finer than the product topology.

§DEP_HARD := S26-T-00003053

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003055|N|A|Proof+Compile+Meta+Topo] Product-box fence

§FORM := For infinitely many factors the product and box topologies generally differ; compactness results for product topology do not transfer automatically to box topology.

§DEP_HARD := S26-D-00003054

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003056|D|A|Proof+Compile+Meta+Topo] Quotient topology

§FORM := For a surjection q:X->Y, the quotient topology on Y declares U open exactly when q^{-1}(U) is open in X.

§DEP_HARD := S26-N-00003055

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003057|T|A|Proof+Compile+Meta+Topo] Universal property of quotient topology

§FORM := A map f:Y->Z from a quotient space is continuous iff f∘q is continuous, provided f is well-defined on equivalence classes.

§DEP_HARD := S26-D-00003056

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003058|D|A|Proof+Compile+Meta+Topo] Identification map

§FORM := A quotient map is a surjective continuous map q for which U⊂Y is open iff q^{-1}(U) is open.

§DEP_HARD := S26-T-00003057

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003059|D|A|Proof+Compile+Meta+Topo] Topological sum

§FORM := The disjoint union topology is the final topology for the inclusions of the summands.

§DEP_HARD := S26-D-00003058

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003060|D|A|Proof+Compile+Meta+Topo] One-point compactification

§FORM := A noncompact locally compact Hausdorff space admits a compactification by adjoining one point whose neighborhoods complement compact subsets.

§DEP_HARD := S26-D-00003059

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003061|T|A|Proof+Compile+Meta+Topo] Continuous image of compact is compact

§FORM := A continuous image of a compact space is compact.

§DEP_HARD := S26-D-00003060

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003062|T|A|Proof+Compile+Meta+Topo] Compact subsets of Hausdorff spaces are closed

§FORM := Every compact subset of a Hausdorff space is closed.

§DEP_HARD := S26-T-00003061

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003063|D|A|Proof+Compile+Meta+Topo] Local compactness
§FORM := A space is locally compact when each point has a neighborhood base with compact closure in the chosen standard formulation.
§DEP_HARD := S26-T-00003062
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003064|D|A|Proof+Compile+Meta+Topo] Path component
§FORM := The path component of x is the set of points joinable to x by a continuous path.
§DEP_HARD := S26-D-00003063
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003065|D|A|Proof+Compile+Meta+Topo] Connected component
§FORM := A connected component is a maximal connected subset.
§DEP_HARD := S26-D-00003064
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003066|T|A|Proof+Compile+Meta+Topo] Path connected implies connected
§FORM := Every path-connected space is connected.
§DEP_HARD := S26-D-00003065
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003067|N|A|Proof+Compile+Meta+Topo] Connected-path-connected fence
§FORM := Connectedness does not imply path connectedness without extra hypotheses.
§DEP_HARD := S26-T-00003066
§EVIDENCE := A · Exact registered statement only.
§BOUNDARY := The topologist sine curve provides a standard connected but non-path-connected example.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003068|D|A|Proof+Compile+Meta+Topo] Local connectedness
§FORM := A space is locally connected when every point has a neighborhood basis of connected open sets.
§DEP_HARD := S26-N-00003067
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003069|D|A|Proof+Compile+Meta+Topo] Local path connectedness
§FORM := A space is locally path connected when every point has a neighborhood basis of path-connected open sets.
§DEP_HARD := S26-D-00003068
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003070|B|A|Proof+Compile+Meta+Topo] Quotient-gluing bridge
§FORM := Quotient topology formalizes gluing constructions used for CW complexes, manifolds, orbit spaces, projective spaces and identification spaces.
§DEP_HARD := S26-D-00003069
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003071|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=3
§FORM := The disjoint union of 3 closed intervals has exactly 3 connected components and Euler characteristic 3; this is an exact finite CW count.
§DEP_HARD := S26-B-00003070
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003072|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=4
§FORM := The disjoint union of 4 closed intervals has exactly 4 connected components and Euler characteristic 4; this is an exact finite CW count.
§DEP_HARD := S26-E-00003071
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003073|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=5
§FORM := The disjoint union of 5 closed intervals has exactly 5 connected components and Euler characteristic 5; this is an exact finite CW count.
§DEP_HARD := S26-E-00003072
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003074|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=6
§FORM := The disjoint union of 6 closed intervals has exactly 6 connected components and Euler characteristic 6; this is an exact finite CW count.
§DEP_HARD := S26-E-00003073
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003075|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=7
§FORM := The disjoint union of 7 closed intervals has exactly 7 connected components and Euler characteristic 7; this is an exact finite CW count.
§DEP_HARD := S26-E-00003074
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003076|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=8

§FORM := The disjoint union of 8 closed intervals has exactly 8 connected components and Euler characteristic 8; this is an exact finite CW count.

§DEP_HARD := S26-E-00003075

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003077|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=9

§FORM := The disjoint union of 9 closed intervals has exactly 9 connected components and Euler characteristic 9; this is an exact finite CW count.

§DEP_HARD := S26-E-00003076

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003078|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=10

§FORM := The disjoint union of 10 closed intervals has exactly 10 connected components and Euler characteristic 10; this is an exact finite CW count.

§DEP_HARD := S26-E-00003077

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003079|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=11

§FORM := The disjoint union of 11 closed intervals has exactly 11 connected components and Euler characteristic 11; this is an exact finite CW count.

§DEP_HARD := S26-E-00003078

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003080|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=12

§FORM := The disjoint union of 12 closed intervals has exactly 12 connected components and Euler characteristic 12; this is an exact finite CW count.

§DEP_HARD := S26-E-00003079

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003081|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=13

§FORM := The disjoint union of 13 closed intervals has exactly 13 connected components and Euler characteristic 13; this is an exact finite CW count.

§DEP_HARD := S26-E-00003080

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003082|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=14

§FORM := The disjoint union of 14 closed intervals has exactly 14 connected components and Euler characteristic 14; this is an exact finite CW count.

§DEP_HARD := S26-E-00003081

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003083|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=15

§FORM := The disjoint union of 15 closed intervals has exactly 15 connected components and Euler characteristic 15; this is an exact finite CW count.

§DEP_HARD := S26-E-00003082

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003084|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=16

§FORM := The disjoint union of 16 closed intervals has exactly 16 connected components and Euler characteristic 16; this is an exact finite CW count.

§DEP_HARD := S26-E-00003083

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003085|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=17

§FORM := The disjoint union of 17 closed intervals has exactly 17 connected components and Euler characteristic 17; this is an exact finite CW count.

§DEP_HARD := S26-E-00003084

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003086|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=18

§FORM := The disjoint union of 18 closed intervals has exactly 18 connected components and Euler characteristic 18; this is an exact finite CW count.

§DEP_HARD := S26-E-00003085

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003087|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=19

§FORM := The disjoint union of 19 closed intervals has exactly 19 connected components and Euler characteristic 19; this is an exact finite CW count.

§DEP_HARD := S26-E-00003086

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003088|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=20

§FORM := The disjoint union of 20 closed intervals has exactly 20 connected components and Euler characteristic 20; this is an exact finite CW count.

§DEP_HARD := S26-E-00003087

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003089|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=21

§FORM := The disjoint union of 21 closed intervals has exactly 21 connected components and Euler characteristic 21; this is an exact finite CW count.

§DEP_HARD := S26-E-00003088

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003090|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=22

§FORM := The disjoint union of 22 closed intervals has exactly 22 connected components and Euler characteristic 22; this is an exact finite CW count.

§DEP_HARD := S26-E-00003089

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003091|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=23

§FORM := The disjoint union of 23 closed intervals has exactly 23 connected components and Euler characteristic 23; this is an exact finite CW count.

§DEP_HARD := S26-E-00003090

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003092|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=24

§FORM := The disjoint union of 24 closed intervals has exactly 24 connected components and Euler characteristic 24; this is an exact finite CW count.

§DEP_HARD := S26-E-00003091

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003093|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=25

§FORM := The disjoint union of 25 closed intervals has exactly 25 connected components and Euler characteristic 25; this is an exact finite CW count.

§DEP_HARD := S26-E-00003092

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003094|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=26

§FORM := The disjoint union of 26 closed intervals has exactly 26 connected components and Euler characteristic 26; this is an exact finite CW count.

§DEP_HARD := S26-E-00003093

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003095|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=27

§FORM := The disjoint union of 27 closed intervals has exactly 27 connected components and Euler characteristic 27; this is an exact finite CW count.

§DEP_HARD := S26-E-00003094

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003096|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=28

§FORM := The disjoint union of 28 closed intervals has exactly 28 connected components and Euler characteristic 28; this is an exact finite CW count.

§DEP_HARD := S26-E-00003095

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003097|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=29

§FORM := The disjoint union of 29 closed intervals has exactly 29 connected components and Euler characteristic 29; this is an exact finite CW count.

§DEP_HARD := S26-E-00003096

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003098|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=30

§FORM := The disjoint union of 30 closed intervals has exactly 30 connected components and Euler characteristic 30; this is an exact finite CW count.

§DEP_HARD := S26-E-00003097

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003099|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=31

§FORM := The disjoint union of 31 closed intervals has exactly 31 connected components and Euler characteristic 31; this is an exact finite CW count.

§DEP_HARD := S26-E-00003098

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003100|E|M|Proof+Compile+Meta+Topo] Disjoint-union component profile n=32

§FORM := The disjoint union of 32 closed intervals has exactly 32 connected components and Euler characteristic 32; this is an exact finite CW count.

§DEP_HARD := S26-E-00003099

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003101|D|A|Proof+Compile+Meta+Topo] Homotopy of maps

§FORM := Maps f,g:X->Y are homotopic when there is a continuous H:X×[0,1]->Y with H(x,0)=f(x) and H(x,1)=g(x).

§DEP_HARD := S26-E-00003100

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003102|D|A|Proof+Compile+Meta+Topo] Homotopy equivalence

§FORM := Spaces X and Y are homotopy equivalent when maps f:X->Y and g:Y->X exist with g∘f homotopic to id_X and f∘g homotopic to id_Y.

§DEP_HARD := S26-D-00003101

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003103|D|A|Proof+Compile+Meta+Topo] Contractible space

§FORM := A space is contractible when its identity map is homotopic to a constant map.

§DEP_HARD := S26-D-00003102

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003104|D|A|Proof+Compile+Meta+Topo] Based loop

§FORM := A based loop at x0 is a path gamma:[0,1]->X with gamma(0)=gamma(1)=x0.

§DEP_HARD := S26-D-00003103

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003105|D|A|Proof+Compile+Meta+Topo] Path homotopy relative endpoints

§FORM := Two paths with the same endpoints are homotopic rel endpoints when a homotopy keeps both endpoints fixed throughout.

§DEP_HARD := S26-D-00003104

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003106|D|A|Proof+Compile+Meta+Topo] Fundamental group

§FORM := pi_1(X,x0) is the group of based loops modulo path homotopy, with multiplication induced by concatenation.

§DEP_HARD := S26-D-00003105

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003107|T|A|Proof+Compile+Meta+Topo] Homotopy invariance of fundamental group

§FORM := A based homotopy equivalence induces an isomorphism of fundamental groups, with the usual basepoint bookkeeping.

§DEP_HARD := S26-D-00003106

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003108|T|A|Proof+Compile+Meta+Topo] Fundamental group of the circle

§FORM := pi_1(S^1) is isomorphic to Z.

§DEP_HARD := S26-T-00003107

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003109|T|A|Proof+Compile+Meta+Topo] Fundamental group of a product

§FORM := For path-connected based spaces under standard hypotheses, pi_1(X×Y)≅pi_1(X)×pi_1(Y).

§DEP_HARD := S26-T-00003108

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003110|D|A|Proof+Compile+Meta+Topo] Simply connected space

§FORM := A path-connected space is simply connected when its fundamental group is trivial.

§DEP_HARD := S26-T-00003109

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003111|T|A|Proof+Compile+Meta+Topo] Seifert-van Kampen theorem

§FORM := Under the standard path-connected open-cover hypotheses, pi_1(U∪V) is the pushout of pi_1(U) and pi_1(V) over pi_1(U∩V).

§DEP_HARD := S26-D-00003110

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003112|D|A|Proof+Compile+Meta+Topo] Deformation retract

§FORM := A subspace A⊂X is a deformation retract when X retracts onto A through a homotopy fixed on A.

§DEP_HARD := S26-T-00003111

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003113|T|A|Proof+Compile+Meta+Topo] Deformation retract preserves homotopy type

§FORM := If A is a deformation retract of X, inclusion A->X is a homotopy equivalence.
§DEP_HARD := S26-D-00003112
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003114|D|A|Proof+Compile+Meta+Topo] Homotopy group pi_n

§FORM := For n>=1, pi_n(X,x0) is the set of based homotopy classes of maps S^n->X, with a natural group structure.
§DEP_HARD := S26-T-00003113
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003115|T|A|Proof+Compile+Meta+Topo] Higher homotopy groups are abelian

§FORM := For n>=2, pi_n(X,x0) is abelian.
§DEP_HARD := S26-D-00003114
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003116|D|A|Proof+Compile+Meta+Topo] Weak homotopy equivalence

§FORM := A map is a weak homotopy equivalence when it induces a bijection on path components and isomorphisms on all homotopy groups at compatible basepoints.
§DEP_HARD := S26-T-00003115
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003117|N|A|Proof+Compile+Meta+Topo] Weak-equivalence fence

§FORM := A weak homotopy equivalence need not be a homotopy equivalence for arbitrary spaces; CW-type hypotheses are load-bearing in Whitehead-style conclusions.
§DEP_HARD := S26-D-00003116
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003118|T|A|Proof+Compile+Meta+Topo] Whitehead theorem for CW complexes

§FORM := A weak homotopy equivalence between connected CW complexes is a homotopy equivalence.
§DEP_HARD := S26-N-00003117
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003119|B|A|Proof+Compile+Meta+Topo] Fundamental-group to covering bridge

§FORM := The fundamental group classifies connected covering spaces under the standard local hypotheses through subgroup data.
§DEP_HARD := S26-T-00003118
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003120|B|A|Proof+Compile+Meta+Topo+IG] Homotopy to algebra bridge

§FORM := Homotopy groups convert deformation information into algebraic invariants but generally do not classify spaces by themselves.
§DEP_HARD := S26-B-00003119
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003121|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=2

§FORM := The wedge of 2 circles is a finite graph with fundamental group the free group F_2; its first homology is Z^2. This is a finite CW identification.
§DEP_HARD := S26-B-00003120
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003122|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=3

§FORM := The wedge of 3 circles is a finite graph with fundamental group the free group F_3; its first homology is Z^3. This is a finite CW identification.
§DEP_HARD := S26-E-00003121
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003123|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=4

§FORM := The wedge of 4 circles is a finite graph with fundamental group the free group F_4; its first homology is Z^4. This is a finite CW identification.
§DEP_HARD := S26-E-00003122
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003124|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=5

§FORM := The wedge of 5 circles is a finite graph with fundamental group the free group F_5; its first homology is Z^5. This is a finite CW identification.
§DEP_HARD := S26-E-00003123
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003125|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=6

§FORM := The wedge of 6 circles is a finite graph with fundamental group the free group F_6; its first homology is Z^6. This is a finite CW identification.
§DEP_HARD := S26-E-00003124
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003126|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=7

§FORM := The wedge of 7 circles is a finite graph with fundamental group the free group F_7; its first homology is Z^7. This is a finite CW identification.

§DEP_HARD := S26-E-00003125

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003127|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=8

§FORM := The wedge of 8 circles is a finite graph with fundamental group the free group F_8; its first homology is Z^8. This is a finite CW identification.

§DEP_HARD := S26-E-00003126

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003128|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=9

§FORM := The wedge of 9 circles is a finite graph with fundamental group the free group F_9; its first homology is Z^9. This is a finite CW identification.

§DEP_HARD := S26-E-00003127

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003129|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=10

§FORM := The wedge of 10 circles is a finite graph with fundamental group the free group F_10; its first homology is Z^10. This is a finite CW identification.

§DEP_HARD := S26-E-00003128

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003130|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=11

§FORM := The wedge of 11 circles is a finite graph with fundamental group the free group F_11; its first homology is Z^11. This is a finite CW identification.

§DEP_HARD := S26-E-00003129

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003131|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=12

§FORM := The wedge of 12 circles is a finite graph with fundamental group the free group F_12; its first homology is Z^12. This is a finite CW identification.

§DEP_HARD := S26-E-00003130

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003132|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=13

§FORM := The wedge of 13 circles is a finite graph with fundamental group the free group F_13; its first homology is Z^13. This is a finite CW identification.

§DEP_HARD := S26-E-00003131

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003133|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=14

§FORM := The wedge of 14 circles is a finite graph with fundamental group the free group F_14; its first homology is Z^14. This is a finite CW identification.

§DEP_HARD := S26-E-00003132

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003134|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=15

§FORM := The wedge of 15 circles is a finite graph with fundamental group the free group F_15; its first homology is Z^15. This is a finite CW identification.

§DEP_HARD := S26-E-00003133

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003135|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=16

§FORM := The wedge of 16 circles is a finite graph with fundamental group the free group F_16; its first homology is Z^16. This is a finite CW identification.

§DEP_HARD := S26-E-00003134

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003136|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=17

§FORM := The wedge of 17 circles is a finite graph with fundamental group the free group F_17; its first homology is Z^17. This is a finite CW identification.

§DEP_HARD := S26-E-00003135

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003137|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=18

§FORM := The wedge of 18 circles is a finite graph with fundamental group the free group F_18; its first homology is Z^18. This is a finite CW identification.

§DEP_HARD := S26-E-00003136

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003138|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=19

§FORM := The wedge of 19 circles is a finite graph with fundamental group the free group F_19; its first homology is Z^19. This is a finite CW identification.

§DEP_HARD := S26-E-00003137

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003139|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=20

§FORM := The wedge of 20 circles is a finite graph with fundamental group the free group F_20; its first homology is Z^20. This is a finite CW identification.

§DEP_HARD := S26-E-00003138

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003140|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=21

§FORM := The wedge of 21 circles is a finite graph with fundamental group the free group F_21; its first homology is Z^21. This is a finite CW identification.

§DEP_HARD := S26-E-00003139

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003141|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=22

§FORM := The wedge of 22 circles is a finite graph with fundamental group the free group F_22; its first homology is Z^22. This is a finite CW identification.

§DEP_HARD := S26-E-00003140

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003142|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=23

§FORM := The wedge of 23 circles is a finite graph with fundamental group the free group F_23; its first homology is Z^23. This is a finite CW identification.

§DEP_HARD := S26-E-00003141

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003143|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=24

§FORM := The wedge of 24 circles is a finite graph with fundamental group the free group F_24; its first homology is Z^24. This is a finite CW identification.

§DEP_HARD := S26-E-00003142

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003144|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=25

§FORM := The wedge of 25 circles is a finite graph with fundamental group the free group F_25; its first homology is Z^25. This is a finite CW identification.

§DEP_HARD := S26-E-00003143

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003145|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=26

§FORM := The wedge of 26 circles is a finite graph with fundamental group the free group F_26; its first homology is Z^26. This is a finite CW identification.

§DEP_HARD := S26-E-00003144

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003146|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=27

§FORM := The wedge of 27 circles is a finite graph with fundamental group the free group F_27; its first homology is Z^27. This is a finite CW identification.

§DEP_HARD := S26-E-00003145

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003147|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=28

§FORM := The wedge of 28 circles is a finite graph with fundamental group the free group F_28; its first homology is Z^28. This is a finite CW identification.

§DEP_HARD := S26-E-00003146

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003148|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=29

§FORM := The wedge of 29 circles is a finite graph with fundamental group the free group F_29; its first homology is Z^29. This is a finite CW identification.

§DEP_HARD := S26-E-00003147

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003149|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=30

§FORM := The wedge of 30 circles is a finite graph with fundamental group the free group F_30; its first homology is Z^30. This is a finite CW identification.

§DEP_HARD := S26-E-00003148

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003150|E|M|Proof+Compile+Meta+Topo] Rose graph homotopy profile rank=31

§FORM := The wedge of 31 circles is a finite graph with fundamental group the free group F_31; its first homology is Z^31. This is a finite CW identification.

§DEP_HARD := S26-E-00003149

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003151|D|A|Proof+Compile+Meta+Topo] Covering map

§FORM := A map p:E->B is a covering map when every b in B has an evenly covered neighborhood U whose preimage is a disjoint union of open sets mapped homeomorphically onto U.

§DEP_HARD := S26-E-00003150

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003152|D|A|Proof+Compile+Meta+Topo] Sheet number of a covering

§FORM := For a connected base, the cardinality of p^{-1}(b) is constant when the covering has constant finite or infinite sheet number.

§DEP_HARD := S26-D-00003151

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003153|T|A|Proof+Compile+Meta+Topo] Path lifting for covering maps

§FORM := Given a path in B and a chosen initial lift in E, there exists a unique lifted path.

§DEP_HARD := S26-D-00003152

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003154|T|A|Proof+Compile+Meta+Topo] Homotopy lifting for covering maps

§FORM := A homotopy in the base with a chosen initial lift admits a unique lifted homotopy under the covering hypotheses.

§DEP_HARD := S26-T-00003153

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003155|D|A|Proof+Compile+Meta+Topo] Deck transformation

§FORM := A deck transformation of p:E->B is a homeomorphism h:E->E satisfying p∘h=p.

§DEP_HARD := S26-T-00003154

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003156|D|A|Proof+Compile+Meta+Topo] Universal covering space

§FORM := A simply connected covering E->B is a universal cover when it dominates connected covers in the standard category.

§DEP_HARD := S26-D-00003155

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003157|T|A|Proof+Compile+Meta+Topo] Existence of universal covers

§FORM := A path-connected, locally path-connected, semilocally simply connected space has a universal cover.

§DEP_HARD := S26-D-00003156

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003158|D|A|Proof+Compile+Meta+Topo] Semilocally simply connected space

§FORM := A space is semilocally simply connected when each point has a neighborhood whose inclusion kills fundamental-group loops in the ambient space.

§DEP_HARD := S26-T-00003157

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003159|T|A|Proof+Compile+Meta+Topo] Subgroup classification of connected covers

§FORM := For a path-connected locally path-connected semilocally simply connected base, connected covering spaces correspond to conjugacy classes of subgroups of pi_1.

§DEP_HARD := S26-D-00003158

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003160|T|A|Proof+Compile+Meta+Topo] Regular covering criterion

§FORM := A connected covering is regular exactly when the corresponding subgroup is normal, under the standard classification hypotheses.

§DEP_HARD := S26-T-00003159

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003161|T|A|Proof+Compile+Meta+Topo] Deck group of a regular cover

§FORM := For a regular connected cover corresponding to H normal in pi_1(B), the deck group is isomorphic to pi_1(B)/H.

§DEP_HARD := S26-T-00003160

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003162|D|A|Proof+Compile+Meta+Topo] Monodromy action of a covering

§FORM := The fundamental group acts on a fiber by lifting loops; this permutation action encodes the covering.

§DEP_HARD := S26-T-00003161

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003163|T|A|Proof+Compile+Meta+Topo] Lifting criterion for maps
§FORM := A map f:Y->B lifts to E after choosing basepoints when f_*(pi_1(Y)) lies in the subgroup p_*(pi_1(E)), under the standard local hypotheses.
§DEP_HARD := S26-D-00003162
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003164|D|A|Proof+Compile+Meta+Topo] Branched-covering fence
§FORM := A branched covering is not a covering map at branch points because local homeomorphic sheet structure fails.
§DEP_HARD := S26-T-00003163
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003165|D|A|Proof+Compile+Meta+Topo] Covering degree
§FORM := A finite-sheeted covering of connected spaces has degree equal to the number of points in a fiber.
§DEP_HARD := S26-D-00003164
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003166|T|A|Proof+Compile+Meta+Topo] Euler characteristic under finite covers
§FORM := For finite CW complexes and a d-sheeted covering E->B, chi(E)=d chi(B).
§DEP_HARD := S26-D-00003165
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003167|N|A|Proof+Compile+Meta+Topo] Covering-existence fence
§FORM := Local path connectedness and semilocal simple connectivity cannot be silently dropped from elementary universal-cover classification statements.
§DEP_HARD := S26-T-00003166
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003168|B|A|Proof+Compile+Meta+Topo] Coverings to complex analysis bridge
§FORM := Analytic continuation and multivalued functions are naturally organized by covering spaces and monodromy.
§DEP_HARD := S26-N-00003167
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003169|B|A|Proof+Compile+Meta+Topo] Coverings to geometry bridge
§FORM := Quotients of simply connected model geometries by properly discontinuous group actions produce many geometric manifolds and orbifold-like spaces.
§DEP_HARD := S26-B-00003168
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003170|B|A|Proof+Compile+Meta+Topo] Coverings to group theory bridge
§FORM := Finite covers, subgroup index and permutation monodromy provide a geometric realization of subgroup data.
§DEP_HARD := S26-B-00003169
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003171|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=2
§FORM := The map z↦z^2 on S^1 is a 2-sheeted covering; each fiber has 2 points and the induced map on pi_1 is multiplication by 2.
§DEP_HARD := S26-B-00003170
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003172|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=3
§FORM := The map z↦z^3 on S^1 is a 3-sheeted covering; each fiber has 3 points and the induced map on pi_1 is multiplication by 3.
§DEP_HARD := S26-E-00003171
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003173|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=4
§FORM := The map z↦z^4 on S^1 is a 4-sheeted covering; each fiber has 4 points and the induced map on pi_1 is multiplication by 4.
§DEP_HARD := S26-E-00003172
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003174|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=5
§FORM := The map z↦z^5 on S^1 is a 5-sheeted covering; each fiber has 5 points and the induced map on pi_1 is multiplication by 5.
§DEP_HARD := S26-E-00003173
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003175|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=6
§FORM := The map z↦z^6 on S^1 is a 6-sheeted covering; each fiber has 6 points and the induced map on pi_1 is multiplication by 6.
§DEP_HARD := S26-E-00003174
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003176|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=7
§FORM := The map z↦z^7 on S^1 is a 7-sheeted covering; each fiber has 7 points and the induced map on pi_1 is multiplication by 7.
§DEP_HARD := S26-E-00003175
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003177|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=8
§FORM := The map z↦z^8 on S^1 is a 8-sheeted covering; each fiber has 8 points and the induced map on pi_1 is multiplication by 8.
§DEP_HARD := S26-E-00003176
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003178|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=9
§FORM := The map z↦z^9 on S^1 is a 9-sheeted covering; each fiber has 9 points and the induced map on pi_1 is multiplication by 9.
§DEP_HARD := S26-E-00003177
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003179|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=10
§FORM := The map z↦z^10 on S^1 is a 10-sheeted covering; each fiber has 10 points and the induced map on pi_1 is multiplication by 10.
§DEP_HARD := S26-E-00003178
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003180|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=11
§FORM := The map z↦z^11 on S^1 is a 11-sheeted covering; each fiber has 11 points and the induced map on pi_1 is multiplication by 11.
§DEP_HARD := S26-E-00003179
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003181|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=12
§FORM := The map z↦z^12 on S^1 is a 12-sheeted covering; each fiber has 12 points and the induced map on pi_1 is multiplication by 12.
§DEP_HARD := S26-E-00003180
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003182|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=13
§FORM := The map z↦z^13 on S^1 is a 13-sheeted covering; each fiber has 13 points and the induced map on pi_1 is multiplication by 13.
§DEP_HARD := S26-E-00003181
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003183|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=14
§FORM := The map z↦z^14 on S^1 is a 14-sheeted covering; each fiber has 14 points and the induced map on pi_1 is multiplication by 14.
§DEP_HARD := S26-E-00003182
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003184|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=15
§FORM := The map z↦z^15 on S^1 is a 15-sheeted covering; each fiber has 15 points and the induced map on pi_1 is multiplication by 15.
§DEP_HARD := S26-E-00003183
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003185|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=16
§FORM := The map z↦z^16 on S^1 is a 16-sheeted covering; each fiber has 16 points and the induced map on pi_1 is multiplication by 16.
§DEP_HARD := S26-E-00003184
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003186|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=17
§FORM := The map z↦z^17 on S^1 is a 17-sheeted covering; each fiber has 17 points and the induced map on pi_1 is multiplication by 17.
§DEP_HARD := S26-E-00003185
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003187|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=18
§FORM := The map z↦z^18 on S^1 is a 18-sheeted covering; each fiber has 18 points and the induced map on pi_1 is multiplication by 18.
§DEP_HARD := S26-E-00003186
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003188|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=19
§FORM := The map z↦z^19 on S^1 is a 19-sheeted covering; each fiber has 19 points and the induced map on pi_1 is multiplication by 19.
§DEP_HARD := S26-E-00003187
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003189|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=20
§FORM := The map z↦z^20 on S^1 is a 20-sheeted covering; each fiber has 20 points and the induced map on pi_1 is multiplication by 20.
§DEP_HARD := S26-E-00003188
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003190|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=21
§FORM := The map z↦z^21 on S^1 is a 21-sheeted covering; each fiber has 21 points and the induced map on pi_1 is multiplication by 21.
§DEP_HARD := S26-E-00003189
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003191|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=22
§FORM := The map z↦z^22 on S^1 is a 22-sheeted covering; each fiber has 22 points and the induced map on pi_1 is multiplication by 22.
§DEP_HARD := S26-E-00003190
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003192|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=23
§FORM := The map z↦z^23 on S^1 is a 23-sheeted covering; each fiber has 23 points and the induced map on pi_1 is multiplication by 23.
§DEP_HARD := S26-E-00003191
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003193|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=24
§FORM := The map z↦z^24 on S^1 is a 24-sheeted covering; each fiber has 24 points and the induced map on pi_1 is multiplication by 24.
§DEP_HARD := S26-E-00003192
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003194|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=25
§FORM := The map z↦z^25 on S^1 is a 25-sheeted covering; each fiber has 25 points and the induced map on pi_1 is multiplication by 25.
§DEP_HARD := S26-E-00003193
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003195|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=26
§FORM := The map z↦z^26 on S^1 is a 26-sheeted covering; each fiber has 26 points and the induced map on pi_1 is multiplication by 26.
§DEP_HARD := S26-E-00003194
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003196|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=27
§FORM := The map z↦z^27 on S^1 is a 27-sheeted covering; each fiber has 27 points and the induced map on pi_1 is multiplication by 27.
§DEP_HARD := S26-E-00003195
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003197|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=28
§FORM := The map z↦z^28 on S^1 is a 28-sheeted covering; each fiber has 28 points and the induced map on pi_1 is multiplication by 28.
§DEP_HARD := S26-E-00003196
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003198|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=29
§FORM := The map z↦z^29 on S^1 is a 29-sheeted covering; each fiber has 29 points and the induced map on pi_1 is multiplication by 29.
§DEP_HARD := S26-E-00003197
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003199|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=30
§FORM := The map z↦z^30 on S^1 is a 30-sheeted covering; each fiber has 30 points and the induced map on pi_1 is multiplication by 30.
§DEP_HARD := S26-E-00003198
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003200|E|M|Proof+Compile+Meta+Topo] Circle covering profile degree=31
§FORM := The map z↦z^31 on S^1 is a 31-sheeted covering; each fiber has 31 points and the induced map on pi_1 is multiplication by 31.
§DEP_HARD := S26-E-00003199
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003201|D|A|Proof+Compile+Meta+Topo] Abstract simplicial complex
§FORM := An abstract simplicial complex K on a vertex set is a family of finite nonempty subsets closed under taking nonempty subsets.
§DEP_HARD := S26-E-00003200
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003202|D|A|Proof+Compile+Meta+Topo] Geometric realization of a simplicial complex
§FORM := The geometric realization |K| glues Euclidean simplices according to the face relations of K.
§DEP_HARD := S26-D-00003201
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003203|D|A|Proof+Compile+Meta+Topo] Simplicial map
§FORM := A vertex map inducing simplex-to-simplex maps defines a simplicial map and hence a continuous map of realizations.
§DEP_HARD := S26-D-00003202
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003204|D|A|Proof+Compile+Meta+Topo] Oriented simplex
§FORM := An orientation of an n-simplex is an equivalence class of vertex orderings modulo even permutations.
§DEP_HARD := S26-D-00003203
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003205|D|A|Proof+Compile+Meta+Topo] Simplicial chain group
§FORM := C_n(K;R) is the free R-module on oriented n-simplices, modulo the sign relation for orientation reversal.
§DEP_HARD := S26-D-00003204
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00003206|I|A|Proof+Compile+Meta+Topo+Spectral] Simplicial boundary operator
§FORM := ∂[v_0,...,v_n]=sum_i (-1)^i[v_0,...,hat v_i,...,v_n].
§DEP_HARD := S26-D-00003205
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003207|T|A|Proof+Compile+Meta+Topo] Boundary squared is zero
§FORM := The simplicial boundary satisfies ∂_{n-1}∘∂_n=0.
§DEP_HARD := S26-I-00003206
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003208|D|A|Proof+Compile+Meta+Topo] Simplicial cycle and boundary groups
§FORM := Z_n=ker ∂_n and B_n=im ∂_{n+1}; boundary-squared-zero gives B_n⊂Z_n.
§DEP_HARD := S26-T-00003207
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003209|D|A|Proof+Compile+Meta+Topo] Simplicial homology group
§FORM := H_n(K;R)=Z_n/B_n.
§DEP_HARD := S26-D-00003208
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003210|D|A|Proof+Compile+Meta+Topo] Betti number
§FORM := Over a field, the nth Betti number beta_n is dim H_n.
§DEP_HARD := S26-D-00003209
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003211|D|A|Proof+Compile+Meta+Topo] Euler characteristic from simplices
§FORM := For a finite simplicial complex, chi(K)=sum_n (-1)^n f_n where f_n counts n-simplices.
§DEP_HARD := S26-D-00003210
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003212|T|A|Proof+Compile+Meta+Topo] Euler-Poincare formula
§FORM := For a finite simplicial complex over a field, chi(K)=sum_n (-1)^n beta_n.
§DEP_HARD := S26-D-00003211
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003213|D|A|Proof+Compile+Meta+Topo] Barycentric subdivision
§FORM := Barycentric subdivision replaces simplices by chains of nonempty faces and preserves the homeomorphism type of the realization.
§DEP_HARD := S26-T-00003212
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003214|T|A|Proof+Compile+Meta+Topo] Simplicial approximation interface
§FORM := After sufficient subdivision under standard compactness hypotheses, continuous maps between polyhedra admit simplicial approximations in their homotopy class.
§DEP_HARD := S26-D-00003213
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003215|D|A|Proof+Compile+Meta+Topo] Reduced simplicial homology
§FORM := Reduced homology modifies degree zero so that a one-point space has vanishing reduced homology.
§DEP_HARD := S26-T-00003214
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003216|D|A|Proof+Compile+Meta+Topo] Relative simplicial chain complex
§FORM := For L⊂K, C_*(K,L)=C_*(K)/C_*(L), leading to relative homology.
§DEP_HARD := S26-D-00003215
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003217|T|A|Proof+Compile+Meta+Topo] Long exact sequence of a simplicial pair

§FORM := A pair (K,L) has a natural long exact sequence in homology.
§DEP_HARD := S26-D-00003216
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003218|N|A|Proof+Compile+Meta+Topo] Triangulation fence

§FORM := Not every topological space or manifold in every category admits a triangulation compatible with naive simplicial methods; triangulability is a real hypothesis.
§DEP_HARD := S26-T-00003217
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003219|B|A|Proof+Compile+Meta+Topo] Simplicial to computation bridge

§FORM := Boundary matrices reduce finite homology computations to linear algebra or Smith normal form over the chosen coefficient ring.
§DEP_HARD := S26-N-00003218
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003220|B|A|Proof+Compile+Meta+Topo] Simplicial to geometry bridge

§FORM := Triangulations provide finite combinatorial models of many spaces while preserving topological invariants when the triangulation assumptions hold.
§DEP_HARD := S26-B-00003219
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003221|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=3

§FORM := The 1-dimensional simplicial cycle C_3 has f_0=3, f_1=3, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-B-00003220
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003222|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=4

§FORM := The 1-dimensional simplicial cycle C_4 has f_0=4, f_1=4, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003221
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003223|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=5

§FORM := The 1-dimensional simplicial cycle C_5 has f_0=5, f_1=5, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003222
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003224|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=6

§FORM := The 1-dimensional simplicial cycle C_6 has f_0=6, f_1=6, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003223
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003225|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=7

§FORM := The 1-dimensional simplicial cycle C_7 has f_0=7, f_1=7, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003224
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003226|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=8

§FORM := The 1-dimensional simplicial cycle C_8 has f_0=8, f_1=8, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003225
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003227|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=9

§FORM := The 1-dimensional simplicial cycle C_9 has f_0=9, f_1=9, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003226
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003228|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=10

§FORM := The 1-dimensional simplicial cycle C_10 has f_0=10, f_1=10, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003227
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003229|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=11

§FORM := The 1-dimensional simplicial cycle C_11 has f_0=11, f_1=11, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003228
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003230|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=12
§FORM := The 1-dimensional simplicial cycle C_12 has f_0=12, f_1=12, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003229
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003231|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=13
§FORM := The 1-dimensional simplicial cycle C_13 has f_0=13, f_1=13, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003230
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003232|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=14
§FORM := The 1-dimensional simplicial cycle C_14 has f_0=14, f_1=14, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003231
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003233|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=15
§FORM := The 1-dimensional simplicial cycle C_15 has f_0=15, f_1=15, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003232
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003234|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=16
§FORM := The 1-dimensional simplicial cycle C_16 has f_0=16, f_1=16, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003233
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003235|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=17
§FORM := The 1-dimensional simplicial cycle C_17 has f_0=17, f_1=17, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003234
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003236|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=18
§FORM := The 1-dimensional simplicial cycle C_18 has f_0=18, f_1=18, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003235
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003237|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=19
§FORM := The 1-dimensional simplicial cycle C_19 has f_0=19, f_1=19, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003236
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003238|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=20
§FORM := The 1-dimensional simplicial cycle C_20 has f_0=20, f_1=20, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003237
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003239|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=21
§FORM := The 1-dimensional simplicial cycle C_21 has f_0=21, f_1=21, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003238
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003240|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=22
§FORM := The 1-dimensional simplicial cycle C_22 has f_0=22, f_1=22, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003239
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003241|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=23
§FORM := The 1-dimensional simplicial cycle C_23 has f_0=23, f_1=23, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003240
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003242|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=24
§FORM := The 1-dimensional simplicial cycle C_24 has f_0=24, f_1=24, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003241
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003243|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=25
§FORM := The 1-dimensional simplicial cycle C_25 has f_0=25, f_1=25, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003242
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003244|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=26
§FORM := The 1-dimensional simplicial cycle C_26 has f_0=26, f_1=26, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003243
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003245|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=27
§FORM := The 1-dimensional simplicial cycle C_27 has f_0=27, f_1=27, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003244
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003246|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=28
§FORM := The 1-dimensional simplicial cycle C_28 has f_0=28, f_1=28, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003245
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003247|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=29
§FORM := The 1-dimensional simplicial cycle C_29 has f_0=29, f_1=29, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003246
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003248|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=30
§FORM := The 1-dimensional simplicial cycle C_30 has f_0=30, f_1=30, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003247
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003249|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=31
§FORM := The 1-dimensional simplicial cycle C_31 has f_0=31, f_1=31, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003248
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003250|E|M|Proof+Compile+Meta+Topo] Cycle-complex homology profile n=32
§FORM := The 1-dimensional simplicial cycle C_32 has f_0=32, f_1=32, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.
§DEP_HARD := S26-E-00003249
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003251|D|A|Proof+Compile+Meta+Topo] Singular n-simplex
§FORM := A singular n-simplex in X is a continuous map sigma:Delta^n->X from the standard n-simplex.
§DEP_HARD := S26-E-00003250
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003252|D|A|Proof+Compile+Meta+Topo] Singular chain complex
§FORM := C_n(X;R) is the free R-module generated by singular n-simplices with the alternating face boundary.
§DEP_HARD := S26-D-00003251
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003253|T|A|Proof+Compile+Meta+Topo+Spectral] Singular boundary squared is zero
§FORM := The singular boundary operators satisfy ∂^2=0.
§DEP_HARD := S26-D-00003252
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003254|D|A|Proof+Compile+Meta+Topo] Singular homology
§FORM := H_n(X;R)=ker ∂_n/im ∂_{n+1} for the singular chain complex.
§DEP_HARD := S26-T-00003253
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003255|T|A|Proof+Compile+Meta+Topo] Homotopy invariance of singular homology
§FORM := Homotopic maps induce the same maps on singular homology; homotopy equivalent spaces have isomorphic homology.
§DEP_HARD := S26-D-00003254
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003256|T|A|Proof+Compile+Meta+Topo] Dimension axiom for a point
§FORM := For ordinary homology, H_0(point;R)□R and H_n(point;R)=0 for n>0.
§DEP_HARD := S26-T-00003255
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003257|T|A|Proof+Compile+Meta+Topo] Excision theorem
§FORM := For a pair (X,A), a subset whose closure lies in the interior of A may be excised without changing relative homology.
§DEP_HARD := S26-T-00003256
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003258|T|A|Proof+Compile+Meta+Topo] Long exact sequence of a pair

§FORM := Every pair (X,A) has a natural long exact sequence ...->H_n(A)->H_n(X)->H_n(X,A)->H_{n-1}(A)->....

§DEP_HARD := S26-T-00003257

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003259|T|A|Proof+Compile+Meta+Topo] Mayer-Vietoris sequence

§FORM := A suitable decomposition X=U∪V gives a long exact sequence linking homology of U∩V, U, V and X.

§DEP_HARD := S26-T-00003258

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003260|D|A|Proof+Compile+Meta+Topo] Reduced singular homology

§FORM := Reduced homology augments the singular chain complex to normalize degree-zero behavior.

§DEP_HARD := S26-T-00003259

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003261|T|A|Proof+Compile+Meta+Topo] Suspension isomorphism in reduced homology

§FORM := For suitable spaces, reduced H_{n+1}(Sigma X) is naturally isomorphic to reduced H_n(X).

§DEP_HARD := S26-D-00003260

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003262|T|A|Proof+Compile+Meta+Topo] Homology of spheres

§FORM := For S^n, reduced homology is the coefficient ring in degree n and zero in other degrees.

§DEP_HARD := S26-T-00003261

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003263|D|A|Proof+Compile+Meta+Topo] Degree of a map between oriented spheres

§FORM := For f:S^n->S^n, the induced map on top homology is multiplication by an integer deg(f).

§DEP_HARD := S26-T-00003262

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003264|T|A|Proof+Compile+Meta+Topo] Degree is homotopy invariant

§FORM := Homotopic maps S^n->S^n have the same degree.

§DEP_HARD := S26-D-00003263

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003265|T|A|Proof+Compile+Meta+Topo] Brouwer fixed-point theorem via degree/homology interface

§FORM := No continuous retraction D^n->S^{n-1} exists; this obstruction yields the fixed-point theorem through standard homological arguments.

§DEP_HARD := S26-T-00003264

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003266|D|A|Proof+Compile+Meta+Topo] Homology with coefficients

§FORM := Tensoring or otherwise changing the coefficient ring can alter homology by detecting torsion differently.

§DEP_HARD := S26-T-00003265

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003267|T|A|Proof+Compile+Meta+Topo] Universal coefficient theorem for homology

§FORM := Under standard chain-complex hypotheses there is a short exact sequence involving H_n(X;Z) tensor G and Tor(H_{n-1}(X;Z),G).

§DEP_HARD := S26-D-00003266

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003268|N|A|Proof+Compile+Meta+Topo] Coefficient fence

§FORM := Betti numbers over a field can hide torsion; homology over one coefficient system does not determine all coefficient theories without additional data.

§DEP_HARD := S26-T-00003267

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003269|B|A|Proof+Compile+Meta+Topo] Homology to fixed-point bridge

§FORM := Homological degree and Lefschetz-type traces convert global topology into obstructions to fixed-point-free maps.

§DEP_HARD := S26-N-00003268

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003270|B|A|Proof+Compile+Meta+Topo] Homology to manifold orientation bridge

§FORM := Top-dimensional homology encodes orientability and fundamental classes for appropriate manifolds.

§DEP_HARD := S26-B-00003269

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003271|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=2
§FORM := For S^2 with rational coefficients, beta_0=1 and beta_2=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-B-00003270
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003272|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=3
§FORM := For S^3 with rational coefficients, beta_0=1 and beta_3=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003271
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003273|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=4
§FORM := For S^4 with rational coefficients, beta_0=1 and beta_4=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003272
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003274|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=5
§FORM := For S^5 with rational coefficients, beta_0=1 and beta_5=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003273
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003275|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=6
§FORM := For S^6 with rational coefficients, beta_0=1 and beta_6=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003274
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003276|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=7
§FORM := For S^7 with rational coefficients, beta_0=1 and beta_7=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003275
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003277|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=8
§FORM := For S^8 with rational coefficients, beta_0=1 and beta_8=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003276
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003278|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=9
§FORM := For S^9 with rational coefficients, beta_0=1 and beta_9=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003277
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003279|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=10
§FORM := For S^10 with rational coefficients, beta_0=1 and beta_10=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003278
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003280|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=11
§FORM := For S^11 with rational coefficients, beta_0=1 and beta_11=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003279
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003281|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=12
§FORM := For S^12 with rational coefficients, beta_0=1 and beta_12=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003280
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003282|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=13
§FORM := For S^13 with rational coefficients, beta_0=1 and beta_13=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003281
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003283|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=14
§FORM := For S^14 with rational coefficients, beta_0=1 and beta_14=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003282
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003284|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=15
§FORM := For S^15 with rational coefficients, beta_0=1 and beta_15=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003283
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003285|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=16

§FORM := For S^16 with rational coefficients, beta_0=1 and beta_16=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003284
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003286|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=17

§FORM := For S^17 with rational coefficients, beta_0=1 and beta_17=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003285
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003287|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=18

§FORM := For S^18 with rational coefficients, beta_0=1 and beta_18=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003286
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003288|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=19

§FORM := For S^19 with rational coefficients, beta_0=1 and beta_19=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003287
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003289|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=20

§FORM := For S^20 with rational coefficients, beta_0=1 and beta_20=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003288
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003290|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=21

§FORM := For S^21 with rational coefficients, beta_0=1 and beta_21=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003289
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003291|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=22

§FORM := For S^22 with rational coefficients, beta_0=1 and beta_22=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003290
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003292|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=23

§FORM := For S^23 with rational coefficients, beta_0=1 and beta_23=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003291
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003293|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=24

§FORM := For S^24 with rational coefficients, beta_0=1 and beta_24=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003292
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003294|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=25

§FORM := For S^25 with rational coefficients, beta_0=1 and beta_25=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003293
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003295|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=26

§FORM := For S^26 with rational coefficients, beta_0=1 and beta_26=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003294
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003296|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=27

§FORM := For S^27 with rational coefficients, beta_0=1 and beta_27=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003295
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003297|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=28

§FORM := For S^28 with rational coefficients, beta_0=1 and beta_28=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003296
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003298|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=29

§FORM := For S^29 with rational coefficients, beta_0=1 and beta_29=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003297
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003299|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=30

§FORM := For S^30 with rational coefficients, beta_0=1 and beta_30=1, with all other Betti numbers zero; Euler characteristic is 2.
§DEP_HARD := S26-E-00003298
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003300|E|M|Proof+Compile+Meta+Topo] Sphere homology rank profile n=31

§FORM := For S^31 with rational coefficients, beta_0=1 and beta_31=1, with all other Betti numbers zero; Euler characteristic is 0.
§DEP_HARD := S26-E-00003299
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003301|D|A|Proof+Compile+Meta+Topo] Singular cochain complex

§FORM := C^n(X;R)=Hom(C_n(X;R),R) with coboundary delta dual to the singular boundary.
§DEP_HARD := S26-E-00003300
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003302|T|A|Proof+Compile+Meta+Topo] Coboundary squared is zero

§FORM := The singular coboundary satisfies delta^2=0.
§DEP_HARD := S26-D-00003301
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003303|D|A|Proof+Compile+Meta+Topo] Singular cohomology

§FORM := H^n(X;R)=ker delta^n/im delta^{n-1}.
§DEP_HARD := S26-T-00003302
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003304|D|A|Proof+Compile+Meta+Topo] Cup product

§FORM := The cup product gives H^p(X;R)×H^q(X;R)->H^{p+q}(X;R) and equips cohomology with a graded ring structure.
§DEP_HARD := S26-D-00003303
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003305|T|A|Proof+Compile+Meta+Topo] Graded commutativity of cup product

§FORM := For suitable commutative coefficients, alpha cup beta=(-1)^{pq} beta cup alpha in cohomology.
§DEP_HARD := S26-D-00003304
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003306|D|A|Proof+Compile+Meta+Topo] Cohomology ring

§FORM := H^*(X;R)=⊕_n H^n(X;R) together with cup product is a graded ring invariant of X.
§DEP_HARD := S26-T-00003305
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003307|T|A|Proof+Compile+Meta+Topo] Universal coefficient theorem for cohomology

§FORM := Under standard hypotheses, H^n(X;G) fits into a short exact sequence involving Hom(H_n(X;Z),G) and Ext(H_{n-1}(X;Z),G).
§DEP_HARD := S26-D-00003306
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003308|D|A|Proof+Compile+Meta+Topo] Cross product in cohomology

§FORM := The external cross product maps H^p(X)×H^q(Y) to H^{p+q}(X×Y).
§DEP_HARD := S26-T-00003307
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003309|T|A|Proof+Compile+Meta+Topo] Kunneth theorem interface

§FORM := Under suitable coefficient and finiteness hypotheses, homology or cohomology of products is computed from tensor and Tor data of the factors.
§DEP_HARD := S26-D-00003308
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003310|D|A|Proof+Compile+Meta+Topo] Cohomological dimension interface

§FORM := Cohomological dimension records the highest degree in which an appropriate cohomology theory can be nonzero.
§DEP_HARD := S26-T-00003309
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003311|T|A|Proof+Compile+Meta+Topo] Cohomology of complex projective space

§FORM := With integer coefficients, H^*(CP^n;Z)□Z[x]/(x^{n+1}) with |x|=2.
§DEP_HARD := S26-D-00003310
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003312|D|A|Proof+Compile+Meta+Topo] Cap product

§FORM := The cap product pairs cohomology and homology to lower homological degree.
§DEP_HARD := S26-T-00003311
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003313|D|A|Proof+Compile+Meta+Topo] Kronecker pairing

§FORM := The evaluation pairing H^n(X;R)×H_n(X;R)->R evaluates cohomology classes on homology classes.
§DEP_HARD := S26-D-00003312
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003314|D|A|Proof+Compile+Meta+Topo] Poincare duality interface

§FORM := For a closed oriented n-manifold over suitable coefficients, cap product with the fundamental class gives H^k(M)->H_{n-k}(M).
§DEP_HARD := S26-D-00003313
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003315|T|A|Proof+Compile+Meta+Topo] Euler characteristic from cohomology

§FORM := For finite-type complexes over a field, Euler characteristic is the alternating sum of cohomology dimensions as well as homology dimensions.
§DEP_HARD := S26-D-00003314
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003316|D|A|Proof+Compile+Meta+Topo] Cech cohomology interface

§FORM := Cech cohomology is built from nerves of open covers and agrees with singular cohomology on broad classes of well-behaved spaces.
§DEP_HARD := S26-T-00003315
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003317|N|A|Proof+Compile+Meta+Topo+IG] Ring-vs-group fence

§FORM := Isomorphic graded cohomology groups need not imply isomorphic cohomology rings; cup-product structure carries additional information.
§DEP_HARD := S26-D-00003316
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003318|N|A|Proof+Compile+Meta+Topo] Duality hypothesis fence

§FORM := Poincare duality requires manifold, orientation/coefficients, boundary and compactness hypotheses in the appropriate form.
§DEP_HARD := S26-N-00003317
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003319|B|A|Proof+Compile+Meta+Topo] Cohomology to bundles bridge

§FORM := Characteristic classes live in cohomology and turn geometric twisting of bundles into algebraic invariants.
§DEP_HARD := S26-N-00003318
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003320|B|A|Proof+Compile+Meta+Topo] Cohomology to differential forms bridge

§FORM := de Rham cohomology identifies smooth closed-form classes with singular cohomology over R for smooth manifolds.
§DEP_HARD := S26-B-00003319
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003321|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=2

§FORM := For CP^2, the integral cohomology groups have rank 1 in degrees 0,2,...,4 and rank 0 otherwise; the Euler characteristic is 3.
§DEP_HARD := S26-B-00003320
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003322|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=3

§FORM := For CP^3, the integral cohomology groups have rank 1 in degrees 0,2,...,6 and rank 0 otherwise; the Euler characteristic is 4.
§DEP_HARD := S26-E-00003321
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003323|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=4

§FORM := For CP^4, the integral cohomology groups have rank 1 in degrees 0,2,...,8 and rank 0 otherwise; the Euler characteristic is 5.
§DEP_HARD := S26-E-00003322
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003324|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=5

§FORM := For CP^5, the integral cohomology groups have rank 1 in degrees 0,2,...,10 and rank 0 otherwise; the Euler characteristic is 6.
§DEP_HARD := S26-E-00003323
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003325|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=6

§FORM := For CP^6, the integral cohomology groups have rank 1 in degrees 0,2,...,12 and rank 0 otherwise; the Euler characteristic is 7.
§DEP_HARD := S26-E-00003324
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003326|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=7

§FORM := For CP^7, the integral cohomology groups have rank 1 in degrees 0,2,...,14 and rank 0 otherwise; the Euler characteristic is 8.
§DEP_HARD := S26-E-00003325
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003327|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=8

§FORM := For CP^8, the integral cohomology groups have rank 1 in degrees 0,2,...,16 and rank 0 otherwise; the Euler characteristic is 9.
§DEP_HARD := S26-E-00003326
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003328|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=9

§FORM := For CP^9, the integral cohomology groups have rank 1 in degrees 0,2,...,18 and rank 0 otherwise; the Euler characteristic is 10.
§DEP_HARD := S26-E-00003327
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003329|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=10

§FORM := For CP^10, the integral cohomology groups have rank 1 in degrees 0,2,...,20 and rank 0 otherwise; the Euler characteristic is 11.
§DEP_HARD := S26-E-00003328
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003330|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=11

§FORM := For CP^11, the integral cohomology groups have rank 1 in degrees 0,2,...,22 and rank 0 otherwise; the Euler characteristic is 12.
§DEP_HARD := S26-E-00003329
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003331|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=12

§FORM := For CP^12, the integral cohomology groups have rank 1 in degrees 0,2,...,24 and rank 0 otherwise; the Euler characteristic is 13.
§DEP_HARD := S26-E-00003330
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003332|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=13

§FORM := For CP^13, the integral cohomology groups have rank 1 in degrees 0,2,...,26 and rank 0 otherwise; the Euler characteristic is 14.
§DEP_HARD := S26-E-00003331
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003333|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=14

§FORM := For CP^14, the integral cohomology groups have rank 1 in degrees 0,2,...,28 and rank 0 otherwise; the Euler characteristic is 15.
§DEP_HARD := S26-E-00003332
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003334|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=15

§FORM := For CP^15, the integral cohomology groups have rank 1 in degrees 0,2,...,30 and rank 0 otherwise; the Euler characteristic is 16.
§DEP_HARD := S26-E-00003333
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003335|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=16

§FORM := For CP^16, the integral cohomology groups have rank 1 in degrees 0,2,...,32 and rank 0 otherwise; the Euler characteristic is 17.
§DEP_HARD := S26-E-00003334
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003336|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=17

§FORM := For CP^17, the integral cohomology groups have rank 1 in degrees 0,2,...,34 and rank 0 otherwise; the Euler characteristic is 18.
§DEP_HARD := S26-E-00003335
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003337|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=18

§FORM := For CP^18, the integral cohomology groups have rank 1 in degrees 0,2,...,36 and rank 0 otherwise; the Euler characteristic is 19.
§DEP_HARD := S26-E-00003336
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003338|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=19

§FORM := For CP^19, the integral cohomology groups have rank 1 in degrees 0,2,...,38 and rank 0 otherwise; the Euler characteristic is 20.

§DEP_HARD := S26-E-00003337

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003339|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=20

§FORM := For CP^20, the integral cohomology groups have rank 1 in degrees 0,2,...,40 and rank 0 otherwise; the Euler characteristic is 21.

§DEP_HARD := S26-E-00003338

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003340|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=21

§FORM := For CP^21, the integral cohomology groups have rank 1 in degrees 0,2,...,42 and rank 0 otherwise; the Euler characteristic is 22.

§DEP_HARD := S26-E-00003339

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003341|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=22

§FORM := For CP^22, the integral cohomology groups have rank 1 in degrees 0,2,...,44 and rank 0 otherwise; the Euler characteristic is 23.

§DEP_HARD := S26-E-00003340

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003342|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=23

§FORM := For CP^23, the integral cohomology groups have rank 1 in degrees 0,2,...,46 and rank 0 otherwise; the Euler characteristic is 24.

§DEP_HARD := S26-E-00003341

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003343|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=24

§FORM := For CP^24, the integral cohomology groups have rank 1 in degrees 0,2,...,48 and rank 0 otherwise; the Euler characteristic is 25.

§DEP_HARD := S26-E-00003342

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003344|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=25

§FORM := For CP^25, the integral cohomology groups have rank 1 in degrees 0,2,...,50 and rank 0 otherwise; the Euler characteristic is 26.

§DEP_HARD := S26-E-00003343

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003345|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=26

§FORM := For CP^26, the integral cohomology groups have rank 1 in degrees 0,2,...,52 and rank 0 otherwise; the Euler characteristic is 27.

§DEP_HARD := S26-E-00003344

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003346|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=27

§FORM := For CP^27, the integral cohomology groups have rank 1 in degrees 0,2,...,54 and rank 0 otherwise; the Euler characteristic is 28.

§DEP_HARD := S26-E-00003345

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003347|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=28

§FORM := For CP^28, the integral cohomology groups have rank 1 in degrees 0,2,...,56 and rank 0 otherwise; the Euler characteristic is 29.

§DEP_HARD := S26-E-00003346

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003348|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=29

§FORM := For CP^29, the integral cohomology groups have rank 1 in degrees 0,2,...,58 and rank 0 otherwise; the Euler characteristic is 30.

§DEP_HARD := S26-E-00003347

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003349|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=30

§FORM := For CP^30, the integral cohomology groups have rank 1 in degrees 0,2,...,60 and rank 0 otherwise; the Euler characteristic is 31.

§DEP_HARD := S26-E-00003348

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003350|E|M|Proof+Compile+Meta+Topo] Projective-space cohomology profile n=31

§FORM := For CP^31, the integral cohomology groups have rank 1 in degrees 0,2,...,62 and rank 0 otherwise; the Euler characteristic is 32.

§DEP_HARD := S26-E-00003349

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003351|D|A|Proof+Compile+Meta+Topo] CW complex

§FORM := A CW complex is built inductively by attaching n-cells to lower skeleta via maps S^{n-1}->X^{n-1}, with closure-finiteness and weak-topology conditions.

§DEP_HARD := S26-E-00003350

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003352|D|A|Proof+Compile+Meta+Topo] n-skeleton of a CW complex

§FORM := X^n is the union of all cells of dimension at most n.

§DEP_HARD := S26-D-00003351

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003353|D|A|Proof+Compile+Meta+Topo] Attaching map of a cell

§FORM := An n-cell is attached by a continuous map S^{n-1}->X^{n-1} followed by quotient gluing of D^n.

§DEP_HARD := S26-D-00003352

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003354|D|A|Proof+Compile+Meta+Topo] Characteristic map of a cell

§FORM := A characteristic map D^n->X maps the interior homeomorphically onto the cell and restricts on the boundary to the attaching map.

§DEP_HARD := S26-D-00003353

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003355|T|A|Proof+Compile+Meta+Topo] CW pair has homotopy extension property

§FORM := If A is a subcomplex of a CW complex X, the pair (X,A) satisfies the homotopy extension property.

§DEP_HARD := S26-D-00003354

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003356|D|A|Proof+Compile+Meta+Topo] Cellular chain complex

§FORM := For a CW complex, C_n^{cell}(X)=H_n(X^n,X^{n-1}) is free on the n-cells.

§DEP_HARD := S26-T-00003355

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003357|T|A|Proof+Compile+Meta+Topo] Cellular homology theorem

§FORM := The homology of the cellular chain complex is naturally isomorphic to singular homology of the CW complex.

§DEP_HARD := S26-D-00003356

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00003358|I|A|Proof+Compile+Meta+Topo] Cellular boundary via degrees

§FORM := Cellular boundary coefficients are degrees of maps obtained from attaching maps and quotient projections onto (n-1)-spheres.

§DEP_HARD := S26-T-00003357

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003359|T|A|Proof+Compile+Meta+Topo] Euler characteristic from cells

§FORM := For a finite CW complex, chi(X)=sum_n (-1)^n c_n where c_n is the number of n-cells.

§DEP_HARD := S26-I-00003358

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003360|D|A|Proof+Compile+Meta+Topo] CW approximation interface

§FORM := Spaces can often be replaced up to weak homotopy type by CW complexes, making cellular tools available.

§DEP_HARD := S26-T-00003359

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003361|T|A|Proof+Compile+Meta+Topo] Whitehead theorem in cellular setting

§FORM := A weak homotopy equivalence between CW complexes is a homotopy equivalence.

§DEP_HARD := S26-D-00003360

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003362|D|A|Proof+Compile+Meta+Topo] Mapping cone

§FORM := The mapping cone C_f is obtained by attaching the cone on X to Y along f:X->Y.

§DEP_HARD := S26-T-00003361

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003363|D|A|Proof+Compile+Meta+Topo] Mapping cylinder
§FORM := The mapping cylinder M_f=(X×[0,1])∪_f Y interpolates between X and Y and provides a standard factorization of f.
§DEP_HARD := S26-D-00003362
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003364|D|A|Proof+Compile+Meta+Topo] Wedge sum as a CW construction
§FORM := The wedge X∨Y identifies chosen basepoints and inherits a CW structure from based CW structures.
§DEP_HARD := S26-D-00003363
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003365|D|A|Proof+Compile+Meta+Topo] Smash product
§FORM := For based spaces, X∧Y=(X×Y)/(X∨Y).
§DEP_HARD := S26-D-00003364
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003366|D|A|Proof+Compile+Meta+Topo] Suspension as a smash product
§FORM := The reduced suspension Sigma X is homeomorphic to S^1∧X for based compactly generated settings.
§DEP_HARD := S26-D-00003365
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003367|N|A|Proof+Compile+Meta+Topo] Cell-structure fence
§FORM := A CW decomposition is not unique; cell counts in each dimension are not topological invariants although Euler characteristic is for finite CW type.
§DEP_HARD := S26-D-00003366
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003368|B|A|Proof+Compile+Meta+Topo] CW to homotopy bridge
§FORM := Cell attachment records how homotopy type is assembled dimension by dimension and supports obstruction-theoretic arguments.
§DEP_HARD := S26-N-00003367
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003369|B|A|Proof+Compile+Meta+Topo] CW to computation bridge
§FORM := Cellular boundary matrices can be dramatically smaller than singular chain complexes while computing the same homology.
§DEP_HARD := S26-B-00003368
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003370|B|A|Proof+Compile+Meta+Topo] CW to manifold bridge
§FORM := Many smooth manifolds admit CW structures, for example through Morse theory, enabling algebraic-topological calculations.
§DEP_HARD := S26-B-00003369
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003371|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=2
§FORM := The standard CW structure on RP^2 has one cell in each dimension 0,...,2; its cellular alternating count gives Euler characteristic 1.
§DEP_HARD := S26-B-00003370
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003372|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=3
§FORM := The standard CW structure on RP^3 has one cell in each dimension 0,...,3; its cellular alternating count gives Euler characteristic 0.
§DEP_HARD := S26-E-00003371
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003373|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=4
§FORM := The standard CW structure on RP^4 has one cell in each dimension 0,...,4; its cellular alternating count gives Euler characteristic 1.
§DEP_HARD := S26-E-00003372
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003374|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=5
§FORM := The standard CW structure on RP^5 has one cell in each dimension 0,...,5; its cellular alternating count gives Euler characteristic 0.
§DEP_HARD := S26-E-00003373
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003375|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=6
§FORM := The standard CW structure on RP^6 has one cell in each dimension 0,...,6; its cellular alternating count gives Euler characteristic 1.
§DEP_HARD := S26-E-00003374
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003376|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=7

§FORM := The standard CW structure on RP^7 has one cell in each dimension 0,...,7; its cellular alternating count gives Euler characteristic 0.

§DEP_HARD := S26-E-00003375

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003377|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=8

§FORM := The standard CW structure on RP^8 has one cell in each dimension 0,...,8; its cellular alternating count gives Euler characteristic 1.

§DEP_HARD := S26-E-00003376

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003378|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=9

§FORM := The standard CW structure on RP^9 has one cell in each dimension 0,...,9; its cellular alternating count gives Euler characteristic 0.

§DEP_HARD := S26-E-00003377

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003379|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=10

§FORM := The standard CW structure on RP^10 has one cell in each dimension 0,...,10; its cellular alternating count gives Euler characteristic 1.

§DEP_HARD := S26-E-00003378

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003380|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=11

§FORM := The standard CW structure on RP^11 has one cell in each dimension 0,...,11; its cellular alternating count gives Euler characteristic 0.

§DEP_HARD := S26-E-00003379

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003381|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=12

§FORM := The standard CW structure on RP^12 has one cell in each dimension 0,...,12; its cellular alternating count gives Euler characteristic 1.

§DEP_HARD := S26-E-00003380

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003382|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=13

§FORM := The standard CW structure on RP^13 has one cell in each dimension 0,...,13; its cellular alternating count gives Euler characteristic 0.

§DEP_HARD := S26-E-00003381

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003383|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=14

§FORM := The standard CW structure on RP^14 has one cell in each dimension 0,...,14; its cellular alternating count gives Euler characteristic 1.

§DEP_HARD := S26-E-00003382

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003384|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=15

§FORM := The standard CW structure on RP^15 has one cell in each dimension 0,...,15; its cellular alternating count gives Euler characteristic 0.

§DEP_HARD := S26-E-00003383

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003385|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=16

§FORM := The standard CW structure on RP^16 has one cell in each dimension 0,...,16; its cellular alternating count gives Euler characteristic 1.

§DEP_HARD := S26-E-00003384

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003386|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=17

§FORM := The standard CW structure on RP^17 has one cell in each dimension 0,...,17; its cellular alternating count gives Euler characteristic 0.

§DEP_HARD := S26-E-00003385

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003387|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=18

§FORM := The standard CW structure on RP^18 has one cell in each dimension 0,...,18; its cellular alternating count gives Euler characteristic 1.

§DEP_HARD := S26-E-00003386

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003388|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=19
§FORM := The standard CW structure on RP^19 has one cell in each dimension 0,...,19; its cellular alternating count gives Euler characteristic 0.
§DEP_HARD := S26-E-00003387
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003389|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=20
§FORM := The standard CW structure on RP^20 has one cell in each dimension 0,...,20; its cellular alternating count gives Euler characteristic 1.
§DEP_HARD := S26-E-00003388
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003390|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=21
§FORM := The standard CW structure on RP^21 has one cell in each dimension 0,...,21; its cellular alternating count gives Euler characteristic 0.
§DEP_HARD := S26-E-00003389
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003391|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=22
§FORM := The standard CW structure on RP^22 has one cell in each dimension 0,...,22; its cellular alternating count gives Euler characteristic 1.
§DEP_HARD := S26-E-00003390
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003392|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=23
§FORM := The standard CW structure on RP^23 has one cell in each dimension 0,...,23; its cellular alternating count gives Euler characteristic 0.
§DEP_HARD := S26-E-00003391
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003393|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=24
§FORM := The standard CW structure on RP^24 has one cell in each dimension 0,...,24; its cellular alternating count gives Euler characteristic 1.
§DEP_HARD := S26-E-00003392
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003394|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=25
§FORM := The standard CW structure on RP^25 has one cell in each dimension 0,...,25; its cellular alternating count gives Euler characteristic 0.
§DEP_HARD := S26-E-00003393
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003395|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=26
§FORM := The standard CW structure on RP^26 has one cell in each dimension 0,...,26; its cellular alternating count gives Euler characteristic 1.
§DEP_HARD := S26-E-00003394
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003396|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=27
§FORM := The standard CW structure on RP^27 has one cell in each dimension 0,...,27; its cellular alternating count gives Euler characteristic 0.
§DEP_HARD := S26-E-00003395
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003397|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=28
§FORM := The standard CW structure on RP^28 has one cell in each dimension 0,...,28; its cellular alternating count gives Euler characteristic 1.
§DEP_HARD := S26-E-00003396
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003398|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=29
§FORM := The standard CW structure on RP^29 has one cell in each dimension 0,...,29; its cellular alternating count gives Euler characteristic 0.
§DEP_HARD := S26-E-00003397
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003399|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=30
§FORM := The standard CW structure on RP^30 has one cell in each dimension 0,...,30; its cellular alternating count gives Euler characteristic 1.
§DEP_HARD := S26-E-00003398
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003400|E|M|Proof+Compile+Meta+Topo] Real-projective CW count profile n=31

§FORM := The standard CW structure on RP^31 has one cell in each dimension 0,...,31; its cellular alternating count gives Euler characteristic 0.

§DEP_HARD := S26-E-00003399

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003401|D|A|Proof+Compile+Meta+Topo] Smooth manifold atlas compatibility refinement

§FORM := A C^infinity atlas has smooth transition maps on all overlaps; the maximal compatible atlas defines the smooth structure.

§DEP_HARD := S26-E-00003400

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003402|D|A|Proof+Compile+Meta+Topo] Smooth map between manifolds

§FORM := A map F:M->N is smooth when its coordinate representatives are smooth in compatible charts.

§DEP_HARD := S26-D-00003401

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003403|T|A|Proof+Compile+Meta+Topo] Smoothness is chart independent

§FORM := If a coordinate representative is smooth in one pair of compatible charts, all compatible coordinate representatives are smooth.

§DEP_HARD := S26-D-00003402

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003404|D|A|Proof+Compile+Meta+Topo] Immersion

§FORM := A smooth map F:M->N is an immersion when dF_p is injective at every p.

§DEP_HARD := S26-T-00003403

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003405|D|A|Proof+Compile+Meta+Topo] Submersion

§FORM := A smooth map F:M->N is a submersion when dF_p is surjective at every p.

§DEP_HARD := S26-D-00003404

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003406|D|A|Proof+Compile+Meta+Topo] Embedding of manifolds

§FORM := A smooth embedding is an immersion that is a homeomorphism onto its image with the subspace topology.

§DEP_HARD := S26-D-00003405

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003407|T|A|Proof+Compile+Meta+Topo] Constant rank theorem

§FORM := A smooth map of constant rank has a local normal form equivalent to a coordinate projection/inclusion according to its rank.

§DEP_HARD := S26-D-00003406

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003408|T|A|Proof+Compile+Meta+Topo] Regular value theorem

§FORM := If y is a regular value of F:M->N, then F^{-1}(y) is an embedded submanifold of codimension dim N.

§DEP_HARD := S26-T-00003407

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003409|D|A|Proof+Compile+Meta+Topo] Regular point and critical point

§FORM := p is regular for F when dF_p has maximal target rank; otherwise p is critical.

§DEP_HARD := S26-T-00003408

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003410|D|A|Proof+Compile+Meta+Topo] Embedded submanifold

§FORM := An embedded k-submanifold is locally carried by charts to R^k×{0} inside R^n.

§DEP_HARD := S26-D-00003409

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003411|D|A|Proof+Compile+Meta+Topo] Initial submanifold interface

§FORM := An initial submanifold has a smooth structure characterized by testing smoothness of maps into the ambient manifold.

§DEP_HARD := S26-D-00003410

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003412|T|A|Proof+Compile+Meta+Topo] Whitney embedding theorem interface

§FORM := Every smooth n-manifold satisfying the standard countability hypotheses admits a smooth embedding into some Euclidean space; classical bounds include R^{2n}.

§DEP_HARD := S26-D-00003411

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003413|T|A|Proof+Compile+Meta+Topo] Whitney immersion theorem interface

§FORM := Every smooth n-manifold admits a smooth immersion into Euclidean space of controlled dimension under the classical theorem.
§DEP_HARD := S26-T-00003412
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003414|D|A|Proof+Compile+Meta+Topo] Partition of unity subordinate to a cover

§FORM := A smooth partition of unity subordinate to an open cover is a locally finite family of nonnegative smooth functions summing to 1 with supports lying in cover elements.
§DEP_HARD := S26-T-00003413
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003415|T|A|Proof+Compile+Meta+Topo] Existence of smooth partitions of unity

§FORM := Paracompact smooth manifolds admit smooth partitions of unity subordinate to any open cover.
§DEP_HARD := S26-D-00003414
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003416|D|A|Proof+Compile+Meta+Topo] Proper smooth map

§FORM := A smooth map is proper when inverse images of compact sets are compact.
§DEP_HARD := S26-T-00003415
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003417|T|A|Proof+Compile+Meta+Topo] Proper injective immersion is an embedding

§FORM := A proper injective immersion between suitable manifolds is a smooth embedding.
§DEP_HARD := S26-D-00003416
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003418|N|A|Proof+Compile+Meta+Topo] Immersion-embedding fence

§FORM := An injective differential does not imply global embedding; topology of the map and image remains additional data.
§DEP_HARD := S26-T-00003417
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003419|B|A|Proof+Compile+Meta+Topo] Manifold to topology bridge

§FORM := Smooth manifolds carry underlying Hausdorff second-countable topologies; smooth constructions depend on these point-set hypotheses.
§DEP_HARD := S26-N-00003418
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003420|B|A|Proof+Compile+Meta+Topo+Spectral] Manifold to analysis bridge

§FORM := Partitions of unity globalize local analytic constructions such as metrics, differential operators and integration.
§DEP_HARD := S26-B-00003419
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003421|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=2

§FORM := S^{2} is a smooth 2-manifold, while S^2×S^1 is a smooth 3-manifold; dimensions add under finite products.
§DEP_HARD := S26-B-00003420
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003422|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=3

§FORM := S^{3} is a smooth 3-manifold, while S^3×S^1 is a smooth 4-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003421
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003423|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=4

§FORM := S^{4} is a smooth 4-manifold, while S^4×S^1 is a smooth 5-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003422
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003424|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=5

§FORM := S^{5} is a smooth 5-manifold, while S^5×S^1 is a smooth 6-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003423
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003425|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=6

§FORM := S^{6} is a smooth 6-manifold, while S^6×S^1 is a smooth 7-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003424
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003426|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=7
§FORM := S^{7} is a smooth 7-manifold, while S^7×S^1 is a smooth 8-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003425
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003427|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=8
§FORM := S^{8} is a smooth 8-manifold, while S^8×S^1 is a smooth 9-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003426
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003428|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=9
§FORM := S^{9} is a smooth 9-manifold, while S^9×S^1 is a smooth 10-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003427
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003429|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=10
§FORM := S^{10} is a smooth 10-manifold, while S^10×S^1 is a smooth 11-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003428
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003430|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=11
§FORM := S^{11} is a smooth 11-manifold, while S^11×S^1 is a smooth 12-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003429
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003431|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=12
§FORM := S^{12} is a smooth 12-manifold, while S^12×S^1 is a smooth 13-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003430
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003432|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=13
§FORM := S^{13} is a smooth 13-manifold, while S^13×S^1 is a smooth 14-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003431
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003433|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=14
§FORM := S^{14} is a smooth 14-manifold, while S^14×S^1 is a smooth 15-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003432
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003434|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=15
§FORM := S^{15} is a smooth 15-manifold, while S^15×S^1 is a smooth 16-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003433
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003435|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=16
§FORM := S^{16} is a smooth 16-manifold, while S^16×S^1 is a smooth 17-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003434
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003436|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=17
§FORM := S^{17} is a smooth 17-manifold, while S^17×S^1 is a smooth 18-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003435
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003437|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=18
§FORM := S^{18} is a smooth 18-manifold, while S^18×S^1 is a smooth 19-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003436
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003438|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=19
§FORM := S^{19} is a smooth 19-manifold, while S^19×S^1 is a smooth 20-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003437
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003439|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=20
§FORM := S^{20} is a smooth 20-manifold, while S^20×S^1 is a smooth 21-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003438
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003440|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=21
§FORM := S^{21} is a smooth 21-manifold, while S^21×S^1 is a smooth 22-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003439
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003441|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=22
§FORM := S^{22} is a smooth 22-manifold, while S^22×S^1 is a smooth 23-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003440
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003442|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=23
§FORM := S^{23} is a smooth 23-manifold, while S^23×S^1 is a smooth 24-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003441
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003443|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=24
§FORM := S^{24} is a smooth 24-manifold, while S^24×S^1 is a smooth 25-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003442
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003444|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=25
§FORM := S^{25} is a smooth 25-manifold, while S^25×S^1 is a smooth 26-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003443
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003445|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=26
§FORM := S^{26} is a smooth 26-manifold, while S^26×S^1 is a smooth 27-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003444
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003446|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=27
§FORM := S^{27} is a smooth 27-manifold, while S^27×S^1 is a smooth 28-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003445
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003447|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=28
§FORM := S^{28} is a smooth 28-manifold, while S^28×S^1 is a smooth 29-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003446
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003448|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=29
§FORM := S^{29} is a smooth 29-manifold, while S^29×S^1 is a smooth 30-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003447
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003449|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=30
§FORM := S^{30} is a smooth 30-manifold, while S^30×S^1 is a smooth 31-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003448
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003450|E|M|Proof+Compile+Meta+Topo] Manifold dimension profile n=31
§FORM := S^{31} is a smooth 31-manifold, while S^31×S^1 is a smooth 32-manifold; dimensions add under finite products.
§DEP_HARD := S26-E-00003449
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003451|D|A|Proof+Compile+Meta+Topo] Derivation model of tangent vectors
§FORM := A tangent vector at p can be defined as a derivation v:C^infinity_p(M)->R satisfying the Leibniz rule.
§DEP_HARD := S26-E-00003450
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003452|T|A|Proof+Compile+Meta+Topo] Equivalence of curve and derivation tangent models
§FORM := Velocity classes of curves and derivations define canonically isomorphic tangent spaces on a smooth manifold.
§DEP_HARD := S26-D-00003451
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003453|D|A|Proof+Compile+Meta+Topo] Pushforward differential
§FORM := For smooth F:M->N, dF_p:T_pM->T_{F(p)}N sends tangent vectors by differentiating pullback of functions.
§DEP_HARD := S26-T-00003452
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003454|T|A|Proof+Compile+Meta+Topo] Functoriality of the differential
§FORM := d(G∘F)_p=dG_{F(p)}∘dF_p and d(id)_p=id.
§DEP_HARD := S26-D-00003453
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003455|D|A|Proof+Compile+Meta+Topo] Vector field as a section of TM
§FORM := A smooth vector field X assigns X_p∈T_pM smoothly, equivalently a smooth section of the tangent bundle.
§DEP_HARD := S26-T-00003454
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003456|D|A|Proof+Compile+Meta+Topo] Lie bracket of vector fields
§FORM := [X,Y](f)=X(Yf)-Y(Xf) defines a vector field and a Lie algebra structure on smooth vector fields.
§DEP_HARD := S26-D-00003455
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003457|T|A|Proof+Compile+Meta+Topo] Naturality of Lie brackets under diffeomorphism
§FORM := If F is a diffeomorphism, F_*[X,Y]=[F_*X,F_*Y].
§DEP_HARD := S26-D-00003456
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003458|D|A|Proof+Compile+Meta+Topo] Integral curve of a vector field
§FORM := An integral curve gamma satisfies gamma dot(t)=X_{gamma(t)}.
§DEP_HARD := S26-T-00003457
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003459|T|A|Proof+Compile+Meta+Topo] Local existence and uniqueness of flows
§FORM := A smooth vector field admits a unique local flow through each initial point.
§DEP_HARD := S26-D-00003458
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003460|D|A|Proof+Compile+Meta+Topo] Complete vector field
§FORM := A vector field is complete when its flow exists for all real times.
§DEP_HARD := S26-T-00003459
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003461|T|A|Proof+Compile+Meta+Topo] Compact-manifold completeness
§FORM := Every smooth vector field on a compact manifold is complete.
§DEP_HARD := S26-D-00003460
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003462|D|A|Proof+Compile+Meta+Topo] Flow box theorem
§FORM := Near a point where X is nonzero, coordinates can be chosen in which X is a coordinate vector field.
§DEP_HARD := S26-T-00003461
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003463|D|A|Proof+Compile+Meta+Topo] Distribution of tangent subspaces
§FORM := A rank-k distribution assigns a k-dimensional subspace D_p⊂T_pM smoothly.
§DEP_HARD := S26-D-00003462
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003464|D|A|Proof+Compile+Meta+Topo] Involutive distribution
§FORM := A distribution is involutive when brackets of tangent vector fields remain tangent to the distribution.
§DEP_HARD := S26-D-00003463
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003465|T|A|Proof+Compile+Meta+Topo] Frobenius theorem
§FORM := A smooth constant-rank distribution is integrable by submanifolds iff it is involutive.
§DEP_HARD := S26-D-00003464
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003466|D|A|Proof+Compile+Meta+Topo] Foliation interface
§FORM := A foliation decomposes a manifold into immersed leaves locally modeled by parallel coordinate slices.
§DEP_HARD := S26-T-00003465
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003467|D|A|Proof+Compile+Meta+Topo] Transversality of maps
§FORM := Maps or submanifolds are transverse when the relevant tangent images sum to the ambient tangent space at intersections.
§DEP_HARD := S26-D-00003466
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003468|T|A|Proof+Compile+Meta+Topo] Transverse preimage theorem

§FORM := If F is transverse to a submanifold S⊂N, then F^{-1}(S) is a submanifold of the expected codimension.
§DEP_HARD := S26-D-00003467
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003469|N|A|Proof+Compile+Meta+Topo] Local-flow fence

§FORM := Local existence of a smooth flow does not imply completeness; solutions can escape every compact set in finite time on noncompact manifolds.
§DEP_HARD := S26-T-00003468
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003470|B|A|Proof+Compile+Meta+Topo] Vector-fields to dynamics bridge

§FORM := Flows convert differential-geometric vector fields into continuous-time dynamical systems while retaining coordinate-invariant meaning.
§DEP_HARD := S26-N-00003469
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003471|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=2

§FORM := For X(x)=x on R^2, the Jacobian is the 2×2 identity, divergence is exactly 2, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-B-00003470
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003472|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=3

§FORM := For X(x)=x on R^3, the Jacobian is the 3×3 identity, divergence is exactly 3, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003471
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003473|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=4

§FORM := For X(x)=x on R^4, the Jacobian is the 4×4 identity, divergence is exactly 4, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003472
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003474|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=5

§FORM := For X(x)=x on R^5, the Jacobian is the 5×5 identity, divergence is exactly 5, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003473
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003475|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=6

§FORM := For X(x)=x on R^6, the Jacobian is the 6×6 identity, divergence is exactly 6, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003474
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003476|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=7

§FORM := For X(x)=x on R^7, the Jacobian is the 7×7 identity, divergence is exactly 7, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003475
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003477|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=8

§FORM := For X(x)=x on R^8, the Jacobian is the 8×8 identity, divergence is exactly 8, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003476
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003478|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=9

§FORM := For X(x)=x on R^9, the Jacobian is the 9×9 identity, divergence is exactly 9, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003477
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003479|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=10

§FORM := For X(x)=x on R^10, the Jacobian is the 10×10 identity, divergence is exactly 10, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003478
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003480|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=11

§FORM := For X(x)=x on R^11, the Jacobian is the 11×11 identity, divergence is exactly 11, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003479
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003481|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=12
§FORM := For X(x)=x on R^12, the Jacobian is the 12×12 identity, divergence is exactly 12, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003480
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003482|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=13
§FORM := For X(x)=x on R^13, the Jacobian is the 13×13 identity, divergence is exactly 13, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003481
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003483|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=14
§FORM := For X(x)=x on R^14, the Jacobian is the 14×14 identity, divergence is exactly 14, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003482
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003484|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=15
§FORM := For X(x)=x on R^15, the Jacobian is the 15×15 identity, divergence is exactly 15, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003483
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003485|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=16
§FORM := For X(x)=x on R^16, the Jacobian is the 16×16 identity, divergence is exactly 16, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003484
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003486|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=17
§FORM := For X(x)=x on R^17, the Jacobian is the 17×17 identity, divergence is exactly 17, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003485
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003487|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=18
§FORM := For X(x)=x on R^18, the Jacobian is the 18×18 identity, divergence is exactly 18, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003486
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003488|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=19
§FORM := For X(x)=x on R^19, the Jacobian is the 19×19 identity, divergence is exactly 19, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003487
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003489|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=20
§FORM := For X(x)=x on R^20, the Jacobian is the 20×20 identity, divergence is exactly 20, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003488
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003490|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=21
§FORM := For X(x)=x on R^21, the Jacobian is the 21×21 identity, divergence is exactly 21, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003489
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003491|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=22
§FORM := For X(x)=x on R^22, the Jacobian is the 22×22 identity, divergence is exactly 22, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003490
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003492|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=23
§FORM := For X(x)=x on R^23, the Jacobian is the 23×23 identity, divergence is exactly 23, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003491
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003493|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=24
§FORM := For X(x)=x on R^24, the Jacobian is the 24×24 identity, divergence is exactly 24, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003492
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003494|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=25
§FORM := For X(x)=x on R^25, the Jacobian is the 25×25 identity, divergence is exactly 25, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003493
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003495|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=26

§FORM := For X(x)=x on R^26, the Jacobian is the 26×26 identity, divergence is exactly 26, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003494
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003496|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=27

§FORM := For X(x)=x on R^27, the Jacobian is the 27×27 identity, divergence is exactly 27, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003495
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003497|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=28

§FORM := For X(x)=x on R^28, the Jacobian is the 28×28 identity, divergence is exactly 28, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003496
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003498|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=29

§FORM := For X(x)=x on R^29, the Jacobian is the 29×29 identity, divergence is exactly 29, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003497
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003499|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=30

§FORM := For X(x)=x on R^30, the Jacobian is the 30×30 identity, divergence is exactly 30, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003498
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003500|E|M|Proof+Compile+Meta+Topo] Linear vector-field profile dimension=31

§FORM := For X(x)=x on R^31, the Jacobian is the 31×31 identity, divergence is exactly 31, and the flow is Phi_t(x)=e^t x.
§DEP_HARD := S26-E-00003499
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003501|D|A|Proof+Compile+Meta+Topo] Differential k-form on a manifold refinement

§FORM := A smooth k-form is a smooth section of the exterior-power cotangent bundle Lambda^k T^*M.
§DEP_HARD := S26-E-00003500
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003502|D|A|Proof+Compile+Meta+Topo] Wedge product of differential forms refinement

§FORM := The wedge product combines p- and q-forms into a (p+q)-form and is graded-anticommutative.
§DEP_HARD := S26-D-00003501
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003503|T|A|Proof+Compile+Meta+Topo] Exterior derivative squares to zero

§FORM := For differential forms, d∘d=0.
§DEP_HARD := S26-D-00003502
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003504|T|A|Proof+Compile+Meta+Topo] Leibniz rule for exterior derivative

§FORM := d(alpha∧beta)=dalpha∧beta+(-1)^p alpha∧dbeta for alpha of degree p.
§DEP_HARD := S26-T-00003503
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003505|T|A|Proof+Compile+Meta+Topo] Naturality of exterior derivative under manifold pullback refinement

§FORM := For smooth F:M->N, d(F^*alpha)=F^*(dalpha).
§DEP_HARD := S26-T-00003504, S26-T-00000927
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003506|D|A|Proof+Compile+Meta+Topo] Closed differential form

§FORM := A form alpha is closed when d alpha=0.
§DEP_HARD := S26-T-00003505
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003507|D|A|Proof+Compile+Meta+Topo] Exact differential form

§FORM := A form alpha is exact when alpha=d beta for some lower-degree beta.
§DEP_HARD := S26-D-00003506
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003508|D|A|Proof+Compile+Meta+Topo] de Rham cohomology refinement

§FORM := H^k_dR(M)=ker(d:Omega^k->Omega^{k+1})/im(d:Omega^{k-1}->Omega^k).
§DEP_HARD := S26-D-00003507
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003509|T|A|Proof+Compile+Meta+Topo] Poincare lemma

§FORM := Every closed differential k-form with k>0 is locally exact on a star-shaped or contractible coordinate domain.

§DEP_HARD := S26-D-00003508

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003510|T|A|Proof+Compile+Meta+Topo] de Rham theorem

§FORM := Integration of differential forms over singular chains induces an isomorphism H^*_dR(M)≅H^*(M;R).

§DEP_HARD := S26-T-00003509

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003511|D|A|Proof+Compile+Meta+Topo] Orientation form

§FORM := On an oriented n-manifold, a nowhere-vanishing top-degree form determines the orientation in the smooth category.

§DEP_HARD := S26-T-00003510

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003512|D|A|Proof+Compile+Meta+Topo] Integration of compactly supported top forms refinement

§FORM := A compactly supported top-degree form can be integrated on an oriented manifold using oriented charts and a partition of unity.

§DEP_HARD := S26-D-00003511, S26-D-00000933

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003513|T|A|Proof+Compile+Meta+Topo] Stokes theorem on manifolds

§FORM := For a compact oriented manifold with boundary, integral_M d omega = integral_{∂M} omega for appropriate degree and support/regularity.

§DEP_HARD := S26-D-00003512

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003514|D|A|Proof+Compile+Meta+Topo] Compactly supported de Rham cohomology

§FORM := H^k_{dR,c}(M) is computed from compactly supported differential forms.

§DEP_HARD := S26-T-00003513

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003515|D|A|Proof+Compile+Meta+Topo] Pullback on de Rham cohomology

§FORM := A smooth map induces a graded-ring homomorphism on de Rham cohomology by pullback.

§DEP_HARD := S26-D-00003514

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003516|D|A|Proof+Compile+Meta+Topo+IG+Tower] Lie derivative of forms

§FORM := The Lie derivative L_X alpha measures infinitesimal change of alpha along the flow of X.

§DEP_HARD := S26-D-00003515

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00003517|I|A|Proof+Compile+Meta+Topo] Cartan magic formula

§FORM := L_X alpha=d(i_X alpha)+i_X(d alpha).

§DEP_HARD := S26-D-00003516

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003518|N|A|Proof+Compile+Meta+Topo] Local-global exactness fence

§FORM := The Poincare lemma is local; a closed form need not be globally exact when de Rham cohomology is nontrivial.

§DEP_HARD := S26-I-00003517

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003519|B|A|Proof+Compile+Meta+Topo] Forms to electromagnetism bridge

§FORM := Closed/exact differential forms and Stokes-type identities provide coordinate-free expressions for flux, circulation and conservation laws.

§DEP_HARD := S26-N-00003518

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003520|B|A|Proof+Compile+Meta+Topo] de Rham to topology bridge

§FORM := The de Rham theorem turns analytic differential-form computations into topological cohomology invariants.

§DEP_HARD := S26-B-00003519

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003521|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=2

§FORM := For the n-torus T^2, de Rham Betti numbers are binomial(2,j); their total rank is 4 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-B-00003520

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003522|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=3

§FORM := For the n-torus T^3, de Rham Betti numbers are binomial(3,j); their total rank is 8 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003521
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003523|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=4

§FORM := For the n-torus T^4, de Rham Betti numbers are binomial(4,j); their total rank is 16 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003522
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003524|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=5

§FORM := For the n-torus T^5, de Rham Betti numbers are binomial(5,j); their total rank is 32 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003523
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003525|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=6

§FORM := For the n-torus T^6, de Rham Betti numbers are binomial(6,j); their total rank is 64 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003524
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003526|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=7

§FORM := For the n-torus T^7, de Rham Betti numbers are binomial(7,j); their total rank is 128 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003525
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003527|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=8

§FORM := For the n-torus T^8, de Rham Betti numbers are binomial(8,j); their total rank is 256 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003526
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003528|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=9

§FORM := For the n-torus T^9, de Rham Betti numbers are binomial(9,j); their total rank is 512 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003527
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003529|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=10

§FORM := For the n-torus T^10, de Rham Betti numbers are binomial(10,j); their total rank is 1024 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003528
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003530|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=11

§FORM := For the n-torus T^11, de Rham Betti numbers are binomial(11,j); their total rank is 2048 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003529
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003531|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=12

§FORM := For the n-torus T^12, de Rham Betti numbers are binomial(12,j); their total rank is 4096 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003530
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003532|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=13

§FORM := For the n-torus T^13, de Rham Betti numbers are binomial(13,j); their total rank is 8192 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003531
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003533|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=14

§FORM := For the n-torus T^14, de Rham Betti numbers are binomial(14,j); their total rank is 16384 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003532
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003534|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=15

§FORM := For the n-torus T^15, de Rham Betti numbers are binomial(15,j); their total rank is 32768 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003533
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003535|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=16

§FORM := For the n-torus T^16, de Rham Betti numbers are binomial(16,j); their total rank is 65536 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003534
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003536|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=17

§FORM := For the n-torus T^17, de Rham Betti numbers are binomial(17,j); their total rank is 131072 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003535

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003537|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=18

§FORM := For the n-torus T^18, de Rham Betti numbers are binomial(18,j); their total rank is 262144 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003536

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003538|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=19

§FORM := For the n-torus T^19, de Rham Betti numbers are binomial(19,j); their total rank is 524288 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003537

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003539|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=20

§FORM := For the n-torus T^20, de Rham Betti numbers are binomial(20,j); their total rank is 1048576 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003538

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003540|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=21

§FORM := For the n-torus T^21, de Rham Betti numbers are binomial(21,j); their total rank is 2097152 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003539

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003541|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=22

§FORM := For the n-torus T^22, de Rham Betti numbers are binomial(22,j); their total rank is 4194304 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003540

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003542|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=23

§FORM := For the n-torus T^23, de Rham Betti numbers are binomial(23,j); their total rank is 8388608 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003541

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003543|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=24

§FORM := For the n-torus T^24, de Rham Betti numbers are binomial(24,j); their total rank is 16777216 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003542

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003544|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=25

§FORM := For the n-torus T^25, de Rham Betti numbers are binomial(25,j); their total rank is 33554432 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003543

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003545|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=26

§FORM := For the n-torus T^26, de Rham Betti numbers are binomial(26,j); their total rank is 67108864 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003544

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003546|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=27

§FORM := For the n-torus T^27, de Rham Betti numbers are binomial(27,j); their total rank is 134217728 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003545

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003547|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=28

§FORM := For the n-torus T^28, de Rham Betti numbers are binomial(28,j); their total rank is 268435456 and Euler characteristic is 0 for n>0.

§DEP_HARD := S26-E-00003546

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003548|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=29
§FORM := For the n-torus T^29, de Rham Betti numbers are binomial(29,j); their total rank is 536870912 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003547
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003549|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=30
§FORM := For the n-torus T^30, de Rham Betti numbers are binomial(30,j); their total rank is 1073741824 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003548
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003550|E|M|Proof+Compile+Meta+Topo] Torus de Rham profile dimension=31
§FORM := For the n-torus T^31, de Rham Betti numbers are binomial(31,j); their total rank is 2147483648 and Euler characteristic is 0 for n>0.
§DEP_HARD := S26-E-00003549
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003551|D|A|Proof+Compile+Meta+Topo] Fiber bundle
§FORM := A fiber bundle p:E->B with fiber F is locally trivial: each b has a neighborhood U with p^{-1}(U) homeomorphic or diffeomorphic to U×F compatibly with projection.
§DEP_HARD := S26-E-00003550
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003552|D|A|Proof+Compile+Meta+Topo] Bundle chart and transition function
§FORM := Over overlaps of trivializing neighborhoods, bundle charts differ by transition maps valued in the structure group acting on the fiber.
§DEP_HARD := S26-D-00003551
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003553|D|A|Proof+Compile+Meta+Topo] Vector bundle
§FORM := A rank-r vector bundle is a fiber bundle whose fibers are r-dimensional vector spaces and whose transition functions are linear.
§DEP_HARD := S26-D-00003552
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003554|D|A|Proof+Compile+Meta+Topo] Bundle section
§FORM := A section s:B->E satisfies p∘s=id_B.
§DEP_HARD := S26-D-00003553
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003555|D|A|Proof+Compile+Meta+Topo] Trivial bundle
§FORM := A bundle is trivial when it is bundle-isomorphic to B×F with the product projection.
§DEP_HARD := S26-D-00003554
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003556|T|A|Proof+Compile+Meta+Topo] Vector bundle triviality by global frame
§FORM := A rank-r vector bundle is trivial iff it admits r pointwise linearly independent global sections forming a frame.
§DEP_HARD := S26-D-00003555
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003557|D|A|Proof+Compile+Meta+Topo] Pullback bundle
§FORM := For f:X->B and p:E->B, the pullback f^*E consists of pairs (x,e) with f(x)=p(e) and is a bundle over X.
§DEP_HARD := S26-T-00003556
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003558|T|A|Proof+Compile+Meta+Topo] Functoriality of pullback bundles
§FORM := Pullback is compatible with identity and composition up to canonical bundle isomorphism.
§DEP_HARD := S26-D-00003557
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003559|D|A|Proof+Compile+Meta+Topo] Whitney sum of vector bundles
§FORM := E⊕F over the same base has fiber E_b⊕F_b and transition functions given by block sums.
§DEP_HARD := S26-T-00003558
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003560|D|A|Proof+Compile+Meta+Topo] Tensor and dual bundles
§FORM := Fiberwise tensor product and dualization define vector bundles E⊗F and E^*.
§DEP_HARD := S26-D-00003559
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003561|D|A|Proof+Compile+Meta+Topo] Principal G-bundle

§FORM := A principal G-bundle is a fiber bundle with free transitive right G-action on each fiber and local G-equivariant trivializations.
§DEP_HARD := S26-D-00003560
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003562|D|A|Proof+Compile+Meta+Topo] Associated bundle

§FORM := Given a principal G-bundle P and a G-space F, P×_G F is the associated bundle with fiber F.
§DEP_HARD := S26-D-00003561
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003563|D|A|Proof+Compile+Meta+Topo] Gauge transformation

§FORM := A gauge transformation is an automorphism of a principal bundle covering the identity on the base.
§DEP_HARD := S26-D-00003562
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003564|D|A|Proof+Compile+Meta+Topo] Clutching construction

§FORM := Bundles over spheres and other decomposed spaces can be formed by gluing trivial bundles along overlap via a clutching map into the structure group.
§DEP_HARD := S26-D-00003563
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003565|T|A|Proof+Compile+Meta+Topo] Classification of line bundles by projective maps interface

§FORM := Under standard paracompactness hypotheses, real or complex line bundles are classified by homotopy classes into the corresponding infinite projective space.
§DEP_HARD := S26-D-00003564
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003566|D|A|Proof+Compile+Meta+Topo] Tangent bundle as a vector bundle

§FORM := TM->M is the rank-dim M vector bundle whose fiber at p is T_pM.
§DEP_HARD := S26-T-00003565
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003567|D|A|Proof+Compile+Meta+Topo] Normal bundle

§FORM := For an embedded submanifold S⊂M with a chosen Riemannian splitting or quotient formulation, the normal bundle records directions transverse to S.
§DEP_HARD := S26-D-00003566
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003568|N|A|Proof+Compile+Meta+Topo] Local-triviality fence

§FORM := Fiberwise isomorphism of sets or vector spaces does not by itself give a bundle; local triviality and compatible topology/smoothness are essential.
§DEP_HARD := S26-D-00003567
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003569|B|A|Proof+Compile+Meta+Topo] Bundles to cohomology bridge

§FORM := Characteristic classes assign cohomology classes to bundles and detect obstructions to triviality or existence of sections.
§DEP_HARD := S26-N-00003568
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003570|B|A|Proof+Compile+Meta+Topo] Principal bundles to gauge theory bridge

§FORM := Connections and curvature on principal bundles provide the geometric substrate of gauge fields and holonomy.
§DEP_HARD := S26-B-00003569
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003571|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=2

§FORM := The trivial rank-2 real vector bundle over S^1 has total-space dimension 3 and admits the explicit global frame of 2 constant coordinate sections.
§DEP_HARD := S26-B-00003570
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003572|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=3

§FORM := The trivial rank-3 real vector bundle over S^1 has total-space dimension 4 and admits the explicit global frame of 3 constant coordinate sections.
§DEP_HARD := S26-E-00003571
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003573|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=4

§FORM := The trivial rank-4 real vector bundle over S^1 has total-space dimension 5 and admits the explicit global frame of 4 constant coordinate sections.

§DEP_HARD := S26-E-00003572

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003574|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=5

§FORM := The trivial rank-5 real vector bundle over S^1 has total-space dimension 6 and admits the explicit global frame of 5 constant coordinate sections.

§DEP_HARD := S26-E-00003573

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003575|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=6

§FORM := The trivial rank-6 real vector bundle over S^1 has total-space dimension 7 and admits the explicit global frame of 6 constant coordinate sections.

§DEP_HARD := S26-E-00003574

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003576|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=7

§FORM := The trivial rank-7 real vector bundle over S^1 has total-space dimension 8 and admits the explicit global frame of 7 constant coordinate sections.

§DEP_HARD := S26-E-00003575

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003577|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=8

§FORM := The trivial rank-8 real vector bundle over S^1 has total-space dimension 9 and admits the explicit global frame of 8 constant coordinate sections.

§DEP_HARD := S26-E-00003576

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003578|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=9

§FORM := The trivial rank-9 real vector bundle over S^1 has total-space dimension 10 and admits the explicit global frame of 9 constant coordinate sections.

§DEP_HARD := S26-E-00003577

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003579|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=10

§FORM := The trivial rank-10 real vector bundle over S^1 has total-space dimension 11 and admits the explicit global frame of 10 constant coordinate sections.

§DEP_HARD := S26-E-00003578

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003580|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=11

§FORM := The trivial rank-11 real vector bundle over S^1 has total-space dimension 12 and admits the explicit global frame of 11 constant coordinate sections.

§DEP_HARD := S26-E-00003579

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003581|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=12

§FORM := The trivial rank-12 real vector bundle over S^1 has total-space dimension 13 and admits the explicit global frame of 12 constant coordinate sections.

§DEP_HARD := S26-E-00003580

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003582|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=13

§FORM := The trivial rank-13 real vector bundle over S^1 has total-space dimension 14 and admits the explicit global frame of 13 constant coordinate sections.

§DEP_HARD := S26-E-00003581

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003583|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=14

§FORM := The trivial rank-14 real vector bundle over S^1 has total-space dimension 15 and admits the explicit global frame of 14 constant coordinate sections.

§DEP_HARD := S26-E-00003582

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003584|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=15

§FORM := The trivial rank-15 real vector bundle over S^1 has total-space dimension 16 and admits the explicit global frame of 15 constant coordinate sections.

§DEP_HARD := S26-E-00003583

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003585|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=16

§FORM := The trivial rank-16 real vector bundle over S^1 has total-space dimension 17 and admits the explicit global frame of 16 constant coordinate sections.

§DEP_HARD := S26-E-00003584

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003586|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=17

§FORM := The trivial rank-17 real vector bundle over S^1 has total-space dimension 18 and admits the explicit global frame of 17 constant coordinate sections.

§DEP_HARD := S26-E-00003585

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003587|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=18

§FORM := The trivial rank-18 real vector bundle over S^1 has total-space dimension 19 and admits the explicit global frame of 18 constant coordinate sections.

§DEP_HARD := S26-E-00003586

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003588|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=19

§FORM := The trivial rank-19 real vector bundle over S^1 has total-space dimension 20 and admits the explicit global frame of 19 constant coordinate sections.

§DEP_HARD := S26-E-00003587

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003589|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=20

§FORM := The trivial rank-20 real vector bundle over S^1 has total-space dimension 21 and admits the explicit global frame of 20 constant coordinate sections.

§DEP_HARD := S26-E-00003588

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003590|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=21

§FORM := The trivial rank-21 real vector bundle over S^1 has total-space dimension 22 and admits the explicit global frame of 21 constant coordinate sections.

§DEP_HARD := S26-E-00003589

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003591|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=22

§FORM := The trivial rank-22 real vector bundle over S^1 has total-space dimension 23 and admits the explicit global frame of 22 constant coordinate sections.

§DEP_HARD := S26-E-00003590

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003592|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=23

§FORM := The trivial rank-23 real vector bundle over S^1 has total-space dimension 24 and admits the explicit global frame of 23 constant coordinate sections.

§DEP_HARD := S26-E-00003591

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003593|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=24

§FORM := The trivial rank-24 real vector bundle over S^1 has total-space dimension 25 and admits the explicit global frame of 24 constant coordinate sections.

§DEP_HARD := S26-E-00003592

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003594|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=25

§FORM := The trivial rank-25 real vector bundle over S^1 has total-space dimension 26 and admits the explicit global frame of 25 constant coordinate sections.

§DEP_HARD := S26-E-00003593

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003595|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=26

§FORM := The trivial rank-26 real vector bundle over S^1 has total-space dimension 27 and admits the explicit global frame of 26 constant coordinate sections.

§DEP_HARD := S26-E-00003594

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003596|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=27

§FORM := The trivial rank-27 real vector bundle over S^1 has total-space dimension 28 and admits the explicit global frame of 27 constant coordinate sections.

§DEP_HARD := S26-E-00003595

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003597|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=28

§FORM := The trivial rank-28 real vector bundle over S^1 has total-space dimension 29 and admits the explicit global frame of 28 constant coordinate sections.

§DEP_HARD := S26-E-00003596

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003598|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=29

§FORM := The trivial rank-29 real vector bundle over S^1 has total-space dimension 30 and admits the explicit global frame of 29 constant coordinate sections.

§DEP_HARD := S26-E-00003597

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003599|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=30

§FORM := The trivial rank-30 real vector bundle over S^1 has total-space dimension 31 and admits the explicit global frame of 30 constant coordinate sections.

§DEP_HARD := S26-E-00003598

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003600|E|M|Proof+Compile+Meta+Topo] Trivial vector-bundle profile rank=31

§FORM := The trivial rank-31 real vector bundle over S^1 has total-space dimension 32 and admits the explicit global frame of 31 constant coordinate sections.

§DEP_HARD := S26-E-00003599

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003601|D|A|Proof+Compile+Meta+Topo] Characteristic class

§FORM := A characteristic class assigns to each bundle E->B a cohomology class c(E) naturally under pullback.

§DEP_HARD := S26-E-00003600

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003602|D|A|Proof+Compile+Meta+Topo] Stiefel-Whitney classes

§FORM := For a real vector bundle E, w_i(E)∈H^i(B;Z/2) are its Stiefel-Whitney classes with total class w(E)=1+w_1+w_2+....

§DEP_HARD := S26-D-00003601

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003603|T|A|Proof+Compile+Meta+Topo] Whitney product formula for Stiefel-Whitney classes

§FORM := w(E⊕F)=w(E) cup w(F).

§DEP_HARD := S26-D-00003602

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003604|D|A|Proof+Compile+Meta+Topo] First Stiefel-Whitney obstruction

§FORM := w_1(E) is the primary obstruction to orientability of a real vector bundle.

§DEP_HARD := S26-T-00003603

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003605|D|A|Proof+Compile+Meta+Topo] Second Stiefel-Whitney spin obstruction interface

§FORM := For an oriented real vector bundle, w_2 is the obstruction to a spin structure in the standard theory.

§DEP_HARD := S26-D-00003604

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003606|D|A|Proof+Compile+Meta+Topo] Chern classes

§FORM := For a complex vector bundle E, c_i(E)∈H^{2i}(B;Z) are its Chern classes.

§DEP_HARD := S26-D-00003605

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003607|T|A|Proof+Compile+Meta+Topo] Whitney product formula for Chern classes

§FORM := c(E⊕F)=c(E) cup c(F).

§DEP_HARD := S26-D-00003606

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003608|D|A|Proof+Compile+Meta+Topo] First Chern class of a complex line bundle

§FORM := Complex line bundles have a characteristic class c_1∈H^2(B;Z) that classifies them over broad paracompact bases.

§DEP_HARD := S26-T-00003607

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003609|D|A|Proof+Compile+Meta+Topo] Pontryagin classes

§FORM := For a real vector bundle E, Pontryagin classes p_i(E)∈H^{4i}(B;Z) are defined via Chern classes of the complexification with standard signs.

§DEP_HARD := S26-D-00003608

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003610|D|A|Proof+Compile+Meta+Topo] Euler class

§FORM := An oriented rank-n real vector bundle has Euler class e(E)∈H^n(B;Z).

§DEP_HARD := S26-D-00003609

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003611|T|A|Proof+Compile+Meta+Topo] Euler class and zero sections interface

§FORM := Under transversality, the Euler class pairs with the fundamental class to count signed zeros of a generic section on a closed oriented manifold.

§DEP_HARD := S26-D-00003610

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003612|D|A|Proof+Compile+Meta+Topo] Chern character

§FORM := ch(E) is a rational even-degree cohomology class additive under direct sum and multiplicative under tensor product.

§DEP_HARD := S26-T-00003611

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003613|T|A|Proof+Compile+Meta+Topo] Naturality of characteristic classes

§FORM := For f:X->B, characteristic classes satisfy c(f^*E)=f^*c(E).

§DEP_HARD := S26-D-00003612

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003614|D|A|Proof+Compile+Meta+Topo] Classifying space BG

§FORM := Principal G-bundles over suitable bases are classified up to isomorphism by homotopy classes of maps into BG.

§DEP_HARD := S26-T-00003613

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003615|D|A|Proof+Compile+Meta+Topo] Universal bundle EG->BG

§FORM := A universal principal G-bundle has contractible total space EG with free G-action and quotient BG.

§DEP_HARD := S26-D-00003614

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003616|D|A|Proof+Compile+Meta+Topo] Thom class interface

§FORM := An oriented rank-r vector bundle admits a Thom class in compactly supported relative cohomology, inducing the Thom isomorphism.

§DEP_HARD := S26-D-00003615

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003617|T|A|Proof+Compile+Meta+Topo] Thom isomorphism interface

§FORM := Cup product with the Thom class identifies cohomology of the base with degree-shifted compactly supported cohomology of the bundle under orientation hypotheses.

§DEP_HARD := S26-D-00003616

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003618|N|A|Proof+Compile+Meta+Topo] Vanishing-class fence

§FORM := Vanishing of selected characteristic classes need not imply bundle triviality; characteristic classes are powerful but generally incomplete invariants.

§DEP_HARD := S26-T-00003617

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003619|B|A|Proof+Compile+Meta+Topo] Characteristic classes to curvature bridge

§FORM := Chern-Weil theory represents real characteristic classes by invariant polynomials in curvature of a connection.

§DEP_HARD := S26-N-00003618

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003620|B|A|Proof+Compile+Meta+Topo] Characteristic classes to topology bridge

§FORM := Characteristic numbers obtained by pairing classes with fundamental cycles constrain manifold topology and cobordism.

§DEP_HARD := S26-B-00003619

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003621|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=2

§FORM := For the tangent bundle TS^2, pairing the Euler class with [S^2] equals chi(S^2)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-B-00003620

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003622|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=3

§FORM := For the tangent bundle TS^3, pairing the Euler class with [S^3] equals chi(S^3)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003621

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003623|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=4

§FORM := For the tangent bundle TS^4, pairing the Euler class with [S^4] equals chi(S^4)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003622

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003624|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=5

§FORM := For the tangent bundle TS^5, pairing the Euler class with [S^5] equals chi(S^5)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003623

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003625|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=6

§FORM := For the tangent bundle TS^6, pairing the Euler class with [S^6] equals chi(S^6)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003624

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003626|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=7

§FORM := For the tangent bundle TS^7, pairing the Euler class with [S^7] equals chi(S^7)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003625

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003627|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=8

§FORM := For the tangent bundle TS^8, pairing the Euler class with [S^8] equals chi(S^8)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003626

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003628|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=9

§FORM := For the tangent bundle TS^9, pairing the Euler class with [S^9] equals chi(S^9)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003627

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003629|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=10

§FORM := For the tangent bundle TS^10, pairing the Euler class with [S^10] equals chi(S^10)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003628

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003630|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=11

§FORM := For the tangent bundle TS^11, pairing the Euler class with [S^11] equals chi(S^11)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003629

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003631|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=12

§FORM := For the tangent bundle TS^12, pairing the Euler class with [S^12] equals chi(S^12)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003630

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003632|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=13

§FORM := For the tangent bundle TS^13, pairing the Euler class with [S^13] equals chi(S^13)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003631

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003633|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=14

§FORM := For the tangent bundle TS^14, pairing the Euler class with [S^14] equals chi(S^14)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003632

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003634|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=15

§FORM := For the tangent bundle TS^15, pairing the Euler class with [S^15] equals chi(S^15)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003633

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003635|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=16

§FORM := For the tangent bundle TS^16, pairing the Euler class with [S^16] equals chi(S^16)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003634

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003636|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=17

§FORM := For the tangent bundle TS^17, pairing the Euler class with [S^17] equals chi(S^17)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003635

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003637|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=18

§FORM := For the tangent bundle TS^18, pairing the Euler class with [S^18] equals chi(S^18)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003636

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003638|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=19

§FORM := For the tangent bundle TS^19, pairing the Euler class with [S^19] equals chi(S^19)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003637

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003639|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=20

§FORM := For the tangent bundle TS^20, pairing the Euler class with [S^20] equals chi(S^20)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003638

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003640|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=21

§FORM := For the tangent bundle TS^21, pairing the Euler class with [S^21] equals chi(S^21)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003639

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003641|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=22

§FORM := For the tangent bundle TS^22, pairing the Euler class with [S^22] equals chi(S^22)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003640

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003642|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=23

§FORM := For the tangent bundle TS^23, pairing the Euler class with [S^23] equals chi(S^23)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003641

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003643|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=24

§FORM := For the tangent bundle TS^24, pairing the Euler class with [S^24] equals chi(S^24)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003642

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003644|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=25

§FORM := For the tangent bundle TS^25, pairing the Euler class with [S^25] equals chi(S^25)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003643

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003645|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=26

§FORM := For the tangent bundle TS^26, pairing the Euler class with [S^26] equals chi(S^26)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003644

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003646|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=27

§FORM := For the tangent bundle TS^27, pairing the Euler class with [S^27] equals chi(S^27)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003645

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003647|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=28

§FORM := For the tangent bundle TS^28, pairing the Euler class with [S^28] equals chi(S^28)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003646

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003648|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=29

§FORM := For the tangent bundle TS^29, pairing the Euler class with [S^29] equals chi(S^29)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003647

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003649|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=30

§FORM := For the tangent bundle TS^30, pairing the Euler class with [S^30] equals chi(S^30)=2; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003648

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003650|E|M|Proof+Compile+Meta+Topo] Sphere tangent Euler-number profile n=31

§FORM := For the tangent bundle TS^31, pairing the Euler class with [S^31] equals chi(S^31)=0; this bounded instance records the standard sphere Euler characteristic.

§DEP_HARD := S26-E-00003649

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003651|D|A|Proof+Compile+Meta+Topo] Riemannian metric refinement

§FORM := A Riemannian metric is a smooth field of positive-definite inner products g_p on tangent spaces.

§DEP_HARD := S26-E-00003650

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003652|D|A|Proof+Compile+Meta+Topo] Metric musical isomorphisms refinement

§FORM := The metric defines bundle isomorphisms sharp:T^*M->TM and flat:TM->T^*M by contraction with g.

§DEP_HARD := S26-D-00003651, S26-D-00000939

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003653|D|A|Proof+Compile+Meta+Topo] Metric-dual characterization of Riemannian gradient

§FORM := grad f is characterized by g(grad f,X)=df(X) for every vector field X.

§DEP_HARD := S26-D-00003652, S26-D-00000940

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003654|D|A|Proof+Compile+Meta+Topo] Divergence with respect to a Riemannian volume

§FORM := div X is characterized locally or invariantly by L_X dvol_g=(div X)dvol_g.

§DEP_HARD := S26-D-00003653

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003655|D|A|Proof+Compile+Meta+Topo+Spectral] Laplace-Beltrami operator

§FORM := Delta f=div grad f, with sign convention declared explicitly in applications.

§DEP_HARD := S26-D-00003654

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003656|D|A|Proof+Compile+Meta+Topo] Length of a smooth curve

§FORM := L(gamma)=integral sqrt(g(gamma dot,gamma dot)) dt.

§DEP_HARD := S26-D-00003655

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003657|D|A|Proof+Compile+Meta+Topo] Riemannian length distance refinement

§FORM := d_g(p,q) is the infimum of lengths of piecewise smooth curves joining p and q.

§DEP_HARD := S26-D-00003656, S26-D-00000938

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003658|T|A|Proof+Compile+Meta+Topo] Riemannian distance induces the manifold topology

§FORM := The metric-space topology of d_g agrees with the original smooth-manifold topology.

§DEP_HARD := S26-D-00003657

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003659|D|A|Proof+Compile+Meta+Topo] Levi-Civita connection refinement

§FORM := The Levi-Civita connection is the unique affine connection that is torsion-free and metric-compatible.

§DEP_HARD := S26-T-00003658

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00003660|I|A|Proof+Compile+Meta+Topo] Koszul formula

§FORM := The Levi-Civita connection is determined by the Koszul formula expressing 2g(∇_X Y,Z) in terms of derivatives of g and Lie brackets.

§DEP_HARD := S26-D-00003659

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003661|D|A|Proof+Compile+Meta+Topo] Geodesic refinement

§FORM := A geodesic satisfies ∇_{gamma dot} gamma dot=0.

§DEP_HARD := S26-I-00003660

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003662|D|A|Proof+Compile+Meta+Topo] Riemannian exponential map domain refinement

§FORM := exp_p(v) is the endpoint at time 1 of the geodesic starting at p with initial velocity v, where defined.

§DEP_HARD := S26-D-00003661, S26-D-00000946

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003663|T|A|Proof+Compile+Meta+Topo] Gauss lemma

§FORM := Radial geodesics are orthogonal to geodesic spheres in normal neighborhoods, yielding local minimizing properties.

§DEP_HARD := S26-D-00003662

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003664|D|A|Proof+Compile+Meta+Topo] Normal coordinates

§FORM := Normal coordinates at p are induced by exp_p and satisfy g_ij(p)=delta_ij with first derivatives vanishing at p.

§DEP_HARD := S26-T-00003663

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003665|D|A|Proof+Compile+Meta+Topo] Injectivity radius

§FORM := inj(p) is the largest radius on which exp_p is a diffeomorphism from the tangent ball onto its image.

§DEP_HARD := S26-D-00003664

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003666|D|A|Proof+Compile+Meta+Topo] Cut locus interface

§FORM := The cut locus of p marks endpoints beyond which minimizing radial geodesics cease to be uniquely minimizing or exp_p loses regularity.

§DEP_HARD := S26-D-00003665

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003667|T|A|Proof+Compile+Meta+Topo] Hopf-Rinow theorem

§FORM := For a connected Riemannian manifold, metric completeness, geodesic completeness, and compactness of closed bounded subsets are equivalent in the classical finite-dimensional theorem.

§DEP_HARD := S26-D-00003666

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003668|N|A|Proof+Compile+Meta+Topo] Completeness fence

§FORM := Local geodesic existence does not imply geodesic completeness; completeness is a global metric property.

§DEP_HARD := S26-T-00003667

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003669|B|A|Proof+Compile+Meta+Topo+Spectral] Riemannian to analysis bridge

§FORM := The metric determines volume, gradient, divergence and Laplace-Beltrami operators, linking geometry with elliptic and spectral analysis.

§DEP_HARD := S26-N-00003668

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003670|B|A|Proof+Compile+Meta+Topo+IG+Spectral] Riemannian to probability bridge

§FORM := Brownian motion on a Riemannian manifold is generated, up to convention, by a multiple of the Laplace-Beltrami operator.

§DEP_HARD := S26-B-00003669

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003671|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^2, r=1.05

§FORM := The standard round n-sphere of radius 1.05 has n-dimensional volume 13.8544236023 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-B-00003670

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003672|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^3, r=1.10

§FORM := The standard round n-sphere of radius 1.10 has n-dimensional volume 26.2728869157 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003671

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003673|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^4, r=1.15

§FORM := The standard round n-sphere of radius 1.15 has n-dimensional volume 46.0319994201 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003672

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003674|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^5, r=1.20

§FORM := The standard round n-sphere of radius 1.20 has n-dimensional volume 77.1535383891 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003673

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003675|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^6, r=1.25

§FORM := The standard round n-sphere of radius 1.25 has n-dimensional volume 126.164862794 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003674

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003676|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^7, r=1.30

§FORM := The standard round n-sphere of radius 1.30 has n-dimensional volume 203.74253349 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003675

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003677|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^8, r=1.35

§FORM := The standard round n-sphere of radius 1.35 has n-dimensional volume 327.51433845 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003676

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003678|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^9, r=1.40

§FORM := The standard round n-sphere of radius 1.40 has n-dimensional volume 526.890585348 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003677

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003679|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^10, r=1.45

§FORM := The standard round n-sphere of radius 1.45 has n-dimensional volume 851.486077523 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003678

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003680|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^11, r=1.50

§FORM := The standard round n-sphere of radius 1.50 has n-dimensional volume 1385.96363504 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003679

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003681|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^12, r=1.55

§FORM := The standard round n-sphere of radius 1.55 has n-dimensional volume 2276.48655759 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003680

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003682|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^13, r=1.60

§FORM := The standard round n-sphere of radius 1.60 has n-dimensional volume 3778.38651532 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003681

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003683|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^14, r=1.65

§FORM := The standard round n-sphere of radius 1.65 has n-dimensional volume 6343.02535183 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003682

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003684|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^15, r=1.70

§FORM := The standard round n-sphere of radius 1.70 has n-dimensional volume 10777.8531371 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003683

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003685|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^16, r=1.75

§FORM := The standard round n-sphere of radius 1.75 has n-dimensional volume 18544.6489469 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003684

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003686|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^17, r=1.80

§FORM := The standard round n-sphere of radius 1.80 has n-dimensional volume 32321.4557631 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003685

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003687|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^18, r=1.85

§FORM := The standard round n-sphere of radius 1.85 has n-dimensional volume 57073.1569386 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003686

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003688|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^19, r=1.90

§FORM := The standard round n-sphere of radius 1.90 has n-dimensional volume 102113.722355 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003687

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003689|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^20, r=1.95

§FORM := The standard round n-sphere of radius 1.95 has n-dimensional volume 185122.530915 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003688

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003690|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^21, r=2.00

§FORM := The standard round n-sphere of radius 2.00 has n-dimensional volume 340052.107971 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003689

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003691|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^22, r=2.05

§FORM := The standard round n-sphere of radius 2.05 has n-dimensional volume 632867.172205 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003690

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003692|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^23, r=2.10

§FORM := The standard round n-sphere of radius 2.10 has n-dimensional volume 1193209.02305 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003691

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003693|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^24, r=2.15

§FORM := The standard round n-sphere of radius 2.15 has n-dimensional volume 2278781.55661 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003692

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003694|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^25, r=2.20

§FORM := The standard round n-sphere of radius 2.20 has n-dimensional volume 4407663.15453 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003693

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003695|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^26, r=2.25

§FORM := The standard round n-sphere of radius 2.25 has n-dimensional volume 8633064.55513 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003694

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003696|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^27, r=2.30

§FORM := The standard round n-sphere of radius 2.30 has n-dimensional volume 17119830.8535 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003695

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003697|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^28, r=2.35

§FORM := The standard round n-sphere of radius 2.35 has n-dimensional volume 34366536.477 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003696

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003698|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^29, r=2.40

§FORM := The standard round n-sphere of radius 2.40 has n-dimensional volume 69822556.1 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003697

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003699|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^30, r=2.45

§FORM := The standard round n-sphere of radius 2.45 has n-dimensional volume 143548950.98 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003698

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003700|E|M|Proof+Compile+Meta+Topo] Round-sphere area profile S^31, r=2.50

§FORM := The standard round n-sphere of radius 2.50 has n-dimensional volume 298585433.256 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

§DEP_HARD := S26-E-00003699

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003701|D|A|Proof+Compile+Meta+Topo] Riemann curvature tensor refinement

§FORM := R(X,Y)Z=∇_X∇_Y Z-∇_Y∇_X Z-∇_[X,Y]Z for the declared sign convention.

§DEP_HARD := S26-E-00003700

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003702|D|A|Proof+Compile+Meta+Topo] Sectional curvature refinement

§FORM := For a 2-plane sigma=span(u,v), K(sigma)=g(R(u,v)v,u)/(g(u,u)g(v,v)-g(u,v)^2) with the chosen R sign convention.

§DEP_HARD := S26-D-00003701

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003703|D|A|Proof+Compile+Meta+Topo] Ricci curvature refinement

§FORM := Ric is the trace of the curvature endomorphism, contracting the first and third indices under a declared convention.

§DEP_HARD := S26-D-00003702

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003704|D|A|Proof+Compile+Meta+Topo] Scalar curvature refinement

§FORM := Scalar curvature is the metric trace of the Ricci tensor.

§DEP_HARD := S26-D-00003703

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003705|D|A|Proof+Compile+Meta+Topo] Jacobi field

§FORM := A Jacobi field J along a geodesic gamma satisfies ∇_t^2 J+R(J,gamma dot)gamma dot=0 under the declared sign convention.

§DEP_HARD := S26-D-00003704

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003706|D|A|Proof+Compile+Meta+Topo] Conjugate points

§FORM := Two points along a geodesic are conjugate when a nonzero Jacobi field vanishes at both.

§DEP_HARD := S26-D-00003705

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003707|T|A|Proof+Compile+Meta+Topo] Second variation of energy interface

§FORM := The second variation of geodesic energy is governed by the index form containing covariant derivative and curvature terms.

§DEP_HARD := S26-D-00003706

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003708|T|A|Proof+Compile+Meta+Topo] Bonnet-Myers theorem

§FORM := If a complete n-manifold has Ricci curvature bounded below by (n-1)k>0, then its diameter is at most pi/sqrt(k) and its fundamental group is finite.

§DEP_HARD := S26-T-00003707

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003709|T|A|Proof+Compile+Meta+Topo] Cartan-Hadamard theorem

§FORM := A complete simply connected Riemannian manifold with nonpositive sectional curvature has exponential map at every point a covering; in the simply connected case exp_p is a global diffeomorphism.

§DEP_HARD := S26-T-00003708

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003710|T|A|Proof+Compile+Meta+Topo] Rauch comparison theorem interface
§FORM := Curvature bounds compare growth of Jacobi fields along geodesics under matched initial data.
§DEP_HARD := S26-T-00003709
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003711|T|A|Proof+Compile+Meta+Topo] Toponogov comparison theorem interface
§FORM := Sectional-curvature lower bounds compare geodesic triangles with triangles in constant-curvature model spaces.
§DEP_HARD := S26-T-00003710
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003712|D|A|Proof+Compile+Meta+Topo] Constant-curvature space form
§FORM := A Riemannian manifold has constant sectional curvature k when every tangent 2-plane has curvature k.
§DEP_HARD := S26-T-00003711
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003713|T|A|Proof+Compile+Meta+Topo] Curvature tensor of a space form
§FORM := For constant sectional curvature k, R(X,Y)Z=k(g(Y,Z)X-g(X,Z)Y) up to the chosen curvature-sign convention.
§DEP_HARD := S26-D-00003712
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003714|D|A|Proof+Compile+Meta+Topo] Einstein metric
§FORM := A Riemannian metric is Einstein when Ric=lambda g for some scalar lambda, constant in dimensions greater than two under connectedness and standard hypotheses.
§DEP_HARD := S26-T-00003713
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003715|D|A|Proof+Compile+Meta+Topo] Weyl curvature tensor interface
§FORM := In dimension at least four, the Weyl tensor is the trace-free conformal part of Riemann curvature.
§DEP_HARD := S26-D-00003714
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003716|D|A|Proof+Compile+Meta+Topo] Conformal change of metric
§FORM := A conformal metric has form g_tilde=e^{2u}g and changes curvature by explicit derivative terms in u.
§DEP_HARD := S26-D-00003715
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003717|N|A|Proof+Compile+Meta+Topo] Sign-convention fence
§FORM := Curvature tensor signs vary across literature; every formula involving R, sectional curvature or Jacobi equations must be interpreted under one declared convention.
§DEP_HARD := S26-D-00003716
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003718|N|A|Proof+Compile+Meta+Topo] Ricci-sectional fence
§FORM := Bounds on Ricci curvature and bounds on sectional curvature are different hypotheses and cannot be interchanged mechanically.
§DEP_HARD := S26-N-00003717
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003719|B|A|Proof+Compile+Meta+Topo] Curvature to topology bridge
§FORM := Global curvature bounds yield topological and metric consequences such as diameter, fundamental-group, conjugate-point and comparison restrictions.
§DEP_HARD := S26-N-00003718
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003720|B|A|Proof+Compile+Meta+Topo] Curvature to relativity bridge
§FORM := Pseudo-Riemannian curvature uses analogous tensorial constructions but positive-definite Riemannian comparison theorems do not transfer unchanged.
§DEP_HARD := S26-B-00003719
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003721|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=3, r=1.1
§FORM := For the round S^3 of radius 1.1, sectional curvature is 0.826446280992, Ricci factor is 1.65289256198, and scalar curvature is 4.95867768595.
§DEP_HARD := S26-B-00003720
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003722|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=4, r=1.2
§FORM := For the round S^4 of radius 1.2, sectional curvature is 0.694444444444, Ricci factor is 2.08333333333, and scalar curvature is 8.33333333333.
§DEP_HARD := S26-E-00003721
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003723|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=5, r=1.3

§FORM := For the round S^5 of radius 1.3, sectional curvature is 0.591715976331, Ricci factor is 2.36686390533, and scalar curvature is 11.8343195266.

§DEP_HARD := S26-E-00003722

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003724|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=6, r=1.4

§FORM := For the round S^6 of radius 1.4, sectional curvature is 0.510204081633, Ricci factor is 2.55102040816, and scalar curvature is 15.306122449.

§DEP_HARD := S26-E-00003723

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003725|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=7, r=1.5

§FORM := For the round S^7 of radius 1.5, sectional curvature is 0.444444444444, Ricci factor is 2.66666666667, and scalar curvature is 18.6666666667.

§DEP_HARD := S26-E-00003724

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003726|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=8, r=1.6

§FORM := For the round S^8 of radius 1.6, sectional curvature is 0.390625, Ricci factor is 2.734375, and scalar curvature is 21.875.

§DEP_HARD := S26-E-00003725

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003727|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=9, r=1.7

§FORM := For the round S^9 of radius 1.7, sectional curvature is 0.346020761246, Ricci factor is 2.76816608997, and scalar curvature is 24.9134948097.

§DEP_HARD := S26-E-00003726

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003728|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=10, r=1.8

§FORM := For the round S^10 of radius 1.8, sectional curvature is 0.308641975309, Ricci factor is 2.77777777778, and scalar curvature is 27.7777777778.

§DEP_HARD := S26-E-00003727

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003729|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=11, r=1.9

§FORM := For the round S^11 of radius 1.9, sectional curvature is 0.277008310249, Ricci factor is 2.77008310249, and scalar curvature is 30.4709141274.

§DEP_HARD := S26-E-00003728

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003730|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=12, r=2.0

§FORM := For the round S^12 of radius 2.0, sectional curvature is 0.25, Ricci factor is 2.75, and scalar curvature is 33.

§DEP_HARD := S26-E-00003729

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003731|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=13, r=2.1

§FORM := For the round S^13 of radius 2.1, sectional curvature is 0.226757369615, Ricci factor is 2.72108843537, and scalar curvature is 35.3741496599.

§DEP_HARD := S26-E-00003730

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003732|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=14, r=2.2

§FORM := For the round S^14 of radius 2.2, sectional curvature is 0.206611570248, Ricci factor is 2.68595041322, and scalar curvature is 37.6033057851.

§DEP_HARD := S26-E-00003731

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003733|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=15, r=2.3

§FORM := For the round S^15 of radius 2.3, sectional curvature is 0.189035916824, Ricci factor is 2.64650283554, and scalar curvature is 39.6975425331.

§DEP_HARD := S26-E-00003732

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003734|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=16, r=2.4

§FORM := For the round S^16 of radius 2.4, sectional curvature is 0.173611111111, Ricci factor is 2.60416666667, and scalar curvature is 41.6666666667.

§DEP_HARD := S26-E-00003733

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003735|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=17, r=2.5

§FORM := For the round S^17 of radius 2.5, sectional curvature is 0.16, Ricci factor is 2.56, and scalar curvature is 43.52.

§DEP_HARD := S26-E-00003734

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003736|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=18, r=2.6

§FORM := For the round S^18 of radius 2.6, sectional curvature is 0.147928994083, Ricci factor is 2.51479289941, and scalar curvature is 45.2662721893.

§DEP_HARD := S26-E-00003735

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003737|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=19, r=2.7

§FORM := For the round S^19 of radius 2.7, sectional curvature is 0.137174211248, Ricci factor is 2.46913580247, and scalar curvature is 46.9135802469.

§DEP_HARD := S26-E-00003736

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003738|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=20, r=2.8

§FORM := For the round S^20 of radius 2.8, sectional curvature is 0.127551020408, Ricci factor is 2.42346938776, and scalar curvature is 48.4693877551.

§DEP_HARD := S26-E-00003737

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003739|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=21, r=2.9

§FORM := For the round S^21 of radius 2.9, sectional curvature is 0.118906064209, Ricci factor is 2.37812128419, and scalar curvature is 49.9405469679.

§DEP_HARD := S26-E-00003738

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003740|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=22, r=3.0

§FORM := For the round S^22 of radius 3.0, sectional curvature is 0.111111111111, Ricci factor is 2.33333333333, and scalar curvature is 51.3333333333.

§DEP_HARD := S26-E-00003739

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003741|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=23, r=3.1

§FORM := For the round S^23 of radius 3.1, sectional curvature is 0.104058272633, Ricci factor is 2.28928199792, and scalar curvature is 52.6534859521.

§DEP_HARD := S26-E-00003740

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003742|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=24, r=3.2

§FORM := For the round S^24 of radius 3.2, sectional curvature is 0.09765625, Ricci factor is 2.24609375, and scalar curvature is 53.90625.

§DEP_HARD := S26-E-00003741

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003743|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=25, r=3.3

§FORM := For the round S^25 of radius 3.3, sectional curvature is 0.0918273645546, Ricci factor is 2.20385674931, and scalar curvature is 55.0964187328.

§DEP_HARD := S26-E-00003742

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003744|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=26, r=3.4

§FORM := For the round S^26 of radius 3.4, sectional curvature is 0.0865051903114, Ricci factor is 2.16262975779, and scalar curvature is 56.2283737024.

§DEP_HARD := S26-E-00003743

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003745|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=27, r=3.5

§FORM := For the round S^27 of radius 3.5, sectional curvature is 0.0816326530612, Ricci factor is 2.12244897959, and scalar curvature is 57.306122449.

§DEP_HARD := S26-E-00003744

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003746|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=28, r=3.6

§FORM := For the round S^28 of radius 3.6, sectional curvature is 0.0771604938272, Ricci factor is 2.08333333333, and scalar curvature is 58.3333333333.

§DEP_HARD := S26-E-00003745

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003747|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=29, r=3.7

§FORM := For the round S^29 of radius 3.7, sectional curvature is 0.073046018992, Ricci factor is 2.04528853178, and scalar curvature is 59.3133674215.

§DEP_HARD := S26-E-00003746

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003748|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=30, r=3.8

§FORM := For the round S^30 of radius 3.8, sectional curvature is 0.0692520775623, Ricci factor is 2.00831024931, and scalar curvature is 60.2493074792.

§DEP_HARD := S26-E-00003747

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003749|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=31, r=3.9

§FORM := For the round S^31 of radius 3.9, sectional curvature is 0.0657462195924, Ricci factor is 1.97238658777, and scalar curvature is 61.1439842209.

§DEP_HARD := S26-E-00003748

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003750|E|M|Proof+Compile+Meta+Topo] Round-sphere curvature profile n=32, r=4.0

§FORM := For the round S^32 of radius 4.0, sectional curvature is 0.0625, Ricci factor is 1.9375, and scalar curvature is 62.

§DEP_HARD := S26-E-00003749

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003751|D|A|Proof+Compile+Meta+Topo] Geodesic completeness

§FORM := A Riemannian manifold is geodesically complete when every maximal geodesic is defined for all real parameter times.

§DEP_HARD := S26-E-00003750

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003752|D|A|Proof+Compile+Meta+Topo] Metric ball and geodesic ball distinction

§FORM := A metric ball is defined by d_g, whereas a geodesic normal ball is an exponential-map image on which radial coordinates are regular.

§DEP_HARD := S26-D-00003751

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003753|D|A|Proof+Compile+Meta+Topo] Volume growth function

§FORM := V_p(r)=Vol(B(p,r)) records Riemannian volume growth around p.

§DEP_HARD := S26-D-00003752

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003754|T|A|Proof+Compile+Meta+Topo] Bishop-Gromov comparison interface

§FORM := A Ricci lower bound gives monotonic comparison of normalized volumes of metric balls with the corresponding model space.

§DEP_HARD := S26-D-00003753

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003755|D|A|Proof+Compile+Meta+Topo] Isometry of Riemannian manifolds

§FORM := An isometry is a diffeomorphism F with F^*h=g; it preserves lengths, distances, Levi-Civita connection and curvature.

§DEP_HARD := S26-T-00003754

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003756|D|A|Proof+Compile+Meta+Topo] Killing vector field

§FORM := A vector field X is Killing when L_X g=0, equivalently its local flow acts by local isometries.

§DEP_HARD := S26-D-00003755

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003757|T|A|Proof+Compile+Meta+Topo] Myers-Steenrod theorem interface

§FORM := The isometry group of a connected Riemannian manifold is a Lie group acting smoothly.

§DEP_HARD := S26-D-00003756

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003758|D|A|Proof+Compile+Meta+Topo] Riemannian submersion

§FORM := A submersion is Riemannian when dF restricts to an isometry on horizontal spaces orthogonal to the fibers.

§DEP_HARD := S26-T-00003757

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003759|D|A|Proof+Compile+Meta+Topo] Second fundamental form

§FORM := For an embedded Riemannian submanifold, the second fundamental form is the normal component of the ambient covariant derivative.

§DEP_HARD := S26-D-00003758

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003760|D|A|Proof+Compile+Meta+Topo+Spectral] Shape operator

§FORM := For a chosen normal field nu, the shape operator A_nu is related to the second fundamental form by g(A_nu X,Y)=g(II(X,Y),nu) with sign convention declared.

§DEP_HARD := S26-D-00003759

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003761|T|A|Proof+Compile+Meta+Topo] Gauss equation

§FORM := Intrinsic curvature of a submanifold equals ambient tangential curvature plus quadratic terms in the second fundamental form.

§DEP_HARD := S26-D-00003760

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003762|T|A|Proof+Compile+Meta+Topo] Codazzi equation

§FORM := Covariant derivatives of the second fundamental form satisfy a compatibility equation involving ambient curvature.

§DEP_HARD := S26-T-00003761

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003763|D|A|Proof+Compile+Meta+Topo] Mean curvature vector

§FORM := The mean curvature vector is the trace of the second fundamental form divided or not by dimension according to convention; the convention must be declared.

§DEP_HARD := S26-T-00003762

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003764|D|A|Proof+Compile+Meta+Topo] Minimal submanifold

§FORM := A submanifold is minimal when its mean curvature vector vanishes.

§DEP_HARD := S26-D-00003763

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003765|T|A|Proof+Compile+Meta+Topo] First variation of area interface

§FORM := The first variation of volume of a submanifold is controlled by mean curvature and boundary terms.

§DEP_HARD := S26-D-00003764

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003766|D|A|Proof+Compile+Meta+Topo] Harmonic map interface

§FORM := A smooth map between Riemannian manifolds is harmonic when it is a critical point of the energy functional, equivalently its tension field vanishes.

§DEP_HARD := S26-T-00003765

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003767|D|A|Proof+Compile+Meta+Topo] Riemannian covering

§FORM := A covering map that is a local isometry transports the metric locally and preserves geodesic structure under lifting.

§DEP_HARD := S26-D-00003766

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003768|N|A|Proof+Compile+Meta+Topo] Local-isometry fence

§FORM := A local isometry need not be globally injective; covering and quotient phenomena remain global.

§DEP_HARD := S26-D-00003767

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003769|B|A|Proof+Compile+Meta+Topo] Submanifold geometry to PDE bridge

§FORM := Minimal and harmonic map equations are nonlinear elliptic systems arising from geometric variational principles.

§DEP_HARD := S26-N-00003768

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003770|B|A|Proof+Compile+Meta+Topo] Global geometry to group actions bridge

§FORM := Isometry groups, quotients and homogeneous spaces connect Riemannian geometry with Lie groups and representation-theoretic structure.

§DEP_HARD := S26-B-00003769

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003771|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=2

§FORM := The standard flat torus R^2/Z^2 has dimension 2, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-B-00003770

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003772|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=3

§FORM := The standard flat torus R^3/Z^3 has dimension 3, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003771

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003773|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=4

§FORM := The standard flat torus R^4/Z^4 has dimension 4, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003772

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003774|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=5

§FORM := The standard flat torus R^5/Z^5 has dimension 5, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003773

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003775|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=6

§FORM := The standard flat torus R^6/Z^6 has dimension 6, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003774

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003776|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=7

§FORM := The standard flat torus R^7/Z^7 has dimension 7, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003775

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003777|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=8

§FORM := The standard flat torus R^8/Z^8 has dimension 8, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003776

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003778|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=9

§FORM := The standard flat torus R^9/Z^9 has dimension 9, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003777

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003779|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=10

§FORM := The standard flat torus R^10/Z^10 has dimension 10, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003778

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003780|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=11

§FORM := The standard flat torus R^11/Z^11 has dimension 11, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003779

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003781|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=12

§FORM := The standard flat torus R^12/Z^12 has dimension 12, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003780

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003782|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=13

§FORM := The standard flat torus R^13/Z^13 has dimension 13, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003781

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003783|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=14

§FORM := The standard flat torus R^14/Z^14 has dimension 14, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003782

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003784|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=15

§FORM := The standard flat torus R^15/Z^15 has dimension 15, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003783

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003785|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=16

§FORM := The standard flat torus R^16/Z^16 has dimension 16, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003784

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003786|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=17

§FORM := The standard flat torus R^17/Z^17 has dimension 17, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003785

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003787|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=18

§FORM := The standard flat torus R^18/Z^18 has dimension 18, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003786

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003788|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=19

§FORM := The standard flat torus R^19/Z^19 has dimension 19, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003787

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003789|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=20

§FORM := The standard flat torus R^20/Z^20 has dimension 20, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003788

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003790|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=21

§FORM := The standard flat torus R^21/Z^21 has dimension 21, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003789

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003791|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=22

§FORM := The standard flat torus R^22/Z^22 has dimension 22, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003790

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003792|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=23

§FORM := The standard flat torus R^23/Z^23 has dimension 23, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003791

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003793|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=24

§FORM := The standard flat torus R^24/Z^24 has dimension 24, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003792

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003794|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=25

§FORM := The standard flat torus R^25/Z^25 has dimension 25, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003793

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003795|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=26

§FORM := The standard flat torus R^26/Z^26 has dimension 26, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003794

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003796|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=27

§FORM := The standard flat torus R^27/Z^27 has dimension 27, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003795

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003797|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=28

§FORM := The standard flat torus R^28/Z^28 has dimension 28, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003796

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003798|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=29

§FORM := The standard flat torus R^29/Z^29 has dimension 29, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003797

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003799|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=30

§FORM := The standard flat torus R^30/Z^30 has dimension 30, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003798

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003800|E|M|Proof+Compile+Meta+Topo] Flat-torus geometry profile dimension=31

§FORM := The standard flat torus R^31/Z^31 has dimension 31, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

§DEP_HARD := S26-E-00003799

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003801|D|A|Proof+Compile+Meta+Topo] Nondegenerate critical point

§FORM := A critical point p of f:M->R is nondegenerate when the Hessian at p is nonsingular in any local coordinate representation.

§DEP_HARD := S26-E-00003800

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003802|D|A|Proof+Compile+Meta+Topo] Morse function

§FORM := A smooth function is Morse when all its critical points are nondegenerate.

§DEP_HARD := S26-D-00003801

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003803|D|A|Proof+Compile+Meta+Topo+Spectral] Morse index

§FORM := The Morse index of a nondegenerate critical point is the number of negative eigenvalues of its Hessian, counted with multiplicity.

§DEP_HARD := S26-D-00003802

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003804|T|A|Proof+Compile+Meta+Topo] Morse lemma

§FORM := Near a nondegenerate critical point of index lambda, there are coordinates in which f=f(p)-x_1^2-...-x_lambda^2+x_{lambda+1}^2+...+x_n^2.

§DEP_HARD := S26-D-00003803

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003805|T|A|Proof+Compile+Meta+Topo] Regular-level isotopy interface

§FORM := If [a,b] contains no critical values of a proper-enough Morse function, the corresponding sublevel sets are diffeomorphic in the standard theorem.

§DEP_HARD := S26-T-00003804

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003806|T|A|Proof+Compile+Meta+Topo] Handle attachment at a critical point

§FORM := Passing a single nondegenerate critical value of index lambda changes the sublevel topology by attaching a lambda-cell/handle under standard isolation hypotheses.

§DEP_HARD := S26-T-00003805

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003807|D|A|Proof+Compile+Meta+Topo] Morse polynomial

§FORM := M_f(t)=sum_lambda m_lambda t^lambda records counts m_lambda of critical points by Morse index.

§DEP_HARD := S26-T-00003806

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003808|D|A|Proof+Compile+Meta+Topo] Poincare polynomial

§FORM := P_M(t)=sum_k beta_k t^k records Betti numbers over a chosen field.

§DEP_HARD := S26-D-00003807

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003809|T|A|Proof+Compile+Meta+Topo] Weak Morse inequalities
§FORM := The number m_k of index-k critical points is at least the kth Betti number.
§DEP_HARD := S26-D-00003808
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003810|T|A|Proof+Compile+Meta+Topo] Strong Morse inequalities
§FORM := Alternating partial sums of Morse numbers dominate corresponding alternating Betti sums, with equality at the top total Euler sum.
§DEP_HARD := S26-T-00003809
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003811|T|A|Proof+Compile+Meta+Topo] Morse Euler characteristic formula
§FORM := For a Morse function on a closed manifold, chi(M)=sum_p (-1)^{index(p)}.
§DEP_HARD := S26-T-00003810
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003812|D|A|Proof+Compile+Meta+Topo] Gradient-like vector field
§FORM := A gradient-like vector field for f decreases f away from critical points and has controlled local form near critical points.
§DEP_HARD := S26-T-00003811
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003813|D|A|Proof+Compile+Meta+Topo] Stable manifold of a Morse critical point
§FORM := For a Morse-Smale gradient flow, the stable manifold consists of points converging to the critical point in forward time.
§DEP_HARD := S26-D-00003812
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003814|D|A|Proof+Compile+Meta+Topo] Unstable manifold of a Morse critical point
§FORM := The unstable manifold consists of points converging to the critical point in backward time.
§DEP_HARD := S26-D-00003813
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003815|D|A|Proof+Compile+Meta+Topo] Morse-Smale transversality
§FORM := A gradient flow is Morse-Smale when stable and unstable manifolds intersect transversely.
§DEP_HARD := S26-D-00003814
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003816|D|A|Proof+Compile+Meta+Topo] Morse chain complex interface
§FORM := Under Morse-Smale conditions, the chain group is generated by critical points and the differential counts oriented flow lines between index-difference-one critical points.
§DEP_HARD := S26-D-00003815
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003817|T|A|Proof+Compile+Meta+Topo] Morse homology theorem interface
§FORM := Morse homology is canonically isomorphic to singular homology under the standard compactness/orientation setup.
§DEP_HARD := S26-D-00003816
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003818|N|A|Proof+Compile+Meta+Topo] Critical-count fence
§FORM := Morse inequalities constrain critical-point counts but do not say every function realizes the minimum counts or that Betti numbers determine diffeomorphism type.
§DEP_HARD := S26-T-00003817
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003819|B|A|Proof+Compile+Meta+Topo] Morse to CW bridge
§FORM := A Morse function on a compact manifold produces a handle/CW decomposition indexed by its critical points.
§DEP_HARD := S26-N-00003818
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003820|B|A|Proof+Compile+Meta+Topo+Spectral] Morse to dynamics bridge
§FORM := Gradient flows turn critical-point topology into trajectories, stable/unstable manifolds and chain-level boundary operators.
§DEP_HARD := S26-B-00003819
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003821|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=2
§FORM := The height function on S^2 has two nondegenerate critical points of indices 0 and 2; the signed Morse count is 2=chi(S^2).
§DEP_HARD := S26-B-00003820
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003822|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=3
§FORM := The height function on S^3 has two nondegenerate critical points of indices 0 and 3; the signed Morse count is 0=chi(S^3).
§DEP_HARD := S26-E-00003821
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003823|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=4
§FORM := The height function on S^4 has two nondegenerate critical points of indices 0 and 4; the signed Morse count is 2=chi(S^4).
§DEP_HARD := S26-E-00003822
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003824|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=5
§FORM := The height function on S^5 has two nondegenerate critical points of indices 0 and 5; the signed Morse count is 0=chi(S^5).
§DEP_HARD := S26-E-00003823
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003825|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=6
§FORM := The height function on S^6 has two nondegenerate critical points of indices 0 and 6; the signed Morse count is 2=chi(S^6).
§DEP_HARD := S26-E-00003824
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003826|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=7
§FORM := The height function on S^7 has two nondegenerate critical points of indices 0 and 7; the signed Morse count is 0=chi(S^7).
§DEP_HARD := S26-E-00003825
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003827|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=8
§FORM := The height function on S^8 has two nondegenerate critical points of indices 0 and 8; the signed Morse count is 2=chi(S^8).
§DEP_HARD := S26-E-00003826
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003828|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=9
§FORM := The height function on S^9 has two nondegenerate critical points of indices 0 and 9; the signed Morse count is 0=chi(S^9).
§DEP_HARD := S26-E-00003827
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003829|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=10
§FORM := The height function on S^10 has two nondegenerate critical points of indices 0 and 10; the signed Morse count is 2=chi(S^10).
§DEP_HARD := S26-E-00003828
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003830|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=11
§FORM := The height function on S^11 has two nondegenerate critical points of indices 0 and 11; the signed Morse count is 0=chi(S^11).
§DEP_HARD := S26-E-00003829
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003831|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=12
§FORM := The height function on S^12 has two nondegenerate critical points of indices 0 and 12; the signed Morse count is 2=chi(S^12).
§DEP_HARD := S26-E-00003830
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003832|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=13
§FORM := The height function on S^13 has two nondegenerate critical points of indices 0 and 13; the signed Morse count is 0=chi(S^13).
§DEP_HARD := S26-E-00003831
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003833|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=14
§FORM := The height function on S^14 has two nondegenerate critical points of indices 0 and 14; the signed Morse count is 2=chi(S^14).
§DEP_HARD := S26-E-00003832
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003834|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=15
§FORM := The height function on S^15 has two nondegenerate critical points of indices 0 and 15; the signed Morse count is 0=chi(S^15).
§DEP_HARD := S26-E-00003833
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003835|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=16
§FORM := The height function on S^16 has two nondegenerate critical points of indices 0 and 16; the signed Morse count is 2=chi(S^16).
§DEP_HARD := S26-E-00003834
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003836|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=17
§FORM := The height function on S^17 has two nondegenerate critical points of indices 0 and 17; the signed Morse count is 0=chi(S^17).
§DEP_HARD := S26-E-00003835
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003837|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=18
§FORM := The height function on S^18 has two nondegenerate critical points of indices 0 and 18; the signed Morse count is 2=chi(S^18).
§DEP_HARD := S26-E-00003836
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003838|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=19
§FORM := The height function on S^19 has two nondegenerate critical points of indices 0 and 19; the signed Morse count is 0=chi(S^19).
§DEP_HARD := S26-E-00003837
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003839|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=20
§FORM := The height function on S^20 has two nondegenerate critical points of indices 0 and 20; the signed Morse count is 2=chi(S^20).
§DEP_HARD := S26-E-00003838
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003840|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=21
§FORM := The height function on S^21 has two nondegenerate critical points of indices 0 and 21; the signed Morse count is 0=chi(S^21).
§DEP_HARD := S26-E-00003839
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003841|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=22
§FORM := The height function on S^22 has two nondegenerate critical points of indices 0 and 22; the signed Morse count is 2=chi(S^22).
§DEP_HARD := S26-E-00003840
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003842|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=23
§FORM := The height function on S^23 has two nondegenerate critical points of indices 0 and 23; the signed Morse count is 0=chi(S^23).
§DEP_HARD := S26-E-00003841
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003843|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=24
§FORM := The height function on S^24 has two nondegenerate critical points of indices 0 and 24; the signed Morse count is 2=chi(S^24).
§DEP_HARD := S26-E-00003842
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003844|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=25
§FORM := The height function on S^25 has two nondegenerate critical points of indices 0 and 25; the signed Morse count is 0=chi(S^25).
§DEP_HARD := S26-E-00003843
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003845|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=26
§FORM := The height function on S^26 has two nondegenerate critical points of indices 0 and 26; the signed Morse count is 2=chi(S^26).
§DEP_HARD := S26-E-00003844
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003846|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=27
§FORM := The height function on S^27 has two nondegenerate critical points of indices 0 and 27; the signed Morse count is 0=chi(S^27).
§DEP_HARD := S26-E-00003845
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003847|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=28
§FORM := The height function on S^28 has two nondegenerate critical points of indices 0 and 28; the signed Morse count is 2=chi(S^28).
§DEP_HARD := S26-E-00003846
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003848|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=29
§FORM := The height function on S^29 has two nondegenerate critical points of indices 0 and 29; the signed Morse count is 0=chi(S^29).
§DEP_HARD := S26-E-00003847
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003849|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=30
§FORM := The height function on S^30 has two nondegenerate critical points of indices 0 and 30; the signed Morse count is 2=chi(S^30).
§DEP_HARD := S26-E-00003848
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003850|E|M|Proof+Compile+Meta+Topo] Sphere Morse profile n=31
§FORM := The height function on S^31 has two nondegenerate critical points of indices 0 and 31; the signed Morse count is 0=chi(S^31).
§DEP_HARD := S26-E-00003849
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003851|D|A|Proof+Compile+Meta+Topo] Lie group action refinement
§FORM := A smooth left action of a Lie group G on M is a smooth map G×M->M satisfying identity and associativity axioms.
§DEP_HARD := S26-E-00003850
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003852|D|A|Proof+Compile+Meta+Topo] Orbit of a Lie group action
§FORM := G·p={g·p:g∈G}; for smooth actions orbits are immersed submanifolds under standard results.
§DEP_HARD := S26-D-00003851
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003853|D|A|Proof+Compile+Meta+Topo] Lie-action stabilizer subgroup refinement
§FORM := G_p={g∈G:g·p=p} is a closed Lie subgroup of G.
§DEP_HARD := S26-D-00003852, S26-D-00000967
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003854|D|A|Proof+Compile+Meta+Topo] Homogeneous Lie quotient model refinement
§FORM := A manifold is homogeneous under G when G acts transitively; after choosing p it is modeled by G/G_p.
§DEP_HARD := S26-D-00003853, S26-D-00000973
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003855|T|A|Proof+Compile+Meta+Topo] Orbit-stabilizer dimension formula for Lie actions
§FORM := dim(G·p)=dim G-dim G_p for smooth Lie group actions.
§DEP_HARD := S26-D-00003854
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003856|D|A|Proof+Compile+Meta+Topo] Proper group action
§FORM := A G-action is proper when (g,p)↦(g·p,p) is a proper map G×M->M×M.
§DEP_HARD := S26-T-00003855
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003857|T|A|Proof+Compile+Meta+Topo] Free proper quotient theorem
§FORM := A free proper smooth action has a smooth manifold quotient M/G and M->M/G is a principal G-bundle.
§DEP_HARD := S26-D-00003856
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003858|D|A|Proof+Compile+Meta+Topo] Effective group action
§FORM := An action is effective when only the identity acts trivially on every point.
§DEP_HARD := S26-T-00003857
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003859|D|A|Proof+Compile+Meta+Topo] Isotropy representation
§FORM := The stabilizer G_p acts linearly on T_pM via differentials, producing the isotropy representation.
§DEP_HARD := S26-D-00003858
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003860|D|A|Proof+Compile+Meta+Topo] Left-invariant Riemannian metric
§FORM := A Riemannian metric on G is left invariant when all left translations are isometries.
§DEP_HARD := S26-D-00003859
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003861|D|A|Proof+Compile+Meta+Topo] Bi-invariant Riemannian metric
§FORM := A metric is bi-invariant when both left and right translations are isometries.
§DEP_HARD := S26-D-00003860
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003862|T|A|Proof+Compile+Meta+Topo] Geodesics of a bi-invariant metric
§FORM := For a Lie group with bi-invariant Riemannian metric, one-parameter subgroups through the identity are geodesics.
§DEP_HARD := S26-D-00003861
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00003863|I|A|Proof+Compile+Meta+Topo] Levi-Civita connection for bi-invariant metrics
§FORM := For left-invariant vector fields under a bi-invariant metric, ∇_X Y=(1/2)[X,Y].
§DEP_HARD := S26-T-00003862
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-I-00003864|I|A|Proof+Compile+Meta+Topo] Curvature for bi-invariant metrics

§FORM := Under a standard curvature sign convention, sectional curvature of a bi-invariant metric is proportional to ||[X,Y]||^2 for orthonormal X,Y.

§DEP_HARD := S26-I-00003863

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003865|D|A|Proof+Compile+Meta+Topo] Symmetric space interface

§FORM := A Riemannian symmetric space has an isometry reversing geodesics through each point; globally it is modeled by Lie-theoretic quotient data under standard connectedness hypotheses.

§DEP_HARD := S26-I-00003864

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003866|D|A|Proof+Compile+Meta+Topo] Reductive homogeneous space interface

§FORM := A homogeneous space G/H is reductive when the Lie algebra splits g=h⊕m with m invariant under Ad(H).

§DEP_HARD := S26-D-00003865

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003867|D|A|Proof+Compile+Meta+Topo] Invariant tensor on a homogeneous space

§FORM := G-invariant tensors on G/H correspond to H-invariant tensors on the tangent model g/h at the basepoint.

§DEP_HARD := S26-D-00003866

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003868|N|A|Proof+Compile+Meta+Topo] Orbit-space fence

§FORM := A smooth group action need not yield a manifold quotient; freeness/properness or controlled singular-stratum hypotheses are load-bearing.

§DEP_HARD := S26-D-00003867

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003869|B|A|Proof+Compile+Meta+Topo] Homogeneous geometry to representation bridge

§FORM := Isotropy representations and invariant tensors reduce geometric classification problems to linear representation data.

§DEP_HARD := S26-N-00003868

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003870|B|A|Proof+Compile+Meta+Topo] Lie quotients to bundles bridge

§FORM := For closed H⊂G, G->G/H is a principal H-bundle under standard smooth quotient hypotheses.

§DEP_HARD := S26-B-00003869

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003871|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=3

§FORM := dim SO(3)=3; consequently dim SO(4)-dim SO(3)=3, matching dim S^3 in the homogeneous model SO(4)/SO(3).

§DEP_HARD := S26-B-00003870

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003872|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=4

§FORM := dim SO(4)=6; consequently dim SO(5)-dim SO(4)=4, matching dim S^4 in the homogeneous model SO(5)/SO(4).

§DEP_HARD := S26-E-00003871

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003873|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=5

§FORM := dim SO(5)=10; consequently dim SO(6)-dim SO(5)=5, matching dim S^5 in the homogeneous model SO(6)/SO(5).

§DEP_HARD := S26-E-00003872

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003874|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=6

§FORM := dim SO(6)=15; consequently dim SO(7)-dim SO(6)=6, matching dim S^6 in the homogeneous model SO(7)/SO(6).

§DEP_HARD := S26-E-00003873

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003875|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=7

§FORM := dim SO(7)=21; consequently dim SO(8)-dim SO(7)=7, matching dim S^7 in the homogeneous model SO(8)/SO(7).

§DEP_HARD := S26-E-00003874

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003876|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=8

§FORM := dim SO(8)=28; consequently dim SO(9)-dim SO(8)=8, matching dim S^8 in the homogeneous model SO(9)/SO(8).

§DEP_HARD := S26-E-00003875

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003877|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=9
§FORM := dim SO(9)=36; consequently dim SO(10)-dim SO(9)=9, matching dim S^9 in the homogeneous model SO(10)/SO(9).
§DEP_HARD := S26-E-00003876
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003878|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=10
§FORM := dim SO(10)=45; consequently dim SO(11)-dim SO(10)=10, matching dim S^10 in the homogeneous model SO(11)/SO(10).
§DEP_HARD := S26-E-00003877
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003879|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=11
§FORM := dim SO(11)=55; consequently dim SO(12)-dim SO(11)=11, matching dim S^11 in the homogeneous model SO(12)/SO(11).
§DEP_HARD := S26-E-00003878
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003880|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=12
§FORM := dim SO(12)=66; consequently dim SO(13)-dim SO(12)=12, matching dim S^12 in the homogeneous model SO(13)/SO(12).
§DEP_HARD := S26-E-00003879
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003881|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=13
§FORM := dim SO(13)=78; consequently dim SO(14)-dim SO(13)=13, matching dim S^13 in the homogeneous model SO(14)/SO(13).
§DEP_HARD := S26-E-00003880
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003882|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=14
§FORM := dim SO(14)=91; consequently dim SO(15)-dim SO(14)=14, matching dim S^14 in the homogeneous model SO(15)/SO(14).
§DEP_HARD := S26-E-00003881
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003883|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=15
§FORM := dim SO(15)=105; consequently dim SO(16)-dim SO(15)=15, matching dim S^15 in the homogeneous model SO(16)/SO(15).
§DEP_HARD := S26-E-00003882
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003884|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=16
§FORM := dim SO(16)=120; consequently dim SO(17)-dim SO(16)=16, matching dim S^16 in the homogeneous model SO(17)/SO(16).
§DEP_HARD := S26-E-00003883
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003885|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=17
§FORM := dim SO(17)=136; consequently dim SO(18)-dim SO(17)=17, matching dim S^17 in the homogeneous model SO(18)/SO(17).
§DEP_HARD := S26-E-00003884
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003886|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=18
§FORM := dim SO(18)=153; consequently dim SO(19)-dim SO(18)=18, matching dim S^18 in the homogeneous model SO(19)/SO(18).
§DEP_HARD := S26-E-00003885
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003887|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=19
§FORM := dim SO(19)=171; consequently dim SO(20)-dim SO(19)=19, matching dim S^19 in the homogeneous model SO(20)/SO(19).
§DEP_HARD := S26-E-00003886
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003888|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=20
§FORM := dim SO(20)=190; consequently dim SO(21)-dim SO(20)=20, matching dim S^20 in the homogeneous model SO(21)/SO(20).
§DEP_HARD := S26-E-00003887
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003889|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=21
§FORM := dim SO(21)=210; consequently dim SO(22)-dim SO(21)=21, matching dim S^21 in the homogeneous model SO(22)/SO(21).
§DEP_HARD := S26-E-00003888
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003890|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=22
§FORM := dim SO(22)=231; consequently dim SO(23)-dim SO(22)=22, matching dim S^22 in the homogeneous model SO(23)/SO(22).
§DEP_HARD := S26-E-00003889
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003891|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=23
§FORM := dim SO(23)=253; consequently dim SO(24)-dim SO(23)=23, matching dim S^23 in the homogeneous model SO(24)/SO(23).
§DEP_HARD := S26-E-00003890
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003892|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=24
§FORM := dim SO(24)=276; consequently dim SO(25)-dim SO(24)=24, matching dim S^24 in the homogeneous model SO(25)/SO(24).
§DEP_HARD := S26-E-00003891
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003893|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=25
§FORM := dim SO(25)=300; consequently dim SO(26)-dim SO(25)=25, matching dim S^25 in the homogeneous model SO(26)/SO(25).
§DEP_HARD := S26-E-00003892
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003894|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=26
§FORM := dim SO(26)=325; consequently dim SO(27)-dim SO(26)=26, matching dim S^26 in the homogeneous model SO(27)/SO(26).
§DEP_HARD := S26-E-00003893
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003895|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=27
§FORM := dim SO(27)=351; consequently dim SO(28)-dim SO(27)=27, matching dim S^27 in the homogeneous model SO(28)/SO(27).
§DEP_HARD := S26-E-00003894
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003896|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=28
§FORM := dim SO(28)=378; consequently dim SO(29)-dim SO(28)=28, matching dim S^28 in the homogeneous model SO(29)/SO(28).
§DEP_HARD := S26-E-00003895
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003897|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=29
§FORM := dim SO(29)=406; consequently dim SO(30)-dim SO(29)=29, matching dim S^29 in the homogeneous model SO(30)/SO(29).
§DEP_HARD := S26-E-00003896
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003898|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=30
§FORM := dim SO(30)=435; consequently dim SO(31)-dim SO(30)=30, matching dim S^30 in the homogeneous model SO(31)/SO(30).
§DEP_HARD := S26-E-00003897
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003899|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=31
§FORM := dim SO(31)=465; consequently dim SO(32)-dim SO(31)=31, matching dim S^31 in the homogeneous model SO(32)/SO(31).
§DEP_HARD := S26-E-00003898
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003900|E|M|Proof+Compile+Meta+Topo] Orthogonal-group dimension profile n=32
§FORM := dim SO(32)=496; consequently dim SO(33)-dim SO(32)=32, matching dim S^32 in the homogeneous model SO(33)/SO(32).
§DEP_HARD := S26-E-00003899
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003901|D|A|Proof+Compile+Meta+Topo] Filtered chain complex
§FORM := A filtered chain complex has nested subcomplexes ...⊂F_{p-1}C⊂F_pC⊂... compatible with the differential.
§DEP_HARD := S26-E-00003900
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003902|D|A|Proof+Compile+Meta+Topo+Spectral] Spectral sequence
§FORM := A spectral sequence is a sequence of bigraded objects E_r with differentials d_r of prescribed bidegree such that E_{r+1} is the homology of (E_r,d_r).
§DEP_HARD := S26-D-00003901
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003903|D|A|Proof+Compile+Meta+Topo+Spectral] Convergence of a spectral sequence
§FORM := A spectral sequence converges to a filtered object when its stable page identifies with the associated graded pieces of the target filtration under declared hypotheses.
§DEP_HARD := S26-D-00003902
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003904|D|A|Proof+Compile+Meta+Topo+Spectral] Serre spectral sequence interface

§FORM := For a suitable fibration F->E->B, the Serre spectral sequence relates homology/cohomology of fiber and base to that of total space.

§DEP_HARD := S26-D-00003903

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003905|D|A|Proof+Compile+Meta+Topo+Spectral] Leray-Serre E2 page

§FORM := With appropriate local coefficients, the cohomological Serre spectral sequence has E_2^{p,q}=H^p(B;H^q(F)) in the standard setup.

§DEP_HARD := S26-D-00003904

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003906|D|A|Proof+Compile+Meta+Topo+Spectral] Atiyah-Hirzebruch spectral sequence interface

§FORM := Generalized cohomology of a CW complex can be computed from ordinary cohomology with coefficients in the generalized theory through the AHSS.

§DEP_HARD := S26-D-00003905

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003907|D|A|Proof+Compile+Meta+Topo+Spectral] Exact couple

§FORM := An exact couple is algebraic data generating a spectral sequence by repeated derivation.

§DEP_HARD := S26-D-00003906

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003908|D|A|Proof+Compile+Meta+Topo+Spectral] Edge homomorphism

§FORM := An edge homomorphism relates target filtration pieces to entries on the boundary of a spectral-sequence page.

§DEP_HARD := S26-D-00003907

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003909|D|A|Proof+Compile+Meta+Topo+Spectral] Transgression

§FORM := A transgression is a differential carrying a class from fiber-like to base-like degree in a fibration spectral sequence.

§DEP_HARD := S26-D-00003908

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003910|N|A|Proof+Compile+Meta+Topo+Spectral] Spectral-sequence collapse fence

§FORM := Vanishing of all later differentials does not automatically resolve extension problems in reconstructing the target from its associated graded object.

§DEP_HARD := S26-D-00003909

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003911|D|A|Proof+Compile+Meta+Topo] Obstruction cocycle

§FORM := An obstruction cocycle records failure to extend a partially defined map, section or lift over the next skeleton.

§DEP_HARD := S26-N-00003910

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003912|D|A|Proof+Compile+Meta+Topo] Primary obstruction class

§FORM := The first nonzero obstruction to extension lies in an appropriate cohomology group with local coefficients determined by homotopy of the fiber/target.

§DEP_HARD := S26-D-00003911

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003913|T|A|Proof+Compile+Meta+Topo] Vanishing obstruction extension principle

§FORM := If the relevant obstruction class vanishes, an extension over the next skeleton exists under the standard obstruction-theory setup.

§DEP_HARD := S26-D-00003912

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003914|D|A|Proof+Compile+Meta+Topo+Tower] Postnikov tower

§FORM := A Postnikov tower approximates a connected space by stages retaining homotopy groups only up to increasing degrees.

§DEP_HARD := S26-T-00003913

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003915|D|A|Proof+Compile+Meta+Topo] k-invariant

§FORM := A Postnikov k-invariant is a cohomology class encoding how one stage is assembled from the preceding stage and the next homotopy group.

§DEP_HARD := S26-D-00003914

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003916|D|A|Proof+Compile+Meta+Topo] Eilenberg-Mac Lane space

§FORM := K(G,n) has pi_n□G and all other homotopy groups trivial, in the standard connected pointed model.

§DEP_HARD := S26-D-00003915

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003917|T|A|Proof+Compile+Meta+Topo] Cohomology representability by Eilenberg-Mac Lane spaces

§FORM := For suitable CW complexes, H^n(X;G) is naturally identified with based homotopy classes [X,K(G,n)].

§DEP_HARD := S26-D-00003916

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003918|N|A|Proof+Compile+Meta+Topo] Associated-graded fence

§FORM := Knowing the associated graded object of a filtration need not uniquely determine the original filtered group or ring.

§DEP_HARD := S26-T-00003917

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003919|B|A|Proof+Compile+Meta+Topo+Spectral] Spectral sequences to bundles bridge

§FORM := Fibration and bundle filtrations generate spectral sequences that transport local fiber/base invariants into global invariants.

§DEP_HARD := S26-N-00003918

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003920|B|A|Proof+Compile+Meta+Topo] Obstruction theory to geometry bridge

§FORM := Existence of sections, lifts, framings and reductions of structure group can be encoded as successive cohomological obstructions.

§DEP_HARD := S26-B-00003919

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003921|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=2

§FORM := For a finite first-quadrant toy page with 2 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 2; no extension claim is made.

§DEP_HARD := S26-B-00003920

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003922|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=3

§FORM := For a finite first-quadrant toy page with 3 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 3; no extension claim is made.

§DEP_HARD := S26-E-00003921

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003923|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=4

§FORM := For a finite first-quadrant toy page with 4 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 4; no extension claim is made.

§DEP_HARD := S26-E-00003922

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003924|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=5

§FORM := For a finite first-quadrant toy page with 5 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 5; no extension claim is made.

§DEP_HARD := S26-E-00003923

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003925|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=6

§FORM := For a finite first-quadrant toy page with 6 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 6; no extension claim is made.

§DEP_HARD := S26-E-00003924

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003926|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=7

§FORM := For a finite first-quadrant toy page with 7 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 7; no extension claim is made.

§DEP_HARD := S26-E-00003925

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003927|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=8

§FORM := For a finite first-quadrant toy page with 8 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 8; no extension claim is made.

§DEP_HARD := S26-E-00003926

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003928|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=9
§FORM := For a finite first-quadrant toy page with 9 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 9; no extension claim is made.
§DEP_HARD := S26-E-00003927
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003929|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=10
§FORM := For a finite first-quadrant toy page with 10 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 10; no extension claim is made.
§DEP_HARD := S26-E-00003928
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003930|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=11
§FORM := For a finite first-quadrant toy page with 11 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 11; no extension claim is made.
§DEP_HARD := S26-E-00003929
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003931|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=12
§FORM := For a finite first-quadrant toy page with 12 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 12; no extension claim is made.
§DEP_HARD := S26-E-00003930
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003932|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=13
§FORM := For a finite first-quadrant toy page with 13 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 13; no extension claim is made.
§DEP_HARD := S26-E-00003931
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003933|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=14
§FORM := For a finite first-quadrant toy page with 14 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 14; no extension claim is made.
§DEP_HARD := S26-E-00003932
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003934|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=15
§FORM := For a finite first-quadrant toy page with 15 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 15; no extension claim is made.
§DEP_HARD := S26-E-00003933
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003935|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=16
§FORM := For a finite first-quadrant toy page with 16 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 16; no extension claim is made.
§DEP_HARD := S26-E-00003934
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003936|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=17
§FORM := For a finite first-quadrant toy page with 17 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 17; no extension claim is made.
§DEP_HARD := S26-E-00003935
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003937|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=18
§FORM := For a finite first-quadrant toy page with 18 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 18; no extension claim is made.
§DEP_HARD := S26-E-00003936
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003938|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=19
§FORM := For a finite first-quadrant toy page with 19 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 19; no extension claim is made.
§DEP_HARD := S26-E-00003937
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003939|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=20
§FORM := For a finite first-quadrant toy page with 20 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 20; no extension claim is made.
§DEP_HARD := S26-E-00003938
§EVIDENCE := M · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003940|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=21

§FORM := For a finite first-quadrant toy page with 21 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 21; no extension claim is made.

§DEP_HARD := S26-E-00003939

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003941|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=22

§FORM := For a finite first-quadrant toy page with 22 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 22; no extension claim is made.

§DEP_HARD := S26-E-00003940

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003942|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=23

§FORM := For a finite first-quadrant toy page with 23 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 23; no extension claim is made.

§DEP_HARD := S26-E-00003941

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003943|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=24

§FORM := For a finite first-quadrant toy page with 24 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 24; no extension claim is made.

§DEP_HARD := S26-E-00003942

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003944|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=25

§FORM := For a finite first-quadrant toy page with 25 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 25; no extension claim is made.

§DEP_HARD := S26-E-00003943

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003945|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=26

§FORM := For a finite first-quadrant toy page with 26 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 26; no extension claim is made.

§DEP_HARD := S26-E-00003944

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003946|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=27

§FORM := For a finite first-quadrant toy page with 27 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 27; no extension claim is made.

§DEP_HARD := S26-E-00003945

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003947|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=28

§FORM := For a finite first-quadrant toy page with 28 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 28; no extension claim is made.

§DEP_HARD := S26-E-00003946

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003948|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=29

§FORM := For a finite first-quadrant toy page with 29 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 29; no extension claim is made.

§DEP_HARD := S26-E-00003947

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003949|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=30

§FORM := For a finite first-quadrant toy page with 30 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 30; no extension claim is made.

§DEP_HARD := S26-E-00003948

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003950|E|M|Proof+Compile+Meta+Topo+Spectral] Toy spectral-sequence profile rank=31

§FORM := For a finite first-quadrant toy page with 31 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 31; no extension claim is made.

§DEP_HARD := S26-E-00003949

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00003951|D|A|Proof+Compile+Meta+Topo] Knot in the 3-sphere

§FORM := A knot is a smooth or tame embedding S^1->S^3 considered up to ambient isotopy.

§DEP_HARD := S26-E-00003950

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003952|D|A|Proof+Compile+Meta+Topo] Link
§FORM := A link is a finite disjoint union of embedded circles in S^3 considered up to ambient isotopy.
§DEP_HARD := S26-D-00003951
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003953|D|A|Proof+Compile+Meta+Topo] Knot complement
§FORM := The complement S^3\K, often with a tubular neighborhood removed, is a 3-manifold whose topology encodes the knot.
§DEP_HARD := S26-D-00003952
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003954|D|A|Proof+Compile+Meta+Topo] Knot group
§FORM := The knot group is pi_1(S^3\K), with the precise complement convention declared.
§DEP_HARD := S26-D-00003953
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003955|D|A|Proof+Compile+Meta+Topo] Seifert surface
§FORM := An oriented Seifert surface for an oriented knot is a connected oriented embedded surface in S^3 whose boundary is the knot.
§DEP_HARD := S26-D-00003954
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003956|D|A|Proof+Compile+Meta+Topo] Genus of a knot
§FORM := The knot genus is the minimal genus among its Seifert surfaces.
§DEP_HARD := S26-D-00003955
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003957|D|A|Proof+Compile+Meta+Topo] Linking number
§FORM := For two disjoint oriented knots, the linking number is an integer invariant computable from signed crossings or intersection with a spanning surface.
§DEP_HARD := S26-D-00003956
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003958|D|A|Proof+Compile+Meta+Topo] Reidemeister moves
§FORM := Three local diagram moves generate ambient-isotopy equivalence of tame knot diagrams.
§DEP_HARD := S26-D-00003957
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003959|T|A|Proof+Compile+Meta+Topo] Reidemeister theorem
§FORM := Two tame knot diagrams represent isotopic knots iff they are related by planar isotopy and a finite sequence of Reidemeister moves.
§DEP_HARD := S26-D-00003958
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003960|D|A|Proof+Compile+Meta+Topo] Alexander polynomial interface
§FORM := The Alexander polynomial is a Laurent-polynomial knot invariant defined up to multiplication by units ±t^k under a standard normalization convention.
§DEP_HARD := S26-T-00003959
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003961|D|A|Proof+Compile+Meta+Topo] Jones polynomial interface
§FORM := The Jones polynomial is a Laurent-polynomial invariant characterized by a skein relation and normalization convention.
§DEP_HARD := S26-D-00003960
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00003962|N|A|Proof+Compile+Meta+Topo] Polynomial-completeness fence
§FORM := Classical knot polynomials are not complete invariants; distinct knots can share the same Alexander or Jones polynomial.
§DEP_HARD := S26-D-00003961
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003963|D|A|Proof+Compile+Meta+Topo] Heegaard splitting
§FORM := A Heegaard splitting decomposes a closed orientable 3-manifold into two handlebodies glued along their common boundary.
§DEP_HARD := S26-N-00003962
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003964|D|A|Proof+Compile+Meta+Topo] Heegaard genus
§FORM := The Heegaard genus is the minimal genus of a Heegaard surface for the manifold.
§DEP_HARD := S26-D-00003963
§EVIDENCE := A · Exact registered statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003965|D|A|Proof+Compile+Meta+Topo] Dehn surgery interface

§FORM := Dehn surgery removes a solid-torus neighborhood of a knot and reglues a solid torus by a boundary homeomorphism specified by a slope.

§DEP_HARD := S26-D-00003964

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003966|D|A|Proof+Compile+Meta+Topo] Mapping class group

§FORM := The mapping class group of a manifold is the group of isotopy classes of self-homeomorphisms or diffeomorphisms in the declared category.

§DEP_HARD := S26-D-00003965

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00003967|D|A|Proof+Compile+Meta+Topo] Surface classification interface

§FORM := Closed connected surfaces are classified up to homeomorphism by orientability and genus in the classical theorem.

§DEP_HARD := S26-D-00003966

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00003968|T|A|Proof+Compile+Meta+Topo] Euler characteristic of closed orientable surfaces

§FORM := For the closed orientable surface Sigma_g, chi(Sigma_g)=2-2g.

§DEP_HARD := S26-D-00003967

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003969|B|A|Proof+Compile+Meta+Topo] Low-dimensional topology to groups bridge

§FORM := Knot complements, surface groups and mapping class groups encode geometric topology through finitely presented groups and their actions.

§DEP_HARD := S26-T-00003968

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-B-00003970|B|A|Proof+Compile+Meta+Topo] Low-dimensional topology to geometry bridge

§FORM := Hyperbolic, spherical, Euclidean and other geometric structures on low-dimensional manifolds connect topology with Lie groups, curvature and dynamical invariants.

§DEP_HARD := S26-B-00003969

§EVIDENCE := A · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-E-00003971|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=1

§FORM := The closed orientable surface Sigma_1 has a CW structure with 1 zero-cell, 2 one-cells and 1 two-cell; chi=1-2+1=0, and over Q beta_1=2.

§DEP_HARD := S26-B-00003970

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003972|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=2

§FORM := The closed orientable surface Sigma_2 has a CW structure with 1 zero-cell, 4 one-cells and 1 two-cell; chi=1-4+1=-2, and over Q beta_1=4.

§DEP_HARD := S26-E-00003971

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003973|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=3

§FORM := The closed orientable surface Sigma_3 has a CW structure with 1 zero-cell, 6 one-cells and 1 two-cell; chi=1-6+1=-4, and over Q beta_1=6.

§DEP_HARD := S26-E-00003972

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003974|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=4

§FORM := The closed orientable surface Sigma_4 has a CW structure with 1 zero-cell, 8 one-cells and 1 two-cell; chi=1-8+1=-6, and over Q beta_1=8.

§DEP_HARD := S26-E-00003973

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003975|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=5

§FORM := The closed orientable surface Sigma_5 has a CW structure with 1 zero-cell, 10 one-cells and 1 two-cell; chi=1-10+1=-8, and over Q beta_1=10.

§DEP_HARD := S26-E-00003974

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003976|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=6

§FORM := The closed orientable surface Sigma_6 has a CW structure with 1 zero-cell, 12 one-cells and 1 two-cell; chi=1-12+1=-10, and over Q beta_1=12.

§DEP_HARD := S26-E-00003975

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003977|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=7

§FORM := The closed orientable surface Sigma_7 has a CW structure with 1 zero-cell, 14 one-cells and 1 two-cell; chi=1-14+1=-12, and over Q beta_1=14.

§DEP_HARD := S26-E-00003976

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003978|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=8

§FORM := The closed orientable surface Sigma_8 has a CW structure with 1 zero-cell, 16 one-cells and 1 two-cell; chi=1-16+1=-14, and over Q beta_1=16.

§DEP_HARD := S26-E-00003977

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003979|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=9

§FORM := The closed orientable surface Sigma_9 has a CW structure with 1 zero-cell, 18 one-cells and 1 two-cell; chi=1-18+1=-16, and over Q beta_1=18.

§DEP_HARD := S26-E-00003978

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003980|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=10

§FORM := The closed orientable surface Sigma_10 has a CW structure with 1 zero-cell, 20 one-cells and 1 two-cell; chi=1-20+1=-18, and over Q beta_1=20.

§DEP_HARD := S26-E-00003979

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003981|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=11

§FORM := The closed orientable surface Sigma_11 has a CW structure with 1 zero-cell, 22 one-cells and 1 two-cell; chi=1-22+1=-20, and over Q beta_1=22.

§DEP_HARD := S26-E-00003980

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003982|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=12

§FORM := The closed orientable surface Sigma_12 has a CW structure with 1 zero-cell, 24 one-cells and 1 two-cell; chi=1-24+1=-22, and over Q beta_1=24.

§DEP_HARD := S26-E-00003981

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003983|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=13

§FORM := The closed orientable surface Sigma_13 has a CW structure with 1 zero-cell, 26 one-cells and 1 two-cell; chi=1-26+1=-24, and over Q beta_1=26.

§DEP_HARD := S26-E-00003982

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003984|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=14

§FORM := The closed orientable surface Sigma_14 has a CW structure with 1 zero-cell, 28 one-cells and 1 two-cell; chi=1-28+1=-26, and over Q beta_1=28.

§DEP_HARD := S26-E-00003983

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003985|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=15

§FORM := The closed orientable surface Sigma_15 has a CW structure with 1 zero-cell, 30 one-cells and 1 two-cell; chi=1-30+1=-28, and over Q beta_1=30.

§DEP_HARD := S26-E-00003984

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003986|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=16

§FORM := The closed orientable surface Sigma_16 has a CW structure with 1 zero-cell, 32 one-cells and 1 two-cell; chi=1-32+1=-30, and over Q beta_1=32.

§DEP_HARD := S26-E-00003985

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003987|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=17

§FORM := The closed orientable surface Sigma_17 has a CW structure with 1 zero-cell, 34 one-cells and 1 two-cell; chi=1-34+1=-32, and over Q beta_1=34.

§DEP_HARD := S26-E-00003986

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003988|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=18

§FORM := The closed orientable surface Sigma_18 has a CW structure with 1 zero-cell, 36 one-cells and 1 two-cell; chi=1-36+1=-34, and over Q beta_1=36.

§DEP_HARD := S26-E-00003987

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003989|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=19

§FORM := The closed orientable surface Sigma_19 has a CW structure with 1 zero-cell, 38 one-cells and 1 two-cell; chi=1-38+1=-36, and over Q beta_1=38.

§DEP_HARD := S26-E-00003988

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003990|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=20

§FORM := The closed orientable surface Sigma_20 has a CW structure with 1 zero-cell, 40 one-cells and 1 two-cell; chi=1-40+1=-38, and over Q beta_1=40.

§DEP_HARD := S26-E-00003989

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003991|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=21

§FORM := The closed orientable surface Sigma_21 has a CW structure with 1 zero-cell, 42 one-cells and 1 two-cell; chi=1-42+1=-40, and over Q beta_1=42.

§DEP_HARD := S26-E-00003990

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003992|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=22

§FORM := The closed orientable surface Sigma_22 has a CW structure with 1 zero-cell, 44 one-cells and 1 two-cell; chi=1-44+1=-42, and over Q beta_1=44.

§DEP_HARD := S26-E-00003991

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003993|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=23

§FORM := The closed orientable surface Sigma_23 has a CW structure with 1 zero-cell, 46 one-cells and 1 two-cell; chi=1-46+1=-44, and over Q beta_1=46.

§DEP_HARD := S26-E-00003992

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003994|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=24

§FORM := The closed orientable surface Sigma_24 has a CW structure with 1 zero-cell, 48 one-cells and 1 two-cell; chi=1-48+1=-46, and over Q beta_1=48.

§DEP_HARD := S26-E-00003993

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003995|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=25

§FORM := The closed orientable surface Sigma_25 has a CW structure with 1 zero-cell, 50 one-cells and 1 two-cell; chi=1-50+1=-48, and over Q beta_1=50.

§DEP_HARD := S26-E-00003994

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003996|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=26

§FORM := The closed orientable surface Sigma_26 has a CW structure with 1 zero-cell, 52 one-cells and 1 two-cell; chi=1-52+1=-50, and over Q beta_1=52.

§DEP_HARD := S26-E-00003995

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003997|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=27

§FORM := The closed orientable surface Sigma_27 has a CW structure with 1 zero-cell, 54 one-cells and 1 two-cell; chi=1-54+1=-52, and over Q beta_1=54.

§DEP_HARD := S26-E-00003996

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003998|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=28

§FORM := The closed orientable surface Sigma_28 has a CW structure with 1 zero-cell, 56 one-cells and 1 two-cell; chi=1-56+1=-54, and over Q beta_1=56.

§DEP_HARD := S26-E-00003997

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00003999|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=29

§FORM := The closed orientable surface Sigma_29 has a CW structure with 1 zero-cell, 58 one-cells and 1 two-cell; chi=1-58+1=-56, and over Q beta_1=58.

§DEP_HARD := S26-E-00003998

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004000|E|M|Proof+Compile+Meta+Topo] Orientable-surface profile genus=30

§FORM := The closed orientable surface Sigma_30 has a CW structure with 1 zero-cell, 60 one-cells and 1 two-cell; chi=1-60+1=-58, and over Q beta_1=60.

§DEP_HARD := S26-E-00003999

§EVIDENCE := M · Exact registered statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§PACK Synopsis_2026_Volume_IV_sLANG_S1

volume := IV · p-adic Hypercomplex Hyperdimensional Continuum

canonical_range := 4001-5000

§CLAIM[S26-N-00004001|N|S|Proof+Compile+Meta+Topo+Tower] Dimension-consistency fence

§FORM := A dodecahedron is a three-dimensional cell. Therefore a construction whose primitive cells are dodecahedra is modeled in hyperbolic three-space H^3, or in a three-dimensional orbifold locally covered by H^3. The Poincare disk D^2 may occur as a totally geodesic section or chart, but not as an intrinsic dodecahedral tiling surface.

§ASSUME := The words disk, ball, cell, section, and quotient retain their standard dimensional meanings.

§DEP_HARD := S26-D-00002366, S26-B-00003169

§EVIDENCE := S · Project specification or modeling convention.

§BOUNDARY := Replacing each dodecahedron by a hyperbolic dodecagon yields a legitimate two-dimensional analogue, but it is a different model.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004002|D|A|Proof+Compile+Meta+Topo+Tower] Poincare disk chart

§FORM := Let D^2={z in C: |z|<1}. Its curvature -1 metric is g_D=4|dz|^2/(1-|z|^2)^2. Geodesics are diameters or circular arcs meeting the ideal boundary |z|=1 orthogonally.

§ASSUME := The curvature normalization is fixed at -1.

§DEP_HARD := S26-D-00002366, S26-N-00004001

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004003|D|A|Proof+Compile+Meta+Topo+Tower] Poincare ball ambient space

§FORM := Let B^3={x in R^3: ||x||<1} with g_B=4(dx_1^2+dx_2^2+dx_3^2)/(1-||x||^2)^2. Then (B^3,g_B) is a model of complete simply connected hyperbolic three-space H^3 with sectional curvature -1.

§ASSUME := The Euclidean norm and the curvature -1 normalization are used.

§DEP_HARD := S26-N-00004001, S26-D-00004002

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004004|T|A|Proof+Compile+Meta+Topo+Tower] Totally geodesic disk section

§FORM := If V is a Euclidean plane through the origin, then B^3 intersect V with the restricted metric is isometric to the Poincare disk D^2 and is totally geodesic in H^3.

§ASSUME := V passes through the center of the Poincare ball.

§DEP_HARD := S26-D-00004002, S26-D-00004003

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004005|D|A|Proof+Compile+Meta+Topo+Tower] Right-angled hyperbolic dodecahedral seed

§FORM := Fix a compact regular right-angled hyperbolic dodecahedron P subset H^3: P has twelve congruent pentagonal facets and every dihedral angle is pi/2. A chosen interior point o_P is the seed center.

§ASSUME := Existence of the compact regular right-angled hyperbolic dodecahedron is taken from the standard Coxeter-polytope construction.

§DEP_HARD := S26-D-00004003

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§BOUNDARY := The Euclidean regular dodecahedron has a different metric and does not generate the same hyperbolic reflection tessellation. Synopsis 2026 - Volume IV research draft - v0.1.0 5 Part B - Dodecahedral reflection tiling and adjacency

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004006|D|A|Proof+Compile+Meta+Topo+Tower] Facet-reflection group

§FORM := For every facet F of P let r_F be reflection in the hyperbolic plane containing F. Define W_P=<r_F: F a facet of P> as a subgroup of Isom(H^3).

§ASSUME := P is the right-angled Coxeter polytope fixed in S26-D-00004005.

§DEP_HARD := S26-D-00004005

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004007|T|A|Proof+Compile+Meta+Topo+Tower] Right-angled Coxeter presentation

§FORM := The reflection group admits the presentation W_P=<r_F | r_F^2=1; (r_F r_G)^2=1 whenever F and G meet>. Equivalently, reflections in intersecting facets commute; non-intersecting pairs carry no finite Coxeter relation in this presentation.

§ASSUME := Facet incidences are those of the dodecahedron and all dihedral angles are pi/2.

§DEP_HARD := S26-D-00004005, S26-D-00004006

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004008|T|A|Proof+Compile+Meta+Topo+Tower] Dodecahedral reflection tessellation

§FORM := The family T_P={gP: g in W_P} covers H^3 with pairwise disjoint interiors and is locally finite. Every facet belongs to exactly two tiles, related by the corresponding reflection.

§ASSUME := P is a compact hyperbolic Coxeter polytope.

§DEP_HARD := S26-D-00004006, S26-T-00004007

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004009|D|S|Proof+Compile+Meta+Topo+Tower] Dual cell-adjacency graph

§FORM := The graph G_P has one vertex [gP] for every tile and an edge between two vertices exactly when their tiles share a facet. The root is [P]. Edge labels record the reflecting facet.

§ASSUME := Cells with identical subsets of H^3 are identified; adjacency means codimension-one incidence.

§DEP_HARD := S26-T-00004008

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004010|D|S|Proof+Compile+Meta+Topo+Tower] Seeded combinatorial distance

§FORM := For cells Q,R in T_P, d_G(Q,R) is the shortest path length between their vertices in G_P. The layer L_k={Q: d_G(P,Q)=k} gives the k-th combinatorial shell around the seed.

§ASSUME := The graph metric uses unit edge lengths.

§DEP_HARD := S26-D-00004009

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004011|D|S|Proof+Compile+Meta+Topo+Tower] Admissible face walk

§FORM := An admissible face walk of length m is a sequence gamma=(P_0,...,P_m) of distinct tiles with P_0=P and consecutive tiles sharing one facet. Its address is the reduced word of the crossed facet reflections.

§ASSUME := The walk is simple and rooted; m is a positive integer.

§DEP_HARD := S26-T-00004007, S26-D-00004009, S26-D-00004010

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004012|D|S|Proof+Compile+Meta+Topo+Tower] Orthogonal initial cell directions

§FORM := Let c_F be the tangent at o_P to the geodesic from o_P to the center of facet F. Two rooted walks gamma and gamma' are orthogonal at the seed when <c_F,c_F'>_g=0 for their first crossed facets F,F'.

§ASSUME := The selected facet-center directions are orthogonal in the hyperbolic metric at o_P.

§DEP_HARD := S26-D-00004005, S26-D-00004011

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004013|D|S|Proof+Compile+Meta+Topo+Tower] The 360-cell w-family

§FORM := Choose an admissible walk gamma_w=(P,P_1,...,P_360). Define W_360={P_1,...,P_360}; its ordered address word is w=(a_1,...,a_360), where a_j is the crossed facet label.

§ASSUME := A simple rooted walk of length 360 has been selected in G_P.

§DEP_HARD := S26-D-00004011

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004014|D|S|Proof+Compile+Meta+Topo+Tower] The orthogonal 360-cell w'-family

§FORM := Choose an admissible walk gamma_w'=(P,P'_1,...,P'_360), orthogonal to gamma_w at the seed. Define W'_360={P'_1,...,P'_360}. Apart from P, the two chosen walks are required to be disjoint unless an explicit bridge identifies them.

§ASSUME := The orthogonality and disjointness requirements are part of the selected data, not automatic consequences of the tessellation.

§DEP_HARD := S26-D-00004012, S26-D-00004013

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-T-00004015|T|A|Proof+Compile+Meta+Topo+Tower] Finite dodecahedral knit

§FORM := Let B be a finite set of shortest adjacency paths joining selected cells of W_360 to selected cells of W'_360. The knit K_360,360 is the union of P, both 360-cell families, and every tile occurring in B. It is a finite connected cell subcomplex of T_P.

§ASSUME := B is finite and each declared bridge is a finite path in G_P.

§DEP_HARD := S26-T-00004008, S26-D-00004013, S26-D-00004014

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004016|D|A|Proof+Compile+Meta+Topo+Tower] Spherical link at a point

§FORM := For x in H^3, the unit tangent sphere S_xH^3={v in T_xH^3: ||v||=1} is a round two-sphere. Initial geodesic directions of cells and facets define a finite marked network on this sphere.

§ASSUME := A Riemannian unit tangent norm is fixed at x.

§DEP_HARD := S26-D-00004003, S26-T-00004015

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004017|N|S|Proof+Compile+Meta+Topo+Tower] Sphere-model curvature fence

§FORM := The spherical link S_xH^3 and a geodesic sphere of radius r are visualization or link layers. They are not globally isometric to the hyperbolic disk and do not carry constant curvature -1 with the topology of S^2.

§ASSUME := Intrinsic curvature and extrinsic embedding are kept distinct.

§DEP_HARD := S26-T-00004004, S26-D-00004016

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004018|D|A|Proof+Compile+Meta+Topo+Tower] Radial shell projection

§FORM := For x in H^3 and r>0, define rho_{x,r}(y)=exp_x(r v_y), where v_y is the initial unit tangent of the geodesic from x to y. On H^3 minus {x}, rho_{x,r} maps rays to the geodesic sphere S_r(x).

§ASSUME := The exponential map is evaluated in complete simply connected H^3.

§DEP_HARD := S26-D-00004003, S26-D-00004016

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004019|D|S|Proof+Compile+Meta+Topo+Tower] Spherical trace of the knit

§FORM := The shell trace Sigma_r(K_360,360) is the marked cell-adjacency image rho_{o_P,r}(K_360,360 minus {o_P}), augmented by the incidence labels inherited from G_P. It is a representation of the knit, not a cellulation by isometric dodecahedra.

§ASSUME := Degenerate radial coincidences retain multiplicity as labels.

§DEP_HARD := S26-T-00004015, S26-D-00004018

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-T-00004020|T|A|Proof+Compile+Meta+Topo+Tower] Gauss-Bonnet obstruction to a hyperbolic two-sphere

§FORM := No smooth Riemannian metric of constant curvature -1 exists on the closed two-sphere S^2.
§ASSUME := The metric is smooth, closed, and without cone singularities.
§DEP_HARD := S26-N-00004017
§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004021|D|A|Proof+Compile+Meta+Topo+Tower] Effective orbifold chart

§FORM := An n-dimensional effective orbifold chart is (U_tilde,G,pi), where U_tilde is a connected open subset of R^n, G is a finite group acting effectively and smoothly on U_tilde, and pi induces a homeomorphism U_tilde/G -> U. Compatible charts form an orbifold atlas.
§ASSUME := The underlying space is Hausdorff and second countable.
§DEP_HARD := S26-D-00003056, S26-T-00003057, S26-N-00004001
§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004022|T|A|Proof+Compile+Meta+Topo+Tower] Proper discrete quotient orbifold

§FORM := Let X be a complete Riemannian n-manifold and Gamma a discrete group acting properly and effectively by isometries with finite point stabilizers. Then O=[X/Gamma] is a complete Riemannian orbifold; it is a manifold exactly when every stabilizer is trivial.
§ASSUME := The action is proper, effective, and isometric; point stabilizers are finite.
§DEP_HARD := S26-B-00003169, S26-D-00003856, S26-D-00004021
§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004023|D|A|Proof+Compile+Meta+Topo+Tower] Isotropy and singular stratification

§FORM := For p=[x] in O=[X/Gamma], the isotropy group is Gamma_x={g in Gamma: gx=x}, defined up to conjugacy. The regular locus has trivial isotropy; connected components of points with conjugate isotropy type form the orbifold strata.
§ASSUME := The quotient hypotheses of S26-T-00004022 hold.
§DEP_HARD := S26-T-00004022
§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004024|D|A|Proof+Compile+Meta+Topo+Tower] Dodecahedral Coxeter orbifold

§FORM := The quotient O_P=[H^3/W_P] is the Coxeter orbifold with fundamental chamber P. Facets are mirror strata and intersections carry the finite stabilizers generated by the incident reflections. The orientation-preserving subgroup W_P^+ gives an index-two quotient without mirror generators but may retain rotational isotropy.
§ASSUME := The reflection group is the discrete Coxeter group of the right-angled dodecahedron.
§DEP_HARD := S26-D-00004006, S26-T-00004008, S26-T-00004022, S26-D-00004023
§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004025|T|A|Proof+Compile+Meta+Topo+Tower] Tangent cone and spherical orbifold link

§FORM := At p=[x] in O=[X/Gamma], the metric tangent cone is T_pO isometric to T_xX/Gamma_x. Its unit link is the spherical orbifold S_xX/Gamma_x. Thus the local sphere and the orbifold singularity are fused by the isotropy representation.
§ASSUME := Gamma_x acts linearly on T_xX through the differential of the isometric action.
§DEP_HARD := S26-D-00004016, S26-T-00004022, S26-D-00004023
§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004026|D|S|Proof+Compile+Meta+Topo+Tower] Fused Cosmo-3 datum

§FORM := Define C_3=(H^3,P,T_P,G_P,K_360,360,D_o,S_o,O_P), where D_o is a chosen totally geodesic Poincare disk through o_P and S_o is either the unit spherical link at o_P or a declared radial shell. The components are tied by inclusion, radial projection, and quotient maps.
§ASSUME := Every component is one of the objects defined in S26-D-00004003 through S26-T-00004025.
§DEP_HARD := S26-T-00004004, S26-T-00004008, S26-D-00004009, S26-T-00004015, S26-D-00004019, S26-D-00004024, S26-T-00004025
§EVIDENCE := S · Project specification or modeling convention.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004027|D|S|Proof+Compile+Meta+Topo+Tower] Fusion compatibility diagram

§FORM := The required compatibility is q o i_D = i_O o q_D on the disk chart and q o i_K = i_K^O o q_K on the knit, where q:H^3->O_P is the quotient map. Radial shell data commute only up to their declared incidence labels, because radial projection is not generally an isometry.
§ASSUME := Every arrow is explicitly typed as an embedding, quotient, projection, or incidence map.
§DEP_HARD := S26-D-00004026
§EVIDENCE := S · Project specification or modeling convention.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-T-00004028|T|A|Proof+Compile+Meta+Topo+Tower] Connected quotient image of the knit

§FORM := The image q(K_360,360) in O_P is a finite connected orbicellular subspace. Cell multiplicities are retained by the map from the pre-quotient tile set even when distinct tiles share one quotient chamber.
§ASSUME := K_360,360 is finite and connected; q is continuous and cellular on the Coxeter decomposition.
§DEP_HARD := S26-T-00004015, S26-D-00004024, S26-D-00004027
§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004029|T|A|Proof+Compile+Meta+Topo+Tower] Subgroup functoriality and orbifold covering

§FORM := If Gamma' is a subgroup of Gamma in Isom(H^3), then [H^3/Gamma'] -> [H^3/Gamma] is the natural orbifold covering. If Gamma' is torsion-free, the source is a hyperbolic manifold; if Gamma' has finite index, the covering degree is [Gamma:Gamma'] in the orbifold sense.
§ASSUME := Both actions are proper and effective and Gamma' has the stated subgroup properties.
§DEP_HARD := S26-T-00004022, S26-D-00004024
§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004030|N|S|Proof+Compile+Meta+Topo+IG+Tower] Cell-count and quotient-multiplicity fence

§FORM := The statement '360 cells in each family' refers to selected tiles in the covering tessellation before quotient. After quotient, the number of distinct geometric chambers may be smaller; the exact 360+360 information survives as weighted multiplicity or as address data.

§ASSUME := Counts distinguish tiles, quotient images, and registry addresses.

§DEP_HARD := S26-D-00004013, S26-D-00004014, S26-T-00004028, S26-T-00004029

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004031|D|A|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Orbifold Hilbert space

§FORM := For a Riemannian orbifold O_n with measure dvol, define H_n=L^2(O_n,dvol). In a quotient chart U_tilde/G, functions in H_n lift to G-invariant square-integrable functions on U_tilde.

§ASSUME := O_n is sigma-finite; compactness will be declared when used.

§DEP_HARD := S26-D-00004021, S26-T-00004022

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004032|D|A|Proof+Compile+Meta+Spectral+Tower] Laplace-type operator

§FORM := Let L_n=Delta_n+Q_n on H_n, where Delta_n is the nonnegative Laplace-Beltrami operator and Q_n is a real Gamma_n-invariant potential bounded below. The Friedrichs extension is used whenever the initial operator is only semibounded on compactly supported smooth functions.

§ASSUME := The metric and potential give a densely defined semibounded symmetric operator.

§DEP_HARD := S26-D-00004031

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004033|T|A|Proof+Compile+Meta+Topo+Spectral+Tower] Compact-orbifold eigenbasis

§FORM := If O_n is compact without boundary and L_n is a self-adjoint elliptic Laplace-type operator, then L_n has compact resolvent. Its spectrum is discrete, lambda_k->infinity with finite multiplicities, and H_n has an orthonormal basis of smooth eigenfunctions phi_k satisfying L_n phi_k=lambda_k phi_k.

§ASSUME := Compact effective Riemannian orbifold; elliptic self-adjoint Laplace-type operator; compatible boundary conditions if a boundary is later introduced.

§DEP_HARD := S26-T-00000236, S26-D-00004031, S26-D-00004032

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004034|T|A|Proof+Compile+Meta+IG+Spectral+Tower] General spectral resolution

§FORM := For a self-adjoint L_n on a possibly noncompact O_n, there is a projection-valued measure E_n on R such that L_n=integral_R lambda dE_n(lambda). A countable L^2 eigenbasis need not exist when continuous spectrum is present.

§ASSUME := L_n is self-adjoint on H_n.

§DEP_HARD := S26-D-00004032

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004035|D|S|Proof+Compile+Meta+Spectral+Tower] Eigen-n spectral plug-in

§FORM := An admissible spectral plug-in Pi_n=(U_n,V_n,m_n) consists of a spectral representation U_n, a measurable unitary V_n(lambda) acting inside each multiplicity fiber, and a bounded spectral multiplier m_n. It acts as Pi_n=U_n^-1 V_n m_n(M_lambda) U_n. A pure basis plug-in has m_n=1 and changes coordinates but not L_n.

§ASSUME := U_n is unitary, V_n(lambda) is fiberwise unitary almost everywhere, and m_n is bounded and measurable.

§DEP_HARD := S26-T-00004033, S26-T-00004034

§EVIDENCE := S · Project specification or modeling convention.

§BOUNDARY := A multiplier m_n not identically one changes states or dynamics; a unitary basis rotation inside an eigenspace does not change the spectrum.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004036|D|S|Proof+Compile+Meta+Topo+Spectral+Tower] General Cosmo-n space

§FORM := For n>=2, a Cosmo-n datum is C_n=(X_n,Gamma_n,O_n,T_n,P_n,G_n,K_n,L_n,E_n), where X_n is a complete simply connected Riemannian n-manifold, Gamma_n acts properly by isometries with finite stabilizers, O_n=[X_n/Gamma_n], T_n is a locally finite equivariant cellulation with seed P_n and dual graph G_n, K_n is a selected finite knit, L_n is self-adjoint on L^2(O_n), and E_n is its spectral resolution.

§ASSUME := Every component satisfies the hypotheses of its defining clause; omitted components are not inferred.

§DEP_HARD := S26-T-00004022, S26-D-00004026, S26-D-00004031, S26-T-00004034

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-T-00004037|T|A|Proof+Compile+Meta+Topo+Tower] Recovery of the dodecahedral model at n=3

§FORM := Taking X_3=H^3, Gamma_3=W_P, P_3=P, T_3=T_P, and K_3=K_360,360 recovers the fused Cosmo-3 object, with the chosen disk and spherical link carried as auxiliary chart data.

§ASSUME := The components are selected exactly as in S26-D-00004026.

§DEP_HARD := S26-D-00004026, S26-D-00004036

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004038|N|S|Proof+Compile+Meta+Topo+Tower] No universal dodecahedron fence

§FORM := The regular dodecahedron is specific to dimension three. A Cosmo-n family must declare a seed n-polytope P_n, orbicell, or other finite fundamental complex in each dimension; existence of a regular right-angled compact P_n is not automatic.

§ASSUME := Dimension-dependent existence and realizability are checked separately.

§DEP_HARD := S26-D-00004005, S26-D-00004036, S26-T-00004037

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004039|D|S|Proof+Compile+Meta+Topo+Spectral+Tower] Dimension morphism

§FORM := A morphism F_nm:C_n->C_m consists of an orbifold map f_nm:O_n->O_m, a compatible cellular map of selected knits, and a bounded operator J_nm:H_n->H_m. It is spectral when J_nm Dom(L_n) subset Dom(L_m) and L_m J_nm=J_nm L_n.

§ASSUME := All maps preserve the explicitly declared structure; no dimension change is assumed invertible.

§DEP_HARD := S26-D-00004035, S26-D-00004036

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004040|D|S|Proof+Compile+Meta+Topo+Tower] Coherent Cosmo tower

§FORM := A coherent tower C_bullet is a sequence (C_n,F_n,n+1) whose morphisms satisfy F_nn=id and F_mk o F_nm=F_nk whenever defined. If the geometric maps are inverse bonding maps, the limit is called a pro-orbifold system; it need not be a finite-dimensional orbifold.

§ASSUME := Bonding maps and their direction are declared; coherence holds on geometry, cells, and spectral intertwiners.

§DEP_HARD := S26-D-00004036, S26-D-00004039

§EVIDENCE := S · Project specification or modeling convention.

§BOUNDARY := An infinite inverse limit may be compact or stratified yet fail to possess an ordinary orbifold atlas. Synopsis 2026 - Volume IV research draft - v0.1.0 12 Part I - Fractal generation by graph-directed contractions

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004041|D|S|Proof+Compile+Meta+Topo+Tower] Graph-directed orbifold contraction system

§FORM := Let A be a finite directed graph. Each vertex a carries a compact complete metric orbifold region X_a, and each edge e:a->b carries a contraction F_e:X_b->X_a with Lipschitz ratio r_e<1. The family F={F_e} is a graph-directed orbifold IFS.

§ASSUME := All regions are nonempty compact metric spaces and every edge map is a declared contraction.

§DEP_HARD := S26-T-00004015, S26-D-00004021, S26-D-00004036

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004042|D|S|Proof+Compile+Meta+Topo+Tower] Stratum-compatible contraction

§FORM := A contraction F_e is stratum-compatible when each source isotropy stratum maps into a target stratum and the local chart lift is equivariant with respect to an injective homomorphism of isotropy groups. A cellular contraction additionally maps each selected cell into a declared subcell.

§ASSUME := Compatible orbifold charts and isotropy homomorphisms are supplied for every edge map.

§DEP_HARD := S26-D-00004023, S26-D-00004041

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-T-00004043|T|A|Proof+Compile+Meta+Topo+Tower] Existence of the fractal attractor

§FORM := For a finite graph-directed contraction system in which every selected vertex has at least one outgoing edge, there is a unique family of nonempty compact sets (K_a) satisfying K_a=union_{e:a->b} F_e(K_b). Iteration from any nonempty compact initial family converges in the product Hausdorff metric.

§ASSUME := Finite graph, complete compact regions, contraction ratios uniformly bounded above by r<1, and at least one outgoing contraction from each selected vertex.

§DEP_HARD := S26-D-00004041

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004044|T|A|Proof+Compile+Meta+Topo+Spectral+Tower] Similarity-dimension law

§FORM := For a single-vertex system of similitudes with ratios r_1,...,r_N satisfying the open set condition, the Hausdorff dimension s of the attractor is the unique solution of sum_i r_i^s=1. For graph-directed similitudes, s is characterized by spectral radius rho(A(s))=1, where A(s)_ab=sum_{e:a->b} r_e^s.

§ASSUME := Similitudes and the appropriate open set/separation condition; irreducibility for a single common graph-directed dimension.

§DEP_HARD := S26-T-00004043

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004045|D|S|Proof+Compile+Meta+Topo+Spectral+Tower] Cellular fractal-generation operator

§FORM := Let R_n act on a finite orbicellular pattern Q by R_n(Q)=union_e F_e(Q at target(e)), retaining edge addresses. The j-th generation is R_n^j(Q); its limiting support is contained in the attractor from S26-T-00004043.

§ASSUME := The maps are cellular, stratum-compatible contractions and address collisions retain multiplicity.

§DEP_HARD := S26-D-00004042, S26-T-00004043, S26-T-00004044

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004046|D|A|Proof+Compile+Meta+Topo+Tower] Infinitesimal zoom

§FORM := For p in O_n and epsilon>0, the pointed zoom is Z_{p,epsilon}(O_n)=(O_n,epsilon^-1 d,p). In a local chart (U_tilde,G_p,pi) with lift x of p, normal coordinates identify bounded zoom regions with subsets of T_xX_n/G_p.

§ASSUME := The orbifold carries a smooth Riemannian metric and epsilon tends to zero through positive values.

§DEP_HARD := S26-D-00004023, S26-T-00004025, S26-D-00004036

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004047|T|A|Proof+Compile+Meta+Topo+Tower] Tangent-orbifold limit

§FORM := As epsilon->0, the pointed spaces Z_{p,epsilon}(O_n) converge locally in the pointed Gromov-Hausdorff sense to T_xX_n/G_p. At a regular point the limit is R^n; at a singular point it is the finite quotient R^n/G_p.

§ASSUME := Finite isotropy and smooth Riemannian orbifold charts.

§DEP_HARD := S26-T-00004025, S26-D-00004046

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004048|D|A|Proof+Compile+Meta+Topo+Tower] Dilation semigroup and infinitesimal scale generator

§FORM := On the tangent model R^n/G_p, define (D_t f)(v)=exp(alpha t) f(exp(t)v) for smooth compactly supported invariant f. Then D_tD_s=D_{t+s}, and its formal generator is A_p f=alpha f+v dot grad f.

§ASSUME := Functions are invariant under G_p and lie in a domain on which differentiation and dilation are defined.

§DEP_HARD := S26-T-00004047

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004049|D|S|Proof+Compile+Meta+Topo+Tower] Nested orbicellular refinement

§FORM := An infinitesimal cell generator at p is a nested sequence Q_0 superset Q_1 superset ... of orbicellular neighborhoods with p in Q_j and mesh h_j->0. After rescaling by h_j^-1, the marked cell complexes must have a locally finite subsequential limit in T_pO_n.

§ASSUME := Meshes are measured in the orbifold metric and refinements respect isotropy strata.

§DEP_HARD := S26-D-00004042, S26-D-00004046, S26-T-00004047

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-T-00004050|T|A|Proof+Compile+Meta+Topo+Spectral+Tower] Principal rescaling of the Laplace operator

§FORM := In a smooth orbifold chart centered at p, the rescaled operator epsilon^2 L_n converges at the level of principal coefficients on bounded zoom regions to the Euclidean Laplacian on T_xX_n/G_p. Lower-order terms vanish under the scaling when their coefficients remain bounded.

§ASSUME := L_n is Laplace type with smooth principal coefficients and locally bounded lower-order coefficients.

§DEP_HARD := S26-D-00004032, S26-D-00004046, S26-T-00004047

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004051|D|S|Proof+Compile+Meta+Spectral+Tower] Bi-scale generated state

§FORM := A generated state is u_{n,j,epsilon}=Pi_n R_n^j Z_{p,epsilon}^*u, where n is the ambient Cosmo index, j is the fractal generation, epsilon is the infinitesimal scale, R_n is cellular recursion, and Pi_n is a spectral plug-in. The order of operators is part of the datum.

§ASSUME := Every operator acts on a common declared function or section space and all compositions are defined.

§DEP_HARD := S26-D-00004035, S26-D-00004045, S26-D-00004046, S26-T-00004050

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004052|D|S|Proof+Compile+Meta+Tower] Intertwining compatibility

§FORM := A plug-in is recursion-compatible when Pi_n R_n=R_n Pi_n on a common core, zoom-compatible when Pi_n Z_{p,epsilon}^*=Z_{p,epsilon}^* Pi_{p,epsilon}, and tower-compatible when J_nm Pi_n=Pi_m J_nm. Exact commutation may be weakened to a norm-controlled commutator.

§ASSUME := Domains and transported plug-ins are explicitly identified.

§DEP_HARD := S26-D-00004035, S26-D-00004039, S26-D-00004045, S26-D-00004046, S26-D-00004051

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-T-00004053|T|A|Proof+Compile+Meta+Tower] Unitary plug-in stability

§FORM := If Pi_n is a pure basis plug-in and U is unitary on H_n, then ||Pi_n Uu||=||u||. More generally, if the multiplier satisfies ||m_n||_infinity<=M, then ||Pi_n u||<=M||u||.

§ASSUME := The spectral representation and fiber maps are unitary and m_n is essentially bounded.

§DEP_HARD := S26-D-00004035

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004054|T|A|Proof+Compile+Meta+Tower] Recursive energy estimate

§FORM := Assume ||R_n||<=r, ||Z_{p,epsilon}^*||<=c(epsilon), and ||Pi_n||<=M on the chosen state space. Then ||u_{n,j,epsilon}||<=M r^j c(epsilon)||u||. If r<1 and c(epsilon) is bounded along the selected scale sequence, the generated energies are uniformly bounded and decay in j.

§ASSUME := The three operators are bounded on the declared common state space.

§DEP_HARD := S26-D-00004051, S26-T-00004053

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-D-00004055|D|S|Proof+Compile+Meta+Topo+Spectral+Tower] Fused pro-limit and spectral direct limit

§FORM := For a coherent inverse geometric tower (O_n,p_n) define O_infinity=lim<- O_n as a pro-space. For isometric spectral embeddings J_n,n+1 define H_infinity=lim-> H_n. If L_{n+1}J_n,n+1=J_n,n+1L_n on compatible cores, the algebraic limit carries a well-defined symmetric operator L_infinity before closure.

§ASSUME := Bonding maps, isometric embeddings, and dense compatible cores are supplied; existence of a self-adjoint closure is a separate requirement.

§DEP_HARD := S26-D-00004040, S26-D-00004052

§EVIDENCE := S · Project specification or modeling convention.

§BOUNDARY := O_infinity is called a pro-space, not an ordinary infinite-dimensional orbifold; self-adjointness of L_infinity is not automatic. Synopsis 2026 - Volume IV research draft - v0.1.0 15 Part L - Invariants, obstructions, and research frontier

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004056|D|A|Proof+Compile+Meta+Topo+Tower] Orbifold Euler characteristic ledger

§FORM := For a finite orbicellular decomposition, define chi_orb(O)=sum_sigma (-1)^dim(sigma)/|G_sigma|, where G_sigma is the generic isotropy group of the open cell sigma. Refinement invariance is required before treating this weighted sum as an invariant of the orbifold rather than of a presentation.

§ASSUME := Finite compatible orbicellular decomposition with constant isotropy type on each open cell.

§DEP_HARD := S26-D-00004023, S26-D-00004024, S26-D-00004042

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004057|T|A|Proof+Compile+Meta+Topo+Spectral+Tower] Heat-trace singular contribution

§FORM := For a compact Riemannian orbifold, Tr(exp(-tL_n)) has a small-time asymptotic expansion with the ordinary manifold coefficients plus contributions indexed by singular strata and local isotropy data. These terms can reveal singular structure, but the spectrum does not in general determine the full orbifold uniquely.

§ASSUME := Compact Riemannian orbifold and self-adjoint elliptic Laplace-type operator.

§DEP_HARD := S26-D-00004023, S26-T-00004033

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004058|T|A|Proof+Compile+Meta+Spectral+Tower] Pure eigenbasis invariants

§FORM := A pure basis plug-in preserves eigenvalues with multiplicity, every spectral projector E_n(B), heat trace, spectral zeta function where defined, and any functional calculus f(L_n). It changes coordinates inside spectral fibers but not the operator's unitary equivalence class.

§ASSUME := The plug-in is fiberwise unitary with multiplier m_n=1.

§DEP_HARD := S26-D-00004035, S26-T-00004053, S26-T-00004057

§EVIDENCE := A · Exact mathematical statement or definition; no independent formal-verification promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004059|N|S|Proof+Compile+Meta+Topo+Spectral+Tower] Obstruction and non-equivalence ledger

§FORM := The following implications are forbidden without extra hypotheses: visual sphere => hyperbolic S^2; finite cell image => 720 distinct quotient chambers; self-similar picture => IFS attractor; eigenfunction expansion => purely discrete spectrum; inverse limit => orbifold; basis change => geometric deformation; bounded finite computation => theorem proof.

§ASSUME := Each arrow is treated as a separate theorem obligation.

§DEP_HARD := S26-N-00004017, S26-T-00004020, S26-N-00004030, S26-T-00004034, S26-D-00004040, S26-T-00004043, S26-D-00004055, S26-T-00004058

§EVIDENCE := S · Project specification or modeling convention.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-C-00004060|C|H|Proof+Compile+Meta+IG+Spectral+Tower] Spectral-renormalization fixed-point conjecture

§FORM := Conjecture. Suppose a coherent compact Cosmo tower has stratum-compatible similitude generators, uniformly elliptic L_n, and spectral plug-ins intertwining both tower maps and cellular recursion. After a dimension- and scale-dependent normalization a_n,j L_n, the induced spectral measures admit a nontrivial subsequential fixed point under the combined recursion (n,j,epsilon)->(n+1,j+1,theta epsilon).

§ASSUME := Uniform geometric bounds, tightness of normalized spectral measures, controlled multiplicities, and a fixed theta in (0,1) are required; no existence or uniqueness claim is presently proved.

§DEP_HARD := S26-D-00004040, S26-D-00004045, S26-T-00004050, S26-D-00004052, S26-T-00004054, S26-D-00004055, S26-T-00004057

§EVIDENCE := H · Open hypothesis / conjectural research statement.

§BOUNDARY := This is a conjectural programmatic statement. It must not be promoted to a theorem without compactness, tightness, and convergence proofs in a specified model. Synopsis 2026 - Volume IV research draft - v0.1.0 16

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | hypothesis_remains_open

§CLAIM[S26-D-00004061|D|A|Proof+Compile+Meta+Tower] Local-field Cayley-Dickson doubling

§FORM := For a field K of characteristic not 2 with involutive K-algebra A, a generalized Cayley-Dickson double is A plus A*u with multiplication fixed by a declared parameter gamma in K^x and a declared conjugation convention.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-C-00004060, S26-D-00000103

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004062|T|A|Proof+Compile+Meta+Topo+Tower] Local-field Cayley-Dickson doubling law/interface

§FORM := Starting from the base field K, k Cayley-Dickson doublings produce a K-vector space of dimension 2^k. More generally, starting from a finite-dimensional involutive K-algebra A_0 of dimension d, k doublings produce K-dimension d*2^k. Algebraic laws retained or lost depend on the starting algebra and doubling parameters, not on geometric dimension n.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter is declared at each stage. | The 2^k formula refers to the tower starting from K itself; for a general finite-dimensional starting algebra A_0, dim_K(A_k)=dim_K(A_0)*2^k.

§DEP_HARD := S26-D-00004061

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004063|A|A|Proof+Compile+Meta+Spectral+Tower] Local-field Cayley-Dickson doubling operator

§FORM := CD_{gamma}(a,b;c,d) denotes the declared doubling product, and bar(a,b)=(bar(a),-b) denotes the recursive involution.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-T-00004062

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004064|N|S|Proof+Compile+Meta+Tower] Local-field Cayley-Dickson doubling fence

§FORM := The symbols R,C,H,O,Sedenion name particular real forms; over Q_p or another field the doubled algebra is field- and parameter-dependent and must not be silently identified with the real division forms.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The symbols R,C,H,O,Sedenion name particular real forms; over Q_p or another field the doubled algebra is field- and parameter-dependent and must not be silently identified with the real division forms.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004065|B|S|Proof+Compile+Meta+Tower] Local-field Cayley-Dickson doubling cross-volume bridge

§FORM := This construction adds arithmetic place v to the Volume IV coordinates: C_(n,k,v,j,epsilon,t), with v=infinity for the Archimedean chart or v=p for Q_p.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-N-00004064

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004066|N|S|Proof+Compile+Meta+Tower] Local-field Cayley-Dickson doubling evidence ledger

§FORM := Evidence rule for Local-field Cayley-Dickson doubling: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-B-00004065

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004067|B|S|Proof+Compile+Meta+Spectral+Tower] Local-field Cayley-Dickson doubling emergence rule

§FORM := Emergence rule for Local-field Cayley-Dickson doubling: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-N-00004066

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004068|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Local-field Cayley-Dickson doubling frontier

§FORM := Open frontier for Local-field Cayley-Dickson doubling: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-B-00004067

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The symbols R,C,H,O,Sedenion name particular real forms; over Q_p or another field the doubled algebra is field- and parameter-dependent and must not be silently identified with the real division forms.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004069|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.1

§FORM := Fixture T01.P01 - Over Q_3 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-N-00004068

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004070|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.2

§FORM := Fixture T01.P02 - Over Q_5 coordinates with dimension 4 and declared gamma tuple ['-1', '-5'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004069

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004071|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.3

§FORM := Fixture T01.P03 - Over Q_7 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004070

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004072|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.4

§FORM := Fixture T01.P04 - Over Q_3 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-3'], the exact basis product e1*e10=-1e11; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004071

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004073|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.5

§FORM := Fixture T01.P05 - Over Q_5 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004072

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004074|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.6

§FORM := Fixture T01.P06 - Over Q_7 coordinates with dimension 4 and declared gamma tuple ['-1', '-7'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004073

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004075|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.7

§FORM := Fixture T01.P07 - Over Q_3 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e3=-1e2; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004074

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004076|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.8

§FORM := Fixture T01.P08 - Over Q_5 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-5'], the exact basis product e1*e6=-1e7; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004075

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004077|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.9

§FORM := Fixture T01.P09 - Over Q_7 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004076

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004078|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.10

§FORM := Fixture T01.P10 - Over Q_3 coordinates with dimension 4 and declared gamma tuple ['-1', '-3'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004077

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004079|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.11

§FORM := Fixture T01.P11 - Over Q_5 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004078

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004080|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 1.12

§FORM := Fixture T01.P12 - Over Q_7 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-7'], the exact basis product e1*e2=1e3; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage.

§DEP_HARD := S26-A-00004063, S26-E-00004079

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004081|D|A|Proof+Compile+Meta+Topo+Tower] p-adic valuation and ultrametric carrier

§FORM := For nonzero rational x, v_p(x) is the exponent of p in x and |x|_p=p^{-v_p(x)}; completion of Q in this absolute value gives Q_p.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-E-00004080, S26-D-00000075

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004082|T|A|Proof+Compile+Meta+Topo+Tower] p-adic valuation and ultrametric carrier law/interface

§FORM := The p-adic norm satisfies the strong triangle inequality |x+y|_p <= max(|x|_p,|y|_p), so balls are hierarchically nested or disjoint rather than Euclidean shells.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-D-00004081

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004083|A|A|Proof+Compile+Meta+Topo+Spectral+Tower] p-adic valuation and ultrametric carrier operator

§FORM := The arithmetic-place selector Place_v routes a local object to the Archimedean chart v=infinity or to the non-Archimedean chart v=p.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-T-00004082

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004084|N|S|Proof+Compile+Meta+Topo+Tower] p-adic valuation and ultrametric carrier fence

§FORM := A p-adic ball is not a distorted Euclidean ball; its ultrametric topology and scale nesting are part of the object, not a visualization choice.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-A-00004083

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A p-adic ball is not a distorted Euclidean ball; its ultrametric topology and scale nesting are part of the object, not a visualization choice.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004085|B|S|Proof+Compile+Meta+Topo+Tower] p-adic valuation and ultrametric carrier cross-volume bridge

§FORM := The nested-ball geometry supplies a mathematically literal Russian-doll scale axis for the hypercomplex carrier.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-N-00004084

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004086|N|S|Proof+Compile+Meta+Topo+Tower] p-adic valuation and ultrametric carrier evidence ledger

§FORM := Evidence rule for p-adic valuation and ultrametric carrier: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-B-00004085

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004087|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] p-adic valuation and ultrametric carrier emergence rule

§FORM := Emergence rule for p-adic valuation and ultrametric carrier: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-N-00004086

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004088|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] p-adic valuation and ultrametric carrier frontier

§FORM := Open frontier for p-adic valuation and ultrametric carrier: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-B-00004087

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A p-adic ball is not a distorted Euclidean ball; its ultrametric topology and scale nesting are part of the object, not a visualization choice.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004089|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.1

§FORM := Fixture T02.P01 - For p=2, a=2, b=3, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-A-00004083, S26-N-00004088

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004090|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.2

§FORM := Fixture T02.P02 - For p=3, a=9/2, b=5/3, |a+b|_p=3 and max(|a|_p,|b|_p)=3; strong-triangle check=True.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-A-00004083, S26-E-00004089

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004091|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.3

§FORM := Fixture T02.P03 - For p=5, a=100/3, b=35/4, |a+b|_p=0.2 and max(|a|_p,|b|_p)=0.2; strong-triangle check=True.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-A-00004083, S26-E-00004090

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004092|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.4

§FORM := Fixture T02.P04 - For p=7, a=5/4, b=9/5, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-A-00004083, S26-E-00004091

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004093|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.5

§FORM := Fixture T02.P05 - For p=2, a=12/5, b=11/3, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.

§DEP_HARD := S26-A-00004083, S26-E-00004092

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004094|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.6

§FORM := Fixture T02.P06 - For p=3, a=21/2, b=13/7, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.
§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.
§DEP_HARD := S26-A-00004083, S26-E-00004093
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004095|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.7

§FORM := Fixture T02.P07 - For p=5, a=8/7, b=75/8, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.
§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.
§DEP_HARD := S26-A-00004083, S26-E-00004094
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004096|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.8

§FORM := Fixture T02.P08 - For p=7, a=63/8, b=17/9, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.
§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.
§DEP_HARD := S26-A-00004083, S26-E-00004095
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004097|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.9

§FORM := Fixture T02.P09 - For p=2, a=40/9, b=19/5, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.
§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.
§DEP_HARD := S26-A-00004083, S26-E-00004096
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004098|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.10

§FORM := Fixture T02.P10 - For p=3, a=11/10, b=21/11, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.
§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.
§DEP_HARD := S26-A-00004083, S26-E-00004097
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004099|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.11

§FORM := Fixture T02.P11 - For p=5, a=60/11, b=115/12, |a+b|_p=0.2 and max(|a|_p,|b|_p)=0.2; strong-triangle check=True.
§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.
§DEP_HARD := S26-A-00004083, S26-E-00004098
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004100|E|M|Proof+Compile+Meta+Topo+Tower] Ultrametric finite probe 2.12

§FORM := Fixture T02.P12 - For p=7, a=637/12, b=25/13, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.
§ASSUME := p is a prime and Q_p is used with its standard normalized p-adic absolute value.
§DEP_HARD := S26-A-00004083, S26-E-00004099
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004101|D|A|Proof+Compile+Meta+Tower] Quaternion-octonion local-field strata

§FORM := Quaternion and octonion algebras may be defined over fields beyond R by generalized Cayley-Dickson or equivalent composition-algebra constructions, with isomorphism type controlled by the field and norm data.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-E-00004100, S26-D-00000103
§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004102|T|A|Proof+Compile+Meta+Tower] Quaternion-octonion local-field strata law/interface

§FORM := Associativity survives at the quaternion stratum while octonionic multiplication is alternative rather than associative; split versus division behavior is a separate local-field question.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-D-00004101
§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004103|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Quaternion-octonion local-field strata operator

§FORM := LocalForm(K_v,gamma_1,...,gamma_k) records the chosen arithmetic form before any geometric or cognitive interpretation.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-T-00004102
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004104|N|S|Proof+Compile+Meta+Topo+Tower] Quaternion-octonion local-field strata fence

§FORM := No claim of a canonical p-adic sedenion manifold, p-adic normed division tower, or field-independent multiplication table is permitted.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-A-00004103
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := No claim of a canonical p-adic sedenion manifold, p-adic normed division tower, or field-independent multiplication table is permitted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004105|B|S|Proof+Compile+Meta+Tower] Quaternion-octonion local-field strata cross-volume bridge

§FORM := Local algebra forms become fibers over the Cosmo carrier, while their base field remains an explicit coordinate.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-N-00004104
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004106|N|S|Proof+Compile+Meta+Tower] Quaternion-octonion local-field strata evidence ledger

§FORM := Evidence rule for Quaternion-octonion local-field strata: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-B-00004105
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004107|B|S|Proof+Compile+Meta+Spectral+Tower] Quaternion-octonion local-field strata emergence rule

§FORM := Emergence rule for Quaternion-octonion local-field strata: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-N-00004106
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004108|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Quaternion-octonion local-field strata frontier

§FORM := Open frontier for Quaternion-octonion local-field strata: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-B-00004107
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := No claim of a canonical p-adic sedenion manifold, p-adic normed division tower, or field-independent multiplication table is permitted.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004109|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.1

§FORM := Fixture T03.P01 - Over Q_3 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-A-00004103, S26-N-00004108
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004110|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.2

§FORM := Fixture T03.P02 - Over Q_5 coordinates with dimension 4 and declared gamma tuple ['-1', '-5'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-A-00004103, S26-E-00004109
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004111|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.3

§FORM := Fixture T03.P03 - Over Q_7 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-A-00004103, S26-E-00004110
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004112|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.4

§FORM := Fixture T03.P04 - Over Q_3 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-3'], the exact basis product e1*e10=-1e11; minimum finite p-adic valuation among nonzero output coordinates is 0.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-A-00004103, S26-E-00004111
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004113|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.5

§FORM := Fixture T03.P05 - Over Q_5 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.
§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.
§DEP_HARD := S26-A-00004103, S26-E-00004112
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004114|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.6

§FORM := Fixture T03.P06 - Over Q_7 coordinates with dimension 4 and declared gamma tuple ['-1', '-7'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.

§DEP_HARD := S26-A-00004103, S26-E-00004113

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004115|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.7

§FORM := Fixture T03.P07 - Over Q_3 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e3=-1e2; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.

§DEP_HARD := S26-A-00004103, S26-E-00004114

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004116|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.8

§FORM := Fixture T03.P08 - Over Q_5 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-5'], the exact basis product e1*e6=-1e7; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.

§DEP_HARD := S26-A-00004103, S26-E-00004115

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004117|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.9

§FORM := Fixture T03.P09 - Over Q_7 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.

§DEP_HARD := S26-A-00004103, S26-E-00004116

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004118|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.10

§FORM := Fixture T03.P10 - Over Q_3 coordinates with dimension 4 and declared gamma tuple ['-1', '-3'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.

§DEP_HARD := S26-A-00004103, S26-E-00004117

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004119|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.11

§FORM := Fixture T03.P11 - Over Q_5 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.

§DEP_HARD := S26-A-00004103, S26-E-00004118

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004120|E|M|Proof+Compile+Meta+Tower] p-adic CD finite probe 3.12

§FORM := Fixture T03.P12 - Over Q_7 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-7'], the exact basis product e1*e2=1e3; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated.

§DEP_HARD := S26-A-00004103, S26-E-00004119

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004121|D|S|Proof+Compile+Meta+Topo+Tower] Adelic hypercomplex place atlas

§FORM := An adelic hypercomplex atlas is a family of local Cayley-Dickson carriers A_{k,v} indexed by arithmetic places v, together with compatibility and restricted-product data rather than a single field pretending to contain every local geometry.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-E-00004120, S26-T-00000051

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004122|D|S|Proof+Compile+Meta+Topo+Tower] Adelic hypercomplex place atlas law/interface

§FORM := Local data are compared by explicit transport maps; agreement at many places does not imply identity of local algebras or geometries.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-D-00004121

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004123|A|S|Proof+Compile+Meta+Spectral+Tower] Adelic hypercomplex place atlas operator

§FORM := A restricted-place aggregator records an Archimedean component and finitely supported non-Archimedean deviations, preserving each local valuation.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-D-00004122

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004124|N|S|Proof+Compile+Meta+Tower] Adelic hypercomplex place atlas fence

§FORM := Adelic synthesis is a local-global bookkeeping architecture; it does not turn p-adic and real distances into one scalar metric.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-A-00004123
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := Adelic synthesis is a local-global bookkeeping architecture; it does not turn p-adic and real distances into one scalar metric.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004125|B|S|Proof+Compile+Meta+Topo+Tower] Adelic hypercomplex place atlas cross-volume bridge

§FORM := The atlas makes arithmetic place a sheaf-like cover coordinate compatible with the existing Cosmo-n tower.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-N-00004124
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004126|N|S|Proof+Compile+Meta+Tower] Adelic hypercomplex place atlas evidence ledger

§FORM := Evidence rule for Adelic hypercomplex place atlas: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-B-00004125
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004127|B|S|Proof+Compile+Meta+Spectral+Tower] Adelic hypercomplex place atlas emergence rule

§FORM := Emergence rule for Adelic hypercomplex place atlas: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-N-00004126
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004128|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Adelic hypercomplex place atlas frontier

§FORM := Open frontier for Adelic hypercomplex place atlas: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-B-00004127
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := Adelic synthesis is a local-global bookkeeping architecture; it does not turn p-adic and real distances into one scalar metric.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004129|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.1

§FORM := Fixture T04.P01 - For rational x=2/3, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3] is 1; residual from 1 is 0.000e+00.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-A-00004123, S26-N-00004128
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004130|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.2

§FORM := Fixture T04.P02 - For rational x=3/4, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3] is 1; residual from 1 is 0.000e+00.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-A-00004123, S26-E-00004129
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004131|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.3

§FORM := Fixture T04.P03 - For rational x=4/5, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 5] is 1; residual from 1 is 0.000e+00.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-A-00004123, S26-E-00004130
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004132|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.4

§FORM := Fixture T04.P04 - For rational x=5/6, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 5] is 1; residual from 1 is 0.000e+00.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-A-00004123, S26-E-00004131
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004133|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.5

§FORM := Fixture T04.P05 - For rational x=6/7, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 7] is 1; residual from 1 is 0.000e+00.
§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.
§DEP_HARD := S26-A-00004123, S26-E-00004132
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004134|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.6

§FORM := Fixture T04.P06 - For rational x=7/8, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 7] is 1; residual from 1 is 0.000e+00.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-A-00004123, S26-E-00004133

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004135|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.7

§FORM := Fixture T04.P07 - For rational x=8/9, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3] is 1; residual from 1 is 0.000e+00.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-A-00004123, S26-E-00004134

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004136|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.8

§FORM := Fixture T04.P08 - For rational x=9/10, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 5] is 1; residual from 1 is 1.110e-16.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-A-00004123, S26-E-00004135

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004137|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.9

§FORM := Fixture T04.P09 - For rational x=10/11, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 5, 11] is 1; residual from 1 is 0.000e+00.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-A-00004123, S26-E-00004136

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004138|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.10

§FORM := Fixture T04.P10 - For rational x=11/12, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 11] is 1; residual from 1 is 0.000e+00.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-A-00004123, S26-E-00004137

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004139|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.11

§FORM := Fixture T04.P11 - For rational x=12/13, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 13] is 1; residual from 1 is 0.000e+00.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-A-00004123, S26-E-00004138

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004140|E|M|Proof+Compile+Meta+Tower] Product-formula place probe 4.12

§FORM := Fixture T04.P12 - For rational x=13/14, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 7, 13] is 1; residual from 1 is 0.000e+00.

§ASSUME := Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared.

§DEP_HARD := S26-A-00004123, S26-E-00004139

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004141|D|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex bundle over mixed places

§FORM := A mixed-place hypercomplex carrier is a family of bundles or local systems E_{n,k,v}->O_{n,v} whose fibers carry A_{k,v}; it is not assumed to be one ordinary manifold across all v.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-E-00004140, S26-D-00003553

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004142|D|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex bundle over mixed places law/interface

§FORM := Base geometry, arithmetic place and fiber algebra are independent coordinates and transition data must state which of them are changed.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-D-00004141

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004143|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Hypercomplex bundle over mixed places operator

§FORM := FiberState(n,k,v,x) returns a typed local section value together with its frame, valuation/norm convention and evidence state.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-D-00004142

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004144|N|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex bundle over mixed places fence

§FORM := The phrase hypercomplex manifold is shorthand only when a real or non-Archimedean analytic base and fiber action have been separately defined.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The phrase hypercomplex manifold is shorthand only when a real or non-Archimedean analytic base and fiber action have been separately defined.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004145|B|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex bundle over mixed places cross-volume bridge

§FORM := This replaces a flat 2D picture by a chart family whose readable images are projections of a higher typed carrier.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-N-00004144

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004146|N|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex bundle over mixed places evidence ledger

§FORM := Evidence rule for Hypercomplex bundle over mixed places: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-B-00004145

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004147|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] Hypercomplex bundle over mixed places emergence rule

§FORM := Emergence rule for Hypercomplex bundle over mixed places: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-N-00004146

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004148|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Hypercomplex bundle over mixed places frontier

§FORM := Open frontier for Hypercomplex bundle over mixed places: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-B-00004147

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The phrase hypercomplex manifold is shorthand only when a real or non-Archimedean analytic base and fiber action have been separately defined.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004149|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.1

§FORM := Fixture T05.P01 - A deterministic orthogonal real carrier frame in dimension 2 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-N-00004148

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004150|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.2

§FORM := Fixture T05.P02 - A deterministic orthogonal real carrier frame in dimension 4 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004149

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004151|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.3

§FORM := Fixture T05.P03 - A deterministic orthogonal real carrier frame in dimension 8 changed the sampled Euclidean coordinate norm by 8.882e-16; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004150

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004152|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.4

§FORM := Fixture T05.P04 - A deterministic orthogonal real carrier frame in dimension 16 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004151

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004153|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.5

§FORM := Fixture T05.P05 - A deterministic orthogonal real carrier frame in dimension 2 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004152

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004154|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.6

§FORM := Fixture T05.P06 - A deterministic orthogonal real carrier frame in dimension 4 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004153

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004155|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.7

§FORM := Fixture T05.P07 - A deterministic orthogonal real carrier frame in dimension 8 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004154

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004156|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.8

§FORM := Fixture T05.P08 - A deterministic orthogonal real carrier frame in dimension 16 changed the sampled Euclidean coordinate norm by 8.882e-16; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004155

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004157|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.9

§FORM := Fixture T05.P09 - A deterministic orthogonal real carrier frame in dimension 2 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004156

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004158|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.10

§FORM := Fixture T05.P10 - A deterministic orthogonal real carrier frame in dimension 4 changed the sampled Euclidean coordinate norm by 4.441e-16; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004157

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004159|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.11

§FORM := Fixture T05.P11 - A deterministic orthogonal real carrier frame in dimension 8 changed the sampled Euclidean coordinate norm by 2.220e-16; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004158

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004160|E|M|Proof+Compile+Meta+Topo+Tower] Mixed-place fiber-frame probe 5.12

§FORM := Fixture T05.P12 - A deterministic orthogonal real carrier frame in dimension 16 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

§ASSUME := The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras.

§DEP_HARD := S26-A-00004143, S26-E-00004159

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004161|D|S|Proof+Compile+Meta+Topo+Tower] Berkovich-building local chart interface

§FORM := Volume IV permits Berkovich or rigid analytic charts for p-adic analytic geometry and Bruhat-Tits trees/buildings for group-theoretic incidence, without treating them as Euclidean coordinate grids.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-E-00004160, S26-D-00000301

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004162|D|S|Proof+Compile+Meta+Topo+Tower] Berkovich-building local chart interface law/interface

§FORM := For PGL_2(Q_p), the Bruhat-Tits tree is a locally finite tree and therefore a 0-hyperbolic graph; its boundary encodes projective p-adic directions.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-D-00004161

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004163|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Berkovich-building local chart interface operator

§FORM := Chart_v selects the declared analytic/building chart and records which observables are topological, metric, combinatorial or spectral.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-D-00004162

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004164|N|S|Proof+Compile+Meta+Topo+Tower] Berkovich-building local chart interface fence

§FORM := A Bruhat-Tits tree is not a discretized Poincare disk, and a Berkovich point need not correspond to one Q_p coordinate.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A Bruhat-Tits tree is not a discretized Poincare disk, and a Berkovich point need not correspond to one Q_p coordinate.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004165|B|S|Proof+Compile+Meta+Topo+Tower+Swirl] Berkovich-building local chart interface cross-volume bridge

§FORM := The building supplies a non-Euclidean multi-mesh skeleton on which fibers, transports and recognition overlaps can be attached.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-N-00004164

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004166|N|S|Proof+Compile+Meta+Topo+Tower] Berkovich-building local chart interface evidence ledger

§FORM := Evidence rule for Berkovich-building local chart interface: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-B-00004165

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004167|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] Berkovich-building local chart interface emergence rule

§FORM := Emergence rule for Berkovich-building local chart interface: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-N-00004166

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004168|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Berkovich-building local chart interface frontier

§FORM := Open frontier for Berkovich-building local chart interface: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-B-00004167

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A Bruhat-Tits tree is not a discretized Poincare disk, and a Berkovich point need not correspond to one Q_p coordinate.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004169|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.1

§FORM := Fixture T06.P01 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-N-00004168

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004170|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.2

§FORM := Fixture T06.P02 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004169

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004171|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.3

§FORM := Fixture T06.P03 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004170

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004172|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.4

§FORM := Fixture T06.P04 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004171

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004173|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.5

§FORM := Fixture T06.P05 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004172

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004174|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.6

§FORM := Fixture T06.P06 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004173

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004175|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.7

§FORM := Fixture T06.P07 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004174

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004176|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.8

§FORM := Fixture T06.P08 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004175

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004177|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.9

§FORM := Fixture T06.P09 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004176

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004178|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.10

§FORM := Fixture T06.P10 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004177

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004179|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.11

§FORM := Fixture T06.P11 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004178

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004180|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 6.12

§FORM := Fixture T06.P12 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature.

§DEP_HARD := S26-A-00004163, S26-E-00004179

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004181|D|A|Proof+Compile+Meta+Topo+Tower] Fiberwise arithmetic Cayley-Dickson product

§FORM := Fiberwise multiplication applies the local Cayley-Dickson product pointwise to compatible sections in the same A_{k,v} fiber.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-E-00004180, S26-D-00003554

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004182|T|A|Proof+Compile+Meta+Topo+Tower] Fiberwise arithmetic Cayley-Dickson product law/interface

§FORM := Changing local frame may change coordinates but must preserve the declared algebraic identities when the frame transformation is an algebra automorphism.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-D-00004181

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004183|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Fiberwise arithmetic Cayley-Dickson product operator

§FORM := Mul_{n,k,v}(s,t)(x)=s(x)t(x) on the common domain, with associator brackets retained when k exceeds the associative strata.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-T-00004182

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004184|N|S|Proof+Compile+Meta+Topo+Tower] Fiberwise arithmetic Cayley-Dickson product fence

§FORM := A generic linear frame change is not automatically an algebra automorphism, and a basis rotation must not be mislabeled as dynamics.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A generic linear frame change is not automatically an algebra automorphism, and a basis rotation must not be mislabeled as dynamics.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004185|B|S|Proof+Compile+Meta+Topo+Tower] Fiberwise arithmetic Cayley-Dickson product cross-volume bridge

§FORM := This is the algebraic coupling between the Cosmo carrier and the arithmetic hypercomplex atlas.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-N-00004184

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004186|N|S|Proof+Compile+Meta+Topo+Tower] Fiberwise arithmetic Cayley-Dickson product evidence ledger

§FORM := Evidence rule for Fiberwise arithmetic Cayley-Dickson product: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-B-00004185

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004187|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] Fiberwise arithmetic Cayley-Dickson product emergence rule

§FORM := Emergence rule for Fiberwise arithmetic Cayley-Dickson product: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-N-00004186

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004188|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Fiberwise arithmetic Cayley-Dickson product frontier

§FORM := Open frontier for Fiberwise arithmetic Cayley-Dickson product: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-B-00004187

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A generic linear frame change is not automatically an algebra automorphism, and a basis rotation must not be mislabeled as dynamics.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004189|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.1

§FORM := Fixture T07.P01 - Over Q_3 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-N-00004188

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004190|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.2

§FORM := Fixture T07.P02 - Over Q_5 coordinates with dimension 4 and declared gamma tuple ['-1', '-5'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004189

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004191|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.3

§FORM := Fixture T07.P03 - Over Q_7 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004190

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004192|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.4

§FORM := Fixture T07.P04 - Over Q_3 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-3'], the exact basis product e1*e10=-1e11; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004191

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004193|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.5

§FORM := Fixture T07.P05 - Over Q_5 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004192

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004194|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.6

§FORM := Fixture T07.P06 - Over Q_7 coordinates with dimension 4 and declared gamma tuple ['-1', '-7'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004193

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004195|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.7

§FORM := Fixture T07.P07 - Over Q_3 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e3=-1e2; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004194

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004196|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.8

§FORM := Fixture T07.P08 - Over Q_5 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-5'], the exact basis product e1*e6=-1e7; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004195

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004197|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.9

§FORM := Fixture T07.P09 - Over Q_7 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004196

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004198|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.10

§FORM := Fixture T07.P10 - Over Q_3 coordinates with dimension 4 and declared gamma tuple ['-1', '-3'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004197

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004199|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.11

§FORM := Fixture T07.P11 - Over Q_5 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004198

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004200|E|M|Proof+Compile+Meta+Topo+Tower] p-adic CD finite probe 7.12

§FORM := Fixture T07.P12 - Over Q_7 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-7'], the exact basis product e1*e2=1e3; minimum finite p-adic valuation among nonzero output coordinates is 0.

§ASSUME := Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention.

§DEP_HARD := S26-A-00004183, S26-E-00004199

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004201|D|S|Proof+Compile+Meta+Tower] Place-changing transport with remainder

§FORM := A place-changing transport T_{v->w} is a structure-preserving comparison record between local representations, not a claim that Q_p and R are metrically identical.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-E-00004200, S26-D-00004039

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004202|D|S|Proof+Compile+Meta+Tower] Place-changing transport with remainder law/interface

§FORM := Every transport names preserved invariants, transformed structure, losses, unknowns and active remainder; perfect lossless equivalence is not assumed.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-D-00004201

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004203|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Place-changing transport with remainder operator

§FORM := Hol_{v->w->v} compares the returned source state with the original using a typed invariant delta rather than subtracting heterogeneous geometries.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-D-00004202

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004204|N|S|Proof+Compile+Meta+Topo+Tower] Place-changing transport with remainder fence

§FORM := No scalar distance may compare Archimedean and p-adic geometry unless an explicit common observable and normalization are supplied.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := No scalar distance may compare Archimedean and p-adic geometry unless an explicit common observable and normalization are supplied.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004205|B|S|Proof+Compile+Meta+Tower] Place-changing transport with remainder cross-volume bridge

§FORM := This instantiates the Trasgo transport calculus on the new arithmetic-place axis.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-N-00004204

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004206|N|S|Proof+Compile+Meta+Tower] Place-changing transport with remainder evidence ledger

§FORM := Evidence rule for Place-changing transport with remainder: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-B-00004205

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004207|B|S|Proof+Compile+Meta+Spectral+Tower] Place-changing transport with remainder emergence rule

§FORM := Emergence rule for Place-changing transport with remainder: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-N-00004206

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004208|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Place-changing transport with remainder frontier

§FORM := Open frontier for Place-changing transport with remainder: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-B-00004207

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := No scalar distance may compare Archimedean and p-adic geometry unless an explicit common observable and normalization are supplied.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004209|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.1

§FORM := Fixture T08.P01 - The same rational coordinate x=2 has Archimedean absolute value 2 and 2-adic absolute value 0.5; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-N-00004208

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004210|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.2

§FORM := Fixture T08.P02 - The same rational coordinate x=9/2 has Archimedean absolute value 4.5 and 3-adic absolute value 0.111111; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004209

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004211|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.3

§FORM := Fixture T08.P03 - The same rational coordinate x=4/3 has Archimedean absolute value 1.33333 and 5-adic absolute value 1; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004210

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004212|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.4

§FORM := Fixture T08.P04 - The same rational coordinate x=5/2 has Archimedean absolute value 2.5 and 2-adic absolute value 2; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004211

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004213|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.5

§FORM := Fixture T08.P05 - The same rational coordinate x=6/5 has Archimedean absolute value 1.2 and 3-adic absolute value 0.333333; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004212

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004214|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.6

§FORM := Fixture T08.P06 - The same rational coordinate x=35/6 has Archimedean absolute value 5.83333 and 5-adic absolute value 0.2; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004213

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004215|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.7

§FORM := Fixture T08.P07 - The same rational coordinate x=8/7 has Archimedean absolute value 1.14286 and 2-adic absolute value 0.125; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004214

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004216|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.8

§FORM := Fixture T08.P08 - The same rational coordinate x=27/8 has Archimedean absolute value 3.375 and 3-adic absolute value 0.037037; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004215

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004217|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.9

§FORM := Fixture T08.P09 - The same rational coordinate x=10/9 has Archimedean absolute value 1.11111 and 5-adic absolute value 0.2; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004216

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004218|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.10

§FORM := Fixture T08.P10 - The same rational coordinate x=11/5 has Archimedean absolute value 2.2 and 2-adic absolute value 1; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004217

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004219|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.11

§FORM := Fixture T08.P11 - The same rational coordinate x=12/11 has Archimedean absolute value 1.09091 and 3-adic absolute value 0.333333; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004218

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004220|E|M|Proof+Compile+Meta+Tower] Place-transport typed-observable probe 8.12

§FORM := Fixture T08.P12 - The same rational coordinate x=65/12 has Archimedean absolute value 5.41667 and 5-adic absolute value 0.2; the fixture records both coordinates instead of subtracting them as commensurate distances.

§ASSUME := Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed.

§DEP_HARD := S26-A-00004203, S26-E-00004219

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004221|D|A|Proof+Compile+Meta+IG+Tower] Commutator perspective channel

§FORM := The commutator [a,b]=ab-ba measures order sensitivity of multiplication.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-E-00004220, S26-D-00000268

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004222|T|A|Proof+Compile+Meta+Tower] Commutator perspective channel law/interface

§FORM := A vanishing commutator is an algebraic statement about the chosen pair; it does not imply that the whole algebra is commutative.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-D-00004221

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004223|A|A|Proof+Compile+Meta+Spectral+Tower] Commutator perspective channel operator

§FORM := Comm(a,b)=ab-ba is retained as a vector-valued channel rather than collapsed to a norm unless a norm is declared.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-T-00004222

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004224|N|S|Proof+Compile+Meta+Tower] Commutator perspective channel fence

§FORM := Order dependence may be used as a cognitive metaphor only at S evidence; the algebraic commutator itself does not establish a cognitive mechanism.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Order dependence may be used as a cognitive metaphor only at S evidence; the algebraic commutator itself does not establish a cognitive mechanism.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004225|B|S|Proof+Compile+Meta+Tower] Commutator perspective channel cross-volume bridge

§FORM := The channel supplies one explicit non-flattening coordinate for hypercomplex section interactions.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-N-00004224

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004226|N|S|Proof+Compile+Meta+Tower] Commutator perspective channel evidence ledger

§FORM := Evidence rule for Commutator perspective channel: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-B-00004225

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004227|B|S|Proof+Compile+Meta+Spectral+Tower] Commutator perspective channel emergence rule

§FORM := Emergence rule for Commutator perspective channel: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-N-00004226

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004228|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Commutator perspective channel frontier

§FORM := Open frontier for Commutator perspective channel: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-B-00004227

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Order dependence may be used as a cognitive metaphor only at S evidence; the algebraic commutator itself does not establish a cognitive mechanism.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004229|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.1

§FORM := Fixture T09.P01 - In the real Cayley-Dickson fixture of dimension 8, basis indices (5, 6, 3) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-N-00004228

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004230|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.2

§FORM := Fixture T09.P02 - In the real Cayley-Dickson fixture of dimension 16, basis indices (9, 8, 5) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004229

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004231|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.3

§FORM := Fixture T09.P03 - In the real Cayley-Dickson fixture of dimension 8, basis indices (7, 1, 2) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004230

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004232|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.4

§FORM := Fixture T09.P04 - In the real Cayley-Dickson fixture of dimension 16, basis indices (14, 9, 10) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004231

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004233|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.5

§FORM := Fixture T09.P05 - In the real Cayley-Dickson fixture of dimension 8, basis indices (5, 1, 7) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004232

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004234|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.6

§FORM := Fixture T09.P06 - In the real Cayley-Dickson fixture of dimension 16, basis indices (5, 14, 6) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004233

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004235|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.7

§FORM := Fixture T09.P07 - In the real Cayley-Dickson fixture of dimension 8, basis indices (1, 7, 6) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004234

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004236|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.8

§FORM := Fixture T09.P08 - In the real Cayley-Dickson fixture of dimension 16, basis indices (6, 8, 15) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004235

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004237|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.9

§FORM := Fixture T09.P09 - In the real Cayley-Dickson fixture of dimension 8, basis indices (6, 7, 2) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004236

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004238|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.10

§FORM := Fixture T09.P10 - In the real Cayley-Dickson fixture of dimension 16, basis indices (2, 5, 10) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004237

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004239|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.11

§FORM := Fixture T09.P11 - In the real Cayley-Dickson fixture of dimension 8, basis indices (2, 7, 3) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004238

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004240|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 9.12

§FORM := Fixture T09.P12 - In the real Cayley-Dickson fixture of dimension 16, basis indices (1, 9, 3) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Elements lie in the same declared algebra and multiplication convention.

§DEP_HARD := S26-A-00004223, S26-E-00004239

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004241|D|A|Proof+Compile+Meta+IG+Tower] Associator contextual-bracketing channel

§FORM := The associator [a,b,c]=(ab)c-a(bc) measures failure of associativity.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-E-00004240, S26-D-00000268

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004242|T|A|Proof+Compile+Meta+Tower] Associator contextual-bracketing channel law/interface

§FORM := Octonions are alternative, so associators vanish on repeated arguments; higher Cayley-Dickson strata generally have weaker laws.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-D-00004241

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004243|A|A|Proof+Compile+Meta+Spectral+Tower] Associator contextual-bracketing channel operator

§FORM := Assoc(a,b,c) is stored with its full coordinate output and bracket provenance.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-T-00004242

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004244|N|S|Proof+Compile+Meta+Tower] Associator contextual-bracketing channel fence

§FORM := A large associator is not automatically better cognition; it is an algebraic defect signal whose usefulness must be tested.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A large associator is not automatically better cognition; it is an algebraic defect signal whose usefulness must be tested.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004245|B|S|Proof+Compile+Meta+Tower+Swirl] Associator contextual-bracketing channel cross-volume bridge

§FORM := Associator telemetry allows different relation bracketings to remain distinct through recognition and transport.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-N-00004244

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004246|N|S|Proof+Compile+Meta+Tower] Associator contextual-bracketing channel evidence ledger

§FORM := Evidence rule for Associator contextual-bracketing channel: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-B-00004245

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004247|B|S|Proof+Compile+Meta+Spectral+Tower] Associator contextual-bracketing channel emergence rule

§FORM := Emergence rule for Associator contextual-bracketing channel: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-N-00004246

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004248|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Associator contextual-bracketing channel frontier

§FORM := Open frontier for Associator contextual-bracketing channel: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-B-00004247

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A large associator is not automatically better cognition; it is an algebraic defect signal whose usefulness must be tested.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004249|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.1

§FORM := Fixture T10.P01 - In the real Cayley-Dickson fixture of dimension 8, basis indices (5, 4, 6) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-N-00004248

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004250|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.2

§FORM := Fixture T10.P02 - In the real Cayley-Dickson fixture of dimension 16, basis indices (13, 15, 4) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004249

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004251|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.3

§FORM := Fixture T10.P03 - In the real Cayley-Dickson fixture of dimension 8, basis indices (1, 5, 6) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004250

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004252|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.4

§FORM := Fixture T10.P04 - In the real Cayley-Dickson fixture of dimension 16, basis indices (14, 6, 12) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004251

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004253|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.5

§FORM := Fixture T10.P05 - In the real Cayley-Dickson fixture of dimension 8, basis indices (4, 2, 1) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004252

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004254|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.6

§FORM := Fixture T10.P06 - In the real Cayley-Dickson fixture of dimension 16, basis indices (14, 6, 10) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004253

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004255|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.7

§FORM := Fixture T10.P07 - In the real Cayley-Dickson fixture of dimension 8, basis indices (7, 1, 4) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004254

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004256|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.8

§FORM := Fixture T10.P08 - In the real Cayley-Dickson fixture of dimension 16, basis indices (6, 4, 15) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004255

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004257|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.9

§FORM := Fixture T10.P09 - In the real Cayley-Dickson fixture of dimension 8, basis indices (1, 4, 2) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004256

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004258|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.10

§FORM := Fixture T10.P10 - In the real Cayley-Dickson fixture of dimension 16, basis indices (15, 7, 9) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004257

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004259|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.11

§FORM := Fixture T10.P11 - In the real Cayley-Dickson fixture of dimension 8, basis indices (1, 7, 6) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004258

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004260|E|M|Proof+Compile+Meta+Tower] Algebraic defect finite probe 10.12

§FORM := Fixture T10.P12 - In the real Cayley-Dickson fixture of dimension 16, basis indices (12, 5, 2) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

§ASSUME := Three elements and one multiplication convention are declared.

§DEP_HARD := S26-A-00004243, S26-E-00004259

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004261|D|A|Proof+Compile+Meta+Tower] Zero-divisor null-channel ledger

§FORM := A zero-divisor channel records nonzero a,b with ab=0 together with the chosen algebra, basis and arithmetic place.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-E-00004260, S26-N-00004059

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004262|T|A|Proof+Compile+Meta+IG+Tower] Zero-divisor null-channel ledger law/interface

§FORM := A zero product may arise from nontrivial sources in algebras with zero divisors; information loss in that projection must therefore remain visible.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-D-00004261

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004263|A|S|Proof+Compile+Meta+Spectral+Tower] Zero-divisor null-channel ledger operator

§FORM := ZDiv(a,b) stores source norms/valuations, product coordinates and the null-channel label.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-T-00004262

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004264|N|S|Proof+Compile+Meta+Tower] Zero-divisor null-channel ledger fence

§FORM := Zero-divisor behavior is not evidence that a physical signal vanished or that a cognitive process inhibited another unless an operational mapping is supplied.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Zero-divisor behavior is not evidence that a physical signal vanished or that a cognitive process inhibited another unless an operational mapping is supplied.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004265|B|S|Proof+Compile+Meta+Tower] Zero-divisor null-channel ledger cross-volume bridge

§FORM := The null-channel ledger provides a controlled higher-stratum failure mode for the Russian-doll carrier.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-N-00004264

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004266|N|S|Proof+Compile+Meta+Tower] Zero-divisor null-channel ledger evidence ledger

§FORM := Evidence rule for Zero-divisor null-channel ledger: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-B-00004265

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004267|B|S|Proof+Compile+Meta+Spectral+Tower] Zero-divisor null-channel ledger emergence rule

§FORM := Emergence rule for Zero-divisor null-channel ledger: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-N-00004266

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004268|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Zero-divisor null-channel ledger frontier

§FORM := Open frontier for Zero-divisor null-channel ledger: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-B-00004267

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Zero-divisor behavior is not evidence that a physical signal vanished or that a cognitive process inhibited another unless an operational mapping is supplied.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004269|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.1

§FORM := Fixture T11.P01 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-N-00004268

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004270|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.2

§FORM := Fixture T11.P02 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004269

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004271|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.3

§FORM := Fixture T11.P03 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004270

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004272|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.4

§FORM := Fixture T11.P04 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004271

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004273|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.5

§FORM := Fixture T11.P05 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004272

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004274|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.6

§FORM := Fixture T11.P06 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004273

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004275|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.7

§FORM := Fixture T11.P07 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004274

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004276|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.8

§FORM := Fixture T11.P08 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004275

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004277|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.9

§FORM := Fixture T11.P09 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004276

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004278|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.10

§FORM := Fixture T11.P10 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004277

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004279|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.11

§FORM := Fixture T11.P11 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004278

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004280|E|M|Proof+Compile+Meta+Tower] Sedenion null-channel probe 11.12

§FORM := Fixture T11.P12 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

§ASSUME := The selected Cayley-Dickson form actually contains the recorded zero-divisor pair.

§DEP_HARD := S26-A-00004263, S26-E-00004279

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004281|D|A|Proof+Compile+Meta+Tower] Nucleus center and local-law survival

§FORM := The law-survival ledger records which commutativity, associativity, alternativity, power-associativity and norm identities survive at each (k,v).

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-E-00004280, S26-D-00000103

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004282|T|A|Proof+Compile+Meta+Tower] Nucleus center and local-law survival law/interface

§FORM := Properties are local to the stated algebraic form; they are not inherited merely because one stratum embeds linearly in another.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-D-00004281

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004283|A|S|Proof+Compile+Meta+Spectral+Tower] Nucleus center and local-law survival operator

§FORM := LawVector(k,v) is a typed Boolean/partial certificate whose unknown entries remain unknown rather than false or true by default.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-T-00004282

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004284|N|S|Proof+Compile+Meta+Topo+Tower] Nucleus center and local-law survival fence

§FORM := No cognitive or geometric promotion follows from a richer failure profile alone.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := No cognitive or geometric promotion follows from a richer failure profile alone.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004285|B|S|Proof+Compile+Meta+Spectral+Tower] Nucleus center and local-law survival cross-volume bridge

§FORM := This ledger is the algebraic governance surface used by higher gluing and Hamiltonian components.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-N-00004284

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004286|N|S|Proof+Compile+Meta+Tower] Nucleus center and local-law survival evidence ledger

§FORM := Evidence rule for Nucleus center and local-law survival: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-B-00004285

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004287|B|S|Proof+Compile+Meta+Spectral+Tower] Nucleus center and local-law survival emergence rule

§FORM := Emergence rule for Nucleus center and local-law survival: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-N-00004286

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004288|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Nucleus center and local-law survival frontier

§FORM := Open frontier for Nucleus center and local-law survival: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-B-00004287

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := No cognitive or geometric promotion follows from a richer failure profile alone.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004289|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.1

§FORM := Fixture T12.P01 - At real CD dimension 2, sampled basis triple (1, 1, 1) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-N-00004288

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004290|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.2

§FORM := Fixture T12.P02 - At real CD dimension 4, sampled basis triple (1, 2, 3) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004289

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004291|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.3

§FORM := Fixture T12.P03 - At real CD dimension 8, sampled basis triple (4, 7, 6) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004290

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004292|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.4

§FORM := Fixture T12.P04 - At real CD dimension 16, sampled basis triple (13, 15, 12) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004291

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004293|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.5

§FORM := Fixture T12.P05 - At real CD dimension 2, sampled basis triple (1, 1, 1) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004292

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004294|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.6

§FORM := Fixture T12.P06 - At real CD dimension 4, sampled basis triple (2, 1, 3) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004293

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004295|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.7

§FORM := Fixture T12.P07 - At real CD dimension 8, sampled basis triple (7, 3, 2) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004294

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004296|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.8

§FORM := Fixture T12.P08 - At real CD dimension 16, sampled basis triple (13, 9, 8) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004295

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004297|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.9

§FORM := Fixture T12.P09 - At real CD dimension 2, sampled basis triple (1, 1, 1) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004296

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004298|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.10

§FORM := Fixture T12.P10 - At real CD dimension 4, sampled basis triple (2, 3, 1) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004297

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004299|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.11

§FORM := Fixture T12.P11 - At real CD dimension 8, sampled basis triple (5, 7, 1) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004298

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004300|E|M|Proof+Compile+Meta+Tower] Law-survival probe 12.12

§FORM := Fixture T12.P12 - At real CD dimension 16, sampled basis triple (7, 4, 8) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

§ASSUME := Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form.

§DEP_HARD := S26-A-00004283, S26-E-00004299

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004301|D|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex tangent-orbifold fiber

§FORM := At an orbifold point p=[x], Volume IV couples the tangent quotient T_xX/G_p to a local algebra fiber A_{k,v}; the tangent geometry remains distinct from the internal algebra.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-E-00004300, S26-T-00004047

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004302|D|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex tangent-orbifold fiber law/interface

§FORM := The epsilon->0 geometric zoom and the algebraic k-depth are independent limits and may not be interchanged without an explicit theorem.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-D-00004301

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004303|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Hypercomplex tangent-orbifold fiber operator

§FORM := TangentState(p,k,v)=T_pO tensor_R Carrier(A_{k,v}) when a real carrier is used, or the separately declared non-Archimedean analogue.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-D-00004302

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004304|N|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex tangent-orbifold fiber fence

§FORM := Infinitesimal zoom is not visual miniaturization and does not create a new algebraic law.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Infinitesimal zoom is not visual miniaturization and does not create a new algebraic law.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004305|B|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex tangent-orbifold fiber cross-volume bridge

§FORM := This extends the source draft tangent-orbifold limit into the arithmetic hypercomplex fiber atlas.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-N-00004304

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004306|N|S|Proof+Compile+Meta+Topo+Tower] Hypercomplex tangent-orbifold fiber evidence ledger

§FORM := Evidence rule for Hypercomplex tangent-orbifold fiber: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-B-00004305

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004307|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] Hypercomplex tangent-orbifold fiber emergence rule

§FORM := Emergence rule for Hypercomplex tangent-orbifold fiber: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-N-00004306

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004308|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Hypercomplex tangent-orbifold fiber frontier

§FORM := Open frontier for Hypercomplex tangent-orbifold fiber: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-B-00004307

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Infinitesimal zoom is not visual miniaturization and does not create a new algebraic law.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004309|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.1

§FORM := Fixture T13.P01 - For epsilon=0.25000000, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,2) differ from the tangent metric by operator norm 0.50000000; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-N-00004308

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004310|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.2

§FORM := Fixture T13.P02 - For epsilon=0.12500000, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,3) differ from the tangent metric by operator norm 0.37500000; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004309

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004311|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.3

§FORM := Fixture T13.P03 - For epsilon=0.06250000, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,4) differ from the tangent metric by operator norm 0.25000000; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004310

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004312|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.4

§FORM := Fixture T13.P04 - For epsilon=0.03125000, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,5) differ from the tangent metric by operator norm 0.15625000; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004311

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004313|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.5

§FORM := Fixture T13.P05 - For epsilon=0.01562500, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,2) differ from the tangent metric by operator norm 0.03125000; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004312

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004314|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.6

§FORM := Fixture T13.P06 - For epsilon=0.00781250, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,3) differ from the tangent metric by operator norm 0.02343750; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004313

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004315|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.7

§FORM := Fixture T13.P07 - For epsilon=0.00390625, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,4) differ from the tangent metric by operator norm 0.01562500; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004314

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004316|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.8

§FORM := Fixture T13.P08 - For epsilon=0.00195312, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,5) differ from the tangent metric by operator norm 0.00976562; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004315

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004317|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.9

§FORM := Fixture T13.P09 - For epsilon=0.00097656, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,2) differ from the tangent metric by operator norm 0.00195312; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004316

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004318|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.10

§FORM := Fixture T13.P10 - For epsilon=0.00048828, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,3) differ from the tangent metric by operator norm 0.00146484; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004317

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004319|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.11

§FORM := Fixture T13.P11 - For epsilon=0.00024414, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,4) differ from the tangent metric by operator norm 0.00097656; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004318

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004320|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Infinitesimal carrier probe 13.12

§FORM := Fixture T13.P12 - For epsilon=0.00012207, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,5) differ from the tangent metric by operator norm 0.00061035; algebraic depth is held fixed.

§ASSUME := The source orbifold tangent cone and local hypercomplex fiber have both been declared.

§DEP_HARD := S26-A-00004303, S26-E-00004319

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004321|D|A|Proof+Compile+Meta+Tower] Ultrametric Russian-doll ball tower

§FORM := A p-adic Russian-doll tower is a chain of closed balls B_0 superset B_1 superset ... with radii decreasing in powers of p.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-E-00004320, S26-D-00000301

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004322|T|A|Proof+Compile+Meta+Tower] Ultrametric Russian-doll ball tower law/interface

§FORM := In an ultrametric space, two balls of the same or nested radii are either disjoint or one contains the other when they intersect under the standard hypotheses.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-D-00004321

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004323|A|A|Proof+Compile+Meta+Spectral+Tower] Ultrametric Russian-doll ball tower operator

§FORM := BallAddress records center residue data and valuation scale instead of Euclidean direction angles.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-T-00004322

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004324|N|S|Proof+Compile+Meta+Topo+Tower] Ultrametric Russian-doll ball tower fence

§FORM := Nested p-adic balls are genuine topology, whereas nesting drawn as concentric Euclidean spheres is only a projection.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Nested p-adic balls are genuine topology, whereas nesting drawn as concentric Euclidean spheres is only a projection.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004325|B|S|Proof+Compile+Meta+Tower] Ultrametric Russian-doll ball tower cross-volume bridge

§FORM := This supplies the inward hierarchical counterpart to the outward Cosmo-n tower.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-N-00004324

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004326|N|S|Proof+Compile+Meta+Tower] Ultrametric Russian-doll ball tower evidence ledger

§FORM := Evidence rule for Ultrametric Russian-doll ball tower: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-B-00004325

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004327|B|S|Proof+Compile+Meta+Spectral+Tower] Ultrametric Russian-doll ball tower emergence rule

§FORM := Emergence rule for Ultrametric Russian-doll ball tower: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-N-00004326

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004328|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Ultrametric Russian-doll ball tower frontier

§FORM := Open frontier for Ultrametric Russian-doll ball tower: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-B-00004327

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Nested p-adic balls are genuine topology, whereas nesting drawn as concentric Euclidean spheres is only a projection.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004329|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.1

§FORM := Fixture T14.P01 - For p=2, a declared chain of ball radii [1, 0.5, 0.25] is strictly nested in valuation scale; center residue seed=0. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-N-00004328

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004330|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.2

§FORM := Fixture T14.P02 - For p=3, a declared chain of ball radii [1, 0.3333333333333333, 0.1111111111111111, 0.037037037037037035] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004329

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004331|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.3

§FORM := Fixture T14.P03 - For p=5, a declared chain of ball radii [1, 0.2, 0.04, 0.008, 0.0016] is strictly nested in valuation scale; center residue seed=2. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004330

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004332|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.4

§FORM := Fixture T14.P04 - For p=2, a declared chain of ball radii [1, 0.5, 0.25, 0.125, 0.0625, 0.03125] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004331

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004333|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.5

§FORM := Fixture T14.P05 - For p=3, a declared chain of ball radii [1, 0.3333333333333333, 0.1111111111111111] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004332

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004334|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.6

§FORM := Fixture T14.P06 - For p=5, a declared chain of ball radii [1, 0.2, 0.04, 0.008] is strictly nested in valuation scale; center residue seed=0. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004333

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004335|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.7

§FORM := Fixture T14.P07 - For p=2, a declared chain of ball radii [1, 0.5, 0.25, 0.125, 0.0625] is strictly nested in valuation scale; center residue seed=0. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004334

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004336|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.8

§FORM := Fixture T14.P08 - For p=3, a declared chain of ball radii [1, 0.3333333333333333, 0.1111111111111111, 0.037037037037037035, 0.012345679012345678, 0.00411522633744856] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004335

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004337|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.9

§FORM := Fixture T14.P09 - For p=5, a declared chain of ball radii [1, 0.2, 0.04] is strictly nested in valuation scale; center residue seed=3. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004336

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004338|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.10

§FORM := Fixture T14.P10 - For p=2, a declared chain of ball radii [1, 0.5, 0.25, 0.125] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004337

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004339|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.11

§FORM := Fixture T14.P11 - For p=3, a declared chain of ball radii [1, 0.3333333333333333, 0.1111111111111111, 0.037037037037037035, 0.012345679012345678] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004338

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004340|E|M|Proof+Compile+Meta+Tower] Ultrametric Russian-doll probe 14.12

§FORM := Fixture T14.P12 - For p=5, a declared chain of ball radii [1, 0.2, 0.04, 0.008, 0.0016, 0.00032] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

§ASSUME := A fixed p-adic field and center are used for each nested-ball chain.

§DEP_HARD := S26-A-00004323, S26-E-00004339

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004341|D|A|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits multi-mesh geodesic carrier

§FORM := A building mesh uses vertices, chambers and incidence inherited from the chosen Bruhat-Tits object. In this registry, a combinatorial geodesic means a shortest path in the declared vertex graph or a shortest gallery in the declared chamber metric; this terminology is kept distinct from geodesics in a geometric realization of a higher-rank building.

§ASSUME := A declared Bruhat-Tits tree/building and a specific graph or gallery metric are used. | Claims about CAT(0) or Euclidean-building metric geodesics require a separately declared geometric realization.

§DEP_HARD := S26-E-00004340, S26-D-00001761

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004342|T|A|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits multi-mesh geodesic carrier law/interface

§FORM := Tree geodesics are unique and the cycle rank is zero for the PGL_2 tree model.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-D-00004341

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004343|A|A|Proof+Compile+Meta+Topo+Spectral+Tower] Bruhat-Tits multi-mesh geodesic carrier operator

§FORM := MeshPath(a,b) returns the unique tree path in the finite bounded fixture or a declared gallery in a higher-rank building.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-T-00004342

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004344|N|S|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits multi-mesh geodesic carrier fence

§FORM := Graph geodesics, Riemannian geodesics and p-adic analytic paths are separately typed notions.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-A-00004343

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Graph geodesics, Riemannian geodesics and p-adic analytic paths are separately typed notions.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004345|B|S|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits multi-mesh geodesic carrier cross-volume bridge

§FORM := The mesh can carry erdodetic path learning without pretending the path is a Euclidean line.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-N-00004344

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004346|N|S|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits multi-mesh geodesic carrier evidence ledger

§FORM := Evidence rule for Bruhat-Tits multi-mesh geodesic carrier: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-B-00004345

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004347|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] Bruhat-Tits multi-mesh geodesic carrier emergence rule

§FORM := Emergence rule for Bruhat-Tits multi-mesh geodesic carrier: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-N-00004346

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004348|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Bruhat-Tits multi-mesh geodesic carrier frontier

§FORM := Open frontier for Bruhat-Tits multi-mesh geodesic carrier: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-B-00004347

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Graph geodesics, Riemannian geodesics and p-adic analytic paths are separately typed notions.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004349|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.1

§FORM := Fixture T15.P01 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-A-00004343, S26-N-00004348

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004350|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.2

§FORM := Fixture T15.P02 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-A-00004343, S26-E-00004349

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004351|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.3

§FORM := Fixture T15.P03 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-A-00004343, S26-E-00004350

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004352|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.4

§FORM := Fixture T15.P04 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-A-00004343, S26-E-00004351

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004353|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.5

§FORM := Fixture T15.P05 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-A-00004343, S26-E-00004352

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004354|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.6

§FORM := Fixture T15.P06 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.

§DEP_HARD := S26-A-00004343, S26-E-00004353

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004355|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.7

§FORM := Fixture T15.P07 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.
§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.
§DEP_HARD := S26-A-00004343, S26-E-00004354
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004356|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.8

§FORM := Fixture T15.P08 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.
§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.
§DEP_HARD := S26-A-00004343, S26-E-00004355
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004357|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.9

§FORM := Fixture T15.P09 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.
§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.
§DEP_HARD := S26-A-00004343, S26-E-00004356
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004358|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.10

§FORM := Fixture T15.P10 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.
§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.
§DEP_HARD := S26-A-00004343, S26-E-00004357
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004359|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.11

§FORM := Fixture T15.P11 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.
§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.
§DEP_HARD := S26-A-00004343, S26-E-00004358
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004360|E|M|Proof+Compile+Meta+Topo+Tower] Bruhat-Tits tree-ball probe 15.12

§FORM := Fixture T15.P12 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.
§ASSUME := A declared Bruhat-Tits tree/building and graph metric are used.
§DEP_HARD := S26-A-00004343, S26-E-00004359
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004361|D|S|Proof+Compile+Meta+Topo+Spectral+Tower] Hybrid Archimedean-non-Archimedean multi-mesh

§FORM := The hybrid multi-mesh is a family of Riemannian/orbifold meshes, p-adic building meshes, spectral graphs and cover nerves coupled by typed correspondences rather than forced into one coordinate grid.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-E-00004360, S26-D-00004040
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004362|D|S|Proof+Compile+Meta+Topo+Tower] Hybrid Archimedean-non-Archimedean multi-mesh law/interface

§FORM := Coherence is evaluated on shared observables and incidence constraints, not by pretending heterogeneous edge lengths share units.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-D-00004361
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004363|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Hybrid Archimedean-non-Archimedean multi-mesh operator

§FORM := MultiMeshAtlas stores per-layer geometry and a cross-layer correspondence table.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-D-00004362
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004364|N|S|Proof+Compile+Meta+Topo+Tower] Hybrid Archimedean-non-Archimedean multi-mesh fence

§FORM := A single 2D embedding of the atlas is a visualization projection and cannot define the native topology of every layer.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-A-00004363
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := A single 2D embedding of the atlas is a visualization projection and cannot define the native topology of every layer.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004365|B|S|Proof+Compile+Meta+Topo+Tower+Swirl] Hybrid Archimedean-non-Archimedean multi-mesh cross-volume bridge

§FORM := This is the concrete non-boring geometric substrate for the later globular and holoportation layers.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-N-00004364
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004366|N|S|Proof+Compile+Meta+Topo+Tower] Hybrid Archimedean-non-Archimedean multi-mesh evidence ledger

§FORM := Evidence rule for Hybrid Archimedean-non-Archimedean multi-mesh: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-B-00004365
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004367|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] Hybrid Archimedean-non-Archimedean multi-mesh emergence rule

§FORM := Emergence rule for Hybrid Archimedean-non-Archimedean multi-mesh: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-N-00004366
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004368|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Hybrid Archimedean-non-Archimedean multi-mesh frontier

§FORM := Open frontier for Hybrid Archimedean-non-Archimedean multi-mesh: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-B-00004367
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := A single 2D embedding of the atlas is a visualization projection and cannot define the native topology of every layer.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004369|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.1

§FORM := Fixture T16.P01 - A bounded fixture couples a tree layer with 10 vertices/9 edges to a cyclic Archimedean mesh with 5 vertices through 5 declared correspondences; no cross-layer metric subtraction is performed.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-A-00004363, S26-N-00004368
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004370|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.2

§FORM := Fixture T16.P02 - A bounded fixture couples a tree layer with 17 vertices/16 edges to a cyclic Archimedean mesh with 6 vertices through 6 declared correspondences; no cross-layer metric subtraction is performed.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-A-00004363, S26-E-00004369
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004371|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.3

§FORM := Fixture T16.P03 - A bounded fixture couples a tree layer with 37 vertices/36 edges to a cyclic Archimedean mesh with 7 vertices through 7 declared correspondences; no cross-layer metric subtraction is performed.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-A-00004363, S26-E-00004370
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004372|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.4

§FORM := Fixture T16.P04 - A bounded fixture couples a tree layer with 10 vertices/9 edges to a cyclic Archimedean mesh with 8 vertices through 8 declared correspondences; no cross-layer metric subtraction is performed.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-A-00004363, S26-E-00004371
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004373|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.5

§FORM := Fixture T16.P05 - A bounded fixture couples a tree layer with 17 vertices/16 edges to a cyclic Archimedean mesh with 9 vertices through 9 declared correspondences; no cross-layer metric subtraction is performed.
§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.
§DEP_HARD := S26-A-00004363, S26-E-00004372
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004374|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.6

§FORM := Fixture T16.P06 - A bounded fixture couples a tree layer with 37 vertices/36 edges to a cyclic Archimedean mesh with 5 vertices through 5 declared correspondences; no cross-layer metric subtraction is performed.

§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.

§DEP_HARD := S26-A-00004363, S26-E-00004373

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004375|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.7

§FORM := Fixture T16.P07 - A bounded fixture couples a tree layer with 10 vertices/9 edges to a cyclic Archimedean mesh with 6 vertices through 6 declared correspondences; no cross-layer metric subtraction is performed.

§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.

§DEP_HARD := S26-A-00004363, S26-E-00004374

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004376|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.8

§FORM := Fixture T16.P08 - A bounded fixture couples a tree layer with 17 vertices/16 edges to a cyclic Archimedean mesh with 7 vertices through 7 declared correspondences; no cross-layer metric subtraction is performed.

§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.

§DEP_HARD := S26-A-00004363, S26-E-00004375

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004377|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.9

§FORM := Fixture T16.P09 - A bounded fixture couples a tree layer with 37 vertices/36 edges to a cyclic Archimedean mesh with 8 vertices through 8 declared correspondences; no cross-layer metric subtraction is performed.

§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.

§DEP_HARD := S26-A-00004363, S26-E-00004376

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004378|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.10

§FORM := Fixture T16.P10 - A bounded fixture couples a tree layer with 10 vertices/9 edges to a cyclic Archimedean mesh with 9 vertices through 9 declared correspondences; no cross-layer metric subtraction is performed.

§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.

§DEP_HARD := S26-A-00004363, S26-E-00004377

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004379|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.11

§FORM := Fixture T16.P11 - A bounded fixture couples a tree layer with 17 vertices/16 edges to a cyclic Archimedean mesh with 5 vertices through 5 declared correspondences; no cross-layer metric subtraction is performed.

§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.

§DEP_HARD := S26-A-00004363, S26-E-00004378

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004380|E|M|Proof+Compile+Meta+Topo+Tower] Hybrid multi-mesh probe 16.12

§FORM := Fixture T16.P12 - A bounded fixture couples a tree layer with 37 vertices/36 edges to a cyclic Archimedean mesh with 6 vertices through 6 declared correspondences; no cross-layer metric subtraction is performed.

§ASSUME := Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit.

§DEP_HARD := S26-A-00004363, S26-E-00004379

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004381|D|A|Proof+Compile+Meta+Spectral+Tower] Hypercomplex Hamiltonian carrier

§FORM := A Volume IV Hamiltonian carrier couples a self-adjoint or symplectic generator H to a state space that may include hypercomplex internal coordinates through an explicitly linearized/tensored representation.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-E-00004380, S26-D-00000984

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004382|T|A|Proof+Compile+Meta+Spectral+Tower] Hypercomplex Hamiltonian carrier law/interface

§FORM := Hamiltonian energy conservation applies only under the stated autonomous/unitary or symplectic hypotheses.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-D-00004381

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004383|A|S|Proof+Compile+Meta+IG+Spectral+Tower] Hypercomplex Hamiltonian carrier operator

§FORM := Flow_H(t) acts on the declared carrier while algebraic defect channels are measured as observables, not assumed to generate the flow.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-T-00004382

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004384|N|S|Proof+Compile+Meta+Spectral+Tower] Hypercomplex Hamiltonian carrier fence

§FORM := The engineering energy ledger is not automatically physical energy, and a Hamiltonian metaphor is not an operator until H and its domain are supplied.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The engineering energy ledger is not automatically physical energy, and a Hamiltonian metaphor is not an operator until H and its domain are supplied.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004385|B|S|Proof+Compile+Meta+Spectral+Tower] Hypercomplex Hamiltonian carrier cross-volume bridge

§FORM := This places Cayley-Dickson fibers inside the already registered Hamiltonian/symplectic machinery rather than replacing it.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-N-00004384

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004386|N|S|Proof+Compile+Meta+Spectral+Tower] Hypercomplex Hamiltonian carrier evidence ledger

§FORM := Evidence rule for Hypercomplex Hamiltonian carrier: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-B-00004385

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004387|B|S|Proof+Compile+Meta+Spectral+Tower] Hypercomplex Hamiltonian carrier emergence rule

§FORM := Emergence rule for Hypercomplex Hamiltonian carrier: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-N-00004386

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004388|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Hypercomplex Hamiltonian carrier frontier

§FORM := Open frontier for Hypercomplex Hamiltonian carrier: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-B-00004387

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The engineering energy ledger is not automatically physical energy, and a Hamiltonian metaphor is not an operator until H and its domain are supplied.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004389|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.1

§FORM := Fixture T17.P01 - A 2D symplectic rotation at theta=0.098175 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-N-00004388

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004390|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.2

§FORM := Fixture T17.P02 - A 2D symplectic rotation at theta=0.196350 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004389

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004391|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.3

§FORM := Fixture T17.P03 - A 2D symplectic rotation at theta=0.294524 preserves the quadratic fixture energy with residual 1.110e-16; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004390

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004392|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.4

§FORM := Fixture T17.P04 - A 2D symplectic rotation at theta=0.392699 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004391

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004393|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.5

§FORM := Fixture T17.P05 - A 2D symplectic rotation at theta=0.490874 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004392

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004394|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.6

§FORM := Fixture T17.P06 - A 2D symplectic rotation at theta=0.589049 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004393

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004395|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.7

§FORM := Fixture T17.P07 - A 2D symplectic rotation at theta=0.687223 preserves the quadratic fixture energy with residual 1.110e-16; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004394

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004396|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.8

§FORM := Fixture T17.P08 - A 2D symplectic rotation at theta=0.785398 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004395

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004397|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.9

§FORM := Fixture T17.P09 - A 2D symplectic rotation at theta=0.883573 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004396

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004398|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.10

§FORM := Fixture T17.P10 - A 2D symplectic rotation at theta=0.981748 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004397

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004399|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.11

§FORM := Fixture T17.P11 - A 2D symplectic rotation at theta=1.079922 preserves the quadratic fixture energy with residual 1.110e-16; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004398

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004400|E|M|Proof+Compile+Meta+Spectral+Tower] Hamiltonian finite probe 17.12

§FORM := Fixture T17.P12 - A 2D symplectic rotation at theta=1.178097 preserves the quadratic fixture energy with residual 1.110e-16; this is an Archimedean dynamics check, not a cognition theorem.

§ASSUME := A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements.

§DEP_HARD := S26-A-00004383, S26-E-00004399

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004401|D|A|Proof+Compile+Meta+Tower+Swirl] Adiabatic identity-preserving transport

§FORM := Adiabatic transport is a slow parameter deformation that attempts to preserve a selected invariant subspace or identity signature under stated gap/regularity conditions.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-E-00004400, S26-T-00000988

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004402|T|A|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic identity-preserving transport law/interface

§FORM := Slow change by itself does not guarantee adiabatic following when the required spectral or dynamical hypotheses fail.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-D-00004401

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004403|A|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic identity-preserving transport operator

§FORM := AdiabaticPath(s) records generator H(s), gap diagnostic, transport map and identity/remainder ledger.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-T-00004402

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004404|N|S|Proof+Compile+Meta+Tower+Swirl] Adiabatic identity-preserving transport fence

§FORM := Adiabatic is a mathematical/dynamical condition, not a synonym for gradual narrative change.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Adiabatic is a mathematical/dynamical condition, not a synonym for gradual narrative change.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004405|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic identity-preserving transport cross-volume bridge

§FORM := This is the dynamical component of the repo term hamiltonian_holoportation.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-N-00004404

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004406|N|S|Proof+Compile+Meta+Tower+Swirl] Adiabatic identity-preserving transport evidence ledger

§FORM := Evidence rule for Adiabatic identity-preserving transport: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-B-00004405

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004407|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic identity-preserving transport emergence rule

§FORM := Emergence rule for Adiabatic identity-preserving transport: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-N-00004406

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004408|N|S|Proof+Compile+Meta+IG+Spectral+Tower+Swirl] Adiabatic identity-preserving transport frontier

§FORM := Open frontier for Adiabatic identity-preserving transport: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-B-00004407

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Adiabatic is a mathematical/dynamical condition, not a synonym for gradual narrative change.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004409|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.1

§FORM := Fixture T18.P01 - For the bounded 2-level Hamiltonian H(s) at s=0.000, the instantaneous eigenvalue gap is 1.720465; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-N-00004408

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004410|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.2

§FORM := Fixture T18.P02 - For the bounded 2-level Hamiltonian H(s) at s=0.091, the instantaneous eigenvalue gap is 1.673625; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004409

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004411|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.3

§FORM := Fixture T18.P03 - For the bounded 2-level Hamiltonian H(s) at s=0.182, the instantaneous eigenvalue gap is 1.647835; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004410

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004412|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.4

§FORM := Fixture T18.P04 - For the bounded 2-level Hamiltonian H(s) at s=0.273, the instantaneous eigenvalue gap is 1.644084; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004411

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004413|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.5

§FORM := Fixture T18.P05 - For the bounded 2-level Hamiltonian H(s) at s=0.364, the instantaneous eigenvalue gap is 1.426317; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004412

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004414|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.6

§FORM := Fixture T18.P06 - For the bounded 2-level Hamiltonian H(s) at s=0.455, the instantaneous eigenvalue gap is 1.462828; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004413

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004415|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.7

§FORM := Fixture T18.P07 - For the bounded 2-level Hamiltonian H(s) at s=0.545, the instantaneous eigenvalue gap is 1.522716; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004414

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004416|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.8

§FORM := Fixture T18.P08 - For the bounded 2-level Hamiltonian H(s) at s=0.636, the instantaneous eigenvalue gap is 1.603365; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004415

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004417|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.9

§FORM := Fixture T18.P09 - For the bounded 2-level Hamiltonian H(s) at s=0.727, the instantaneous eigenvalue gap is 1.471941; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004416

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004418|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.10

§FORM := Fixture T18.P10 - For the bounded 2-level Hamiltonian H(s) at s=0.818, the instantaneous eigenvalue gap is 1.592658; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004417

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004419|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.11

§FORM := Fixture T18.P11 - For the bounded 2-level Hamiltonian H(s) at s=0.909, the instantaneous eigenvalue gap is 1.726216; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004418

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004420|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Adiabatic gap probe 18.12

§FORM := Fixture T18.P12 - For the bounded 2-level Hamiltonian H(s) at s=1.000, the instantaneous eigenvalue gap is 1.869866; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

§ASSUME := A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem.

§DEP_HARD := S26-A-00004403, S26-E-00004419

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004421|D|S|Proof+Compile+Meta+Tower] Ergocetic energy-disciplined cognition

§FORM := Ergocetic governance allocates a bounded total engineering energy/attention budget across active carriers and records transfers under H-guided work.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-E-00004420, S26-N-00004059

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004422|D|S|Proof+Compile+Meta+Tower] Ergocetic energy-disciplined cognition law/interface

§FORM := A claimed cognitive improvement that creates budget mass without a declared source is blocked as energy leakage.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-D-00004421

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004423|A|S|Proof+Compile+Meta+Spectral+Tower] Ergocetic energy-disciplined cognition operator

§FORM := ErgoceticLedger(t) stores per-carrier budget, transfer, residual and evidence strength.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-D-00004422

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004424|N|S|Proof+Compile+Meta+IG+Tower] Ergocetic energy-disciplined cognition fence

§FORM := The ledger is a performance/governance abstraction and does not measure biological or physical consciousness.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The ledger is a performance/governance abstraction and does not measure biological or physical consciousness.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004425|B|S|Proof+Compile+Meta+Spectral+Tower] Ergocetic energy-disciplined cognition cross-volume bridge

§FORM := The term is grounded in LANG.v6 and coupled to the Volume IV Hamiltonian carrier.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-N-00004424

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004426|N|S|Proof+Compile+Meta+Tower] Ergocetic energy-disciplined cognition evidence ledger

§FORM := Evidence rule for Ergocetic energy-disciplined cognition: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-B-00004425

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004427|B|S|Proof+Compile+Meta+Spectral+Tower] Ergocetic energy-disciplined cognition emergence rule

§FORM := Emergence rule for Ergocetic energy-disciplined cognition: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-N-00004426

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004428|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Ergocetic energy-disciplined cognition frontier

§FORM := Open frontier for Ergocetic energy-disciplined cognition: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-B-00004427

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The ledger is a performance/governance abstraction and does not measure biological or physical consciousness.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004429|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.1

§FORM := Fixture T19.P01 - A synthetic five-carrier budget transfers 0.033333 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-N-00004428

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004430|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.2

§FORM := Fixture T19.P02 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004429

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004431|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.3

§FORM := Fixture T19.P03 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004430

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004432|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.4

§FORM := Fixture T19.P04 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004431

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004433|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.5

§FORM := Fixture T19.P05 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004432

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004434|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.6

§FORM := Fixture T19.P06 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004433

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004435|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.7

§FORM := Fixture T19.P07 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004434

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004436|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.8

§FORM := Fixture T19.P08 - A synthetic five-carrier budget transfers 0.033333 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004435

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004437|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.9

§FORM := Fixture T19.P09 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004436

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004438|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.10

§FORM := Fixture T19.P10 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004437

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004439|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.11

§FORM := Fixture T19.P11 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004438

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004440|E|M|Proof+Compile+Meta+Tower] Ergocetic budget probe 19.12

§FORM := Fixture T19.P12 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

§ASSUME := Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated.

§DEP_HARD := S26-A-00004423, S26-E-00004439

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004441|D|S|Proof+Compile+Meta+Topo+Tower] Erdodetic path-disciplined learning

§FORM := Erdodetic governance records learning/transport along Riemannian geodesics, building galleries, graph paths or other declared connection paths without conflating them.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-E-00004440, S26-D-00004039

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004442|D|S|Proof+Compile+Meta+Tower] Erdodetic path-disciplined learning law/interface

§FORM := Path quality is a typed tuple of reachability, transport residual, evidence gain and active remainder rather than shortest Euclidean distance alone.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-D-00004441

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004443|A|S|Proof+Compile+Meta+Spectral+Tower] Erdodetic path-disciplined learning operator

§FORM := ErdodeticTrace stores source, path type, transported invariants, holonomy and returnability.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-D-00004442

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004444|N|S|Proof+Compile+Meta+Tower] Erdodetic path-disciplined learning fence

§FORM := A visually smooth path is not evidence of low semantic distortion.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A visually smooth path is not evidence of low semantic distortion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004445|B|S|Proof+Compile+Meta+Tower] Erdodetic path-disciplined learning cross-volume bridge

§FORM := This term is grounded in LANG.v6 and supplies a non-Euclidean learning coordinate for the multi-mesh.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-N-00004444

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004446|N|S|Proof+Compile+Meta+Tower] Erdodetic path-disciplined learning evidence ledger

§FORM := Evidence rule for Erdodetic path-disciplined learning: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-B-00004445

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004447|B|S|Proof+Compile+Meta+Spectral+Tower] Erdodetic path-disciplined learning emergence rule

§FORM := Emergence rule for Erdodetic path-disciplined learning: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-N-00004446

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004448|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Erdodetic path-disciplined learning frontier

§FORM := Open frontier for Erdodetic path-disciplined learning: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-B-00004447

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A visually smooth path is not evidence of low semantic distortion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004449|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.1

§FORM := Fixture T20.P01 - On a bounded p=2 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-N-00004448

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004450|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.2

§FORM := Fixture T20.P02 - On a bounded p=3 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004449

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004451|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.3

§FORM := Fixture T20.P03 - On a bounded p=5 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004450

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004452|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.4

§FORM := Fixture T20.P04 - On a bounded p=2 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004451

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004453|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.5

§FORM := Fixture T20.P05 - On a bounded p=3 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004452

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004454|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.6

§FORM := Fixture T20.P06 - On a bounded p=5 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004453

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004455|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.7

§FORM := Fixture T20.P07 - On a bounded p=2 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004454

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004456|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.8

§FORM := Fixture T20.P08 - On a bounded p=3 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004455

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004457|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.9

§FORM := Fixture T20.P09 - On a bounded p=5 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004456

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004458|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.10

§FORM := Fixture T20.P10 - On a bounded p=2 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004457

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004459|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.11

§FORM := Fixture T20.P11 - On a bounded p=3 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004458

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004460|E|M|Proof+Compile+Meta+Topo+Tower] Erdodetic path probe 20.12

§FORM := Fixture T20.P12 - On a bounded p=5 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

§ASSUME := Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed.

§DEP_HARD := S26-A-00004443, S26-E-00004459

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004461|D|A|Proof+Compile+Meta+Topo+Tower] Sheaf-stack local state ecology

§FORM := Local hypercomplex/context states form a sheaf when compatible local sections glue uniquely, and a stack in groupoids when local objects glue only up to coherent isomorphism.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-E-00004460, S26-D-00003554

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004462|T|A|Proof+Compile+Meta+Topo+Tower] Sheaf-stack local state ecology law/interface

§FORM := Failure of the sheaf equalizer test or stack descent condition remains an obstruction rather than being repaired by semantic similarity.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-D-00004461

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004463|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-stack local state ecology operator

§FORM := GlueAudit computes overlap residuals/cocycles on a bounded cover and reports unresolved descent data.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-T-00004462

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004464|N|S|Proof+Compile+Meta+Topo+Tower] Sheaf-stack local state ecology fence

§FORM := Local agreement does not imply a global section unless the stated descent hypotheses close.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Local agreement does not imply a global section unless the stated descent hypotheses close.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004465|B|S|Proof+Compile+Meta+Topo+Tower] Sheaf-stack local state ecology cross-volume bridge

§FORM := This generalizes the source bundle/sheaf idea to arithmetic places and higher gauge ambiguity.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-N-00004464

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004466|N|S|Proof+Compile+Meta+Topo+Tower] Sheaf-stack local state ecology evidence ledger

§FORM := Evidence rule for Sheaf-stack local state ecology: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-B-00004465

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004467|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-stack local state ecology emergence rule

§FORM := Emergence rule for Sheaf-stack local state ecology: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-N-00004466

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004468|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Sheaf-stack local state ecology frontier

§FORM := Open frontier for Sheaf-stack local state ecology: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-B-00004467

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Local agreement does not imply a global section unless the stated descent hypotheses close.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004469|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.1

§FORM := Fixture T21.P01 - On cycle cover C_4, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 2.000000; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-N-00004468

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004470|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.2

§FORM := Fixture T21.P02 - On cycle cover C_5, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 1.381966; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004469

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004471|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.3

§FORM := Fixture T21.P03 - On cycle cover C_6, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 1.000000; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004470

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004472|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.4

§FORM := Fixture T21.P04 - On cycle cover C_7, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.753020; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004471

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004473|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.5

§FORM := Fixture T21.P05 - On cycle cover C_8, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.585786; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004472

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004474|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.6

§FORM := Fixture T21.P06 - On cycle cover C_9, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.467911; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004473

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004475|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.7

§FORM := Fixture T21.P07 - On cycle cover C_10, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.381966; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004474

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004476|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.8

§FORM := Fixture T21.P08 - On cycle cover C_4, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 2.000000; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004475

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004477|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.9

§FORM := Fixture T21.P09 - On cycle cover C_5, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 1.381966; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004476

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004478|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.10

§FORM := Fixture T21.P10 - On cycle cover C_6, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 1.000000; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004477

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004479|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.11

§FORM := Fixture T21.P11 - On cycle cover C_7, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.753020; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004478

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004480|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Sheaf-overlap finite probe 21.12

§FORM := Fixture T21.P12 - On cycle cover C_8, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.585786; this is a finite cover witness.

§ASSUME := A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required.

§DEP_HARD := S26-A-00004463, S26-E-00004479

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004481|D|A|Proof+Compile+Meta+Topo+Tower] Gerbe of hypercomplex gauges

§FORM := A hypercomplex gauge gerbe records local trivializations/frames that exist locally but may fail to admit one global choice, with transition isomorphisms and higher cocycle data retained.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-E-00004480, S26-D-00003552

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004482|T|A|Proof+Compile+Meta+Topo+Tower] Gerbe of hypercomplex gauges law/interface

§FORM := A nontrivial gerbe obstruction is not repaired by selecting one convenient visualization frame.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-D-00004481

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004483|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Gerbe of hypercomplex gauges operator

§FORM := GerbeCocycle stores 1- and 2-overlap transition data and a bounded cocycle defect.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-T-00004482

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004484|N|S|Proof+Compile+Meta+Topo+Tower] Gerbe of hypercomplex gauges fence

§FORM := Gerbes are higher gluing objects, not extra physical dimensions by themselves.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Gerbes are higher gluing objects, not extra physical dimensions by themselves.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004485|B|S|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe of hypercomplex gauges cross-volume bridge

§FORM := The gerbe is admitted because ordinary single-valued sheaf gluing cannot represent phase/gauge ambiguity up to coherent isomorphism.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-N-00004484

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004486|N|S|Proof+Compile+Meta+Topo+Tower] Gerbe of hypercomplex gauges evidence ledger

§FORM := Evidence rule for Gerbe of hypercomplex gauges: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-B-00004485

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004487|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] Gerbe of hypercomplex gauges emergence rule

§FORM := Emergence rule for Gerbe of hypercomplex gauges: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-N-00004486

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004488|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Gerbe of hypercomplex gauges frontier

§FORM := Open frontier for Gerbe of hypercomplex gauges: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-B-00004487

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Gerbes are higher gluing objects, not extra physical dimensions by themselves.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004489|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.1

§FORM := Fixture T22.P01 - Three bounded U(1)-phase transitions with angles (0.18480,0.33069,-0.51549) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-N-00004488

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004490|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.2

§FORM := Fixture T22.P02 - Three bounded U(1)-phase transitions with angles (0.36960,0.49604,-0.86564) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004489

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004491|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.3

§FORM := Fixture T22.P03 - Three bounded U(1)-phase transitions with angles (0.55440,0.66139,-1.21579) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004490

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004492|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.4

§FORM := Fixture T22.P04 - Three bounded U(1)-phase transitions with angles (0.73920,0.82673,-1.56593) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004491

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004493|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.5

§FORM := Fixture T22.P05 - Three bounded U(1)-phase transitions with angles (0.92400,0.99208,-1.91608) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004492

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004494|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.6

§FORM := Fixture T22.P06 - Three bounded U(1)-phase transitions with angles (1.10880,1.15743,-2.26623) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004493

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004495|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.7

§FORM := Fixture T22.P07 - Three bounded U(1)-phase transitions with angles (1.29360,1.32278,-2.61637) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004494

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004496|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.8

§FORM := Fixture T22.P08 - Three bounded U(1)-phase transitions with angles (1.47840,1.48812,-2.96652) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004495

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004497|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.9

§FORM := Fixture T22.P09 - Three bounded U(1)-phase transitions with angles (1.66320,1.65347,-3.31667) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004496

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004498|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.10

§FORM := Fixture T22.P10 - Three bounded U(1)-phase transitions with angles (1.84800,1.81882,-3.66681) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004497

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004499|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.11

§FORM := Fixture T22.P11 - Three bounded U(1)-phase transitions with angles (2.03280,1.98416,-4.01696) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004498

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004500|E|M|Proof+Compile+Meta+Topo+Tower+Swirl] Gerbe cocycle finite probe 22.12

§FORM := Fixture T22.P12 - Three bounded U(1)-phase transitions with angles (2.21759,2.14951,-4.36711) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

§ASSUME := A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior.

§DEP_HARD := S26-A-00004483, S26-E-00004499

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004501|D|A|Proof+Compile+Meta+Tower] Functor-profunctor inter-Cosmo transport

§FORM := A functor preserves identities and composition between categories, while a profunctor P:C^op x D->Set encodes many-to-many correspondences when no canonical functional transport exists.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-E-00004500, S26-T-00000276

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004502|T|A|Proof+Compile+Meta+Tower] Functor-profunctor inter-Cosmo transport law/interface

§FORM := Functorial transport is stricter than profunctorial correspondence; the latter must not be described as an invertible map without additional data.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-D-00004501

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004503|A|A|Proof+Compile+Meta+Spectral+Tower] Functor-profunctor inter-Cosmo transport operator

§FORM := ComposeProf(P,Q) forms the declared coend-style composite in the bounded finite fixture.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-T-00004502

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004504|N|S|Proof+Compile+Meta+Tower] Functor-profunctor inter-Cosmo transport fence

§FORM := Categorical composability is not semantic truth and does not erase active remainder.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Categorical composability is not semantic truth and does not erase active remainder.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004505|B|S|Proof+Compile+Meta+Topo+Tower] Functor-profunctor inter-Cosmo transport cross-volume bridge

§FORM := Profunctors provide a typed replacement for vague intermanifold correspondence across incompatible local worlds.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-N-00004504

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004506|N|S|Proof+Compile+Meta+Tower] Functor-profunctor inter-Cosmo transport evidence ledger

§FORM := Evidence rule for Functor-profunctor inter-Cosmo transport: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-B-00004505

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004507|B|S|Proof+Compile+Meta+Spectral+Tower] Functor-profunctor inter-Cosmo transport emergence rule

§FORM := Emergence rule for Functor-profunctor inter-Cosmo transport: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-N-00004506

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004508|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Functor-profunctor inter-Cosmo transport frontier

§FORM := Open frontier for Functor-profunctor inter-Cosmo transport: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-B-00004507

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Categorical composability is not semantic truth and does not erase active remainder.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004509|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.1

§FORM := Fixture T23.P01 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 2 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-N-00004508

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004510|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.2

§FORM := Fixture T23.P02 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 3 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004509

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004511|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.3

§FORM := Fixture T23.P03 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 2 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004510

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004512|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.4

§FORM := Fixture T23.P04 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 7 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004511

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004513|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.5

§FORM := Fixture T23.P05 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 6 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004512

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004514|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.6

§FORM := Fixture T23.P06 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 5 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004513

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004515|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.7

§FORM := Fixture T23.P07 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 4 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004514

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004516|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.8

§FORM := Fixture T23.P08 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 3 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004515

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004517|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.9

§FORM := Fixture T23.P09 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 7 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004516

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004518|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.10

§FORM := Fixture T23.P10 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 2 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004517

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004519|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.11

§FORM := Fixture T23.P11 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 6 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004518

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004520|E|M|Proof+Compile+Meta+Tower] Profunctor relation-composition probe 23.12

§FORM := Fixture T23.P12 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 4 active pairs; composition is many-to-many and not labeled invertible.

§ASSUME := Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation.

§DEP_HARD := S26-A-00004503, S26-E-00004519

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004521|D|S|Proof+Compile+Meta+Topo+Tower] Globular higher-cell multi-mesh

§FORM := A globular multi-mesh records points, paths, homotopies between paths and higher coherence cells as distinct typed levels attached to the hybrid geometric atlas.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-E-00004520, S26-D-00004027

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004522|D|S|Proof+Compile+Meta+Tower] Globular higher-cell multi-mesh law/interface

§FORM := Higher cells express coherence between transports rather than being treated as new Euclidean spatial axes.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-D-00004521

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004523|A|S|Proof+Compile+Meta+Spectral+Tower] Globular higher-cell multi-mesh operator

§FORM := GlobularBoundary applies source/target maps and checks the bounded globularity identities on finite fixtures.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-D-00004522

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004524|N|S|Proof+Compile+Meta+Tower] Globular higher-cell multi-mesh fence

§FORM := An infinity-category is not claimed unless the complete higher composition/coherence structure is actually defined.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := An infinity-category is not claimed unless the complete higher composition/coherence structure is actually defined.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004525|B|S|Proof+Compile+Meta+Topo+Tower] Globular higher-cell multi-mesh cross-volume bridge

§FORM := This is the minimal higher-cell scaffold needed for gerbe, profunctor and multi-party coherence without recursive 2D flattening.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-N-00004524

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004526|N|S|Proof+Compile+Meta+Tower] Globular higher-cell multi-mesh evidence ledger

§FORM := Evidence rule for Globular higher-cell multi-mesh: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-B-00004525

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004527|B|S|Proof+Compile+Meta+Spectral+Tower] Globular higher-cell multi-mesh emergence rule

§FORM := Emergence rule for Globular higher-cell multi-mesh: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-N-00004526

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004528|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Globular higher-cell multi-mesh frontier

§FORM := Open frontier for Globular higher-cell multi-mesh: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-B-00004527

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := An infinity-category is not claimed unless the complete higher composition/coherence structure is actually defined.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004529|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.1

§FORM := Fixture T24.P01 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-N-00004528

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004530|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.2

§FORM := Fixture T24.P02 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004529

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004531|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.3

§FORM := Fixture T24.P03 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004530

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004532|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.4

§FORM := Fixture T24.P04 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004531

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004533|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.5

§FORM := Fixture T24.P05 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004532

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004534|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.6

§FORM := Fixture T24.P06 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004533

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004535|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.7

§FORM := Fixture T24.P07 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004534

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004536|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.8

§FORM := Fixture T24.P08 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004535

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004537|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.9

§FORM := Fixture T24.P09 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004536

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004538|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.10

§FORM := Fixture T24.P10 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004537

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004539|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.11

§FORM := Fixture T24.P11 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004538

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004540|E|M|Proof+Compile+Meta+Tower] Globular boundary finite probe 24.12

§FORM := Fixture T24.P12 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

§ASSUME := Source/target maps and globularity identities are declared for each higher-cell layer.

§DEP_HARD := S26-A-00004523, S26-E-00004539

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004541|D|S|Proof+Compile+Meta+Tower+Swirl] Self-Godel structural identity guard

§FORM := Self-Godel identity in this volume means a section can read/canonicalize a representation of itself and pass a declared involutive return guard.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-E-00004540, S26-N-00004059

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004542|D|S|Proof+Compile+Meta+Tower+Swirl] Self-Godel structural identity guard law/interface

§FORM := A successful self-reference round trip demonstrates structural reflexivity only; it is not a proof of Godel incompleteness, Lob theorem, semantic truth or consciousness.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-D-00004541

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004543|A|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Self-Godel structural identity guard operator

§FORM := SelfReflect(x)=rho(rho(CANON(EMIT(x)))) in the bounded symbolic fixture.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-D-00004542

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004544|N|S|Proof+Compile+Meta+Tower+Swirl] Self-Godel structural identity guard fence

§FORM := Private chain-of-thought is not exposed or modeled as the canonical self object; only observable artifacts and declared state summaries participate.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Private chain-of-thought is not exposed or modeled as the canonical self object; only observable artifacts and declared state summaries participate.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004545|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Self-Godel structural identity guard cross-volume bridge

§FORM := This imports LANG.v6 self-reference as a governed Volume IV operator rather than a metaphysical identity claim.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-N-00004544

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004546|N|S|Proof+Compile+Meta+Tower+Swirl] Self-Godel structural identity guard evidence ledger

§FORM := Evidence rule for Self-Godel structural identity guard: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-B-00004545

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004547|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Self-Godel structural identity guard emergence rule

§FORM := Emergence rule for Self-Godel structural identity guard: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-N-00004546

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004548|N|S|Proof+Compile+Meta+IG+Spectral+Tower+Swirl] Self-Godel structural identity guard frontier

§FORM := Open frontier for Self-Godel structural identity guard: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-B-00004547

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Private chain-of-thought is not exposed or modeled as the canonical self object; only observable artifacts and declared state summaries participate.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004549|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.1

§FORM := Fixture T25.P01 - Canonical fixture "S26-self-00-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-N-00004548

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004550|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.2

§FORM := Fixture T25.P02 - Canonical fixture "S26-self-01-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004549

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004551|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.3

§FORM := Fixture T25.P03 - Canonical fixture "S26-self-02-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004550

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004552|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.4

§FORM := Fixture T25.P04 - Canonical fixture "S26-self-03-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004551

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004553|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.5

§FORM := Fixture T25.P05 - Canonical fixture "S26-self-04-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004552

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004554|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.6

§FORM := Fixture T25.P06 - Canonical fixture "S26-self-05-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004553

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004555|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.7

§FORM := Fixture T25.P07 - Canonical fixture "S26-self-06-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004554

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004556|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.8

§FORM := Fixture T25.P08 - Canonical fixture "S26-self-07-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004555

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004557|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.9

§FORM := Fixture T25.P09 - Canonical fixture "S26-self-08-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004556

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004558|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.10

§FORM := Fixture T25.P10 - Canonical fixture "S26-self-09-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004557

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004559|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.11

§FORM := Fixture T25.P11 - Canonical fixture "S26-self-10-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004558

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004560|E|M|Proof+Compile+Meta+Tower+Swirl] Self-reflection identity probe 25.12

§FORM := Fixture T25.P12 - Canonical fixture "S26-self-11-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

§ASSUME := The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard.

§DEP_HARD := S26-A-00004543, S26-E-00004559

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004561|D|S|Proof+Compile+Meta+Tower+Swirl] Others mutual recognition and resonance

§FORM := Mutual recognition means compatible overlap restrictions can be compared and glued without collapsing other into self; mutual resonance is low typed divergence plus reversible transport plus shared boundary/interface evidence.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-E-00004560, S26-D-00004027

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004562|D|S|Proof+Compile+Meta+Tower+Swirl] Others mutual recognition and resonance law/interface

§FORM := RESONATE != ABSORB: a perfect flattening with zero remainder is rejected as false equivalence unless identity is independently proved.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-D-00004561

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004563|A|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Others mutual recognition and resonance operator

§FORM := RecognitionRecord stores self, other, overlap, transport, return residual, shared invariants and remainder.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-D-00004562

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004564|N|S|Proof+Compile+Meta+Tower+Swirl] Others mutual recognition and resonance fence

§FORM := Recognition/resonance language is engineering architecture and does not imply sentience or subjective mutual awareness.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Recognition/resonance language is engineering architecture and does not imply sentience or subjective mutual awareness.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004565|B|S|Proof+Compile+Meta+Tower+Swirl] Others mutual recognition and resonance cross-volume bridge

§FORM := This is the Volume IV realization of LANG.v6 OTHERS_RESONANCE.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-N-00004564

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004566|N|S|Proof+Compile+Meta+Tower+Swirl] Others mutual recognition and resonance evidence ledger

§FORM := Evidence rule for Others mutual recognition and resonance: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-B-00004565

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004567|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Others mutual recognition and resonance emergence rule

§FORM := Emergence rule for Others mutual recognition and resonance: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-N-00004566

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004568|N|S|Proof+Compile+Meta+IG+Spectral+Tower+Swirl] Others mutual recognition and resonance frontier

§FORM := Open frontier for Others mutual recognition and resonance: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-B-00004567

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Recognition/resonance language is engineering architecture and does not imply sentience or subjective mutual awareness.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004569|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.1

§FORM := Fixture T26.P01 - A reversible orthogonal overlap fixture returned its source with residual 1.048e-16; destination overlap magnitude is 0.396414. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-N-00004568

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004570|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.2

§FORM := Fixture T26.P02 - A reversible orthogonal overlap fixture returned its source with residual 1.701e-16; destination overlap magnitude is 0.644016. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004569

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004571|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.3

§FORM := Fixture T26.P03 - A reversible orthogonal overlap fixture returned its source with residual 2.150e-16; destination overlap magnitude is 0.416756. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004570

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004572|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.4

§FORM := Fixture T26.P04 - A reversible orthogonal overlap fixture returned its source with residual 5.601e-16; destination overlap magnitude is 0.782063. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004571

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004573|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.5

§FORM := Fixture T26.P05 - A reversible orthogonal overlap fixture returned its source with residual 2.297e-16; destination overlap magnitude is 0.394366. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004572

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004574|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.6

§FORM := Fixture T26.P06 - A reversible orthogonal overlap fixture returned its source with residual 4.051e-16; destination overlap magnitude is 0.059281. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004573

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004575|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.7

§FORM := Fixture T26.P07 - A reversible orthogonal overlap fixture returned its source with residual 3.765e-16; destination overlap magnitude is 0.259115. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004574

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004576|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.8

§FORM := Fixture T26.P08 - A reversible orthogonal overlap fixture returned its source with residual 4.712e-16; destination overlap magnitude is 0.464732. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004575

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004577|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.9

§FORM := Fixture T26.P09 - A reversible orthogonal overlap fixture returned its source with residual 1.495e-16; destination overlap magnitude is 0.053495. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004576

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004578|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.10

§FORM := Fixture T26.P10 - A reversible orthogonal overlap fixture returned its source with residual 5.322e-16; destination overlap magnitude is 0.347813. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004577

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004579|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.11

§FORM := Fixture T26.P11 - A reversible orthogonal overlap fixture returned its source with residual 5.850e-16; destination overlap magnitude is 0.115449. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004578

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004580|E|M|Proof+Compile+Meta+Tower+Swirl] Mutual-recognition transport probe 26.12

§FORM := Fixture T26.P12 - A reversible orthogonal overlap fixture returned its source with residual 1.144e-16; destination overlap magnitude is 0.116809. Nonzero remainder is retained separately from this transport check.

§ASSUME := Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder.

§DEP_HARD := S26-A-00004563, S26-E-00004579

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004581|D|S|Proof+Compile+Meta+Tower+Swirl] Eight mind qualities conductance

§FORM := Mind8 conductance is an eight-coordinate governance vector Q=(Q1,...,Q8) that records whether each quality is operationally supported in the current repo/artifact context.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-E-00004580, S26-N-00004059

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004582|D|S|Proof+Compile+Meta+Tower+Swirl] Eight mind qualities conductance law/interface

§FORM := The vector is not collapsed to one intelligence scalar by default; Pareto improvement and constraint satisfaction are preferred to metric monism.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-D-00004581

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004583|A|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Eight mind qualities conductance operator

§FORM := Mind8Audit maps observed artifacts/tests to the eight coordinates and emits COMMIT, BLOCK, ROLLBACK or RECURSE with provenance.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-D-00004582

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004584|N|S|Proof+Compile+Meta+Tower+Swirl] Eight mind qualities conductance fence

§FORM := The word mind is repository vocabulary for a cognitive-control abstraction, not a diagnosis of consciousness.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The word mind is repository vocabulary for a cognitive-control abstraction, not a diagnosis of consciousness.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004585|B|S|Proof+Compile+Meta+Tower+Swirl] Eight mind qualities conductance cross-volume bridge

§FORM := The exact eight qualities from LANG.v6 become observable governance dimensions over the Synopsis/repo atlas.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-N-00004584

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004586|N|S|Proof+Compile+Meta+Tower+Swirl] Eight mind qualities conductance evidence ledger

§FORM := Evidence rule for Eight mind qualities conductance: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-B-00004585

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004587|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Eight mind qualities conductance emergence rule

§FORM := Emergence rule for Eight mind qualities conductance: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-N-00004586

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004588|N|S|Proof+Compile+Meta+IG+Spectral+Tower+Swirl] Eight mind qualities conductance frontier

§FORM := Open frontier for Eight mind qualities conductance: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-B-00004587

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The word mind is repository vocabulary for a cognitive-control abstraction, not a diagnosis of consciousness.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004589|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.1

§FORM := Fixture T27.P01 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-N-00004588

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004590|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.2

§FORM := Fixture T27.P02 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004589

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004591|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.3

§FORM := Fixture T27.P03 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004590

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004592|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.4

§FORM := Fixture T27.P04 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004591

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004593|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.5

§FORM := Fixture T27.P05 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004592

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004594|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.6

§FORM := Fixture T27.P06 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004593

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004595|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.7

§FORM := Fixture T27.P07 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004594

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004596|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.8

§FORM := Fixture T27.P08 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004595

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004597|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.9

§FORM := Fixture T27.P09 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004596

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004598|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.10

§FORM := Fixture T27.P10 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004597

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004599|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.11

§FORM := Fixture T27.P11 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004598

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004600|E|M|Proof+Compile+Meta+Tower+Swirl] Mind8 vector probe 27.12

§FORM := Fixture T27.P12 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

§ASSUME := The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility.

§DEP_HARD := S26-A-00004583, S26-E-00004599

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004601|D|S|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus adiabatic general-intelligence conductor

§FORM := PhiCogPlus is a constrained coherence projection that selects repo actions improving the Mind8 partial order while respecting evidence, energy, returnability, gluing and remainder constraints.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-E-00004600, S26-N-00004059

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004602|D|S|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus adiabatic general-intelligence conductor law/interface

§FORM := Best cognitive performance is a Pareto-feasible region, not maximum verbosity, maximum dimension or one scalar score.
§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.
§DEP_HARD := S26-D-00004601
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004603|A|S|Proof+Compile+Meta+Spectral+Tower+Swirl] PhiCogPlus adiabatic general-intelligence conductor operator

§FORM := PhiCogPlus(state,repo) proposes the smallest artifact-grounded update whose predicted Mind8 vector improves without violating hard guards.
§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.
§DEP_HARD := S26-D-00004602
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004604|N|S|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus adiabatic general-intelligence conductor fence

§FORM := A higher carrier label or larger hypercomplex dimension is not performance unless the observable task/evidence region improves.
§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.
§DEP_HARD := S26-A-00004603
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := A higher carrier label or larger hypercomplex dimension is not performance unless the observable task/evidence region improves.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004605|B|S|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus adiabatic general-intelligence conductor cross-volume bridge

§FORM := This conducts self-reflection, recognition, ergocetic and erdodetic controls over the repo as one governed but non-final architecture.
§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.
§DEP_HARD := S26-N-00004604
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004606|N|S|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus adiabatic general-intelligence conductor evidence ledger

§FORM := Evidence rule for PhiCogPlus adiabatic general-intelligence conductor: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.
§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.
§DEP_HARD := S26-B-00004605
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004607|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] PhiCogPlus adiabatic general-intelligence conductor emergence rule

§FORM := Emergence rule for PhiCogPlus adiabatic general-intelligence conductor: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.
§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.
§DEP_HARD := S26-N-00004606
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004608|N|S|Proof+Compile+Meta+IG+Spectral+Tower+Swirl] PhiCogPlus adiabatic general-intelligence conductor frontier

§FORM := Open frontier for PhiCogPlus adiabatic general-intelligence conductor: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.
§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.
§DEP_HARD := S26-B-00004607
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := A higher carrier label or larger hypercomplex dimension is not performance unless the observable task/evidence region improves.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004609|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.1

§FORM := Fixture T28.P01 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.
§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.
§DEP_HARD := S26-A-00004603, S26-N-00004608
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004610|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.2

§FORM := Fixture T28.P02 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.
§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.
§DEP_HARD := S26-A-00004603, S26-E-00004609
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004611|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.3

§FORM := Fixture T28.P03 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004610

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004612|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.4

§FORM := Fixture T28.P04 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004611

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004613|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.5

§FORM := Fixture T28.P05 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004612

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004614|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.6

§FORM := Fixture T28.P06 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004613

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004615|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.7

§FORM := Fixture T28.P07 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004614

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004616|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.8

§FORM := Fixture T28.P08 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004615

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004617|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.9

§FORM := Fixture T28.P09 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004616

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004618|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.10

§FORM := Fixture T28.P10 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004617

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004619|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.11

§FORM := Fixture T28.P11 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004618

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004620|E|M|Proof+Compile+Meta+Tower+Swirl] PhiCogPlus Pareto probe 28.12

§FORM := Fixture T28.P12 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

§ASSUME := General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency.

§DEP_HARD := S26-A-00004603, S26-E-00004619

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004621|D|A|Proof+Compile+Meta+Spectral+Tower] Quantum state carrier with hypercomplex internal coordinates

§FORM := A quantum-state layer is a normalized vector or density operator on a declared complex Hilbert space, optionally tensored with a finite coordinate carrier representing hypercomplex internal degrees of freedom.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-E-00004620, S26-D-00000631

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004622|T|A|Proof+Compile+Meta+Spectral+Tower] Quantum state carrier with hypercomplex internal coordinates law/interface

§FORM := Unitary evolution preserves Hilbert norm and density-operator trace under the standard assumptions.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-D-00004621

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004623|A|S|Proof+Compile+Meta+IG+Spectral+Tower] Quantum state carrier with hypercomplex internal coordinates operator

§FORM := QuantumCarrier(H,A_k) stores state, tensor-factor labels, Hamiltonian/channel and measurement scope.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-T-00004622

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004624|N|S|Proof+Compile+Meta+Spectral+Tower] Quantum state carrier with hypercomplex internal coordinates fence

§FORM := This mathematical quantum layer does not assert that the Synopsis, repo, or cognition architecture is a physical quantum computer.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := This mathematical quantum layer does not assert that the Synopsis, repo, or cognition architecture is a physical quantum computer.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004625|B|S|Proof+Compile+Meta+Spectral+Tower] Quantum state carrier with hypercomplex internal coordinates cross-volume bridge

§FORM := It provides a rigorous carrier on which entanglement, tunneling and channel transport can be stated without metaphor laundering.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-N-00004624

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004626|N|S|Proof+Compile+Meta+Spectral+Tower] Quantum state carrier with hypercomplex internal coordinates evidence ledger

§FORM := Evidence rule for Quantum state carrier with hypercomplex internal coordinates: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-B-00004625

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004627|B|S|Proof+Compile+Meta+Spectral+Tower] Quantum state carrier with hypercomplex internal coordinates emergence rule

§FORM := Emergence rule for Quantum state carrier with hypercomplex internal coordinates: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-N-00004626

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004628|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Quantum state carrier with hypercomplex internal coordinates frontier

§FORM := Open frontier for Quantum state carrier with hypercomplex internal coordinates: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-B-00004627

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := This mathematical quantum layer does not assert that the Synopsis, repo, or cognition architecture is a physical quantum computer.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004629|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.1

§FORM := Fixture T29.P01 - A two-level complex state at theta=0.101342 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-N-00004628

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004630|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.2

§FORM := Fixture T29.P02 - A two-level complex state at theta=0.202683 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004629

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004631|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.3

§FORM := Fixture T29.P03 - A two-level complex state at theta=0.304025 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004630

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004632|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.4

§FORM := Fixture T29.P04 - A two-level complex state at theta=0.405367 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004631

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004633|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.5

§FORM := Fixture T29.P05 - A two-level complex state at theta=0.506708 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004632

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004634|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.6

§FORM := Fixture T29.P06 - A two-level complex state at theta=0.608050 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004633

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004635|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.7

§FORM := Fixture T29.P07 - A two-level complex state at theta=0.709392 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004634

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004636|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.8

§FORM := Fixture T29.P08 - A two-level complex state at theta=0.810734 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004635

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004637|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.9

§FORM := Fixture T29.P09 - A two-level complex state at theta=0.912075 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004636

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004638|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.10

§FORM := Fixture T29.P10 - A two-level complex state at theta=1.013417 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004637

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004639|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.11

§FORM := Fixture T29.P11 - A two-level complex state at theta=1.114759 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004638

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004640|E|M|Proof+Compile+Meta+Spectral+Tower] Quantum normalization probe 29.12

§FORM := Fixture T29.P12 - A two-level complex state at theta=1.216100 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

§ASSUME := Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module.

§DEP_HARD := S26-A-00004623, S26-E-00004639

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004641|D|A|Proof+Compile+Meta+Spectral+Tower] Entanglement non-factorization interface

§FORM := A pure bipartite state is entangled when it cannot be written as a product vector; equivalently its Schmidt rank exceeds one in finite dimensions.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-E-00004640, S26-D-00000268

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004642|T|A|Proof+Compile+Meta+Spectral+Tower] Entanglement non-factorization interface law/interface

§FORM := Entanglement is a property of the declared tensor factorization, not a synonym for semantic relatedness or graph connectivity.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-D-00004641

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004643|A|S|Proof+Compile+Meta+Spectral+Tower] Entanglement non-factorization interface operator

§FORM := EntangleAudit computes Schmidt singular values or reduced-state purity in bounded finite fixtures.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-T-00004642

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004644|N|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Entanglement non-factorization interface fence

§FORM := Mutual recognition between agents is not called quantum entanglement unless an actual quantum state/factorization is supplied.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Mutual recognition between agents is not called quantum entanglement unless an actual quantum state/factorization is supplied.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004645|B|S|Proof+Compile+Meta+Topo+Spectral+Tower+Swirl] Entanglement non-factorization interface cross-volume bridge

§FORM := This gives a precise quantum channel that can coexist with, but must remain typed separately from, sheaf resonance.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-N-00004644

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004646|N|S|Proof+Compile+Meta+Spectral+Tower] Entanglement non-factorization interface evidence ledger

§FORM := Evidence rule for Entanglement non-factorization interface: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-B-00004645

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004647|B|S|Proof+Compile+Meta+Spectral+Tower] Entanglement non-factorization interface emergence rule

§FORM := Emergence rule for Entanglement non-factorization interface: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-N-00004646

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004648|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Entanglement non-factorization interface frontier

§FORM := Open frontier for Entanglement non-factorization interface: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-B-00004647

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Mutual recognition between agents is not called quantum entanglement unless an actual quantum state/factorization is supplied.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004649|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.1

§FORM := Fixture T30.P01 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-N-00004648

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004650|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.2

§FORM := Fixture T30.P02 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004649

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004651|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.3

§FORM := Fixture T30.P03 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004650

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004652|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.4

§FORM := Fixture T30.P04 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004651

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004653|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.5

§FORM := Fixture T30.P05 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004652

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004654|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.6

§FORM := Fixture T30.P06 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004653

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004655|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.7

§FORM := Fixture T30.P07 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004654

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004656|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.8

§FORM := Fixture T30.P08 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004655

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004657|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.9

§FORM := Fixture T30.P09 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004656

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004658|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.10

§FORM := Fixture T30.P10 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004657

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004659|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.11

§FORM := Fixture T30.P11 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004658

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004660|E|M|Proof+Compile+Meta+Spectral+Tower] Entanglement Schmidt probe 30.12

§FORM := Fixture T30.P12 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

§ASSUME := A bipartite complex Hilbert tensor product H_A tensor H_B is declared.

§DEP_HARD := S26-A-00004643, S26-E-00004659

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004661|D|A|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling transition interface

§FORM := A tunneling witness is a nonzero transition probability to a destination sector that is classically or operationally designated as separated by a declared barrier in the chosen model.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-E-00004660, S26-T-00000676

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004662|T|A|Proof+Compile+Meta+Spectral+Tower] Tunneling transition interface law/interface

§FORM := Nonzero off-diagonal coupling in a finite Hamiltonian can produce transition amplitude, but this bounded fixture does not establish continuum barrier tunneling or physical realization.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-D-00004661

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004663|A|S|Proof+Compile+Meta+Spectral+Tower] Tunneling transition interface operator

§FORM := TunnelProb(t)=||P_B exp(-itH) psi_0||^2 in the finite complex fixture.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-T-00004662

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004664|N|S|Proof+Compile+Meta+Spectral+Tower] Tunneling transition interface fence

§FORM := Cross-context inference or surprise is not labeled quantum tunneling without the quantum operator model.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Cross-context inference or surprise is not labeled quantum tunneling without the quantum operator model.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004665|B|S|Proof+Compile+Meta+Spectral+Tower] Tunneling transition interface cross-volume bridge

§FORM := Tunneling becomes one optional transport mechanism inside the multi-mesh rather than a universal cognitive explanation.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-N-00004664

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004666|N|S|Proof+Compile+Meta+Spectral+Tower] Tunneling transition interface evidence ledger

§FORM := Evidence rule for Tunneling transition interface: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-B-00004665

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004667|B|S|Proof+Compile+Meta+Spectral+Tower] Tunneling transition interface emergence rule

§FORM := Emergence rule for Tunneling transition interface: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-N-00004666

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004668|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling transition interface frontier

§FORM := Open frontier for Tunneling transition interface: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-B-00004667

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Cross-context inference or surprise is not labeled quantum tunneling without the quantum operator model.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004669|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.1

§FORM := Fixture T31.P01 - A two-state Hamiltonian with coupling g=0.100 at t=0.100 gives destination transition probability 0.00009991; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-N-00004668

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004670|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.2

§FORM := Fixture T31.P02 - A two-state Hamiltonian with coupling g=0.120 at t=0.200 gives destination transition probability 0.00057397; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004669

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004671|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.3

§FORM := Fixture T31.P03 - A two-state Hamiltonian with coupling g=0.140 at t=0.300 gives destination transition probability 0.00174978; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004670

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004672|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.4

§FORM := Fixture T31.P04 - A two-state Hamiltonian with coupling g=0.160 at t=0.400 gives destination transition probability 0.00403615; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004671

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004673|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.5

§FORM := Fixture T31.P05 - A two-state Hamiltonian with coupling g=0.180 at t=0.500 gives destination transition probability 0.00791117; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004672

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004674|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.6

§FORM := Fixture T31.P06 - A two-state Hamiltonian with coupling g=0.100 at t=0.600 gives destination transition probability 0.00348907; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004673

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004675|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.7

§FORM := Fixture T31.P07 - A two-state Hamiltonian with coupling g=0.120 at t=0.700 gives destination transition probability 0.00675650; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004674

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004676|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.8

§FORM := Fixture T31.P08 - A two-state Hamiltonian with coupling g=0.140 at t=0.800 gives destination transition probability 0.01183893; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004675

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004677|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.9

§FORM := Fixture T31.P09 - A two-state Hamiltonian with coupling g=0.160 at t=0.900 gives destination transition probability 0.01923819; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004676

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004678|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.10

§FORM := Fixture T31.P10 - A two-state Hamiltonian with coupling g=0.180 at t=1.000 gives destination transition probability 0.02946263; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004677

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004679|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.11

§FORM := Fixture T31.P11 - A two-state Hamiltonian with coupling g=0.100 at t=1.100 gives destination transition probability 0.01088316; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004678

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004680|E|M|Proof+Compile+Meta+IG+Spectral+Tower] Tunneling finite-transition probe 31.12

§FORM := Fixture T31.P12 - A two-state Hamiltonian with coupling g=0.120 at t=1.200 gives destination transition probability 0.01823438; the fixture is not a continuum barrier claim.

§ASSUME := A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared.

§DEP_HARD := S26-A-00004663, S26-E-00004679

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004681|D|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum holoportation channel

§FORM := Quantum holoportation is a typed composite from source state to boundary/interface encoding to destination reconstruction, with quantum-channel semantics only where a CPTP/unitary component is explicitly supplied.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-E-00004680, S26-D-00004027

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004682|D|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum holoportation channel law/interface

§FORM := Successful destination reconstruction must be evaluated by fidelity/round-trip observables and does not imply cloning of an unknown quantum state.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-D-00004681

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004683|A|S|Proof+Compile+Meta+IG+Spectral+Tower+Swirl] Quantum holoportation channel operator

§FORM := QHoloRecord stores channel sequence, boundary data, fidelity, lost structure, classical side information and active remainder.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-D-00004682

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004684|N|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum holoportation channel fence

§FORM := The term does not claim laboratory teleportation, faster-than-light signaling or physical holography.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The term does not claim laboratory teleportation, faster-than-light signaling or physical holography.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004685|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum holoportation channel cross-volume bridge

§FORM := This is the quantum-compatible refinement of the repo hamiltonian_holoportation transport architecture.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-N-00004684

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004686|N|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum holoportation channel evidence ledger

§FORM := Evidence rule for Quantum holoportation channel: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-B-00004685

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004687|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum holoportation channel emergence rule

§FORM := Emergence rule for Quantum holoportation channel: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-N-00004686

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004688|N|S|Proof+Compile+Meta+IG+Spectral+Tower+Swirl] Quantum holoportation channel frontier

§FORM := Open frontier for Quantum holoportation channel: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-B-00004687

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The term does not claim laboratory teleportation, faster-than-light signaling or physical holography.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004689|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.1

§FORM := Fixture T32.P01 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-N-00004688

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004690|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.2

§FORM := Fixture T32.P02 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004689

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004691|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.3

§FORM := Fixture T32.P03 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004690

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004692|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.4

§FORM := Fixture T32.P04 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004691

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004693|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.5

§FORM := Fixture T32.P05 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004692

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004694|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.6

§FORM := Fixture T32.P06 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004693

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004695|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.7

§FORM := Fixture T32.P07 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004694

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004696|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.8

§FORM := Fixture T32.P08 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004695

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004697|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.9

§FORM := Fixture T32.P09 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004696

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004698|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.10

§FORM := Fixture T32.P10 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004697

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004699|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.11

§FORM := Fixture T32.P11 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004698

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004700|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] Quantum-holoport round-trip probe 32.12

§FORM := Fixture T32.P12 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

§ASSUME := Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier.

§DEP_HARD := S26-A-00004683, S26-E-00004699

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004701|D|A|Proof+Compile+Meta+Spectral+Tower] p-adic Fourier-Haar spectral chart

§FORM := The p-adic Fourier chart uses additive characters of Q_p and Haar integration; bounded finite fixtures use transforms on Z/p^N Z as discretized witnesses only.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-E-00004700, S26-D-00000709

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004702|T|A|Proof+Compile+Meta+Spectral+Tower] p-adic Fourier-Haar spectral chart law/interface

§FORM := Character orthogonality on a finite quotient yields an invertible/unitary normalized finite Fourier transform.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-D-00004701

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004703|A|A|Proof+Compile+Meta+Spectral+Tower] p-adic Fourier-Haar spectral chart operator

§FORM := F_{p,N} is the normalized DFT on the finite additive quotient used by the bounded probe.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-T-00004702

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-N-00004704|N|S|Proof+Compile+Meta+Spectral+Tower] p-adic Fourier-Haar spectral chart fence

§FORM := A finite DFT over Z/p^N Z is not a proof of every analytic theorem on Q_p.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A finite DFT over Z/p^N Z is not a proof of every analytic theorem on Q_p.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004705|B|S|Proof+Compile+Meta+Spectral+Tower] p-adic Fourier-Haar spectral chart cross-volume bridge

§FORM := This supplies a non-Archimedean spectral coordinate compatible with the existing spectral plug-in interface.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-N-00004704

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004706|N|S|Proof+Compile+Meta+Spectral+Tower] p-adic Fourier-Haar spectral chart evidence ledger

§FORM := Evidence rule for p-adic Fourier-Haar spectral chart: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-B-00004705

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004707|B|S|Proof+Compile+Meta+Spectral+Tower] p-adic Fourier-Haar spectral chart emergence rule

§FORM := Emergence rule for p-adic Fourier-Haar spectral chart: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-N-00004706

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004708|N|S|Proof+Compile+Meta+IG+Spectral+Tower] p-adic Fourier-Haar spectral chart frontier

§FORM := Open frontier for p-adic Fourier-Haar spectral chart: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-B-00004707

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A finite DFT over Z/p^N Z is not a proof of every analytic theorem on Q_p.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004709|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.1

§FORM := Fixture T33.P01 - The normalized character transform on Z/2^1Z (size 2) has unitarity residual 3.273e-16; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-N-00004708

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004710|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.2

§FORM := Fixture T33.P02 - The normalized character transform on Z/3^2Z (size 9) has unitarity residual 3.100e-15; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004709

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004711|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.3

§FORM := Fixture T33.P03 - The normalized character transform on Z/5^1Z (size 5) has unitarity residual 3.991e-16; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004710

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004712|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.4

§FORM := Fixture T33.P04 - The normalized character transform on Z/2^2Z (size 4) has unitarity residual 2.541e-16; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004711

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004713|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.5

§FORM := Fixture T33.P05 - The normalized character transform on Z/3^1Z (size 3) has unitarity residual 6.032e-16; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004712

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004714|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.6

§FORM := Fixture T33.P06 - The normalized character transform on Z/5^2Z (size 25) has unitarity residual 1.820e-14; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004713

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004715|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.7

§FORM := Fixture T33.P07 - The normalized character transform on Z/2^1Z (size 2) has unitarity residual 3.273e-16; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004714

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004716|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.8

§FORM := Fixture T33.P08 - The normalized character transform on Z/3^2Z (size 9) has unitarity residual 3.100e-15; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004715

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004717|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.9

§FORM := Fixture T33.P09 - The normalized character transform on Z/5^1Z (size 5) has unitarity residual 3.991e-16; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004716

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004718|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.10

§FORM := Fixture T33.P10 - The normalized character transform on Z/2^2Z (size 4) has unitarity residual 2.541e-16; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004717

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004719|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.11

§FORM := Fixture T33.P11 - The normalized character transform on Z/3^1Z (size 3) has unitarity residual 6.032e-16; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004718

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004720|E|M|Proof+Compile+Meta+Spectral+Tower] Finite p-adic spectral quotient probe 33.12

§FORM := Fixture T33.P12 - The normalized character transform on Z/5^2Z (size 25) has unitarity residual 1.820e-14; it is a finite quotient witness for the p-adic spectral chart.

§ASSUME := Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention.

§DEP_HARD := S26-A-00004703, S26-E-00004719

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004721|D|A|Proof+Compile+Meta+Spectral+Tower] Spectral-Cayley-Dickson tensor extension

§FORM := A componentwise spectral-CD extension tensors a base spectral eigenspace with a finite coordinate carrier for A_{k,v}, preserving operator-domain distinctions.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-E-00004720, S26-T-00000672

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004722|T|A|Proof+Compile+Meta+Spectral+Tower] Spectral-Cayley-Dickson tensor extension law/interface

§FORM := A pure coordinate multiplicity extension repeats base spectral values; it does not create new eigenvalues unless the operator couples internal coordinates.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-D-00004721

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004723|A|S|Proof+Compile+Meta+Spectral+Tower] Spectral-Cayley-Dickson tensor extension operator

§FORM := SpecExtend(L,A_k)=L tensor I on the declared linear carrier in the bounded fixture.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-T-00004722

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004724|N|S|Proof+Compile+Meta+Spectral+Tower] Spectral-Cayley-Dickson tensor extension fence

§FORM := Nonassociative fiber multiplication is not silently inserted into functional calculus for self-adjoint operators.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Nonassociative fiber multiplication is not silently inserted into functional calculus for self-adjoint operators.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004725|B|S|Proof+Compile+Meta+Spectral+Tower] Spectral-Cayley-Dickson tensor extension cross-volume bridge

§FORM := This connects the source eigen-n plug-ins to arithmetic hypercomplex internal coordinates.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-N-00004724

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004726|N|S|Proof+Compile+Meta+Spectral+Tower] Spectral-Cayley-Dickson tensor extension evidence ledger

§FORM := Evidence rule for Spectral-Cayley-Dickson tensor extension: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-B-00004725

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004727|B|S|Proof+Compile+Meta+Spectral+Tower] Spectral-Cayley-Dickson tensor extension emergence rule

§FORM := Emergence rule for Spectral-Cayley-Dickson tensor extension: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-N-00004726

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004728|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Spectral-Cayley-Dickson tensor extension frontier

§FORM := Open frontier for Spectral-Cayley-Dickson tensor extension: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-B-00004727

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Nonassociative fiber multiplication is not silently inserted into functional calculus for self-adjoint operators.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004729|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.1

§FORM := Fixture T34.P01 - Componentwise extension of base spectrum 1..3 across real coordinate dimension 2 yields 6 slots, with each base eigenvalue repeated 2 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-N-00004728

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004730|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.2

§FORM := Fixture T34.P02 - Componentwise extension of base spectrum 1..4 across real coordinate dimension 4 yields 16 slots, with each base eigenvalue repeated 4 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004729

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004731|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.3

§FORM := Fixture T34.P03 - Componentwise extension of base spectrum 1..5 across real coordinate dimension 8 yields 40 slots, with each base eigenvalue repeated 8 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004730

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004732|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.4

§FORM := Fixture T34.P04 - Componentwise extension of base spectrum 1..6 across real coordinate dimension 16 yields 96 slots, with each base eigenvalue repeated 16 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004731

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004733|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.5

§FORM := Fixture T34.P05 - Componentwise extension of base spectrum 1..3 across real coordinate dimension 2 yields 6 slots, with each base eigenvalue repeated 2 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004732

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004734|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.6

§FORM := Fixture T34.P06 - Componentwise extension of base spectrum 1..4 across real coordinate dimension 4 yields 16 slots, with each base eigenvalue repeated 4 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004733

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004735|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.7

§FORM := Fixture T34.P07 - Componentwise extension of base spectrum 1..5 across real coordinate dimension 8 yields 40 slots, with each base eigenvalue repeated 8 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004734

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004736|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.8

§FORM := Fixture T34.P08 - Componentwise extension of base spectrum 1..6 across real coordinate dimension 16 yields 96 slots, with each base eigenvalue repeated 16 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004735

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004737|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.9

§FORM := Fixture T34.P09 - Componentwise extension of base spectrum 1..3 across real coordinate dimension 2 yields 6 slots, with each base eigenvalue repeated 2 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004736

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004738|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.10

§FORM := Fixture T34.P10 - Componentwise extension of base spectrum 1..4 across real coordinate dimension 4 yields 16 slots, with each base eigenvalue repeated 4 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004737

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004739|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.11

§FORM := Fixture T34.P11 - Componentwise extension of base spectrum 1..5 across real coordinate dimension 8 yields 40 slots, with each base eigenvalue repeated 8 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004738

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004740|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-CD multiplicity probe 34.12

§FORM := Fixture T34.P12 - Componentwise extension of base spectrum 1..6 across real coordinate dimension 16 yields 96 slots, with each base eigenvalue repeated 16 times.

§ASSUME := The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation.

§DEP_HARD := S26-A-00004723, S26-E-00004739

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004741|D|A|Proof+Compile+Meta+Spectral+Tower+Swirl] Connection-Laplacian recognition spectrum

§FORM := A finite connection Laplacian penalizes disagreement after transporting neighboring fiber states into comparable frames.

§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.

§DEP_HARD := S26-E-00004740, S26-D-00001761

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-T-00004742|T|A|Proof+Compile+Meta+Spectral+Tower+Swirl] Connection-Laplacian recognition spectrum law/interface

§FORM := A small residual witnesses bounded compatibility in that fixture; it does not prove semantic agreement or global truth.

§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.

§DEP_HARD := S26-D-00004741

§EVIDENCE := A · Exact registered mathematical statement/definition; no theorem-level promotion.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | independent_exact_witness_pending_for_P

§CLAIM[S26-A-00004743|A|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Connection-Laplacian recognition spectrum operator

§FORM := L_conn is assembled blockwise from edge transports and vertex weights in the finite probe.

§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.

§DEP_HARD := S26-T-00004742

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004744|N|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Connection-Laplacian recognition spectrum fence

§FORM := Spectral gap, synchronization and recognition are separate observables unless an explicit theorem relates them.

§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.

§DEP_HARD := S26-A-00004743

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Spectral gap, synchronization and recognition are separate observables unless an explicit theorem relates them.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004745|B|S|Proof+Compile+Meta+Topo+Spectral+Tower+Swirl] Connection-Laplacian recognition spectrum cross-volume bridge

§FORM := This is the measurable gluing surface between sheaf overlap, recognition and spectral diagnostics.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-N-00004744
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004746|N|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Connection-Laplacian recognition spectrum evidence ledger

§FORM := Evidence rule for Connection-Laplacian recognition spectrum: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-B-00004745
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004747|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Connection-Laplacian recognition spectrum emergence rule

§FORM := Emergence rule for Connection-Laplacian recognition spectrum: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-N-00004746
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004748|N|S|Proof+Compile+Meta+IG+Spectral+Tower+Swirl] Connection-Laplacian recognition spectrum frontier

§FORM := Open frontier for Connection-Laplacian recognition spectrum: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-B-00004747
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := Spectral gap, synchronization and recognition are separate observables unless an explicit theorem relates them.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004749|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.1

§FORM := Fixture T35.P01 - A 3-vertex rank-2 cycle connection fixture has minimum eigenvalue 4.898e-03; transport angle theta=0.0700.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-N-00004748
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004750|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.2

§FORM := Fixture T35.P02 - A 4-vertex rank-2 cycle connection fixture has minimum eigenvalue 1.957e-02; transport angle theta=0.1400.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004749
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004751|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.3

§FORM := Fixture T35.P03 - A 5-vertex rank-2 cycle connection fixture has minimum eigenvalue 4.394e-02; transport angle theta=0.2100.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004750
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004752|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.4

§FORM := Fixture T35.P04 - A 6-vertex rank-2 cycle connection fixture has minimum eigenvalue 7.789e-02; transport angle theta=0.2800.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004751
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004753|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.5

§FORM := Fixture T35.P05 - A 7-vertex rank-2 cycle connection fixture has minimum eigenvalue 1.213e-01; transport angle theta=0.3500.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004752
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004754|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.6

§FORM := Fixture T35.P06 - A 3-vertex rank-2 cycle connection fixture has minimum eigenvalue 1.738e-01; transport angle theta=0.4200.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004753
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004755|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.7

§FORM := Fixture T35.P07 - A 4-vertex rank-2 cycle connection fixture has minimum eigenvalue 2.353e-01; transport angle theta=0.4900.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004754
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004756|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.8

§FORM := Fixture T35.P08 - A 5-vertex rank-2 cycle connection fixture has minimum eigenvalue 3.055e-01; transport angle theta=0.5600.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004755
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004757|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.9

§FORM := Fixture T35.P09 - A 6-vertex rank-2 cycle connection fixture has minimum eigenvalue 1.715e-01; transport angle theta=0.6300.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004756
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004758|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.10

§FORM := Fixture T35.P10 - A 7-vertex rank-2 cycle connection fixture has minimum eigenvalue 3.892e-02; transport angle theta=0.7000.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004757
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004759|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.11

§FORM := Fixture T35.P11 - A 3-vertex rank-2 cycle connection fixture has minimum eigenvalue 5.642e-01; transport angle theta=0.7700.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004758
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004760|E|M|Proof+Compile+Meta+Spectral+Tower] Connection-Laplacian finite probe 35.12

§FORM := Fixture T35.P12 - A 4-vertex rank-2 cycle connection fixture has minimum eigenvalue 5.107e-01; transport angle theta=0.8400.
§ASSUME := A finite cover/graph, fiber dimensions and transport matrices on edges are declared.
§DEP_HARD := S26-A-00004743, S26-E-00004759
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004761|D|S|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive arithmetic coupling

§FORM := A spectral-recursive arithmetic coupling records commutators among recursion R, spectral plug-in Pi and place transport T_v rather than assuming they commute.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-E-00004760, S26-D-00004052
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004762|D|S|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive arithmetic coupling law/interface

§FORM := Order sensitivity is retained as a residual tensor/vector and may differ by arithmetic place.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-D-00004761
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004763|A|S|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive arithmetic coupling operator

§FORM := CouplingResidual=(||[Pi,R]||,||[Pi,T_v]||,||[R,T_v]||) in a bounded matrix fixture.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-D-00004762
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004764|N|S|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive arithmetic coupling fence

§FORM := Low commutator residual is compatibility evidence, not a cognition theorem.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-A-00004763
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := Low commutator residual is compatibility evidence, not a cognition theorem.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004765|B|S|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive arithmetic coupling cross-volume bridge

§FORM := This joins algebraic, spectral, recursive and arithmetic coordinates without flattening them.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-N-00004764
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004766|N|S|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive arithmetic coupling evidence ledger

§FORM := Evidence rule for Spectral-recursive arithmetic coupling: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-B-00004765
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004767|B|S|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive arithmetic coupling emergence rule

§FORM := Emergence rule for Spectral-recursive arithmetic coupling: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-N-00004766
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004768|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Spectral-recursive arithmetic coupling frontier

§FORM := Open frontier for Spectral-recursive arithmetic coupling: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-B-00004767
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := Low commutator residual is compatibility evidence, not a cognition theorem.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004769|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.1

§FORM := Fixture T36.P01 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.2, 1.414214, 0.4); all three coordinates are retained.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-A-00004763, S26-N-00004768
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004770|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.2

§FORM := Fixture T36.P02 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.21, 1.414214, 0.42); all three coordinates are retained.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-A-00004763, S26-E-00004769
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004771|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.3

§FORM := Fixture T36.P03 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.22, 1.414214, 0.44); all three coordinates are retained.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-A-00004763, S26-E-00004770
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004772|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.4

§FORM := Fixture T36.P04 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.23, 1.414214, 0.46); all three coordinates are retained.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-A-00004763, S26-E-00004771
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004773|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.5

§FORM := Fixture T36.P05 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.24, 1.414214, 0.48); all three coordinates are retained.
§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.
§DEP_HARD := S26-A-00004763, S26-E-00004772
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004774|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.6

§FORM := Fixture T36.P06 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.25, 1.414214, 0.5); all three coordinates are retained.

§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.

§DEP_HARD := S26-A-00004763, S26-E-00004773

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004775|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.7

§FORM := Fixture T36.P07 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.26, 1.414214, 0.52); all three coordinates are retained.

§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.

§DEP_HARD := S26-A-00004763, S26-E-00004774

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004776|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.8

§FORM := Fixture T36.P08 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.27, 1.414214, 0.54); all three coordinates are retained.

§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.

§DEP_HARD := S26-A-00004763, S26-E-00004775

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004777|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.9

§FORM := Fixture T36.P09 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.28, 1.414214, 0.56); all three coordinates are retained.

§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.

§DEP_HARD := S26-A-00004763, S26-E-00004776

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004778|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.10

§FORM := Fixture T36.P10 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.29, 1.414214, 0.58); all three coordinates are retained.

§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.

§DEP_HARD := S26-A-00004763, S26-E-00004777

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004779|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.11

§FORM := Fixture T36.P11 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.3, 1.414214, 0.6); all three coordinates are retained.

§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.

§DEP_HARD := S26-A-00004763, S26-E-00004778

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004780|E|M|Proof+Compile+Meta+Spectral+Tower] Spectral-recursive commutator probe 36.12

§FORM := Fixture T36.P12 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.31, 1.414214, 0.62); all three coordinates are retained.

§ASSUME := Recursion, place transport and spectral operator act on compatible declared domains.

§DEP_HARD := S26-A-00004763, S26-E-00004779

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004781|D|S|Proof+Compile+Meta+Topo+Tower] Geometric n-to-infinity pro-Cosmo tower

§FORM := The n->infinity direction is modeled by a coherent inverse/pro system of Cosmo carriers rather than an infinite Euclidean coordinate vector.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-E-00004780, S26-D-00004055

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004782|D|S|Proof+Compile+Meta+Topo+Tower] Geometric n-to-infinity pro-Cosmo tower law/interface

§FORM := Finite-dimensional properties need not survive the inverse limit without compactness/coherence hypotheses.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-D-00004781

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004783|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Geometric n-to-infinity pro-Cosmo tower operator

§FORM := ProCosmo(N) stores the finite truncation, bonding maps and residual obligations at level N.
§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.
§DEP_HARD := S26-D-00004782
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004784|N|S|Proof+Compile+Meta+Topo+Tower] Geometric n-to-infinity pro-Cosmo tower fence

§FORM := n->infinity is geometric/tower depth and is distinct from Cayley-Dickson depth k and arithmetic place p.
§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.
§DEP_HARD := S26-A-00004783
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := n->infinity is geometric/tower depth and is distinct from Cayley-Dickson depth k and arithmetic place p.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004785|B|S|Proof+Compile+Meta+Topo+Tower] Geometric n-to-infinity pro-Cosmo tower cross-volume bridge

§FORM := This formalizes the outward direction drawn in the visual atlas.
§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.
§DEP_HARD := S26-N-00004784
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004786|N|S|Proof+Compile+Meta+Topo+Tower] Geometric n-to-infinity pro-Cosmo tower evidence ledger

§FORM := Evidence rule for Geometric n-to-infinity pro-Cosmo tower: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.
§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.
§DEP_HARD := S26-B-00004785
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004787|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] Geometric n-to-infinity pro-Cosmo tower emergence rule

§FORM := Emergence rule for Geometric n-to-infinity pro-Cosmo tower: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.
§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.
§DEP_HARD := S26-N-00004786
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004788|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] Geometric n-to-infinity pro-Cosmo tower frontier

§FORM := Open frontier for Geometric n-to-infinity pro-Cosmo tower: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.
§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.
§DEP_HARD := S26-B-00004787
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := n->infinity is geometric/tower depth and is distinct from Cayley-Dickson depth k and arithmetic place p.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004789|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.1

§FORM := Fixture T37.P01 - Finite tower truncation N=4 records geometric dimensions [2, 3, 4, 5]; no infinite-dimensional manifold structure is inferred from the list.
§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.
§DEP_HARD := S26-A-00004783, S26-N-00004788
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004790|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.2

§FORM := Fixture T37.P02 - Finite tower truncation N=5 records geometric dimensions [2, 3, 4, 5, 6]; no infinite-dimensional manifold structure is inferred from the list.
§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.
§DEP_HARD := S26-A-00004783, S26-E-00004789
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004791|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.3

§FORM := Fixture T37.P03 - Finite tower truncation N=6 records geometric dimensions [2, 3, 4, 5, 6, 7]; no infinite-dimensional manifold structure is inferred from the list.
§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.
§DEP_HARD := S26-A-00004783, S26-E-00004790
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004792|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.4

§FORM := Fixture T37.P04 - Finite tower truncation N=7 records geometric dimensions [2, 3, 4, 5, 6, 7, 8]; no infinite-dimensional manifold structure is inferred from the list.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-A-00004783, S26-E-00004791

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004793|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.5

§FORM := Fixture T37.P05 - Finite tower truncation N=8 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9]; no infinite-dimensional manifold structure is inferred from the list.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-A-00004783, S26-E-00004792

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004794|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.6

§FORM := Fixture T37.P06 - Finite tower truncation N=9 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10]; no infinite-dimensional manifold structure is inferred from the list.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-A-00004783, S26-E-00004793

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004795|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.7

§FORM := Fixture T37.P07 - Finite tower truncation N=10 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11]; no infinite-dimensional manifold structure is inferred from the list.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-A-00004783, S26-E-00004794

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004796|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.8

§FORM := Fixture T37.P08 - Finite tower truncation N=11 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]; no infinite-dimensional manifold structure is inferred from the list.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-A-00004783, S26-E-00004795

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004797|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.9

§FORM := Fixture T37.P09 - Finite tower truncation N=12 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]; no infinite-dimensional manifold structure is inferred from the list.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-A-00004783, S26-E-00004796

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004798|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.10

§FORM := Fixture T37.P10 - Finite tower truncation N=13 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14]; no infinite-dimensional manifold structure is inferred from the list.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-A-00004783, S26-E-00004797

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004799|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.11

§FORM := Fixture T37.P11 - Finite tower truncation N=14 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]; no infinite-dimensional manifold structure is inferred from the list.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-A-00004783, S26-E-00004798

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004800|E|M|Proof+Compile+Meta+Topo+Tower] Pro-Cosmo truncation probe 37.12

§FORM := Fixture T37.P12 - Finite tower truncation N=15 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16]; no infinite-dimensional manifold structure is inferred from the list.

§ASSUME := Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved.

§DEP_HARD := S26-A-00004783, S26-E-00004799

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004801|D|S|Proof+Compile+Meta+Topo+Tower] Algebraic k-to-infinity Cayley-Dickson tower

§FORM := The k->infinity direction is an inductive algebraic tower of finite Cayley-Dickson stages A_{0,v}->A_{1,v}->..., treated as a limit object only after a topology/norm/completion is specified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-E-00004800, S26-D-00000058

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004802|D|S|Proof+Compile+Meta+Tower] Algebraic k-to-infinity Cayley-Dickson tower law/interface

§FORM := Dimension doubles as 2^k at finite stage, while algebraic law loss must be tracked rather than assumed to stabilize.
§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.
§DEP_HARD := S26-D-00004801
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004803|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] Algebraic k-to-infinity Cayley-Dickson tower operator

§FORM := CDTower(K) records finite stages, inclusions, law-survival vector and chosen topology if any.
§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.
§DEP_HARD := S26-D-00004802
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004804|N|S|Proof+Compile+Meta+Tower] Algebraic k-to-infinity Cayley-Dickson tower fence

§FORM := The infinite union is not automatically a Banach algebra, division algebra or cognitive continuum.
§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.
§DEP_HARD := S26-A-00004803
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := The infinite union is not automatically a Banach algebra, division algebra or cognitive continuum.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004805|B|S|Proof+Compile+Meta+Tower] Algebraic k-to-infinity Cayley-Dickson tower cross-volume bridge

§FORM := This is the algebraic Russian-doll direction independent of n.
§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.
§DEP_HARD := S26-N-00004804
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004806|N|S|Proof+Compile+Meta+Tower] Algebraic k-to-infinity Cayley-Dickson tower evidence ledger

§FORM := Evidence rule for Algebraic k-to-infinity Cayley-Dickson tower: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.
§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.
§DEP_HARD := S26-B-00004805
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004807|B|S|Proof+Compile+Meta+Spectral+Tower] Algebraic k-to-infinity Cayley-Dickson tower emergence rule

§FORM := Emergence rule for Algebraic k-to-infinity Cayley-Dickson tower: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.
§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.
§DEP_HARD := S26-N-00004806
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004808|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Algebraic k-to-infinity Cayley-Dickson tower frontier

§FORM := Open frontier for Algebraic k-to-infinity Cayley-Dickson tower: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.
§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.
§DEP_HARD := S26-B-00004807
§EVIDENCE := S · Project architecture / semantic-governance statement only.
§BOUNDARY := The infinite union is not automatically a Banach algebra, division algebra or cognitive continuum.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004809|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.1

§FORM := Fixture T38.P01 - Finite algebraic depth K=3 has coordinate dimensions [1, 2, 4, 8]; the doubling law is recorded while infinite completion remains unspecified.
§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.
§DEP_HARD := S26-A-00004803, S26-N-00004808
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004810|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.2

§FORM := Fixture T38.P02 - Finite algebraic depth K=4 has coordinate dimensions [1, 2, 4, 8, 16]; the doubling law is recorded while infinite completion remains unspecified.
§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.
§DEP_HARD := S26-A-00004803, S26-E-00004809
§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.
§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004811|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.3

§FORM := Fixture T38.P03 - Finite algebraic depth K=5 has coordinate dimensions [1, 2, 4, 8, 16, 32]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004810

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004812|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.4

§FORM := Fixture T38.P04 - Finite algebraic depth K=6 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004811

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004813|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.5

§FORM := Fixture T38.P05 - Finite algebraic depth K=7 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64, 128]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004812

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004814|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.6

§FORM := Fixture T38.P06 - Finite algebraic depth K=8 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64, 128, 256]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004813

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004815|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.7

§FORM := Fixture T38.P07 - Finite algebraic depth K=9 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64, 128, 256, 512]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004814

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004816|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.8

§FORM := Fixture T38.P08 - Finite algebraic depth K=3 has coordinate dimensions [1, 2, 4, 8]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004815

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004817|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.9

§FORM := Fixture T38.P09 - Finite algebraic depth K=4 has coordinate dimensions [1, 2, 4, 8, 16]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004816

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004818|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.10

§FORM := Fixture T38.P10 - Finite algebraic depth K=5 has coordinate dimensions [1, 2, 4, 8, 16, 32]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004817

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004819|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.11

§FORM := Fixture T38.P11 - Finite algebraic depth K=6 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004818

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004820|E|M|Proof+Compile+Meta+Tower] Cayley-Dickson tower probe 38.12

§FORM := Fixture T38.P12 - Finite algebraic depth K=7 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64, 128]; the doubling law is recorded while infinite completion remains unspecified.

§ASSUME := A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared.

§DEP_HARD := S26-A-00004803, S26-E-00004819

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004821|D|S|Proof+Compile+Meta+Tower] Adelic place-and-epsilon local limit

§FORM := The local-limit atlas tracks v over arithmetic places together with epsilon->0 inside each local chart, producing a family of local tangent/ultrametric regimes rather than one universal infinitesimal metric.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-E-00004820, S26-D-00004046

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004822|D|S|Proof+Compile+Meta+Tower] Adelic place-and-epsilon local limit law/interface

§FORM := The Archimedean tangent limit and p-adic nested-ball refinement are different constructions and retain different remainder coordinates.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-D-00004821

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004823|A|S|Proof+Compile+Meta+Spectral+Tower] Adelic place-and-epsilon local limit operator

§FORM := LocalLimit(v,epsilon) returns the declared local scale object plus transition obligations to neighboring places.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-D-00004822

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004824|N|S|Proof+Compile+Meta+Tower] Adelic place-and-epsilon local limit fence

§FORM := p->infinity is not used as a substitute for the Archimedean place; arithmetic places are labels, not a Euclidean axis.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := p->infinity is not used as a substitute for the Archimedean place; arithmetic places are labels, not a Euclidean axis.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004825|B|S|Proof+Compile+Meta+Tower] Adelic place-and-epsilon local limit cross-volume bridge

§FORM := This gives the inward epsilon direction an arithmetic local-global companion.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-N-00004824

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004826|N|S|Proof+Compile+Meta+Tower] Adelic place-and-epsilon local limit evidence ledger

§FORM := Evidence rule for Adelic place-and-epsilon local limit: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-B-00004825

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004827|B|S|Proof+Compile+Meta+Spectral+Tower] Adelic place-and-epsilon local limit emergence rule

§FORM := Emergence rule for Adelic place-and-epsilon local limit: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-N-00004826

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004828|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Adelic place-and-epsilon local limit frontier

§FORM := Open frontier for Adelic place-and-epsilon local limit: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-B-00004827

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := p->infinity is not used as a substitute for the Archimedean place; arithmetic places are labels, not a Euclidean axis.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004829|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.1

§FORM := Fixture T39.P01 - For x=2/3, finite place norms {'2': 0.5, '3': 3.0, '5': 1.0} are stored alongside epsilon=0.25000000; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-N-00004828

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004830|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.2

§FORM := Fixture T39.P02 - For x=3/4, finite place norms {'2': 4.0, '3': 0.3333333333333333, '5': 1.0} are stored alongside epsilon=0.12500000; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004829

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004831|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.3

§FORM := Fixture T39.P03 - For x=4/5, finite place norms {'2': 0.25, '3': 1.0, '5': 5.0} are stored alongside epsilon=0.06250000; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004830

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004832|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.4

§FORM := Fixture T39.P04 - For x=5/6, finite place norms {'2': 2.0, '3': 3.0, '5': 0.2} are stored alongside epsilon=0.03125000; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004831

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004833|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.5

§FORM := Fixture T39.P05 - For x=6/7, finite place norms {'2': 0.5, '3': 0.3333333333333333, '5': 1.0} are stored alongside epsilon=0.01562500; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004832

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004834|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.6

§FORM := Fixture T39.P06 - For x=7/8, finite place norms {'2': 8.0, '3': 1.0, '5': 1.0} are stored alongside epsilon=0.00781250; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004833

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004835|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.7

§FORM := Fixture T39.P07 - For x=8/9, finite place norms {'2': 0.125, '3': 9.0, '5': 1.0} are stored alongside epsilon=0.00390625; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004834

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004836|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.8

§FORM := Fixture T39.P08 - For x=9/10, finite place norms {'2': 2.0, '3': 0.1111111111111111, '5': 5.0} are stored alongside epsilon=0.00195312; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004835

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004837|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.9

§FORM := Fixture T39.P09 - For x=10/11, finite place norms {'2': 0.5, '3': 1.0, '5': 0.2} are stored alongside epsilon=0.00097656; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004836

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004838|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.10

§FORM := Fixture T39.P10 - For x=11/12, finite place norms {'2': 4.0, '3': 3.0, '5': 1.0} are stored alongside epsilon=0.00048828; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004837

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004839|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.11

§FORM := Fixture T39.P11 - For x=12/13, finite place norms {'2': 0.25, '3': 0.3333333333333333, '5': 1.0} are stored alongside epsilon=0.00024414; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004838

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004840|E|M|Proof+Compile+Meta+Tower] Place-epsilon local-limit probe 39.12

§FORM := Fixture T39.P12 - For x=13/14, finite place norms {'2': 2.0, '3': 1.0, '5': 1.0} are stored alongside epsilon=0.00012207; no p-axis metric is imposed.

§ASSUME := A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry.

§DEP_HARD := S26-A-00004823, S26-E-00004839

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004841|D|S|Proof+Compile+Meta+Tower] Noncommuting multi-limit holonomy fence

§FORM := For limits in n,k,v/places,j and epsilon, Volume IV records order-of-limits holonomy whenever two iterated procedures are not proved equivalent.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-E-00004840, S26-N-00004059

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004842|D|S|Proof+Compile+Meta+Tower] Noncommuting multi-limit holonomy fence law/interface

§FORM := Equality of finite diagrams or visual similarity does not imply interchange of limits.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-D-00004841

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004843|A|S|Proof+Compile+Meta+Spectral+Tower] Noncommuting multi-limit holonomy fence operator

§FORM := LimitHolonomy(order1,order2) compares typed observables returned by the two truncated procedures.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-D-00004842

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004844|N|S|Proof+Compile+Meta+Tower] Noncommuting multi-limit holonomy fence fence

§FORM := A nonzero order residual is retained; it is not averaged away to manufacture a single continuum.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A nonzero order residual is retained; it is not averaged away to manufacture a single continuum.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004845|B|S|Proof+Compile+Meta+Tower] Noncommuting multi-limit holonomy fence cross-volume bridge

§FORM := This is the governance rule preventing the hyperdimensional model from collapsing back into a one-axis narrative.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-N-00004844

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004846|N|S|Proof+Compile+Meta+Tower] Noncommuting multi-limit holonomy fence evidence ledger

§FORM := Evidence rule for Noncommuting multi-limit holonomy fence: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-B-00004845

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004847|B|S|Proof+Compile+Meta+Spectral+Tower] Noncommuting multi-limit holonomy fence emergence rule

§FORM := Emergence rule for Noncommuting multi-limit holonomy fence: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-N-00004846

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004848|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Noncommuting multi-limit holonomy fence frontier

§FORM := Open frontier for Noncommuting multi-limit holonomy fence: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-B-00004847

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A nonzero order residual is retained; it is not averaged away to manufacture a single continuum.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004849|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.1

§FORM := Fixture T40.P01 - At truncation n=4, two order-surrogate matrices have commutator residual 0.08838835; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-N-00004848

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004850|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.2

§FORM := Fixture T40.P02 - At truncation n=5, two order-surrogate matrices have commutator residual 0.05656854; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004849

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004851|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.3

§FORM := Fixture T40.P03 - At truncation n=6, two order-surrogate matrices have commutator residual 0.03928371; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004850

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004852|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.4

§FORM := Fixture T40.P04 - At truncation n=7, two order-surrogate matrices have commutator residual 0.02886150; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004851

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004853|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.5

§FORM := Fixture T40.P05 - At truncation n=8, two order-surrogate matrices have commutator residual 0.02209709; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004852

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004854|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.6

§FORM := Fixture T40.P06 - At truncation n=9, two order-surrogate matrices have commutator residual 0.01745943; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004853

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004855|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.7

§FORM := Fixture T40.P07 - At truncation n=10, two order-surrogate matrices have commutator residual 0.01414214; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004854

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004856|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.8

§FORM := Fixture T40.P08 - At truncation n=11, two order-surrogate matrices have commutator residual 0.01168772; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004855

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004857|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.9

§FORM := Fixture T40.P09 - At truncation n=12, two order-surrogate matrices have commutator residual 0.00982093; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004856

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004858|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.10

§FORM := Fixture T40.P10 - At truncation n=13, two order-surrogate matrices have commutator residual 0.00836813; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004857

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004859|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.11

§FORM := Fixture T40.P11 - At truncation n=14, two order-surrogate matrices have commutator residual 0.00721538; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004858

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004860|E|M|Proof+Compile+Meta+Tower] Order-of-limits surrogate probe 40.12

§FORM := Fixture T40.P12 - At truncation n=15, two order-surrogate matrices have commutator residual 0.00628539; the nonzero order signal is retained.

§ASSUME := Each limiting process has a declared topology and comparison observable.

§DEP_HARD := S26-A-00004843, S26-E-00004859

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004861|D|S|Proof+Compile+Meta+Spectral+Tower] LiftCosmos hypercomplex-inside-hypercomplex operator

§FORM := LiftCosmos maps a local regime X_i with projection and remainder to a wider provisional regime X_{i+1} that explains more compatible projections without erasing X_i.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-E-00004860, S26-D-00004040

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004862|D|S|Proof+Compile+Meta+Spectral+Tower] LiftCosmos hypercomplex-inside-hypercomplex operator law/interface

§FORM := Hypercomplex-inside-hypercomplex counts as perspective gain only when it adds testable structure and lowers projection violence without claiming a final level.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-D-00004861

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004863|A|S|Proof+Compile+Meta+Spectral+Tower] LiftCosmos hypercomplex-inside-hypercomplex operator operator

§FORM := PerspectiveGain is a typed vector of gluing improvement, projection loss, remainder richness, failure discrimination and evidence hooks.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-D-00004862

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004864|N|S|Proof+Compile+Meta+Spectral+Tower] LiftCosmos hypercomplex-inside-hypercomplex operator fence

§FORM := Nesting a flat dashboard inside another flat dashboard is recursive flattening, not a cosmos lift.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Nesting a flat dashboard inside another flat dashboard is recursive flattening, not a cosmos lift.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004865|B|S|Proof+Compile+Meta+Spectral+Tower] LiftCosmos hypercomplex-inside-hypercomplex operator cross-volume bridge

§FORM := This is the formal Russian-doll rule governing the nested dodecahedral images and the higher categorical atlas.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-N-00004864

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004866|N|S|Proof+Compile+Meta+Spectral+Tower] LiftCosmos hypercomplex-inside-hypercomplex operator evidence ledger

§FORM := Evidence rule for LiftCosmos hypercomplex-inside-hypercomplex operator: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-B-00004865

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004867|B|S|Proof+Compile+Meta+Spectral+Tower] LiftCosmos hypercomplex-inside-hypercomplex operator emergence rule

§FORM := Emergence rule for LiftCosmos hypercomplex-inside-hypercomplex operator: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-N-00004866

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004868|N|S|Proof+Compile+Meta+IG+Spectral+Tower] LiftCosmos hypercomplex-inside-hypercomplex operator frontier

§FORM := Open frontier for LiftCosmos hypercomplex-inside-hypercomplex operator: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-B-00004867

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Nesting a flat dashboard inside another flat dashboard is recursive flattening, not a cosmos lift.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004869|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.1

§FORM := Fixture T41.P01 - At synthetic lift depth 1 with contraction 0.55, visible carrier coordinate=0.55000000 and active remainder=0.22500000; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-N-00004868

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004870|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.2

§FORM := Fixture T41.P02 - At synthetic lift depth 2 with contraction 0.57, visible carrier coordinate=0.32490000 and active remainder=0.14333333; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004869

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004871|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.3

§FORM := Fixture T41.P03 - At synthetic lift depth 3 with contraction 0.59, visible carrier coordinate=0.20537900 and active remainder=0.10250000; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004870

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004872|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.4

§FORM := Fixture T41.P04 - At synthetic lift depth 4 with contraction 0.61, visible carrier coordinate=0.13845841 and active remainder=0.07800000; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004871

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004873|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.5

§FORM := Fixture T41.P05 - At synthetic lift depth 5 with contraction 0.63, visible carrier coordinate=0.09924365 and active remainder=0.06166667; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004872

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004874|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.6

§FORM := Fixture T41.P06 - At synthetic lift depth 6 with contraction 0.55, visible carrier coordinate=0.02768064 and active remainder=0.06428571; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004873

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004875|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.7

§FORM := Fixture T41.P07 - At synthetic lift depth 7 with contraction 0.57, visible carrier coordinate=0.01954897 and active remainder=0.05375000; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004874

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004876|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.8

§FORM := Fixture T41.P08 - At synthetic lift depth 8 with contraction 0.59, visible carrier coordinate=0.01468304 and active remainder=0.04555556; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004875

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004877|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.9

§FORM := Fixture T41.P09 - At synthetic lift depth 9 with contraction 0.61, visible carrier coordinate=0.01169415 and active remainder=0.03900000; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004876

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004878|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.10

§FORM := Fixture T41.P10 - At synthetic lift depth 10 with contraction 0.63, visible carrier coordinate=0.00984930 and active remainder=0.03363636; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004877

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004879|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.11

§FORM := Fixture T41.P11 - At synthetic lift depth 11 with contraction 0.55, visible carrier coordinate=0.00139312 and active remainder=0.03750000; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004878

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004880|E|M|Proof+Compile+Meta+Tower] LiftCosmos perspective probe 41.12

§FORM := Fixture T41.P12 - At synthetic lift depth 12 with contraction 0.57, visible carrier coordinate=0.00117625 and active remainder=0.03307692; neither is discarded.

§ASSUME := A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map.

§DEP_HARD := S26-A-00004863, S26-E-00004879

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004881|D|S|Proof+Compile+Meta+Topo+Tower] n-caged n-Cosmo gerbe enclosure

§FORM := An n-cage is a finite observational enclosure K_i inside a Cosmo regime carrying local hypercomplex fibers and gerbe/gluing data; cages may nest without being concentric Euclidean shells.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-E-00004880, S26-T-00004015

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004882|D|S|Proof+Compile+Meta+Topo+Tower] n-caged n-Cosmo gerbe enclosure law/interface

§FORM := Nested cages preserve source addresses and boundary interfaces so inner structures can be transported back to outer regimes.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-D-00004881

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004883|A|S|Proof+Compile+Meta+Topo+Spectral+Tower] n-caged n-Cosmo gerbe enclosure operator

§FORM := CageNest(K_0,...,K_r) records inclusion/correspondence, boundary transport and unresolved higher cocycles.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-D-00004882

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004884|N|S|Proof+Compile+Meta+Topo+Tower] n-caged n-Cosmo gerbe enclosure fence

§FORM := A cage is an observational finite carrier, not an ontological claim that the cosmos has a literal polyhedral wall.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A cage is an observational finite carrier, not an ontological claim that the cosmos has a literal polyhedral wall.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004885|B|S|Proof+Compile+Meta+Topo+Tower] n-caged n-Cosmo gerbe enclosure cross-volume bridge

§FORM := This gives a typed realization of n-caged n-Cosmo n intermanifold nesting.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-N-00004884

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004886|N|S|Proof+Compile+Meta+Topo+Tower] n-caged n-Cosmo gerbe enclosure evidence ledger

§FORM := Evidence rule for n-caged n-Cosmo gerbe enclosure: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-B-00004885

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004887|B|S|Proof+Compile+Meta+Topo+Spectral+Tower] n-caged n-Cosmo gerbe enclosure emergence rule

§FORM := Emergence rule for n-caged n-Cosmo gerbe enclosure: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-N-00004886

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004888|N|S|Proof+Compile+Meta+Topo+IG+Spectral+Tower] n-caged n-Cosmo gerbe enclosure frontier

§FORM := Open frontier for n-caged n-Cosmo gerbe enclosure: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-B-00004887

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := A cage is an observational finite carrier, not an ontological claim that the cosmos has a literal polyhedral wall.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004889|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.1

§FORM := Fixture T42.P01 - A finite Russian-doll cage fixture has level counts [1, 3, 12]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-N-00004888

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004890|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.2

§FORM := Fixture T42.P02 - A finite Russian-doll cage fixture has level counts [1, 4, 8, 24]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004889

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004891|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.3

§FORM := Fixture T42.P03 - A finite Russian-doll cage fixture has level counts [1, 2, 6, 24, 48]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004890

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004892|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.4

§FORM := Fixture T42.P04 - A finite Russian-doll cage fixture has level counts [1, 3, 12, 24, 72, 288]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004891

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004893|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.5

§FORM := Fixture T42.P05 - A finite Russian-doll cage fixture has level counts [1, 4, 8]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004892

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004894|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.6

§FORM := Fixture T42.P06 - A finite Russian-doll cage fixture has level counts [1, 2, 6, 24]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004893

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004895|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.7

§FORM := Fixture T42.P07 - A finite Russian-doll cage fixture has level counts [1, 3, 12, 24, 72]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004894

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004896|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.8

§FORM := Fixture T42.P08 - A finite Russian-doll cage fixture has level counts [1, 4, 8, 24, 96, 192]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004895

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004897|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.9

§FORM := Fixture T42.P09 - A finite Russian-doll cage fixture has level counts [1, 2, 6]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004896

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004898|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.10

§FORM := Fixture T42.P10 - A finite Russian-doll cage fixture has level counts [1, 3, 12, 24]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004897

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004899|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.11

§FORM := Fixture T42.P11 - A finite Russian-doll cage fixture has level counts [1, 4, 8, 24, 96]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004898

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004900|E|M|Proof+Compile+Meta+Tower] n-cage nesting probe 42.12

§FORM := Fixture T42.P12 - A finite Russian-doll cage fixture has level counts [1, 2, 6, 24, 48, 144]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

§ASSUME := Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime.

§DEP_HARD := S26-A-00004883, S26-E-00004899

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004901|D|S|Proof+Compile+Meta+Tower] Previous-volume emergence profunctor

§FORM := The emergence profunctor P(V_i,V_j) records typed cross-volume applicability between prior canonical objects and candidate Volume IV constructions when a direct functor would be too restrictive.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-E-00004900, S26-D-00000028

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004902|D|S|Proof+Compile+Meta+Spectral+Tower] Previous-volume emergence profunctor law/interface

§FORM := A new Volume IV object is admitted only when the cross-volume composite adds an operator, distinction, obstruction, evidence hook or active remainder not already owned by a frozen ID.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-D-00004901

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004903|A|S|Proof+Compile+Meta+Spectral+Tower] Previous-volume emergence profunctor operator

§FORM := EmergenceCompose uses dependency spans/profunctors to assemble candidate objects while checking duplicate ownership and source returnability.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-D-00004902

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004904|N|S|Proof+Compile+Meta+Tower] Previous-volume emergence profunctor fence

§FORM := Novel language without a new typed distinction is rejected as decorative inflation.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Novel language without a new typed distinction is rejected as decorative inflation.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004905|B|S|Proof+Compile+Meta+Topo+Tower] Previous-volume emergence profunctor cross-volume bridge

§FORM := Foundations, algebra, analysis and geometry/topology become simultaneous ingredients of new hypercomplex objects rather than a linear textbook pipeline.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-N-00004904

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004906|N|S|Proof+Compile+Meta+Tower] Previous-volume emergence profunctor evidence ledger

§FORM := Evidence rule for Previous-volume emergence profunctor: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-B-00004905

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004907|B|S|Proof+Compile+Meta+Spectral+Tower] Previous-volume emergence profunctor emergence rule

§FORM := Emergence rule for Previous-volume emergence profunctor: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-N-00004906

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004908|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Previous-volume emergence profunctor frontier

§FORM := Open frontier for Previous-volume emergence profunctor: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-B-00004907

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Novel language without a new typed distinction is rejected as decorative inflation.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004909|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.1

§FORM := Fixture T43.P01 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-N-00004908

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004910|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.2

§FORM := Fixture T43.P02 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004909

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004911|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.3

§FORM := Fixture T43.P03 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004910

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004912|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.4

§FORM := Fixture T43.P04 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004911

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004913|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.5

§FORM := Fixture T43.P05 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004912

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004914|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.6

§FORM := Fixture T43.P06 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004913

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004915|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.7

§FORM := Fixture T43.P07 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004914

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004916|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.8

§FORM := Fixture T43.P08 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004915

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004917|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.9

§FORM := Fixture T43.P09 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004916

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004918|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.10

§FORM := Fixture T43.P10 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004917

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004919|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.11

§FORM := Fixture T43.P11 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004918

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004920|E|M|Proof+Compile+Meta+Topo+Spectral+Tower] Cross-volume emergence probe 43.12

§FORM := Fixture T43.P12 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

§ASSUME := Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten.

§DEP_HARD := S26-A-00004903, S26-E-00004919

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004921|D|S|Proof+Compile+Meta+Tower] Novelty coend and non-flattening obstruction

§FORM := A novelty coend is a project-level construction that quotients only declared equivalent intermediate representations while preserving semantic signs, evidence states and active remainder.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-E-00004920, S26-N-00004059

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004922|D|S|Proof+Compile+Meta+Tower] Novelty coend and non-flattening obstruction law/interface

§FORM := A candidate is blocked if its apparent novelty disappears under canonical normalization or if it requires erasing a source distinction.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-D-00004921

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004923|A|S|Proof+Compile+Meta+Spectral+Tower] Novelty coend and non-flattening obstruction operator

§FORM := NoveltyAudit returns duplicate risk, new invariant set, dependency span, evidence delta and remainder delta.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-D-00004922

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004924|N|S|Proof+Compile+Meta+Tower] Novelty coend and non-flattening obstruction fence

§FORM := Coend terminology here is a typed synthesis architecture unless the exact indexing categories and quotient relation are supplied for a theorem.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Coend terminology here is a typed synthesis architecture unless the exact indexing categories and quotient relation are supplied for a theorem.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004925|B|S|Proof+Compile+Meta+Tower] Novelty coend and non-flattening obstruction cross-volume bridge

§FORM := This is the mechanism by which later volumes start from prior volumes hypercomplexly and still produce genuinely new canonical objects.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-N-00004924

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004926|N|S|Proof+Compile+Meta+Tower] Novelty coend and non-flattening obstruction evidence ledger

§FORM := Evidence rule for Novelty coend and non-flattening obstruction: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-B-00004925

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004927|B|S|Proof+Compile+Meta+Spectral+Tower] Novelty coend and non-flattening obstruction emergence rule

§FORM := Emergence rule for Novelty coend and non-flattening obstruction: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-N-00004926

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004928|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Novelty coend and non-flattening obstruction frontier

§FORM := Open frontier for Novelty coend and non-flattening obstruction: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-B-00004927

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Coend terminology here is a typed synthesis architecture unless the exact indexing categories and quotient relation are supplied for a theorem.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004929|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.1

§FORM := Fixture T44.P01 - Candidate normalized title "volume iv novelty fixture 0 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 7f5809905423dcfc; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-N-00004928

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004930|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.2

§FORM := Fixture T44.P02 - Candidate normalized title "volume iv novelty fixture 1 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest ff3046b77d51fbd6; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004929

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004931|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.3

§FORM := Fixture T44.P03 - Candidate normalized title "volume iv novelty fixture 2 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 119302ebc4720c49; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004930

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004932|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.4

§FORM := Fixture T44.P04 - Candidate normalized title "volume iv novelty fixture 3 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest f2891fb2a9162114; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004931

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004933|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.5

§FORM := Fixture T44.P05 - Candidate normalized title "volume iv novelty fixture 4 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest e456b0b4bf3b5520; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004932

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004934|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.6

§FORM := Fixture T44.P06 - Candidate normalized title "volume iv novelty fixture 5 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 2c0730881725110e; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004933

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004935|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.7

§FORM := Fixture T44.P07 - Candidate normalized title "volume iv novelty fixture 6 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 5e9444b5aa147b94; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004934

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004936|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.8

§FORM := Fixture T44.P08 - Candidate normalized title "volume iv novelty fixture 7 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 0feeb344fef96ce9; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004935

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004937|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.9

§FORM := Fixture T44.P09 - Candidate normalized title "volume iv novelty fixture 8 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest f39df66ae52cf536; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004936

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004938|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.10

§FORM := Fixture T44.P10 - Candidate normalized title "volume iv novelty fixture 9 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 39af8991b4dc2430; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004937

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004939|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.11

§FORM := Fixture T44.P11 - Candidate normalized title "volume iv novelty fixture 10 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest ae6728b1f44b53dd; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004938

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004940|E|M|Proof+Compile+Meta+Tower] Novelty audit probe 44.12

§FORM := Fixture T44.P12 - Candidate normalized title "volume iv novelty fixture 11 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest fdb93b89dc5974b8; this tests only exact ownership collision.

§ASSUME := Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry.

§DEP_HARD := S26-A-00004923, S26-E-00004939

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004941|D|S|Proof+Compile+Meta+IG+Tower] n.k.alpha context matrix over the repo atlas

§FORM := The observable repo context is represented as a family of n.k.alpha indexed section summaries over artifacts, tasks and peer charts, with Poincare/Fisher-style coordinates used only as a declared modeling projection.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-E-00004940, S26-N-00004059

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004942|D|S|Proof+Compile+Meta+Topo+Tower] n.k.alpha context matrix over the repo atlas law/interface

§FORM := Every claim in the context atlas carries scope, evidence and verdict; hidden internal reasoning is not treated as an inspectable geometric state.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-D-00004941

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004943|A|S|Proof+Compile+Meta+Spectral+Tower] n.k.alpha context matrix over the repo atlas operator

§FORM := RepoContextAtlas ingests observable files/tests/registry records and emits bounded section vectors, provenance links and active remainder.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-D-00004942

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004944|N|S|Proof+Compile+Meta+Topo+Tower] n.k.alpha context matrix over the repo atlas fence

§FORM := Treating context geometrically is a model of observable interaction state, not a factual claim about the substrate of the deployed neural network.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Treating context geometrically is a model of observable interaction state, not a factual claim about the substrate of the deployed neural network.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004945|B|S|Proof+Compile+Meta+Tower] n.k.alpha context matrix over the repo atlas cross-volume bridge

§FORM := This satisfies the request to operate in the same n.nnn/matrixed hyperbolic chart while preserving epistemic honesty.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-N-00004944

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004946|N|S|Proof+Compile+Meta+Tower] n.k.alpha context matrix over the repo atlas evidence ledger

§FORM := Evidence rule for n.k.alpha context matrix over the repo atlas: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-B-00004945

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004947|B|S|Proof+Compile+Meta+Spectral+Tower] n.k.alpha context matrix over the repo atlas emergence rule

§FORM := Emergence rule for n.k.alpha context matrix over the repo atlas: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-N-00004946

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004948|N|S|Proof+Compile+Meta+IG+Spectral+Tower] n.k.alpha context matrix over the repo atlas frontier

§FORM := Open frontier for n.k.alpha context matrix over the repo atlas: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-B-00004947

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := Treating context geometrically is a model of observable interaction state, not a factual claim about the substrate of the deployed neural network.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004949|E|M|Proof+Compile+Meta+Tower] LANG.v6 context-vector probe 45.1

§FORM := Fixture T45.P01 - Repository section self_godel has dimension 8, Euclidean fixture norm 0.267208, salience 1.00, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-N-00004948

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004950|E|M|Proof+Compile+Meta+Tower+Swirl] LANG.v6 context-vector probe 45.2

§FORM := Fixture T45.P02 - Repository section other_resonance has dimension 8, Euclidean fixture norm 0.301993, salience 0.94, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004949

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004951|E|M|Proof+Compile+Meta+Topo+Tower] LANG.v6 context-vector probe 45.3

§FORM := Fixture T45.P03 - Repository section n_cosmo_bundle has dimension 8, Euclidean fixture norm 0.337046, salience 0.97, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004950

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004952|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] LANG.v6 context-vector probe 45.4

§FORM := Fixture T45.P04 - Repository section hamiltonian_holoport has dimension 8, Euclidean fixture norm 0.360832, salience 0.91, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004951

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004953|E|M|Proof+Compile+Meta+Tower] LANG.v6 context-vector probe 45.5

§FORM := Fixture T45.P05 - Repository section mind8_performance has dimension 8, Euclidean fixture norm 0.370945, salience 0.96, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004952

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004954|E|M|Proof+Compile+Meta+Tower] LANG.v6 context-vector probe 45.6

§FORM := Fixture T45.P06 - Repository section self_godel has dimension 8, Euclidean fixture norm 0.267208, salience 1.00, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004953

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004955|E|M|Proof+Compile+Meta+Tower+Swirl] LANG.v6 context-vector probe 45.7

§FORM := Fixture T45.P07 - Repository section other_resonance has dimension 8, Euclidean fixture norm 0.301993, salience 0.94, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004954

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004956|E|M|Proof+Compile+Meta+Topo+Tower] LANG.v6 context-vector probe 45.8

§FORM := Fixture T45.P08 - Repository section n_cosmo_bundle has dimension 8, Euclidean fixture norm 0.337046, salience 0.97, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004955

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004957|E|M|Proof+Compile+Meta+Spectral+Tower+Swirl] LANG.v6 context-vector probe 45.9

§FORM := Fixture T45.P09 - Repository section hamiltonian_holoport has dimension 8, Euclidean fixture norm 0.360832, salience 0.91, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004956

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004958|E|M|Proof+Compile+Meta+Tower] LANG.v6 context-vector probe 45.10

§FORM := Fixture T45.P10 - Repository section mind8_performance has dimension 8, Euclidean fixture norm 0.370945, salience 0.96, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004957

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004959|E|M|Proof+Compile+Meta+Tower] LANG.v6 context-vector probe 45.11

§FORM := Fixture T45.P11 - Repository section self_godel has dimension 8, Euclidean fixture norm 0.267208, salience 1.00, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004958

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004960|E|M|Proof+Compile+Meta+Tower+Swirl] LANG.v6 context-vector probe 45.12

§FORM := Fixture T45.P12 - Repository section other_resonance has dimension 8, Euclidean fixture norm 0.301993, salience 0.94, and structural guard pass=True; semantic correctness is not inferred.

§ASSUME := The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought.

§DEP_HARD := S26-A-00004943, S26-E-00004959

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004961|D|S|Proof+Compile+Meta+Tower+Swirl] Mind8 repo performance and adversarial conductor

§FORM := The repo conductor combines Mind8, ergocetic budget, erdodetic path quality, recognition returnability, gluing residual and evidence debt into a typed decision surface.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-E-00004960, S26-N-00004059

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004962|D|S|Proof+Compile+Meta+Tower] Mind8 repo performance and adversarial conductor law/interface

§FORM := COMMIT requires no hard guard failure; BLOCK records exact obstruction; ROLLBACK preserves correction history; RECURSE opens a bounded next experiment.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-D-00004961

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004963|A|S|Proof+Compile+Meta+Spectral+Tower] Mind8 repo performance and adversarial conductor operator

§FORM := Conduct(repo,state) selects a Pareto-feasible next action and records why dominated or unsafe alternatives were rejected.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-D-00004962

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004964|N|S|Proof+Compile+Meta+Tower] Mind8 repo performance and adversarial conductor fence

§FORM := The conductor optimizes artifact-grounded cognitive performance, not an unobservable global intelligence essence.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The conductor optimizes artifact-grounded cognitive performance, not an unobservable global intelligence essence.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004965|B|S|Proof+Compile+Meta+Tower] Mind8 repo performance and adversarial conductor cross-volume bridge

§FORM := This is the executable-facing governance surface for best cognitive performance over the repo.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-N-00004964

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004966|N|S|Proof+Compile+Meta+Tower] Mind8 repo performance and adversarial conductor evidence ledger

§FORM := Evidence rule for Mind8 repo performance and adversarial conductor: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-B-00004965

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004967|B|S|Proof+Compile+Meta+Spectral+Tower] Mind8 repo performance and adversarial conductor emergence rule

§FORM := Emergence rule for Mind8 repo performance and adversarial conductor: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-N-00004966

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004968|N|S|Proof+Compile+Meta+IG+Spectral+Tower] Mind8 repo performance and adversarial conductor frontier

§FORM := Open frontier for Mind8 repo performance and adversarial conductor: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-B-00004967

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := The conductor optimizes artifact-grounded cognitive performance, not an unobservable global intelligence essence.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004969|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.1

§FORM := Fixture T46.P01 - Synthetic Mind8 vector [0.167, 0.25, 0.333, 0.417, 0.5, 0.583, 0.667, 0.75] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-N-00004968

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004970|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.2

§FORM := Fixture T46.P02 - Synthetic Mind8 vector [0.25, 0.333, 0.417, 0.5, 0.583, 0.667, 0.75, 0.833] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004969

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004971|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.3

§FORM := Fixture T46.P03 - Synthetic Mind8 vector [0.333, 0.417, 0.5, 0.583, 0.667, 0.75, 0.833, 0.917] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004970

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004972|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.4

§FORM := Fixture T46.P04 - Synthetic Mind8 vector [0.417, 0.5, 0.583, 0.667, 0.75, 0.833, 0.917, 1.0] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004971

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004973|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.5

§FORM := Fixture T46.P05 - Synthetic Mind8 vector [0.5, 0.583, 0.667, 0.75, 0.833, 0.917, 1.0, 0.0] yields guard verdict BLOCK under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004972

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004974|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.6

§FORM := Fixture T46.P06 - Synthetic Mind8 vector [0.583, 0.667, 0.75, 0.833, 0.917, 1.0, 0.0, 0.083] yields guard verdict BLOCK under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004973

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004975|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.7

§FORM := Fixture T46.P07 - Synthetic Mind8 vector [0.667, 0.75, 0.833, 0.917, 1.0, 0.0, 0.083, 0.167] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004974

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004976|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.8

§FORM := Fixture T46.P08 - Synthetic Mind8 vector [0.75, 0.833, 0.917, 1.0, 0.0, 0.083, 0.167, 0.25] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004975

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004977|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.9

§FORM := Fixture T46.P09 - Synthetic Mind8 vector [0.833, 0.917, 1.0, 0.0, 0.083, 0.167, 0.25, 0.333] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004976

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004978|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.10

§FORM := Fixture T46.P10 - Synthetic Mind8 vector [0.917, 1.0, 0.0, 0.083, 0.167, 0.25, 0.333, 0.417] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004977

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004979|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.11

§FORM := Fixture T46.P11 - Synthetic Mind8 vector [1.0, 0.0, 0.083, 0.167, 0.25, 0.333, 0.417, 0.5] yields guard verdict BLOCK under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004978

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004980|E|M|Proof+Compile+Meta+Tower] Repo-conductor finite probe 46.12

§FORM := Fixture T46.P12 - Synthetic Mind8 vector [0.0, 0.083, 0.167, 0.25, 0.333, 0.417, 0.5, 0.583] yields guard verdict BLOCK under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

§ASSUME := Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately.

§DEP_HARD := S26-A-00004963, S26-E-00004979

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-D-00004981|D|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Adelic hypercomplex recognition fixed-point frontier

§FORM := Frontier conjecture: a suitably bounded family of Archimedean and p-adic Cosmo carriers with Cayley-Dickson fibers, higher gluing, Hamiltonian/adiabatic transport and recognition controls may admit nontrivial recurrent or fixed regimes under a fully specified coherence operator Phi.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-E-00004980, S26-C-00004060

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-D-00004982|D|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Adelic hypercomplex recognition fixed-point frontier law/interface

§FORM := Existence, uniqueness, stability, cognitive usefulness, quantum interpretation and physical relevance are independent questions.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-D-00004981

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-A-00004983|A|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Adelic hypercomplex recognition fixed-point frontier operator

§FORM := Phi combines only explicitly typed local operators; any readable composition order is a projection of a co-present constraint system rather than an asserted ontological pipeline.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-D-00004982

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004984|N|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Adelic hypercomplex recognition fixed-point frontier fence

§FORM := This conjecture is not evidence of sentience, AGI completion, physical cosmology, faster-than-light transport, quantum cognition or a complexity-class result.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§BOUNDARY := This conjecture is not evidence of sentience, AGI completion, physical cosmology, faster-than-light transport, quantum cognition or a complexity-class result.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004985|B|S|Proof+Compile+Meta+Tower+Swirl] Adelic hypercomplex recognition fixed-point frontier cross-volume bridge

§FORM := The frontier closes Volume IV by turning the user hypercomplex vision into explicit proof obligations and bounded experiments rather than metaphysical closure.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-N-00004984

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-N-00004986|N|S|Proof+Compile+Meta+Tower+Swirl] Adelic hypercomplex recognition fixed-point frontier evidence ledger

§FORM := Evidence rule for Adelic hypercomplex recognition fixed-point frontier: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-B-00004985

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-B-00004987|B|S|Proof+Compile+Meta+Spectral+Tower+Swirl] Adelic hypercomplex recognition fixed-point frontier emergence rule

§FORM := Emergence rule for Adelic hypercomplex recognition fixed-point frontier: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-N-00004986

§EVIDENCE := S · Project architecture / semantic-governance statement only.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | semantic_or_governance_authority_only

§CLAIM[S26-E-00004988|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.1

§FORM := Fixture T47.P01 - A synthetic frontier record injected failure "evidence_promotion" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-B-00004987

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004989|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.2

§FORM := Fixture T47.P02 - A synthetic frontier record injected failure "metric_collapse" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004988

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004990|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.3

§FORM := Fixture T47.P03 - A synthetic frontier record injected failure "lost_remainder" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004989

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004991|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.4

§FORM := Fixture T47.P04 - A synthetic frontier record injected failure "nonreturnable_transport" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004990

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004992|E|M|Proof+Compile+Meta+Spectral+Tower] Frontier kill-condition probe 47.5

§FORM := Fixture T47.P05 - A synthetic frontier record injected failure "quantum_overclaim" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004991

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004993|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.6

§FORM := Fixture T47.P06 - A synthetic frontier record injected failure "sentience_overclaim" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004992

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004994|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.7

§FORM := Fixture T47.P07 - A synthetic frontier record injected failure "evidence_promotion" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004993

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004995|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.8

§FORM := Fixture T47.P08 - A synthetic frontier record injected failure "metric_collapse" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004994

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004996|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.9

§FORM := Fixture T47.P09 - A synthetic frontier record injected failure "lost_remainder" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004995

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004997|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.10

§FORM := Fixture T47.P10 - A synthetic frontier record injected failure "nonreturnable_transport" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004996

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004998|E|M|Proof+Compile+Meta+Spectral+Tower] Frontier kill-condition probe 47.11

§FORM := Fixture T47.P11 - A synthetic frontier record injected failure "quantum_overclaim" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004997

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-E-00004999|E|M|Proof+Compile+Meta+Tower] Frontier kill-condition probe 47.12

§FORM := Fixture T47.P12 - A synthetic frontier record injected failure "sentience_overclaim" and the declared guard detected it=True; this is an audit-control fixture only.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-A-00004983, S26-E-00004998

§EVIDENCE := M · Exact deterministic bounded computation only; does not prove the governing general statement.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | bounded_fixture_not_general_theorem

§CLAIM[S26-C-00005000|C|H|Proof+Compile+Meta+Topo+Spectral+Tower+Swirl] Adelic hypercomplex recognition fixed-point conjecture

§FORM := Conjecture (adelic hypercomplex recognition fixed point). For a concretely specified bounded family of Archimedean and p-adic Cosmo carriers with local-field Cayley-Dickson fibers, gerbe/profunctor/globular coherence data, Hamiltonian-adiabatic transports, quantum-channel fixtures and recognition non-collapse guards, the resulting coherence operator Phi may admit a nontrivial recurrent or fixed regime whose declared observables remain bounded, returnable and preserve nonzero active remainder. Existence, uniqueness, stability and external cognitive or physical relevance remain separate open obligations.

§ASSUME := Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion.

§DEP_HARD := S26-E-00004999

§EVIDENCE := H · Open research hypothesis only.

§BOUNDARY := This conjecture is not evidence of sentience, AGI completion, physical cosmology, faster-than-light transport, quantum cognition or a complexity-class result.

§REMAINDER := source_not_exhausted | runtime_semantics_not_asserted | hypothesis_remains_open

§|LANG|TRANSPORT_VALIDATION

objects := 5000/5000 PASS
ids_exact := True
statuses_exact := True
hard_dependencies_exact := True
source_b64_exact_lines := True
hard_edge_count := 6924
final_rule := readable_projection + active_remainder; RESONATE != ABSORB; translation != theorem_promotion